ETSI TS 104 152-1 V1.1.1 (2026-02)



TECHNICAL SPECIFICATION

Mission Critical (MC) services; Mission Critical Video (MCVideo) Application Server (AS) Protocol conformance specification for server-to-server interface; Part 1: Test structure, configurations, conformance requirement and test purposes

Reference
DTS/TCCE-00263

Keywords
LTE

ETSI

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Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee TETRA and Critical Communications Evolution (TCCE).

The present document is based on 3GPP TS 36.579-8 [28] produced by the 3rd Generation Partnership Project (3GPP).

The present document is part 1 of a multi-part deliverable covering the Mission Critical (MC) services; Mission Critical Video (MCVideo) Application Server (AS) Protocol conformance specification for server-to-server interface, as identified below:

Part 1: "Test structure, configurations, conformance requirement and test purposes";

Part 2: "Test Applicability and Implementation Conformance Statement (ICS) pro forma specification".

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

Introduction

The ETSI Technical Specification used as the reference document for the present document is part 8 of a multi-part deliverable covering conformance test specification for Mission Critical Services over LTE consisting of:

- ETSI TS 136 579-1 [1]: "LTE; Mission Critical (MC) services over LTE; Part 1: Common test environment (3GPP TS 36.579-1)";
- ETSI TS 136 579-2 [i.2]: "LTE; Mission Critical (MC) services over LTE; Part 2: Mission Critical Push To Talk (MCPTT) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-2)";
- ETSI TS 136 579-3 [i.3]: "LTE; Mission Critical (MC) services over LTE; Part 3: Mission Critical Push To Talk (MCPTT) Server Application conformance specification (3GPP TS 36.579-3)";
- ETSI TS 136 579-4 [2]: "LTE; Mission Critical (MC) services over LTE; Part 4: Test Applicability and Implementation Conformance Statement (ICS) pro forma specification (3GPP TS 36.579-4)";
- ETSI TS 136 579-5 [i.4]: "LTE; Mission Critical (MC) services over LTE; Part 5: Abstract test suite (ATS) (3GPP TS 36.579-5)";
- ETSI TS 136 579-6 [i.5]: "LTE; Mission Critical (MC) services over LTE; Part 6: Mission Critical Video (MCVideo) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-6)";
- ETSI TS 136 579-7 [i.6]: "LTE; Mission Critical (MC) services over LTE; Part 7: Mission Critical Data (MCData) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-7)";
- **3GPP TS 36.579-8 [i.8]: "Mission Critical (MC) services over LTE; Part 8: Mission Critical Video (MCVideo) Server Application conformance specification";**
- 3GPP TS 36.579-9 [i.7]: "Mission Critical (MC) services over LTE; Part 9: Mission Critical Data (MCData) Server Application conformance specification".

1 Scope

The present document specifies the protocol conformance testing for testing a MCVideo Server for compliance to the Mission Critical Video (MCVideo) over LTE protocol requirements defined by 3GPP. The present document addresses only MCVideo Server-Server communication scenarios. It does not cover e.g. MCVideo Server-EPS, MCVideo Server-SIP Core, etc. scenarios which involve interfaces which implementation may widely vary.

In particular the present specification contains:

- the overall test structure;
- the test configurations;
- the conformance requirement and reference to the core specifications;
- the test purposes; and
- a brief description of the test procedure, the specific test requirements and short message exchange table.

The present document is valid for MCVideo Servers implemented according to 3GPP releases starting from Release 14 up to the Release indicated on the cover page of the present document.

The following information relevant to testing specified in the present document could be found in accompanying specifications:

- default setting of the test parameters ETSI TS 136 579-1 [1] (3GPP TS 36.579-1);
- Implementation Conformance Statement (ICS) ETSI TS 136 579-4 [2] (3GPP TS 36.579-4) and Implementation extra Information for Testing (IXIT) ETSI TS 136 579-5 [i.4] (3GPP TS 36.579-5);
- the applicability of each test case ETSI TS 136 579-4 [2] (3GPP TS 36.579-4).

The present document does not specify the protocol conformance testing for the EPS (LTE) bearers which carry the MCVideo data sent or received by the MCVideo Server. The specification of such testing is out of the scope of RAN5.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found in the ETSI docbox.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 136 579-1: "LTE; Mission Critical (MC) services over LTE; Part 1: Common test environment (3GPP TS 36.579-1)".

[2] ETSI TS 136 579-4: "LTE; Mission Critical (MC) services over LTE; Part 4: Test Applicability and Implementation Conformance Statement (ICS) proforma specification (3GPP TS 36.579-4)".

[3] ETSI TS 124 379: "LTE; Mission Critical Push To Talk (MCPTT) call control; Protocol specification (3GPP TS 24.379)".

[4] ETSI TS 133 180: "LTE; Security of the Mission Critical (MC) service (3GPP TS 33.180)".

[5] ETSI TS 124 281: "LTE; Mission Critical Video (MCVideo) signalling control; Protocol specification (3GPP TS 24.281)".

[6] ETSI TS 124 581: "LTE; Mission Critical Video (MCVideo) media plane control; Protocol specification (3GPP TS 24.581)".

[7] IETF RFC 3261 (June 2002): "SIP: Session Initiation Protocol".

[8] IETF RFC 4028 (April 2005): "Session Timers in the Session Initiation Protocol (SIP)".

[9] ETSI TS 126 114: "Universal Mobile Telecommunications System (UMTS); LTE; 5G; IP Multimedia Subsystem (IMS); Multimedia Telephony; Media handling and interaction (3GPP TS 26.114)".

[10] IETF RFC 4867 (April 2007): "RTP Payload Format and File Storage Format for the Adaptive Multi-Rate (AMR) and Adaptive Multi-Rate Wideband (AMR-WB) Audio Codecs".

[11] IETF RFC 6335 (August 2011): "Internet Assigned Numbers Authority (IANA) Procedures for the Management of the Service Name and Transport Protocol Port Number Registry".

[12] IETF RFC 3556 (July 2003): "Session Description Protocol (SDP) Bandwidth Modifiers for RTP Control Protocol (RTCP) Bandwidth".

[13] IETF RFC 4412 (February 2006): "Communications Resource Priority for the Session Initiation Protocol (SIP)".

[14] IETF RFC 8101 (March 2017): "IANA Registration of New Session Initiation Protocol (SIP) Resource-Priority Namespace for Mission Critical Push To Talk service".

[15] IETF RFC 7134 (March 2014): "The Management Policy of the Resource Priority Header (RPH) Registry Changed to "IETF Review"".

[16] IETF RFC 5621 (September 2009): "Message Body Handling in the Session Initiation Protocol (SIP)".

[17] IETF RFC 3903 (October 2004): "Session Initiation Protocol (SIP) Extension for Event State Publication".

[18] IETF RFC 6184 (May 2011): "RTP Payload Format for H.264 Video".

[19] ETSI TS 124 282: "LTE; Mission Critical Data (MCData) signalling control; Protocol specification (3GPP TS 24.282)".

[20] ETSI TS 136 509: "LTE; Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); Special conformance testing functions for User Equipment (UE) (3GPP TS 36.509)".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ETSI TR 121 905: "Digital cellular telecommunications system (Phase 2+) (GSM); Universal Mobile Telecommunications System (UMTS); LTE; 5G; Vocabulary for 3GPP Specifications (3GPP TR 21.905)".

[i.2] ETSI TS 136 579-2: "LTE; Mission Critical (MC) services over LTE; Part 2: Mission Critical Push To Talk (MCPTT) User Equipment (UE) Protocol conformance specification (3GPP TS 6.579-2)".

[i.3] ETSI TS 136 579-3: "LTE; Mission Critical (MC) services over LTE; Part 3: Mission Critical Push To Talk (MCPTT) Server Application conformance specification (3GPP TS 36.579-3)".

[i.4] ETSI TS 136 579-5: "LTE; Mission Critical (MC) services over LTE; Part 5: Abstract test suite (ATS) (3GPP TS 36.579-5)".

[i.5] ETSI TS 136 579-6: "LTE; Mission Critical (MC) services over LTE; Part 6: Mission Critical Video (MCVideo) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-6)".

[i.6] ETSI TS 136 579-7: "LTE; Mission Critical (MC) services over LTE; Part 7: Mission Critical Data (MCData) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-7)".

[i.7] 3GPP TS 36.579-9: "Mission Critical (MC) services over LTE; Part 9: Mission Critical Data (MCData) Server Application conformance specification".

[i.8] 3GPP TS 36.579-8: "Mission Critical (MC) services over LTE; Part 8: Mission Critical Video (MCVideo) Server Application conformance specification".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI TR 121 905 [i.1] (3GPP TR 21.905) and the following apply:

NOTE: A term defined in the present document takes precedence over the definition of the same term, if any, in ETSI TR 121 905 [i.1] (3GPP TR 21.905).

For the purpose of the present document, the following terms given in ETSI TS 124 281 [5] (3GPP TS 24.281) apply:

- An MCVideo user is affiliated to an MCVideo group
- An MCVideo user is affiliated to an MCVideo group at an MCVideo client
- Affiliation status
- Group identity
- MCVideo client ID
- MCVideo emergency alert state
- MCVideo emergency group state
- MCVideo emergency group call state
- MCVideo emergency private call
- MCVideo emergency private call state
- MCVideo emergency private priority state
- MCVideo imminent peril group call state
- MCVideo imminent peril group state
- MCVideo private call

- MCVideo private emergency alert state
- MCVideo video media
- Media-transmission control entity
- Private call
- Remote change of an MCVideo user's selected group
- Temporary MCVideo group identity
- Trusted mutual aid
- Untrusted mutual aid

For the purpose of the present document, the following terms given in ETSI TS 133 180 [4] (3GPP TS 33.180) apply:

- Client Server Key (CSK)
- Private Call Key (PCK)
- Signalling Protection Key (SPK)
- XML Protection Key (XPK)

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI TR 121 905 [i.1] (3GPP TR 21.905) and the following apply:

NOTE: An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in ETSI TR 121 905 [i.1] (3GPP TR 21.905).

CSK	Client-Server Key
GKTP	Group Key Transport Payload
GMOP	Group Management OPeration
ICS	Implementation Conformance Statement
IUT	Implementation Under Test
IXIT	Implementation eXtra Information for Testing
MBMS	Multimedia Broadcast and Multicast Service
MC	Mission Critical
MCData	Mission Critical Data
MCPTT	Mission Critical Push To Talk
MCS	Mission Critical Service
MCVideo group ID	MCVideo group Identity
MCVideo	Mission Critical Video
MIME	Multipurpose Internet Mail Extensions
MVEA	MCVideo Emergency Alert
MVEG	MCVideo Emergency Group
MVEGC	MCVideo Emergency Group Call
MVIG	MCVideo Imminent peril Group
MVIGC	MCVideo Imminent peril Group Call
NAT	Network Address Translation
PCK	Private Call Key
PIDF	Presence Information Data Format
RTP	Real-time Transport Protocol
SDP	Session Description Protocol

SIP	Session Initiation Protocol
SPK	Signalling Protection Key
SS	System Simulator
SSRC	Synchronization SouRCe
TGI	Temporary MCVideo Group Identity
TP	Test Purpose
TP	Transmission Point
UE	User Equipment
URI	Uniform Resource Identifier
XCAP	eXtensible Markup Language (XML) Access Configuration Protocol
XPK	XML Protection Key

4 General

4.1 Test methodology

4.1.1 Testing of optional functions and procedures

Any function or procedure which is optional, may be subject to a conformance test if it is implemented in the MCVideo Server.

A declaration by the MCVideo Server supplier (to use the Implementation Conformance Statement (ICS) proforma specified in ETSI TS 136 579-4 [2] (3GPP TS 36.579-4)) is used to determine whether an optional function/procedure has been implemented.

4.1.2 Test interfaces and facilities

Detailed descriptions of the MCVideo test interfaces and special facilities for testing are provided in ETSI TS 136 509 [20] (3GPP TS 36.509).

4.2 Implicit testing

For some 3GPP MCVideo protocol features conformance is not verified explicitly in the present document. This does not imply that correct functioning of these features is not essential, but that these are implicitly tested to a sufficient degree in tests which are not explicitly dedicated to test the feature.

4.3 Repetition of tests

As a general rule, the test cases specified in the present document are highly reproducible and do not need to be repeated unless otherwise stated.

4.4 Handling of differences between conformance requirements in different releases of core specifications

The conformance requirements which determine the scope of each test case are explicitly copy-pasted from relevant core specifications in the especially dedicated for this clause of each test with the title 'Conformance requirements'.

NOTE: When in the copy/pasted text there are references to other specifications the reference numbers will not match the reference numbers used in the present document. This approach has been taken in order to allow easy copy and then search for conformance requirements in those specifications.

When differences between conformance requirements in different releases of the cores specifications have impact on the Pre-test conditions, Test procedure sequence or/and the Specific message contents, the Conformance requirements related to different releases are specified separately with clear indication of the Release of the spec from which they were copied.

When there is no Release indicated for a conformance requirement text, this should be understood either as the Conformance requirements in the latest version of the spec with release = the TC Applicability release (which can be found in ETSI TS 136 579-4 [2] (3GPP TS 36.579-4), Table 4-1: Applicability of tests and additional information for testing, column 'Release'), or, as the Conformance requirements in the latest version of the spec of the release when the feature was introduced to the core specs.

4.5 Reference conditions

The reference environments used by all signalling and protocol tests is specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1). Where a test requires an environment that is different, this will be specified in the test itself.

For all test cases in the present document unless specified otherwise the condition MCVIDEO applies for all message contents.

4.6 Generic setup procedures

A set of basic generic procedures for MCVideo Client-Server communication are described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1).

These procedures will be used in numerous test cases throughout the present document.

5 MCVideo Server - MCVideo Server operation

5.1 Group Calls

5.1.1 Pre-arranged Group Call

5.1.1.1 MCVideo Server - MCVideo Server / Pre-arranged Group Call / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group call / Cancel Imminent Peril State / Controlling server

5.1.1.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function initiates a pre-
arranged group call with implicit transmission control }
    then { IUT (MCVideo Server) accepts the call from initiator and establishes the call with all
affiliated users by sending call establishment requests to SS (MCVideo Server) providing the
terminating participating function and sends transmission granted to SS-UE1 (MCVideo Client) and
sends media transmission notification to SS-UE2 (MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests to end the transmission control }
    then { IUT (MCVideo Server) sends transmission control idle to SS-UE1 (MCVideo Client) and sends
transmission control idle to SS-UE2 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control }
    then { IUT (MCVideo Server) sends transmission granted to SS-UE1 (MCVideo Client) and sends
media transmission notification to SS-UE2 (MCVideo Client) }
            }
```

(4)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function upgrades the call
to emergency group call with implicit transmission control }
    then { IUT (MCVideo Server) accepts the upgrade from the initiator and updates the call for all
participants and sends transmission granted to SS-UE1 (MCVideo Client) and sends media transmission
notification to SS-UE2 (MCVideo Client) }
            }
```

(5)

```
with { IUT (MCVideo Server) having upgraded to an emergency pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function requests to cancel
the ongoing emergency state }
    then { IUT (MCVideo Server) accepts the emergency state cancellation and updates the call for
all participants }
            }
```

(6)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function upgrades the call
to imminent peril group call with implicit transmission control }
    then { IUT (MCVideo Server) accepts the upgrade from the initiator and updates the call for all
participants and sends transmission granted to SS-UE1 (MCVideo Client) and sends media transmission
notification to SS-UE2 (MCVideo Client) }
            }
```

(7)

```
with { IUT (MCVideo Server) having upgraded to an imminent peril pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function requests to cancel
the ongoing imminent peril state }
    then { IUT (MCVideo Server) accepts the imminent peril state cancellation and updates the call
for all participants }
            }
```

(8)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function ends the pre-
arranged group call }
    then { IUT (MCVideo Server) responds by sending a SIP 200 (OK) message to SS (MCVideo Server)
ending the call and sends a SIP BYE message to SS (MCVideo Server) providing the terminating
participating function }
            }
```

5.1.1.1.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.1.4.2, 9.2.1.4.3, 9.2.1.4.7, 9.2.1.4.8, 9.2.1.4.1.1, 9.2.1.4.1.2, 6.3.3.1.6, 6.3.3.1.11, 6.3.3.1.12.2, 6.3.3.1.12.3, 6.3.3.1.12.4, 6.3.3.1.12.5, 6.3.3.2.1, 6.3.3.2.2, 6.3.3.2.4, 6.3.3.4 and 6.3.5.5, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.3.1, 6.3.2.2, 9.2.1, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.2]

"In the procedures in this clause:

1) *MCVideo ID in an incoming SIP INVITE request refers to the MCVideo ID of the originating user from the <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;*

2) *group identity in an incoming SIP INVITE request refers to the group identity from the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;*

3) *MCVideo ID in an outgoing SIP INVITE request refers to the MCVideo ID of the called user in the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the outgoing SIP INVITE request;*

4) *indication of required group members in a SIP 183 (Session Progress) response refers to the <required> element of the application/vnd.3gpp.mcvideo-info+xml MIME body set to "true" in a SIP 183 (Session Progress) sent by the non-controlling MCVideo function of an MCVideo group;*

5) *emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and*

6) *imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.*

Upon receipt of a "SIP INVITE request for controlling MCVideo function of an MCVideo group", the controlling MCVideo function:

…

2) *shall determine if the media parameters are acceptable and the MCVideo codecs are offered in the SDP offer and if not reject the request with a SIP 488 (Not Acceptable Here) response and skip the rest of the steps;*

3) *shall reject the SIP request with a SIP 403 (Forbidden) response and not process the remaining steps if:*

   a) *an Accept-Contact header field does not include the g.3gpp.mcvideo media feature tag; or*

   b) *an Accept-Contact header field does not include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo";*

4) *if received SIP INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.3.1.17;*

5) *shall retrieve the necessary group document(s) from the group management server for the group identity contained in the SIP INVITE request and carry out initial processing as specified in clause 6.3.5.2;*

6) *if the result of the initial processing in clause 6.3.5.2 was:*

   a) *that authorization of the MCVideo ID is required at a non-controlling MCVideo function of an MCVideo group is required, perform the actions in clause 6.3.3.1.13.7 and do not continue with the rest of the steps in this clause; and*

   b) *that a SIP 3xx, 4xx, 5xx or 6xx response to the "SIP INVITE request for controlling MCVideo function of an MCVideo group" has been sent, do not continue with the rest of the steps in this clause;*

7) *shall perform the actions as described in clause 6.3.3.2.2;*

8) *shall maintain a local counter of the number of SIP 200 (OK) responses received from invited members and shall initialise this local counter to zero;*

9) *shall determine if an MCVideo group call for the group identity is already ongoing by determining if an MCVideo session identity has already been allocated for the group call and the MCVideo session is active;*

…

12) if a Resource-Priority header field is included in the SIP INVITE request:

a) if the Resource-Priority header field is set to the value indicated for emergency calls and the SIP INVITE request does not contain an emergency indication and the in-progress emergency state of the group is set to a value of "false", shall reject the SIP INVITE request with a SIP 403 (Forbidden) response and skip the rest of the steps; or

b) if the Resource-Priority header field is set to the value indicated for imminent peril calls and the SIP INVITE request does not contain an imminent peril indication and the in-progress imminent peril state of the group is set to a value of "false", shall reject the SIP INVITE request with a SIP 403 (Forbidden) response and skip the rest of the steps;

13) if the MCVideo group call is not ongoing then:

…

c) if the received SIP INVITE request contains an an authorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2 or MCVideo imminent peril group call as determined by clause 6.3.3.1.13.5 and the MCVideo user is eligible to be implicitly affiliated with the MCVideo group as determined as determined in step 13) a) iv) above, shall perform the implicit affiliation as specified in clause 8.2.2.3.7;

d) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [11];

e) shall create a prearranged group session and allocate an MCVideo session identity for the prearranged group call, and shall handle timer TNG3 (group call timer) as specified in clause 6.3.3.5;

f) if the group identity in the "SIP INVITE request for controlling MCVideo function of an MCVideo group" is a TGI:

i) shall for each of the constituent MCVideo groups homed on the primary MCVideo system:

…;

B) if the controlling MCVideo function owns the MCVideo group identified by the MCVideo group ID then:

I) determine the members to invite to the prearranged MCVideo group call as specified in clause 6.3.5.5;

II) invite each group member determined in step A) above, to the group session, as specified in clause 9.2.1.4.1.1; and

III) interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3; and

ii) shall for each of the constituent MCVideo groups homed on the partner MCVideo system generate a SIP INVITE request for the MCVideo group identity homed on the partner MCVideo system as specified in clause 9.2.1.4.1.2; and

g) if the group identity in the SIP INVITE request for controlling MCVideo function of an MCVideo group is an MCVideo group ID:

i) shall determine the members to invite to the prearranged MCVideo group call as specified in clause 6.3.5.5;

ii) if necessary, shall start timer TNG1 (acknowledged call setup timer) according to the conditions stated in clause 6.3.3.3;

iii) if the received SIP INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element set to a value of "true":

A) shall cache the information that this MCVideo user has initiated an MCVideo emergency call;

B) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the information that this MCVideo user has initiated an MCVideo emergency alert; and

C) if the in-progress emergency state of the group is set to a value of "false":

I) shall set the value of the in-progress emergency state of the group to "true"; and

II) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16;

iv) if the in-progress emergency state of the group is set to a value of "false" and if the received SIP INVITE request contains an imminent peril indication set to a value of "true", the controlling MCVideo function shall:

A) shall cache the information that this MCVideo user has initiated an MCVideo imminent peril call; and

B) if the in-progress imminent peril state of the group is set to a value of "false", shall set the in-progress imminent peril state of the group to a value of "true";

v) shall invite each group member determined in step 13)g)i) above, to the group session, as specified in clause 9.2.1.4.1.1; and

vi) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3; and

14) if the MCVideo group call is ongoing then:

…

c) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [11];

…

e) if the received SIP INVITE request contains an an authorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2 or MCVideo imminent peril group call as determined by clause 6.3.3.1.13.5 and the MCVideo user is eligible to be implicitly affiliated with the MCVideo group as determined in step 14) a) iv) above, shall perform the implicit affiliation as specified in clause 8.2.2.3.7;

f) if the received SIP INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element set to a value of "true":

i) shall cache the information that this MCVideo user has initiated an MCVideo emergency call;

ii) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the information that this MCVideo user has initiated an MCVideo emergency alert;

iii) if the in-progress emergency state of the group is set to a value of "false":

A) shall set the value of the in-progress emergency state of the group to "true";

B) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16; and

C) shall generate SIP re-INVITE requests for the MCVideo emergency group call to the other call participants of the MCVideo group as specified in clause 6.3.3.1.6;

iv) if the in-progress imminent peril state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's imminent peril indication as specified in clause 6.3.3.1.11, setting the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11]; and

v) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5];

g) if the in-progress emergency state of the group is set to a value of "false" and if the SIP INVITE request contains an imminent peril indication set to a value of "true", the controlling MCVideo function:

i) shall cache the information that this MCVideo user has initiated an MCVideo imminent peril call; and

ii) if the in-progress imminent peril state of the group is set to a value of "false":

A) shall set the in-progress imminent peril state of the group to a value of "true";

B) shall generate SIP re-INVITE requests for the MCVideo imminent peril group call to the other call participants of the MCVideo group as specified in clause 6.3.3.1.15; and

C) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

iii) if the in-progress imminent peril state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's imminent peril indication as specified in clause 6.3.3.1.11, setting the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11];

h) shall generate a SIP 200 (OK) response as specified in the clause 6.3.3.2.4.2;

i) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

j) shall include in the SIP 200 (OK) response with the warning text set to "123 MCVideo session already exists" as specified in clause 4.4;

k) if the received SIP re-INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

l) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 4: In this case, the request was for an imminent peril call but a higher priority MCVideo emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

m) shall interact with media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 5: Resulting media plane processing is completed before the next step is performed.

n) shall send the SIP 200 (OK) response towards the inviting MCVideo client or inviting non-controlling MCVideo function according to 3GPP TS 24.229 [11];

o) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo User has joined in the MCVideo group session, as specified in clause 6.3.3.4;

NOTE 6: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

p) shall send a SIP NOTIFY request to each MCVideo client according to 3GPP TS 24.229 [11];

q) Upon receiving a SIP ACK to the above SIP 200 (OK) response and the SIP 200 (OK) response contained a Warning header field as specified in clause 4.4 with the warning text containing the mcvideo-warn-code set to "149", shall follow the procedures in clause 6.3.3.1.18; and

r) shall not continue with the rest of the clause.

Upon receiving a SIP 183 (Session Progress) response to the SIP INVITE request specified in clause 9.2.1.4.1 containing a P-Answer-State header field with the value "Unconfirmed" as specified in IETF RFC 4964 [30], the timer TNG1 (acknowledged call setup timer) is not running, the controlling MCVideo function supports media buffering and the SIP final response is not yet sent to the inviting MCVideo client:

1) shall generate a SIP 200 (OK) response to SIP INVITE request as specified in the clause 6.3.3.2.3.2;

2) shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if the prearranged MCVideo group has more than <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [24];

3) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

4) shall include a P-Answer-State header field with the value "Unconfirmed";

5) if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

6) if the received SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

7) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 7: Resulting user plane processing is completed before the next step is performed.

8) shall send the SIP 200 (OK) response towards the inviting MCVideo client according to 3GPP TS 24.229 [11];

9) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo User has joined in the MCVideo group session, as specified in clause 6.3.3.4; and

NOTE 8: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

10) shall send a SIP NOTIFY request to each MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 183 (Session Progress) response for a SIP INVITE request as specified in clause 9.2.1.4.1.2 containing an indication of required group members, the timer TNG1 (acknowledged call setup timer) is running and all SIP 200 (OK) responses have been received to all SIP INVITE requests sent to MCVideo clients specified in clause 9.2.1.4.1.1, then the controlling MCVideo function shall wait until the SIP 200 (OK) response has been received to the SIP INVITE request specified in clause 9.2.1.4.1.2 before generating a SIP 200 (OK) response to the "SIP INVITE request for controlling MCVideo function of an MCVideo group".

Upon receiving a SIP 200 (OK) response for a SIP INVITE request as specified in clause 9.2.1.4.1 that was sent to an affiliated and <on-network-required> group member as specified in 3GPP TS 24.481 [24]; and

1) *if the MCVideo ID in the SIP 200 (OK) response matches to the MCVideo ID in the corresponding SIP INVITE request;*

2) *there are no outstanding SIP 200 (OK) responses to SIP INVITE requests which were sent to affiliated and <on-network-required> group members as specified in 3GPP TS 24.481 [24]; and*

3) *there is no outstanding SIP 200 (OK) response to a SIP INVITE request sent in clause 9.2.1.4.1.2 where the SIP 183 (Session Progress) response contained an indication of required group members;*

the controlling MCVideo function:

1) *shall stop timer TNG1 (acknowledged call setup timer) as described in clause 6.3.3.3;*

2) *shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;*

3) *shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if all members were not invited because the prearranged MCVideo group has been exceeded the <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [24];*

4) *shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;*

5) *shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;*

NOTE 9: Resulting media plane processing is completed before the next step is performed.

6) *shall send a SIP 200 (OK) response to the inviting MCVideo client according to 3GPP TS 24.229 [11];*

7) *shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo user has joined in the MCVideo group session, as specified in clause 6.3.3.4; and*

NOTE 10: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

8) *shall send the SIP NOTIFY request to the MCVideo clients according to 3GPP TS 24.229 [11].*

Upon:

1) *receiving a SIP 200 (OK) response for a SIP INVITE request as specified in clause 9.2.1.4.1;*

2) *the timer TNG1 (acknowledged call setup timer) is not running;*

3) *the local counter of the number of SIP 200 (OK) responses received from invited members is equal to the value of the <on-network-minimum-number-to-start> element of the group document;*

4) *the controlling MCVideo function supports media buffering; and*

5) *the SIP final response has not yet been sent to the inviting MCVideo client;*

the controlling MCVideo function according to local policy:

1) *shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.2;*

2) *shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if all members were not invited because the prearranged MCVideo group has exceeded the <max-participant-count> members as specified in 3GPP TS 24.481 [24];*

3) *shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;*

4) if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

5) if the received SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

6) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 11: Resulting media plane processing is completed before the next step is performed.

7) shall send a SIP 200 (OK) response to the inviting MCVideo client according to 3GPP TS 24.229 [11];

8) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo user has joined in the MCVideo group session, as specified in clause 6.3.3.4; and

NOTE 12: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

9) shall send the SIP NOTIFY request to the MCVideo clients according to 3GPP TS 24.229 [11].

Upon expiry of timer TNG1 (acknowledged call setup timer), if there are outstanding SIP 200 (OK) responses to SIP INVITE requests sent to affiliated and <on-network-required> group members as specified in 3GPP TS 24.481 [24], the controlling MCVideo function shall follow the procedures specified in clause 6.3.3.3.

If timer TNG1 (acknowledged call setup timer) is running and a final SIP 4xx, 5xx or 6xx response is received from an affiliated and <on-network-required> group member as specified in 3GPP TS 24.481 [24], the controlling MCVideo function shall follow the relevant procedures specified in clause 6.3.3.3.

If:

1) timer TNG1 (acknowledged call setup timer) is not running;

2) the local counter of the number of SIP 200 (OK) responses received from invited members is equal to the value of the <on-network-minimum-number-to-start> element of the group document; and

3) a final SIP 4xx, 5xx or 6xx response is received from an invited MCVideo client;

then the controlling MCVideo function shall perform one of the following based on policy:

1) send the SIP final response towards the inviting MCVideo client, according to 3GPP TS 24.229 [11], if a SIP final response was received from all the other invited MCVideo clients and the SIP 200 (OK) response is not yet sent; or

2) remove the invited MCVideo client from the MCVideo Session as specified in clause 6.3.3.1.5, if a SIP final response other than 2xx or 3xx was received from all the invited MCVideo clients and the SIP 200 (OK) response is already sent. The controlling MCVideo function may invite an additional member of the prearranged MCVideo group as specified in clause 9.2.1.4.1 that has not already been invited, if the prearranged MCVideo group has more than <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [24], and all members have not yet been invited.

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCVideo client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.18."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.3]

"Upon receiving a SIP BYE request the controlling MCVideo function shall follow the procedures as specified in clause 6.3.3.2.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.7]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

Upon receipt of a SIP re-INVITE request for an MCVideo session identity identifying an on-demand prearranged MCVideo group session, the controlling MCVideo function:

…

2) if received SIP re-INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.3.1.17;

…

4) if the received SIP re-INVITE request contains an imminent peril indication set to "true" for an MCVideo imminent peril group call and this is an unauthorised request for an MCVideo imminent peril group call as determined by clause 6.3.3.1.13.6, shall reject the SIP re-INVITE request …

5) if a Resource-Priority header field is included in the received SIP re-INVITE request:

a) if the Resource-Priority header field is set to the value indicated for emergency calls and the SIP re-INVITE request does not contain an emergency indication and the in-progress emergency state of the group is set to a value of "false", shall reject the SIP re-INVITE request … and skip the rest of the steps; and

b) if the Resource-Priority header field is set to the value indicated for imminent peril calls and the SIP re-INVITE request does not contain an imminent peril indication and the in-progress imminent peril state of the group is set to a value of "false", shall reject the SIP INVITE request …and skip the rest of the steps;

6) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "true" and is an authorised request to initiate an MCVideo emergency group call as determined by clause 6.3.3.1.13.2, the controlling MCVideo function shall:

i) shall cache the MCVideo ID of the MCVideo user that has initiated an MCVideo emergency call;

ii) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the MCVideo ID of the MCVideo user that has initiated an MCVideo emergency alert;

iii) if the in-progress emergency state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's emergency indication as specified in clause 6.3.3.1.11, setting the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true";

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11]; and

C) if the in-progress imminent peril state of the group is set to a value of "true", shall set it to a value of "false"; and

iv) if the in-progress emergency state of the group is set to a value of "false":

A) shall set the value of the in-progress emergency state of the group to "true";

B) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16;

NOTE 2: The interactions of TNG2 with the TNG3 (group call timer) are explained in clause 6.3.3.5.2.

Editor's Note:timers need to be defined.

C) shall generate SIP re-INVITE requests for the MCVideo emergency group call to the other participants of the MCVideo group call as specified in clause 6.3.3.1.6;

D) shall send the SIP re-INVITEs towards the other participants of the MCVideo group call; and

E) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5];

7) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "false" and is an unauthorised request for an MCVideo emergency group call cancellation as determined by clause 6.3.3.1.13.4:

a) shall reject the SIP re-INVITE request.

...

c) if an <alert-ind> element of the mcvideoinfo MIME body is included in the SIP re-INVITE request set to "false", and there is an outstanding MCVideo emergency alert for this MCVideo user, shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <alert-ind> element set to a value of "true"; and

d) shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [11] and skip the rest of the steps;

8) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "false" and is determined to be an authorised request for an MCVideo emergency call cancellation as specified in clause 6.3.3.1.16 and the in-progress emergency state of the group to is set to a value of "true" the controlling MCVideo function:

a) shall set the in-progress emergency group state of the group to a value of "false";

b) shall clear the cache of the MCVideo ID of the MCVideo user as having an outstanding MCVideo emergency group call;

c) if an <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body is included and set to "false" and is determined to be an authorised request for an MCVideo emergency alert cancellation as specified in clause 6.3.3.1.13.3 and there is an outstanding MCVideo emergency alert for this MCVideo user shall:

i) if the received SIP re-INVITE request contains an <originated-by> element in the application/vnd.3gpp.mcvideo-info+xml MIME body, clear the cache of the MCVideo ID of the MCVideo user identified by the <originated-by> element as having an outstanding MCVideo emergency alert; or

ii) if the received SIP re-INVITE request does not contain an <originated-by> element in the application/vnd.3gpp.mcvideo-info+xml MIME body, clear the cache of the MCVideo ID of the sender of the SIP re-INVITE request as having an outstanding MCVideo emergency alert;

d) shall generate SIP re-INVITE requests to the participants in the group call as specified in clause 6.3.3.1.6. The MCVideo controlling function:

i) for each of the other participants in the group call shall send the SIP re-INVITE request towards the MCVideo client as specified in 3GPP TS 24.229 [11]; and

ii) Upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5];

NOTE 3: Clause 6.3.3.1.5 will inform the group call participants of the cancellation of the MCVideo group's in-progress emergency state and the cancellation of the MCVideo emergency alert if applicable.

e) shall stop timer TNG2 (in-progress emergency group call timer); and

NOTE 4: The interactions of TNG2 with the TNG3 (group call timer) are explained in clause 6.3.3.5.2;

f) for each of the affiliated members of the group that are not participating in the call:

i) generate a SIP MESSAGE request notification of the cancellation of the MCVideo user's emergency call as specified in clause 6.3.3.1.11;

ii) set the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "false";

iii) if indicated above in step 8) c), set the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "false"; and

iv) send the SIP MESSAGE request according to 3GPP TS 24.229 [11];

9) if the received SIP re-INVITE request contains an imminent peril indication and the in-progress emergency group state of the group is set to a value of "false", shall perform the procedures specified in clause 9.2.1.4.8 and skip the rest of the steps.

Upon receiving a SIP 200 (OK) response to a SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5];

1) shall generate a SIP 200 (OK) response according to rules and procedures of 3GPP TS 24.229 [11];

2) shall include in the SIP 200 (OK) response an SDP answer according to 3GPP TS 24.229 [11] with the clarifications specified in clause 6.3.3.2.1;

3) shall include the "norefersub" option tag in a Supported header field according to IETF RFC 4488 [31];

4) shall include the "tdialog" option tag in a Supported header field according to IETF RFC 4538 [32];

5) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <alert-ind> element set to a value of "true" and if this is an unauthorised request for an MCVideo emergency alert as determined by clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

6) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <alert-ind> element set to a value of "false" and if this is an unauthorised request for an MCVideo emergency alert cancellation as determined by clause 6.3.3.1.13.3, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

7) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true", this is an authorised request for an MCVideo imminent peril group call and the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 5: In this case, the request was for an imminent peril call but a higher priority MCVideo emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

8) shall interact with media plane as specified in 3GPP TS 24.581 [5]; and

9) shall send the SIP 200 (OK) response towards the MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCVideo client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.18.

Upon receipt of a SIP 2xx response for an outgoing SIP MESSAGE request, shall handle according to 3GPP TS 24.229 [11]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.8]

"This procedure is initiated by the controlling MCVideo function as the result of an action in clause 9.2.1.4.7.

In the procedures in this clause:

1) imminent peril indication in an incoming SIP re-INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

When the controlling function receives a SIP re-INVITE request with an imminent peril indication set to "true", the controlling function:

1) if the in-progress emergency state of the group is set to a value of "false" and if the SIP re-INVITE request contains an imminent peril indication set to a value of "true" or the in-progress imminent peril state of the group to "true", the controlling MCVideo function shall:

NOTE: 1 The calling procedure has already determined that this is not an unauthorised request for an MCVideo imminent peril call, therefore that check does not need to be repeated in the current procedure.

a) if the in-progress imminent peril state of the group is set to a value of "true" and this MCVideo user is indicating a new imminent peril indication:

i) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's imminent peril indication as specified in clause 6.3.3.1.11 with the following clarifications;

A) set the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11];

b) if the in-progress imminent peril state of the group is set to a value of "false";

i) set the value of the in-progress imminent peril state of the group to "true";

ii) generate SIP re-INVITE requests for the MCVideo imminent peril group call to participants in the MCVideo group call as specified in clause 6.3.3.1.15;

iii) send the SIP re-INVITES to all of the other participants in the MCVideo group call;

iv) for each of the affiliated members of the group not participating in the group call, generate a SIP MESSAGE request notification of the MCVideo user's imminent peril indication as specified in clause 6.3.3.1.11 with the following clarifications;

A) set the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11]; and

c) cache the information that this MCVideo user has initiated an MCVideo imminent peril call;

2) if the SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "false" and is an unauthorised request for an MCVideo imminent peril group call cancellation as determined by clause 6.3.3.1.13.6 shall:

a) reject the SIP re-INVITE request with a SIP 403 (Forbidden) response to the SIP re-INVITE request; and

b) include in the SIP 403 (Forbidden) response an application/vnd.3gpp.mcvideo-info+xml MIME body as specified in Annex F.1 with the <mcvideoinfo> element containing the <mcvideo-Params> element with the <imminentperil-ind> element set to a value of "false";

c) send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [11]; and

d) skip the rest of the steps;

3) if the SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "false" and is determined to be an authorised request for an MCVideo imminent peril call cancellation as specified in clause 6.3.3.1.13.6 and the in-progress imminent peril state of the group to is set to a value of "true" the controlling MCVideo function shall:

a) set the in-progress imminent peril state of the group to a value of "false";

b) cache the information that this MCVideo user no longer has an outstanding MCVideo imminent peril group call;

c) generate SIP re-INVITES requests to the other participants in the MCVideo group call as specified in clause 6.3.3.1.15. The MCVideo controlling function:

i) for each participant shall send the SIP re-INVITE request towards the MCVideo client as specified in 3GPP TS 24.229 [11]; and

ii) Upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function interact with the media plane as specified in 3GPP TS 24.581 [5]; and

NOTE 2: Clause 6.3.3.1.14 will inform the affiliated and joined members of the cancellation of the MCVideo group's in-progress emergency state and the cancellation of the MCVideo emergency alert if applicable.

d) for each of the affiliated members of the group not participating in the call shall:

i) generate a SIP MESSAGE request notification of the cancellation of the MCVideo user's imminent peril call as specified in clause 6.3.3.1.11;

ii) set the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "false"; and

iii) send the SIP MESSAGE request according to 3GPP TS 24.229 [11];

4) shall include in the SIP 200 (OK) response an SDP answer according to 3GPP TS 24.229 [11] with the clarifications specified in clause 6.3.3.2.1;

5) shall include the "norefersub" option tag in a Supported header field according to IETF RFC 4488 [31];

6) shall include the "tdialog" option tag in a Supported header field according to IETF RFC 4538 [32];

7) shall interact with media plane as specified in 3GPP TS 24.581 [5]; and

8) shall send the SIP 200 (OK) response towards the MCVideo client according to 3GPP TS 24.229 [11].

Upon receipt of a SIP 2xx response for an outgoing SIP MESSAGE request, shall handle according to 3GPP TS 24.229 [11]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.1.1]

"This clause describes the procedures for inviting an MCVideo user to an MCVideo session. The procedure is initiated by the controlling MCVideo function as the result of an action in clause 9.2.1.4.2 or as the result of receiving a SIP 403 (Forbidden) response as described in this clause.

The controlling MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the public service identity of the terminating participating MCVideo function associated to the MCVideo user to be invited.;

NOTE 1: How the controlling MCVideo function finds the address of the terminating MCVideo participating function is out of the scope of the current release.

NOTE 2: If the terminating MCVideo user is part of a partner MCVideo system, then the public service identity can identify an entry point in the partner network that is able to identify the terminating participating MCVideo function.

3) shall set the P-Asserted-Identity header field to the public service identity of the controlling MCVideo function;

4) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcvideo-request-uri> element set to the MCVideo ID of the terminating user; and

b) the <mcvideo-calling-group-id> element set to the group identity;

NOTE 3: The <mcvideo-calling-user-id> is already included in the MIME body as a result of calling clause 6.3.3.1.2 in step 1).

5) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1;

6) if the in-progress emergency state of the group is set to a value of "true" the controlling MCVideo function:

a) shall include a Resource-Priority header field populated with the values for an MCVideo emergency group call as specified in clause 6.3.3.1.19;

b) if the received SIP INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "true":

i) shall include in the outgoing SIP INVITE request in the application/vnd.3gpp.mcvideo-info+xml MIME body an <emergency-ind> element set to a value of "true"; and

ii) if the <alert-ind> element is set to "true" in the received SIP INVITE request and the requesting MCVideo user and MCVideo group are authorised for the initiation of MCVideo emergency alerts as determined by the procedures of clause 6.3.3.1.13.1, shall populate the application/vnd.3gpp.mcvideo-info+xml MIME body and the application/vnd.3gpp.location-info+xml MIME body as specified in clause 6.3.3.1.12. Otherwise, shall set the <alert-ind> element to a value of "false"; and

c) if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <imminentperil-ind> element set to a value of "false";

7) if the in-progress emergency state of the group is set to a value of "false" and the in-progress imminent peril state of the group is set to a value of "true", the controlling MCVideo function:

a) shall include a Resource-Priority header field populated with the values for an MCVideo imminent peril group call as specified in clause 6.3.3.1.19; and

b) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true";

8) if:

a) an MCVideo GKTP document exists for the group identity contained in the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request; and

b) the MCVideo GKTP document contains a <MKFC-GKTPs> element;

then:

a) for each instance of <GKTP> element of the <MKFC-GKTPs> element of the MCVideo GKTP document:

i) shall perform the procedure in clause 6.3.3.6.2 to re-generate an I_MESSAGE; and

ii) if the procedure in clause 6.3.3.6.2 was successful, shall include the I_MESSAGE in a <GKTP> element of the <MKFC-GKTPs> element of an application/vnd.3gpp.mcvideo-info+xml MIME body included in the outgoing SIP INVITE request; and

9) shall send the SIP INVITE request towards the terminating network in accordance with 3GPP TS 24.229 [11].

Upon receiving a SIP 183 (Session Progress) response containing a Require header field with the option tag "100rel" and containing a P-Answer-State header field with the value "Unconfirmed" in response to the SIP INVITE request the controlling MCVideo function:

1) shall send a SIP PRACK request towards the MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCVideo function:

1) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

2) shall send a SIP NOTIFY request to all participants with a subscription to the conference event package as specified in clause 9.2.3.4; and

3) shall increment the local counter of the number of SIP 200 (OK) responses received from invited members, by 1.

NOTE 4: The notifications above could be sent prior to the SIP 200 (OK) response being sent to the inviting MCVideo client. These notifications received by MCVideo clients that are group members do not mean that the group session will be successfully established.

NOTE 5: The procedures executed by the controlling MCVideo function prior to sending a response to the inviting MCVideo client are specified in clause 9.2.1.4.2."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.1.2]

"The controlling MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the public service identity of the non-controlling MCVideo function serving the group identity of the MCVideo group owned by the partner MCVideo system;

3) shall set the P-Asserted-Identity to the public service identity of the controlling MCVideo function;

4) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcvideo-request-uri> element set to the group identity of the MCVideo group hosted by the non-controlling MCVideo function in the partner MCVideo system; and

b) the <mcvideo-calling-group-id> element set to the group identity of the group served by the controlling MCVideo function;

5) shall include the Recv-Info header field set to g.3gpp.mcvideo-transmission-request;

6) if:

a) an MCVideo GKTP document exists for the group identity contained in the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request; and

b) the MCVideo GKTP document contains a <MKFC-GKTPs> element;

then:

a) for each instance of <GKTP> element of the <MKFC-GKTPs> element of the MCVideo GKTP document:

i) shall perform the procedure in clause 6.3.3.6.2 to re-generate an I_MESSAGE; and

ii) if the procedure in clause 6.3.3.6.2 was successful, shall include the I_MESSAGE in a <GKTP> element of the <MKFC-GKTPs> element of an application/vnd.3gpp.mcvideo-info+xml MIME body included in the outgoing SIP INVITE request;

7) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1; and

8) shall send the SIP INVITE request towards the partner MCVideo system in accordance with 3GPP TS 24.229 [11].

Upon receiving SIP 403 (Forbidden) response for the SIP INVITE request, if according to local policy and if:

1) the response contains a Warning header field with the MCVideo warning code "128"; and

2) the response contains a P-Refused-URI-List header field and an application/resource-lists+xml MIME body as specified in IETF RFC 5318 [28];

NOTE 1: The application/resource-lists+xml MIME body contains MCVideo IDs identifying MCVideo users in a partner MCVideo system that needs to be invited to the prearranged group call in case of group regrouping using interrogating method.

Editor's Note:The above note currently isn't defined in the 23.280 and 23.281.

then the controlling MCVideo function:

1) shall check if the number of members of the MCVideo group exceeds the value contained in the <on-network-max-participant-count> element of the group document as specified in 3GPP TS 24.481 [24]. If exceeded, the controlling MCVideo function shall invite only <on-network-max-participant-count> members from the application/resource-lists+xml MIME body; and

NOTE 2: The <on-network-max-participant-count> element indicates the maximum number of participants allowed in the prearranged group session It is operator policy that determines which participants in the application/resource-lists+xml MIME body are invited to the group call.

2) shall invite MCVideo users as specified in this clause using the list of MCVideo IDs in URI-List.

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCVideo function:

1) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3; and

NOTE 3: The procedures executed by the controlling MCVideo function prior to sending a response to the inviting MCVideo client are specified in clause 9.2.1.4.2.

2) if at least one of the invited MCVideo clients has subscribed to the conference package, shall subscribe to the conference event package in the non-controlling MCVideo function as specified in clause 9.2.3.4.3."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.6]

"This clause is referenced from other procedures.

This clause describes the procedures for inviting an MCVideo user to an MCVideo session associated with an MCVideo emergency group call or MCVideo imminent peril group call. The procedure is initiated by the controlling MCVideo function as the result of an action in clause 9.2.2.4.1.1.

The controlling MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the address of the terminating participating MCVideo function associated with the MCVideo ID of the targeted MCVideo user;

3) shall include an application/vnd.3gpp.mcvideo-info+xml MIME body with the <mcvideoinfo> element containing the <mcvideo-Params> element populated as follows:

a) the <mcvideo-request-uri> element set to the value of the MCVideo ID of the targeted MCVideo user;

b) the <mcvideo-calling-user-id> element set to the value of the MCVideo ID of the calling MCVideo user; and

c) the <mcvideo-calling-group-id> element set to the value of the MCVideo group ID of the emergency group call.

4) shall include in the P-Asserted-Identity header field the public service identity of the controlling MCVideo function;

5) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1; and

6) if the in-progress emergency group state of the group is set to a value of "true" the controlling MCVideo function:

a) shall include a Resource-Priority header field populated with the values for an MCVideo emergency group call as specified in clause 6.3.3.1.18;

b) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <emergency-ind> element set to a value of "true";

c) if the <alert-ind> element is set to "true" in the received SIP INVITE request and the requesting MCVideo user and MCVideo group are authorised for the initiation of MCVideo emergency alerts as determined by the procedures of clause 6.3.3.1.12.1, shall populate the application/vnd.3gpp.mcvideo-info+xml MIME body and the application/vnd.3gpp.mcvideo-location-info+xml MIME body as specified in clause 6.3.3.1.11. Otherwise, shall set the <alert-ind> element to a value of "false" in the application/vnd.3gpp.mcvideo-info+xml MIME body; and

d) if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <imminentperil-ind> element set to a value of "false";

NOTE: If the imminent peril state of the group is true at this point, the controlling function will set it to false as part of the calling procedure.

7) if the in-progress emergency state of the group is set to a value of "false" and the in-progress imminent peril state of the group is set to a value of "true", the controlling MCVideo function:

a) shall include a Resource-Priority header field populated with the values for an MCVideo imminent peril group call as specified in clause 6.3.3.1.18; and

b) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true"; and

8) if:

a) an MCVideo GKTP document exists for the group identity contained in the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request; and

b) the MCVideo GKTP document contains an <MKFC-GKTPs> element;

then:

a) for each instance of <GKTP> element of the <MKFC-GKTPs> element of the MCVideo GKTP document:

i) shall perform the procedure in clause 6.3.3.6.2 to re-generate an I_MESSAGE; and

ii) if the procedure in clause 6.3.3.6.2 was successful, shall include the I_MESSAGE in a <GKTP> element of the <MKFC-GKTPs> element of an application/vnd.3gpp.mcvideo-info+xml MIME body included in the outgoing SIP INVITE request."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.11]

"This clause is referenced from other procedures.

This clause describes the procedures for populating the application/vnd.3gpp.mcvideo-info+xml and application/vnd.3gpp.mcvideo-location-info+xml MIME bodies for an MCVideo emergency alert. The procedure is initiated by the controlling MCVideo function when it has received a SIP request initiating an MCVideo emergency alert and generates a message containing the MCVideo emergency alert information required by 3GPP TS 23.281 [26].

The controlling MCVideo function:

1) *shall include, if not already present, an application/vnd.3gpp.mcvideo-info+xml MIME body as specified in Annex F.1, and set the <alert-ind> element to a value of "true";*

2) *shall determine the value of the MCVideo user's Mission Critical Organization from the <MissionCriticalOrganization> element, of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]);*

3) *shall include in the <mcvideoinfo> element containing the <mcvideo-Params> element containing an <mc-org> element set to the value of the MCVideo user's Mission Critical Organization; and*

4) *shall copy the contents of the application/vnd.3gpp.mcvideo-location-info+xml MIME body in the received SIP request into an application/vnd.3gpp.mcvideo-location-info+xml MIME body included in the outgoing SIP request."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.12.2]

"If the controlling MCVideo function has received a SIP request for an MCVideo group call with the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body set to a value of "true" and:

1) *if the <allow-emergency-group-call> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true" and:*

   a) *if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> element contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "DedicatedGroup" and:*

      i) *if the <uri-entry> element of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) contains the identity of the MCVideo group targeted by the calling MCVideo user; and*

      ii) *if the <allow-MCVideo-emergency-call> element of the <list-service> element of the group document identified by the targeted MCVideo group identity is set to a value of "true" as specified in 3GPP TS 24.481 [24];*

   then the controlling MCVideo function shall consider the MCVideo emergency group call request to be an authorised request for an MCVideo emergency group call and skip the remaining steps; or;

   b) *if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> element contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "UseCurrentlySelectedGroup" and if the <allow-MCVideo-emergency-call> element of the <list-service> element of the group document identified by the targeted MCVideo group identity is set to a value of "true" as specified in 3GPP TS 24.481 [24];*

   then the controlling MCVideo function shall consider the MCVideo emergency group call request to be an authorised request for an MCVideo emergency group call and skip the remaining steps; or

2) *if the controlling MCVideo function does not consider the MCVideo emergency group call request to be an authorised request for an MCVideo emergency group call by step 1) above, then the controlling MCVideo function shall consider the MCVideo emergency group call request to be an unauthorised request for an MCVideo emergency group call.*

If the controlling MCVideo function has received a SIP request for an MCVideo private call with the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body set to a value of "true" and:

1) *if the <allow-emergency-private-call> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true"; and*

   a) *if the "entry-info" attribute of the <entry> element of the <MCVideoPrivateRecipient> element of the <EmergencyCall> element contained within the <PrivateCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "UsePreConfigured" and if the MCVideo ID targeted for the call is contained in the <uri-entry> element of the <entry> element of the <MCVideoPrivateRecipient> element of the <EmergencyCall> element contained within the <PrivateCall> element (see the MCVideo user profile document in 3GPP TS 24.484 [25]); or*

   b) *if the "entry-info" attribute of the <entry> element of the <MCVideoPrivateRecipient> element of the <EmergencyCall> element contained within the <PrivateCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "LocallyDetermined";*

then the controlling MCVideo function shall consider the MCVideo emergency private call request to be an authorised request for an MCVideo emergency private call and skip step 2) below; or

2) *if the controlling MCVideo function does not consider the MCVideo emergency private call request to be an authorised request for an MCVideo emergency private call by step 1) above, then the controlling MCVideo function shall consider the MCVideo emergency private call request to be an unauthorised request for an MCVideo emergency private call."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.12.3]

"If the controlling MCVideo function has received a SIP request with the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body set to a value of "false" and:

1) *if the <allow-cancel-emergency-alert> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true", then the MCVideo emergency alert cancellation request shall be considered to be an authorised request for an MCVideo emergency alert cancellation; and*

2) *if the <allow-cancel-emergency-alert> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "false", then the MCVideo emergency alert cancellation request shall be considered to be an unauthorised request for an MCVideo emergency alert cancellation."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.12.4]

"If the controlling MCVideo function has received a SIP request with the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body set to a value of "false" and:

1) *if the <allow-cancel-emergency-alert> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true", then the MCVideo emergency alert cancellation request shall be considered to be an authorised request for an MCVideo emergency alert cancellation; and*

2) *if the <allow-cancel-emergency-alert> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "false", then the MCVideo emergency alert cancellation request shall be considered to be an unauthorised request for an MCVideo emergency alert cancellation."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.12.5]

"If the controlling MCVideo function has received a SIP request with the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body set to a value of "true" and:

1) *if the <allow-imminent-peril-call> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value other than "true" the request for initiating an MCVideo imminent peril call shall be considered to be an unauthorised request for an MCVideo imminent peril call and skip the remaining steps;*

2) *if the <allow-imminent-peril-call> element of the <list-service> element of the group document identified by the targeted MCVideo group identity is set to a value other than "true" as specified in 3GPP TS 24.481 [24], the request for initiating an MCVideo imminent peril call shall be considered to be an unauthorised request for an MCVideo imminent peril call and skip the remaining steps;*

3) *if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element contained within the <ImminentPerilCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "DedicatedGroup" and if the MCVideo group identity targeted for the call is contained in the <uri-entry> element of the <entry> element of the <MCVideoGroupInitiation> element contained within the <ImminentPerilCall> element (see the MCVideo user profile document in 3GPP TS 24.484 [25]); or*

4) *if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element contained within the <ImminentPerilCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "UseCurrentlySelectedGroup".*

then the MCVideo imminent peril call request shall be considered to be an authorised request for an MCVideo imminent peril call. In all other cases, it shall be considered to be an unauthorised request for an MCVideo imminent peril call."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [11], the controlling MCVideo function:

1) *for the accepted media stream in the received SDP offer:*

   a) *shall replace the IP address and port number in the received SDP offer with the IP address and port number of the controlling MCVideo function; and*

2) *for the accepted media-transmission control entity, if present in the received SDP offer:*

   a) *shall replace the IP address and port number in the received SDP offer with the IP address and port number of the controlling MCVideo function, for the accepted media-transmission control entity, if present in the received SDP offer; and*

   b) *shall include 'fmtp' attributes as specified in 3GPP TS 24.581 clause 14."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.2.2]

"On receipt of an initial SIP INVITE request the controlling MCVideo function shall cache SIP feature tags, if received in the Contact header field and if the specific feature tags are supported."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.2.4]

"Upon receiving a SIP BYE request the controlling MCVideo function:

1) *shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.581 [5] for releasing the media plane resource associated with the SIP session towards the MCVideo client;*

NOTE: The non-controlling MCVideo function is also regarded as a MCVideo client in a temporary MCVideo group session.

2) *shall generate a SIP 200 (OK) response and send the SIP response towards the MCVideo client according to 3GPP TS 24.229 [11];*

3) shall check the MCVideo session release policy as specified in clause 6.3.8.1 and clause 6.3.8.2 whether the MCVideo session needs to be released for each participant of the MCVideo session;

4) if release of the MCVideo session is required:

a) shall perform the procedures as specified in the clause 6.3.3.1.4 with the clarification that if the received SIP BYE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body, copy the application/vnd.3gpp.mcvideo-info+xml MIME body into the outgoing SIP BYE request; and

5) if a release of the MCVideo session is not required, shall send a SIP NOTIFY request to all remaining MCVideo clients in the MCVideo session with a subscription to the conference event package as specified in clause 9.2.3.4.2.

Upon receiving a SIP 200 (OK) response to the SIP BYE request the controlling MCVideo function shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.581 [5] for releasing media plane resources associated with the SIP session with the MCVideo participant."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.4]

"The controlling MCVideo function shall generate a SIP NOTIFY request according to 3GPP TS 24.229 [11] with the clarification in this clause.

In the SIP NOTIFY request, the controlling MCVideo function:

1) shall set the P-Asserted-Identity header field to the public service identity of the controlling MCVideo function;

2) shall include an Event header field set to the "conference" event package;

3) shall include an Expires header field set to 3600 seconds according to IETF RFC 4575 [57], as default value;

4) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcvideo" (coded as specified in 3GPP TS 24.229 [11]), in a P-Preferred-Service header field according to IETF RFC 6050 [14]; and

5) shall include an application/vnd.3gpp.mcvideo-info+xml MIME body with the <mcvideoinfo> element containing the <mcvideo-Params> element with:

a) the <mcvideo-calling-group-id> set to the value of the MCVideo group ID;

b) if the target is a MCVideo user, the value of <mcvideo-request-uri> element set to the value of MCVideo ID of the targeted MCVideo user; and

c) if the target is the non-controlling MCVideo function, the value of <mcvideo-request-uri> element set to the constituent MCVideo group ID.

In the SIP NOTIFY request, the controlling MCVideo function shall include an application/conference-info+xml MIME body according to IETF RFC 4575 [57] with the following limitations:

1) the controlling MCVideo function shall include the MCVideo group ID of the MCVideo group in the "entity" attribute of the <conference-info> element;

2) for each participant in the MCVideo session with the exception of non-controlling MCVideo functions, the controlling MCVideo function shall include a <user> element. The <user> element shall:

NOTE: Non-controlling MCVideo functions will appear as a participant in temporary group sessions.

a) include the "entity" attribute. The "entity" attribute:

i) shall for the MCVideo client, which initiated, joined or re-joined an MCVideo session, include the MCVideo ID of the MCVideo user which originates SIP INVITE request; and

ii) shall for an invited MCVideo client include the MCVideo ID of the invited MCVideo user in case of a prearranged group call or chat group call;

b) shall include a single <endpoint> element. The <endpoint> element:

i) shall include the "entity" attribute; and

ii) shall include the <status> element indicating the status of the MCVideo session according to RFC 4575; and

c) may include <roles> element.

NOTE: The usage of <roles> is only applicable for human consumption."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.5.5]

"The MCVideo server shall only invite affiliated group members to a group session. The MCVideo server determines the affiliated members from the entries contained in the <list> element of the group document by following the procedures specified in clause 6.3.6.

NOTE 1: The term "affiliated group members" used above also includes those members that are implicitly affiliated by the controlling MCVideo function.

If the number of members of the MCVideo group exceeds the value contained in the <on-network-max-participant-count> element the MCVideo server shall invite only <on-network-max-participant-count> members from the list, but shall prioritise inviting those group members to the group session that have an <entry> element in the <list> element with a <on-network-required> element present.

NOTE 2: The <on-network-max-participant-count> element indicates the maximum number of participants allowed in the group session. The <on-network-required> element is used to determine which group members need to acknowledge the group call before audio transmission can proceed.

NOTE 3: Other requirements for how the controlling MCVideo function selects which of the <on-network-max-participant-count> members to invite is outside the scope of this specification.

NOTE 4: It is assumed that validation checks are performed at the group management server to ensure that the <on-network-max-participant-count> cannot be less than the number of <on-network-required> users."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.1]

"The transmission control server arbitration logic in the transmission control server shall support the procedures in clauses 6.3.2 and 6.3.3 and shall behave according to the transmission control server state transition diagram for 'general transmission control operation' in clause 6.3.4.

The transmission control interface towards the MCVideo client in the transmission control server shall behave according to the transmission control server state transition diagram for 'basic transmission control operation towards the transmission participant' as specified in clause 6.3.5.

The reception control arbitration logic in the transmission control server shall support the procedures in clauses 6.3.2 and 6.3.3 and shall behave according to the reception control server state transition diagram for 'general reception control operation' in clause 6.3.6.

The transmission control interface towards the MCVideo client in the transmission control server shall behave according to the transmission control server state transition diagram for 'basic reception control operation towards the transmission participant' as specified in clause 6.3.7."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.2.2]

"When an MCVideo call is established a new instance of the transmission control server state machine for 'general transmission control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' is added.

When an MCVideo call is established a new instance of the transmission control server state machine for 'general reception control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic reception control operation towards the transmission participant' is added.

If the optional "mc_queueing" feature is supported and has been negotiated as specified in clause 14, the transmission control server could queue the implicit transmission control request for the media-transmission control entity.

The original initial SIP INVITE request or SIP REFER request to establish an MCVideo chat group call or to rejoin an ongoing MCVideo call is not handled as an implicit transmission control request message by the transmission control server unless explicitly stated in the SIP INVITE request or in the SIP REFER request.

The permission to send media to the inviting MCVideo client due to implicit transmission control request is applicable to both confirmed indication and unconfirmed indication.

When the first unconfirmed indication is received from the invited participating MCVideo function (see 3GPP TS 24.281 [2]) the transmission control server optionally can give an early indication to send RTP media packets, to the inviting MCVideo client.

If an early indication to send RTP media packets is given to the inviting MCVideo client, the transmission participant is granted the permission to send media and the MCVideo server buffers RTP media packets received from the MCVideo client at least until the first invited MCVideo client accepts the invitation or until the RTP media packet buffer exceeds it maximum limit to store RTP media packets.

If the MCVideo server does not support or does not allow media buffering then when an early indication to send RTP media packets is not given to the inviting MCVideo client, the transmission participant is granted the permission to send media when the first invited MCVideo client accepts the media.

Before the transmission control server sends the first transmission control message in the MCVideo call, the transmission control server has to assign itself a SSRC identifier to be included in media transmission control messages and quality feedback messages if the MCVideo server is supporting that option. A suitable algorithm to generate the SSRC identifier is described in IETF RFC 3550 [3].

The transmission participant and the transmission control server can negotiate the maximum priority level that the transmission participant is permitted to request. The transmission control server can pre-empt the current sender based on the negotiated maximum priority level that the transmission participant is permitted to request and the priority level included in the Transmission Media Request message.

NOTE: The maximum priority level that a transmission participant can use is negotiated as specified in clause 14.3.3 and is based on group configuration data retrieved by the controlling MCVideo function from the group management server as described in 3GPP TS 24.481 [12] and service configuration data retrieved by the controlling MCVideo function from the configuration management server as described in 3GPP TS 24.484 [13].

The transmission participant and the transmission control server can negotiate queueing of Transmission requests using the "mc_queueing" fmtp attribute as described in clause 14. If queueing is supported and negotiated, the transmission control server queues the transmission control request if a Transmission Media Request message is received when another transmission participant has the transmission and the priority of the current speaker is the same or higher."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.1]

"The transmission control messages are coded as described in clause 9.1.2 where the transmission control message is part of the application-dependent data.

For the transmission control protocol the ASCII name string is: MCVx (Mission Critical Video x) with x=0, 1 or 2 as specified in clause 9.1.2.

A list of transmission control messages can be found in clause 9.2.2.1.

The transmission control specific fields are specified in clause 9.2.3."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.

Step 2 When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

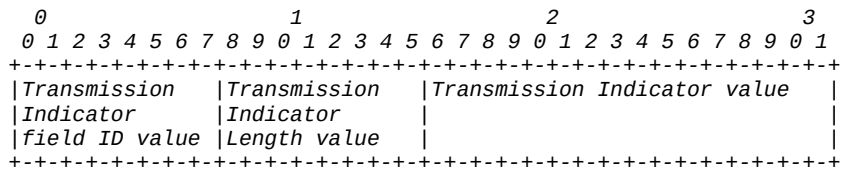
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

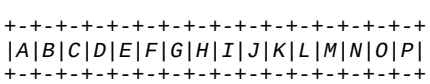


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. *if the active Transmission request queue is empty the transmission control server:*
    a. *shall send Transmission Idle message to all transmission participants. The Transmission Idle message:*
        i. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*
        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;*
    b. *shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;*
    c. *shall start timer T1 (Inactivity);*
    d. *shall set the general state to the 'G: Transmit Idle' state; and*
    e. *shall initialise counter Cx (Simultaneous transmission video) to 0.*
2. *if the active Transmission request queue is not empty the transmission control server:*
    a. *shall select a queued Transmission request from the top of the active Transmission request queue;*
    b. *shall remove that queued Transmission request from the active Transmission request queue; and*
    c. *shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:*
    a. *shall include the granted priority in the Transmission priority field;*
    b. *shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;*
    c. *if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*
    d. *shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;*
2. *shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;*
3. *shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:*
    a. *shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;*
    b. *shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;*
    c. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;*

d. if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';

e. if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

g. shall initiate the 'general reception control operation' state machine;and

4. shall enter the 'G: Transmit Taken' state."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. if counter Cx (Simultaneous transmission video) has not reached its upper limit:

a. shall perform the actions specified in the clause 6.3.4.4.2;

2. if counter Cx (Simultaneous transmission video) has reached its upper limit:

a. select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;

b. shall stop timer T4 (Transmission Grant), if running;

c. shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);

d. shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;

e. shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and

f. shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:

i. shall include the queue position and transmission priority in the Queue Info field; and

ii. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:

a. shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and

b. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

2. shall set the general state to 'G: pending Transmission Revoke'."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

b. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.1.1.3 Test description

5.1.1.1.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the participating MCVideo functions.
- SS-UE1 (MCVideo Client) and SS-UE2 (MCVideo Client), simulated by the SS (MCVideo Server).

IUT:

- IUT (MCVideo Server), providing the controlling MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of all sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Transmission Control Server.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server).
- The SS-UE1 (MCVideo client) and the SS-UE2 (MCVideo client) are registered and authorized to use the MCVideo system.
- The SS-UE1 (MCVideo client) and the SS-UE2 (MCVideo client) are affiliated to the pre-arranged group and are authorized to initiate group calls.
- The IUT (MCVideo Server) is the controlling server of the pre-arranged group.

5.1.1.1.3.2 Test procedure sequence

Table 5.1.1.1.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E group call establishment' as described in Annex A, Table A.4.1.4.3-1 to establish a pre-arranged group call with implicit transmission control request according to option b.ii of note in Annex A, Table A.4.1.4.3-1, initiated by SS-UE1 (MCVideo Client)?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1 to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E session establishment/modification without provisional responses other than 100 Trying', as described in Annex A, Table A.4.1.1.3-1 to upgrade the call to emergency group call with implicit transmission request according to option b.ii of note in Annex A, Table A.4.1.1.3-1?	-	-	4	P
4	Check: Does the IUT (MCVideo Server) correctly perform procedure ' MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
5	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	3	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
6	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
7	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E session establishment/modification without provisional responses other than 100 Trying', as described in Annex A, Table A.4.1.1.3-1 to downgrade the call without implicit transmission request, with option a of note on Table A.4.1.1.3-1?	-	-	5	P
8	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	3	P
9	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
10	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E session establishment/modification without provisional responses other than 100 Trying', as described in Annex A, Table A.4.1.1.3-1 to upgrade the call to imminent peril group call with implicit transmission request according to option b.ii of note in Annex A, Table A.4.1.1.3-1?	-	-	6	P
11	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
12	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	3	P
13	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
14	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E session establishment/modification without provisional responses other than 100 Trying', as described in Annex A, Table A.4.1.1.3-1 to downgrade the call without implicit transmission request, with option a of note on Table A.4.1.1.3-1?	-	-	7	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
15	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.1.5.3-1, to release the call?	-	-	8	P

5.1.1.1.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

Table 5.1.1.1.3.3-1: SIP INVITE from SS (MCVideo Server)
(Step 1, Table 5.1.1.1.3.2-1; Step 1, Annex A, Table A.4.1.4.3-1)

Derivation Path: Annex A, Table A.4.1.4.4-1

Table 5.1.1.1.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.1.1.3.2-1; Step 2, Annex A, Table A.4.1.4.3-1)

Derivation Path: Annex A, Table A.4.1.4.4-2

Table 5.1.1.1.3.3-3: SIP 200 (OK) from SS (MCVideo Server)
(Steps 1, 3, 7, 10 and 14, Table 5.1.1.1.3.2-1; Step 8, Annex A, Table A.4.1.4.3-1; Step 5, Annex A, Table A.4.1.1.3-1)

Derivation Path: Annex A, Table A.4.1.4.4-6

Table 5.1.1.1.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Steps 1, 3, 7, 10 and 14, Table 5.1.1.1.3.2-1; Step 9, Annex A, Table A.4.1.4.3-1; Step 6, Annex A, Table A.4.1.1.3-1)

Derivation Path: Annex A, Table A.4.1.4.4-7

Table 5.1.1.1.3.3-5: SIP re-INVITE from SS (MCVideo Server)
(Step 3, Table 5.1.1.1.3.2-1; Step 1, Annex A, Table A.4.1.1.3-1)

Derivation Path: Annex A, Table A.4.1.1.4-1, with conditions GROUP-CALL, re_INVITE and EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.1.1.3.3-6: SIP re-INVITE from IUT (MCVideo Server)
(Step 3, Table 5.1.1.1.3.2-1; Step 2, Annex A, Table A.4.1.1.3-1)

Derivation Path: Annex A, Table A.4.1.1.4-4, with condition GROUP-CALL, re_INVITE and EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.1.1.3.3-7: SIP re-INVITE from SS (MCVideo Server)
(Steps 7 and 14, Table 5.1.1.1.3.2-1; Step 1, Annex A, Table A.4.1.1.3-1)

Derivation Path: Annex A, Table A.4.1.1.4-1, with conditions GROUP-CALL and re_INVITE.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.1.3.3-8: SIP re-INVITE from IUT (MCVideo Server)
(Steps 7 and 14, Table 5.1.1.1.3.2-1; Step 2, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.4.1.1.4-4, with condition GROUP-CALL and re_INVITE.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.1.3.3-9: SIP re-INVITE from SS (MCVideo Server)
(Step 10, Table 5.1.1.1.3.2-1; Step 1, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.4.1.1.4-1, with conditions GROUP-CALL, re_INVITE and IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.1.3.3-10: SIP re-INVITE from IUT (MCVideo Server)
(Step 10, Table 5.1.1.1.3.2-1; Step 2, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.4.1.1.4-4, with conditions GROUP-CALL, re_INVITE and IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.1.3.3-11: Transmission Idle from IUT (MCVideo Server)
(Steps 4 and 6, Table 5.1.1.1.3.2-1; Steps 4 and 5, Annex A, Table A.3.8.3-1)**

Derivation Path: Annex A, Table A.3.8.4-6, with condition EMERGENCY-CALL.

**Table 5.1.1.1.3.3-12: Transmission Request from SS (MCVideo Server)
(Step 5, Table 5.1.1.1.3.2-1; Step 1, Annex A, Table A.3.7.3-1)**

Derivation Path: Annex A, Table A.3.7.4-1, with condition EMERGENCY-CALL.

**Table 5.1.1.1.3.3-13: Transmission Granted from IUT (MCVideo Server)
(Steps 3 and 5, Table 5.1.1.1.3.2-1;
Step 3a1 or 3b1, Annex A, Table A.3.7.3-1; Step 9a1, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.3.7.4-2 or A.3.7.4-3 (note), with condition EMERGENCY-CALL.
NOTE: Depends on server implementation.

**Table 5.1.1.1.3.3-14: Media Transmission Notification from IUT (MCVideo Server)
(Steps 3 and 5, Table 5.1.1.1.3.2-1; Step 3a3 or 3b2, Annex A, Table A.3.7.3-1;
Step 10a1, Annex A, Table 4.1.1.3-1)**

Derivation Path: Annex A, Table A.3.7.4-4 or A.3.7.4-5 (note), with condition EMERGENCY-CALL.
NOTE: Depends on server implementation.

**Table 5.1.1.1.3.3-15: Transmission Idle from IUT (MCVideo Server)
(Steps 11 and 13, Table 5.1.1.1.3.2-1; Steps 4 and 5, Annex A, Table A.3.8.3-1)**

Derivation Path: Annex A, Table A.3.8.4-6, with condition IMMPERIL-CALL.

**Table 5.1.1.1.3.3-16: Transmission Request from SS (MCVideo Server)
(Step 12, Table 5.1.1.1.3.2-1; Step 1, Annex A, Table A.3.7.3-1)**

Derivation Path: Annex A, Table A.3.7.4-1, with condition IMMPERIL-CALL.

Table 5.1.1.1.3.3-17: Transmission Granted from IUT (MCVideo Server) (Steps 10 and 12, Table 5.1.1.1.3.2-1; Step 3a1 or 3b1, Annex A, Table A.3.7.3-1; Step 9a1, Annex A, Table A.4.1.1.3-1)

Derivation Path: Annex A, Table A.3.7.4-2 or A.3.7.4-3 (note), with condition IMMPERIL-CALL.
NOTE: Depends on server implementation.

Table 5.1.1.1.3.3-18: Media Transmission Notification from IUT (MCVideo Server) (Steps 10 and 12, Table 5.1.1.1.3.2-1; Step 3a3 or 3b2, Annex A, Table A.3.7.3-1; Step 10a1, Annex A, Table 4.1.1.3-1)

Derivation Path: Annex A, Table A.3.7.4-4 or A.3.7.4-5 (note), with condition IMMPERIL-CALL.
NOTE: Depends on server implementation.

5.1.1.2 MCVideo Server - MCVideo Server / Pre-arranged Group Call / Automatic Commencement / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group call / Cancel Imminent Peril State / Terminating Participating server

5.1.1.2.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the controlling function initiates a pre-arranged group
call with automatic commencement }
    then { IUT (MCVideo Server) accepts the call and establishes the call with SS-UE2 (MCVideo
Client) and forwards the media transmission notification received from SS (MCVideo Server) to SS-UE2
(MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) ends the media transmission }
    then { IUT (MCVideo Server) forwards the transmission end notify received from SS (MCVideo
Server) to SS-UE2 (MCVideo Client) and forwards the transmission idle received from SS (MCVideo
Server) to SS-UE2 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) providing the controlling function upgrades the call to emergency
group call }
    then { IUT (MCVideo Server) accepts the upgrade and updates the call for SS-UE2 (MCVideo Client)
and sends media transmission notification to SS-UE2 (MCVideo Client) }
            }
```

(4)

```
with { IUT (MCVideo Server) having upgraded to an emergency pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) providing the controlling function cancels the ongoing emergency
state }
    then { IUT (MCVideo Server) accepts the emergency state cancellation and updates the call for
SS-UE2 (MCVideo Client) }
            }
```

(5)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) providing the controlling function upgrades the call to imminent
peril group call }
    then { IUT (MCVideo Server) accepts the upgrade and updates the call for SS-UE2 (MCVideo Client)
and sends media transmission notification to SS-UE2 (MCVideo Client) }
           }
```

(6)

```
with { IUT (MCVideo Server) having upgraded to an imminent peril pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) providing the controlling function cancels the ongoing imminent
peril state }
    then { IUT (MCVideo Server) accepts the imminent peril state cancellation and updates the call
for SS-UE2 (MCVideo Client) }
           }
```

(7)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) providing the controlling function ends the pre-arranged group call
}
    then { IUT (MCVideo Server) sends a SIP BYE message to SS-UE2 (MCVideo Client) and responds by
sending a SIP 200 (OK) message to SS (MCVideo Server) }
           }
```

5.1.1.2.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.1.3.2, 9.2.1.3.4, 9.2.1.3.6, 6.3.2.3.2, 6.3.2.2.5.1, 6.3.2.2.6.1, 6.3.2.2.8.1, 6.3.2.2.10, 6.3.2.2.4.2, 6.3.2.2.2.1 and 6.3.2.1.2.1, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.4.1, 6.4.2, 6.4.3, 6.4.4, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.2]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

NOTE 1: This clause covers on-demand session.

Upon receipt of a "SIP INVITE request for terminating participating MCVideo function", the participating MCVideo function:

…

2) shall check the presence of the isfocus media feature tag in the URI of the Contact header field and if it is not present then the participating MCVideo function shall reject the request …, and shall not continue with the rest of the steps;

3) if the Answer-Mode Indication in the application/poc-settings+xml MIME body has not yet been received from the invited MCVideo client as defined in clause 7.3.3 or clause 7.3.4, shall reject the request …, and shall not continue with the rest of the steps;

4) shall use the MCVideo ID present in the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCVideo ID and public user identity;

5) if the binding between the MCVideo ID and public user identity does not exist, then the participating MCVideo function shall reject the SIP INVITE request with a SIP 404 (Not Found) response. Otherwise, continue with the rest of the steps;

6) if the SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with an <MKFC-GKTPs> element, shall perform the procedures in clause 6.3.2.3.2;

7) shall perform the automatic commencement procedures specified in clause 6.3.2.2.5.1 and according to IETF RFC 5373 [27] if the "SIP INVITE request for terminating participating MCVideo function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCVideo client as per clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; and

8) shall perform the manual commencement procedures specified in clause 6.3.2.2.6.1 and according to IETF RFC 5373 [27] if the "SIP INVITE request for terminating participating MCVideo function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCVideo client as per clause 7.3.3 or clause 7.3.4 is set to "manual-answer"."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.4]

"Upon receiving a SIP BYE request from the controlling MCVideo function, the participating MCVideo function shall follow the procedures as specified in clause 6.3.2.2.8.1."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.6]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

Upon receipt of a SIP re-INVITE request for a terminating MCVideo client of a MCVideo group containing an emergency indication or imminent peril indication, the participating MCVideo function:

1) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may perform further actions outside the scope of this specification to act upon an included Resource-Priority header field as specified in 3GPP TS 24.229 [11];

2) shall generate an outgoing SIP re-INVITE request as specified in clause 6.3.2.2.10;

3) shall include in the SIP re-INVITE request an SDP offer based on the SDP offer in the received SIP re-INVITE request as specified in clause 6.3.2.2.1; and

4) shall send the SIP re-INVITE request towards the MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response to the above SIP re-INVITE request sent to the MCVideo client, the participating MCVideo function:

1) shall generate a SIP 200 (OK) response as described in the clause 6.3.2.2.4.2;

2) shall include in the SIP 200 (OK) response an SDP answer based on the SDP answer in the received SIP 200 (OK) response as specified in clause 6.3.2.2.2.1;

3) shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

4) shall forward the SIP 200 (OK) response according to 3GPP TS 24.229 [11]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.3.2]

"The participating MCVideo function:

1) shall extract the URI from the initiator field (IDRi) of the I_MESSAGE and use it together with the timer related parameter to check the signature of the I_MESSAGE as described in 3GPP TS 33.180 [8];

NOTE: If the terminating participating MCVideo function receives the SIP INVITE request from a controlling MCVideo function, then the URI in the IDRi is the controlling MCVideo function URI. If the terminating participating MCVideo function receives the SIP INVITE request from a non-controlling MCVideo function, then the URI in the IDRi is the non-controlling MCVideo function URI.

2) if the signature is not valid, shall exit this procedure. Otherwise shall validate that the contents of the recipient field (IDRr) of the I_MESSAGE to ensure it matches to the URI of the participating MCVideo function; and

3) if the contents of the IDRr do not match to the participating MCVideo function URI, shall exit this procedure. Otherwise, shall use the contents of the IDRr to decrypt the I_MESSAGE and extract the MKFC and MKFC-ID.

After the MKFC(s) and MKFC-ID(s) have been extracted from the I_MESSAGE(s), the participating MCVideo function shall provide the media-transmission control entity with the MKFC(s) and MKFC-ID(s), by interacting with the media plane as specified in 3GPP TS 24.581 [5]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.5.1]

"When receiving a "SIP INVITE request for terminating participating MCVideo function" that requires automatic commencement mode:

1) if:

a) the invited MCVideo client has one or more pre-established sessions without an associated MCVideo session;

b) the media-level sections for the offered MCVideo video media stream are the same as the media-level sections for MCVideo video media stream in an existing pre-established session; and

c) the media-level section of the offered media-transmission control entity is the same as the media-level section for media-transmission control entity in an existing pre-established session;

then the participating MCVideo function shall perform the actions specified in clause 6.3.2.2.5.3; or

2) otherwise the participating MCVideo function shall perform the actions specified in clause 6.3.2.2.5.2."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.6.1]

"When receiving a "SIP INVITE request for terminating participating MCVideo function" that requires manual commencement mode:

1) if:

a) the invited MCVideo client has one or more pre-established sessions without an associated MCVideo session;

b) the media-level sections for the offered MCVideo video media stream are the same as the media-level section for MCVideo video media stream in the existing pre-established session; and

c) the media-level section of the offered media-transmission control entity is the same as the media-level section for media-transmission control entity in the existing pre-established session;

then the participating MCVideo function shall perform the actions specified in clause 6.3.2.2.6.3; or

2) otherwise the participating MCVideo function shall perform the actions specified in clause 6.3.2.2.6.2."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.8.1]

"Upon receiving a SIP BYE request from the controlling MCVideo function, the participating MCVideo function:

1) shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.581 [5] for releasing media plane resource associated with the SIP session with the controlling MCVideo function;

2) shall generate a SIP BYE request according to 3GPP TS 24.229 [11];

3) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;

4) if the received SIP BYE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body into the outgoing SIP BYE request; and

5) shall send the SIP BYE request to the MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response to the SIP BYE request the participating MCVideo function:

1) shall send a SIP 200 (OK) response to the SIP BYE request received from the controlling MCVideo function according to 3GPP TS 24.229 [11]; and

2) shall interact with the media plane as specified in 3GPP TS 24.581 [5] for releasing media plane resources associated with the SIP session with the MCVideo client."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.10]

"This clause is referenced from other procedures.

The participating MCVideo function shall generate a SIP re-INVITE request according to 3GPP TS 24.229 [11] and:

1) shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [32];

2) may include a Resource-Share header field in accordance with clause 5.7.1.20.3 in 3GPP TS 24.229 [11];

3) shall perform the procedures specified in clause 6.3.2.2.9 to copy any MIME bodies in the received SIP re-INVITE request to the outgoing SIP re-INVITE request; and

4) if the received SIP re-INVITE request contains a Resource-Priority header field, shall include a Resource-Priority header field with the contents set as in the received Resource-Priority header field."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.4.2]

"This clause is referenced from other procedures.

When sending SIP 200 (OK) responses, the participating MCVideo function shall generate a SIP 200 (OK) response according to 3GPP TS 24.229 [11] and:

1) shall include the option tag "timer" in a Require header field;

2) shall include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [23], "UAS Behavior". If no "refresher" parameter was included in the SIP INVITE request, the "refresher" parameter in the Session-Expires header field shall be set to "uas";

3) shall include the following in the Contact header field:

a) the g.3gpp.mcvideo media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo"; and

c) an MCVideo session identity mapped to the MCVideo session identity provided in the Contact header field of the received SIP INVITE request from the controlling MCVideo function;

4) shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [32]; and

5) if the incoming SIP response contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body to the outgoing SIP 200 (OK) response."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.2.1]

"The participating MCVideo function shall follow the procedures in clause 6.3.2.1.2.1."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [11], the participating MCVideo function:

1) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCVideo function, for the accepted media stream in the received SDP offer, if required; and

NOTE 1: Requirements can exist for the participating MCVideo function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCVideo function, for the accepted media-transmission control entity, if present in the received SDP offer.

NOTE 2: If the participating MCVideo function and the controlling MCVideo function are in the same MCVideo server, and the participating MCVideo function does not have a dedicated IP address or a dedicated port number for media transmission control or media stream, the replacement of the IP address or the port number is omitted."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.1]

"Once an on-demand MCVideo session is established or a pre-established session is in use when the participating MCVideo function receives transmission control messages from the transmission participant in the MCVideo client or from the transmission control server in the controlling MCVideo function, the behaviour of the participating MCVideo function is described in the following clauses."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.2]

"Upon receiving a transmission control message the participating MCVideo function:

1. shall immediately forward the transmission control message to the transmission control server if the message is received from the transmission participant;

2. if an MBMS subchannel is not used for a transmission in the session the transmission control message is associated with, shall immediately forward the transmission control message to the transmission participant if the message is received from the transmission control server; and

3. if an MBMS subchannel is used for a transmission in the session the transmission control message is associated with:

a. if

i. the transmission control message is not a Transmission Idle message or a Media Transmission Notify message;

ii. the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or

iii. the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report;

shall immediately forward the transmission control message to the transmission participant; and

b. if

i. the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report; and

ii if the transmission control message is the Transmission Idle message or the Media Transmission Notify message,

shall perform actions as specified in clause 10.2.

NOTE: When the Transmit Idle or Media Transmission Notify messages are discarded the messages are sent to the MCVideo clients over the MBMS subchannel allocated for the transmission as specified in clause 10.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.3]

"Upon receiving RTP media packets the participating MCVideo function:

1. *shall immediately forward the RTP media packet to the controlling MCVideo function if the RTP packet is from an MCVideo client; and*

2. *if an MBMS subchannel is not used for a transmission in the session the RTP media packets are associated with, shall immediately forward the RTP media packets to the MCVideo client if the RTP packet is from the controlling MCVideo function or the non-controlling MCVideo function.*

3. *if an MBMS subchannel is used for a transmission in the session the RTP media packets are associated with and if RTP media packets are received from the controlling MCVideo function or the non-controlling MCVideo function:*

    a. *if*

        i. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

        ii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report,*

        shall immediately forward the RTP media packets to the MCVideo client; and

    b. *if the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report, shall perform actions as specified in clause 10.2."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.4]

"When the participating function receives an indication from the application and signalling plane that session release is initiated, the participating MCVideo function:

1. *shall stop sending transmission control messages towards the transmission participant and the transmission control server; and*

2. *shall stop sending RTP media packets towards the MCVideo client and towards the controlling MCVideo function.*

When the participating MCVideo function receives an indication from the application and signalling plane that the session is released, the participating MCVideo function:

1. *in case of a pre-established session, shall perform the actions in clause 9.3.2; and*

2. *in case of an on-demand session, shall release the media resources associated with the session. "*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.

Step 2 When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

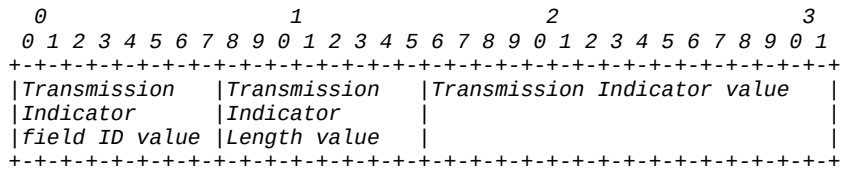
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

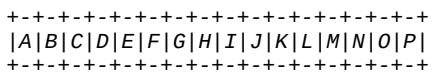


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. *if the active Transmission request queue is empty the transmission control server:*
    a. *shall send Transmission Idle message to all transmission participants. The Transmission Idle message:*
        i. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*
        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;*
    b. *shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;*
    c. *shall start timer T1 (Inactivity);*
    d. *shall set the general state to the 'G: Transmit Idle' state; and*
    e. *shall initialise counter Cx (Simultaneous transmission video) to 0.*
2. *if the active Transmission request queue is not empty the transmission control server:*
    a. *shall select a queued Transmission request from the top of the active Transmission request queue;*
    b. *shall remove that queued Transmission request from the active Transmission request queue; and*
    c. *shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:*
    a. *shall include the granted priority in the Transmission priority field;*
    b. *shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;*
    c. *if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*
    d. *shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;*
2. *shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;*
3. *shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:*
    a. *shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;*
    b. *shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;*
    c. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;*

d. if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';

e. if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

g. shall initiate the 'general reception control operation' state machine;and

4. shall enter the 'G: Transmit Taken' state."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. if counter Cx (Simultaneous transmission video) has not reached its upper limit:

a. shall perform the actions specified in the clause 6.3.4.4.2;

2. if counter Cx (Simultaneous transmission video) has reached its upper limit:

a. select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;

b. shall stop timer T4 (Transmission Grant), if running;

c. shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);

d. shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;

e. shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and

f. shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:

i. shall include the queue position and transmission priority in the Queue Info field; and

ii. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:

a. shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and

b. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

2. shall set the general state to 'G: pending Transmission Revoke'."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

b. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.1.2.3 Test description

5.1.1.2.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the controlling MCVideo function.
- SS-UE2 (MCVideo Client).

IUT:

- IUT (MCVideo Server), providing the terminating participating MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of some sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server and the HTTP Client.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server) and SS-UE2 (MCVideo Client).
- The SS-UE2 (MCVideo client) is registered and authorized to use the MCVideo system.
- The SS-UE2 (MCVideo client) is affiliated to the pre-arranged group and is authorized to initiate group calls.

5.1.1.2.3.2 Test procedure sequence

Table 5.1.1.2.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CT group/private call establishment, automatic commencement' as described in Annex A, Table A.4.3.3.3-1 to establish a pre-arranged group call with automatic commencement mode and notify a media transmission from SS-UE1 (MCVideo Client)?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Notification - Idle' as described in Annex A, Table A.4.3.8.3-1 to notify the end of SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CT session establishment/modification without provisional responses other than 100 Trying', as described in Annex A, Table A.4.3.1.3-1 to upgrade the call to emergency group call and notify a media transmission from SS-UE1 (MCVideo Client)?	-	-	3	P
4	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Notification - Idle' as described in Annex A, Table A.4.3.8.3-1 to notify the end of SS-UE1 (MCVideo Client) transmission?	-	-	2	P
5	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CT session establishment/modification without provisional responses other than 100 Trying', as described in Annex A, Table A.4.3.1.3-1 to downgrade the call and without media transmission notification?	-	-	4	P
6	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CT session establishment/modification without provisional responses other than 100 Trying', as described in Annex A, Table A.4.3.1.3-1 to upgrade the call to imminent peril group call and notify a media transmission from SS-UE1 (MCVideo Client)?	-	-	5	P
7	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Notification - Idle' as described in Annex A, Table A.4.3.8.3-1 to notify the end of SS-UE1 (MCVideo Client) transmission?	-	-	2	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
8	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CT session establishment/modification without provisional responses other than 100 Trying', as described in Annex A, Table A.4.3.1.3-1 to downgrade the call and without media transmission notification?	-	-	6	P
9	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.3.5.3-1, to release the call?	-	-	7	P

5.1.1.2.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

Table 5.1.1.2.3.3-1: SIP INVITE from SS (MCVideo Server)
(Step 1, Table 5.1.1.2.3.2-1; Step 1, Annex A, Table A.4.3.3.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-1, with condition GROUP CALL.

Table 5.1.1.2.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.1.2.3.2-1; Step 2, Annex A, Table A.4.3.3.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-2, with condition GROUP CALL.

Table 5.1.1.2.3.3-3: SIP 200 (OK) from SS-UE2 (MCVideo Client)
(Steps 1, 3, 5, 6 and 8, Table 5.1.1.2.3.2-1; Step 6, Annex A, Table A.4.3.3.3-1; Step 5, Annex A, Table A.4.3.1.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-7, with condition GROUP CALL.

Table 5.1.1.2.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Steps 1, 3, 5, 6 and 8, Table 5.1.1.2.3.2-1; Step 7, Annex A, Table A.4.3.3.3-1; Step 6, Annex A, Table A.4.3.1.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-8, with condition GROUP CALL.

Table 5.1.1.2.3.3-5: SIP re-INVITE from SS (MCVideo Server)
(Step 3, Table 5.1.1.2.3.2-1; Step 1, Annex A, Table A.4.3.1.3-1)

Derivation Path: Annex A, Table A.4.3.1.4-1, with conditions GROUP-CALL, re_INVITE and EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.1.2.3.3-6: SIP re-INVITE from IUT (MCVideo Server)
(Step 3, Table 5.1.1.2.3.2-1; Step 2, Annex A, Table A.4.3.1.3-1)

Derivation Path: Annex A, Table A.4.3.1.4-2, with condition GROUP-CALL, re_INVITE and EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.2.3.3-7: SIP re-INVITE from SS (MCVideo Server)
(Steps 5 and 8, Table 5.1.1.2.3.2-1; Step 1, Annex A, Table A.4.3.1.3-1)**

Derivation Path: Annex A, Table A.4.3.1.4-1, with conditions GROUP-CALL, re_INVITE.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.2.3.3-8: SIP re-INVITE from IUT (MCVideo Server)
(Steps 5 and 8, Table 5.1.1.2.3.2-1; Step 2, Annex A, Table A.4.3.1.3-1)**

Derivation Path: Annex A, Table A.4.3.1.4-2, with condition GROUP-CALL, re_INVITE.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.2.3.3-9: SIP re-INVITE from SS (MCVideo Server)
(Step 6, Table 5.1.1.2.3.2-1; Step 1, Annex A, Table A.4.3.1.3-1)**

Derivation Path: Annex A, Table A.4.3.1.4-1, with conditions GROUP-CALL, re_INVITE and IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.2.3.3-10: SIP re-INVITE from IUT (MCVideo Server)
(Step 6, Table 5.1.1.2.3.2-1; Step 2, Annex A, Table A.4.3.1.3-1)**

Derivation Path: Annex A, Table A.4.3.1.4-2, with condition GROUP-CALL, re_INVITE and IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.2.3.3-11: Transmission Idle
(Step 4, Table 5.1.1.2.3.2-1; Steps 13 and 14, Annex A, Table A.4.3.8.3-1)**

Derivation Path: Annex A, Table A.4.3.8.4-6, with condition EMERGENCY-CALL.

**Table 5.1.1.2.3.3-12: Media Transmission Notification
(Step 3, Table 5.1.1.2.3.2-1; Steps 9a1 and 9a2, Annex A, Table 4.3.1.3-1)**

Derivation Path: Annex A, Table A.4.3.1.4-7, with condition EMERGENCY-CALL.

**Table 5.1.1.2.3.3-13: Transmission Idle
(Step 7, Table 5.1.1.2.3.2-1; Steps 13 and 14, Annex A, Table A.4.3.8.3-1)**

Derivation Path: Annex A, Table A.4.3.8.4-6, with condition IMMPERIL-CALL.

**Table 5.1.1.2.3.3-14: Media Transmission Notification
(Step 6, Table 5.1.1.2.3.2-1; Steps 9a1 and 9a2, Annex A, Table 4.3.1.3-1)**

Derivation Path: Annex A, Table A.4.3.1.4-7, with condition IMMPERIL-CALL.

5.1.1.3 MCVideo Server - MCVideo Server / Pre-arranged Group Call / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group call / Cancel Imminent Peril State / Originating Participating server

5.1.1.3.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS-UE1 (MCVideo Client) initiates a pre-arranged group call with implicit transmission
control }
    then { IUT (MCVideo Server) accepts the call from initiator and establishes the call with the SS
(MCVideo Server) providing the controlling function and forwards the transmission granted received
from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests to end the transmission control }
    then { IUT (MCVideo Server) forwards the transmission end request to SS (MCVideo Server) and
forwards the transmission end response to SS-UE1 (MCVideo Client) and forwards the transmission idle
received from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control }
    then { IUT (MCVideo Server) forwards the transmission request to SS (MCVideo Server) and
forwards the transmission granted received from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(4)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) upgrades the call to emergency group call with implicit
transmission request }
    then { IUT (MCVideo Server) forwards the upgrade request to SS (MCVideo Server) and forwards
upgrade request confirmation from SS (MCVideo Server) to SS-UE1 (MCVideo Client) and forwards the
transmission granted received from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(5)

```
with { IUT (MCVideo Server) having upgraded to an emergency pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) cancels the ongoing emergency state }
    then { IUT (MCVideo Server) forwards the emergency cancellation request to SS (MCVideo Server)
and forwards the emergency state cancellation confirmation from SS (MCVideo Server) to SS-UE1
(MCVideo Client) }
            }
```

(6)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) upgrades the call to imminent peril group call with implicit
transmission request }
    then { IUT (MCVideo Server) forwards the upgrade request to SS (MCVideo Server) and forwards
upgrade request confirmation from SS (MCVideo Server) to SS-UE1 (MCVideo Client) and forwards the
transmission granted received from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(7)

```
with { IUT (MCVideo Server) having upgraded to an imminent peril pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) cancels the ongoing imminent peril state }
    then { IUT (MCVideo Server) forwards the imminent peril cancellation request to SS (MCVideo
Server) and forwards the imminent peril state cancellation confirmation from SS (MCVideo Server) to
SS-UE1 (MCVideo Client) }
            }
```

(8)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) ends the pre-arranged group call }
    then { IUT (MCVideo Server) sends a SIP BYE message to SS (MCVideo Server) providing the
controlling function and responds by sending a SIP 200 (OK) message to SS-UE1 (MCVideo Client) }
            }
```

5.1.1.3.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.1.3.1.1, 9.2.1.3.1.3, 9.2.1.3.3, 6.3.1.1, 6.3.2.1.3, 6.3.2.1.1.1, 6.3.2.1.6.1 and 6.3.2.1.7, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.4.1, 6.4.2, 6.4.3, 6.4.4, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.1.1]

"In the procedures in this clause:

1) group identity in an incoming SIP INVITE request refers to the group identity from the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;

2) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

3) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

Upon receipt of a "SIP INVITE request for originating participating MCVideo function" containing an application/vnd.3gpp.mcvideo-info+xml MIME body with the <session-type> element set to a value of "prearranged", the participating MCVideo function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP INVITE request with a SIP 500 (Server Internal Error) response. The participating MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15]. Otherwise, continue with the rest of the steps;

NOTE 1: if the SIP INVITE request contains an emergency indication or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCVideo function can according to local policy choose to accept the request.

2) shall determine the MCVideo ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request, and shall authorise the calling user;

NOTE 2: The MCVideo ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

3) if through local policy in the participating MCVideo function, the user identified by the MCVideo ID is not authorised to initiate prearranged group calls, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response to the SIP INVITE request, with warning text set to "109 user not authorised to make prearranged group calls" in a Warning header field as specified in clause 4.4;

4) shall validate the media parameters and if the MCVideo codecs are not offered in the SIP INVITE request shall reject the request with a SIP 488 (Not Acceptable Here) response. Otherwise, continue with the rest of the steps;

5) shall check if the number of maximum simultaneous MCVideo group calls supported for the MCVideo user as specified in the <MaxSimultaneousCallsN6> element of the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) has been exceeded. If exceeded, the participating MCVideo function shall respond with a SIP 486 (Busy Here) response with the warning text set to "103 maximum simultaneous MCVideo group calls reached" in a Warning header field as specified in clause 4.4. Otherwise, continue with the rest of the steps;

NOTE 3: If the SIP INVITE request contains an emergency indication or an imminent peril indication, the participating MCVideo function can by means beyond the scope of this specification choose to allow for an exception to the limit for the maximum simultaneous MCVideo sessions supported for the MCVideo user. Alternatively, a lower priority session of the MCVideo user could be terminated to allow for the new session.

6) if the user identified by the MCVideo ID is not affiliated to the group identified in the "SIP INVITE request for originating participating MCVideo function" as determined by clause 8.2.2.2.11 and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, shall perform the actions specified in clause 8.2.2.2.12 for implicit affiliation;

7) if the actions for implicit affiliation specified in step 6) above were performed but not successful in affiliating the MCVideo user due to the MCVideo user already having N2 simultaneous affiliations, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 486 (Busy Here) response with the warning text set to "102 too many simultaneous affiliations" in a Warning header field as specified in clause 4.4. and skip the rest of the steps.

NOTE 4: N2 is the total number of MCVideo groups that an MCVideo user can be affiliated to simultaneously as specified in 3GPP TS 23.281 [26].

NOTE 5: if the SIP INVITE request contains an emergency indication set to a value of "true" or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCVideo function can according to local policy choose to allow an exception to the N2 limit. Alternatively, a lower priority affiliation of the MCVideo user could be cancelled to allow for the new affiliation.

8) shall determine the public service identity of the controlling MCVideo function associated with the group identity in the SIP INVITE request;

NOTE 6: The public service identity can identify the controlling MCVideo function in the primary MCVideo system or a partner MCVideo system.

NOTE 7: How the participating MCVideo server discovers the public service identity of the controlling MCVideo function associated with the group identity is out of scope of the current release.

9) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3;

10) shall set the Request-URI to the public service identity of the controlling MCVideo function associated with the group identity which was present in the incoming SIP INVITE request;

11) shall not copy the following header fields from the incoming SIP INVITE request to the outgoing SIP INVITE request, if they were present in the incoming SIP INVITE request:

a) Answer-Mode header field as specified in IETF RFC 5373 [27]; and

b) Priv-Answer-Mode header field as specified in IETF RFC 5373 [27];

12) shall set the <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the SIP INVITE request to the MCVideo ID of the calling user;

13) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the MCVideo client as specified in clause 6.3.2.1.1.1;

14) if the received SIP INVITE request contains an application/vnd.3gpp.location-info+xml MIME body and if not already copied, shall copy the contents of the application/vnd.3gpp.location-info+xml MIME body received in the SIP INVITE request into an application/vnd.3gpp.location-info+xml MIME body included in the outgoing SIP request;

15) if a Resource-Priority header field was included in the received SIP INVITE request, shall include a Resource-Priority header field according to rules and procedures of 3GPP TS 24.229 [11] set to the value indicated in the Resource-Priority header field of the SIP INVITE request from the MCVideo client; and

NOTE 8: The participating MCVideo function will leave verification of the Resource-Priority header field to the controlling MCVideo function.

16) shall forward the SIP INVITE request, according to 3GPP TS 24.229 [11].

Upon receipt of a SIP 302 (Moved Temporarily) response to the above SIP INVITE request in step 15), the participating MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.2.1.10;

2) shall include an SDP offer based upon the SDP offer in the received SIP INVITE request from the MCVideo client as specified in clause 6.3.2.1.1.1; and

3) shall forward the SIP INVITE request according to 3GPP TS 24.229 [11].

Upon receipt of a SIP 2xx response in response to the above SIP INVITE request in step 15), the participating MCVideo function:

1) if the received SIP 2xx response contains an application/vnd.3gpp.mcvideo-info+xml MIME body with an <MKFC-GKTPs> element, shall perform the procedures in clause 6.3.2.3.2;

2) shall generate a SIP 200 (OK) response as in clause 6.3.2.1.5.2;

3) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

4) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response;

5) shall include the public service identity received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response;

6) shall include an MCVideo session identity mapped to the MCVideo session identity provided in the Contact header field of the received SIP 200 (OK) response;

7) if the procedures of clause 8.2.2.2.12 for implicit affiliation were performed in the present clause, shall complete the implicit affiliation by performing the procedures of clause 8.2.2.2.13;

8) shall send the SIP 200 (OK) response to the MCVideo client according to 3GPP TS 24.229 [11];

9) shall interact with Media Plane as specified in 3GPP TS 24.581 [5]; and

10) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [23].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the above SIP INVITE request in step 14) the participating MCVideo function:

1) shall generate a SIP response according to 3GPP TS 24.229 [11];

2) shall include Warning header field(s) that were received in the incoming SIP response;

3) shall forward the SIP response to the MCVideo client according to 3GPP TS 24.229 [11]; and

4) if the implicit affiliation procedures of clause 8.2.2.2.12 were invoked in this procedure, shall perform the procedures of clause 8.2.2.2.14;"

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.1.3]

"This clause covers on-demand session.

Upon receipt of a SIP re-INVITE request for an MCVideo session identifying an on-demand prearranged MCVideo group session, the participating MCVideo function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP re-INVITE request with a SIP 500 (Server Internal Error) response. The participating MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15] and skip the rest of the steps;

NOTE 1: If the SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "true", the participating MCVideo function can choose to accept the request.

2) shall determine if the media parameters are acceptable and the MCVideo codecs are offered in the SDP offer and if not, reject the request with a SIP 488 (Not Acceptable Here) response. Otherwise, continue with the rest of the steps;

3) shall generate an outgoing SIP re-INVITE request as specified in clause 6.3.2.1.9;

4) shall, if the SIP re-INVITE request was received within an on-demand session, include in the SIP re-INVITE request an SDP offer based on the SDP offer in the received SIP re-INVITE request as specified in clause 6.3.2.1.1;

5) if the received SIP re-INVITE request contains a Resource-Priority header field, shall include a Resource-Priority header field with the contents set as in the received Resource-Priority header field; and

NOTE 3: The controlling MCVideo function will determine the validity of the Resource-Priority header field.

6) shall forward the SIP re-INVITE request according to 3GPP TS 24.229 [11].

Upon receipt of a SIP 2xx response to the above SIP re-INVITE request in step 7) the participating MCVideo function:

1) shall generate a SIP 200 (OK) response as specified in the clause 6.3.2.1.5.2;

2) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

3) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response;

4) shall copy the contents received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response;

5) shall send the SIP 200 (OK) response towards the MCVideo client according to 3GPP TS 24.229 [11]; and

6) shall interact with the media plane as specified in 3GPP TS 24.581 [5].

Upon receipt of a SIP 403 (Forbidden) response to the above SIP INVITE request in step 7) the participating MCVideo function:

1) shall generate a SIP 403 (Forbidden) response according to 3GPP TS 24.229 [11];

2) shall copy, if included in the received SIP 403 (Forbidden) response, the application/vnd.3gpp.mcvideo-info+xml MIME body MIME body to the outgoing SIP (Forbidden) response;

3) shall include Warning header field(s) that were received in the incoming SIP 403 (Forbidden) response;

4) shall forward the SIP 403 (Forbidden) response to the MCVideo client according to 3GPP TS 24.229 [11]; and

5) shall interact with the media plane as specified in 3GPP TS 24.581 [5]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.3]

"Upon receiving from the MCVideo client a SIP BYE request the participating MCVideo function shall follow the procedures as specified in clause 6.3.2.1.6."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.1.1]

"The MCVideo server needs to distinguish between the following initial SIP INVITE requests for originations and terminations:

- *SIP INVITE requests routed to the participating MCVideo function as a result of processing initial filter criteria at the S-CSCF in accordance with the origination procedures as specified in 3GPP TS 24.229 [11] and the Request-URI is set to a public service identity of the participating MCVideo function that does not identify the pre-established session set-up. Such requests are known as "SIP INVITE request for originating participating MCVideo function" in the procedures in the present document;*

- *SIP INVITE requests routed to the participating MCVideo function as a result of processing initial filter criteria at the S-CSCF in accordance with the termination procedures as specified in 3GPP TS 24.229 [11] and the Request-URI contains a PSI of the terminating participating MCVideo function. Such requests are known as "SIP INVITE request for terminating participating MCVideo function" in the procedures in the present document;*

- *SIP INVITE requests routed to the controlling MCVideo function as a result of PSI routing on the originating side in accordance with the originating procedures as specified in 3GPP TS 24.229 [11], or as a result of direct PSI routing, in accordance with the termination procedures as specified in 3GPP TS 24.229 [11], the Request-URI is set to a public service identity for MCVideo private call and the Contact header field does not contain the isfocus media feature tag specified in IETF RFC 3840 [16]. Such requests are known as "SIP INVITE request for controlling MCVideo function of a private call" in the procedures in the present document;*

- *SIP INVITE requests routed to the controlling MCVideo function as a result of PSI routing on the originating side in accordance with the originating procedures as specified in 3GPP TS 24.229 [11], or as a result of direct PSI routing, in accordance with the termination procedures as specified in 3GPP TS 24.229 [11], the Request-URI is set to a public service identity serving an MCVideo group and the Contact header field does not contain the isfocus media feature tag specified in IETF RFC 3840 [16]. Such requests are known as "SIP INVITE request for controlling MCVideo function of an MCVideo group" in the procedures in the present document; and*

- *SIP INVITE requests routed to the controlling MCVideo function as a result of PSI routing on the originating side in accordance with the originating procedures as specified in 3GPP TS 24.229 [11], or as a result of direct PSI routing, in accordance with the termination procedures as specified in 3GPP TS 24.229 [11], the Request-URI is set to a public service identity for MCVideo ambient viewing call and the Contact header field does not contain the isfocus media feature tag specified in IETF RFC 3840 [22]. Such requests are known as "SIP INVITE request for controlling MCVideo function of an ambient viewing call" in the procedures in the present document."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.3]

"This clause is referenced from other procedures.

When generating an initial SIP INVITE request according to 3GPP TS 24.229 [11], on receipt of an incoming SIP INVITE request, the participating MCVideo function:

1) *shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [20] if included in the incoming SIP INVITE request;*

2) *should include the Session-Expires header field according to IETF RFC 4028 [23]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";*

3) *shall include the option tag "timer" in the Supported header field;*

4) *shall copy the contents of the P-Asserted-Identity header field of the incoming SIP INVITE request to the P-Asserted-Identity header field of the outgoing SIP INVITE request;*

5) *shall include the g.3gpp.mcvideo media feature tag into the Contact header field of the outgoing SIP INVITE request;*

6) *shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcvideo" (coded as specified in 3GPP TS 24.229 [11]), into the P-Asserted-Service header field of the outgoing SIP INVITE request;*

7) if the incoming SIP INVITE request contained a MIME resource-lists body with the MCVideo ID of the invited MCVideo user, shall copy the MIME resource-lists body, according to rules and procedures of IETF RFC 5366 [37];

8) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the contents of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request to the outgoing SIP INVITE request; and

9) if the incoming SIP INVITE request contained an application/vnd.3gpp.location-info+xml MIME body, shall copy the contents of the application/vnd.3gpp.location-info+xml MIME body of the incoming SIP INVITE request to the outgoing SIP INVITE request."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.1.1]

"This clause is referenced from other clauses.

The SDP offer is generated based on the received SDP offer. The SDP offer generated by the participating MCVideo function:

1) shall contain two SDP media-level sections for MCVideo video media as contained in the received SDP offer; and

2) shall contain an SDP media-level section for one media-transmission control entity, if present in the received SDP offer.

When composing the SDP offer according to 3GPP TS 24.229 [11], the participating MCVideo function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the participating MCVideo function, if required;

NOTE 1: Requirements can exist for the participating MCVideo function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number for the offered media transmission control entity, if any, in the received SDP offer with the IP address and port number of the participating MCVideo function; and

NOTE 2: If the participating MCVideo function and the controlling MCVideo function are in the same MCVideo server, and the participating MCVideo function does not have a dedicated IP address or a dedicated port number for media transmission control or media stream, the replacement of the IP address or the port number is omitted.

3) shall contain an "a=key-mgmt" attribute field with a "mikey" attribute value, if present in the received SDP offer."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.6.1]

"When the participating MCVideo function receives a request from the MCVideo client to originate an MCVideo emergency group call and needs to determine if the request is an authorised request for an MCVideo emergency call, the participating MCVideo function shall check the following:

1) if the <allow-emergency-group-call> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true" and:

a) if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> element contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "DedicatedGroup" and if the <uri-entry> element of the <entry> element of the <MCVideoGroupInitiation> element contains the identity of the MCVideo group targeted by the calling MCVideo user; or

b) if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "UseCurrentlySelectedGroup";

then the participating MCVideo function shall consider the MCVideo emergency group call request to be an authorised request for an MCVideo emergency group call;

In all other cases, the participating MCVideo function shall consider the request to originate an MCVideo emergency group call to be an unauthorised request to originate an MCVideo emergency group call.

When the participating MCVideo function receives a request from the MCVideo client to originate an MCVideo imminent peril group call and needs to determine if the request is an authorised request for an MCVideo imminent peril group call the participating MCVideo function shall check the following:

1) *if the <allow-imminent-peril-call> element of <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true"; and*

   a) *if the "entry-info" attribute of the <MCVideoGroupInitiation> element contained within the <ImminentPerilCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "DedicatedGroup" and if the <uri-entry> element of the <entry> element of the <MCVideoGroupInitiation> element contains the identity of the MCVideo group targeted by the calling MCVideo user; or*

   b) *if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element contained within the <ImminentPerilCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "UseCurrentlySelectedGroup";*

   then the participating MCVideo function shall consider the MCVideo imminent peril group call request to be an authorised request for an MCVideo emergency group call;

In all other cases, the participating MCVideo function shall consider the request to originate an MCVideo imminent peril group call to be an unauthorised request to originate an MCVideo imminent peril call."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.7]

"This clause is referenced from other procedures.

When generating a SIP re-INVITE request according to 3GPP TS 24.229 [11] on receipt of an incoming SIP re-INVITE request, the participating MCVideo function:

1) *if the incoming SIP re-INVITE request contained a MIME resource-lists body with the MCVideo ID of the invited MCVideo user, shall copy the MIME resource-lists body, according to rules and procedures of IETF RFC 5366 [37];*

2) *if the incoming SIP re-INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body; and*

3) *if the incoming SIP re-INVITE request contained an application/vnd.3gpp.location-info+xml MIME body, shall copy the application/vnd.3gpp.location-info+xml MIME body."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.1]

"Once an on-demand MCVideo session is established or a pre-established session is in use when the participating MCVideo function receives transmission control messages from the transmission participant in the MCVideo client or from the transmission control server in the controlling MCVideo function, the behaviour of the participating MCVideo function is described in the following clauses."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.2]

"Upon receiving a transmission control message the participating MCVideo function:

1. *shall immediately forward the transmission control message to the transmission control server if the message is received from the transmission participant;*

2. *if an MBMS subchannel is not used for a transmission in the session the transmission control message is associated with, shall immediately forward the transmission control message to the transmission participant if the message is received from the transmission control server; and*

3. *if an MBMS subchannel is used for a transmission in the session the transmission control message is associated with:*

   a. *if*

      i. *the transmission control message is not a Transmission Idle message or a Media Transmission Notify message;*

      ii. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

      iii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report;*

      shall immediately forward the transmission control message to the transmission participant; and

   b. *if*

      i. *the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report; and*

      ii *if the transmission control message is the Transmission Idle message or the Media Transmission Notify message,*

   shall perform actions as specified in clause 10.2.

NOTE: When the Transmit Idle or Media Transmission Notify messages are discarded the messages are sent to the MCVideo clients over the MBMS subchannel allocated for the transmission as specified in clause 10.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.3]

"Upon receiving RTP media packets the participating MCVideo function:

1. *shall immediately forward the RTP media packet to the controlling MCVideo function if the RTP packet is from an MCVideo client; and*

2. *if an MBMS subchannel is not used for a transmission in the session the RTP media packets are associated with, shall immediately forward the RTP media packets to the MCVideo client if the RTP packet is from the controlling MCVideo function or the non-controlling MCVideo function.*

3. *if an MBMS subchannel is used for a transmission in the session the RTP media packets are associated with and if RTP media packets are received from the controlling MCVideo function or the non-controlling MCVideo function:*

   a. *if*

      i. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

      ii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report,*

      shall immediately forward the RTP media packets to the MCVideo client; and

   b. *if the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report, shall perform actions as specified in clause 10.2."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.4]

"When the participating function receives an indication from the application and signalling plane that session release is initiated, the participating MCVideo function:

1. *shall stop sending transmission control messages towards the transmission participant and the transmission control server; and*

2. *shall stop sending RTP media packets towards the MCVideo client and towards the controlling MCVideo function.*

When the participating MCVideo function receives an indication from the application and signalling plane that the session is released, the participating MCVideo function:

1. *in case of a pre-established session, shall perform the actions in clause 9.3.2; and*

2. *in case of an on-demand session, shall release the media resources associated with the session."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

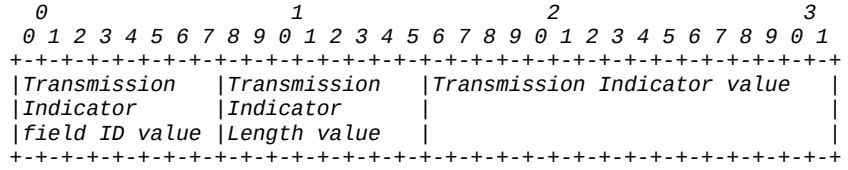
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

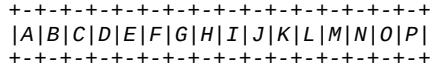


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. *if the active Transmission request queue is empty the transmission control server:*
    a. *shall send Transmission Idle message to all transmission participants. The Transmission Idle message:*
        i. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*
        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;*
    b. *shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;*
    c. *shall start timer T1 (Inactivity);*
    d. *shall set the general state to the 'G: Transmit Idle' state; and*
    e. *shall initialise counter Cx (Simultaneous transmission video) to 0.*
2. *if the active Transmission request queue is not empty the transmission control server:*
    a. *shall select a queued Transmission request from the top of the active Transmission request queue;*
    b. *shall remove that queued Transmission request from the active Transmission request queue; and*
    c. *shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:*
    a. *shall include the granted priority in the Transmission priority field;*
    b. *shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;*

c. if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

d. shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;

2. shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;

3. shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:

a. shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;

b. shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;

c. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;

d. if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';

e. if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

g. shall initiate the 'general reception control operation' state machine;and

4. shall enter the 'G: Transmit Taken' state."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. if counter Cx (Simultaneous transmission video) has not reached its upper limit:

a. shall perform the actions specified in the clause 6.3.4.4.2;

2. if counter Cx (Simultaneous transmission video) has reached its upper limit:

a. select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;

b. shall stop timer T4 (Transmission Grant), if running;

c. shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);

d. shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;

e. shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and

f. shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:

i. shall include the queue position and transmission priority in the Queue Info field; and

ii. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:*

   a. *shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and*

   b. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

2. *shall set the general state to 'G: pending Transmission Revoke'."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. *if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:*

   a. *shall initialize a general state machine as specified in clause 6.3.4.2.2; and*

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

   b. *shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;*

2. *if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and*

   a. *if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:*

      i. *should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:*

         A. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*

         B. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

      ii. *shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;*

   b. *if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:*

      i. *shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and*

      ii. *shall initialize a general state machine as specified in clause 6.3.4.2.2; and*

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.1.3.3 Test description

5.1.1.3.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the controlling MCVideo function.
- SS-UE1 (MCVideo Client).

IUT:

- IUT (MCVideo Server), providing the originating participating MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of some sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server and the HTTP Client.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server) and SS-UE1 (MCVideo Client).
- The SS-UE1 (MCVideo client) is registered and authorized to use the MCVideo system.
- The SS-UE1 (MCVideo client) is affiliated to the pre-arranged group and is authorized to initiate group calls.

5.1.1.3.3.2 Test procedure sequence

Table 5.1.1.3.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.4.2.1.3-1 to establish a pre-arranged group call with implicit transmission control request according to option b.ii of note in Annex A, Table A.4.2.1.3-1, initiated by SS-UE1 (MCVideo Client?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO Transmission End Request - Transmission End Response' as described in Annex A, Table A.4.2.5.3-1 to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO session establishment/modification without provisional responses other than 100 Trying', as described in Annex A, Table A.4.2.1.3-1 to upgrade the call to emergency group call with implicit transmission request according to option b.ii of note in Annex A, Table A.4.2.1.3-1?	-	-	4	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
4	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO Transmission End Request - Transmission End Response' as described in Annex A, Table A.4.2.5.3-1 to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
5	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Transmission Granted' as described in Annex A, Table A.4.2.4.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	3	P
6	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO Transmission End Request - Transmission End Response' as described in Annex A, Table A.4.2.5.3-1 to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
7	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO session establishment/modification without provisional responses other than 100 Trying', as described in Annex A, Table A.4.2.1.3-1 to downgrade the call without implicit transmission request, with option a of note on Table A.4.2.1.3-1?	-	-	5	P
8	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Transmission Granted' as described in Annex A, Table A.4.2.4.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	3	P
9	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO Transmission End Request - Transmission End Response' as described in Annex A, Table A.4.2.5.3-1 to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
10	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO session establishment/modification without provisional responses other than 100 Trying', as described in Annex A, Table A.4.2.1.3-1 to upgrade the call to imminent peril group call with implicit transmission request according to option b.ii of note in Annex A, Table A.4.2.1.3-1?	-	-	6	P
11	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO Transmission End Request - Transmission End Response' as described in Annex A, Table A.4.2.5.3-1 to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
12	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Transmission Granted' as described in Annex A, Table A.4.2.4.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	3	P
13	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO Transmission End Request - Transmission End Response' as described in Annex A, Table A.4.2.5.3-1 to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
14	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO session establishment/modification without provisional responses other than 100 Trying', as described in Annex A, Table A.4.2.1.3-1 to downgrade the call without implicit transmission request, with option a of note on Table A.4.2.1.3-1?	-	-	7	P
15	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO call release' as described in Annex A, Table A.4.2.3.3-1, to release the call?	-	-	8	P

5.1.1.3.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

**Table 5.1.1.3.3.3-1: SIP INVITE from SS-UE1 (MCVideo Client)
(Step 1, Table 5.1.1.3.3.2-1; Step 1, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-1, with condition GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.3.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.1.3.3.2-1; Step 2, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-2, with condition GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.3.3.3-3: SIP 200 (OK) from SS (MCVideo Server)
(Steps 1, 3, 7, 10 and 14, Table 5.1.1.3.3.2-1; Step 5, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-3, with condition GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.3.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Steps 1, 3, 7, 10 and 14, Table 5.1.1.3.3.2-1; Step 6, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-4, with condition GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.3.3.3-5: SIP re-INVITE from SS-UE1 (MCVideo Client)
(Step 3, Table 5.1.1.3.3.2-1; Step 1, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-1, with conditions GROUP-CALL, re_INVITE and EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.3.3.3-6: SIP re-INVITE from IUT (MCVideo Server)
(Step 3, Table 5.1.1.3.3.2-1; Step 2, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-2, with condition GROUP-CALL, re_INVITE and EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.3.3.3-7: SIP re-INVITE from SS-UE1 (MCVideo Client)
(Steps 7 and 14, Table 5.1.1.3.3.2-1; Step 1, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-1, with conditions GROUP-CALL and re_INVITE.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.3.3.3-8: SIP re-INVITE from IUT (MCVideo Server)
(Steps 7 and 14, Table 5.1.1.3.3.2-1; Step 2, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-2, with condition GROUP-CALL and re_INVITE.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.3.3.3-9: SIP re-INVITE from SS-UE1 (MCVideo Client)
(Step 10, Table 5.1.1.3.3.2-1; Step 1, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-1, with conditions GROUP-CALL, re_INVITE and IMMPERIL -CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.3.3.3-10: SIP re-INVITE from IUT (MCVideo Server)
(Step 10, Table 5.1.1.3.3.2-1; Step 2, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-2, with conditions GROUP-CALL, re_INVITE and IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.3.3.3-11: Transmission Idle
(Steps 4 and 6, Table 5.1.1.3.3.2-1; Steps 9 and 10, Annex A, Table A.4.2.5.3-1)**

Derivation Path: Annex A, Table A.4.2.5.4-3, with condition EMERGENCY-CALL.

**Table 5.1.1.3.3.3-12: Transmission Request
(Step 5, Table 5.1.1.3.3.2-1; Steps 1 and 2, Annex A, Table A.4.2.4.3-1)**

Derivation Path: Annex A, Table A.4.2.4.4-1, with condition EMERGENCY-CALL.

**Table 5.1.1.3.3.3-13: Transmission Granted
(Steps 3 and 5, Table 5.1.1.3.3.2-1;
Steps 5 and 6, Annex A, Table A.4.2.4.3-1; Steps 9a1 and 9a2, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.4.4-2, with condition EMERGENCY-CALL.

**Table 5.1.1.3.3.3-14: Transmission Idle
(Steps 11 and 13, Table 5.1.1.3.3.2-1; Steps 9 and 10, Annex A, Table A.4.2.5.3-1)**

Derivation Path: Annex A, Table A.4.2.5.4-3, with condition IMMPERIL-CALL.

**Table 5.1.1.3.3.3-15: Transmission Request
(Step 12, Table 5.1.1.3.3.2-1; Steps 1 and 2, Annex A, Table A.4.2.4.3-1)**

Derivation Path: Annex A, Table A.4.2.4.4-1, with condition IMMPERIL-CALL.

**Table 5.1.1.3.3.3-16: Transmission Granted
(Steps 10 and 12, Table 5.1.1.3.3.2-1;
Steps 5 and 6, Annex A, Table A.4.2.4.3-1; Steps 9a1 and 9a2, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.4.4-2, with condition IMMPERIL-CALL.

5.1.1.4 MCVideo Server - MCVideo Server / Pre-arranged Group Call / Manual Commencement / Terminating Participating server

5.1.1.4.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the controlling function initiates a pre-arranged group
call with manual commencement }
    then { IUT (MCVideo Server) accepts the call and establishes the call with SS-UE2 (MCVideo
Client) and forwards the media transmission notification received from SS (MCVideo Server) to SS-UE2
(MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) ends the media transmission }
    then { IUT (MCVideo Server) forwards the transmission end notify received from SS (MCVideo
Server) to SS-UE2 (MCVideo Client) and forwards the transmission idle received from SS (MCVideo
Server) to SS-UE2 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) providing the controlling function ends the pre-arranged group call
}
    then { IUT (MCVideo Server) sends a SIP BYE message to SS-UE2 (MCVideo Client) and responds by
sending a SIP 200 (OK) message to SS (MCVideo Server) }
            }
```

5.1.1.4.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.1.3.2, 9.2.1.3.4, 6.3.2.3.2, 6.3.2.2.6.1, 6.3.2.2.6.2, 6.3.2.2.8.1, 6.3.2.2.4.2, 6.3.2.2.2.1, 6.3.2.1.2.1 and 6.3.2.2.3, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.4.1, 6.4.2, 6.4.3, 6.4.4, 6.3.3, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.2]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

NOTE 1: This clause covers on-demand session.

Upon receipt of a "SIP INVITE request for terminating participating MCVideo function", the participating MCVideo function:

…

2) shall check the presence of the isfocus media feature tag in the URI of the Contact header field and if it is not present then the participating MCVideo function shall reject the request …, and shall not continue with the rest of the steps;

3) if the Answer-Mode Indication in the application/poc-settings+xml MIME body has not yet been received from the invited MCVideo client as defined in clause 7.3.3 or clause 7.3.4, shall reject the request …, and shall not continue with the rest of the steps;

4) shall use the MCVideo ID present in the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCVideo ID and public user identity;

5) if the binding between the MCVideo ID and public user identity does not exist, then the participating MCVideo function shall reject the SIP INVITE request with a SIP 404 (Not Found) response. Otherwise, continue with the rest of the steps;

6) if the SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with an <MKFC-GKTPs> element, shall perform the procedures in clause 6.3.2.3.2;

7) shall perform the automatic commencement procedures specified in clause 6.3.2.2.5.1 and according to IETF RFC 5373 [27] if the "SIP INVITE request for terminating participating MCVideo function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCVideo client as per clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; and

8) shall perform the manual commencement procedures specified in clause 6.3.2.2.6.1 and according to IETF RFC 5373 [27] if the "SIP INVITE request for terminating participating MCVideo function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCVideo client as per clause 7.3.3 or clause 7.3.4 is set to "manual-answer"."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.4]

"Upon receiving a SIP BYE request from the controlling MCVideo function, the participating MCVideo function shall follow the procedures as specified in clause 6.3.2.2.8.1."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.6.1]

"When receiving a "SIP INVITE request for terminating participating MCVideo function" that requires manual commencement mode:

1) if:

a) the invited MCVideo client has one or more pre-established sessions without an associated MCVideo session;

b) the media-level sections for the offered MCVideo video media stream are the same as the media-level section for MCVideo video media stream in the existing pre-established session; and

c) the media-level section of the offered media-transmission control entity is the same as the media-level section for media-transmission control entity in the existing pre-established session;

then the participating MCVideo function shall perform the actions specified in clause 6.3.2.2.6.3; or

2) otherwise the participating MCVideo function shall perform the actions specified in clause 6.3.2.2.6.2."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.6.2]

"When receiving a "SIP INVITE request for terminating participating MCVideo function" for an on-demand session that requires manual commencement mode the participating MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.2.2.3;

2) shall set the Request-URI to the public user identity associated to the MCVideo ID of the MCVideo user to be invited;

3) shall perform the procedures specified in clause 6.3.2.2.9 to include any MIME bodies in the received SIP INVITE request;

4) if the Answer-Mode header field is present in the incoming SIP INVITE request, participating MCVideo function, shall include an Answer-Mode header field with the value "Manual";

5) if no Answer-Mode header field was received in the incoming SIP INVITE request and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCVideo client as defined in clause 7.3.3 or clause 7.3.4 is set to "manual-answer", shall set the Answer-Mode header field to "Manual" in the outgoing SIP INVITE request;

6) if the Priv-Answer-Mode header field is present in the incoming SIP INVITE request, the participating MCVideo function shall include a Priv-Answer-Mode header field with the value "Manual";

7) shall copy the contents of the P-Asserted-Identity header field of the incoming "SIP INVITE request for terminating participating MCVideo function" to the P-Asserted-Identity header field of the outgoing SIP INVITE request;

8) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received "SIP INVITE request for terminating participating MCVideo function" as specified in clause 6.3.2.2.1;

9) if the received SIP INVITE request contains a Resource-Priority header field, shall include a Resource-Priority header field with the contents set as in the received Resource-Priority header field; and

10) shall send the SIP INVITE request towards the MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 180 (Ringing) response to the above SIP INVITE request, the participating MCVideo function:

NOTE 1: A SIP 180 (Ringing) response is received from a terminating MCVideo client in the case of a private call.

1) shall generate a SIP 180 (Ringing) response as specified in clause 6.3.2.2.4.1;

2) shall include the P-Asserted-Identity header field as received in the incoming SIP 180 (Ringing) response; and

3) shall forward the SIP 180 (Ringing) response according to 3GPP TS 24.229 [11].

Upon receiving a SIP 183 (Session Progress) response to the above SIP INVITE request, the participating MCVideo function:

NOTE 2: A SIP 183 (Session Progress) response can be received from a terminating MCVideo client in the case of a group call.

1) shall generate a SIP 183 (Session Progress) response as specified in clause 6.3.2.2.4.1;

2) shall include the P-Asserted-Identity header field as received in the incoming SIP 183 (Session Progress) response;

3) shall include the P-Answer-State header field if received in the incoming SIP 183 (Session Progress) request; and

4) shall forward the SIP 183 (Session Progress) response according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response to the SIP INVITE request sent to the MCVideo client, the participating MCVideo function:

When the participating MCVideo function sends the SIP 200 (OK) response the participating MCVideo function:

1) shall generate a SIP 200 (OK) response as described in the clause 6.3.2.2.4.2;

2) shall include in the SIP 200 (OK) response an SDP answer based on the SDP answer in the received SIP 200 (OK) response as specified in clause 6.3.2.2.2.1;

3) shall copy the P-Asserted-Identity header field from the incoming SIP 200 (OK) response to the outgoing SIP 200 (OK) response;

4) shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

5) shall forward the SIP 200 (OK) response according to 3GPP TS 24.229 [11].

The participating MCVideo function shall forward any other SIP response that does not contain SDP along the signalling path to the originating network according to 3GPP TS 24.229 [11]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.8.1]

"Upon receiving a SIP BYE request from the controlling MCVideo function, the participating MCVideo function:

1) *shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.581 [5] for releasing media plane resource associated with the SIP session with the controlling MCVideo function;*

2) *shall generate a SIP BYE request according to 3GPP TS 24.229 [11];*

3) *shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;*

4) *if the received SIP BYE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body into the outgoing SIP BYE request; and*

5) *shall send the SIP BYE request to the MCVideo client according to 3GPP TS 24.229 [11].*

Upon receiving a SIP 200 (OK) response to the SIP BYE request the participating MCVideo function:

1) *shall send a SIP 200 (OK) response to the SIP BYE request received from the controlling MCVideo function according to 3GPP TS 24.229 [11]; and*

2) *shall interact with the media plane as specified in 3GPP TS 24.581 [5] for releasing media plane resources associated with the SIP session with the MCVideo client."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.4.2]

"This clause is referenced from other procedures.

When sending SIP 200 (OK) responses, the participating MCVideo function shall generate a SIP 200 (OK) response according to 3GPP TS 24.229 [11] and:

1) *shall include the option tag "timer" in a Require header field;*

2) *shall include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [23], "UAS Behavior". If no "refresher" parameter was included in the SIP INVITE request, the "refresher" parameter in the Session-Expires header field shall be set to "uas";*

3) *shall include the following in the Contact header field:*

   a) *the g.3gpp.mcvideo media feature tag;*

   b) *the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo"; and*

   c) *an MCVideo session identity mapped to the MCVideo session identity provided in the Contact header field of the received SIP INVITE request from the controlling MCVideo function;*

4) *shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [32]; and*

5) *if the incoming SIP response contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body to the outgoing SIP 200 (OK) response."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.2.1]

"The participating MCVideo function shall follow the procedures in clause 6.3.2.1.2.1."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [11], the participating MCVideo function:

1) *shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCVideo function, for the accepted media stream in the received SDP offer, if required; and*

NOTE 1: Requirements can exist for the participating MCVideo function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCVideo function, for the accepted media-transmission control entity, if present in the received SDP offer.

NOTE 2: If the participating MCVideo function and the controlling MCVideo function are in the same MCVideo server, and the participating MCVideo function does not have a dedicated IP address or a dedicated port number for media transmission control or media stream, the replacement of the IP address or the port number is omitted."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.3]

"The participating MCVideo function shall generate an initial SIP INVITE request according to 3GPP TS 24.229 [11] and:

1) shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [20] if included in the incoming SIP INVITE request;

2) should include the Session-Expires header field according to IETF RFC 4028 [23]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

3) shall include the option tag "timer" in the Supported header field;

4) shall include the following in the Contact header field:

a) the g.3gpp.mcvideo media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo";

c) the isfocus media feature tag;

d) an MCVideo session identity mapped to the MCVideo session identity provided in the Contact header field of the incoming SIP INVITE request; and

e) any other uri-parameter provided in the Contact header field of the incoming SIP INVITE request;

5) shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [32];

6) shall include the option tag "norefersub" in a Supported header field according to rules and procedures of IETF RFC 4488 [31];

7) may include a Resource-Share header field in accordance with clause 5.7.1.20.3 in 3GPP TS 24.229 [11]; and

8) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body to the outgoing SIP INVITE request."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.1]

"Once an on-demand MCVideo session is established or a pre-established session is in use when the participating MCVideo function receives transmission control messages from the transmission participant in the MCVideo client or from the transmission control server in the controlling MCVideo function, the behaviour of the participating MCVideo function is described in the following clauses."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.2]

"Upon receiving a transmission control message the participating MCVideo function:

1. shall immediately forward the transmission control message to the transmission control server if the message is received from the transmission participant;

2. *if an MBMS subchannel is not used for a transmission in the session the transmission control message is associated with, shall immediately forward the transmission control message to the transmission participant if the message is received from the transmission control server; and*

3. *if an MBMS subchannel is used for a transmission in the session the transmission control message is associated with:*

   a. *if*

      i. *the transmission control message is not a Transmission Idle message or a Media Transmission Notify message;*

      ii. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

      iii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report;*

      shall immediately forward the transmission control message to the transmission participant; and

   b. *if*

      i. *the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report; and*

      ii *if the transmission control message is the Transmission Idle message or the Media Transmission Notify message,*

   shall perform actions as specified in clause 10.2.

NOTE: When the Transmit Idle or Media Transmission Notify messages are discarded the messages are sent to the MCVideo clients over the MBMS subchannel allocated for the transmission as specified in clause 10.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.3]

"Upon receiving RTP media packets the participating MCVideo function:

1. *shall immediately forward the RTP media packet to the controlling MCVideo function if the RTP packet is from an MCVideo client; and*

2. *if an MBMS subchannel is not used for a transmission in the session the RTP media packets are associated with, shall immediately forward the RTP media packets to the MCVideo client if the RTP packet is from the controlling MCVideo function or the non-controlling MCVideo function.*

3. *if an MBMS subchannel is used for a transmission in the session the RTP media packets are associated with and if RTP media packets are received from the controlling MCVideo function or the non-controlling MCVideo function:*

   a. *if*

      i. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

      ii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report,*

      shall immediately forward the RTP media packets to the MCVideo client; and

   b. *if the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report, shall perform actions as specified in clause 10.2."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.4]

"When the participating function receives an indication from the application and signalling plane that session release is initiated, the participating MCVideo function:

1. *shall stop sending transmission control messages towards the transmission participant and the transmission control server; and*

2. *shall stop sending RTP media packets towards the MCVideo client and towards the controlling MCVideo function.*

When the participating MCVideo function receives an indication from the application and signalling plane that the session is released, the participating MCVideo function:

1. *in case of a pre-established session, shall perform the actions in clause 9.3.2; and*

2. *in case of an on-demand session, shall release the media resources associated with the session."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. *if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:*

   a. *shall initialize a general state machine as specified in clause 6.3.4.2.2; and*

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

   b. *shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;*

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.1.4.3 Test description

5.1.1.4.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the controlling MCVideo function.
- SS-UE2 (MCVideo Client).

IUT:

- IUT (MCVideo Server), providing the terminating participating MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of some sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server and the HTTP Client.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server) and SS-UE2 (MCVideo Client).
- The SS-UE2 (MCVideo client) is registered and authorized to use the MCVideo system.

- The SS-UE2 (MCVideo client) is affiliated to the pre-arranged group and is authorized to initiate group calls.

5.1.1.4.3.2 Test procedure sequence

Table 5.1.1.4.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CT group call establishment, manual commencement' as described in Annex A, Table A.4.3.4.3-1 to establish a pre-arranged group call with manual commencement mode and notify a media transmission from SS-UE1 (MCVideo Client)?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Notification - Idle' as described in Annex A, Table A.4.3.8.3-1 to notify the end of SS-UE1 (MCVideo Client) transmission?	-	-	2	P
9	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.3.5.3-1, to release the call?	-	-	3	P

5.1.1.4.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step.

5.1.1.5 MCVideo Server - MCVideo Server / Pre-arranged Group Call / Emergency Group Call / Controlling server

5.1.1.5.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function initiates an
emergency pre-arranged group call with implicit transmission control }
    then { IUT (MCVideo Server) accepts the call from initiator and establishes the call with all
affiliated users by sending call establishment requests to SS (MCVideo Server) providing the
terminating participating function and sends transmission granted to SS-UE1 (MCVideo Client) and
sends media transmission notification to SS-UE2 (MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established an emergency pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests to end the transmission control }
    then { IUT (MCVideo Server) sends transmission control idle to SS-UE1 (MCVideo Client) and sends
transmission control idle to SS-UE2 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established an emergency pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control }
    then { IUT (MCVideo Server) sends transmission granted to SS-UE1 (MCVideo Client) and sends
media transmission notification to SS-UE2 (MCVideo Client) }
            }
```

(4)

```
with { IUT (MCVideo Server) having established an emergency pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function ends the pre-
arranged group call }
    then { IUT (MCVideo Server) responds by sending a SIP 200 (OK) message to SS (MCVideo Server)
ending the call and sends a SIP BYE message to SS (MCVideo Server) providing the terminating
participating function }
            }
```

5.1.1.5.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.1.4.2, 9.2.1.4.3, 9.2.1.4.1.1, 9.2.1.4.1.2, 6.3.3.1.6, 6.3.3.1.12.2, 6.3.3.1.12.4, 6.3.3.2.1 and 6.3.3.2.2, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.3.1, 6.3.2.2, 9.2.1, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.2]

"In the procedures in this clause:

1) *MCVideo ID in an incoming SIP INVITE request refers to the MCVideo ID of the originating user from the <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;*

2) *group identity in an incoming SIP INVITE request refers to the group identity from the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;*

3) *MCVideo ID in an outgoing SIP INVITE request refers to the MCVideo ID of the called user in the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the outgoing SIP INVITE request;*

4) *indication of required group members in a SIP 183 (Session Progress) response refers to the <required> element of the application/vnd.3gpp.mcvideo-info+xml MIME body set to "true" in a SIP 183 (Session Progress) sent by the non-controlling MCVideo function of an MCVideo group;*

5) *emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and*

6) *imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.*

Upon receipt of a "SIP INVITE request for controlling MCVideo function of an MCVideo group", the controlling MCVideo function:

…

2) *shall determine if the media parameters are acceptable and the MCVideo codecs are offered in the SDP offer and if not reject the request with a SIP 488 (Not Acceptable Here) response and skip the rest of the steps;*

3) *shall reject the SIP request with a SIP 403 (Forbidden) response and not process the remaining steps if:*

   a) *an Accept-Contact header field does not include the g.3gpp.mcvideo media feature tag; or*

   b) *an Accept-Contact header field does not include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo";*

4) *if received SIP INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.3.1.17;*

5) *shall retrieve the necessary group document(s) from the group management server for the group identity contained in the SIP INVITE request and carry out initial processing as specified in clause 6.3.5.2;*

6) if the result of the initial processing in clause 6.3.5.2 was:

a) that authorization of the MCVideo ID is required at a non-controlling MCVideo function of an MCVideo group is required, perform the actions in clause 6.3.3.1.13.7 and do not continue with the rest of the steps in this clause; and

b) that a SIP 3xx, 4xx, 5xx or 6xx response to the "SIP INVITE request for controlling MCVideo function of an MCVideo group" has been sent, do not continue with the rest of the steps in this clause;

7) shall perform the actions as described in clause 6.3.3.2.2;

8) shall maintain a local counter of the number of SIP 200 (OK) responses received from invited members and shall initialise this local counter to zero;

9) shall determine if an MCVideo group call for the group identity is already ongoing by determining if an MCVideo session identity has already been allocated for the group call and the MCVideo session is active;

…

12) if a Resource-Priority header field is included in the SIP INVITE request:

a) if the Resource-Priority header field is set to the value indicated for emergency calls and the SIP INVITE request does not contain an emergency indication and the in-progress emergency state of the group is set to a value of "false", shall reject the SIP INVITE request with a SIP 403 (Forbidden) response and skip the rest of the steps; or

b) if the Resource-Priority header field is set to the value indicated for imminent peril calls and the SIP INVITE request does not contain an imminent peril indication and the in-progress imminent peril state of the group is set to a value of "false", shall reject the SIP INVITE request with a SIP 403 (Forbidden) response and skip the rest of the steps;

13) if the MCVideo group call is not ongoing then:

…

c) if the received SIP INVITE request contains an an authorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2 or MCVideo imminent peril group call as determined by clause 6.3.3.1.13.5 and the MCVideo user is eligible to be implicitly affiliated with the MCVideo group as determined as determined in step 13) a) iv) above, shall perform the implicit affiliation as specified in clause 8.2.2.3.7;

d) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [11];

e) shall create a prearranged group session and allocate an MCVideo session identity for the prearranged group call, and shall handle timer TNG3 (group call timer) as specified in clause 6.3.3.5;

f) if the group identity in the "SIP INVITE request for controlling MCVideo function of an MCVideo group" is a TGI:

i) shall for each of the constituent MCVideo groups homed on the primary MCVideo system:

…;

B) if the controlling MCVideo function owns the MCVideo group identified by the MCVideo group ID then:

I) determine the members to invite to the prearranged MCVideo group call as specified in clause 6.3.5.5;

II) invite each group member determined in step A) above, to the group session, as specified in clause 9.2.1.4.1.1; and

III) interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3; and

ii) shall for each of the constituent MCVideo groups homed on the partner MCVideo system generate a SIP INVITE request for the MCVideo group identity homed on the partner MCVideo system as specified in clause 9.2.1.4.1.2; and

g) if the group identity in the SIP INVITE request for controlling MCVideo function of an MCVideo group is an MCVideo group ID:

i) shall determine the members to invite to the prearranged MCVideo group call as specified in clause 6.3.5.5;

ii) if necessary, shall start timer TNG1 (acknowledged call setup timer) according to the conditions stated in clause 6.3.3.3;

iii) if the received SIP INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element set to a value of "true":

A) shall cache the information that this MCVideo user has initiated an MCVideo emergency call;

B) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the information that this MCVideo user has initiated an MCVideo emergency alert; and

C) if the in-progress emergency state of the group is set to a value of "false":

I) shall set the value of the in-progress emergency state of the group to "true"; and

II) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16;

iv) if the in-progress emergency state of the group is set to a value of "false" and if the received SIP INVITE request contains an imminent peril indication set to a value of "true", the controlling MCVideo function shall:

A) shall cache the information that this MCVideo user has initiated an MCVideo imminent peril call; and

B) if the in-progress imminent peril state of the group is set to a value of "false", shall set the in-progress imminent peril state of the group to a value of "true";

v) shall invite each group member determined in step 13)g)i) above, to the group session, as specified in clause 9.2.1.4.1.1; and

vi) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3; and

14) if the MCVideo group call is ongoing then:

…

c) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [11];

…

e) if the received SIP INVITE request contains an an authorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2 or MCVideo imminent peril group call as determined by clause 6.3.3.1.13.5 and the MCVideo user is eligible to be implicitly affiliated with the MCVideo group as determined in step 14) a) iv) above, shall perform the implicit affiliation as specified in clause 8.2.2.3.7;

f) if the received SIP INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element set to a value of "true":

i) shall cache the information that this MCVideo user has initiated an MCVideo emergency call;

ii) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the information that this MCVideo user has initiated an MCVideo emergency alert;

iii) if the in-progress emergency state of the group is set to a value of "false":

A) shall set the value of the in-progress emergency state of the group to "true";

B) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16; and

C) shall generate SIP re-INVITE requests for the MCVideo emergency group call to the other call participants of the MCVideo group as specified in clause 6.3.3.1.6;

iv) if the in-progress imminent peril state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's imminent peril indication as specified in clause 6.3.3.1.11, setting the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11]; and

v) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5];

g) if the in-progress emergency state of the group is set to a value of "false" and if the SIP INVITE request contains an imminent peril indication set to a value of "true", the controlling MCVideo function:

i) shall cache the information that this MCVideo user has initiated an MCVideo imminent peril call; and

ii) if the in-progress imminent peril state of the group is set to a value of "false":

A) shall set the in-progress imminent peril state of the group to a value of "true";

B) shall generate SIP re-INVITE requests for the MCVideo imminent peril group call to the other call participants of the MCVideo group as specified in clause 6.3.3.1.15; and

C) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

iii) if the in-progress imminent peril state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's imminent peril indication as specified in clause 6.3.3.1.11, setting the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11];

h) shall generate a SIP 200 (OK) response as specified in the clause 6.3.3.2.4.2;

i) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

j) shall include in the SIP 200 (OK) response with the warning text set to "123 MCVideo session already exists" as specified in clause 4.4;

k) if the received SIP re-INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

l) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 4: In this case, the request was for an imminent peril call but a higher priority MCVideo emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

m) shall interact with media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 5: Resulting media plane processing is completed before the next step is performed.

n) shall send the SIP 200 (OK) response towards the inviting MCVideo client or inviting non-controlling MCVideo function according to 3GPP TS 24.229 [11];

o) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo User has joined in the MCVideo group session, as specified in clause 6.3.3.4;

NOTE 6: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

p) shall send a SIP NOTIFY request to each MCVideo client according to 3GPP TS 24.229 [11];

q) Upon receiving a SIP ACK to the above SIP 200 (OK) response and the SIP 200 (OK) response contained a Warning header field as specified in clause 4.4 with the warning text containing the mcvideo-warn-code set to "149", shall follow the procedures in clause 6.3.3.1.18; and

r) shall not continue with the rest of the clause.

Upon receiving a SIP 183 (Session Progress) response to the SIP INVITE request specified in clause 9.2.1.4.1 containing a P-Answer-State header field with the value "Unconfirmed" as specified in IETF RFC 4964 [30], the timer TNG1 (acknowledged call setup timer) is not running, the controlling MCVideo function supports media buffering and the SIP final response is not yet sent to the inviting MCVideo client:

1) shall generate a SIP 200 (OK) response to SIP INVITE request as specified in the clause 6.3.3.2.3.2;

2) shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if the prearranged MCVideo group has more than <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [24];

3) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

4) shall include a P-Answer-State header field with the value "Unconfirmed";

5) if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

6) if the received SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

7) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 7: Resulting user plane processing is completed before the next step is performed.

8) shall send the SIP 200 (OK) response towards the inviting MCVideo client according to 3GPP TS 24.229 [11];

9) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo User has joined in the MCVideo group session, as specified in clause 6.3.3.4; and

NOTE 8: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

10) shall send a SIP NOTIFY request to each MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 183 (Session Progress) response for a SIP INVITE request as specified in clause 9.2.1.4.1.2 containing an indication of required group members, the timer TNG1 (acknowledged call setup timer) is running and all SIP 200 (OK) responses have been received to all SIP INVITE requests sent to MCVideo clients specified in clause 9.2.1.4.1.1, then the controlling MCVideo function shall wait until the SIP 200 (OK) response has been received to the SIP INVITE request specified in clause 9.2.1.4.1.2 before generating a SIP 200 (OK) response to the "SIP INVITE request for controlling MCVideo function of an MCVideo group".

Upon receiving a SIP 200 (OK) response for a SIP INVITE request as specified in clause 9.2.1.4.1 that was sent to an affiliated and <on-network-required> group member as specified in 3GPP TS 24.481 [24]; and

1) if the MCVideo ID in the SIP 200 (OK) response matches to the MCVideo ID in the corresponding SIP INVITE request;

2) there are no outstanding SIP 200 (OK) responses to SIP INVITE requests which were sent to affiliated and <on-network-required> group members as specified in 3GPP TS 24.481 [24]; and

3) there is no outstanding SIP 200 (OK) response to a SIP INVITE request sent in clause 9.2.1.4.1.2 where the SIP 183 (Session Progress) response contained an indication of required group members;

the controlling MCVideo function:

1) shall stop timer TNG1 (acknowledged call setup timer) as described in clause 6.3.3.3;

2) shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;

3) shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if all members were not invited because the prearranged MCVideo group has been exceeded the <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [24];

4) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

5) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 9: Resulting media plane processing is completed before the next step is performed.

6) shall send a SIP 200 (OK) response to the inviting MCVideo client according to 3GPP TS 24.229 [11];

7) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo user has joined in the MCVideo group session, as specified in clause 6.3.3.4; and

NOTE 10: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

8) shall send the SIP NOTIFY request to the MCVideo clients according to 3GPP TS 24.229 [11].

Upon:

1) receiving a SIP 200 (OK) response for a SIP INVITE request as specified in clause 9.2.1.4.1;

2) the timer TNG1 (acknowledged call setup timer) is not running;

3) the local counter of the number of SIP 200 (OK) responses received from invited members is equal to the value of the <on-network-minimum-number-to-start> element of the group document;

4) the controlling MCVideo function supports media buffering; and

5) the SIP final response has not yet been sent to the inviting MCVideo client;

the controlling MCVideo function according to local policy:

1) shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.2;

2) shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if all members were not invited because the prearranged MCVideo group has exceeded the <max-participant-count> members as specified in 3GPP TS 24.481 [24];

3) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

4) if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

5) if the received SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

6) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 11: Resulting media plane processing is completed before the next step is performed.

7) shall send a SIP 200 (OK) response to the inviting MCVideo client according to 3GPP TS 24.229 [11];

8) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo user has joined in the MCVideo group session, as specified in clause 6.3.3.4; and

NOTE 12: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

9) shall send the SIP NOTIFY request to the MCVideo clients according to 3GPP TS 24.229 [11].

Upon expiry of timer TNG1 (acknowledged call setup timer), if there are outstanding SIP 200 (OK) responses to SIP INVITE requests sent to affiliated and <on-network-required> group members as specified in 3GPP TS 24.481 [24], the controlling MCVideo function shall follow the procedures specified in clause 6.3.3.3.

If timer TNG1 (acknowledged call setup timer) is running and a final SIP 4xx, 5xx or 6xx response is received from an affiliated and <on-network-required> group member as specified in 3GPP TS 24.481 [24], the controlling MCVideo function shall follow the relevant procedures specified in clause 6.3.3.3.

If:

1) timer TNG1 (acknowledged call setup timer) is not running;

2) the local counter of the number of SIP 200 (OK) responses received from invited members is equal to the value of the <on-network-minimum-number-to-start> element of the group document; and

3) a final SIP 4xx, 5xx or 6xx response is received from an invited MCVideo client;

then the controlling MCVideo function shall perform one of the following based on policy:

1) send the SIP final response towards the inviting MCVideo client, according to 3GPP TS 24.229 [11], if a SIP final response was received from all the other invited MCVideo clients and the SIP 200 (OK) response is not yet sent; or

2) remove the invited MCVideo client from the MCVideo Session as specified in clause 6.3.3.1.5, if a SIP final response other than 2xx or 3xx was received from all the invited MCVideo clients and the SIP 200 (OK) response is already sent. The controlling MCVideo function may invite an additional member of the prearranged MCVideo group as specified in clause 9.2.1.4.1 that has not already been invited, if the prearranged MCVideo group has more than <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [24], and all members have not yet been invited.

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCVideo client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.18."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.3]

"Upon receiving a SIP BYE request the controlling MCVideo function shall follow the procedures as specified in clause 6.3.3.2.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.7]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

Upon receipt of a SIP re-INVITE request for an MCVideo session identity identifying an on-demand prearranged MCVideo group session, the controlling MCVideo function:

…

2) if received SIP re-INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.3.1.17;

…

4) if the received SIP re-INVITE request contains an imminent peril indication set to "true" for an MCVideo imminent peril group call and this is an unauthorised request for an MCVideo imminent peril group call as determined by clause 6.3.3.1.13.6, shall reject the SIP re-INVITE request …

5) if a Resource-Priority header field is included in the received SIP re-INVITE request:

a) if the Resource-Priority header field is set to the value indicated for emergency calls and the SIP re-INVITE request does not contain an emergency indication and the in-progress emergency state of the group is set to a value of "false", shall reject the SIP re-INVITE request … and skip the rest of the steps; and

b) if the Resource-Priority header field is set to the value indicated for imminent peril calls and the SIP re-INVITE request does not contain an imminent peril indication and the in-progress imminent peril state of the group is set to a value of "false", shall reject the SIP INVITE request …and skip the rest of the steps;

6) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "true" and is an authorised request to initiate an MCVideo emergency group call as determined by clause 6.3.3.1.13.2, the controlling MCVideo function shall:

i) shall cache the MCVideo ID of the MCVideo user that has initiated an MCVideo emergency call;

ii) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the MCVideo ID of the MCVideo user that has initiated an MCVideo emergency alert;

iii) if the in-progress emergency state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's emergency indication as specified in clause 6.3.3.1.11, setting the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true";

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11]; and

C) if the in-progress imminent peril state of the group is set to a value of "true", shall set it to a value of "false"; and

iv) if the in-progress emergency state of the group is set to a value of "false":

A) shall set the value of the in-progress emergency state of the group to "true";

B) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16;

NOTE 2: The interactions of TNG2 with the TNG3 (group call timer) are explained in clause 6.3.3.5.2.

Editor's Note:timers need to be defined.

C) shall generate SIP re-INVITE requests for the MCVideo emergency group call to the other participants of the MCVideo group call as specified in clause 6.3.3.1.6;

D) shall send the SIP re-INVITEs towards the other participants of the MCVideo group call; and

E) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5];

7) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "false" and is an unauthorised request for an MCVideo emergency group call cancellation as determined by clause 6.3.3.1.13.4:

a) shall reject the SIP re-INVITE request;

…

c) if an <alert-ind> element of the mcvideoinfo MIME body is included in the SIP re-INVITE request set to "false", and there is an outstanding MCVideo emergency alert for this MCVideo user, shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <alert-ind> element set to a value of "true"; and

d) shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [11] and skip the rest of the steps;

8) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "false" and is determined to be an authorised request for an MCVideo emergency call cancellation as specified in clause 6.3.3.1.16 and the in-progress emergency state of the group to is set to a value of "true" the controlling MCVideo function:

a) shall set the in-progress emergency group state of the group to a value of "false";

b) shall clear the cache of the MCVideo ID of the MCVideo user as having an outstanding MCVideo emergency group call;

c) if an <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body is included and set to "false" and is determined to be an authorised request for an MCVideo emergency alert cancellation as specified in clause 6.3.3.1.13.3 and there is an outstanding MCVideo emergency alert for this MCVideo user shall:

i) if the received SIP re-INVITE request contains an <originated-by> element in the application/vnd.3gpp.mcvideo-info+xml MIME body, clear the cache of the MCVideo ID of the MCVideo user identified by the <originated-by> element as having an outstanding MCVideo emergency alert; or

ii) if the received SIP re-INVITE request does not contain an <originated-by> element in the application/vnd.3gpp.mcvideo-info+xml MIME body, clear the cache of the MCVideo ID of the sender of the SIP re-INVITE request as having an outstanding MCVideo emergency alert;

d) shall generate SIP re-INVITE requests to the participants in the group call as specified in clause 6.3.3.1.6. The MCVideo controlling function:

i) for each of the other participants in the group call shall send the SIP re-INVITE request towards the MCVideo client as specified in 3GPP TS 24.229 [11]; and

ii) Upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5];

NOTE 3: Clause 6.3.3.1.5 will inform the group call participants of the cancellation of the MCVideo group's in-progress emergency state and the cancellation of the MCVideo emergency alert if applicable.

e) shall stop timer TNG2 (in-progress emergency group call timer); and

NOTE 4: The interactions of TNG2 with the TNG3 (group call timer) are explained in clause 6.3.3.5.2;

f) for each of the affiliated members of the group that are not participating in the call:

i) generate a SIP MESSAGE request notification of the cancellation of the MCVideo user's emergency call as specified in clause 6.3.3.1.11;

ii) set the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "false";

iii) if indicated above in step 8) c), set the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "false"; and

iv) send the SIP MESSAGE request according to 3GPP TS 24.229 [11];

9) if the received SIP re-INVITE request contains an imminent peril indication and the in-progress emergency group state of the group is set to a value of "false", shall perform the procedures specified in clause 9.2.1.4.8 and skip the rest of the steps.

Upon receiving a SIP 200 (OK) response to a SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5];

1) shall generate a SIP 200 (OK) response according to rules and procedures of 3GPP TS 24.229 [11];

2) shall include in the SIP 200 (OK) response an SDP answer according to 3GPP TS 24.229 [11] with the clarifications specified in clause 6.3.3.2.1;

3) shall include the "norefersub" option tag in a Supported header field according to IETF RFC 4488 [31];

4) shall include the "tdialog" option tag in a Supported header field according to IETF RFC 4538 [32];

5) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <alert-ind> element set to a value of "true" and if this is an unauthorised request for an MCVideo emergency alert as determined by clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

6) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <alert-ind> element set to a value of "false" and if this is an unauthorised request for an MCVideo emergency alert cancellation as determined by clause 6.3.3.1.13.3, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

7) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true", this is an authorised request for an MCVideo imminent peril group call and the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 5: In this case, the request was for an imminent peril call but a higher priority MCVideo emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

8) shall interact with media plane as specified in 3GPP TS 24.581 [5]; and

9) shall send the SIP 200 (OK) response towards the MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCVideo client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.18.

Upon receipt of a SIP 2xx response for an outgoing SIP MESSAGE request, shall handle according to 3GPP TS 24.229 [11]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.1.1]

"This clause describes the procedures for inviting an MCVideo user to an MCVideo session. The procedure is initiated by the controlling MCVideo function as the result of an action in clause 9.2.1.4.2 or as the result of receiving a SIP 403 (Forbidden) response as described in this clause.

The controlling MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the public service identity of the terminating participating MCVideo function associated to the MCVideo user to be invited.;

NOTE 1: How the controlling MCVideo function finds the address of the terminating MCVideo participating function is out of the scope of the current release.

NOTE 2: If the terminating MCVideo user is part of a partner MCVideo system, then the public service identity can identify an entry point in the partner network that is able to identify the terminating participating MCVideo function.

3) shall set the P-Asserted-Identity header field to the public service identity of the controlling MCVideo function;

4) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcvideo-request-uri> element set to the MCVideo ID of the terminating user; and

b) the <mcvideo-calling-group-id> element set to the group identity;

NOTE 3: The <mcvideo-calling-user-id> is already included in the MIME body as a result of calling clause 6.3.3.1.2 in step 1).

5) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1;

6) if the in-progress emergency state of the group is set to a value of "true" the controlling MCVideo function:

a) shall include a Resource-Priority header field populated with the values for an MCVideo emergency group call as specified in clause 6.3.3.1.19;

b) if the received SIP INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "true":

i) shall include in the outgoing SIP INVITE request in the application/vnd.3gpp.mcvideo-info+xml MIME body an <emergency-ind> element set to a value of "true"; and

ii) *if the <alert-ind> element is set to "true" in the received SIP INVITE request and the requesting MCVideo user and MCVideo group are authorised for the initiation of MCVideo emergency alerts as determined by the procedures of clause 6.3.3.1.13.1, shall populate the application/vnd.3gpp.mcvideo-info+xml MIME body and the application/vnd.3gpp.location-info+xml MIME body as specified in clause 6.3.3.1.12. Otherwise, shall set the <alert-ind> element to a value of "false"; and*

c) *if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <imminentperil-ind> element set to a value of "false";*

7) *if the in-progress emergency state of the group is set to a value of "false" and the in-progress imminent peril state of the group is set to a value of "true", the controlling MCVideo function:*

a) *shall include a Resource-Priority header field populated with the values for an MCVideo imminent peril group call as specified in clause 6.3.3.1.19; and*

b) *shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true";*

8) *if:*

a) *an MCVideo GKTP document exists for the group identity contained in the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request; and*

b) *the MCVideo GKTP document contains a <MKFC-GKTPs> element;*

then:

a) *for each instance of <GKTP> element of the <MKFC-GKTPs> element of the MCVideo GKTP document:*

i) *shall perform the procedure in clause 6.3.3.6.2 to re-generate an I_MESSAGE; and*

ii) *if the procedure in clause 6.3.3.6.2 was successful, shall include the I_MESSAGE in a <GKTP> element of the <MKFC-GKTPs> element of an application/vnd.3gpp.mcvideo-info+xml MIME body included in the outgoing SIP INVITE request; and*

9) *shall send the SIP INVITE request towards the terminating network in accordance with 3GPP TS 24.229 [11].*

Upon receiving a SIP 183 (Session Progress) response containing a Require header field with the option tag "100rel" and containing a P-Answer-State header field with the value "Unconfirmed" in response to the SIP INVITE request the controlling MCVideo function:

1) *shall send a SIP PRACK request towards the MCVideo client according to 3GPP TS 24.229 [11].*

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCVideo function:

1) *shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;*

2) *shall send a SIP NOTIFY request to all participants with a subscription to the conference event package as specified in clause 9.2.3.4; and*

3) *shall increment the local counter of the number of SIP 200 (OK) responses received from invited members, by 1.*

NOTE 4: The notifications above could be sent prior to the SIP 200 (OK) response being sent to the inviting MCVideo client. These notifications received by MCVideo clients that are group members do not mean that the group session will be successfully established.

NOTE 5: The procedures executed by the controlling MCVideo function prior to sending a response to the inviting MCVideo client are specified in clause 9.2.1.4.2."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.1.2]

"The controlling MCVideo function:

1) *shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;*

2) *shall set the Request-URI to the public service identity of the non-controlling MCVideo function serving the group identity of the MCVideo group owned by the partner MCVideo system;*

3) *shall set the P-Asserted-Identity to the public service identity of the controlling MCVideo function;*

4) *shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body in the outgoing SIP INVITE request:*

   a) *the <mcvideo-request-uri> element set to the group identity of the MCVideo group hosted by the non-controlling MCVideo function in the partner MCVideo system; and*

   b) *the <mcvideo-calling-group-id> element set to the group identity of the group served by the controlling MCVideo function;*

5) *shall include the Recv-Info header field set to g.3gpp.mcvideo-transmission-request;*

6) *if:*

   a) *an MCVideo GKTP document exists for the group identity contained in the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request; and*

   b) *the MCVideo GKTP document contains a <MKFC-GKTPs> element;*

   then:

   a) *for each instance of <GKTP> element of the <MKFC-GKTPs> element of the MCVideo GKTP document:*

      i) *shall perform the procedure in clause 6.3.3.6.2 to re-generate an I_MESSAGE; and*

      ii) *if the procedure in clause 6.3.3.6.2 was successful, shall include the I_MESSAGE in a <GKTP> element of the <MKFC-GKTPs> element of an application/vnd.3gpp.mcvideo-info+xml MIME body included in the outgoing SIP INVITE request;*

7) *shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1; and*

8) *shall send the SIP INVITE request towards the partner MCVideo system in accordance with 3GPP TS 24.229 [11].*

Upon receiving SIP 403 (Forbidden) response for the SIP INVITE request, if according to local policy and if:

1) *the response contains a Warning header field with the MCVideo warning code "128"; and*

2) *the response contains a P-Refused-URI-List header field and an application/resource-lists+xml MIME body as specified in IETF RFC 5318 [28];*

NOTE 1: The application/resource-lists+xml MIME body contains MCVideo IDs identifying MCVideo users in a partner MCVideo system that needs to be invited to the prearranged group call in case of group regrouping using interrogating method.

Editor's Note:The above note currently isn't defined in the 23.280 and 23.281.

then the controlling MCVideo function:

1) *shall check if the number of members of the MCVideo group exceeds the value contained in the <on-network-max-participant-count> element of the group document as specified in 3GPP TS 24.481 [24]. If exceeded, the controlling MCVideo function shall invite only <on-network-max-participant-count> members from the application/resource-lists+xml MIME body; and*

NOTE 2: The <on-network-max-participant-count> element indicates the maximum number of participants allowed in the prearranged group session It is operator policy that determines which participants in the application/resource-lists+xml MIME body are invited to the group call.

2) shall invite MCVideo users as specified in this clause using the list of MCVideo IDs in URI-List.

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCVideo function:

1) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3; and

NOTE 3: The procedures executed by the controlling MCVideo function prior to sending a response to the inviting MCVideo client are specified in clause 9.2.1.4.2.

2) if at least one of the invited MCVideo clients has subscribed to the conference package, shall subscribe to the conference event package in the non-controlling MCVideo function as specified in clause 9.2.3.4.3."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.6]

"This clause is referenced from other procedures.

This clause describes the procedures for inviting an MCVideo user to an MCVideo session associated with an MCVideo emergency group call or MCVideo imminent peril group call. The procedure is initiated by the controlling MCVideo function as the result of an action in clause 9.2.2.4.1.1.

The controlling MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the address of the terminating participating MCVideo function associated with the MCVideo ID of the targeted MCVideo user;

3) shall include an application/vnd.3gpp.mcvideo-info+xml MIME body with the <mcvideoinfo> element containing the <mcvideo-Params> element populated as follows:

a) the <mcvideo-request-uri> element set to the value of the MCVideo ID of the targeted MCVideo user;

b) the <mcvideo-calling-user-id> element set to the value of the MCVideo ID of the calling MCVideo user; and

c) the <mcvideo-calling-group-id> element set to the value of the MCVideo group ID of the emergency group call.

4) shall include in the P-Asserted-Identity header field the public service identity of the controlling MCVideo function;

5) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1; and

6) if the in-progress emergency group state of the group is set to a value of "true" the controlling MCVideo function:

a) shall include a Resource-Priority header field populated with the values for an MCVideo emergency group call as specified in clause 6.3.3.1.18;

b) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <emergency-ind> element set to a value of "true";

c) if the <alert-ind> element is set to "true" in the received SIP INVITE request and the requesting MCVideo user and MCVideo group are authorised for the initiation of MCVideo emergency alerts as determined by the procedures of clause 6.3.3.1.12.1, shall populate the application/vnd.3gpp.mcvideo-info+xml MIME body and the application/vnd.3gpp.mcvideo-location-info+xml MIME body as specified in clause 6.3.3.1.11. Otherwise, shall set the <alert-ind> element to a value of "false" in the application/vnd.3gpp.mcvideo-info+xml MIME body; and

d) if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <imminentperil-ind> element set to a value of "false";

NOTE: If the imminent peril state of the group is true at this point, the controlling function will set it to false as part of the calling procedure.

7) if the in-progress emergency state of the group is set to a value of "false" and the in-progress imminent peril state of the group is set to a value of "true", the controlling MCVideo function:

a) shall include a Resource-Priority header field populated with the values for an MCVideo imminent peril group call as specified in clause 6.3.3.1.18; and

b) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true"; and

8) if:

a) an MCVideo GKTP document exists for the group identity contained in the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request; and

b) the MCVideo GKTP document contains an <MKFC-GKTPs> element;

then:

a) for each instance of <GKTP> element of the <MKFC-GKTPs> element of the MCVideo GKTP document:

i) shall perform the procedure in clause 6.3.3.6.2 to re-generate an I_MESSAGE; and

ii) if the procedure in clause 6.3.3.6.2 was successful, shall include the I_MESSAGE in a <GKTP> element of the <MKFC-GKTPs> element of an application/vnd.3gpp.mcvideo-info+xml MIME body included in the outgoing SIP INVITE request."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.12.2]

"If the controlling MCVideo function has received a SIP request for an MCVideo group call with the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body set to a value of "true" and:

1) if the <allow-emergency-group-call> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true" and:

a) if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> element contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "DedicatedGroup" and:

i) if the <uri-entry> element of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) contains the identity of the MCVideo group targeted by the calling MCVideo user; and

ii) if the <allow-MCVideo-emergency-call> element of the <list-service> element of the group document identified by the targeted MCVideo group identity is set to a value of "true" as specified in 3GPP TS 24.481 [24];

then the controlling MCVideo function shall consider the MCVideo emergency group call request to be an authorised request for an MCVideo emergency group call and skip the remaining steps; or;

b) if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> element contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "UseCurrentlySelectedGroup" and if the <allow-MCVideo-emergency-call> element of the <list-service> element of the group document identified by the targeted MCVideo group identity is set to a value of "true" as specified in 3GPP TS 24.481 [24];

then the controlling MCVideo function shall consider the MCVideo emergency group call request to be an authorised request for an MCVideo emergency group call and skip the remaining steps; or

2) if the controlling MCVideo function does not consider the MCVideo emergency group call request to be an authorised request for an MCVideo emergency group call by step 1) above, then the controlling MCVideo function shall consider the MCVideo emergency group call request to be an unauthorised request for an MCVideo emergency group call.

If the controlling MCVideo function has received a SIP request for an MCVideo private call with the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body set to a value of "true" and:

1) if the <allow-emergency-private-call> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true"; and

a) if the "entry-info" attribute of the <entry> element of the <MCVideoPrivateRecipient> element of the <EmergencyCall> element contained within the <PrivateCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "UsePreConfigured" and if the MCVideo ID targeted for the call is contained in the <uri-entry> element of the <entry> element of the <MCVideoPrivateRecipient> element of the <EmergencyCall> element contained within the <PrivateCall> element (see the MCVideo user profile document in 3GPP TS 24.484 [25]); or

b) if the "entry-info" attribute of the <entry> element of the <MCVideoPrivateRecipient> element of the <EmergencyCall> element contained within the <PrivateCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "LocallyDetermined";

then the controlling MCVideo function shall consider the MCVideo emergency private call request to be an authorised request for an MCVideo emergency private call and skip step 2) below; or

2) if the controlling MCVideo function does not consider the MCVideo emergency private call request to be an authorised request for an MCVideo emergency private call by step 1) above, then the controlling MCVideo function shall consider the MCVideo emergency private call request to be an unauthorised request for an MCVideo emergency private call."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.12.4]

"If the controlling MCVideo function has received a SIP request with the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body set to a value of "false" and:

1) if the <allow-cancel-emergency-alert> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true", then the MCVideo emergency alert cancellation request shall be considered to be an authorised request for an MCVideo emergency alert cancellation; and

2) if the <allow-cancel-emergency-alert> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "false", then the MCVideo emergency alert cancellation request shall be considered to be an unauthorised request for an MCVideo emergency alert cancellation."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [11], the controlling MCVideo function:

1) for the accepted media stream in the received SDP offer:

a) shall replace the IP address and port number in the received SDP offer with the IP address and port number of the controlling MCVideo function; and

2) for the accepted media-transmission control entity, if present in the received SDP offer:

a) shall replace the IP address and port number in the received SDP offer with the IP address and port number of the controlling MCVideo function, for the accepted media-transmission control entity, if present in the received SDP offer; and

b) shall include 'fmtp' attributes as specified in 3GPP TS 24.581 clause 14."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.2.2]

"On receipt of an initial SIP INVITE request the controlling MCVideo function shall cache SIP feature tags, if received in the Contact header field and if the specific feature tags are supported."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.1]

"The transmission control server arbitration logic in the transmission control server shall support the procedures in clauses 6.3.2 and 6.3.3 and shall behave according to the transmission control server state transition diagram for 'general transmission control operation' in clause 6.3.4.

The transmission control interface towards the MCVideo client in the transmission control server shall behave according to the transmission control server state transition diagram for 'basic transmission control operation towards the transmission participant' as specified in clause 6.3.5.

The reception control arbitration logic in the transmission control server shall support the procedures in clauses 6.3.2 and 6.3.3 and shall behave according to the reception control server state transition diagram for 'general reception control operation' in clause 6.3.6.

The transmission control interface towards the MCVideo client in the transmission control server shall behave according to the transmission control server state transition diagram for 'basic reception control operation towards the transmission participant' as specified in clause 6.3.7."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.2.2]

"When an MCVideo call is established a new instance of the transmission control server state machine for 'general transmission control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' is added.

When an MCVideo call is established a new instance of the transmission control server state machine for 'general reception control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic reception control operation towards the transmission participant' is added.

If the optional "mc_queueing" feature is supported and has been negotiated as specified in clause 14, the transmission control server could queue the implicit transmission control request for the media-transmission control entity.

The original initial SIP INVITE request or SIP REFER request to establish an MCVideo chat group call or to rejoin an ongoing MCVideo call is not handled as an implicit transmission control request message by the transmission control server unless explicitly stated in the SIP INVITE request or in the SIP REFER request.

The permission to send media to the inviting MCVideo client due to implicit transmission control request is applicable to both confirmed indication and unconfirmed indication.

When the first unconfirmed indication is received from the invited participating MCVideo function (see 3GPP TS 24.281 [2]) the transmission control server optionally can give an early indication to send RTP media packets, to the inviting MCVideo client.

If an early indication to send RTP media packets is given to the inviting MCVideo client, the transmission participant is granted the permission to send media and the MCVideo server buffers RTP media packets received from the MCVideo client at least until the first invited MCVideo client accepts the invitation or until the RTP media packet buffer exceeds it maximum limit to store RTP media packets.

If the MCVideo server does not support or does not allow media buffering then when an early indication to send RTP media packets is not given to the inviting MCVideo client, the transmission participant is granted the permission to send media when the first invited MCVideo client accepts the media.

Before the transmission control server sends the first transmission control message in the MCVideo call, the transmission control server has to assign itself a SSRC identifier to be included in media transmission control messages and quality feedback messages if the MCVideo server is supporting that option. A suitable algorithm to generate the SSRC identifier is described in IETF RFC 3550 [3].

The transmission participant and the transmission control server can negotiate the maximum priority level that the transmission participant is permitted to request. The transmission control server can pre-empt the current sender based on the negotiated maximum priority level that the transmission participant is permitted to request and the priority level included in the Transmission Media Request message.

NOTE: The maximum priority level that a transmission participant can use is negotiated as specified in clause 14.3.3 and is based on group configuration data retrieved by the controlling MCVideo function from the group management server as described in 3GPP TS 24.481 [12] and service configuration data retrieved by the controlling MCVideo function from the configuration management server as described in 3GPP TS 24.484 [13].

The transmission participant and the transmission control server can negotiate queueing of Transmission requests using the "mc_queueing" fmtp attribute as described in clause 14. If queueing is supported and negotiated, the transmission control server queues the transmission control request if a Transmission Media Request message is received when another transmission participant has the transmission and the priority of the current speaker is the same or higher."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.1]

"The transmission control messages are coded as described in clause 9.1.2 where the transmission control message is part of the application-dependent data.

For the transmission control protocol the ASCII name string is: MCVx (Mission Critical Video x) with x=0, 1 or 2 as specified in clause 9.1.2.

A list of transmission control messages can be found in clause 9.2.2.1.

The transmission control specific fields are specified in clause 9.2.3."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

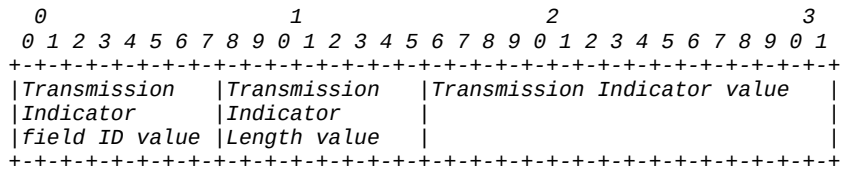
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

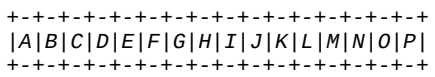


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. *if the active Transmission request queue is empty the transmission control server:*
    a. *shall send Transmission Idle message to all transmission participants. The Transmission Idle message:*
        i. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*
        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;*
    b. *shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;*
    c. *shall start timer T1 (Inactivity);*
    d. *shall set the general state to the 'G: Transmit Idle' state; and*
    e. *shall initialise counter Cx (Simultaneous transmission video) to 0.*
2. *if the active Transmission request queue is not empty the transmission control server:*
    a. *shall select a queued Transmission request from the top of the active Transmission request queue;*
    b. *shall remove that queued Transmission request from the active Transmission request queue; and*
    c. *shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:*
    a. *shall include the granted priority in the Transmission priority field;*
    b. *shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;*
    c. *if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*
    d. *shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;*
2. *shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;*
3. *shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:*
    a. *shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;*
    b. *shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;*
    c. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;*

d. if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';

e. if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

g. shall initiate the 'general reception control operation' state machine;and

4. shall enter the 'G: Transmit Taken' state."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. if counter Cx (Simultaneous transmission video) has not reached its upper limit:

a. shall perform the actions specified in the clause 6.3.4.4.2;

2. if counter Cx (Simultaneous transmission video) has reached its upper limit:

a. select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;

b. shall stop timer T4 (Transmission Grant), if running;

c. shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);

d. shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;

e. shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and

f. shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:

i. shall include the queue position and transmission priority in the Queue Info field; and

ii. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:

a. shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and

b. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

2. shall set the general state to 'G: pending Transmission Revoke'."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

b. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.1.5.3 Test description

5.1.1.5.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the participating MCVideo functions.
- SS-UE1 (MCVideo Client) and SS-UE2 (MCVideo Client), simulated by the SS (MCVideo Server).

IUT:

- IUT (MCVideo Server), providing the controlling MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of all sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Transmission Control Server.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server).
- The SS-UE1 (MCVideo client) and the SS-UE2 (MCVideo client) are registered and authorized to use the MCVideo system.
- The SS-UE1 (MCVideo client) and the SS-UE2 (MCVideo client) are affiliated to the pre-arranged group and are authorized to initiate group calls.
- The IUT (MCVideo Server) is the controlling server of the pre-arranged group.

5.1.1.5.3.2 Test procedure sequence

Table 5.1.1.5.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E group call establishment' as described in Annex A, Table A.4.1.4.3-1 to establish an emergency pre-arranged group call with implicit transmission control request according to option b.ii of note in Annex A, Table A.4.1.4.3-1, initiated by SS-UE1 (MCVideo Client)?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure ' MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	3	P
4	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.1.5.3-1, to release the call?	-	-	4	P

5.1.1.5.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

**Table 5.1.1.5.3.3-1: SIP INVITE from SS (MCVideo Server)
(Step 1, Table 5.1.1.5.3.2-1; Step 1, Annex A, Table A.4.1.4.3-1)**

Derivation Path: Annex A, Table A.4.1.4.4-1, with condition EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.5.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.1.5.3.2-1; Step 2, Annex A, Table A.4.1.4.3-1)**

Derivation Path: Annex A, Table A.4.1.4.4-2, with condition EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.5.3.3-3: SIP 200 (OK) from SS (MCVideo Server)
(Step 1, Table 5.1.1.5.3.2-1; Step 8, Annex A, Table A.4.1.4.3-1)**

Derivation Path: Annex A, Table A.4.1.4.4-6

**Table 5.1.1.5.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.1.1.5.3.2-1; Step 9, Annex A, Table A.4.1.4.3-1)**

Derivation Path: Annex A, Table A.4.1.4.4-7

**Table 5.1.1.5.3.3-5: Transmission Idle from IUT (MCVideo Server)
(Step 2, Table 5.1.1.5.3.2-1; Steps 4 and 5, Annex A, Table A.3.8.3-1)**

Derivation Path: Annex A, Table A.3.8.4-6, with condition EMERGENCY-CALL.

**Table 5.1.1.5.3.3-6: Transmission Request from SS (MCVideo Server)
(Step 3, Table 5.1.1.5.3.2-1; Step 1, Annex A, Table A.3.7.3-1)**

Derivation Path: Annex A, Table A.3.7.4-1, with condition EMERGENCY-CALL.

**Table 5.1.1.5.3.3-7: Transmission Granted from IUT (MCVideo Server)
(Steps 1 and 3, Table 5.1.1.5.3.2-1; Step 3a1 or 3b1, Annex A, Table A.3.7.3-1;
Step 9a1, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.3.7.4-2 or A.3.7.4-3 (note), with condition EMERGENCY-CALL.
NOTE: Depends on server implementation.

**Table 5.1.1.5.3.3-8: Media Transmission Notification from IUT (MCVideo Server)
(Steps 1 and 3, Table 5.1.1.5.3.2-1; Step 3a3 or 3b2, Annex A, Table A.3.7.3-1;
Step 10a1, Annex A, Table 4.1.1.3-1)**

Derivation Path: Annex A, Table A.3.7.4-4 or A.3.7.4-5 (note), with condition EMERGENCY-CALL.
NOTE: Depends on server implementation.

5.1.1.6 MCVideo Server - MCVideo Server / Pre-arranged Group Call / Emergency Group Call / Terminating Participating server

5.1.1.6.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the controlling function initiates an emergency pre-
arranged group call with automatic commencement }
    then { IUT (MCVideo Server) accepts the call and establishes the call with SS-UE2 (MCVideo
Client) and forwards the media transmission notification received from SS (MCVideo Server) to SS-UE2
(MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established an emergency pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) ends the media transmission }
    then { IUT (MCVideo Server) forwards the transmission end notify received from SS (MCVideo
Server) to SS-UE2 (MCVideo Client) and forwards the transmission idle received from SS (MCVideo
Server) to SS-UE2 (MCVideo Client) }
           }
```

(3)

```
with { IUT (MCVideo Server) having established an emergency pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) providing the controlling function ends the pre-arranged group call
}
    then { IUT (MCVideo Server) sends a SIP BYE message to SS-UE2 (MCVideo Client) and responds by
sending a SIP 200 (OK) message to SS (MCVideo Server) }
           }
```

5.1.1.6.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.1.3.2, 9.2.1.3.4, 6.3.2.1.6.0, 6.3.2.1.6.1, 6.3.3.1.6, 6.3.2.2.5.1, 6.3.2.2.6.1, 6.3.2.2.8.1, 6.3.2.2.4.2, 6.3.2.2.2.1 and 6.3.2.1.2.1, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.4.1, 6.4.2, 6.4.3, 6.4.4, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.2]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

NOTE 1: This clause covers on-demand session.

Upon receipt of a "SIP INVITE request for terminating participating MCVideo function", the participating MCVideo function:

…

2) shall check the presence of the isfocus media feature tag in the URI of the Contact header field and if it is not present then the participating MCVideo function shall reject the request …, and shall not continue with the rest of the steps;

3) if the Answer-Mode Indication in the application/poc-settings+xml MIME body has not yet been received from the invited MCVideo client as defined in clause 7.3.3 or clause 7.3.4, shall reject the request …, and shall not continue with the rest of the steps;

4) shall use the MCVideo ID present in the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCVideo ID and public user identity;

5) if the binding between the MCVideo ID and public user identity does not exist, then the participating MCVideo function shall reject the SIP INVITE request with a SIP 404 (Not Found) response. Otherwise, continue with the rest of the steps;

6) if the SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with an <MKFC-GKTPs> element, shall perform the procedures in clause 6.3.2.3.2;

7) shall perform the automatic commencement procedures specified in clause 6.3.2.2.5.1 and according to IETF RFC 5373 [27] if the "SIP INVITE request for terminating participating MCVideo function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCVideo client as per clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; and

8) shall perform the manual commencement procedures specified in clause 6.3.2.2.6.1 and according to IETF RFC 5373 [27] if the "SIP INVITE request for terminating participating MCVideo function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCVideo client as per clause 7.3.3 or clause 7.3.4 is set to "manual-answer"."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.4]

"Upon receiving a SIP BYE request from the controlling MCVideo function, the participating MCVideo function shall follow the procedures as specified in clause 6.3.2.2.8.1."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.6.0]

"This clause contains common procedures to be used for MCVideo emergency group calls and MCVideo imminent peril group calls."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.6.1]

"When the participating MCVideo function receives a request from the MCVideo client to originate an MCVideo emergency group call and needs to determine if the request is an authorised request for an MCVideo emergency call, the participating MCVideo function shall check the following:

1) if the <allow-emergency-group-call> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true" and:

a) if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> element contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "DedicatedGroup" and if the <uri-entry> element of the <entry> element of the <MCVideoGroupInitiation> element contains the identity of the MCVideo group targeted by the calling MCVideo user; or

b) if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "UseCurrentlySelectedGroup";

then the participating MCVideo function shall consider the MCVideo emergency group call request to be an authorised request for an MCVideo emergency group call;

In all other cases, the participating MCVideo function shall consider the request to originate an MCVideo emergency group call to be an unauthorised request to originate an MCVideo emergency group call.

When the participating MCVideo function receives a request from the MCVideo client to originate an MCVideo imminent peril group call and needs to determine if the request is an authorised request for an MCVideo imminent peril group call the participating MCVideo function shall check the following:

1) if the <allow-imminent-peril-call> element of <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true"; and

a) if the "entry-info" attribute of the <MCVideoGroupInitiation> element contained within the <ImminentPerilCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "DedicatedGroup" and if the <uri-entry> element of the <entry> element of the <MCVideoGroupInitiation> element contains the identity of the MCVideo group targeted by the calling MCVideo user; or

b) if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element contained within the <ImminentPerilCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "UseCurrentlySelectedGroup";

then the participating MCVideo function shall consider the MCVideo imminent peril group call request to be an authorised request for an MCVideo emergency group call;

In all other cases, the participating MCVideo function shall consider the request to originate an MCVideo imminent peril group call to be an unauthorised request to originate an MCVideo imminent peril call."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.6]

"This clause is referenced from other procedures.

This clause describes the procedures for inviting an MCVideo user to an MCVideo session associated with an MCVideo emergency group call or MCVideo imminent peril group call. The procedure is initiated by the controlling MCVideo function as the result of an action in clause 9.2.2.4.1.1.

The controlling MCVideo function:

1) *shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;*

2) *shall set the Request-URI to the address of the terminating participating MCVideo function associated with the MCVideo ID of the targeted MCVideo user;*

3) *shall include an application/vnd.3gpp.mcvideo-info+xml MIME body with the <mcvideoinfo> element containing the <mcvideo-Params> element populated as follows:*

    a) *the <mcvideo-request-uri> element set to the value of the MCVideo ID of the targeted MCVideo user;*

    b) *the <mcvideo-calling-user-id> element set to the value of the MCVideo ID of the calling MCVideo user; and*

    c) *the <mcvideo-calling-group-id> element set to the value of the MCVideo group ID of the emergency group call.*

4) *shall include in the P-Asserted-Identity header field the public service identity of the controlling MCVideo function;*

5) *shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1; and*

6) *if the in-progress emergency group state of the group is set to a value of "true" the controlling MCVideo function:*

    a) *shall include a Resource-Priority header field populated with the values for an MCVideo emergency group call as specified in clause 6.3.3.1.18;*

    b) *shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <emergency-ind> element set to a value of "true";*

    c) *if the <alert-ind> element is set to "true" in the received SIP INVITE request and the requesting MCVideo user and MCVideo group are authorised for the initiation of MCVideo emergency alerts as determined by the procedures of clause 6.3.3.1.12.1, shall populate the application/vnd.3gpp.mcvideo-info+xml MIME body and the application/vnd.3gpp.mcvideo-location-info+xml MIME body as specified in clause 6.3.3.1.11. Otherwise, shall set the <alert-ind> element to a value of "false" in the application/vnd.3gpp.mcvideo-info+xml MIME body; and*

    d) *if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <imminentperil-ind> element set to a value of "false";*

NOTE: If the imminent peril state of the group is true at this point, the controlling function will set it to false as part of the calling procedure.

7) *if the in-progress emergency state of the group is set to a value of "false" and the in-progress imminent peril state of the group is set to a value of "true", the controlling MCVideo function:*

    a) *shall include a Resource-Priority header field populated with the values for an MCVideo imminent peril group call as specified in clause 6.3.3.1.18; and*

    b) *shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true"; and*

8) if:

a) an MCVideo GKTP document exists for the group identity contained in the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request; and

b) the MCVideo GKTP document contains an <MKFC-GKTPs> element;

then:

a) for each instance of <GKTP> element of the <MKFC-GKTPs> element of the MCVideo GKTP document:

i) shall perform the procedure in clause 6.3.3.6.2 to re-generate an I_MESSAGE; and

ii) if the procedure in clause 6.3.3.6.2 was successful, shall include the I_MESSAGE in a <GKTP> element of the <MKFC-GKTPs> element of an application/vnd.3gpp.mcvideo-info+xml MIME body included in the outgoing SIP INVITE request."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.5.1]

"When receiving a "SIP INVITE request for terminating participating MCVideo function" that requires automatic commencement mode:

1) if:

a) the invited MCVideo client has one or more pre-established sessions without an associated MCVideo session;

b) the media-level sections for the offered MCVideo video media stream are the same as the media-level sections for MCVideo video media stream in an existing pre-established session; and

c) the media-level section of the offered media-transmission control entity is the same as the media-level section for media-transmission control entity in an existing pre-established session;

then the participating MCVideo function shall perform the actions specified in clause 6.3.2.2.5.3; or

2) otherwise the participating MCVideo function shall perform the actions specified in clause 6.3.2.2.5.2."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.6.1]

"When receiving a "SIP INVITE request for terminating participating MCVideo function" that requires manual commencement mode:

1) if:

a) the invited MCVideo client has one or more pre-established sessions without an associated MCVideo session;

b) the media-level sections for the offered MCVideo video media stream are the same as the media-level section for MCVideo video media stream in the existing pre-established session; and

c) the media-level section of the offered media-transmission control entity is the same as the media-level section for media-transmission control entity in the existing pre-established session;

then the participating MCVideo function shall perform the actions specified in clause 6.3.2.2.6.3; or

2) otherwise the participating MCVideo function shall perform the actions specified in clause 6.3.2.2.6.2."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.8.1]

"Upon receiving a SIP BYE request from the controlling MCVideo function, the participating MCVideo function:

1) shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.581 [5] for releasing media plane resource associated with the SIP session with the controlling MCVideo function;

2) shall generate a SIP BYE request according to 3GPP TS 24.229 [11];

3) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;

4) if the received SIP BYE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body into the outgoing SIP BYE request; and

5) shall send the SIP BYE request to the MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response to the SIP BYE request the participating MCVideo function:

1) shall send a SIP 200 (OK) response to the SIP BYE request received from the controlling MCVideo function according to 3GPP TS 24.229 [11]; and

2) shall interact with the media plane as specified in 3GPP TS 24.581 [5] for releasing media plane resources associated with the SIP session with the MCVideo client."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.4.2]

"This clause is referenced from other procedures.

When sending SIP 200 (OK) responses, the participating MCVideo function shall generate a SIP 200 (OK) response according to 3GPP TS 24.229 [11] and:

1) shall include the option tag "timer" in a Require header field;

2) shall include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [23], "UAS Behavior". If no "refresher" parameter was included in the SIP INVITE request, the "refresher" parameter in the Session-Expires header field shall be set to "uas";

3) shall include the following in the Contact header field:

a) the g.3gpp.mcvideo media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo"; and

c) an MCVideo session identity mapped to the MCVideo session identity provided in the Contact header field of the received SIP INVITE request from the controlling MCVideo function;

4) shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [32]; and

5) if the incoming SIP response contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body to the outgoing SIP 200 (OK) response."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.2.1]

"The participating MCVideo function shall follow the procedures in clause 6.3.2.1.2.1."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [11], the participating MCVideo function:

1) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCVideo function, for the accepted media stream in the received SDP offer, if required; and

NOTE 1: Requirements can exist for the participating MCVideo function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCVideo function, for the accepted media-transmission control entity, if present in the received SDP offer.

NOTE 2: If the participating MCVideo function and the controlling MCVideo function are in the same MCVideo server, and the participating MCVideo function does not have a dedicated IP address or a dedicated port number for media transmission control or media stream, the replacement of the IP address or the port number is omitted."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.1]

"Once an on-demand MCVideo session is established or a pre-established session is in use when the participating MCVideo function receives transmission control messages from the transmission participant in the MCVideo client or from the transmission control server in the controlling MCVideo function, the behaviour of the participating MCVideo function is described in the following clauses."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.2]

"Upon receiving a transmission control message the participating MCVideo function:

1. *shall immediately forward the transmission control message to the transmission control server if the message is received from the transmission participant;*

2. *if an MBMS subchannel is not used for a transmission in the session the transmission control message is associated with, shall immediately forward the transmission control message to the transmission participant if the message is received from the transmission control server; and*

3. *if an MBMS subchannel is used for a transmission in the session the transmission control message is associated with:*

   a. *if*

      i. *the transmission control message is not a Transmission Idle message or a Media Transmission Notify message;*

      ii. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

      iii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report;*

      shall immediately forward the transmission control message to the transmission participant; and

   b. *if*

      i. *the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report; and*

      ii *if the transmission control message is the Transmission Idle message or the Media Transmission Notify message,*

   shall perform actions as specified in clause 10.2.

NOTE: When the Transmit Idle or Media Transmission Notify messages are discarded the messages are sent to the MCVideo clients over the MBMS subchannel allocated for the transmission as specified in clause 10.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.3]

"Upon receiving RTP media packets the participating MCVideo function:

1. *shall immediately forward the RTP media packet to the controlling MCVideo function if the RTP packet is from an MCVideo client; and*

2. *if an MBMS subchannel is not used for a transmission in the session the RTP media packets are associated with, shall immediately forward the RTP media packets to the MCVideo client if the RTP packet is from the controlling MCVideo function or the non-controlling MCVideo function.*

3. *if an MBMS subchannel is used for a transmission in the session the RTP media packets are associated with and if RTP media packets are received from the controlling MCVideo function or the non-controlling MCVideo function:*

   a. *if*

      i. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

      ii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report,*

      shall immediately forward the RTP media packets to the MCVideo client; and

   b. *if the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report, shall perform actions as specified in clause 10.2."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.4]

"When the participating function receives an indication from the application and signalling plane that session release is initiated, the participating MCVideo function:

1. *shall stop sending transmission control messages towards the transmission participant and the transmission control server; and*

2. *shall stop sending RTP media packets towards the MCVideo client and towards the controlling MCVideo function.*

When the participating MCVideo function receives an indication from the application and signalling plane that the session is released, the participating MCVideo function:

1. *in case of a pre-established session, shall perform the actions in clause 9.3.2; and*

2. *in case of an on-demand session, shall release the media resources associated with the session."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

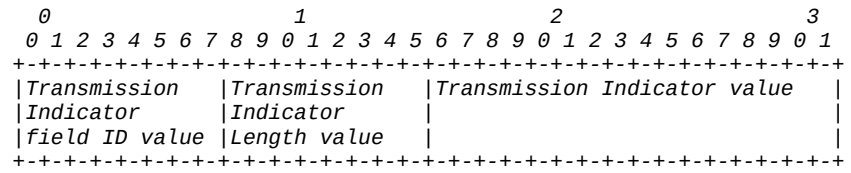
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

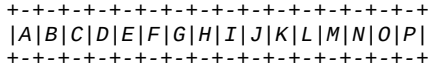


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. if the active Transmission request queue is empty the transmission control server:

a. shall send Transmission Idle message to all transmission participants. The Transmission Idle message:

i. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

ii. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

b. shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;

c. shall start timer T1 (Inactivity);

d. shall set the general state to the 'G: Transmit Idle' state; and

e. shall initialise counter Cx (Simultaneous transmission video) to 0.

2. if the active Transmission request queue is not empty the transmission control server:

a. shall select a queued Transmission request from the top of the active Transmission request queue;

b. shall remove that queued Transmission request from the active Transmission request queue; and

c. shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:

a. shall include the granted priority in the Transmission priority field;

b. shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;

c. if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

d. shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;

2. shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;

3. shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:

a. shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;

b. shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;

c. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;

d. if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';

e. if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

g. shall initiate the 'general reception control operation' state machine;and

4. shall enter the 'G: Transmit Taken' state."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. if counter Cx (Simultaneous transmission video) has not reached its upper limit:

a. shall perform the actions specified in the clause 6.3.4.4.2;

2. *if counter Cx (Simultaneous transmission video) has reached its upper limit:*

   a. *select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;*

   b. *shall stop timer T4 (Transmission Grant), if running;*

   c. *shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);*

   d. *shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;*

   e. *shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and*

   f. *shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:*

      i. *shall include the queue position and transmission priority in the Queue Info field; and*

      ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:*

   a. *shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and*

   b. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

2. *shall set the general state to 'G: pending Transmission Revoke'."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. *if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:*

   a. *shall initialize a general state machine as specified in clause 6.3.4.2.2; and*

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

   b. *shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;*

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.1.6.3 Test description

5.1.1.6.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the controlling MCVideo function.
- SS-UE2 (MCVideo Client).

IUT:

- IUT (MCVideo Server), providing the terminating participating MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of some sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server and the HTTP Client.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server) and SS-UE2 (MCVideo Client).
- The SS-UE2 (MCVideo client) is registered and authorized to use the MCVideo system.

- The SS-UE2 (MCVideo client) is affiliated to the pre-arranged group and is authorized to initiate group calls.

5.1.1.6.3.2 Test procedure sequence

Table 5.1.1.6.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CT group/private call establishment, automatic commencement' as described in Annex A, Table A.4.3.3.3-1 to establish an emergency pre-arranged group call with automatic commencement mode and notify a media transmission from SS-UE1 (MCVideo Client)?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Notification - Idle' as described in Annex A, Table A.4.3.8.3-1 to notify the end of SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.3.5.3-1, to release the call?	-	-	3	P

5.1.1.6.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

Table 5.1.1.6.3.3-1: SIP INVITE from SS (MCVideo Server)
(Step 1, Table 5.1.1.6.3.2-1; Step 1, Annex A, Table A.4.3.3.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-1, with conditions GROUP-CALL and EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.1.6.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.1.6.3.2-1; Step 2, Annex A, Table A.4.3.3.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-2, with conditions GROUP-CALL and EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.1.6.3.3-3: SIP 200 (OK) from SS-UE2 (MCVideo Client)
(Step 1, Table 5.1.1.6.3.2-1; Step 6, Annex A, Table A.4.3.3.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-7, with condition GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.1.6.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.1.1.6.3.2-1; Step 7, Annex A, Table A.4.3.3.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-8, with condition GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.1.6.3.3-5: Transmission Idle
(Step 2, Table 5.1.1.6.3.2-1; Steps 13 and 14, Annex A, Table A.4.3.8.3-1)

Derivation Path: Annex A, Table A.4.3.8.4-6, with condition EMERGENCY-CALL.

Table 5.1.1.6.3.3-6: Media Transmission Notification
(Step 1, Table 5.1.1.6.3.2-1; Steps 10a1 and 10a2, Annex A, Table 4.3.3.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-11, with condition EMERGENCY-CALL.

5.1.1.7 MCVideo Server - MCVideo Server / Pre-arranged Group Call / Emergency Group Call / Originating Participating server

5.1.1.7.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS-UE1 (MCVideo Client) initiates an emergency pre-arranged group call with implicit
transmission control }
    then { IUT (MCVideo Server) accepts the call from initiator and establishes the call with the SS
(MCVideo Server) providing the controlling function and forwards the transmission granted received
from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established an emergency pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests to end the transmission control }
    then { IUT (MCVideo Server) forwards the transmission end request to SS (MCVideo Server) and
forwards the transmission end response to SS-UE1 (MCVideo Client) and forwards the transmission idle
received from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established an emergency pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control }
    then { IUT (MCVideo Server) forwards the transmission request to SS (MCVideo Server) and
forwards the transmission granted received from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(4)

```
with { IUT (MCVideo Server) having established an emergency pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) ends the pre-arranged group call }
    then { IUT (MCVideo Server) sends a SIP BYE message to SS (MCVideo Server) providing the
controlling function and responds by sending a SIP 200 (OK) message to SS-UE1 (MCVideo Client) }
            }
```

5.1.1.7.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.1.3.1.1, 9.2.1.3.3, 6.2.8.1.1, 6.3.1.1, 6.3.2.1.3, 6.3.2.1.1.1 and 6.3.2.1.6.1, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.4.1, 6.4.2, 6.4.3, 6.4.4, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.1.1]

"In the procedures in this clause:

1) group identity in an incoming SIP INVITE request refers to the group identity from the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;

2) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

3) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

Upon receipt of a "SIP INVITE request for originating participating MCVideo function" containing an application/vnd.3gpp.mcvideo-info+xml MIME body with the <session-type> element set to a value of "prearranged", the participating MCVideo function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP INVITE request with a SIP 500 (Server Internal Error) response. The participating MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15]. Otherwise, continue with the rest of the steps;

NOTE 1: if the SIP INVITE request contains an emergency indication or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCVideo function can according to local policy choose to accept the request.

2) shall determine the MCVideo ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request, and shall authorise the calling user;

NOTE 2: The MCVideo ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

3) if through local policy in the participating MCVideo function, the user identified by the MCVideo ID is not authorised to initiate prearranged group calls, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response to the SIP INVITE request, with warning text set to "109 user not authorised to make prearranged group calls" in a Warning header field as specified in clause 4.4;

4) shall validate the media parameters and if the MCVideo codecs are not offered in the SIP INVITE request shall reject the request with a SIP 488 (Not Acceptable Here) response. Otherwise, continue with the rest of the steps;

5) shall check if the number of maximum simultaneous MCVideo group calls supported for the MCVideo user as specified in the <MaxSimultaneousCallsN6> element of the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) has been exceeded. If exceeded, the participating MCVideo function shall respond with a SIP 486 (Busy Here) response with the warning text set to "103 maximum simultaneous MCVideo group calls reached" in a Warning header field as specified in clause 4.4. Otherwise, continue with the rest of the steps;

NOTE 3: If the SIP INVITE request contains an emergency indication or an imminent peril indication, the participating MCVideo function can by means beyond the scope of this specification choose to allow for an exception to the limit for the maximum simultaneous MCVideo sessions supported for the MCVideo user. Alternatively, a lower priority session of the MCVideo user could be terminated to allow for the new session.

6) if the user identified by the MCVideo ID is not affiliated to the group identified in the "SIP INVITE request for originating participating MCVideo function" as determined by clause 8.2.2.2.11 and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, shall perform the actions specified in clause 8.2.2.2.12 for implicit affiliation;

7) if the actions for implicit affiliation specified in step 6) above were performed but not successful in affiliating the MCVideo user due to the MCVideo user already having N2 simultaneous affiliations, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 486 (Busy Here) response with the warning text set to "102 too many simultaneous affiliations" in a Warning header field as specified in clause 4.4. and skip the rest of the steps.

NOTE 4: N2 is the total number of MCVideo groups that an MCVideo user can be affiliated to simultaneously as specified in 3GPP TS 23.281 [26].

NOTE 5: if the SIP INVITE request contains an emergency indication set to a value of "true" or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCVideo function can according to local policy choose to allow an exception to the N2 limit. Alternatively, a lower priority affiliation of the MCVideo user could be cancelled to allow for the new affiliation.

8) shall determine the public service identity of the controlling MCVideo function associated with the group identity in the SIP INVITE request;

NOTE 6: The public service identity can identify the controlling MCVideo function in the primary MCVideo system or a partner MCVideo system.

NOTE 7: How the participating MCVideo server discovers the public service identity of the controlling MCVideo function associated with the group identity is out of scope of the current release.

9) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3;

10) shall set the Request-URI to the public service identity of the controlling MCVideo function associated with the group identity which was present in the incoming SIP INVITE request;

11) shall not copy the following header fields from the incoming SIP INVITE request to the outgoing SIP INVITE request, if they were present in the incoming SIP INVITE request:

a) Answer-Mode header field as specified in IETF RFC 5373 [27]; and

b) Priv-Answer-Mode header field as specified in IETF RFC 5373 [27];

12) shall set the <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the SIP INVITE request to the MCVideo ID of the calling user;

13) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the MCVideo client as specified in clause 6.3.2.1.1.1;

14) if the received SIP INVITE request contains an application/vnd.3gpp.location-info+xml MIME body and if not already copied, shall copy the contents of the application/vnd.3gpp.location-info+xml MIME body received in the SIP INVITE request into an application/vnd.3gpp.location-info+xml MIME body included in the outgoing SIP request;

15) if a Resource-Priority header field was included in the received SIP INVITE request, shall include a Resource-Priority header field according to rules and procedures of 3GPP TS 24.229 [11] set to the value indicated in the Resource-Priority header field of the SIP INVITE request from the MCVideo client; and

NOTE 8: The participating MCVideo function will leave verification of the Resource-Priority header field to the controlling MCVideo function.

16) shall forward the SIP INVITE request, according to 3GPP TS 24.229 [11].

Upon receipt of a SIP 302 (Moved Temporarily) response to the above SIP INVITE request in step 15), the participating MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.2.1.10;

2) shall include an SDP offer based upon the SDP offer in the received SIP INVITE request from the MCVideo client as specified in clause 6.3.2.1.1.1; and

3) shall forward the SIP INVITE request according to 3GPP TS 24.229 [11].

Upon receipt of a SIP 2xx response in response to the above SIP INVITE request in step 15), the participating MCVideo function:

1) if the received SIP 2xx response contains an application/vnd.3gpp.mcvideo-info+xml MIME body with an <MKFC-GKTPs> element, shall perform the procedures in clause 6.3.2.3.2;

2) shall generate a SIP 200 (OK) response as in clause 6.3.2.1.5.2;

3) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

4) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response;

5) shall include the public service identity received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response;

6) shall include an MCVideo session identity mapped to the MCVideo session identity provided in the Contact header field of the received SIP 200 (OK) response;

7) if the procedures of clause 8.2.2.2.12 for implicit affiliation were performed in the present clause, shall complete the implicit affiliation by performing the procedures of clause 8.2.2.2.13;

8) shall send the SIP 200 (OK) response to the MCVideo client according to 3GPP TS 24.229 [11];

9) shall interact with Media Plane as specified in 3GPP TS 24.581 [5]; and

10) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [23].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the above SIP INVITE request in step 14) the participating MCVideo function:

1) shall generate a SIP response according to 3GPP TS 24.229 [11];

2) shall include Warning header field(s) that were received in the incoming SIP response;

3) shall forward the SIP response to the MCVideo client according to 3GPP TS 24.229 [11]; and

4) if the implicit affiliation procedures of clause 8.2.2.2.12 were invoked in this procedure, shall perform the procedures of clause 8.2.2.2.14;"

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.3]

"Upon receiving from the MCVideo client a SIP BYE request the participating MCVideo function shall follow the procedures as specified in clause 6.3.2.1.6."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.2.8.1.1]

"This clause is referenced from other procedures.

When the MCVideo emergency state is set and this MCVideo user and MCVideo group are authorised to initiate MCVideo emergency group calls as determined by the procedures of clause 6.2.8.1.8, the MCVideo client:

1) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body in the SIP INVITE request, an <emergency-ind> element set to "true" and if the MCVideo emergency group call state is set to "MVEGC 1: emergency-gc-capable", shall set the MCVideo emergency group call state to "MVEGC 2: emergency-call-requested";

2) if the MCVideo user has also requested an MCVideo emergency alert to be sent and this is an authorised request for MCVideo emergency alert as determined by the procedures of clause 6.2.8.1.6, and the MCVideo emergency alert state is set to "MVEA 1: no-alert", shall:

a) set the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to "true" and set the MCVideo emergency alert state to "MVEA 2: emergency-alert-confirm-pending"; and

b) perform the procedures specified in clause 6.2.9.1 for the MCVideo emergency alert trigger;

3) if the MCVideo user has not requested an MCVideo emergency alert to be sent and the MCVideo emergency alert state is set to "MVEA 1: no-alert", shall set the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to "false"; and

4) if the MCVideo client emergency group state of the group is set to a value other than "MVEG 2: in-progress" set the MCVideo client emergency group state of the MCVideo group to "MVEG 3: confirm-pending".

NOTE 1: This is the case of an MCVideo user already being in the MCVideo emergency state it initiated previously while originating an MCVideo emergency group call or MCVideo emergency alert. All group calls the MCVideo user originates while in MCVideo emergency state will be MCVideo emergency group calls.

When the MCVideo emergency state is clear and the MCVideo emergency group call state is set to "MVEGC 1: emergency-gc-capable" and the MCVideo user is authorised to initiate an MCVideo emergency group call on the targetted MCVideo group as determined by the procedures of clause 6.2.8.1.8, the MCVideo client shall set the MCVideo emergency state and perform the following actions:

1) *shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body in the SIP INVITE request an <emergency-ind> element set to "true" and set the MCVideo emergency group call state to "MVEGC 2: emergency-call-requested" state;*

2) *if the MCVideo user has also requested an MCVideo emergency alert to be sent and this is an authorised request for MCVideo emergency alert as determined by the procedures of clause 6.2.8.1.6, shall:*

   a) *include in the application/vnd.3gpp.mcvideo-info+xml MIME body the <alert-ind> element set to "true" and set the MCVideo emergency alert state to "MVEA 2: emergency-alert-confirm-pending"; and*

   b) *perform the procedures specified in clause 6.2.9.1 for the MCVideo emergency alert trigger;*

3) *if the MCVideo user has not requested an MCVideo emergency alert to be sent, shall set the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to "false"; and*

4) *if the MCVideo client emergency group state of the group is set to a value other than "MVEG 2: in-progress" shall set the MCVideo client emergency group state of the MCVideo group to "MVEG 3: confirm-pending".*

NOTE 2: This is the case of an initial MCVideo emergency group call and optionally an MCVideo emergency alert being sent. As the MCVideo emergency state is not sent, there is no MCVideo emergency alert outstanding.

NOTE 3: An MCVideo group call originated by an affiliated member of an MCVideo group which is in an in-progress emergency state (as tracked on the MCVideo client by the MCVideo client emergency group state) but is not in an MCVideo emergency state of their own will also be an MCVideo emergency group call. The <emergency-ind> and <alert-ind> elements of the application/vnd.3gpp.mcvideo-info+xml MIME body do not need to be included in this case and hence no action needs to be taken in this clause."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.1.1]

"The MCVideo server needs to distinguish between the following initial SIP INVITE requests for originations and terminations:

- *SIP INVITE requests routed to the participating MCVideo function as a result of processing initial filter criteria at the S-CSCF in accordance with the origination procedures as specified in 3GPP TS 24.229 [11] and the Request-URI is set to a public service identity of the participating MCVideo function that does not identify the pre-established session set-up. Such requests are known as "SIP INVITE request for originating participating MCVideo function" in the procedures in the present document;*

- *SIP INVITE requests routed to the participating MCVideo function as a result of processing initial filter criteria at the S-CSCF in accordance with the termination procedures as specified in 3GPP TS 24.229 [11] and the Request-URI contains a PSI of the terminating participating MCVideo function. Such requests are known as "SIP INVITE request for terminating participating MCVideo function" in the procedures in the present document;*

- *SIP INVITE requests routed to the controlling MCVideo function as a result of PSI routing on the originating side in accordance with the originating procedures as specified in 3GPP TS 24.229 [11], or as a result of direct PSI routing, in accordance with the termination procedures as specified in 3GPP TS 24.229 [11], the Request-URI is set to a public service identity for MCVideo private call and the Contact header field does not contain the isfocus media feature tag specified in IETF RFC 3840 [16]. Such requests are known as "SIP INVITE request for controlling MCVideo function of a private call" in the procedures in the present document;*

- *SIP INVITE requests routed to the controlling MCVideo function as a result of PSI routing on the originating side in accordance with the originating procedures as specified in 3GPP TS 24.229 [11], or as a result of direct PSI routing, in accordance with the termination procedures as specified in 3GPP TS 24.229 [11], the Request-URI is set to a public service identity serving an MCVideo group and the Contact header field does not contain the isfocus media feature tag specified in IETF RFC 3840 [16]. Such requests are known as "SIP INVITE request for controlling MCVideo function of an MCVideo group" in the procedures in the present document; and*

- *SIP INVITE requests routed to the controlling MCVideo function as a result of PSI routing on the originating side in accordance with the originating procedures as specified in 3GPP TS 24.229 [11], or as a result of direct PSI routing, in accordance with the termination procedures as specified in 3GPP TS 24.229 [11], the Request-URI is set to a public service identity for MCVideo ambient viewing call and the Contact header field does not contain the isfocus media feature tag specified in IETF RFC 3840 [22]. Such requests are known as "SIP INVITE request for controlling MCVideo function of an ambient viewing call" in the procedures in the present document."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.3]

"This clause is referenced from other procedures.

When generating an initial SIP INVITE request according to 3GPP TS 24.229 [11], on receipt of an incoming SIP INVITE request, the participating MCVideo function:

1) *shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [20] if included in the incoming SIP INVITE request;*

2) *should include the Session-Expires header field according to IETF RFC 4028 [23]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";*

3) *shall include the option tag "timer" in the Supported header field;*

4) *shall copy the contents of the P-Asserted-Identity header field of the incoming SIP INVITE request to the P-Asserted-Identity header field of the outgoing SIP INVITE request;*

5) *shall include the g.3gpp.mcvideo media feature tag into the Contact header field of the outgoing SIP INVITE request;*

6) *shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcvideo" (coded as specified in 3GPP TS 24.229 [11]), into the P-Asserted-Service header field of the outgoing SIP INVITE request;*

7) *if the incoming SIP INVITE request contained a MIME resource-lists body with the MCVideo ID of the invited MCVideo user, shall copy the MIME resource-lists body, according to rules and procedures of IETF RFC 5366 [37];*

8) *if the incoming SIP INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the contents of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request to the outgoing SIP INVITE request; and*

9) *if the incoming SIP INVITE request contained an application/vnd.3gpp.location-info+xml MIME body, shall copy the contents of the application/vnd.3gpp.location-info+xml MIME body of the incoming SIP INVITE request to the outgoing SIP INVITE request."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.1.1]

"This clause is referenced from other clauses.

The SDP offer is generated based on the received SDP offer. The SDP offer generated by the participating MCVideo function:

1) *shall contain two SDP media-level sections for MCVideo video media as contained in the received SDP offer; and*

2) *shall contain an SDP media-level section for one media-transmission control entity, if present in the received SDP offer.*

When composing the SDP offer according to 3GPP TS 24.229 [11], the participating MCVideo function:

1) *shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the participating MCVideo function, if required;*

NOTE 1: Requirements can exist for the participating MCVideo function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number for the offered media transmission control entity, if any, in the received SDP offer with the IP address and port number of the participating MCVideo function; and

NOTE 2: If the participating MCVideo function and the controlling MCVideo function are in the same MCVideo server, and the participating MCVideo function does not have a dedicated IP address or a dedicated port number for media transmission control or media stream, the replacement of the IP address or the port number is omitted.

3) shall contain an "a=key-mgmt" attribute field with a "mikey" attribute value, if present in the received SDP offer."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.6.1]

"When the participating MCVideo function receives a request from the MCVideo client to originate an MCVideo emergency group call and needs to determine if the request is an authorised request for an MCVideo emergency call, the participating MCVideo function shall check the following:

1) if the <allow-emergency-group-call> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true" and:

a) if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> element contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "DedicatedGroup" and if the <uri-entry> element of the <entry> element of the <MCVideoGroupInitiation> element contains the identity of the MCVideo group targeted by the calling MCVideo user; or

b) if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "UseCurrentlySelectedGroup";

then the participating MCVideo function shall consider the MCVideo emergency group call request to be an authorised request for an MCVideo emergency group call;

In all other cases, the participating MCVideo function shall consider the request to originate an MCVideo emergency group call to be an unauthorised request to originate an MCVideo emergency group call.

When the participating MCVideo function receives a request from the MCVideo client to originate an MCVideo imminent peril group call and needs to determine if the request is an authorised request for an MCVideo imminent peril group call the participating MCVideo function shall check the following:

1) if the <allow-imminent-peril-call> element of <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true"; and

a) if the "entry-info" attribute of the <MCVideoGroupInitiation> element contained within the <ImminentPerilCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "DedicatedGroup" and if the <uri-entry> element of the <entry> element of the <MCVideoGroupInitiation> element contains the identity of the MCVideo group targeted by the calling MCVideo user; or

b) if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element contained within the <ImminentPerilCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "UseCurrentlySelectedGroup";

then the participating MCVideo function shall consider the MCVideo imminent peril group call request to be an authorised request for an MCVideo emergency group call;

In all other cases, the participating MCVideo function shall consider the request to originate an MCVideo imminent peril group call to be an unauthorised request to originate an MCVideo imminent peril call."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.1]

"Once an on-demand MCVideo session is established or a pre-established session is in use when the participating MCVideo function receives transmission control messages from the transmission participant in the MCVideo client or from the transmission control server in the controlling MCVideo function, the behaviour of the participating MCVideo function is described in the following clauses."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.2]

"Upon receiving a transmission control message the participating MCVideo function:

1. *shall immediately forward the transmission control message to the transmission control server if the message is received from the transmission participant;*
2. *if an MBMS subchannel is not used for a transmission in the session the transmission control message is associated with, shall immediately forward the transmission control message to the transmission participant if the message is received from the transmission control server; and*
3. *if an MBMS subchannel is used for a transmission in the session the transmission control message is associated with:*
    a. *if*
        i. *the transmission control message is not a Transmission Idle message or a Media Transmission Notify message;*
        ii. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*
        iii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report;*

        shall immediately forward the transmission control message to the transmission participant; and
    b. *if*
        i. *the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report; and*
        ii *if the transmission control message is the Transmission Idle message or the Media Transmission Notify message,*

    shall perform actions as specified in clause 10.2.

NOTE: When the Transmit Idle or Media Transmission Notify messages are discarded the messages are sent to the MCVideo clients over the MBMS subchannel allocated for the transmission as specified in clause 10.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.3]

"Upon receiving RTP media packets the participating MCVideo function:

1. *shall immediately forward the RTP media packet to the controlling MCVideo function if the RTP packet is from an MCVideo client; and*
2. *if an MBMS subchannel is not used for a transmission in the session the RTP media packets are associated with, shall immediately forward the RTP media packets to the MCVideo client if the RTP packet is from the controlling MCVideo function or the non-controlling MCVideo function.*

3. *if an MBMS subchannel is used for a transmission in the session the RTP media packets are associated with and if RTP media packets are received from the controlling MCVideo function or the non-controlling MCVideo function:*

   a. *if*

      i. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

      ii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report,*

      shall immediately forward the RTP media packets to the MCVideo client; and

   b. *if the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report, shall perform actions as specified in clause 10.2."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.4]

"When the participating function receives an indication from the application and signalling plane that session release is initiated, the participating MCVideo function:

1. *shall stop sending transmission control messages towards the transmission participant and the transmission control server; and*

2. *shall stop sending RTP media packets towards the MCVideo client and towards the controlling MCVideo function.*

When the participating MCVideo function receives an indication from the application and signalling plane that the session is released, the participating MCVideo function:

1. *in case of a pre-established session, shall perform the actions in clause 9.3.2; and*

2. *in case of an on-demand session, shall release the media resources associated with the session."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

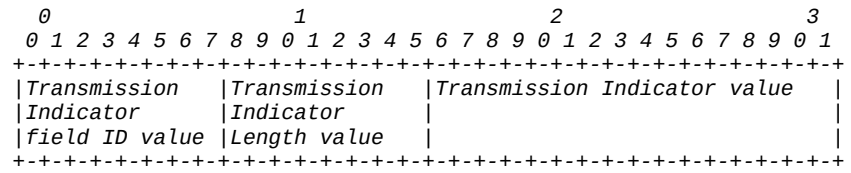
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

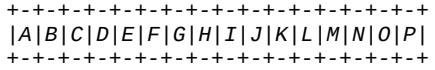


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. if the active Transmission request queue is empty the transmission control server:

a. shall send Transmission Idle message to all transmission participants. The Transmission Idle message:

i. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

ii. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

b. shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;

c. shall start timer T1 (Inactivity);

d. shall set the general state to the 'G: Transmit Idle' state; and

e. shall initialise counter Cx (Simultaneous transmission video) to 0.

2. if the active Transmission request queue is not empty the transmission control server:

a. shall select a queued Transmission request from the top of the active Transmission request queue;

b. shall remove that queued Transmission request from the active Transmission request queue; and

c. shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:

a. shall include the granted priority in the Transmission priority field;

b. shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;

c. if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

d. shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;

2. shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;

3. shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:

a. shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;

b. shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;

c. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;

d. if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';

e. if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

g. shall initiate the 'general reception control operation' state machine;and

4. shall enter the 'G: Transmit Taken' state."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. if counter Cx (Simultaneous transmission video) has not reached its upper limit:

a. shall perform the actions specified in the clause 6.3.4.4.2;

2. *if counter Cx (Simultaneous transmission video) has reached its upper limit:*

    a. *select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;*

    b. *shall stop timer T4 (Transmission Grant), if running;*

    c. *shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);*

    d. *shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;*

    e. *shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and*

    f. *shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:*

        i. *shall include the queue position and transmission priority in the Queue Info field; and*

        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:*

    a. *shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and*

    b. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

2. *shall set the general state to 'G: pending Transmission Revoke'."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. *if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:*

    a. *shall initialize a general state machine as specified in clause 6.3.4.2.2; and*

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

    b. *shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;*

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.1.7.3 Test description

5.1.1.7.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the controlling MCVideo function.
- SS-UE1 (MCVideo Client).

IUT:

- IUT (MCVideo Server), providing the originating participating MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of some sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server and the HTTP Client.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server) and SS-UE1 (MCVideo Client).
- The SS-UE1 (MCVideo client) is registered and authorized to use the MCVideo system.

- The SS-UE1 (MCVideo client) is affiliated to the pre-arranged group and is authorized to initiate group calls.

5.1.1.7.3.2 Test procedure sequence

Table 5.1.1.7.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.4.2.1.3-1 to establish an emergency pre-arranged group call with implicit transmission control request according to option b.ii of note in Annex A, Table A.4.2.1.3-1, initiated by SS-UE1 (MCVideo Client?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO Transmission End Request - Transmission End Response' as described in Annex A, Table A.4.2.5.3-1 to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Transmission Granted' as described in Annex A, Table A.4.2.4.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	3	P
4	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO call release' as described in Annex A, Table A.4.2.3.3-1, to release the call?	-	-	4	P

5.1.1.7.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

Table 5.1.1.7.3.3-1: SIP INVITE from SS-UE1 (MCVideo Client)
(Step 1, Table 5.1.1.7.3.2-1; Step 1, Annex A, Table A.4.2.1.3-1)

Derivation Path: Annex A, Table A.4.2.1.4-1, with conditions GROUP-CALL and EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.1.7.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.1.7.3.2-1; Step 2, Annex A, Table A.4.2.1.3-1)

Derivation Path: Annex A, Table A.4.2.1.4-2, with conditions GROUP-CALL and EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.1.7.3.3-3: SIP 200 (OK) from SS (MCVideo Server)
(Step 1, Table 5.1.1.7.3.2-1; Step 5, Annex A, Table A.4.2.1.3-1)

Derivation Path: Annex A, Table A.4.2.1.4-3, with condition GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.7.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.1.1.7.3.2-1; Step 6, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-4, with condition GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.7.3.3-5: Transmission Idle
(Steps 2, Table 5.1.1.7.3.2-1; Steps 9 and 10, Annex A, Table A.4.2.5.3-1)**

Derivation Path: Annex A, Table A.4.2.5.4-3, with condition EMERGENCY-CALL.

**Table 5.1.1.7.3.3-6: Transmission Request
(Step 3, Table 5.1.1.7.3.2-1; Steps 1 and 2, Annex A, Table A.4.2.4.3-1)**

Derivation Path: Annex A, Table A.4.2.4.4-1, with condition EMERGENCY-CALL.

**Table 5.1.1.7.3.3-7: Transmission Granted
(Steps 1 and 3, Table 5.1.1.7.3.2-1;
Steps 5 and 6, Annex A, Table A.4.2.4.3-1; Steps 9a1 and 9a2, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.4.4-2, with condition EMERGENCY-CALL.

5.1.1.8 MCVideo Server - MCVideo Server / Pre-arranged Group Call / Imminent Peril Group Call / Controlling server

5.1.1.8.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function initiates an
imminent peril pre-arranged group call with implicit transmission control }
    then { IUT (MCVideo Server) accepts the call from initiator and establishes the call with all
affiliated users by sending call establishment requests to SS (MCVideo Server) providing the
terminating participating function and sends transmission granted to SS-UE1 (MCVideo Client) and
sends media transmission notification to SS-UE2 (MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established an imminent peril pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests to end the transmission control }
    then { IUT (MCVideo Server) sends transmission control idle to SS-UE1 (MCVideo Client) and sends
transmission control idle to SS-UE2 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established an imminent peril pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control }
    then { IUT (MCVideo Server) sends transmission granted to SS-UE1 (MCVideo Client) and sends
media transmission notification to SS-UE2 (MCVideo Client) }
            }
```

(4)

```
with { IUT (MCVideo Server) having established an imminent peril pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function ends the pre-
arranged group call }
    then { IUT (MCVideo Server) responds by sending a SIP 200 (OK) message to SS (MCVideo Server)
ending the call and sends a SIP BYE message to SS (MCVideo Server) providing the terminating
participating function }
            }
```

5.1.1.8.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.1.4.2, 9.2.1.4.3, 9.2.1.4.1.1, 9.2.1.4.1.2, 6.3.3.1.6, 6.3.3.1.12.2, 6.3.3.1.12.4, 6.3.3.2.1 and 6.3.3.2.2, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.3.1, 6.3.2.2, 9.2.1, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.2]

"In the procedures in this clause:

1) *MCVideo ID in an incoming SIP INVITE request refers to the MCVideo ID of the originating user from the <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;*

2) *group identity in an incoming SIP INVITE request refers to the group identity from the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;*

3) *MCVideo ID in an outgoing SIP INVITE request refers to the MCVideo ID of the called user in the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the outgoing SIP INVITE request;*

4) *indication of required group members in a SIP 183 (Session Progress) response refers to the <required> element of the application/vnd.3gpp.mcvideo-info+xml MIME body set to "true" in a SIP 183 (Session Progress) sent by the non-controlling MCVideo function of an MCVideo group;*

5) *emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and*

6) *imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.*

Upon receipt of a "SIP INVITE request for controlling MCVideo function of an MCVideo group", the controlling MCVideo function:

…

2) *shall determine if the media parameters are acceptable and the MCVideo codecs are offered in the SDP offer and if not reject the request with a SIP 488 (Not Acceptable Here) response and skip the rest of the steps;*

3) *shall reject the SIP request with a SIP 403 (Forbidden) response and not process the remaining steps if:*

   a) *an Accept-Contact header field does not include the g.3gpp.mcvideo media feature tag; or*

   b) *an Accept-Contact header field does not include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo";*

4) *if received SIP INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.3.1.17;*

5) *shall retrieve the necessary group document(s) from the group management server for the group identity contained in the SIP INVITE request and carry out initial processing as specified in clause 6.3.5.2;*

6) if the result of the initial processing in clause 6.3.5.2 was:

a) that authorization of the MCVideo ID is required at a non-controlling MCVideo function of an MCVideo group is required, perform the actions in clause 6.3.3.1.13.7 and do not continue with the rest of the steps in this clause; and

b) that a SIP 3xx, 4xx, 5xx or 6xx response to the "SIP INVITE request for controlling MCVideo function of an MCVideo group" has been sent, do not continue with the rest of the steps in this clause;

7) shall perform the actions as described in clause 6.3.3.2.2;

8) shall maintain a local counter of the number of SIP 200 (OK) responses received from invited members and shall initialise this local counter to zero;

9) shall determine if an MCVideo group call for the group identity is already ongoing by determining if an MCVideo session identity has already been allocated for the group call and the MCVideo session is active;

…

12) if a Resource-Priority header field is included in the SIP INVITE request:

a) if the Resource-Priority header field is set to the value indicated for emergency calls and the SIP INVITE request does not contain an emergency indication and the in-progress emergency state of the group is set to a value of "false", shall reject the SIP INVITE request with a SIP 403 (Forbidden) response and skip the rest of the steps; or

b) if the Resource-Priority header field is set to the value indicated for imminent peril calls and the SIP INVITE request does not contain an imminent peril indication and the in-progress imminent peril state of the group is set to a value of "false", shall reject the SIP INVITE request with a SIP 403 (Forbidden) response and skip the rest of the steps;

13) if the MCVideo group call is not ongoing then:

…

c) if the received SIP INVITE request contains an an authorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2 or MCVideo imminent peril group call as determined by clause 6.3.3.1.13.5 and the MCVideo user is eligible to be implicitly affiliated with the MCVideo group as determined as determined in step 13) a) iv) above, shall perform the implicit affiliation as specified in clause 8.2.2.3.7;

d) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [11];

e) shall create a prearranged group session and allocate an MCVideo session identity for the prearranged group call, and shall handle timer TNG3 (group call timer) as specified in clause 6.3.3.5;

f) if the group identity in the "SIP INVITE request for controlling MCVideo function of an MCVideo group" is a TGI:

i) shall for each of the constituent MCVideo groups homed on the primary MCVideo system:

…;

B) if the controlling MCVideo function owns the MCVideo group identified by the MCVideo group ID then:

I) determine the members to invite to the prearranged MCVideo group call as specified in clause 6.3.5.5;

II) invite each group member determined in step A) above, to the group session, as specified in clause 9.2.1.4.1.1; and

III) interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3; and

ii) shall for each of the constituent MCVideo groups homed on the partner MCVideo system generate a SIP INVITE request for the MCVideo group identity homed on the partner MCVideo system as specified in clause 9.2.1.4.1.2; and

g) if the group identity in the SIP INVITE request for controlling MCVideo function of an MCVideo group is an MCVideo group ID:

i) shall determine the members to invite to the prearranged MCVideo group call as specified in clause 6.3.5.5;

ii) if necessary, shall start timer TNG1 (acknowledged call setup timer) according to the conditions stated in clause 6.3.3.3;

iii) if the received SIP INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element set to a value of "true":

A) shall cache the information that this MCVideo user has initiated an MCVideo emergency call;

B) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the information that this MCVideo user has initiated an MCVideo emergency alert; and

C) if the in-progress emergency state of the group is set to a value of "false":

I) shall set the value of the in-progress emergency state of the group to "true"; and

II) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16;

iv) if the in-progress emergency state of the group is set to a value of "false" and if the received SIP INVITE request contains an imminent peril indication set to a value of "true", the controlling MCVideo function shall:

A) shall cache the information that this MCVideo user has initiated an MCVideo imminent peril call; and

B) if the in-progress imminent peril state of the group is set to a value of "false", shall set the in-progress imminent peril state of the group to a value of "true";

v) shall invite each group member determined in step 13)g)i) above, to the group session, as specified in clause 9.2.1.4.1.1; and

vi) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3; and

14) if the MCVideo group call is ongoing then:

…

c) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [11];

…

e) if the received SIP INVITE request contains an an authorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2 or MCVideo imminent peril group call as determined by clause 6.3.3.1.13.5 and the MCVideo user is eligible to be implicitly affiliated with the MCVideo group as determined in step 14) a) iv) above, shall perform the implicit affiliation as specified in clause 8.2.2.3.7;

f) if the received SIP INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element set to a value of "true":

i) shall cache the information that this MCVideo user has initiated an MCVideo emergency call;

ii) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the information that this MCVideo user has initiated an MCVideo emergency alert;

iii) if the in-progress emergency state of the group is set to a value of "false":

A) shall set the value of the in-progress emergency state of the group to "true";

B) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16; and

C) shall generate SIP re-INVITE requests for the MCVideo emergency group call to the other call participants of the MCVideo group as specified in clause 6.3.3.1.6;

iv) if the in-progress imminent peril state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's imminent peril indication as specified in clause 6.3.3.1.11, setting the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11]; and

v) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5];

g) if the in-progress emergency state of the group is set to a value of "false" and if the SIP INVITE request contains an imminent peril indication set to a value of "true", the controlling MCVideo function:

i) shall cache the information that this MCVideo user has initiated an MCVideo imminent peril call; and

ii) if the in-progress imminent peril state of the group is set to a value of "false":

A) shall set the in-progress imminent peril state of the group to a value of "true";

B) shall generate SIP re-INVITE requests for the MCVideo imminent peril group call to the other call participants of the MCVideo group as specified in clause 6.3.3.1.15; and

C) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

iii) if the in-progress imminent peril state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's imminent peril indication as specified in clause 6.3.3.1.11, setting the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11];

h) shall generate a SIP 200 (OK) response as specified in the clause 6.3.3.2.4.2;

i) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

j) shall include in the SIP 200 (OK) response with the warning text set to "123 MCVideo session already exists" as specified in clause 4.4;

k) if the received SIP re-INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

l) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 4: In this case, the request was for an imminent peril call but a higher priority MCVideo emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

m) shall interact with media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 5: Resulting media plane processing is completed before the next step is performed.

n) shall send the SIP 200 (OK) response towards the inviting MCVideo client or inviting non-controlling MCVideo function according to 3GPP TS 24.229 [11];

o) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo User has joined in the MCVideo group session, as specified in clause 6.3.3.4;

NOTE 6: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

p) shall send a SIP NOTIFY request to each MCVideo client according to 3GPP TS 24.229 [11];

q) Upon receiving a SIP ACK to the above SIP 200 (OK) response and the SIP 200 (OK) response contained a Warning header field as specified in clause 4.4 with the warning text containing the mcvideo-warn-code set to "149", shall follow the procedures in clause 6.3.3.1.18; and

r) shall not continue with the rest of the clause.

Upon receiving a SIP 183 (Session Progress) response to the SIP INVITE request specified in clause 9.2.1.4.1 containing a P-Answer-State header field with the value "Unconfirmed" as specified in IETF RFC 4964 [30], the timer TNG1 (acknowledged call setup timer) is not running, the controlling MCVideo function supports media buffering and the SIP final response is not yet sent to the inviting MCVideo client:

1) shall generate a SIP 200 (OK) response to SIP INVITE request as specified in the clause 6.3.3.2.3.2;

2) shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if the prearranged MCVideo group has more than <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [24];

3) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

4) shall include a P-Answer-State header field with the value "Unconfirmed";

5) if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

6) if the received SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

7) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 7: Resulting user plane processing is completed before the next step is performed.

8) shall send the SIP 200 (OK) response towards the inviting MCVideo client according to 3GPP TS 24.229 [11];

9) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo User has joined in the MCVideo group session, as specified in clause 6.3.3.4; and

NOTE 8: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

10) shall send a SIP NOTIFY request to each MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 183 (Session Progress) response for a SIP INVITE request as specified in clause 9.2.1.4.1.2 containing an indication of required group members, the timer TNG1 (acknowledged call setup timer) is running and all SIP 200 (OK) responses have been received to all SIP INVITE requests sent to MCVideo clients specified in clause 9.2.1.4.1.1, then the controlling MCVideo function shall wait until the SIP 200 (OK) response has been received to the SIP INVITE request specified in clause 9.2.1.4.1.2 before generating a SIP 200 (OK) response to the "SIP INVITE request for controlling MCVideo function of an MCVideo group".

Upon receiving a SIP 200 (OK) response for a SIP INVITE request as specified in clause 9.2.1.4.1 that was sent to an affiliated and <on-network-required> group member as specified in 3GPP TS 24.481 [24]; and

1) if the MCVideo ID in the SIP 200 (OK) response matches to the MCVideo ID in the corresponding SIP INVITE request;

2) there are no outstanding SIP 200 (OK) responses to SIP INVITE requests which were sent to affiliated and <on-network-required> group members as specified in 3GPP TS 24.481 [24]; and

3) there is no outstanding SIP 200 (OK) response to a SIP INVITE request sent in clause 9.2.1.4.1.2 where the SIP 183 (Session Progress) response contained an indication of required group members;

the controlling MCVideo function:

1) shall stop timer TNG1 (acknowledged call setup timer) as described in clause 6.3.3.3;

2) shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;

3) shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if all members were not invited because the prearranged MCVideo group has been exceeded the <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [24];

4) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

5) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 9: Resulting media plane processing is completed before the next step is performed.

6) shall send a SIP 200 (OK) response to the inviting MCVideo client according to 3GPP TS 24.229 [11];

7) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo user has joined in the MCVideo group session, as specified in clause 6.3.3.4; and

NOTE 10: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

8) shall send the SIP NOTIFY request to the MCVideo clients according to 3GPP TS 24.229 [11].

Upon:

1) receiving a SIP 200 (OK) response for a SIP INVITE request as specified in clause 9.2.1.4.1;

2) the timer TNG1 (acknowledged call setup timer) is not running;

3) the local counter of the number of SIP 200 (OK) responses received from invited members is equal to the value of the <on-network-minimum-number-to-start> element of the group document;

4) the controlling MCVideo function supports media buffering; and

5) the SIP final response has not yet been sent to the inviting MCVideo client;

the controlling MCVideo function according to local policy:

1) shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.2;

2) shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if all members were not invited because the prearranged MCVideo group has exceeded the <max-participant-count> members as specified in 3GPP TS 24.481 [24];

3) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

4) if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

5) if the received SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

6) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 11: Resulting media plane processing is completed before the next step is performed.

7) shall send a SIP 200 (OK) response to the inviting MCVideo client according to 3GPP TS 24.229 [11];

8) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo user has joined in the MCVideo group session, as specified in clause 6.3.3.4; and

NOTE 12: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

9) shall send the SIP NOTIFY request to the MCVideo clients according to 3GPP TS 24.229 [11].

Upon expiry of timer TNG1 (acknowledged call setup timer), if there are outstanding SIP 200 (OK) responses to SIP INVITE requests sent to affiliated and <on-network-required> group members as specified in 3GPP TS 24.481 [24], the controlling MCVideo function shall follow the procedures specified in clause 6.3.3.3.

If timer TNG1 (acknowledged call setup timer) is running and a final SIP 4xx, 5xx or 6xx response is received from an affiliated and <on-network-required> group member as specified in 3GPP TS 24.481 [24], the controlling MCVideo function shall follow the relevant procedures specified in clause 6.3.3.3.

If:

1) timer TNG1 (acknowledged call setup timer) is not running;

2) the local counter of the number of SIP 200 (OK) responses received from invited members is equal to the value of the <on-network-minimum-number-to-start> element of the group document; and

3) a final SIP 4xx, 5xx or 6xx response is received from an invited MCVideo client;

then the controlling MCVideo function shall perform one of the following based on policy:

1) send the SIP final response towards the inviting MCVideo client, according to 3GPP TS 24.229 [11], if a SIP final response was received from all the other invited MCVideo clients and the SIP 200 (OK) response is not yet sent; or

2) remove the invited MCVideo client from the MCVideo Session as specified in clause 6.3.3.1.5, if a SIP final response other than 2xx or 3xx was received from all the invited MCVideo clients and the SIP 200 (OK) response is already sent. The controlling MCVideo function may invite an additional member of the prearranged MCVideo group as specified in clause 9.2.1.4.1 that has not already been invited, if the prearranged MCVideo group has more than <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [24], and all members have not yet been invited.

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCVideo client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.18."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.3]

"Upon receiving a SIP BYE request the controlling MCVideo function shall follow the procedures as specified in clause 6.3.3.2.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.7]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

Upon receipt of a SIP re-INVITE request for an MCVideo session identity identifying an on-demand prearranged MCVideo group session, the controlling MCVideo function:

…

2) if received SIP re-INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.3.1.17;

…

4) if the received SIP re-INVITE request contains an imminent peril indication set to "true" for an MCVideo imminent peril group call and this is an unauthorised request for an MCVideo imminent peril group call as determined by clause 6.3.3.1.13.6, shall reject the SIP re-INVITE request …

5) if a Resource-Priority header field is included in the received SIP re-INVITE request:

a) if the Resource-Priority header field is set to the value indicated for emergency calls and the SIP re-INVITE request does not contain an emergency indication and the in-progress emergency state of the group is set to a value of "false", shall reject the SIP re-INVITE request … and skip the rest of the steps; and

b) if the Resource-Priority header field is set to the value indicated for imminent peril calls and the SIP re-INVITE request does not contain an imminent peril indication and the in-progress imminent peril state of the group is set to a value of "false", shall reject the SIP INVITE request …and skip the rest of the steps;

6) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "true" and is an authorised request to initiate an MCVideo emergency group call as determined by clause 6.3.3.1.13.2, the controlling MCVideo function shall:

i) shall cache the MCVideo ID of the MCVideo user that has initiated an MCVideo emergency call;

ii) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the MCVideo ID of the MCVideo user that has initiated an MCVideo emergency alert;

iii) if the in-progress emergency state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's emergency indication as specified in clause 6.3.3.1.11, setting the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true";

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11]; and

C) if the in-progress imminent peril state of the group is set to a value of "true", shall set it to a value of "false"; and

iv) if the in-progress emergency state of the group is set to a value of "false":

A) shall set the value of the in-progress emergency state of the group to "true";

B) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16;

NOTE 2: The interactions of TNG2 with the TNG3 (group call timer) are explained in clause 6.3.3.5.2.

Editor's Note:timers need to be defined.

C) shall generate SIP re-INVITE requests for the MCVideo emergency group call to the other participants of the MCVideo group call as specified in clause 6.3.3.1.6;

D) shall send the SIP re-INVITEs towards the other participants of the MCVideo group call; and

E) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5];

7) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "false" and is an unauthorised request for an MCVideo emergency group call cancellation as determined by clause 6.3.3.1.13.4:

a) shall reject the SIP re-INVITE request.

…

c) if an <alert-ind> element of the mcvideoinfo MIME body is included in the SIP re-INVITE request set to "false", and there is an outstanding MCVideo emergency alert for this MCVideo user, shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <alert-ind> element set to a value of "true"; and

d) shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [11] and skip the rest of the steps;

8) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "false" and is determined to be an authorised request for an MCVideo emergency call cancellation as specified in clause 6.3.3.1.16 and the in-progress emergency state of the group to is set to a value of "true" the controlling MCVideo function:

a) shall set the in-progress emergency group state of the group to a value of "false";

b) shall clear the cache of the MCVideo ID of the MCVideo user as having an outstanding MCVideo emergency group call;

c) if an <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body is included and set to "false" and is determined to be an authorised request for an MCVideo emergency alert cancellation as specified in clause 6.3.3.1.13.3 and there is an outstanding MCVideo emergency alert for this MCVideo user shall:

i) if the received SIP re-INVITE request contains an <originated-by> element in the application/vnd.3gpp.mcvideo-info+xml MIME body, clear the cache of the MCVideo ID of the MCVideo user identified by the <originated-by> element as having an outstanding MCVideo emergency alert; or

ii) if the received SIP re-INVITE request does not contain an <originated-by> element in the application/vnd.3gpp.mcvideo-info+xml MIME body, clear the cache of the MCVideo ID of the sender of the SIP re-INVITE request as having an outstanding MCVideo emergency alert;

d) shall generate SIP re-INVITE requests to the participants in the group call as specified in clause 6.3.3.1.6. The MCVideo controlling function:

i) for each of the other participants in the group call shall send the SIP re-INVITE request towards the MCVideo client as specified in 3GPP TS 24.229 [11]; and

ii) Upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5];

NOTE 3: Clause 6.3.3.1.5 will inform the group call participants of the cancellation of the MCVideo group's in-progress emergency state and the cancellation of the MCVideo emergency alert if applicable.

e) shall stop timer TNG2 (in-progress emergency group call timer); and

NOTE 4: The interactions of TNG2 with the TNG3 (group call timer) are explained in clause 6.3.3.5.2;

f) for each of the affiliated members of the group that are not participating in the call:

i) generate a SIP MESSAGE request notification of the cancellation of the MCVideo user's emergency call as specified in clause 6.3.3.1.11;

ii) set the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "false";

iii) if indicated above in step 8) c), set the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "false"; and

iv) send the SIP MESSAGE request according to 3GPP TS 24.229 [11];

9) if the received SIP re-INVITE request contains an imminent peril indication and the in-progress emergency group state of the group is set to a value of "false", shall perform the procedures specified in clause 9.2.1.4.8 and skip the rest of the steps.

Upon receiving a SIP 200 (OK) response to a SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5];

1) shall generate a SIP 200 (OK) response according to rules and procedures of 3GPP TS 24.229 [11];

2) shall include in the SIP 200 (OK) response an SDP answer according to 3GPP TS 24.229 [11] with the clarifications specified in clause 6.3.3.2.1;

3) shall include the "norefersub" option tag in a Supported header field according to IETF RFC 4488 [31];

4) shall include the "tdialog" option tag in a Supported header field according to IETF RFC 4538 [32];

5) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <alert-ind> element set to a value of "true" and if this is an unauthorised request for an MCVideo emergency alert as determined by clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

6) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <alert-ind> element set to a value of "false" and if this is an unauthorised request for an MCVideo emergency alert cancellation as determined by clause 6.3.3.1.13.3, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

7) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true", this is an authorised request for an MCVideo imminent peril group call and the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 5: In this case, the request was for an imminent peril call but a higher priority MCVideo emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

8) shall interact with media plane as specified in 3GPP TS 24.581 [5]; and

9) shall send the SIP 200 (OK) response towards the MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCVideo client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.18.

Upon receipt of a SIP 2xx response for an outgoing SIP MESSAGE request, shall handle according to 3GPP TS 24.229 [11]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.1.1]

"This clause describes the procedures for inviting an MCVideo user to an MCVideo session. The procedure is initiated by the controlling MCVideo function as the result of an action in clause 9.2.1.4.2 or as the result of receiving a SIP 403 (Forbidden) response as described in this clause.

The controlling MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the public service identity of the terminating participating MCVideo function associated to the MCVideo user to be invited.;

NOTE 1: How the controlling MCVideo function finds the address of the terminating MCVideo participating function is out of the scope of the current release.

NOTE 2: If the terminating MCVideo user is part of a partner MCVideo system, then the public service identity can identify an entry point in the partner network that is able to identify the terminating participating MCVideo function.

3) shall set the P-Asserted-Identity header field to the public service identity of the controlling MCVideo function;

4) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcvideo-request-uri> element set to the MCVideo ID of the terminating user; and

b) the <mcvideo-calling-group-id> element set to the group identity;

NOTE 3: The <mcvideo-calling-user-id> is already included in the MIME body as a result of calling clause 6.3.3.1.2 in step 1).

5) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1;

6) if the in-progress emergency state of the group is set to a value of "true" the controlling MCVideo function:

a) shall include a Resource-Priority header field populated with the values for an MCVideo emergency group call as specified in clause 6.3.3.1.19;

b) if the received SIP INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "true":

i) shall include in the outgoing SIP INVITE request in the application/vnd.3gpp.mcvideo-info+xml MIME body an <emergency-ind> element set to a value of "true"; and

ii) if the <alert-ind> element is set to "true" in the received SIP INVITE request and the requesting MCVideo user and MCVideo group are authorised for the initiation of MCVideo emergency alerts as determined by the procedures of clause 6.3.3.1.13.1, shall populate the application/vnd.3gpp.mcvideo-info+xml MIME body and the application/vnd.3gpp.location-info+xml MIME body as specified in clause 6.3.3.1.12. Otherwise, shall set the <alert-ind> element to a value of "false"; and

c) if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <imminentperil-ind> element set to a value of "false";

7) if the in-progress emergency state of the group is set to a value of "false" and the in-progress imminent peril state of the group is set to a value of "true", the controlling MCVideo function:

a) shall include a Resource-Priority header field populated with the values for an MCVideo imminent peril group call as specified in clause 6.3.3.1.19; and

b) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true";

8) if:

a) an MCVideo GKTP document exists for the group identity contained in the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request; and

b) the MCVideo GKTP document contains a <MKFC-GKTPs> element;

then:

a) for each instance of <GKTP> element of the <MKFC-GKTPs> element of the MCVideo GKTP document:

i) shall perform the procedure in clause 6.3.3.6.2 to re-generate an I_MESSAGE; and

ii) if the procedure in clause 6.3.3.6.2 was successful, shall include the I_MESSAGE in a <GKTP> element of the <MKFC-GKTPs> element of an application/vnd.3gpp.mcvideo-info+xml MIME body included in the outgoing SIP INVITE request; and

9) shall send the SIP INVITE request towards the terminating network in accordance with 3GPP TS 24.229 [11].

Upon receiving a SIP 183 (Session Progress) response containing a Require header field with the option tag "100rel" and containing a P-Answer-State header field with the value "Unconfirmed" in response to the SIP INVITE request the controlling MCVideo function:

1) shall send a SIP PRACK request towards the MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCVideo function:

1) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

2) shall send a SIP NOTIFY request to all participants with a subscription to the conference event package as specified in clause 9.2.3.4; and

3) shall increment the local counter of the number of SIP 200 (OK) responses received from invited members, by 1.

NOTE 4: The notifications above could be sent prior to the SIP 200 (OK) response being sent to the inviting MCVideo client. These notifications received by MCVideo clients that are group members do not mean that the group session will be successfully established.

NOTE 5: The procedures executed by the controlling MCVideo function prior to sending a response to the inviting MCVideo client are specified in clause 9.2.1.4.2."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.1.2]

"The controlling MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the public service identity of the non-controlling MCVideo function serving the group identity of the MCVideo group owned by the partner MCVideo system;

3) shall set the P-Asserted-Identity to the public service identity of the controlling MCVideo function;

4) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcvideo-request-uri> element set to the group identity of the MCVideo group hosted by the non-controlling MCVideo function in the partner MCVideo system; and

b) the <mcvideo-calling-group-id> element set to the group identity of the group served by the controlling MCVideo function;

5) shall include the Recv-Info header field set to g.3gpp.mcvideo-transmission-request;

6) if:

a) an MCVideo GKTP document exists for the group identity contained in the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request; and

b) the MCVideo GKTP document contains a <MKFC-GKTPs> element;

then:

a) for each instance of <GKTP> element of the <MKFC-GKTPs> element of the MCVideo GKTP document:

i) shall perform the procedure in clause 6.3.3.6.2 to re-generate an I_MESSAGE; and

ii) if the procedure in clause 6.3.3.6.2 was successful, shall include the I_MESSAGE in a <GKTP> element of the <MKFC-GKTPs> element of an application/vnd.3gpp.mcvideo-info+xml MIME body included in the outgoing SIP INVITE request;

7) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1; and

8) shall send the SIP INVITE request towards the partner MCVideo system in accordance with 3GPP TS 24.229 [11].

Upon receiving SIP 403 (Forbidden) response for the SIP INVITE request, if according to local policy and if:

1) the response contains a Warning header field with the MCVideo warning code "128"; and

2) the response contains a P-Refused-URI-List header field and an application/resource-lists+xml MIME body as specified in IETF RFC 5318 [28];

NOTE 1: The application/resource-lists+xml MIME body contains MCVideo IDs identifying MCVideo users in a partner MCVideo system that needs to be invited to the prearranged group call in case of group regrouping using interrogating method.

Editor's Note:The above note currently isn't defined in the 23.280 and 23.281.

then the controlling MCVideo function:

1) shall check if the number of members of the MCVideo group exceeds the value contained in the <on-network-max-participant-count> element of the group document as specified in 3GPP TS 24.481 [24]. If exceeded, the controlling MCVideo function shall invite only <on-network-max-participant-count> members from the application/resource-lists+xml MIME body; and

NOTE 2: The <on-network-max-participant-count> element indicates the maximum number of participants allowed in the prearranged group session It is operator policy that determines which participants in the application/resource-lists+xml MIME body are invited to the group call.

2) shall invite MCVideo users as specified in this clause using the list of MCVideo IDs in URI-List.

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCVideo function:

1) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3; and

NOTE 3: The procedures executed by the controlling MCVideo function prior to sending a response to the inviting MCVideo client are specified in clause 9.2.1.4.2.

2) if at least one of the invited MCVideo clients has subscribed to the conference package, shall subscribe to the conference event package in the non-controlling MCVideo function as specified in clause 9.2.3.4.3."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.6]

"This clause is referenced from other procedures.

This clause describes the procedures for inviting an MCVideo user to an MCVideo session associated with an MCVideo emergency group call or MCVideo imminent peril group call. The procedure is initiated by the controlling MCVideo function as the result of an action in clause 9.2.2.4.1.1.

The controlling MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the address of the terminating participating MCVideo function associated with the MCVideo ID of the targeted MCVideo user;

3) shall include an application/vnd.3gpp.mcvideo-info+xml MIME body with the <mcvideoinfo> element containing the <mcvideo-Params> element populated as follows:

a) the <mcvideo-request-uri> element set to the value of the MCVideo ID of the targeted MCVideo user;

b) the <mcvideo-calling-user-id> element set to the value of the MCVideo ID of the calling MCVideo user; and

c) the <mcvideo-calling-group-id> element set to the value of the MCVideo group ID of the emergency group call.

4) shall include in the P-Asserted-Identity header field the public service identity of the controlling MCVideo function;

5) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1; and

6) if the in-progress emergency group state of the group is set to a value of "true" the controlling MCVideo function:

a) shall include a Resource-Priority header field populated with the values for an MCVideo emergency group call as specified in clause 6.3.3.1.18;

b) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <emergency-ind> element set to a value of "true";

c) if the <alert-ind> element is set to "true" in the received SIP INVITE request and the requesting MCVideo user and MCVideo group are authorised for the initiation of MCVideo emergency alerts as determined by the procedures of clause 6.3.3.1.12.1, shall populate the application/vnd.3gpp.mcvideo-info+xml MIME body and the application/vnd.3gpp.mcvideo-location-info+xml MIME body as specified in clause 6.3.3.1.11. Otherwise, shall set the <alert-ind> element to a value of "false" in the application/vnd.3gpp.mcvideo-info+xml MIME body; and

d) if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <imminentperil-ind> element set to a value of "false";

NOTE: If the imminent peril state of the group is true at this point, the controlling function will set it to false as part of the calling procedure.

7) if the in-progress emergency state of the group is set to a value of "false" and the in-progress imminent peril state of the group is set to a value of "true", the controlling MCVideo function:

a) shall include a Resource-Priority header field populated with the values for an MCVideo imminent peril group call as specified in clause 6.3.3.1.18; and

b) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true"; and

8) if:

a) an MCVideo GKTP document exists for the group identity contained in the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request; and

b) the MCVideo GKTP document contains an <MKFC-GKTPs> element;

then:

a) for each instance of <GKTP> element of the <MKFC-GKTPs> element of the MCVideo GKTP document:

i) shall perform the procedure in clause 6.3.3.6.2 to re-generate an I_MESSAGE; and

ii) if the procedure in clause 6.3.3.6.2 was successful, shall include the I_MESSAGE in a <GKTP> element of the <MKFC-GKTPs> element of an application/vnd.3gpp.mcvideo-info+xml MIME body included in the outgoing SIP INVITE request."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.12.2]

"If the controlling MCVideo function has received a SIP request for an MCVideo group call with the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body set to a value of "true" and:

1) if the <allow-emergency-group-call> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true" and:

a) if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> element contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "DedicatedGroup" and:

i) if the <uri-entry> element of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) contains the identity of the MCVideo group targeted by the calling MCVideo user; and

ii) if the <allow-MCVideo-emergency-call> element of the <list-service> element of the group document identified by the targeted MCVideo group identity is set to a value of "true" as specified in 3GPP TS 24.481 [24];

then the controlling MCVideo function shall consider the MCVideo emergency group call request to be an authorised request for an MCVideo emergency group call and skip the remaining steps; or;

b) if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> element contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "UseCurrentlySelectedGroup" and if the <allow-MCVideo-emergency-call> element of the <list-service> element of the group document identified by the targeted MCVideo group identity is set to a value of "true" as specified in 3GPP TS 24.481 [24];

then the controlling MCVideo function shall consider the MCVideo emergency group call request to be an authorised request for an MCVideo emergency group call and skip the remaining steps; or

2) if the controlling MCVideo function does not consider the MCVideo emergency group call request to be an authorised request for an MCVideo emergency group call by step 1) above, then the controlling MCVideo function shall consider the MCVideo emergency group call request to be an unauthorised request for an MCVideo emergency group call.

If the controlling MCVideo function has received a SIP request for an MCVideo private call with the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body set to a value of "true" and:

1) if the <allow-emergency-private-call> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true"; and

a) if the "entry-info" attribute of the <entry> element of the <MCVideoPrivateRecipient> element of the <EmergencyCall> element contained within the <PrivateCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "UsePreConfigured" and if the MCVideo ID targeted for the call is contained in the <uri-entry> element of the <entry> element of the <MCVideoPrivateRecipient> element of the <EmergencyCall> element contained within the <PrivateCall> element (see the MCVideo user profile document in 3GPP TS 24.484 [25]); or

b) if the "entry-info" attribute of the <entry> element of the <MCVideoPrivateRecipient> element of the <EmergencyCall> element contained within the <PrivateCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "LocallyDetermined";

then the controlling MCVideo function shall consider the MCVideo emergency private call request to be an authorised request for an MCVideo emergency private call and skip step 2) below; or

2) if the controlling MCVideo function does not consider the MCVideo emergency private call request to be an authorised request for an MCVideo emergency private call by step 1) above, then the controlling MCVideo function shall consider the MCVideo emergency private call request to be an unauthorised request for an MCVideo emergency private call."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.12.4]

"If the controlling MCVideo function has received a SIP request with the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body set to a value of "false" and:

1) if the <allow-cancel-emergency-alert> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true", then the MCVideo emergency alert cancellation request shall be considered to be an authorised request for an MCVideo emergency alert cancellation; and

2) if the <allow-cancel-emergency-alert> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "false", then the MCVideo emergency alert cancellation request shall be considered to be an unauthorised request for an MCVideo emergency alert cancellation."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [11], the controlling MCVideo function:

1) for the accepted media stream in the received SDP offer:

a) shall replace the IP address and port number in the received SDP offer with the IP address and port number of the controlling MCVideo function; and

2) for the accepted media-transmission control entity, if present in the received SDP offer:

a) shall replace the IP address and port number in the received SDP offer with the IP address and port number of the controlling MCVideo function, for the accepted media-transmission control entity, if present in the received SDP offer; and

b) shall include 'fmtp' attributes as specified in 3GPP TS 24.581 clause 14."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.2.2]

"On receipt of an initial SIP INVITE request the controlling MCVideo function shall cache SIP feature tags, if received in the Contact header field and if the specific feature tags are supported."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.2.1]

"The transmission control server arbitration logic in the transmission control server shall support the procedures in clauses 6.3.2 and 6.3.3 and shall behave according to the transmission control server state transition diagram for 'general transmission control operation' in clause 6.3.4.

The transmission control interface towards the MCVideo client in the transmission control server shall behave according to the transmission control server state transition diagram for 'basic transmission control operation towards the transmission participant' as specified in clause 6.3.5.

The reception control arbitration logic in the transmission control server shall support the procedures in clauses 6.3.2 and 6.3.3 and shall behave according to the reception control server state transition diagram for 'general reception control operation' in clause 6.3.6.

The transmission control interface towards the MCVideo client in the transmission control server shall behave according to the transmission control server state transition diagram for 'basic reception control operation towards the transmission participant' as specified in clause 6.3.7."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.2.2]

"When an MCVideo call is established a new instance of the transmission control server state machine for 'general transmission control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' is added.

When an MCVideo call is established a new instance of the transmission control server state machine for 'general reception control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic reception control operation towards the transmission participant' is added.

If the optional "mc_queueing" feature is supported and has been negotiated as specified in clause 14, the transmission control server could queue the implicit transmission control request for the media-transmission control entity.

The original initial SIP INVITE request or SIP REFER request to establish an MCVideo chat group call or to rejoin an ongoing MCVideo call is not handled as an implicit transmission control request message by the transmission control server unless explicitly stated in the SIP INVITE request or in the SIP REFER request.

The permission to send media to the inviting MCVideo client due to implicit transmission control request is applicable to both confirmed indication and unconfirmed indication.

When the first unconfirmed indication is received from the invited participating MCVideo function (see 3GPP TS 24.281 [2]) the transmission control server optionally can give an early indication to send RTP media packets, to the inviting MCVideo client.

If an early indication to send RTP media packets is given to the inviting MCVideo client, the transmission participant is granted the permission to send media and the MCVideo server buffers RTP media packets received from the MCVideo client at least until the first invited MCVideo client accepts the invitation or until the RTP media packet buffer exceeds it maximum limit to store RTP media packets.

If the MCVideo server does not support or does not allow media buffering then when an early indication to send RTP media packets is not given to the inviting MCVideo client, the transmission participant is granted the permission to send media when the first invited MCVideo client accepts the media.

Before the transmission control server sends the first transmission control message in the MCVideo call, the transmission control server has to assign itself a SSRC identifier to be included in media transmission control messages and quality feedback messages if the MCVideo server is supporting that option. A suitable algorithm to generate the SSRC identifier is described in IETF RFC 3550 [3].

The transmission participant and the transmission control server can negotiate the maximum priority level that the transmission participant is permitted to request. The transmission control server can pre-empt the current sender based on the negotiated maximum priority level that the transmission participant is permitted to request and the priority level included in the Transmission Media Request message.

NOTE: The maximum priority level that a transmission participant can use is negotiated as specified in clause 14.3.3 and is based on group configuration data retrieved by the controlling MCVideo function from the group management server as described in 3GPP TS 24.481 [12] and service configuration data retrieved by the controlling MCVideo function from the configuration management server as described in 3GPP TS 24.484 [13].

The transmission participant and the transmission control server can negotiate queueing of Transmission requests using the "mc_queueing" fmtp attribute as described in clause 14. If queueing is supported and negotiated, the transmission control server queues the transmission control request if a Transmission Media Request message is received when another transmission participant has the transmission and the priority of the current speaker is the same or higher."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.1]

"The transmission control messages are coded as described in clause 9.1.2 where the transmission control message is part of the application-dependent data.

For the transmission control protocol the ASCII name string is: MCVx (Mission Critical Video x) with x=0, 1 or 2 as specified in clause 9.1.2.

A list of transmission control messages can be found in clause 9.2.2.1.

The transmission control specific fields are specified in clause 9.2.3."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

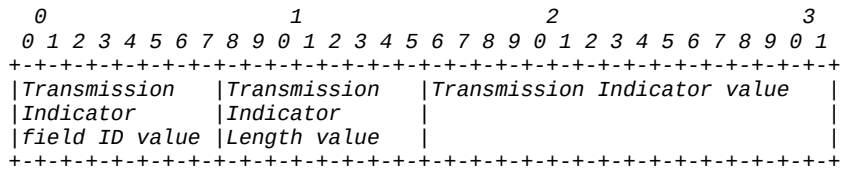
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

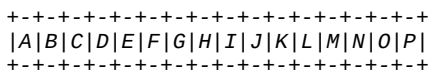


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. *if the active Transmission request queue is empty the transmission control server:*
    a. *shall send Transmission Idle message to all transmission participants. The Transmission Idle message:*
        i. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*
        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;*
    b. *shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;*
    c. *shall start timer T1 (Inactivity);*
    d. *shall set the general state to the 'G: Transmit Idle' state; and*
    e. *shall initialise counter Cx (Simultaneous transmission video) to 0.*
2. *if the active Transmission request queue is not empty the transmission control server:*
    a. *shall select a queued Transmission request from the top of the active Transmission request queue;*
    b. *shall remove that queued Transmission request from the active Transmission request queue; and*
    c. *shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:*
    a. *shall include the granted priority in the Transmission priority field;*
    b. *shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;*
    c. *if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*
    d. *shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;*
2. *shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;*
3. *shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:*
    a. *shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;*
    b. *shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;*
    c. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;*

d. if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';

e. if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

g. shall initiate the 'general reception control operation' state machine;and

4. shall enter the 'G: Transmit Taken' state."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. if counter Cx (Simultaneous transmission video) has not reached its upper limit:

a. shall perform the actions specified in the clause 6.3.4.4.2;

2. if counter Cx (Simultaneous transmission video) has reached its upper limit:

a. select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;

b. shall stop timer T4 (Transmission Grant), if running;

c. shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);

d. shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;

e. shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and

f. shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:

i. shall include the queue position and transmission priority in the Queue Info field; and

ii. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:

a. shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and

b. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

2. shall set the general state to 'G: pending Transmission Revoke'."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

b. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.1.8.3 Test description

5.1.1.8.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the participating MCVideo functions.
- SS-UE1 (MCVideo Client) and SS-UE2 (MCVideo Client), simulated by the SS (MCVideo Server).

IUT:

- IUT (MCVideo Server), providing the controlling MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of all sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Transmission Control Server.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server).
- The SS-UE1 (MCVideo client) and the SS-UE2 (MCVideo client) are registered and authorized to use the MCVideo system.
- The SS-UE1 (MCVideo client) and the SS-UE2 (MCVideo client) are affiliated to the pre-arranged group and are authorized to initiate group calls.
- The IUT (MCVideo Server) is the controlling server of the pre-arranged group.

5.1.1.8.3.2 Test procedure sequence

Table 5.1.1.8.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E group call establishment' as described in Annex A, Table A.4.1.4.3-1 to establish an imminent peril pre-arranged group call with implicit transmission control request according to option b.ii of note in Annex A, Table A.4.1.4.3-1, initiated by SS-UE1 (MCVideo Client)?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure ' MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	3	P
4	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.1.5.3-1, to release the call?	-	-	4	P

5.1.1.8.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

**Table 5.1.1.8.3.3-1: SIP INVITE from SS (MCVideo Server)
(Step 1, Table 5.1.1.8.3.2-1; Step 1, Annex A, Table A.4.1.4.3-1)**

Derivation Path: Annex A, Table A.4.1.4.4-1, with condition IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.8.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.1.8.3.2-1; Step 2, Annex A, Table A.4.1.4.3-1)**

Derivation Path: Annex A, Table A.4.1.4.4-2, with condition IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.8.3.3-3: SIP 200 (OK) from SS (MCVideo Server)
(Step 1, Table 5.1.1.8.3.2-1; Step 8, Annex A, Table A.4.1.4.3-1)**

Derivation Path: Annex A, Table A.4.1.4.4-6

**Table 5.1.1.8.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.1.1.8.3.2-1; Step 9, Annex A, Table A.4.1.4.3-1)**

Derivation Path: Annex A, Table A.4.1.4.4-7

**Table 5.1.1.8.3.3-5: Transmission Idle from IUT (MCVideo Server)
(Step 2, Table 5.1.1.8.3.2-1; Steps 4 and 5, Annex A, Table A.3.8.3-1)**

Derivation Path: Annex A, Table A.3.8.4-6, with condition IMMPERIL-CALL.

**Table 5.1.1.8.3.3-6: Transmission Request from SS (MCVideo Server)
(Step 3, Table 5.1.1.8.3.2-1; Step 1, Annex A, Table A.3.7.3-1)**

Derivation Path: Annex A, Table A.3.7.4-1, with condition IMMPERIL-CALL.

**Table 5.1.1.8.3.3-7: Transmission Granted from IUT (MCVideo Server)
(Steps 1 and 3, Table 5.1.1.8.3.2-1; Step 3a1 or 3b1, Annex A, Table A.3.7.3-1;
Step 9a1, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.3.7.4-2 or A.3.7.4-3 (note), with condition IMMPERIL-CALL.
NOTE: Depends on server implementation.

**Table 5.1.1.8.3.3-8: Media Transmission Notification from IUT (MCVideo Server)
(Steps 1 and 3, Table 5.1.1.8.3.2-1; Step 3a3 or 3b2, Annex A, Table A.3.7.3-1;
Step 10a1, Annex A, Table 4.1.1.3-1)**

Derivation Path: Annex A, Table A.3.7.4-4 or A.3.7.4-5 (note), with condition IMMPERIL-CALL.
NOTE: Depends on server implementation.

5.1.1.9 MCVideo Server - MCVideo Server / Pre-arranged Group Call / Imminent Peril Group Call / Terminating Participating server

5.1.1.9.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the controlling function initiates an imminent peril pre-
arranged group call with automatic commencement }
    then { IUT (MCVideo Server) accepts the call and establishes the call with SS-UE2 (MCVideo
Client) and forwards the media transmission notification received from SS (MCVideo Server) to SS-UE2
(MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established an imminent peril pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) ends the media transmission }
    then { IUT (MCVideo Server) forwards the transmission end notify received from SS (MCVideo
Server) to SS-UE2 (MCVideo Client) and forwards the transmission idle received from SS (MCVideo
Server) to SS-UE2 (MCVideo Client) }
           }
```

(3)

```
with { IUT (MCVideo Server) having established an imminent peril pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) providing the controlling function ends the pre-arranged group call
}
    then { IUT (MCVideo Server) sends a SIP BYE message to SS-UE2 (MCVideo Client) and responds by
sending a SIP 200 (OK) message to SS (MCVideo Server) }
           }
```

5.1.1.9.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.1.3.2, 9.2.1.3.4, 6.3.2.1.6.0, 6.3.2.1.6.1, 6.3.3.1.6, 6.3.2.2.5.1, 6.3.2.2.6.1, 6.3.2.2.8.1, 6.3.2.2.4.2, 6.3.2.2.2.1 and 6.3.2.1.2.1, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.4.1, 6.4.2, 6.4.3, 6.4.4, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.2]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

NOTE 1: This clause covers on-demand session.

Upon receipt of a "SIP INVITE request for terminating participating MCVideo function", the participating MCVideo function:

…

2) shall check the presence of the isfocus media feature tag in the URI of the Contact header field and if it is not present then the participating MCVideo function shall reject the request …, and shall not continue with the rest of the steps;

3) if the Answer-Mode Indication in the application/poc-settings+xml MIME body has not yet been received from the invited MCVideo client as defined in clause 7.3.3 or clause 7.3.4, shall reject the request …, and shall not continue with the rest of the steps;

4) shall use the MCVideo ID present in the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCVideo ID and public user identity;

5) if the binding between the MCVideo ID and public user identity does not exist, then the participating MCVideo function shall reject the SIP INVITE request with a SIP 404 (Not Found) response. Otherwise, continue with the rest of the steps;

6) if the SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with an <MKFC-GKTPs> element, shall perform the procedures in clause 6.3.2.3.2;

7) shall perform the automatic commencement procedures specified in clause 6.3.2.2.5.1 and according to IETF RFC 5373 [27] if the "SIP INVITE request for terminating participating MCVideo function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCVideo client as per clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; and

8) shall perform the manual commencement procedures specified in clause 6.3.2.2.6.1 and according to IETF RFC 5373 [27] if the "SIP INVITE request for terminating participating MCVideo function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCVideo client as per clause 7.3.3 or clause 7.3.4 is set to "manual-answer"."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.4]

"Upon receiving a SIP BYE request from the controlling MCVideo function, the participating MCVideo function shall follow the procedures as specified in clause 6.3.2.2.8.1."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.6.0]

"This clause contains common procedures to be used for MCVideo emergency group calls and MCVideo imminent peril group calls."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.6.1]

"When the participating MCVideo function receives a request from the MCVideo client to originate an MCVideo emergency group call and needs to determine if the request is an authorised request for an MCVideo emergency call, the participating MCVideo function shall check the following:

1) if the <allow-emergency-group-call> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true" and:

a) if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> element contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "DedicatedGroup" and if the <uri-entry> element of the <entry> element of the <MCVideoGroupInitiation> element contains the identity of the MCVideo group targeted by the calling MCVideo user; or

b) if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "UseCurrentlySelectedGroup";

then the participating MCVideo function shall consider the MCVideo emergency group call request to be an authorised request for an MCVideo emergency group call;

In all other cases, the participating MCVideo function shall consider the request to originate an MCVideo emergency group call to be an unauthorised request to originate an MCVideo emergency group call.

When the participating MCVideo function receives a request from the MCVideo client to originate an MCVideo imminent peril group call and needs to determine if the request is an authorised request for an MCVideo imminent peril group call the participating MCVideo function shall check the following:

1) if the <allow-imminent-peril-call> element of <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true"; and

a) if the "entry-info" attribute of the <MCVideoGroupInitiation> element contained within the <ImminentPerilCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "DedicatedGroup" and if the <uri-entry> element of the <entry> element of the <MCVideoGroupInitiation> element contains the identity of the MCVideo group targeted by the calling MCVideo user; or

b) if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element contained within the <ImminentPerilCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "UseCurrentlySelectedGroup";

then the participating MCVideo function shall consider the MCVideo imminent peril group call request to be an authorised request for an MCVideo emergency group call;

In all other cases, the participating MCVideo function shall consider the request to originate an MCVideo imminent peril group call to be an unauthorised request to originate an MCVideo imminent peril call."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.6]

"This clause is referenced from other procedures.

This clause describes the procedures for inviting an MCVideo user to an MCVideo session associated with an MCVideo emergency group call or MCVideo imminent peril group call. The procedure is initiated by the controlling MCVideo function as the result of an action in clause 9.2.2.4.1.1.

The controlling MCVideo function:

1) *shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;*

2) *shall set the Request-URI to the address of the terminating participating MCVideo function associated with the MCVideo ID of the targeted MCVideo user;*

3) *shall include an application/vnd.3gpp.mcvideo-info+xml MIME body with the <mcvideoinfo> element containing the <mcvideo-Params> element populated as follows:*

    a) *the <mcvideo-request-uri> element set to the value of the MCVideo ID of the targeted MCVideo user;*

    b) *the <mcvideo-calling-user-id> element set to the value of the MCVideo ID of the calling MCVideo user; and*

    c) *the <mcvideo-calling-group-id> element set to the value of the MCVideo group ID of the emergency group call.*

4) *shall include in the P-Asserted-Identity header field the public service identity of the controlling MCVideo function;*

5) *shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1; and*

6) *if the in-progress emergency group state of the group is set to a value of "true" the controlling MCVideo function:*

    a) *shall include a Resource-Priority header field populated with the values for an MCVideo emergency group call as specified in clause 6.3.3.1.18;*

    b) *shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <emergency-ind> element set to a value of "true";*

    c) *if the <alert-ind> element is set to "true" in the received SIP INVITE request and the requesting MCVideo user and MCVideo group are authorised for the initiation of MCVideo emergency alerts as determined by the procedures of clause 6.3.3.1.12.1, shall populate the application/vnd.3gpp.mcvideo-info+xml MIME body and the application/vnd.3gpp.mcvideo-location-info+xml MIME body as specified in clause 6.3.3.1.11. Otherwise, shall set the <alert-ind> element to a value of "false" in the application/vnd.3gpp.mcvideo-info+xml MIME body; and*

    d) *if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <imminentperil-ind> element set to a value of "false";*

NOTE: If the imminent peril state of the group is true at this point, the controlling function will set it to false as part of the calling procedure.

7) *if the in-progress emergency state of the group is set to a value of "false" and the in-progress imminent peril state of the group is set to a value of "true", the controlling MCVideo function:*

    a) *shall include a Resource-Priority header field populated with the values for an MCVideo imminent peril group call as specified in clause 6.3.3.1.18; and*

    b) *shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true"; and*

8) if:

a) an MCVideo GKTP document exists for the group identity contained in the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request; and

b) the MCVideo GKTP document contains an <MKFC-GKTPs> element;

then:

a) for each instance of <GKTP> element of the <MKFC-GKTPs> element of the MCVideo GKTP document:

i) shall perform the procedure in clause 6.3.3.6.2 to re-generate an I_MESSAGE; and

ii) if the procedure in clause 6.3.3.6.2 was successful, shall include the I_MESSAGE in a <GKTP> element of the <MKFC-GKTPs> element of an application/vnd.3gpp.mcvideo-info+xml MIME body included in the outgoing SIP INVITE request."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.5.1]

"When receiving a "SIP INVITE request for terminating participating MCVideo function" that requires automatic commencement mode:

1) if:

a) the invited MCVideo client has one or more pre-established sessions without an associated MCVideo session;

b) the media-level sections for the offered MCVideo video media stream are the same as the media-level sections for MCVideo video media stream in an existing pre-established session; and

c) the media-level section of the offered media-transmission control entity is the same as the media-level section for media-transmission control entity in an existing pre-established session;

then the participating MCVideo function shall perform the actions specified in clause 6.3.2.2.5.3; or

2) otherwise the participating MCVideo function shall perform the actions specified in clause 6.3.2.2.5.2."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.6.1]

"When receiving a "SIP INVITE request for terminating participating MCVideo function" that requires manual commencement mode:

1) if:

a) the invited MCVideo client has one or more pre-established sessions without an associated MCVideo session;

b) the media-level sections for the offered MCVideo video media stream are the same as the media-level section for MCVideo video media stream in the existing pre-established session; and

c) the media-level section of the offered media-transmission control entity is the same as the media-level section for media-transmission control entity in the existing pre-established session;

then the participating MCVideo function shall perform the actions specified in clause 6.3.2.2.6.3; or

2) otherwise the participating MCVideo function shall perform the actions specified in clause 6.3.2.2.6.2."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.8.1]

"Upon receiving a SIP BYE request from the controlling MCVideo function, the participating MCVideo function:

1) shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.581 [5] for releasing media plane resource associated with the SIP session with the controlling MCVideo function;

2) shall generate a SIP BYE request according to 3GPP TS 24.229 [11];

3) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;

4) if the received SIP BYE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body into the outgoing SIP BYE request; and

5) shall send the SIP BYE request to the MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response to the SIP BYE request the participating MCVideo function:

1) shall send a SIP 200 (OK) response to the SIP BYE request received from the controlling MCVideo function according to 3GPP TS 24.229 [11]; and

2) shall interact with the media plane as specified in 3GPP TS 24.581 [5] for releasing media plane resources associated with the SIP session with the MCVideo client."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.4.2]

"This clause is referenced from other procedures.

When sending SIP 200 (OK) responses, the participating MCVideo function shall generate a SIP 200 (OK) response according to 3GPP TS 24.229 [11] and:

1) shall include the option tag "timer" in a Require header field;

2) shall include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [23], "UAS Behavior". If no "refresher" parameter was included in the SIP INVITE request, the "refresher" parameter in the Session-Expires header field shall be set to "uas";

3) shall include the following in the Contact header field:

a) the g.3gpp.mcvideo media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo"; and

c) an MCVideo session identity mapped to the MCVideo session identity provided in the Contact header field of the received SIP INVITE request from the controlling MCVideo function;

4) shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [32]; and

5) if the incoming SIP response contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body to the outgoing SIP 200 (OK) response."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.2.1]

"The participating MCVideo function shall follow the procedures in clause 6.3.2.1.2.1."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [11], the participating MCVideo function:

1) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCVideo function, for the accepted media stream in the received SDP offer, if required; and

NOTE 1: Requirements can exist for the participating MCVideo function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCVideo function, for the accepted media-transmission control entity, if present in the received SDP offer.

NOTE 2: If the participating MCVideo function and the controlling MCVideo function are in the same MCVideo server, and the participating MCVideo function does not have a dedicated IP address or a dedicated port number for media transmission control or media stream, the replacement of the IP address or the port number is omitted."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.1]

"Once an on-demand MCVideo session is established or a pre-established session is in use when the participating MCVideo function receives transmission control messages from the transmission participant in the MCVideo client or from the transmission control server in the controlling MCVideo function, the behaviour of the participating MCVideo function is described in the following clauses."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.2]

"Upon receiving a transmission control message the participating MCVideo function:

1. *shall immediately forward the transmission control message to the transmission control server if the message is received from the transmission participant;*

2. *if an MBMS subchannel is not used for a transmission in the session the transmission control message is associated with, shall immediately forward the transmission control message to the transmission participant if the message is received from the transmission control server; and*

3. *if an MBMS subchannel is used for a transmission in the session the transmission control message is associated with:*

    a. *if*

        i. *the transmission control message is not a Transmission Idle message or a Media Transmission Notify message;*

        ii. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

        iii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report;*

        shall immediately forward the transmission control message to the transmission participant; and

    b. *if*

        i. *the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report; and*

        ii *if the transmission control message is the Transmission Idle message or the Media Transmission Notify message,*

    shall perform actions as specified in clause 10.2.

NOTE: When the Transmit Idle or Media Transmission Notify messages are discarded the messages are sent to the MCVideo clients over the MBMS subchannel allocated for the transmission as specified in clause 10.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.3]

"Upon receiving RTP media packets the participating MCVideo function:

1. *shall immediately forward the RTP media packet to the controlling MCVideo function if the RTP packet is from an MCVideo client; and*

2. *if an MBMS subchannel is not used for a transmission in the session the RTP media packets are associated with, shall immediately forward the RTP media packets to the MCVideo client if the RTP packet is from the controlling MCVideo function or the non-controlling MCVideo function.*

3. *if an MBMS subchannel is used for a transmission in the session the RTP media packets are associated with and if RTP media packets are received from the controlling MCVideo function or the non-controlling MCVideo function:*

   a. *if*

      i. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

      ii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report,*

      shall immediately forward the RTP media packets to the MCVideo client; and

   b. *if the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report, shall perform actions as specified in clause 10.2."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.4]

"When the participating function receives an indication from the application and signalling plane that session release is initiated, the participating MCVideo function:

1. *shall stop sending transmission control messages towards the transmission participant and the transmission control server; and*

2. *shall stop sending RTP media packets towards the MCVideo client and towards the controlling MCVideo function.*

When the participating MCVideo function receives an indication from the application and signalling plane that the session is released, the participating MCVideo function:

1. *in case of a pre-established session, shall perform the actions in clause 9.3.2; and*

2. *in case of an on-demand session, shall release the media resources associated with the session."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

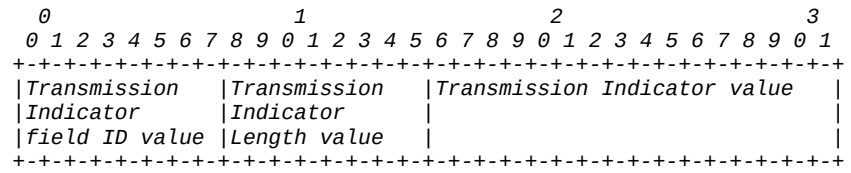
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

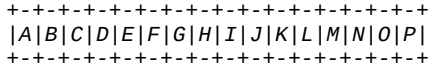


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. if the active Transmission request queue is empty the transmission control server:

a. shall send Transmission Idle message to all transmission participants. The Transmission Idle message:

i. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

ii. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

b. shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;

c. shall start timer T1 (Inactivity);

d. shall set the general state to the 'G: Transmit Idle' state; and

e. shall initialise counter Cx (Simultaneous transmission video) to 0.

2. if the active Transmission request queue is not empty the transmission control server:

a. shall select a queued Transmission request from the top of the active Transmission request queue;

b. shall remove that queued Transmission request from the active Transmission request queue; and

c. shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:

a. shall include the granted priority in the Transmission priority field;

b. shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;

c. if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

d. shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;

2. shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;

3. shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:

a. shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;

b. shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;

c. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;

d. if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';

e. if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

g. shall initiate the 'general reception control operation' state machine;and

4. shall enter the 'G: Transmit Taken' state."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. if counter Cx (Simultaneous transmission video) has not reached its upper limit:

a. shall perform the actions specified in the clause 6.3.4.4.2;

2. *if counter Cx (Simultaneous transmission video) has reached its upper limit:*

    a. *select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;*

    b. *shall stop timer T4 (Transmission Grant), if running;*

    c. *shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);*

    d. *shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;*

    e. *shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and*

    f. *shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:*

        i. *shall include the queue position and transmission priority in the Queue Info field; and*

        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:*

    a. *shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and*

    b. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

2. *shall set the general state to 'G: pending Transmission Revoke'."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. *if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:*

    a. *shall initialize a general state machine as specified in clause 6.3.4.2.2; and*

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

    b. *shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;*

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.1.9.3 Test description

5.1.1.9.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the controlling MCVideo function.
- SS-UE2 (MCVideo Client).

IUT:

- IUT (MCVideo Server), providing the terminating participating MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of some sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server and the HTTP Client.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server) and SS-UE2 (MCVideo Client).
- The SS-UE2 (MCVideo client) is registered and authorized to use the MCVideo system.

- The SS-UE2 (MCVideo client) is affiliated to the pre-arranged group and is authorized to initiate group calls.

5.1.1.9.3.2 Test procedure sequence

Table 5.1.1.9.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CT group/private call establishment, automatic commencement' as described in Annex A, Table A.4.3.3.3-1 to establish an imminent peril pre-arranged group call with automatic commencement mode and notify a media transmission from SS-UE1 (MCVideo Client)?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Notification - Idle' as described in Annex A, Table A.4.3.8.3-1 to notify the end of SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.3.5.3-1, to release the call?	-	-	3	P

5.1.1.9.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

**Table 5.1.1.9.3.3-1: SIP INVITE from SS (MCVideo Server)
(Step 1, Table 5.1.1.9.3.2-1; Step 1, Annex A, Table A.4.3.3.3-1)**

Derivation Path: Annex A, Table A.4.3.3.4-1, with conditions GROUP-CALL and IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.9.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.1.9.3.2-1; Step 2, Annex A, Table A.4.3.3.3-1)**

Derivation Path: Annex A, Table A.4.3.3.4-2, with conditions GROUP-CALL and IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.9.3.3-3: SIP 200 (OK) from SS-UE2 (MCVideo Client)
(Step 1, Table 5.1.1.9.3.2-1; Step 6, Annex A, Table A.4.3.3.3-1)**

Derivation Path: Annex A, Table A.4.3.3.4-7, with condition GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.9.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.1.1.9.3.2-1; Step 7, Annex A, Table A.4.3.3.3-1)**

Derivation Path: Annex A, Table A.4.3.3.4-8, with condition GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.1.9.3.3-5: Transmission Idle
(Step 2, Table 5.1.1.9.3.2-1; Steps 13 and 14, Annex A, Table A.4.3.8.3-1)

Derivation Path: Annex A, Table A.4.3.8.4-6, with condition IMMPERIL-CALL.

Table 5.1.1.9.3.3-6: Media Transmission Notification
(Step 1, Table 5.1.1.9.3.2-1; Steps 10a1 and 10a2, Annex A, Table 4.3.3.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-11, with condition IMMPERIL-CALL.

5.1.1.10 MCVideo Server - MCVideo Server / Pre-arranged Group Call / Imminent Peril Group Call / Originating Participating server

5.1.1.10.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS-UE1 (MCVideo Client) initiates an imminent peril pre-arranged group call with
implicit transmission control }
    then { IUT (MCVideo Server) accepts the call from initiator and establishes the call with the SS
(MCVideo Server) providing the controlling function and forwards the transmission granted received
from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established an imminent peril pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests to end the transmission control }
    then { IUT (MCVideo Server) forwards the transmission end request to SS (MCVideo Server) and
forwards the transmission end response to SS-UE1 (MCVideo Client) and forwards the transmission idle
received from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established an imminent peril pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control }
    then { IUT (MCVideo Server) forwards the transmission request to SS (MCVideo Server) and
forwards the transmission granted received from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(4)

```
with { IUT (MCVideo Server) having established an imminent peril pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) ends the pre-arranged group call }
    then { IUT (MCVideo Server) sends a SIP BYE message to SS (MCVideo Server) providing the
controlling function and responds by sending a SIP 200 (OK) message to SS-UE1 (MCVideo Client) }
            }
```

5.1.1.10.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.1.3.1.1, 9.2.1.3.3, 6.2.8.1.1, 6.3.1.1, 6.3.2.1.3, 6.3.2.1.1.1 and 6.3.2.1.6.1, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.4.1, 6.4.2, 6.4.3, 6.4.4, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.1.1]

"In the procedures in this clause:

1) group identity in an incoming SIP INVITE request refers to the group identity from the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;

2) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

3) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

Upon receipt of a "SIP INVITE request for originating participating MCVideo function" containing an application/vnd.3gpp.mcvideo-info+xml MIME body with the <session-type> element set to a value of "prearranged", the participating MCVideo function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP INVITE request with a SIP 500 (Server Internal Error) response. The participating MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15]. Otherwise, continue with the rest of the steps;

NOTE 1: if the SIP INVITE request contains an emergency indication or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCVideo function can according to local policy choose to accept the request.

2) shall determine the MCVideo ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request, and shall authorise the calling user;

NOTE 2: The MCVideo ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

3) if through local policy in the participating MCVideo function, the user identified by the MCVideo ID is not authorised to initiate prearranged group calls, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response to the SIP INVITE request, with warning text set to "109 user not authorised to make prearranged group calls" in a Warning header field as specified in clause 4.4;

4) shall validate the media parameters and if the MCVideo codecs are not offered in the SIP INVITE request shall reject the request with a SIP 488 (Not Acceptable Here) response. Otherwise, continue with the rest of the steps;

5) shall check if the number of maximum simultaneous MCVideo group calls supported for the MCVideo user as specified in the <MaxSimultaneousCallsN6> element of the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) has been exceeded. If exceeded, the participating MCVideo function shall respond with a SIP 486 (Busy Here) response with the warning text set to "103 maximum simultaneous MCVideo group calls reached" in a Warning header field as specified in clause 4.4. Otherwise, continue with the rest of the steps;

NOTE 3: If the SIP INVITE request contains an emergency indication or an imminent peril indication, the participating MCVideo function can by means beyond the scope of this specification choose to allow for an exception to the limit for the maximum simultaneous MCVideo sessions supported for the MCVideo user. Alternatively, a lower priority session of the MCVideo user could be terminated to allow for the new session.

6) if the user identified by the MCVideo ID is not affiliated to the group identified in the "SIP INVITE request for originating participating MCVideo function" as determined by clause 8.2.2.2.11 and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, shall perform the actions specified in clause 8.2.2.2.12 for implicit affiliation;

7) if the actions for implicit affiliation specified in step 6) above were performed but not successful in affiliating the MCVideo user due to the MCVideo user already having N2 simultaneous affiliations, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 486 (Busy Here) response with the warning text set to "102 too many simultaneous affiliations" in a Warning header field as specified in clause 4.4. and skip the rest of the steps.

NOTE 4: N2 is the total number of MCVideo groups that an MCVideo user can be affiliated to simultaneously as specified in 3GPP TS 23.281 [26].

NOTE 5: if the SIP INVITE request contains an emergency indication set to a value of "true" or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCVideo function can according to local policy choose to allow an exception to the N2 limit. Alternatively, a lower priority affiliation of the MCVideo user could be cancelled to allow for the new affiliation.

8) shall determine the public service identity of the controlling MCVideo function associated with the group identity in the SIP INVITE request;

NOTE 6: The public service identity can identify the controlling MCVideo function in the primary MCVideo system or a partner MCVideo system.

NOTE 7: How the participating MCVideo server discovers the public service identity of the controlling MCVideo function associated with the group identity is out of scope of the current release.

9) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3;

10) shall set the Request-URI to the public service identity of the controlling MCVideo function associated with the group identity which was present in the incoming SIP INVITE request;

11) shall not copy the following header fields from the incoming SIP INVITE request to the outgoing SIP INVITE request, if they were present in the incoming SIP INVITE request:

a) Answer-Mode header field as specified in IETF RFC 5373 [27]; and

b) Priv-Answer-Mode header field as specified in IETF RFC 5373 [27];

12) shall set the <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the SIP INVITE request to the MCVideo ID of the calling user;

13) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the MCVideo client as specified in clause 6.3.2.1.1.1;

14) if the received SIP INVITE request contains an application/vnd.3gpp.location-info+xml MIME body and if not already copied, shall copy the contents of the application/vnd.3gpp.location-info+xml MIME body received in the SIP INVITE request into an application/vnd.3gpp.location-info+xml MIME body included in the outgoing SIP request;

15) if a Resource-Priority header field was included in the received SIP INVITE request, shall include a Resource-Priority header field according to rules and procedures of 3GPP TS 24.229 [11] set to the value indicated in the Resource-Priority header field of the SIP INVITE request from the MCVideo client; and

NOTE 8: The participating MCVideo function will leave verification of the Resource-Priority header field to the controlling MCVideo function.

16) shall forward the SIP INVITE request, according to 3GPP TS 24.229 [11].

Upon receipt of a SIP 302 (Moved Temporarily) response to the above SIP INVITE request in step 15), the participating MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.2.1.10;

2) shall include an SDP offer based upon the SDP offer in the received SIP INVITE request from the MCVideo client as specified in clause 6.3.2.1.1.1; and

3) shall forward the SIP INVITE request according to 3GPP TS 24.229 [11].

Upon receipt of a SIP 2xx response in response to the above SIP INVITE request in step 15), the participating MCVideo function:

1) if the received SIP 2xx response contains an application/vnd.3gpp.mcvideo-info+xml MIME body with an <MKFC-GKTPs> element, shall perform the procedures in clause 6.3.2.3.2;

2) shall generate a SIP 200 (OK) response as in clause 6.3.2.1.5.2;

3) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

4) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response;

5) shall include the public service identity received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response;

6) shall include an MCVideo session identity mapped to the MCVideo session identity provided in the Contact header field of the received SIP 200 (OK) response;

7) if the procedures of clause 8.2.2.2.12 for implicit affiliation were performed in the present clause, shall complete the implicit affiliation by performing the procedures of clause 8.2.2.2.13;

8) shall send the SIP 200 (OK) response to the MCVideo client according to 3GPP TS 24.229 [11];

9) shall interact with Media Plane as specified in 3GPP TS 24.581 [5]; and

10) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [23].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the above SIP INVITE request in step 14) the participating MCVideo function:

1) shall generate a SIP response according to 3GPP TS 24.229 [11];

2) shall include Warning header field(s) that were received in the incoming SIP response;

3) shall forward the SIP response to the MCVideo client according to 3GPP TS 24.229 [11]; and

4) if the implicit affiliation procedures of clause 8.2.2.2.12 were invoked in this procedure, shall perform the procedures of clause 8.2.2.2.14;"

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.3]

"Upon receiving from the MCVideo client a SIP BYE request the participating MCVideo function shall follow the procedures as specified in clause 6.3.2.1.6."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.2.8.1.1]

"This clause is referenced from other procedures.

When the MCVideo emergency state is set and this MCVideo user and MCVideo group are authorised to initiate MCVideo emergency group calls as determined by the procedures of clause 6.2.8.1.8, the MCVideo client:

1) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body in the SIP INVITE request, an <emergency-ind> element set to "true" and if the MCVideo emergency group call state is set to "MVEGC 1: emergency-gc-capable", shall set the MCVideo emergency group call state to "MVEGC 2: emergency-call-requested";

2) if the MCVideo user has also requested an MCVideo emergency alert to be sent and this is an authorised request for MCVideo emergency alert as determined by the procedures of clause 6.2.8.1.6, and the MCVideo emergency alert state is set to "MVEA 1: no-alert", shall:

a) set the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to "true" and set the MCVideo emergency alert state to "MVEA 2: emergency-alert-confirm-pending"; and

b) perform the procedures specified in clause 6.2.9.1 for the MCVideo emergency alert trigger;

3) if the MCVideo user has not requested an MCVideo emergency alert to be sent and the MCVideo emergency alert state is set to "MVEA 1: no-alert", shall set the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to "false"; and

4) if the MCVideo client emergency group state of the group is set to a value other than "MVEG 2: in-progress" set the MCVideo client emergency group state of the MCVideo group to "MVEG 3: confirm-pending".

NOTE 1: This is the case of an MCVideo user already being in the MCVideo emergency state it initiated previously while originating an MCVideo emergency group call or MCVideo emergency alert. All group calls the MCVideo user originates while in MCVideo emergency state will be MCVideo emergency group calls.

When the MCVideo emergency state is clear and the MCVideo emergency group call state is set to "MVEGC 1: emergency-gc-capable" and the MCVideo user is authorised to initiate an MCVideo emergency group call on the targetted MCVideo group as determined by the procedures of clause 6.2.8.1.8, the MCVideo client shall set the MCVideo emergency state and perform the following actions:

1) *shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body in the SIP INVITE request an <emergency-ind> element set to "true" and set the MCVideo emergency group call state to "MVEGC 2: emergency-call-requested" state;*

2) *if the MCVideo user has also requested an MCVideo emergency alert to be sent and this is an authorised request for MCVideo emergency alert as determined by the procedures of clause 6.2.8.1.6, shall:*

   a) *include in the application/vnd.3gpp.mcvideo-info+xml MIME body the <alert-ind> element set to "true" and set the MCVideo emergency alert state to "MVEA 2: emergency-alert-confirm-pending"; and*

   b) *perform the procedures specified in clause 6.2.9.1 for the MCVideo emergency alert trigger;*

3) *if the MCVideo user has not requested an MCVideo emergency alert to be sent, shall set the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to "false"; and*

4) *if the MCVideo client emergency group state of the group is set to a value other than "MVEG 2: in-progress" shall set the MCVideo client emergency group state of the MCVideo group to "MVEG 3: confirm-pending".*

NOTE 2: This is the case of an initial MCVideo emergency group call and optionally an MCVideo emergency alert being sent. As the MCVideo emergency state is not sent, there is no MCVideo emergency alert outstanding.

NOTE 3: An MCVideo group call originated by an affiliated member of an MCVideo group which is in an in-progress emergency state (as tracked on the MCVideo client by the MCVideo client emergency group state) but is not in an MCVideo emergency state of their own will also be an MCVideo emergency group call. The <emergency-ind> and <alert-ind> elements of the application/vnd.3gpp.mcvideo-info+xml MIME body do not need to be included in this case and hence no action needs to be taken in this clause."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.1.1]

"The MCVideo server needs to distinguish between the following initial SIP INVITE requests for originations and terminations:

- *SIP INVITE requests routed to the participating MCVideo function as a result of processing initial filter criteria at the S-CSCF in accordance with the origination procedures as specified in 3GPP TS 24.229 [11] and the Request-URI is set to a public service identity of the participating MCVideo function that does not identify the pre-established session set-up. Such requests are known as "SIP INVITE request for originating participating MCVideo function" in the procedures in the present document;*

- *SIP INVITE requests routed to the participating MCVideo function as a result of processing initial filter criteria at the S-CSCF in accordance with the termination procedures as specified in 3GPP TS 24.229 [11] and the Request-URI contains a PSI of the terminating participating MCVideo function. Such requests are known as "SIP INVITE request for terminating participating MCVideo function" in the procedures in the present document;*

- *SIP INVITE requests routed to the controlling MCVideo function as a result of PSI routing on the originating side in accordance with the originating procedures as specified in 3GPP TS 24.229 [11], or as a result of direct PSI routing, in accordance with the termination procedures as specified in 3GPP TS 24.229 [11], the Request-URI is set to a public service identity for MCVideo private call and the Contact header field does not contain the isfocus media feature tag specified in IETF RFC 3840 [16]. Such requests are known as "SIP INVITE request for controlling MCVideo function of a private call" in the procedures in the present document;*

- *SIP INVITE requests routed to the controlling MCVideo function as a result of PSI routing on the originating side in accordance with the originating procedures as specified in 3GPP TS 24.229 [11], or as a result of direct PSI routing, in accordance with the termination procedures as specified in 3GPP TS 24.229 [11], the Request-URI is set to a public service identity serving an MCVideo group and the Contact header field does not contain the isfocus media feature tag specified in IETF RFC 3840 [16]. Such requests are known as "SIP INVITE request for controlling MCVideo function of an MCVideo group" in the procedures in the present document; and*

- *SIP INVITE requests routed to the controlling MCVideo function as a result of PSI routing on the originating side in accordance with the originating procedures as specified in 3GPP TS 24.229 [11], or as a result of direct PSI routing, in accordance with the termination procedures as specified in 3GPP TS 24.229 [11], the Request-URI is set to a public service identity for MCVideo ambient viewing call and the Contact header field does not contain the isfocus media feature tag specified in IETF RFC 3840 [22]. Such requests are known as "SIP INVITE request for controlling MCVideo function of an ambient viewing call" in the procedures in the present document."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.3]

"This clause is referenced from other procedures.

When generating an initial SIP INVITE request according to 3GPP TS 24.229 [11], on receipt of an incoming SIP INVITE request, the participating MCVideo function:

1) *shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [20] if included in the incoming SIP INVITE request;*

2) *should include the Session-Expires header field according to IETF RFC 4028 [23]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";*

3) *shall include the option tag "timer" in the Supported header field;*

4) *shall copy the contents of the P-Asserted-Identity header field of the incoming SIP INVITE request to the P-Asserted-Identity header field of the outgoing SIP INVITE request;*

5) *shall include the g.3gpp.mcvideo media feature tag into the Contact header field of the outgoing SIP INVITE request;*

6) *shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcvideo" (coded as specified in 3GPP TS 24.229 [11]), into the P-Asserted-Service header field of the outgoing SIP INVITE request;*

7) *if the incoming SIP INVITE request contained a MIME resource-lists body with the MCVideo ID of the invited MCVideo user, shall copy the MIME resource-lists body, according to rules and procedures of IETF RFC 5366 [37];*

8) *if the incoming SIP INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the contents of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request to the outgoing SIP INVITE request; and*

9) *if the incoming SIP INVITE request contained an application/vnd.3gpp.location-info+xml MIME body, shall copy the contents of the application/vnd.3gpp.location-info+xml MIME body of the incoming SIP INVITE request to the outgoing SIP INVITE request."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.1.1]

"This clause is referenced from other clauses.

The SDP offer is generated based on the received SDP offer. The SDP offer generated by the participating MCVideo function:

1) *shall contain two SDP media-level sections for MCVideo video media as contained in the received SDP offer; and*

2) *shall contain an SDP media-level section for one media-transmission control entity, if present in the received SDP offer.*

When composing the SDP offer according to 3GPP TS 24.229 [11], the participating MCVideo function:

1) *shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the participating MCVideo function, if required;*

NOTE 1: Requirements can exist for the participating MCVideo function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number for the offered media transmission control entity, if any, in the received SDP offer with the IP address and port number of the participating MCVideo function; and

NOTE 2: If the participating MCVideo function and the controlling MCVideo function are in the same MCVideo server, and the participating MCVideo function does not have a dedicated IP address or a dedicated port number for media transmission control or media stream, the replacement of the IP address or the port number is omitted.

3) shall contain an "a=key-mgmt" attribute field with a "mikey" attribute value, if present in the received SDP offer."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.6.1]

"When the participating MCVideo function receives a request from the MCVideo client to originate an MCVideo emergency group call and needs to determine if the request is an authorised request for an MCVideo emergency call, the participating MCVideo function shall check the following:

1) if the <allow-emergency-group-call> element of the <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true" and:

a) if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> element contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "DedicatedGroup" and if the <uri-entry> element of the <entry> element of the <MCVideoGroupInitiation> element contains the identity of the MCVideo group targeted by the calling MCVideo user; or

b) if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element of the <EmergencyCall> contained within the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "UseCurrentlySelectedGroup";

then the participating MCVideo function shall consider the MCVideo emergency group call request to be an authorised request for an MCVideo emergency group call;

In all other cases, the participating MCVideo function shall consider the request to originate an MCVideo emergency group call to be an unauthorised request to originate an MCVideo emergency group call.

When the participating MCVideo function receives a request from the MCVideo client to originate an MCVideo imminent peril group call and needs to determine if the request is an authorised request for an MCVideo imminent peril group call the participating MCVideo function shall check the following:

1) if the <allow-imminent-peril-call> element of <ruleset> element of the MCVideo user profile document identified by the MCVideo ID of the calling user (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "true"; and

a) if the "entry-info" attribute of the <MCVideoGroupInitiation> element contained within the <ImminentPerilCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "DedicatedGroup" and if the <uri-entry> element of the <entry> element of the <MCVideoGroupInitiation> element contains the identity of the MCVideo group targeted by the calling MCVideo user; or

b) if the "entry-info" attribute of the <entry> element of the <MCVideoGroupInitiation> element contained within the <ImminentPerilCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) is set to a value of "UseCurrentlySelectedGroup";

then the participating MCVideo function shall consider the MCVideo imminent peril group call request to be an authorised request for an MCVideo emergency group call;

In all other cases, the participating MCVideo function shall consider the request to originate an MCVideo imminent peril group call to be an unauthorised request to originate an MCVideo imminent peril call."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.1]

"Once an on-demand MCVideo session is established or a pre-established session is in use when the participating MCVideo function receives transmission control messages from the transmission participant in the MCVideo client or from the transmission control server in the controlling MCVideo function, the behaviour of the participating MCVideo function is described in the following clauses."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.2]

"Upon receiving a transmission control message the participating MCVideo function:

1. *shall immediately forward the transmission control message to the transmission control server if the message is received from the transmission participant;*
2. *if an MBMS subchannel is not used for a transmission in the session the transmission control message is associated with, shall immediately forward the transmission control message to the transmission participant if the message is received from the transmission control server; and*
3. *if an MBMS subchannel is used for a transmission in the session the transmission control message is associated with:*
    a. *if*
        i. *the transmission control message is not a Transmission Idle message or a Media Transmission Notify message;*
        ii. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*
        iii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report;*

        shall immediately forward the transmission control message to the transmission participant; and
    b. *if*
        i. *the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report; and*
        ii *if the transmission control message is the Transmission Idle message or the Media Transmission Notify message,*

    shall perform actions as specified in clause 10.2.

NOTE: When the Transmit Idle or Media Transmission Notify messages are discarded the messages are sent to the MCVideo clients over the MBMS subchannel allocated for the transmission as specified in clause 10.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.3]

"Upon receiving RTP media packets the participating MCVideo function:

1. *shall immediately forward the RTP media packet to the controlling MCVideo function if the RTP packet is from an MCVideo client; and*
2. *if an MBMS subchannel is not used for a transmission in the session the RTP media packets are associated with, shall immediately forward the RTP media packets to the MCVideo client if the RTP packet is from the controlling MCVideo function or the non-controlling MCVideo function.*

3. *if an MBMS subchannel is used for a transmission in the session the RTP media packets are associated with and if RTP media packets are received from the controlling MCVideo function or the non-controlling MCVideo function:*

   a. *if*

      i. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

      ii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report,*

      shall immediately forward the RTP media packets to the MCVideo client; and

   b. *if the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report, shall perform actions as specified in clause 10.2."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.4]

"When the participating function receives an indication from the application and signalling plane that session release is initiated, the participating MCVideo function:

1. *shall stop sending transmission control messages towards the transmission participant and the transmission control server; and*

2. *shall stop sending RTP media packets towards the MCVideo client and towards the controlling MCVideo function.*

When the participating MCVideo function receives an indication from the application and signalling plane that the session is released, the participating MCVideo function:

1. *in case of a pre-established session, shall perform the actions in clause 9.3.2; and*

2. *in case of an on-demand session, shall release the media resources associated with the session."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

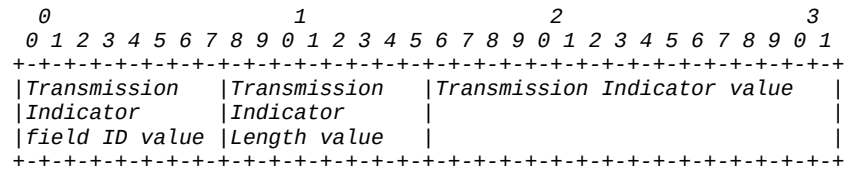
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

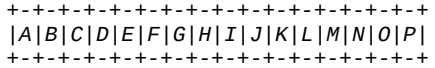


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of *queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:*

1. *if the active Transmission request queue is empty the transmission control server:*
    a. *shall send Transmission Idle message to all transmission participants. The Transmission Idle message:*
        i. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*
        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;*
    b. *shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;*
    c. *shall start timer T1 (Inactivity);*

d. shall set the general state to the 'G: Transmit Idle' state; and

e. shall initialise counter Cx (Simultaneous transmission video) to 0.

2. if the active Transmission request queue is not empty the transmission control server:

a. shall select a queued Transmission request from the top of the active Transmission request queue;

b. shall remove that queued Transmission request from the active Transmission request queue; and

c. shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:

a. shall include the granted priority in the Transmission priority field;

b. shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;

c. if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

d. shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;

2. shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;

3. shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:

a. shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;

b. shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;

c. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;

d. if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';

e. if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

g. shall initiate the 'general reception control operation' state machine;and

4. shall enter the 'G: Transmit Taken' state."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. if counter Cx (Simultaneous transmission video) has not reached its upper limit:

a. shall perform the actions specified in the clause 6.3.4.4.2;

2. *if counter Cx (Simultaneous transmission video) has reached its upper limit:*

    a. *select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;*

    b. *shall stop timer T4 (Transmission Grant), if running;*

    c. *shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);*

    d. *shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;*

    e. *shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and*

    f. *shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:*

        i. *shall include the queue position and transmission priority in the Queue Info field; and*

        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:*

    a. *shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and*

    b. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

2. *shall set the general state to 'G: pending Transmission Revoke'."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. *if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:*

    a. *shall initialize a general state machine as specified in clause 6.3.4.2.2; and*

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

    b. *shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;*

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.1.10.3 Test description

5.1.1.10.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the controlling MCVideo function.
- SS-UE1 (MCVideo Client).

IUT:

- IUT (MCVideo Server), providing the originating participating MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of some sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server and the HTTP Client.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server) and SS-UE1 (MCVideo Client).
- The SS-UE1 (MCVideo client) is registered and authorized to use the MCVideo system.

- The SS-UE1 (MCVideo client) is affiliated to the pre-arranged group and is authorized to initiate group calls.

5.1.1.10.3.2 Test procedure sequence

Table 5.1.1.10.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.4.2.1.3-1 to establish an imminent peril pre-arranged group call with implicit transmission control request according to option b.ii of note in Annex A, Table A.4.2.1.3-1, initiated by SS-UE1 (MCVideo Client?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO Transmission End Request - Transmission End Response' as described in Annex A, Table A.4.2.5.3-1 to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Transmission Granted' as described in Annex A, Table A.4.2.4.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	3	P
4	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO call release' as described in Annex A, Table A.4.2.3.3-1, to release the call?	-	-	4	P

5.1.1.10.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

Table 5.1.1.10.3.3-1: SIP INVITE from SS-UE1 (MCVideo Client)
(Step 1, Table 5.1.1.10.3.2-1; Step 1, Annex A, Table A.4.2.1.3-1)

Derivation Path: Annex A, Table A.4.2.1.4-1, with conditions GROUP-CALL and IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.1.10.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.1.10.3.2-1; Step 2, Annex A, Table A.4.2.1.3-1)

Derivation Path: Annex A, Table A.4.2.1.4-2, with conditions GROUP-CALL and IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.1.10.3.3-3: SIP 200 (OK) from SS (MCVideo Server)
(Step 1, Table 5.1.1.10.3.2-1; Step 5, Annex A, Table A.4.2.1.3-1)

Derivation Path: Annex A, Table A.4.2.1.4-3, with condition GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.1.10.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.1.1.10.3.2-1; Step 6, Annex A, Table A.4.2.1.3-1)

Derivation Path: Annex A, Table A.4.2.1.4-4, with condition GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.1.10.3.3-5: Transmission Idle
(Steps 2, Table 5.1.1.10.3.2-1; Steps 9 and 10, Annex A, Table A.4.2.5.3-1)

Derivation Path: Annex A, Table A.4.2.5.4-3, with condition IMMPERIL-CALL.

Table 5.1.1.10.3.3-6: Transmission Request
(Step 3, Table 5.1.1.10.3.2-1; Steps 1 and 2, Annex A, Table A.4.2.4.3-1)

Derivation Path: Annex A, Table A.4.2.4.4-1, with condition IMMPERIL-CALL.

Table 5.1.1.10.3.3-7: Transmission Granted
(Steps 1 and 3, Table 5.1.1.10.3.2-1;
Steps 5 and 6, Annex A, Table A.4.2.4.3-1; Steps 9a1 and 9a2, Annex A, Table A.4.2.1.3-1)

Derivation Path: Annex A, Table A.4.2.4.4-2, with condition IMMPERIL-CALL.

5.1.1.11 MCVideo Server - MCVideo Server / Broadcast Group Call / Controlling server

5.1.1.11.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function initiates a
broadcast group call with implicit transmission control }
    then { IUT (MCVideo Server) accepts the call from initiator and establishes the call with all
affiliated users by sending call establishment requests to SS (MCVideo Server) providing the
terminating participating function and sends transmission granted to SS-UE1 (MCVideo Client) and
sends media transmission notification to SS-UE2 (MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established a broadcast group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests to end the transmission control }
    then { IUT (MCVideo Server) sends transmission control idle to SS-UE1 (MCVideo Client) and sends
transmission control idle to SS-UE2 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established a broadcast group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control }
    then { IUT (MCVideo Server) sends transmission granted to SS-UE1 (MCVideo Client) and sends
media transmission notification to SS-UE2 (MCVideo Client) }
            }
```

(4)

```
with { IUT (MCVideo Server) having established a broadcast group call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function ends the broadcast
group call }
    then { IUT (MCVideo Server) responds by sending a SIP 200 (OK) message to SS (MCVideo Server)
ending the call and sends a SIP BYE message to SS (MCVideo Server) providing the terminating
participating function }
            }
```

5.1.1.11.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.1.2.1.1, 6.2.8.2, 9.2.1.4.2, 9.2.1.4.3 and 9.2.1.4.4.2, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.3.1, 6.3.2.2, 9.2.1, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.2.1.1]

"Upon receiving a request from an MCVideo user to establish an MCVideo prearranged group session the MCVideo client shall generate an initial SIP INVITE request by following the UE originating session procedures specified in 3GPP TS 24.229 [11], with the clarifications given below.

The MCVideo client:

1) if the MCVideo user has requested the origination of an MCVideo emergency group call or is originating an MCVideo prearranged group call and the MCVideo emergency state is already set, the MCVideo client shall comply with the procedures in clause 6.2.8.1.1;

2) if the MCVideo user has requested the origination of an MCVideo imminent peril group call, the MCVideo client shall comply with the procedures in clause 6.2.8.1.9;

3) if the MCVideo user has requested the origination of a broadcast group call, the MCVideo client shall comply with the procedures in clause 6.2.8.2;

4) shall include the g.3gpp.mcvideo media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo" in the Contact header field of the SIP INVITE request according to IETF RFC 3840 [22];

5) shall include an Accept-Contact header field containing the g.3gpp.mcvideo media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [20];

6) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcvideo" (coded as specified in 3GPP TS 24.229 [11]), in a P-Preferred-Service header field according to IETF RFC 6050 [14] in the SIP INVITE request;

7) shall include an Accept-Contact header field with the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [20];

8) should include the "timer" option tag in the Supported header field;

9) should include the Session-Expires header field according to IETF RFC 4028 [23]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

10) shall set the Request-URI of the SIP INVITE request to the public service identity identifying the participating MCVideo function serving the MCVideo user;

NOTE 1: The MCVideo client is configured with public service identity identifying the participating MCVideo function serving the MCVideo user.

11) may include a P-Preferred-Identity header field in the SIP INVITE request containing a public user identity as specified in 3GPP TS 24.229 [11];

12) if the MCVideo emergency state is already set or the MCVideo client emergency group state for this group is set to "MVEG 2: in-progress", the MCVideo client shall include the Resource-Priority header field and comply with the procedures in clause 6.2.8.1.2;

13) if the MCVideo client imminent peril group state for this group is set to "MVIG 2: in-progress" or "MVIG 4: confirm-pending" shall include the Resource-Priority header field and comply with the procedures in clause 6.2.8.1.12;

14) shall contain in an application/vnd.3gpp.mcvideo-info+xml MIME body with the <mcvideoinfo> element containing the <mcvideo-Params> element with:

a) the <session-type> element set to a value of "prearranged";

b) the <mcvideo-request-uri> element set to the group identity;

c) the <mcvideo-client-id> element set to the MCVideo client ID of the originating MCVideo client; and

NOTE 2: The MCVideo client does not include the MCVideo ID of the originating MCVideo user in the body, as this will be inserted into the body of the SIP INVITE request that is sent from the originating participating MCVideo function.

d) if the group identity can be determined to be a TGI and if the MCVideo client can associate the TGI with a MCVideo group ID, the <associated-group-id> element set to the MCVideo group ID;

NOTE 3: The text "can associate the TGI with a MCVideo group ID" means that the MCVideo client is able to determine that there is a constituent group of the temporary group that it is a member of.

NOTE 4: The MCVideo client is informed about temporary groups and regouping of MCVideo groups that the user is a member of as specified in 3GPP TS 24.481 [24].

NOTE 5: If the MCVideo user selected a TGI where there are several MCVideo groups where the MCVideo user is a member, the MCVideo client selects one of those MCVideo groups.

15) shall include an SDP offer according to 3GPP TS 24.229 [11] with the clarifications given in clause 6.2.1;

16) if an implicit transmission request is required, shall indicate this as specified in clause 6.4; and

17) shall send the SIP INVITE request towards the MCVideo server according to 3GPP TS 24.229 [11].

On receiving a SIP 2xx response to the SIP INVITE request, the MCVideo client:

1) shall interact with the user plane as specified in 3GPP TS 24.581 [5];

2) if the MCVideo emergency group call state is set to "MVEGC 2: emergency-call-requested" or "MVEGC 3: emergency-call-granted" or the MCVideo imminent peril group call state is set to "MVIGC 2: imminent-peril-call-requested" or "MVIGC 3: imminent-peril-call-granted", the MCVideo client shall perform the actions specified in clause 6.2.8.1.4; and

3) may subscribe to the conference event package as specified in clause 9.1.3.1.

On receiving a SIP 4xx response, a SIP 5xx response or a SIP 6xx response to the SIP INVITE request:

1) if the MCVideo emergency group call state is set to "MVEGC 2: emergency-call-requested" or "MVEGC 3: emergency-call-granted"; or

2) if the MCVideo imminent peril group call state is set to "MVIGC 2: imminent-peril-call-requested" or "MVIGC 3: imminent-peril-call-granted";

the MCVideo client shall perform the actions specified in clause 6.2.8.1.5.

On receiving a SIP INFO request where the Request-URI contains an MCVideo session ID identifying an ongoing group session, the MCVideo client shall follow the actions specified in clause 6.2.8.1.13."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.2.8.2]

"NOTE: This clause is referenced from other procedures.

When the MCVideo user initiates a broadcast group call, the MCVideo client:

1) in the case of the prearranged group call is initiated on-demand, shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body the <broadcast-ind> element set to "true" as defined in clause F.1; and

2) in the case the prearranged group call is initiated using a pre-established session, shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body in the hname "body" parameter in the headers portion of the SIP URI in the Refer-To header field the <broadcast-ind> element set to "true" as defined in clause F.1."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.2]

"In the procedures in this clause:

1) MCVideo ID in an incoming SIP INVITE request refers to the MCVideo ID of the originating user from the <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;

2) group identity in an incoming SIP INVITE request refers to the group identity from the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;

3) MCVideo ID in an outgoing SIP INVITE request refers to the MCVideo ID of the called user in the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the outgoing SIP INVITE request;

4) indication of required group members in a SIP 183 (Session Progress) response refers to the <required> element of the application/vnd.3gpp.mcvideo-info+xml MIME body set to "true" in a SIP 183 (Session Progress) sent by the non-controlling MCVideo function of an MCVideo group;

5) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

6) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

Upon receipt of a "SIP INVITE request for controlling MCVideo function of an MCVideo group", the controlling MCVideo function:

…

2) shall determine if the media parameters are acceptable and the MCVideo codecs are offered in the SDP offer and if not reject the request with a SIP 488 (Not Acceptable Here) response and skip the rest of the steps;

3) shall reject the SIP request with a SIP 403 (Forbidden) response and not process the remaining steps if:

a) an Accept-Contact header field does not include the g.3gpp.mcvideo media feature tag; or

b) an Accept-Contact header field does not include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo";

4) if received SIP INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.3.1.17;

5) shall retrieve the necessary group document(s) from the group management server for the group identity contained in the SIP INVITE request and carry out initial processing as specified in clause 6.3.5.2;

6) if the result of the initial processing in clause 6.3.5.2 was:

a) that authorization of the MCVideo ID is required at a non-controlling MCVideo function of an MCVideo group is required, perform the actions in clause 6.3.3.1.13.7 and do not continue with the rest of the steps in this clause; and

b) that a SIP 3xx, 4xx, 5xx or 6xx response to the "SIP INVITE request for controlling MCVideo function of an MCVideo group" has been sent, do not continue with the rest of the steps in this clause;

7) shall perform the actions as described in clause 6.3.3.2.2;

8) shall maintain a local counter of the number of SIP 200 (OK) responses received from invited members and shall initialise this local counter to zero;

9) shall determine if an MCVideo group call for the group identity is already ongoing by determining if an MCVideo session identity has already been allocated for the group call and the MCVideo session is active;

…

12) if a Resource-Priority header field is included in the SIP INVITE request:

a) if the Resource-Priority header field is set to the value indicated for emergency calls and the SIP INVITE request does not contain an emergency indication and the in-progress emergency state of the group is set to a value of "false", shall reject the SIP INVITE request with a SIP 403 (Forbidden) response and skip the rest of the steps; or

b) if the Resource-Priority header field is set to the value indicated for imminent peril calls and the SIP INVITE request does not contain an imminent peril indication and the in-progress imminent peril state of the group is set to a value of "false", shall reject the SIP INVITE request with a SIP 403 (Forbidden) response and skip the rest of the steps;

13) if the MCVideo group call is not ongoing then:

…

c) if the received SIP INVITE request contains an an authorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2 or MCVideo imminent peril group call as determined by clause 6.3.3.1.13.5 and the MCVideo user is eligible to be implicitly affiliated with the MCVideo group as determined as determined in step 13) a) iv) above, shall perform the implicit affiliation as specified in clause 8.2.2.3.7;

d) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [11];

e) shall create a prearranged group session and allocate an MCVideo session identity for the prearranged group call, and shall handle timer TNG3 (group call timer) as specified in clause 6.3.3.5;

f) if the group identity in the "SIP INVITE request for controlling MCVideo function of an MCVideo group" is a TGI:

i) shall for each of the constituent MCVideo groups homed on the primary MCVideo system:

…;

B) if the controlling MCVideo function owns the MCVideo group identified by the MCVideo group ID then:

I) determine the members to invite to the prearranged MCVideo group call as specified in clause 6.3.5.5;

II) invite each group member determined in step A) above, to the group session, as specified in clause 9.2.1.4.1.1; and

III) interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3; and

ii) shall for each of the constituent MCVideo groups homed on the partner MCVideo system generate a SIP INVITE request for the MCVideo group identity homed on the partner MCVideo system as specified in clause 9.2.1.4.1.2; and

g) if the group identity in the SIP INVITE request for controlling MCVideo function of an MCVideo group is an MCVideo group ID:

i) shall determine the members to invite to the prearranged MCVideo group call as specified in clause 6.3.5.5;

ii) if necessary, shall start timer TNG1 (acknowledged call setup timer) according to the conditions stated in clause 6.3.3.3;

iii) if the received SIP INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element set to a value of "true":

A) shall cache the information that this MCVideo user has initiated an MCVideo emergency call;

B) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the information that this MCVideo user has initiated an MCVideo emergency alert; and

C) if the in-progress emergency state of the group is set to a value of "false":

I) shall set the value of the in-progress emergency state of the group to "true"; and

II) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16;

iv) if the in-progress emergency state of the group is set to a value of "false" and if the received SIP INVITE request contains an imminent peril indication set to a value of "true", the controlling MCVideo function shall:

A) shall cache the information that this MCVideo user has initiated an MCVideo imminent peril call; and

B) if the in-progress imminent peril state of the group is set to a value of "false", shall set the in-progress imminent peril state of the group to a value of "true";

v) shall invite each group member determined in step 13)g)i) above, to the group session, as specified in clause 9.2.1.4.1.1; and

vi) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3; and

14) if the MCVideo group call is ongoing then:

…

c) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [11];

…

e) if the received SIP INVITE request contains an an authorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2 or MCVideo imminent peril group call as determined by clause 6.3.3.1.13.5 and the MCVideo user is eligible to be implicitly affiliated with the MCVideo group as determined in step 14) a) iv) above, shall perform the implicit affiliation as specified in clause 8.2.2.3.7;

f) if the received SIP INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element set to a value of "true":

i) shall cache the information that this MCVideo user has initiated an MCVideo emergency call;

ii) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the information that this MCVideo user has initiated an MCVideo emergency alert;

iii) if the in-progress emergency state of the group is set to a value of "false":

A) shall set the value of the in-progress emergency state of the group to "true";

B) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16; and

C) shall generate SIP re-INVITE requests for the MCVideo emergency group call to the other call participants of the MCVideo group as specified in clause 6.3.3.1.6;

iv) if the in-progress imminent peril state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's imminent peril indication as specified in clause 6.3.3.1.11, setting the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11]; and

v) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5];

g) if the in-progress emergency state of the group is set to a value of "false" and if the SIP INVITE request contains an imminent peril indication set to a value of "true", the controlling MCVideo function:

i) shall cache the information that this MCVideo user has initiated an MCVideo imminent peril call; and

ii) if the in-progress imminent peril state of the group is set to a value of "false":

A) shall set the in-progress imminent peril state of the group to a value of "true";

B) shall generate SIP re-INVITE requests for the MCVideo imminent peril group call to the other call participants of the MCVideo group as specified in clause 6.3.3.1.15; and

C) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

iii) if the in-progress imminent peril state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's imminent peril indication as specified in clause 6.3.3.1.11, setting the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11];

h) shall generate a SIP 200 (OK) response as specified in the clause 6.3.3.2.4.2;

i) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

j) shall include in the SIP 200 (OK) response with the warning text set to "123 MCVideo session already exists" as specified in clause 4.4;

k) if the received SIP re-INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

l) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 4: In this case, the request was for an imminent peril call but a higher priority MCVideo emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

m) shall interact with media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 5: Resulting media plane processing is completed before the next step is performed.

n) shall send the SIP 200 (OK) response towards the inviting MCVideo client or inviting non-controlling MCVideo function according to 3GPP TS 24.229 [11];

o) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo User has joined in the MCVideo group session, as specified in clause 6.3.3.4;

NOTE 6: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

p) shall send a SIP NOTIFY request to each MCVideo client according to 3GPP TS 24.229 [11];

q) Upon receiving a SIP ACK to the above SIP 200 (OK) response and the SIP 200 (OK) response contained a Warning header field as specified in clause 4.4 with the warning text containing the mcvideo-warn-code set to "149", shall follow the procedures in clause 6.3.3.1.18; and

r) shall not continue with the rest of the clause.

Upon receiving a SIP 183 (Session Progress) response to the SIP INVITE request specified in clause 9.2.1.4.1 containing a P-Answer-State header field with the value "Unconfirmed" as specified in IETF RFC 4964 [30], the timer TNG1 (acknowledged call setup timer) is not running, the controlling MCVideo function supports media buffering and the SIP final response is not yet sent to the inviting MCVideo client:

1) shall generate a SIP 200 (OK) response to SIP INVITE request as specified in the clause 6.3.3.2.3.2;

2) shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if the prearranged MCVideo group has more than <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [24];

3) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

4) shall include a P-Answer-State header field with the value "Unconfirmed";

5) if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

6) if the received SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

7) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 7: Resulting user plane processing is completed before the next step is performed.

8) shall send the SIP 200 (OK) response towards the inviting MCVideo client according to 3GPP TS 24.229 [11];

9) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo User has joined in the MCVideo group session, as specified in clause 6.3.3.4; and

NOTE 8: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

10) shall send a SIP NOTIFY request to each MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 183 (Session Progress) response for a SIP INVITE request as specified in clause 9.2.1.4.1.2 containing an indication of required group members, the timer TNG1 (acknowledged call setup timer) is running and all SIP 200 (OK) responses have been received to all SIP INVITE requests sent to MCVideo clients specified in clause 9.2.1.4.1.1, then the controlling MCVideo function shall wait until the SIP 200 (OK) response has been received to the SIP INVITE request specified in clause 9.2.1.4.1.2 before generating a SIP 200 (OK) response to the "SIP INVITE request for controlling MCVideo function of an MCVideo group".

Upon receiving a SIP 200 (OK) response for a SIP INVITE request as specified in clause 9.2.1.4.1 that was sent to an affiliated and <on-network-required> group member as specified in 3GPP TS 24.481 [24]; and

1) *if the MCVideo ID in the SIP 200 (OK) response matches to the MCVideo ID in the corresponding SIP INVITE request;*

2) *there are no outstanding SIP 200 (OK) responses to SIP INVITE requests which were sent to affiliated and <on-network-required> group members as specified in 3GPP TS 24.481 [24]; and*

3) *there is no outstanding SIP 200 (OK) response to a SIP INVITE request sent in clause 9.2.1.4.1.2 where the SIP 183 (Session Progress) response contained an indication of required group members;*

the controlling MCVideo function:

1) *shall stop timer TNG1 (acknowledged call setup timer) as described in clause 6.3.3.3;*

2) *shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;*

3) *shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if all members were not invited because the prearranged MCVideo group has been exceeded the <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [24];*

4) *shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;*

5) *shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;*

NOTE 9: Resulting media plane processing is completed before the next step is performed.

6) *shall send a SIP 200 (OK) response to the inviting MCVideo client according to 3GPP TS 24.229 [11];*

7) *shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo user has joined in the MCVideo group session, as specified in clause 6.3.3.4; and*

NOTE 10: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

8) *shall send the SIP NOTIFY request to the MCVideo clients according to 3GPP TS 24.229 [11].*

Upon:

1) *receiving a SIP 200 (OK) response for a SIP INVITE request as specified in clause 9.2.1.4.1;*

2) *the timer TNG1 (acknowledged call setup timer) is not running;*

3) *the local counter of the number of SIP 200 (OK) responses received from invited members is equal to the value of the <on-network-minimum-number-to-start> element of the group document;*

4) *the controlling MCVideo function supports media buffering; and*

5) *the SIP final response has not yet been sent to the inviting MCVideo client;*

the controlling MCVideo function according to local policy:

1) *shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.2;*

2) *shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if all members were not invited because the prearranged MCVideo group has exceeded the <max-participant-count> members as specified in 3GPP TS 24.481 [24];*

3) *shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;*

4) if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

5) if the received SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

6) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 11: Resulting media plane processing is completed before the next step is performed.

7) shall send a SIP 200 (OK) response to the inviting MCVideo client according to 3GPP TS 24.229 [11];

8) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo user has joined in the MCVideo group session, as specified in clause 6.3.3.4; and

NOTE 12: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

9) shall send the SIP NOTIFY request to the MCVideo clients according to 3GPP TS 24.229 [11].

Upon expiry of timer TNG1 (acknowledged call setup timer), if there are outstanding SIP 200 (OK) responses to SIP INVITE requests sent to affiliated and <on-network-required> group members as specified in 3GPP TS 24.481 [24], the controlling MCVideo function shall follow the procedures specified in clause 6.3.3.3.

If timer TNG1 (acknowledged call setup timer) is running and a final SIP 4xx, 5xx or 6xx response is received from an affiliated and <on-network-required> group member as specified in 3GPP TS 24.481 [24], the controlling MCVideo function shall follow the relevant procedures specified in clause 6.3.3.3.

If:

1) timer TNG1 (acknowledged call setup timer) is not running;

2) the local counter of the number of SIP 200 (OK) responses received from invited members is equal to the value of the <on-network-minimum-number-to-start> element of the group document; and

3) a final SIP 4xx, 5xx or 6xx response is received from an invited MCVideo client;

then the controlling MCVideo function shall perform one of the following based on policy:

1) send the SIP final response towards the inviting MCVideo client, according to 3GPP TS 24.229 [11], if a SIP final response was received from all the other invited MCVideo clients and the SIP 200 (OK) response is not yet sent; or

2) remove the invited MCVideo client from the MCVideo Session as specified in clause 6.3.3.1.5, if a SIP final response other than 2xx or 3xx was received from all the invited MCVideo clients and the SIP 200 (OK) response is already sent. The controlling MCVideo function may invite an additional member of the prearranged MCVideo group as specified in clause 9.2.1.4.1 that has not already been invited, if the prearranged MCVideo group has more than <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [24], and all members have not yet been invited.

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCVideo client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.18."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.3]

"Upon receiving a SIP BYE request the controlling MCVideo function shall follow the procedures as specified in clause 6.3.3.2.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.4.2]

"When the MCVideo session for group call needs to be released as specified in clause 6.3.8.1, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.5."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.1]

"The transmission control server arbitration logic in the transmission control server shall support the procedures in clauses 6.3.2 and 6.3.3 and shall behave according to the transmission control server state transition diagram for 'general transmission control operation' in clause 6.3.4.

The transmission control interface towards the MCVideo client in the transmission control server shall behave according to the transmission control server state transition diagram for 'basic transmission control operation towards the transmission participant' as specified in clause 6.3.5.

The reception control arbitration logic in the transmission control server shall support the procedures in clauses 6.3.2 and 6.3.3 and shall behave according to the reception control server state transition diagram for 'general reception control operation' in clause 6.3.6.

The transmission control interface towards the MCVideo client in the transmission control server shall behave according to the transmission control server state transition diagram for 'basic reception control operation towards the transmission participant' as specified in clause 6.3.7."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.2.2]

"When an MCVideo call is established a new instance of the transmission control server state machine for 'general transmission control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' is added.

When an MCVideo call is established a new instance of the transmission control server state machine for 'general reception control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic reception control operation towards the transmission participant' is added.

If the optional "mc_queueing" feature is supported and has been negotiated as specified in clause 14, the transmission control server could queue the implicit transmission control request for the media-transmission control entity.

The original initial SIP INVITE request or SIP REFER request to establish an MCVideo chat group call or to rejoin an ongoing MCVideo call is not handled as an implicit transmission control request message by the transmission control server unless explicitly stated in the SIP INVITE request or in the SIP REFER request.

The permission to send media to the inviting MCVideo client due to implicit transmission control request is applicable to both confirmed indication and unconfirmed indication.

When the first unconfirmed indication is received from the invited participating MCVideo function (see 3GPP TS 24.281 [2]) the transmission control server optionally can give an early indication to send RTP media packets, to the inviting MCVideo client.

If an early indication to send RTP media packets is given to the inviting MCVideo client, the transmission participant is granted the permission to send media and the MCVideo server buffers RTP media packets received from the MCVideo client at least until the first invited MCVideo client accepts the invitation or until the RTP media packet buffer exceeds it maximum limit to store RTP media packets.

If the MCVideo server does not support or does not allow media buffering then when an early indication to send RTP media packets is not given to the inviting MCVideo client, the transmission participant is granted the permission to send media when the first invited MCVideo client accepts the media.

Before the transmission control server sends the first transmission control message in the MCVideo call, the transmission control server has to assign itself a SSRC identifier to be included in media transmission control messages and quality feedback messages if the MCVideo server is supporting that option. A suitable algorithm to generate the SSRC identifier is described in IETF RFC 3550 [3].

The transmission participant and the transmission control server can negotiate the maximum priority level that the transmission participant is permitted to request. The transmission control server can pre-empt the current sender based on the negotiated maximum priority level that the transmission participant is permitted to request and the priority level included in the Transmission Media Request message.

NOTE: The maximum priority level that a transmission participant can use is negotiated as specified in clause 14.3.3 and is based on group configuration data retrieved by the controlling MCVideo function from the group management server as described in 3GPP TS 24.481 [12] and service configuration data retrieved by the controlling MCVideo function from the configuration management server as described in 3GPP TS 24.484 [13].

The transmission participant and the transmission control server can negotiate queueing of Transmission requests using the "mc_queueing" fmtp attribute as described in clause 14. If queueing is supported and negotiated, the transmission control server queues the transmission control request if a Transmission Media Request message is received when another transmission participant has the transmission and the priority of the current speaker is the same or higher."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.1]

"The transmission control messages are coded as described in clause 9.1.2 where the transmission control message is part of the application-dependent data.

For the transmission control protocol the ASCII name string is: MCVx (Mission Critical Video x) with x=0, 1 or 2 as specified in clause 9.1.2.

A list of transmission control messages can be found in clause 9.2.2.1.

The transmission control specific fields are specified in clause 9.2.3."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

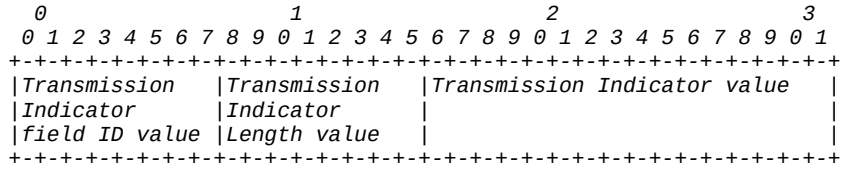
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

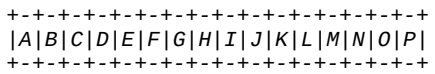


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. *if the active Transmission request queue is empty the transmission control server:*
    a. *shall send Transmission Idle message to all transmission participants. The Transmission Idle message:*
        i. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*
        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;*
    b. *shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;*
    c. *shall start timer T1 (Inactivity);*
    d. *shall set the general state to the 'G: Transmit Idle' state; and*
    e. *shall initialise counter Cx (Simultaneous transmission video) to 0.*

2. *if the active Transmission request queue is not empty the transmission control server:*

    a. *shall select a queued Transmission request from the top of the active Transmission request queue;*

    b. *shall remove that queued Transmission request from the active Transmission request queue; and*

    c. *shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:*

    a. *shall include the granted priority in the Transmission priority field;*

    b. *shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;*

    c. *if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

    d. *shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;*

2. *shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;*

3. *shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:*

    a. *shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;*

    b. *shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;*

    c. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;*

    d. *if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';*

    e. *if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';*

    f. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

    g. *shall initiate the 'general reception control operation' state machine;and*

4. *shall enter the 'G: Transmit Taken' state."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. *if counter Cx (Simultaneous transmission video) has not reached its upper limit:*

    a. *shall perform the actions specified in the clause 6.3.4.4.2;*

2. *if counter Cx (Simultaneous transmission video) has reached its upper limit:*

    a. *select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;*

    b. *shall stop timer T4 (Transmission Grant), if running;*

    c. *shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);*

    d. *shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;*

    e. *shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and*

    f. *shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:*

        i. *shall include the queue position and transmission priority in the Queue Info field; and*

        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:*

    a. *shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and*

    b. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

2. *shall set the general state to 'G: pending Transmission Revoke'."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. *if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:*

    a. *shall initialize a general state machine as specified in clause 6.3.4.2.2; and*

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

    b. *shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;*

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.1.11.3 Test description

5.1.1.11.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the participating MCVideo functions.
- SS-UE1 (MCVideo Client) and SS-UE2 (MCVideo Client), simulated by the SS (MCVideo Server).

IUT:

- IUT (MCVideo Server), providing the controlling MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of all sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Transmission Control Server.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server).

- The SS-UE1 (MCVideo client) and the SS-UE2 (MCVideo client) are registered and authorized to use the MCVideo system.
- The SS-UE1 (MCVideo client) and the SS-UE2 (MCVideo client) are affiliated to the pre-arranged group and are authorized to initiate broadcast group calls.
- The IUT (MCVideo Server) is the controlling server of the pre-arranged group.

5.1.1.11.3.2 Test procedure sequence

Table 5.1.1.11.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E group call establishment' as described in Annex A, Table A.4.1.4.3-1 to establish a broadcast group call with implicit transmission control request according to option b.ii of note in Annex A, Table A.4.1.4.3-1, initiated by SS-UE1 (MCVideo Client)?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure ' MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	3	P
4	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.1.5.3-1, to release the call?	-	-	4	P

5.1.1.11.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

**Table 5.1.1.11.3.3-1: SIP INVITE from SS (MCVideo Server)
(Step 1, Table 5.1.1.11.3.2-1; Step 1, Annex A, Table A.4.1.4.3-1)**

Derivation Path: Annex A, Table A.4.1.4.4-1, with condition BROADCAST-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.11.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.1.11.3.2-1; Step 2, Annex A, Table A.4.1.4.3-1)**

Derivation Path: Annex A, Table A.4.1.4.4-2, with condition BROADCAST-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.11.3.3-3: SIP 200 (OK) from SS (MCVideo Server)
(Step 1, Table 5.1.1.11.3.2-1; Step 8, Annex A, Table A.4.1.4.3-1)**

Derivation Path: Annex A, Table A.4.1.4.4-6

**Table 5.1.1.11.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.1.1.11.3.2-1; Step 9, Annex A, Table A.4.1.4.3-1)**

Derivation Path: Annex A, Table A.4.1.4.4-7

**Table 5.1.1.11.3.3-5: Transmission Idle from IUT (MCVideo Server)
(Step 2, Table 5.1.1.11.3.2-1; Steps 4 and 5, Annex A, Table A.3.8.3-1)**

Derivation Path: Annex A, Table A.3.8.4-6, with condition BROADCAST-CALL.

**Table 5.1.1.11.3.3-6: Transmission Request from SS (MCVideo Server)
(Step 3, Table 5.1.1.11.3.2-1; Step 1, Annex A, Table A.3.7.3-1)**

Derivation Path: Annex A, Table A.3.7.4-1, with condition BROADCAST-CALL.

**Table 5.1.1.11.3.3-7: Transmission Granted from IUT (MCVideo Server)
(Steps 1 and 3, Table 5.1.1.11.3.2-1; Step 3a1 or 3b1, Annex A, Table A.3.7.3-1;
Step 9a1, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.3.7.4-2 or A.3.7.4-3 (note), with condition BROADCAST-CALL.
NOTE: Depends on server implementation.

**Table 5.1.1.11.3.3-8: Media Transmission Notification from IUT (MCVideo Server)
(Steps 1 and 3, Table 5.1.1.11.3.2-1; Step 3a3 or 3b2, Annex A, Table A.3.7.3-1;
Step 10a1, Annex A, Table 4.1.1.3-1)**

Derivation Path: Annex A, Table A.3.7.4-4 or A.3.7.4-5 (note), with condition BROADCAST-CALL.
NOTE: Depends on server implementation.

5.1.1.12 MCVideo Server - MCVideo Server / Broadcast Group Call / Terminating Participating server

5.1.1.12.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the controlling function initiates a broadcast group call
with automatic commencement }
    then { IUT (MCVideo Server) accepts the call and establishes the call with SS-UE2 (MCVideo
Client) and forwards the media transmission notification received from SS (MCVideo Server) to SS-UE2
(MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established a broadcast group call }
ensure that {
  when { the SS (MCVideo Server) ends the media transmission }
    then { IUT (MCVideo Server) forwards the transmission end notify received from SS (MCVideo
Server) to SS-UE2 (MCVideo Client) and forwards the transmission idle received from SS (MCVideo
Server) to SS-UE2 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established a broadcast group call }
ensure that {
  when { the SS (MCVideo Server) providing the controlling function ends the broadcast group call }
    then { IUT (MCVideo Server) sends a SIP BYE message to SS-UE2 (MCVideo Client) and responds by
sending a SIP 200 (OK) message to SS (MCVideo Server) }
           }
```

5.1.1.12.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.1.2.1.1, 6.2.8.2, 9.2.1.3.2 and 9.2.1.3.4.1, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.4.1, 6.4.2, 6.4.3, 6.4.4, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.2.1.1]

"Upon receiving a request from an MCVideo user to establish an MCVideo prearranged group session the MCVideo client shall generate an initial SIP INVITE request by following the UE originating session procedures specified in 3GPP TS 24.229 [11], with the clarifications given below.

The MCVideo client:

1) if the MCVideo user has requested the origination of an MCVideo emergency group call or is originating an MCVideo prearranged group call and the MCVideo emergency state is already set, the MCVideo client shall comply with the procedures in clause 6.2.8.1.1;

2) if the MCVideo user has requested the origination of an MCVideo imminent peril group call, the MCVideo client shall comply with the procedures in clause 6.2.8.1.9;

3) if the MCVideo user has requested the origination of a broadcast group call, the MCVideo client shall comply with the procedures in clause 6.2.8.2;

4) shall include the g.3gpp.mcvideo media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo" in the Contact header field of the SIP INVITE request according to IETF RFC 3840 [22];

5) shall include an Accept-Contact header field containing the g.3gpp.mcvideo media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [20];

6) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcvideo" (coded as specified in 3GPP TS 24.229 [11]), in a P-Preferred-Service header field according to IETF RFC 6050 [14] in the SIP INVITE request;

7) shall include an Accept-Contact header field with the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [20];

8) should include the "timer" option tag in the Supported header field;

9) should include the Session-Expires header field according to IETF RFC 4028 [23]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

10) shall set the Request-URI of the SIP INVITE request to the public service identity identifying the participating MCVideo function serving the MCVideo user;

NOTE 1: The MCVideo client is configured with public service identity identifying the participating MCVideo function serving the MCVideo user.

11) may include a P-Preferred-Identity header field in the SIP INVITE request containing a public user identity as specified in 3GPP TS 24.229 [11];

12) if the MCVideo emergency state is already set or the MCVideo client emergency group state for this group is set to "MVEG 2: in-progress", the MCVideo client shall include the Resource-Priority header field and comply with the procedures in clause 6.2.8.1.2;

13) if the MCVideo client imminent peril group state for this group is set to "MVIG 2: in-progress" or "MVIG 4: confirm-pending" shall include the Resource-Priority header field and comply with the procedures in clause 6.2.8.1.12;

14) shall contain in an application/vnd.3gpp.mcvideo-info+xml MIME body with the <mcvideoinfo> element containing the <mcvideo-Params> element with:

a) the <session-type> element set to a value of "prearranged";

b) the <mcvideo-request-uri> element set to the group identity;

c) the <mcvideo-client-id> element set to the MCVideo client ID of the originating MCVideo client; and

NOTE 2: The MCVideo client does not include the MCVideo ID of the originating MCVideo user in the body, as this will be inserted into the body of the SIP INVITE request that is sent from the originating participating MCVideo function.

d) if the group identity can be determined to be a TGI and if the MCVideo client can associate the TGI with a MCVideo group ID, the <associated-group-id> element set to the MCVideo group ID;

NOTE 3: The text "can associate the TGI with a MCVideo group ID" means that the MCVideo client is able to determine that there is a constituent group of the temporary group that it is a member of.

NOTE 4: The MCVideo client is informed about temporary groups and regouping of MCVideo groups that the user is a member of as specified in 3GPP TS 24.481 [24].

NOTE 5: If the MCVideo user selected a TGI where there are several MCVideo groups where the MCVideo user is a member, the MCVideo client selects one of those MCVideo groups.

15) shall include an SDP offer according to 3GPP TS 24.229 [11] with the clarifications given in clause 6.2.1;

16) if an implicit transmission request is required, shall indicate this as specified in clause 6.4; and

17) shall send the SIP INVITE request towards the MCVideo server according to 3GPP TS 24.229 [11].

On receiving a SIP 2xx response to the SIP INVITE request, the MCVideo client:

1) shall interact with the user plane as specified in 3GPP TS 24.581 [5] ;

2) if the MCVideo emergency group call state is set to "MVEGC 2: emergency-call-requested" or "MVEGC 3: emergency-call-granted" or the MCVideo imminent peril group call state is set to "MVIGC 2: imminent-peril-call-requested" or "MVIGC 3: imminent-peril-call-granted", the MCVideo client shall perform the actions specified in clause 6.2.8.1.4; and

3) may subscribe to the conference event package as specified in clause 9.1.3.1.

On receiving a SIP 4xx response, a SIP 5xx response or a SIP 6xx response to the SIP INVITE request:

1) if the MCVideo emergency group call state is set to "MVEGC 2: emergency-call-requested" or "MVEGC 3: emergency-call-granted"; or

2) if the MCVideo imminent peril group call state is set to "MVIGC 2: imminent-peril-call-requested" or "MVIGC 3: imminent-peril-call-granted";

the MCVideo client shall perform the actions specified in clause 6.2.8.1.5.

On receiving a SIP INFO request where the Request-URI contains an MCVideo session ID identifying an ongoing group session, the MCVideo client shall follow the actions specified in clause 6.2.8.1.13."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.2.8.2]

"NOTE: This clause is referenced from other procedures.

When the MCVideo user initiates a broadcast group call, the MCVideo client:

1) in the case of the prearranged group call is initiated on-demand, shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body the <broadcast-ind> element set to "true" as defined in clause F.1; and

2) in the case the prearranged group call is initiated using a pre-established session, shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body in the hname "body" parameter in the headers portion of the SIP URI in the Refer-To header field the <broadcast-ind> element set to "true" as defined in clause F.1."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.2]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

NOTE 1: This clause covers on-demand session.

Upon receipt of a "SIP INVITE request for terminating participating MCVideo function", the participating MCVideo function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP INVITE request with a SIP 500 (Server Internal Error) response. The participating MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15], and shall not continue with the rest of the steps;

NOTE 2: if the SIP INVITE request contains an emergency indication or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCVideo function can according to local policy choose to accept the request.

2) shall check the presence of the isfocus media feature tag in the URI of the Contact header field and if it is not present then the participating MCVideo function shall reject the request with a SIP 403 (Forbidden) response with the warning text set to "104 isfocus not assigned" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

3) if the Answer-Mode Indication in the application/poc-settings+xml MIME body has not yet been received from the invited MCVideo client as defined in clause 7.3.3 or clause 7.3.4, shall reject the request with a SIP 480 (Temporarily Unavailable) response with the warning text set to "146 T-PF unable to determine the service settings for the called user" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

4) shall use the MCVideo ID present in the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCVideo ID and public user identity;

5) if the binding between the MCVideo ID and public user identity does not exist, then the participating MCVideo function shall reject the SIP INVITE request with a SIP 404 (Not Found) response. Otherwise, continue with the rest of the steps;

6) if the SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with an <MKFC-GKTPs> element, shall perform the procedures in clause 6.3.2.3.2;

7) shall perform the automatic commencement procedures specified in clause 6.3.2.2.5.1 and according to IETF RFC 5373 [27] if the "SIP INVITE request for terminating participating MCVideo function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCVideo client as per clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; and

8) shall perform the manual commencement procedures specified in clause 6.3.2.2.6.1 and according to IETF RFC 5373 [27] if the "SIP INVITE request for terminating participating MCVideo function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCVideo client as per clause 7.3.3 or clause 7.3.4 is set to "manual-answer"."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.4.1]

"In the procedures in this clause:

1) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

2) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

NOTE 1: This clause covers on-demand session.

Upon receipt of a "SIP INVITE request for terminating participating MCVideo function", the participating MCVideo function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP INVITE request with a SIP 500 (Server Internal Error) response. The participating MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15], and shall not continue with the rest of the steps;

NOTE 2: if the SIP INVITE request contains an emergency indication or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCVideo function can according to local policy choose to accept the request.

2) shall check the presence of the isfocus media feature tag in the URI of the Contact header field and if it is not present then the participating MCVideo function shall reject the request with a SIP 403 (Forbidden) response with the warning text set to "104 isfocus not assigned" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

3) if the Answer-Mode Indication in the application/poc-settings+xml MIME body has not yet been received from the invited MCVideo client as defined in clause 7.3.3 or clause 7.3.4, shall reject the request with a SIP 480 (Temporarily Unavailable) response with the warning text set to "146 T-PF unable to determine the service settings for the called user" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

4) shall use the MCVideo ID present in the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCVideo ID and public user identity;

5) if the binding between the MCVideo ID and public user identity does not exist, then the participating MCVideo function shall reject the SIP INVITE request with a SIP 404 (Not Found) response. Otherwise, continue with the rest of the steps;

6) if the SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with an <MKFC-GKTPs> element, shall perform the procedures in clause 6.3.2.3.2;

7) shall perform the automatic commencement procedures specified in clause 6.3.2.2.5.1 and according to IETF RFC 5373 [27] if the "SIP INVITE request for terminating participating MCVideo function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCVideo client as per clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; and

8) shall perform the manual commencement procedures specified in clause 6.3.2.2.6.1 and according to IETF RFC 5373 [27] if the "SIP INVITE request for terminating participating MCVideo function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCVideo client as per clause 7.3.3 or clause 7.3.4 is set to "manual-answer"."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.1]

"Once an on-demand MCVideo session is established or a pre-established session is in use when the participating MCVideo function receives transmission control messages from the transmission participant in the MCVideo client or from the transmission control server in the controlling MCVideo function, the behaviour of the participating MCVideo function is described in the following clauses."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.2]

"Upon receiving a transmission control message the participating MCVideo function:

1. *shall immediately forward the transmission control message to the transmission control server if the message is received from the transmission participant;*

2. *if an MBMS subchannel is not used for a transmission in the session the transmission control message is associated with, shall immediately forward the transmission control message to the transmission participant if the message is received from the transmission control server; and*

3. *if an MBMS subchannel is used for a transmission in the session the transmission control message is associated with:*

    a. *if*

        i. *the transmission control message is not a Transmission Idle message or a Media Transmission Notify message;*

        ii. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

        iii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report;*

        shall immediately forward the transmission control message to the transmission participant; and

    b. *if*

        i. *the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report; and*

        ii *if the transmission control message is the Transmission Idle message or the Media Transmission Notify message,*

    shall perform actions as specified in clause 10.2.

NOTE: When the Transmit Idle or Media Transmission Notify messages are discarded the messages are sent to the MCVideo clients over the MBMS subchannel allocated for the transmission as specified in clause 10.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.3]

"Upon receiving RTP media packets the participating MCVideo function:

1. *shall immediately forward the RTP media packet to the controlling MCVideo function if the RTP packet is from an MCVideo client; and*

2. *if an MBMS subchannel is not used for a transmission in the session the RTP media packets are associated with, shall immediately forward the RTP media packets to the MCVideo client if the RTP packet is from the controlling MCVideo function or the non-controlling MCVideo function.*

3. *if an MBMS subchannel is used for a transmission in the session the RTP media packets are associated with and if RTP media packets are received from the controlling MCVideo function or the non-controlling MCVideo function:*

    a. *if*

        i. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

ii. the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report,

shall immediately forward the RTP media packets to the MCVideo client; and

b. if the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report, shall perform actions as specified in clause 10.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.4]

"When the participating function receives an indication from the application and signalling plane that session release is initiated, the participating MCVideo function:

1. *shall stop sending transmission control messages towards the transmission participant and the transmission control server; and*
2. *shall stop sending RTP media packets towards the MCVideo client and towards the controlling MCVideo function.*

When the participating MCVideo function receives an indication from the application and signalling plane that the session is released, the participating MCVideo function:

1. *in case of a pre-established session, shall perform the actions in clause 9.3.2; and*
2. *in case of an on-demand session, shall release the media resources associated with the session."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

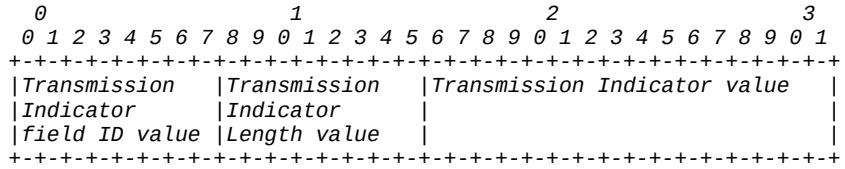
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

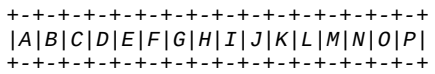


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. *if the active Transmission request queue is empty the transmission control server:*
    a. *shall send Transmission Idle message to all transmission participants. The Transmission Idle message:*
        i. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*
        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;*
    b. *shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;*
    c. *shall start timer T1 (Inactivity);*
    d. *shall set the general state to the 'G: Transmit Idle' state; and*
    e. *shall initialise counter Cx (Simultaneous transmission video) to 0.*

2. *if the active Transmission request queue is not empty the transmission control server:*

    a. *shall select a queued Transmission request from the top of the active Transmission request queue;*

    b. *shall remove that queued Transmission request from the active Transmission request queue; and*

    c. *shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:*

    a. *shall include the granted priority in the Transmission priority field;*

    b. *shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;*

    c. *if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

    d. *shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;*

2. *shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;*

3. *shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:*

    a. *shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;*

    b. *shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;*

    c. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;*

    d. *if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';*

    e. *if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';*

    f. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

    g. *shall initiate the 'general reception control operation' state machine; and*

4. *shall enter the 'G: Transmit Taken' state."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. *if counter Cx (Simultaneous transmission video) has not reached its upper limit:*

    a. *shall perform the actions specified in the clause 6.3.4.4.2;*

2. *if counter Cx (Simultaneous transmission video) has reached its upper limit:*

    a. *select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;*

    b. *shall stop timer T4 (Transmission Grant), if running;*

    c. *shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);*

    d. *shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;*

    e. *shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and*

    f. *shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:*

        i. *shall include the queue position and transmission priority in the Queue Info field; and*

        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:*

    a. *shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and*

    b. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

2. *shall set the general state to 'G: pending Transmission Revoke'."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. *if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:*

    a. *shall initialize a general state machine as specified in clause 6.3.4.2.2; and*

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

    b. *shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;*

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.1.12.3 Test description

5.1.1.12.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the controlling MCVideo function.
- SS-UE2 (MCVideo Client).

IUT:

- IUT (MCVideo Server), providing the terminating participating MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of some sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server and the HTTP Client.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server) and SS-UE2 (MCVideo Client).
- The SS-UE2 (MCVideo client) is registered and authorized to use the MCVideo system.

- The SS-UE2 (MCVideo client) is affiliated to the pre-arranged group and is authorized to initiate broadcast group calls.

5.1.1.12.3.2 Test procedure sequence

Table 5.1.1.12.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CT group/private call establishment, automatic commencement' as described in Annex A, Table A.4.3.3.3-1 to establish a broadcast group call with automatic commencement mode and notify a media transmission from SS-UE1 (MCVideo Client)?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Notification - Idle' as described in Annex A, Table A.4.3.8.3-1 to notify the end of SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.3.5.3-1, to release the call?	-	-	3	P

5.1.1.12.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

Table 5.1.1.12.3.3-1: SIP INVITE from SS (MCVideo Server)
(Step 1, Table 5.1.1.12.3.2-1; Step 1, Annex A, Table A.4.3.3.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-1, with conditions GROUP-CALL and BROADCAST-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.1.12.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.1.12.3.2-1; Step 2, Annex A, Table A.4.3.3.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-2, with conditions GROUP-CALL and BROADCAST-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.1.12.3.3-3: SIP 200 (OK) from SS-UE2 (MCVideo Client)
(Step 1, Table 5.1.1.12.3.2-1; Step 6, Annex A, Table A.4.3.3.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-7, with condition GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.1.12.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.1.1.12.3.2-1; Step 7, Annex A, Table A.4.3.3.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-8, with condition GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.1.12.3.3-5: Transmission Idle
(Step 2, Table 5.1.1.12.3.2-1; Steps 13 and 14, Annex A, Table A.4.3.8.3-1)

Derivation Path: Annex A, Table A.4.3.8.4-6, with condition BROADCAST-CALL.

Table 5.1.1.12.3.3-6: Media Transmission Notification
(Step 1, Table 5.1.1.12.3.2-1; Steps 10a1 and 10a2, Annex A, Table 4.3.3.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-11, with condition BROADCAST-CALL.

5.1.1.13 MCVideo Server - MCVideo Server / Broadcast Group Call / Originating Participating server

5.1.1.13.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS-UE1 (MCVideo Client) initiates a broadcast group call with implicit transmission
control }
    then { IUT (MCVideo Server) accepts the call from initiator and establishes the call with the SS
(MCVideo Server) providing the controlling function and forwards the transmission granted received
from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established a broadcast group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests to end the transmission control }
    then { IUT (MCVideo Server) forwards the transmission end request to SS (MCVideo Server) and
forwards the transmission end response to SS-UE1 (MCVideo Client) and forwards the transmission idle
received from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established a broadcast group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control }
    then { IUT (MCVideo Server) forwards the transmission request to SS (MCVideo Server) and
forwards the transmission granted received from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(4)

```
with { IUT (MCVideo Server) having established a broadcast group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) ends the broadcast group call }
    then { IUT (MCVideo Server) sends a SIP BYE message to SS (MCVideo Server) providing the
controlling function and responds by sending a SIP 200 (OK) message to SS-UE1 (MCVideo Client) }
            }
```

5.1.1.13.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.1.2.1.1, 6.2.8.2, 9.2.1.3.1.1 and 9.2.1.3.3.1, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.4.1, 6.4.2, 6.4.3, 6.4.4, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.2.1.1]

"Upon receiving a request from an MCVideo user to establish an MCVideo prearranged group session the MCVideo client shall generate an initial SIP INVITE request by following the UE originating session procedures specified in 3GPP TS 24.229 [11], with the clarifications given below.

The MCVideo client:

1) *if the MCVideo user has requested the origination of an MCVideo emergency group call or is originating an MCVideo prearranged group call and the MCVideo emergency state is already set, the MCVideo client shall comply with the procedures in clause 6.2.8.1.1;*

2) *if the MCVideo user has requested the origination of an MCVideo imminent peril group call, the MCVideo client shall comply with the procedures in clause 6.2.8.1.9;*

3) *if the MCVideo user has requested the origination of a broadcast group call, the MCVideo client shall comply with the procedures in clause 6.2.8.2;*

4) *shall include the g.3gpp.mcvideo media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo" in the Contact header field of the SIP INVITE request according to IETF RFC 3840 [22];*

5) *shall include an Accept-Contact header field containing the g.3gpp.mcvideo media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [20];*

6) *shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcvideo" (coded as specified in 3GPP TS 24.229 [11]), in a P-Preferred-Service header field according to IETF RFC 6050 [14] in the SIP INVITE request;*

7) *shall include an Accept-Contact header field with the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [20];*

8) *should include the "timer" option tag in the Supported header field;*

9) *should include the Session-Expires header field according to IETF RFC 4028 [23]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";*

10) *shall set the Request-URI of the SIP INVITE request to the public service identity identifying the participating MCVideo function serving the MCVideo user;*

NOTE 1: The MCVideo client is configured with public service identity identifying the participating MCVideo function serving the MCVideo user.

11) *may include a P-Preferred-Identity header field in the SIP INVITE request containing a public user identity as specified in 3GPP TS 24.229 [11];*

12) *if the MCVideo emergency state is already set or the MCVideo client emergency group state for this group is set to "MVEG 2: in-progress", the MCVideo client shall include the Resource-Priority header field and comply with the procedures in clause 6.2.8.1.2;*

13) *if the MCVideo client imminent peril group state for this group is set to "MVIG 2: in-progress" or "MVIG 4: confirm-pending" shall include the Resource-Priority header field and comply with the procedures in clause 6.2.8.1.12;*

14) *shall contain in an application/vnd.3gpp.mcvideo-info+xml MIME body with the <mcvideoinfo> element containing the <mcvideo-Params> element with:*

    a) *the <session-type> element set to a value of "prearranged";*

    b) *the <mcvideo-request-uri> element set to the group identity;*

    c) *the <mcvideo-client-id> element set to the MCVideo client ID of the originating MCVideo client; and*

NOTE 2: The MCVideo client does not include the MCVideo ID of the originating MCVideo user in the body, as this will be inserted into the body of the SIP INVITE request that is sent from the originating participating MCVideo function.

    d) *if the group identity can be determined to be a TGI and if the MCVideo client can associate the TGI with a MCVideo group ID, the <associated-group-id> element set to the MCVideo group ID;*

NOTE 3: The text "can associate the TGI with a MCVideo group ID" means that the MCVideo client is able to determine that there is a constituent group of the temporary group that it is a member of.

NOTE 4: The MCVideo client is informed about temporary groups and regouping of MCVideo groups that the user is a member of as specified in 3GPP TS 24.481 [24].

NOTE 5: If the MCVideo user selected a TGI where there are several MCVideo groups where the MCVideo user is a member, the MCVideo client selects one of those MCVideo groups.

15) shall include an SDP offer according to 3GPP TS 24.229 [11] with the clarifications given in clause 6.2.1;

16) if an implicit transmission request is required, shall indicate this as specified in clause 6.4; and

17) shall send the SIP INVITE request towards the MCVideo server according to 3GPP TS 24.229 [11].

On receiving a SIP 2xx response to the SIP INVITE request, the MCVideo client:

1) shall interact with the user plane as specified in 3GPP TS 24.581 [5] ;

2) if the MCVideo emergency group call state is set to "MVEGC 2: emergency-call-requested" or "MVEGC 3: emergency-call-granted" or the MCVideo imminent peril group call state is set to "MVIGC 2: imminent-peril-call-requested" or "MVIGC 3: imminent-peril-call-granted", the MCVideo client shall perform the actions specified in clause 6.2.8.1.4; and

3) may subscribe to the conference event package as specified in clause 9.1.3.1.

On receiving a SIP 4xx response, a SIP 5xx response or a SIP 6xx response to the SIP INVITE request:

1) if the MCVideo emergency group call state is set to "MVEGC 2: emergency-call-requested" or "MVEGC 3: emergency-call-granted"; or

2) if the MCVideo imminent peril group call state is set to "MVIGC 2: imminent-peril-call-requested" or "MVIGC 3: imminent-peril-call-granted";

the MCVideo client shall perform the actions specified in clause 6.2.8.1.5.

On receiving a SIP INFO request where the Request-URI contains an MCVideo session ID identifying an ongoing group session, the MCVideo client shall follow the actions specified in clause 6.2.8.1.13."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.2.8.2]

"NOTE: This clause is referenced from other procedures.

When the MCVideo user initiates a broadcast group call, the MCVideo client:

1) in the case of the prearranged group call is initiated on-demand, shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body the <broadcast-ind> element set to "true" as defined in clause F.1; and

2) in the case the prearranged group call is initiated using a pre-established session, shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body in the hname "body" parameter in the headers portion of the SIP URI in the Refer-To header field the <broadcast-ind> element set to "true" as defined in clause F.1."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.1.1]

"In the procedures in this clause:

1) group identity in an incoming SIP INVITE request refers to the group identity from the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;

2) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

3) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

Upon receipt of a "SIP INVITE request for originating participating MCVideo function" containing an application/vnd.3gpp.mcvideo-info+xml MIME body with the <session-type> element set to a value of "prearranged", the participating MCVideo function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP INVITE request with a SIP 500 (Server Internal Error) response. The participating MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15]. Otherwise, continue with the rest of the steps;

NOTE 1: if the SIP INVITE request contains an emergency indication or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCVideo function can according to local policy choose to accept the request.

2) shall determine the MCVideo ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request, and shall authorise the calling user;

NOTE 2: The MCVideo ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

3) if through local policy in the participating MCVideo function, the user identified by the MCVideo ID is not authorised to initiate prearranged group calls, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response to the SIP INVITE request, with warning text set to "109 user not authorised to make prearranged group calls" in a Warning header field as specified in clause 4.4;

4) shall validate the media parameters and if the MCVideo codecs are not offered in the SIP INVITE request shall reject the request with a SIP 488 (Not Acceptable Here) response. Otherwise, continue with the rest of the steps;

5) shall check if the number of maximum simultaneous MCVideo group calls supported for the MCVideo user as specified in the <MaxSimultaneousCallsN6> element of the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) has been exceeded. If exceeded, the participating MCVideo function shall respond with a SIP 486 (Busy Here) response with the warning text set to "103 maximum simultaneous MCVideo group calls reached" in a Warning header field as specified in clause 4.4. Otherwise, continue with the rest of the steps;

NOTE 3: If the SIP INVITE request contains an emergency indication or an imminent peril indication, the participating MCVideo function can by means beyond the scope of this specification choose to allow for an exception to the limit for the maximum simultaneous MCVideo sessions supported for the MCVideo user. Alternatively, a lower priority session of the MCVideo user could be terminated to allow for the new session.

6) if the user identified by the MCVideo ID is not affiliated to the group identified in the "SIP INVITE request for originating participating MCVideo function" as determined by clause 8.2.2.2.11 and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, shall perform the actions specified in clause 8.2.2.2.12 for implicit affiliation;

7) if the actions for implicit affiliation specified in step 6) above were performed but not successful in affiliating the MCVideo user due to the MCVideo user already having N2 simultaneous affiliations, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 486 (Busy Here) response with the warning text set to "102 too many simultaneous affiliations" in a Warning header field as specified in clause 4.4. and skip the rest of the steps.

NOTE 4: N2 is the total number of MCVideo groups that an MCVideo user can be affiliated to simultaneously as specified in 3GPP TS 23.281 [26].

NOTE 5: if the SIP INVITE request contains an emergency indication set to a value of "true" or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCVideo function can according to local policy choose to allow an exception to the N2 limit. Alternatively, a lower priority affiliation of the MCVideo user could be cancelled to allow for the new affiliation.

8) shall determine the public service identity of the controlling MCVideo function associated with the group identity in the SIP INVITE request;

NOTE 6: The public service identity can identify the controlling MCVideo function in the primary MCVideo system or a partner MCVideo system.

NOTE 7: How the participating MCVideo server discovers the public service identity of the controlling MCVideo function associated with the group identity is out of scope of the current release.

9) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3;

10) shall set the Request-URI to the public service identity of the controlling MCVideo function associated with the group identity which was present in the incoming SIP INVITE request;

11) shall not copy the following header fields from the incoming SIP INVITE request to the outgoing SIP INVITE request, if they were present in the incoming SIP INVITE request:

a) Answer-Mode header field as specified in IETF RFC 5373 [27]; and

b) Priv-Answer-Mode header field as specified in IETF RFC 5373 [27];

12) shall set the <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the SIP INVITE request to the MCVideo ID of the calling user;

13) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the MCVideo client as specified in clause 6.3.2.1.1.1;

14) if the received SIP INVITE request contains an application/vnd.3gpp.location-info+xml MIME body and if not already copied, shall copy the contents of the application/vnd.3gpp.location-info+xml MIME body received in the SIP INVITE request into an application/vnd.3gpp.location-info+xml MIME body included in the outgoing SIP request;

15) if a Resource-Priority header field was included in the received SIP INVITE request, shall include a Resource-Priority header field according to rules and procedures of 3GPP TS 24.229 [11] set to the value indicated in the Resource-Priority header field of the SIP INVITE request from the MCVideo client; and

NOTE 8: The participating MCVideo function will leave verification of the Resource-Priority header field to the controlling MCVideo function.

16) shall forward the SIP INVITE request, according to 3GPP TS 24.229 [11].

Upon receipt of a SIP 302 (Moved Temporarily) response to the above SIP INVITE request in step 15), the participating MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.2.1.10;

2) shall include an SDP offer based upon the SDP offer in the received SIP INVITE request from the MCVideo client as specified in clause 6.3.2.1.1.1; and

3) shall forward the SIP INVITE request according to 3GPP TS 24.229 [11].

Upon receipt of a SIP 2xx response in response to the above SIP INVITE request in step 15), the participating MCVideo function:

1) if the received SIP 2xx response contains an application/vnd.3gpp.mcvideo-info+xml MIME body with an <MKFC-GKTPs> element, shall perform the procedures in clause 6.3.2.3.2;

2) shall generate a SIP 200 (OK) response as in clause 6.3.2.1.5.2;

3) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

4) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response;

5) shall include the public service identity received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response;

6) shall include an MCVideo session identity mapped to the MCVideo session identity provided in the Contact header field of the received SIP 200 (OK) response;

7) if the procedures of clause 8.2.2.2.12 for implicit affiliation were performed in the present clause, shall complete the implicit affiliation by performing the procedures of clause 8.2.2.2.13;

8) shall send the SIP 200 (OK) response to the MCVideo client according to 3GPP TS 24.229 [11];

9) shall interact with Media Plane as specified in 3GPP TS 24.581 [5]; and

10) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [23].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the above SIP INVITE request in step 14) the participating MCVideo function:

1) shall generate a SIP response according to 3GPP TS 24.229 [11];

2) shall include Warning header field(s) that were received in the incoming SIP response;

3) shall forward the SIP response to the MCVideo client according to 3GPP TS 24.229 [11]; and

4) if the implicit affiliation procedures of clause 8.2.2.2.12 were invoked in this procedure, shall perform the procedures of clause 8.2.2.2.14;"

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.3.3.1]

"Upon receiving from the MCVideo client a SIP BYE request the participating MCVideo function shall follow the procedures as specified in clause 6.3.2.1.6."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.1]

"Once an on-demand MCVideo session is established or a pre-established session is in use when the participating MCVideo function receives transmission control messages from the transmission participant in the MCVideo client or from the transmission control server in the controlling MCVideo function, the behaviour of the participating MCVideo function is described in the following clauses."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.2]

"Upon receiving a transmission control message the participating MCVideo function:

1. *shall immediately forward the transmission control message to the transmission control server if the message is received from the transmission participant;*
2. *if an MBMS subchannel is not used for a transmission in the session the transmission control message is associated with, shall immediately forward the transmission control message to the transmission participant if the message is received from the transmission control server; and*
3. *if an MBMS subchannel is used for a transmission in the session the transmission control message is associated with:*
    a. *if*
        i. *the transmission control message is not a Transmission Idle message or a Media Transmission Notify message;*
        ii. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*
        iii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report;*

        shall immediately forward the transmission control message to the transmission participant; and
    b. *if*
        i. *the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report; and*

ii if the transmission control message is the Transmission Idle message or the Media Transmission Notify message,

shall perform actions as specified in clause 10.2.

NOTE: When the Transmit Idle or Media Transmission Notify messages are discarded the messages are sent to the MCVideo clients over the MBMS subchannel allocated for the transmission as specified in clause 10.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.3]

"Upon receiving RTP media packets the participating MCVideo function:

1. *shall immediately forward the RTP media packet to the controlling MCVideo function if the RTP packet is from an MCVideo client; and*
2. *if an MBMS subchannel is not used for a transmission in the session the RTP media packets are associated with, shall immediately forward the RTP media packets to the MCVideo client if the RTP packet is from the controlling MCVideo function or the non-controlling MCVideo function.*
3. *if an MBMS subchannel is used for a transmission in the session the RTP media packets are associated with and if RTP media packets are received from the controlling MCVideo function or the non-controlling MCVideo function:*
   a. *if*
      i. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*
      ii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report,*

      shall immediately forward the RTP media packets to the MCVideo client; and
   b. *if the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report, shall perform actions as specified in clause 10.2."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.4]

"When the participating function receives an indication from the application and signalling plane that session release is initiated, the participating MCVideo function:

1. *shall stop sending transmission control messages towards the transmission participant and the transmission control server; and*
2. *shall stop sending RTP media packets towards the MCVideo client and towards the controlling MCVideo function.*

When the participating MCVideo function receives an indication from the application and signalling plane that the session is released, the participating MCVideo function:

1. *in case of a pre-established session, shall perform the actions in clause 9.3.2; and*
2. *in case of an on-demand session, shall release the media resources associated with the session."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

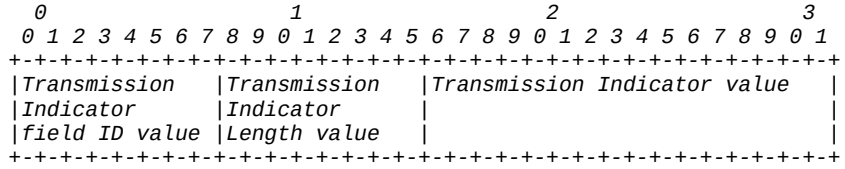
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

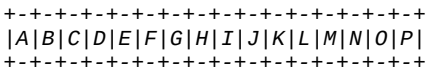


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. *if the active Transmission request queue is empty the transmission control server:*
    a. *shall send Transmission Idle message to all transmission participants. The Transmission Idle message:*
        i. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*
        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;*
    b. *shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;*
    c. *shall start timer T1 (Inactivity);*
    d. *shall set the general state to the 'G: Transmit Idle' state; and*
    e. *shall initialise counter Cx (Simultaneous transmission video) to 0.*
2. *if the active Transmission request queue is not empty the transmission control server:*
    a. *shall select a queued Transmission request from the top of the active Transmission request queue;*
    b. *shall remove that queued Transmission request from the active Transmission request queue; and*
    c. *shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:*
    a. *shall include the granted priority in the Transmission priority field;*
    b. *shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;*
    c. *if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*
    d. *shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;*
2. *shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;*
3. *shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:*
    a. *shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;*
    b. *shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;*

c. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;

d. if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';

e. if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

g. shall initiate the 'general reception control operation' state machine;and

4. shall enter the 'G: Transmit Taken' state."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. if counter Cx (Simultaneous transmission video) has not reached its upper limit:

a. shall perform the actions specified in the clause 6.3.4.4.2;

2. if counter Cx (Simultaneous transmission video) has reached its upper limit:

a. select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;

b. shall stop timer T4 (Transmission Grant), if running;

c. shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);

d. shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;

e. shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and

f. shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:

i. shall include the queue position and transmission priority in the Queue Info field; and

ii. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:

a. shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and

b. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

2. shall set the general state to 'G: pending Transmission Revoke'."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

b. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.1.13.3 Test description

5.1.1.13.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the controlling MCVideo function.
- SS-UE1 (MCVideo Client).

IUT:

- IUT (MCVideo Server), providing the originating participating MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of some sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server and the HTTP Client.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server) and SS-UE1 (MCVideo Client).
- The SS-UE1 (MCVideo client) is registered and authorized to use the MCVideo system.
- The SS-UE1 (MCVideo client) is affiliated to the pre-arranged group and is authorized to initiate broadcast group calls.

5.1.1.13.3.2 Test procedure sequence

Table 5.1.1.13.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.4.2.1.3-1 to establish a broadcast group call with implicit transmission control request according to option b.ii of note in Annex A, Table A.4.2.1.3-1, initiated by SS-UE1 (MCVideo Client?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO Transmission End Request - Transmission End Response' as described in Annex A, Table A.4.2.5.3-1 to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Transmission Granted' as described in Annex A, Table A.4.2.4.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	3	P
4	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO call release' as described in Annex A, Table A.4.2.3.3-1, to release the call?	-	-	4	P

5.1.1.13.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

Table 5.1.1.13.3.3-1: SIP INVITE from SS-UE1 (MCVideo Client)
(Step 1, Table 5.1.1.13.3.2-1; Step 1, Annex A, Table A.4.2.1.3-1)

Derivation Path: Annex A, Table A.4.2.1.4-1, with conditions GROUP-CALL and BROADCAST-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.13.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.1.13.3.2-1; Step 2, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-2, with conditions GROUP-CALL and BROADCAST-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.13.3.3-3: SIP 200 (OK) from SS (MCVideo Server)
(Step 1, Table 5.1.1.13.3.2-1; Step 5, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-3, with condition GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.13.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.1.1.13.3.2-1; Step 6, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-4, with condition GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.1.13.3.3-5: Transmission Idle
(Steps 2, Table 5.1.1.13.3.2-1; Steps 9 and 10, Annex A, Table A.4.2.5.3-1)**

Derivation Path: Annex A, Table A.4.2.5.4-3, with condition BROADCAST-CALL.

**Table 5.1.1.13.3.3-6: Transmission Request
(Step 3, Table 5.1.1.13.3.2-1; Steps 1 and 2, Annex A, Table A.4.2.4.3-1)**

Derivation Path: Annex A, Table A.4.2.4.4-1, with condition BROADCAST-CALL.

**Table 5.1.1.13.3.3-7: Transmission Granted
(Steps 1 and 3, Table 5.1.1.13.3.2-1;
Steps 5 and 6, Annex A, Table A.4.2.4.3-1; Steps 9a1 and 9a2, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.4.4-2, with condition BROADCAST-CALL.

5.1.1.14 MCVideo Server - MCVideo Server / Pre-arranged Group Call / Transmission Control State Transitions / Controlling server

5.1.1.14.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function initiates a pre-
arranged group call without implicit transmission control }
    then { IUT (MCVideo Server) accepts the call from initiator and establishes the call with all
affiliated users by sending call establishment requests to SS (MCVideo Server) providing the
terminating participating function and sends transmission idle to SS-UE1 (MCVideo Client) and SS-UE2
(MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control }
    then { IUT (MCVideo Server) sends transmission granted to SS-UE1 (MCVideo Client) and sends
media transmission notification to SS-UE2 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE2 (MCVideo Client) requests the transmission control with higher priority than SS-
UE1 (MCVideo Client) while SS-UE1 is transmitting media }
    then { IUT (MCVideo Server) revokes SS-UE1 (MCVideo Client) media transmission permission and
grants the transmission to SS-UE2 (MCVideo Client) by sending a transmission granted message and
sends media transmission notification to SS-UE1 (MCVideo Client) }
            }
```

(4)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests to end the transmission control }
    then { IUT (MCVideo Server) sends transmission control idle to SS-UE1 (MCVideo Client) and SS-
UE2 (MCVideo Client) and stops forwarding media from SS-UE1 (MCVideo Client) }
            }
```

(5)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE2 (MCVideo Client) requests to end the transmission control }
    then { IUT (MCVideo Server) sends transmission control idle to SS-UE1 (MCVideo Client) and SS-
UE2 (MCVideo Client) and stops forwarding media from SS-UE2 (MCVideo Client) }
            }
```

(6)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE2 (MCVideo Client) requests the transmission control with high priority }
    then { IUT (MCVideo Server) sends transmission granted to SS-UE2 (MCVideo Client) and sends
media transmission notification to SS-UE1 (MCVideo Client) }
            }
```

(7)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control with lower priority than SS-
UE2 (MCVideo Client), while SS-UE2 is transmitting media }
    then { IUT (MCVideo Server) queues SS-UE1 (MCVideo Client) media transmission request and
responds with a queue position info message to SS-UE1 (MCVideo Client) }
            }
```

(8)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests queue position information }
    then { IUT (MCVideo Server) responds with a queue position info message to SS-UE1 (MCVideo
Client) }
            }
```

(9)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests to cancel queued transmission request }
    then { IUT (MCVideo Server) accepts SS-UE1 (MCVideo client) transmission request cancellation
and removes the request from the queue }
            }
```

(10)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE2 (MCVideo Client) requests to end the transmission control while SS-UE1 (MCVideo
Client) transmission request is queued }
    then { IUT (MCVideo Server) accepts SS-UE2 (MCVideo Client) transmission end request and stops
forwarding media from SS-UE2 (MCVideo Client) and sends transmission granted to SS-UE1 (MCVideo
Client) and sends media transmission notification to SS-UE2 (MCVideo Client) }
            }
```

(11)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control with lower priority than SS-
UE2 (MCVideo Client), while SS-UE2 is transmitting media }
    then { IUT (MCVideo Server) rejects SS-UE1 (MCVideo Client) media transmission request by
sending a transmission rejected message }
            }
```

(12)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) transmits for a longer period than allowed in the transmission
granted }
    then { IUT (MCVideo Server) revokes SS-UE1 (MCVideo Client) media transmission permission and
stops forwarding media }
            }
```

(13)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function ends the pre-
arranged group call }
    then { IUT (MCVideo Server) responds by sending a SIP 200 (OK) message to SS (MCVideo Server)
ending the call and sends a SIP BYE message to SS (MCVideo Server) providing the terminating
participating function }
            }
```

5.1.1.14.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.1.4.1.1, 9.2.1.4.1.2, 9.2.1.4.2, 9.2.1.4.3, 6.3.3.1.2 and 6.3.3.1.1, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.3.2.2, 6.3.3, 6.4.2, 6.4.3, 6.4.4, 9.2.1, 6.2.1, 6.2.2, 6.2.4, 6.3.3 and 6.3.5.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.1.1]

"This clause describes the procedures for inviting an MCVideo user to an MCVideo session. The procedure is initiated by the controlling MCVideo function as the result of an action in clause 9.2.1.4.2 or as the result of receiving a SIP 403 (Forbidden) response as described in this clause.

The controlling MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the public service identity of the terminating participating MCVideo function associated to the MCVideo user to be invited.;

NOTE 1: How the controlling MCVideo function finds the address of the terminating MCVideo participating function is out of the scope of the current release.

NOTE 2: If the terminating MCVideo user is part of a partner MCVideo system, then the public service identity can identify an entry point in the partner network that is able to identify the terminating participating MCVideo function.

3) shall set the P-Asserted-Identity header field to the public service identity of the controlling MCVideo function;

4) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcvideo-request-uri> element set to the MCVideo ID of the terminating user; and

b) the <mcvideo-calling-group-id> element set to the group identity;

NOTE 3: The <mcvideo-calling-user-id> is already included in the MIME body as a result of calling clause 6.3.3.1.2 in step 1).

5) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1;

6) if the in-progress emergency state of the group is set to a value of "true" the controlling MCVideo function:

a) shall include a Resource-Priority header field populated with the values for an MCVideo emergency group call as specified in clause 6.3.3.1.19;

b) if the received SIP INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "true":

i) shall include in the outgoing SIP INVITE request in the application/vnd.3gpp.mcvideo-info+xml MIME body an <emergency-ind> element set to a value of "true"; and

ii) if the <alert-ind> element is set to "true" in the received SIP INVITE request and the requesting MCVideo user and MCVideo group are authorised for the initiation of MCVideo emergency alerts as determined by the procedures of clause 6.3.3.1.13.1, shall populate the application/vnd.3gpp.mcvideo-info+xml MIME body and the application/vnd.3gpp.location-info+xml MIME body as specified in clause 6.3.3.1.12. Otherwise, shall set the <alert-ind> element to a value of "false"; and

c) if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <imminentperil-ind> element set to a value of "false";

7) if the in-progress emergency state of the group is set to a value of "false" and the in-progress imminent peril state of the group is set to a value of "true", the controlling MCVideo function:

a) shall include a Resource-Priority header field populated with the values for an MCVideo imminent peril group call as specified in clause 6.3.3.1.19; and

b) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true";

8) if:

a) an MCVideo GKTP document exists for the group identity contained in the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request; and

b) the MCVideo GKTP document contains a <MKFC-GKTPs> element;

then:

a) for each instance of <GKTP> element of the <MKFC-GKTPs> element of the MCVideo GKTP document:

i) shall perform the procedure in clause 6.3.3.6.2 to re-generate an I_MESSAGE; and

ii) if the procedure in clause 6.3.3.6.2 was successful, shall include the I_MESSAGE in a <GKTP> element of the <MKFC-GKTPs> element of an application/vnd.3gpp.mcvideo-info+xml MIME body included in the outgoing SIP INVITE request; and

9) shall send the SIP INVITE request towards the terminating network in accordance with 3GPP TS 24.229 [11].

Upon receiving a SIP 183 (Session Progress) response containing a Require header field with the option tag "100rel" and containing a P-Answer-State header field with the value "Unconfirmed" in response to the SIP INVITE request the controlling MCVideo function:

1) shall send a SIP PRACK request towards the MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCVideo function:

1) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

2) shall send a SIP NOTIFY request to all participants with a subscription to the conference event package as specified in clause 9.2.3.4; and

3) shall increment the local counter of the number of SIP 200 (OK) responses received from invited members, by 1.

NOTE 4: The notifications above could be sent prior to the SIP 200 (OK) response being sent to the inviting MCVideo client. These notifications received by MCVideo clients that are group members do not mean that the group session will be successfully established.

NOTE 5: The procedures executed by the controlling MCVideo function prior to sending a response to the inviting MCVideo client are specified in clause 9.2.1.4.2."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.1.2]

"The controlling MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the public service identity of the non-controlling MCVideo function serving the group identity of the MCVideo group owned by the partner MCVideo system;

3) shall set the P-Asserted-Identity to the public service identity of the controlling MCVideo function;

4) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcvideo-request-uri> element set to the group identity of the MCVideo group hosted by the non-controlling MCVideo function in the partner MCVideo system; and

b) the <mcvideo-calling-group-id> element set to the group identity of the group served by the controlling MCVideo function;

5) shall include the Recv-Info header field set to g.3gpp.mcvideo-transmission-request;

6) if:

a) an MCVideo GKTP document exists for the group identity contained in the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request; and

b) the MCVideo GKTP document contains a <MKFC-GKTPs> element;

then:

a) for each instance of <GKTP> element of the <MKFC-GKTPs> element of the MCVideo GKTP document:

i) shall perform the procedure in clause 6.3.3.6.2 to re-generate an I_MESSAGE; and

ii) if the procedure in clause 6.3.3.6.2 was successful, shall include the I_MESSAGE in a <GKTP> element of the <MKFC-GKTPs> element of an application/vnd.3gpp.mcvideo-info+xml MIME body included in the outgoing SIP INVITE request;

7) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1; and

8) shall send the SIP INVITE request towards the partner MCVideo system in accordance with 3GPP TS 24.229 [11].

Upon receiving SIP 403 (Forbidden) response for the SIP INVITE request, if according to local policy and if:

1) the response contains a Warning header field with the MCVideo warning code "128"; and

2) the response contains a P-Refused-URI-List header field and an application/resource-lists+xml MIME body as specified in IETF RFC 5318 [28];

NOTE 1: The application/resource-lists+xml MIME body contains MCVideo IDs identifying MCVideo users in a partner MCVideo system that needs to be invited to the prearranged group call in case of group regrouping using interrogating method.

Editor's Note:The above note currently isn't defined in the 23.280 and 23.281.

then the controlling MCVideo function:

1) shall check if the number of members of the MCVideo group exceeds the value contained in the <on-network-max-participant-count> element of the group document as specified in 3GPP TS 24.481 [24]. If exceeded, the controlling MCVideo function shall invite only <on-network-max-participant-count> members from the application/resource-lists+xml MIME body; and

NOTE 2: The <on-network-max-participant-count> element indicates the maximum number of participants allowed in the prearranged group session It is operator policy that determines which participants in the application/resource-lists+xml MIME body are invited to the group call.

2) shall invite MCVideo users as specified in this clause using the list of MCVideo IDs in URI-List.

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCVideo function:

1) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3; and

NOTE 3: The procedures executed by the controlling MCVideo function prior to sending a response to the inviting MCVideo client are specified in clause 9.2.1.4.2.

2) if at least one of the invited MCVideo clients has subscribed to the conference package, shall subscribe to the conference event package in the non-controlling MCVideo function as specified in clause 9.2.3.4.3."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.2]

"In the procedures in this clause:

1) MCVideo ID in an incoming SIP INVITE request refers to the MCVideo ID of the originating user from the <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;

2) group identity in an incoming SIP INVITE request refers to the group identity from the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;

3) MCVideo ID in an outgoing SIP INVITE request refers to the MCVideo ID of the called user in the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the outgoing SIP INVITE request;

4) indication of required group members in a SIP 183 (Session Progress) response refers to the <required> element of the application/vnd.3gpp.mcvideo-info+xml MIME body set to "true" in a SIP 183 (Session Progress) sent by the non-controlling MCVideo function of an MCVideo group;

5) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

6) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

Upon receipt of a "SIP INVITE request for controlling MCVideo function of an MCVideo group", the controlling MCVideo function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP INVITE request with a SIP 500 (Server Internal Error) response. The controlling MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15] and skip the rest of the steps;

NOTE 1: if the SIP INVITE request contains an emergency indication or an imminent peril indication set to a value of "true" and this is an authorised request for originating an MCVideo emergency group call as determined by clause 6.3.3.1.13.2, or for originating an MCVideo imminent peril group call as determined by clause 6.3.3.1.13.5, the controlling MCVideo function can according to local policy choose to accept the request.

2) shall determine if the media parameters are acceptable and the MCVideo codecs are offered in the SDP offer and if not reject the request with a SIP 488 (Not Acceptable Here) response and skip the rest of the steps;

3) shall reject the SIP request with a SIP 403 (Forbidden) response and not process the remaining steps if:

a) an Accept-Contact header field does not include the g.3gpp.mcvideo media feature tag; or

b) an Accept-Contact header field does not include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo";

4) if received SIP INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.3.1.17;

5) shall retrieve the necessary group document(s) from the group management server for the group identity contained in the SIP INVITE request and carry out initial processing as specified in clause 6.3.5.2;

6) if the result of the initial processing in clause 6.3.5.2 was:

a) that authorization of the MCVideo ID is required at a non-controlling MCVideo function of an MCVideo group is required, perform the actions in clause 6.3.3.1.13.7 and do not continue with the rest of the steps in this clause; and

b) that a SIP 3xx, 4xx, 5xx or 6xx response to the "SIP INVITE request for controlling MCVideo function of an MCVideo group" has been sent, do not continue with the rest of the steps in this clause;

7) shall perform the actions as described in clause 6.3.3.2.2;

8) shall maintain a local counter of the number of SIP 200 (OK) responses received from invited members and shall initialise this local counter to zero;

9) shall determine if an MCVideo group call for the group identity is already ongoing by determining if an MCVideo session identity has already been allocated for the group call and the MCVideo session is active;

10) if the SIP INVITE request contains an unauthorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2:

a) shall reject the SIP INVITE request with a SIP 403 (Forbidden) response as specified in clause 6.3.3.1.14; and

b) shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [11] and skip the rest of the steps;

11) if the SIP INVITE request contains an unauthorised request for an MCVideo imminent peril group call as determined by clause 6.3.3.1.13.5, shall reject the SIP INVITE request with a SIP 403 (Forbidden) response with the following clarifications:

a) shall include in the SIP 403 (Forbidden) response an application/vnd.3gpp.mcvideo-info+xml MIME body as specified in clause F.1 with the <mcvideoinfo> element containing the <mcvideo-Params> element with the <imminentperil-ind> element set to a value of "false"; and

b) shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [11] and skip the rest of the *steps;*

12) if a Resource-Priority header field is included in the SIP INVITE request:

a) if the Resource-Priority header field is set to the value indicated for emergency calls and the SIP INVITE request does not contain an emergency indication and the in-progress emergency state of the group is set to a value of "false", shall reject the SIP INVITE request with a SIP 403 (Forbidden) response and skip the rest of the steps; or

b) if the Resource-Priority header field is set to the value indicated for imminent peril calls and the SIP INVITE request does not contain an imminent peril indication and the in-progress imminent peril state of the group is set to a value of "false", shall reject the SIP INVITE request with a SIP 403 (Forbidden) response and skip the rest of the steps;

13) if the MCVideo group call is not ongoing then:

a) if:

i) the user identified by the MCVideo ID is not affiliated to the group identity contained in the SIP INVITE request as specified in clause 6.3.6;

ii) the group identity contained in the SIP INVITE request is not a constituent MCVideo group ID;

iii) the received SIP INVITE request does not contain an emergency indication or imminent peril indication; or

iv) the received SIP INVITE request is an authorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2 or MCVideo imminent peril group call as determined by steps clause 6.3.3.1.13.5 and is determined to not be eligible for implicit affiliation as specified in clause 8.2.2.3.6;

then shall return a SIP 403 (Forbidden) response with the warning text set to "120 user is not affiliated to this group" in a Warning header field as specified in clause 4.4, and skip the rest of the steps below;

b) if the user identified by the MCVideo ID is not authorised to initiate the prearranged group session as specified in clause 6.3.5.4, shall send a SIP 403 (Forbidden) response with the warning text set to: "119 user is not authorised to initiate the group call" in a Warning header field as specified in clause 4.4 and skip the rest of the steps below;

c) if the received SIP INVITE request contains an an authorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2 or MCVideo imminent peril group call as determined by clause 6.3.3.1.13.5 and the MCVideo user is eligible to be implicitly affiliated with the MCVideo group as determined as determined in step 13) a) iv) above, shall perform the implicit affiliation as specified in clause 8.2.2.3.7;

d) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [11];

e) shall create a prearranged group session and allocate an MCVideo session identity for the prearranged group call, and shall handle timer TNG3 (group call timer) as specified in clause 6.3.3.5;

f) if the group identity in the "SIP INVITE request for controlling MCVideo function of an MCVideo group" is a TGI:

i) shall for each of the constituent MCVideo groups homed on the primary MCVideo system:

A) if the controlling MCVideo function does not own the MCVideo group identified by the MCVideo group ID, then generate a SIP INVITE request towards the MCVideo server that owns the MCVideo group identity by following the procedures in clause 9.2.1.4.1.2; and

NOTE 2: The MCVideo server that the SIP INVITE request is sent to acts as a non-controlling MCVideo function;

B) if the controlling MCVideo function owns the MCVideo group identified by the MCVideo group ID then:

I) determine the members to invite to the prearranged MCVideo group call as specified in clause 6.3.5.5;

II) invite each group member determined in step A) above, to the group session, as specified in clause 9.2.1.4.1.1; and

III) interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3; and

ii) shall for each of the constituent MCVideo groups homed on the partner MCVideo system generate a SIP INVITE request for the MCVideo group identity homed on the partner MCVideo system as specified in clause 9.2.1.4.1.2; and

g) if the group identity in the SIP INVITE request for controlling MCVideo function of an MCVideo group is an MCVideo group ID:

i) shall determine the members to invite to the prearranged MCVideo group call as specified in clause 6.3.5.5;

ii) if necessary, shall start timer TNG1 (acknowledged call setup timer) according to the conditions stated in clause 6.3.3.3;

iii) if the received SIP INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element set to a value of "true":

A) shall cache the information that this MCVideo user has initiated an MCVideo emergency call;

B) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the information that this MCVideo user has initiated an MCVideo emergency alert; and

C) if the in-progress emergency state of the group is set to a value of "false":

I) shall set the value of the in-progress emergency state of the group to "true"; and

II) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16;

iv) if the in-progress emergency state of the group is set to a value of "false" and if the received SIP INVITE request contains an imminent peril indication set to a value of "true", the controlling MCVideo function shall:

A) shall cache the information that this MCVideo user has initiated an MCVideo imminent peril call; and

B) if the in-progress imminent peril state of the group is set to a value of "false", shall set the in-progress imminent peril state of the group to a value of "true";

v) shall invite each group member determined in step 13)g)i) above, to the group session, as specified in clause 9.2.1.4.1.1; and

vi) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3; and

14) if the MCVideo group call is ongoing then:

a) if:

i) the user identified by the MCVideo ID in the SIP INVITE request is not affiliated to the group identity contained in the SIP INVITE request as specified in clause 6.3.6;

ii) the group identity contained in the SIP INVITE request is not a constituent MCVideo group ID;

iii) the received SIP INVITE request does not contain an emergency indication or imminent peril indication; or

iv) the received SIP INVITE request is an authorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2 or MCVideo imminent peril group call as determined clause 6.3.3.1.13.5 and is determined to not be eligible for implicit affiliation as specified in clause 8.2.2.3.6;

then shall return a SIP 403 (Forbidden) response with the warning text set to "120 user is not affiliated to this group" in a Warning header field as specified in clause 4.4, and skip the rest of the steps below;

b) if the user identified by the MCVideo ID in the SIP INVITE request is not authorised to join the prearranged group session as specified in clause 6.3.5.3, shall send a SIP 403 (Forbidden) response with the warning text set to "121 user is not allowed to join the group call" in a Warning header field as specified in clause 4.4 and skip the rest of the steps below;

c) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [11];

d) if <on-network-max-participant-count> as specified in 3GPP TS 24.481 [24] is already reached:

i) if, according to local policy, the user identified by the MCVideo ID in the SIP INVITE request is deemed to have a higher priority than an existing user in the group session, may remove a participant from the session by following clause 9.2.1.4.4.3, and skip the next step; and

NOTE 3: The local policy for deciding whether to admit a user to a call that has reached its maximum amount of participants can include the <user-priority> and the <participant-type> of the user as well as other information of the user from the group document as specified in 3GPP TS 24.481 [24]. The local policy decisions can also include taking into account whether the imminent-peril indicator or emergency indicator was received in the SIP INVITE request.

ii) shall return a SIP 486 (Busy Here) response with the warning text set to "122 too many participants" to the originating network as specified in clause 4.4 and skip the rest of the steps;

e) if the received SIP INVITE request contains an an authorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2 or MCVideo imminent peril group call as determined by clause 6.3.3.1.13.5 and the MCVideo user is eligible to be implicitly affiliated with the MCVideo group as determined in step 14) a) iv) above, shall perform the implicit affiliation as specified in clause 8.2.2.3.7;

f) if the received SIP INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element set to a value of "true":

i) shall cache the information that this MCVideo user has initiated an MCVideo emergency call;

ii) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the information that this MCVideo user has initiated an MCVideo emergency alert;

iii) if the in-progress emergency state of the group is set to a value of "false":

A) shall set the value of the in-progress emergency state of the group to "true";

B) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16; and

C) shall generate SIP re-INVITE requests for the MCVideo emergency group call to the other call participants of the MCVideo group as specified in clause 6.3.3.1.6;

iv) if the in-progress imminent peril state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's imminent peril indication as specified in clause 6.3.3.1.11, setting the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11]; and

v) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5];

g) if the in-progress emergency state of the group is set to a value of "false" and if the SIP INVITE request contains an imminent peril indication set to a value of "true", the controlling MCVideo function:

i) shall cache the information that this MCVideo user has initiated an MCVideo imminent peril call; and

ii) if the in-progress imminent peril state of the group is set to a value of "false":

A) shall set the in-progress imminent peril state of the group to a value of "true";

B) shall generate SIP re-INVITE requests for the MCVideo imminent peril group call to the other call participants of the MCVideo group as specified in clause 6.3.3.1.15; and

C) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

iii) if the in-progress imminent peril state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's imminent peril indication as specified in clause 6.3.3.1.11, setting the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11];

h) shall generate a SIP 200 (OK) response as specified in the clause 6.3.3.2.4.2;

i) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

j) shall include in the SIP 200 (OK) response with the warning text set to "123 MCVideo session already exists" as specified in clause 4.4;

k) if the received SIP re-INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

l) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 4: In this case, the request was for an imminent peril call but a higher priority MCVideo emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

m) shall interact with media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 5: Resulting media plane processing is completed before the next step is performed.

n) shall send the SIP 200 (OK) response towards the inviting MCVideo client or inviting non-controlling MCVideo function according to 3GPP TS 24.229 [11];

o) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo User has joined in the MCVideo group session, as specified in clause 6.3.3.4;

NOTE 6: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

p) shall send a SIP NOTIFY request to each MCVideo client according to 3GPP TS 24.229 [11];

q) Upon receiving a SIP ACK to the above SIP 200 (OK) response and the SIP 200 (OK) response contained a Warning header field as specified in clause 4.4 with the warning text containing the mcvideo-warn-code set to "149", shall follow the procedures in clause 6.3.3.1.18; and

r) shall not continue with the rest of the clause.

Upon receiving a SIP 183 (Session Progress) response to the SIP INVITE request specified in clause 9.2.1.4.1 containing a P-Answer-State header field with the value "Unconfirmed" as specified in IETF RFC 4964 [30], the timer TNG1 (acknowledged call setup timer) is not running, the controlling MCVideo function supports media buffering and the SIP final response is not yet sent to the inviting MCVideo client:

1) *shall generate a SIP 200 (OK) response to SIP INVITE request as specified in the clause 6.3.3.2.3.2;*

2) *shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if the prearranged MCVideo group has more than <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [24];*

3) *shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;*

4) *shall include a P-Answer-State header field with the value "Unconfirmed";*

5) *if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;*

6) *if the received SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;*

7) *shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;*

NOTE 7: Resulting user plane processing is completed before the next step is performed.

8) *shall send the SIP 200 (OK) response towards the inviting MCVideo client according to 3GPP TS 24.229 [11];*

9) *shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo User has joined in the MCVideo group session, as specified in clause 6.3.3.4; and*

NOTE 8: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

10) *shall send a SIP NOTIFY request to each MCVideo client according to 3GPP TS 24.229 [11].*

Upon receiving a SIP 183 (Session Progress) response for a SIP INVITE request as specified in clause 9.2.1.4.1.2 containing an indication of required group members, the timer TNG1 (acknowledged call setup timer) is running and all SIP 200 (OK) responses have been received to all SIP INVITE requests sent to MCVideo clients specified in clause 9.2.1.4.1.1, then the controlling MCVideo function shall wait until the SIP 200 (OK) response has been received to the SIP INVITE request specified in clause 9.2.1.4.1.2 before generating a SIP 200 (OK) response to the "SIP INVITE request for controlling MCVideo function of an MCVideo group".

Upon receiving a SIP 200 (OK) response for a SIP INVITE request as specified in clause 9.2.1.4.1 that was sent to an affiliated and <on-network-required> group member as specified in 3GPP TS 24.481 [24]; and

1) *if the MCVideo ID in the SIP 200 (OK) response matches to the MCVideo ID in the corresponding SIP INVITE request;*

2) *there are no outstanding SIP 200 (OK) responses to SIP INVITE requests which were sent to affiliated and <on-network-required> group members as specified in 3GPP TS 24.481 [24]; and*

3) *there is no outstanding SIP 200 (OK) response to a SIP INVITE request sent in clause 9.2.1.4.1.2 where the SIP 183 (Session Progress) response contained an indication of required group members;*

the controlling MCVideo function:

1) *shall stop timer TNG1 (acknowledged call setup timer) as described in clause 6.3.3.3;*

2) shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;

3) shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if all members were not invited because the prearranged MCVideo group has been exceeded the <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [24];

4) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

5) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 9: Resulting media plane processing is completed before the next step is performed.

6) shall send a SIP 200 (OK) response to the inviting MCVideo client according to 3GPP TS 24.229 [11];

7) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo user has joined in the MCVideo group session, as specified in clause 6.3.3.4; and

NOTE 10: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

8) shall send the SIP NOTIFY request to the MCVideo clients according to 3GPP TS 24.229 [11].

Upon:

1) receiving a SIP 200 (OK) response for a SIP INVITE request as specified in clause 9.2.1.4.1;

2) the timer TNG1 (acknowledged call setup timer) is not running;

3) the local counter of the number of SIP 200 (OK) responses received from invited members is equal to the value of the <on-network-minimum-number-to-start> element of the group document;

4) the controlling MCVideo function supports media buffering; and

5) the SIP final response has not yet been sent to the inviting MCVideo client;

the controlling MCVideo function according to local policy:

1) shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.2;

2) shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if all members were not invited because the prearranged MCVideo group has exceeded the <max-participant-count> members as specified in 3GPP TS 24.481 [24];

3) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

4) if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

5) if the received SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

6) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 11: Resulting media plane processing is completed before the next step is performed.

7) shall send a SIP 200 (OK) response to the inviting MCVideo client according to 3GPP TS 24.229 [11];

8) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo user has joined in the MCVideo group session, as specified in clause 6.3.3.4; and

NOTE 12: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

9) shall send the SIP NOTIFY request to the MCVideo clients according to 3GPP TS 24.229 [11].

Upon expiry of timer TNG1 (acknowledged call setup timer), if there are outstanding SIP 200 (OK) responses to SIP INVITE requests sent to affiliated and <on-network-required> group members as specified in 3GPP TS 24.481 [24], the controlling MCVideo function shall follow the procedures specified in clause 6.3.3.3.

If timer TNG1 (acknowledged call setup timer) is running and a final SIP 4xx, 5xx or 6xx response is received from an affiliated and <on-network-required> group member as specified in 3GPP TS 24.481 [24], the controlling MCVideo function shall follow the relevant procedures specified in clause 6.3.3.3.

If:

1) timer TNG1 (acknowledged call setup timer) is not running;

2) the local counter of the number of SIP 200 (OK) responses received from invited members is equal to the value of the <on-network-minimum-number-to-start> element of the group document; and

3) a final SIP 4xx, 5xx or 6xx response is received from an invited MCVideo client;

then the controlling MCVideo function shall perform one of the following based on policy:

1) send the SIP final response towards the inviting MCVideo client, according to 3GPP TS 24.229 [11], if a SIP final response was received from all the other invited MCVideo clients and the SIP 200 (OK) response is not yet sent; or

2) remove the invited MCVideo client from the MCVideo Session as specified in clause 6.3.3.1.5, if a SIP final response other than 2xx or 3xx was received from all the invited MCVideo clients and the SIP 200 (OK) response is already sent. The controlling MCVideo function may invite an additional member of the prearranged MCVideo group as specified in clause 9.2.1.4.1 that has not already been invited, if the prearranged MCVideo group has more than <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [24], and all members have not yet been invited.

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCVideo client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.18."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.3]

"Upon receiving a SIP BYE request the controlling MCVideo function shall follow the procedures as specified in clause 6.3.3.2.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.2]

"This clause is referenced from other procedures.

The controlling MCVideo function shall generate an initial SIP INVITE request according to 3GPP TS 24.229 [11].

The controlling MCVideo function:

1) shall include in the Contact header field an MCVideo session identity for the MCVideo session with the g.3gpp.mcvideo media feature tag, the isfocus media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo" according to IETF RFC 3840 [22];

2) shall include an Accept-Contact header field containing the g.3gpp.mcvideo media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [20];

3) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcvideo" (coded as specified in 3GPP TS 24.229 [11]), in a P-Asserted-Service-Id header field according to IETF RFC 6050 [14] in the SIP INVITE request;

4) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo" along with parameters "require" and "explicit" according to IETF RFC 3841 [20];

5) shall include a Referred-By header field with the public user identity of the inviting MCVideo client;

6) should include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [7]. The refresher parameter shall be omitted;

7) shall include the Supported header field set to "timer";

8) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body to the outgoing INVITE request;

9) if the received SIP INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body containing an <ambient-viewing-type> element and:

a) if the Priv-Answer-Mode header field specified in IETF RFC 5373 [27] was included in the received SIP INVITE request with a value of "Auto" or if no Priv-Answer-Mode header field was received in the received SIP INVITE request; or

b) a Priv-Answer-Mode header field was received containing a value other than "Auto";

shall include a Priv-Answer-Mode header field set to a value of "Auto" in the outgoing SIP INVITE request; and

10) if the received SIP INVITE request did not contain an application/vnd.3gpp.mcvideo-info+xml MIME body containing an <ambient-viewing-type> element, shall include an unmodified Answer-Mode header field if present in the incoming SIP INVITE request."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.1]

"The SDP offer is generated based on the received SDP offer. The SDP offer generated by the controlling MCVideo function:

1) when initiating a new MCVideo session the SDP offer;

a) shall contain only one SDP media-level section for MCVideo video media stream as contained in the received SDP offer; and

b) shall contain an SDP media-level section for one media-transmission control entity, if present in the received SDP offer; and

2) when adding a new MCVideo user to an existing MCVideo Session, the SDP offer shall contain the media stream currently used in the MCVideo session.

When composing the SDP offer according to 3GPP TS 24.229 [11], the controlling MCVideo function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the controlling MCVideo function;

2) for the MCVideo video media stream, shall include all media-level attributes from the received SDP offer;

3) shall replace the IP address and port number for the offered media transmission control entity, if any, in the received SDP offer with the IP address and port number of the controlling MCVideo function; and

4) for the offered media transmission control entity, shall include the offered media transmission control entity 'fmtp' attributes as specified in 3GPP TS 24.581 [5] clause 14."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.2.2]

"When an MCVideo call is established a new instance of the transmission control server state machine for 'general transmission control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' is added.

When an MCVideo call is established a new instance of the transmission control server state machine for 'general reception control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic reception control operation towards the transmission participant' is added.

If the optional "mc_queueing" feature is supported and has been negotiated as specified in clause 14, the transmission control server could queue the implicit transmission control request for the media-transmission control entity.

The original initial SIP INVITE request or SIP REFER request to establish an MCVideo chat group call or to rejoin an ongoing MCVideo call is not handled as an implicit transmission control request message by the transmission control server unless explicitly stated in the SIP INVITE request or in the SIP REFER request.

The permission to send media to the inviting MCVideo client due to implicit transmission control request is applicable to both confirmed indication and unconfirmed indication.

When the first unconfirmed indication is received from the invited participating MCVideo function (see 3GPP TS 24.281 [2]) the transmission control server optionally can give an early indication to send RTP media packets, to the inviting MCVideo client.

If an early indication to send RTP media packets is given to the inviting MCVideo client, the transmission participant is granted the permission to send media and the MCVideo server buffers RTP media packets received from the MCVideo client at least until the first invited MCVideo client accepts the invitation or until the RTP media packet buffer exceeds it maximum limit to store RTP media packets.

If the MCVideo server does not support or does not allow media buffering then when an early indication to send RTP media packets is not given to the inviting MCVideo client, the transmission participant is granted the permission to send media when the first invited MCVideo client accepts the media.

Before the transmission control server sends the first transmission control message in the MCVideo call, the transmission control server has to assign itself a SSRC identifier to be included in media transmission control messages and quality feedback messages if the MCVideo server is supporting that option. A suitable algorithm to generate the SSRC identifier is described in IETF RFC 3550 [3].

The transmission participant and the transmission control server can negotiate the maximum priority level that the transmission participant is permitted to request. The transmission control server can pre-empt the current sender based on the negotiated maximum priority level that the transmission participant is permitted to request and the priority level included in the Transmission Media Request message.

NOTE: The maximum priority level that a transmission participant can use is negotiated as specified in clause 14.3.3 and is based on group configuration data retrieved by the controlling MCVideo function from the group management server as described in 3GPP TS 24.481 [12] and service configuration data retrieved by the controlling MCVideo function from the configuration management server as described in 3GPP TS 24.484 [13].

The transmission participant and the transmission control server can negotiate queueing of Transmission requests using the "mc_queueing" fmtp attribute as described in clause 14. If queueing is supported and negotiated, the transmission control server queues the transmission control request if a Transmission Media Request message is received when another transmission participant has the transmission and the priority of the current speaker is the same or higher."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.2]

"Upon receiving a transmission control message the participating MCVideo function:

1. *shall immediately forward the transmission control message to the transmission control server if the message is received from the transmission participant;*
2. *if an MBMS subchannel is not used for a transmission in the session the transmission control message is associated with, shall immediately forward the transmission control message to the transmission participant if the message is received from the transmission control server; and*
3. *if an MBMS subchannel is used for a transmission in the session the transmission control message is associated with:*

    a. *if*

        i. *the transmission control message is not a Transmission Idle message or a Media Transmission Notify message;*

        ii. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

        iii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report;*

    shall immediately forward the transmission control message to the transmission participant; and

    b. *if*

        i. *the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report; and*

        ii *if the transmission control message is the Transmission Idle message or the Media Transmission Notify message,*

shall perform actions as specified in clause 10.2.

NOTE: When the Transmit Idle or Media Transmission Notify messages are discarded the messages are sent to the MCVideo clients over the MBMS subchannel allocated for the transmission as specified in clause 10.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.3]

"Upon receiving RTP media packets the participating MCVideo function:

1. *shall immediately forward the RTP media packet to the controlling MCVideo function if the RTP packet is from an MCVideo client; and*
2. *if an MBMS subchannel is not used for a transmission in the session the RTP media packets are associated with, shall immediately forward the RTP media packets to the MCVideo client if the RTP packet is from the controlling MCVideo function or the non-controlling MCVideo function.*

3. *if an MBMS subchannel is used for a transmission in the session the RTP media packets are associated with and if RTP media packets are received from the controlling MCVideo function or the non-controlling MCVideo function:*

   a. *if*

      i. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

      ii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report,*

   shall immediately forward the RTP media packets to the MCVideo client; and

   b. *if the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report, shall perform actions as specified in clause 10.2."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.4]

"When the participating function receives an indication from the application and signalling plane that session release is initiated, the participating MCVideo function:

1. *shall stop sending transmission control messages towards the transmission participant and the transmission control server; and*

2. *shall stop sending RTP media packets towards the MCVideo client and towards the controlling MCVideo function.*

When the participating MCVideo function receives an indication from the application and signalling plane that the session is released, the participating MCVideo function:

1. *in case of a pre-established session, shall perform the actions in clause 9.3.2; and*

2. *in case of an on-demand session, shall release the media resources associated with the session."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.1]

"The transmission control messages are coded as described in clause 9.1.2 where the transmission control message is part of the application-dependent data.

For the transmission control protocol the ASCII name string is: MCVx (Mission Critical Video x) with x=0, 1 or 2 as specified in clause 9.1.2.

A list of transmission control messages can be found in clause 9.2.2.1.

The transmission control specific fields are specified in clause 9.2.3."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.2.1]

"Based on the negotiations during the call establishment specified in 3GPP TS 24.281 [2], a new instance of the 'Transmission participant state transition diagram for basic transmission control operation', as specified in clause 6.2.4 and a new instance of the 'Transmission participant state transition diagram for basic reception control operation' as specified in clause 6.2.5, shall be created for this call.

The SIP INVITE request sent by the application and signalling plane:

1. *shall be regarded an implicit Transmission request when an implicit Transmission request is negotiated; and*

2. *shall not be regarded as an implicit Transmission request in case of a rejoin to an already on-going group call.*

NOTE: The transmission participant can negotiate the use of prioritization of the Transmission Media Request message. In that case, the transmission participant can request permission to send media at a priority level that is either the same as or lower than the highest priority that was permitted to the participant in the MCVideo call initialization. If a transmission participant is authorized for pre-emptive priority in the MCVideo call it is good practise to always request permission to send RTP media packets at a priority level that is lower than pre-emptive priority unless the user explicitly requests to pre-empt the current RTP media packets sender."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.2.2]

"The MCVideo call release (whether it is initiated by the transmission participant or transmission control server) is a two-step procedure.

Step 1 The transmission participant stops sending transmission control and reception control messages and the MCVideo client stops sending and receiving RTP media packets.

Step 2 When the application and signalling plane has determined that the MCVideo call is released, the corresponding instance of the 'Transmission participant state transition diagram for basic transmission control operation' as specified in clause 6.2.4 and the corresponding instance of the 'Transmission participant state transition diagram for basic reception control operation' as specified in clause 6.2.5 are terminated and the transmission participant releases all the used resources.

The user plane can initiate the release step 1, but the application and signalling plane always initiates the release step 2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.2.4]

"The transmission participant shall behave according to the state diagram and the state transitions specified in this clause.

Figure 6.2.4.1-1 shows the state diagram for 'Transmission participant state transition diagram for basic transmission control operation'.

…"

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.

Step 2 When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.

Step 2 When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.

The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

b. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

5.1.1.14.3 Test description

5.1.1.14.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the participating MCVideo functions.
- SS-UE1 (MCVideo Client) and SS-UE2 (MCVideo Client), simulated by the SS (MCVideo Server).

IUT:

- IUT (MCVideo Server), providing the controlling MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of all sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Transmission Control Server.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server).
- The SS-UE1 (MCVideo client) and the SS-UE2 (MCVideo client) are registered and authorized to use the MCVideo system.
- The SS-UE1 (MCVideo client) and the SS-UE2 (MCVideo client) are affiliated to the pre-arranged group and are authorized to initiate group calls.
- The IUT (MCVideo Server) is the controlling server of the pre-arranged group.

5.1.1.14.3.2 Test procedure sequence

Table 5.1.1.14.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E group call establishment' as described in Annex A, Table A.4.1.4.3-1 to establish a pre-arranged group call without implicit transmission control request according to option a of note in Annex A, Table A.4.1.4.3-1, initiated by SS-UE1 (MCVideo Client)?	-	-	1	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	2	P
-	EXCEPTION: Steps 3a - 3b describe behaviour that depends on pc_MCVideo_TransmissionRequestQueueing.	-	-	-	-
-	IF pc_MCVideo_TransmissionRequestQueueing THEN:	-	-	-	-
3a	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Queue Position Info' as described in Annex A, Table A.3.12.3-1 to request permission to transmit for SS-UE2 (MCVideo Client) with higher priority than SS-UE1 (MCVideo Client)?	-	-	3	P
-	ELSE:	-	-	-	-
3b	Check: Does the IUT (MCVideo Server) correctly perform steps 1-2 of procedure 'MCVideo Transmission Request - Queue Position Info' as described in Annex A, Table A.3.12.3-1 to request permission to transmit for SS-UE2 (MCVideo Client) with higher priority than SS-UE1 (MCVideo Client)?	-	-	3	P
4	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Revoked - Transmission End Request' as described in Annex A, Table A.3.14.3-1, to revoke SS-UE1 (MCVideo Client) media transmission, with revoke reason #4 (Media Burst pre-empted)?	-	-	3	P
5	Check: Does the IUT (MCVideo Server) send a Transmission End Notify message to SS-UE2 (MCVideo Client) informing that SS-UE1 (MCVideo Client) has stopped to transmit?	<--	Transmission End Notify	3	P
6	IF the IUT (MCVideo Server) requests acknowledgement, the SS-UE2 (MCVideo Client) sends a Transmission Control Ack message in response to the Transmission End Notify message.	-->	Transmission Control Ack	-	-
7	Check: Does the IUT (MCVideo Server) correctly perform steps 3a1-3b2 of procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1 to grant the transmission to SS-UE2 (MCVideo Client)?	-	-	3	P
8	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1 to end the transmission of SS-UE2 (MCVideo Client)?	-	-	5	P
9	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1 to request permission to transmit for SS-UE2 (MCVideo Client) with high priority?	-	-	6	P
-	EXCEPTION: Steps 10a1 - 10b2 describe behaviour that depends on pc_MCVideo_TransmissionRequestQueueing.	-	-	-	-
-	IF pc_MCVideo_TransmissionRequestQueueing THEN:	-	-	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
10a1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Queue Position Info' as described in Annex A, Table A.3.12.3-1 to request permission to transmit for SS-UE1 (MCVideo Client) with lower priority than SS-UE2 (MCVideo Client)?	-	-	7	P
10a2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Queue Position Request - Queue Position Info', as described in Annex A, Table A.3.13.3-1 to request queue position info for SS-UE1 (MCVideo Client)?	-	-	8	P
10a3	The SS-UE1 (MCVideo Client) sends a Transmission End Request message, with acknowledgement, to cancel queued transmission.	-->	Transmission End Request	-	-
10a4	Check: Does the IUT (MCVideo Server) send a Transmission Control Ack message?	<--	Transmission Control Ack	9	P
10a5	Check: Does the IUT (MCVideo Server) send a Transmission End Response message to SS-UE1 (MCVideo Client)?	<--	Transmission End Response	9	P
10a6	IF the IUT (MCVideo Server) requests acknowledgement, the SS-UE1 (MCVideo Client) sends a Transmission Control Ack message in response to the Transmission End Response message.	-->	Transmission Control Ack	-	-
10a7	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Queue Position Info' as described in Annex A, Table A.3.12.3-1 1 to request permission to transmit for SS-UE1 (MCVideo Client) with lower priority than SS-UE2 (MCVideo Client)	-	-	7	P
10a8	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify', except steps 4-5, as described in Annex A, Table A.3.8.3-1 to end the transmission of SS-UE2 (MCVideo Client)?	-	-	10	P
10a9	Check: Does the IUT (MCVideo Server) correctly perform steps 3a1-3b2 of procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1 to grant transmission to SS-UE1 (MCVideo Client)?	-	-	10	P
10a11	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1 to end the transmission of SS-UE1 (MCVideo Client)??	-	-	4	P
-	ELSE:	-	-	-	-
10b1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Rejected' as described in Annex A, Table A.3.11.3-1 to request permission to transmit for SS-UE1 (MCVideo Client) with lower priority than SS-UE2 (MCVideo Client)?	-	-	11	P
10b2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify', as described in Annex A, Table A.3.8.3-1 to end the transmission of SS-UE2 (MCVideo Client)?			5	

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
11	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	2	P
12	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Revoked - Transmission End Request' as described in Annex A, Table A.3.14.3-1, to revoke SS-UE1 (MCVideo Client) media transmission, with revoke reason #2 (Media Burst too long)?	-	-	12	P
13	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, except steps 1-3a2 and 3b1, to inform SS-UE2 (MCVideo Client) that SS-UE1 (MCVideo Client) transmission has ended?	-	-	12	P
14	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.1.5.3-1, to release the call?	-	-	13	P

5.1.1.14.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

Table 5.1.1.14.3.3-1: SIP INVITE from SS (MCVideo Server)
(Step 1, Table 5.1.1.14.3.2-1; Step 1, Annex A, Table A.4.1.4.3-1)

Derivation Path: Annex A, Table A.4.1.4.4-1

Table 5.1.1.14.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.1.14.3.2-1; Step 2, Annex A, Table A.4.1.4.3-1)

Derivation Path: Annex A, Table A.4.1.4.4-2

Table 5.1.1.14.3.3-3: SIP 200 (OK) from SS (MCVideo Server)
(Step 1, Table 5.1.1.14.3.2-1; Step 8, Annex A, Table A.4.1.4.3-1)

Derivation Path: Annex A, Table A.4.1.4.4-6

Table 5.1.1.14.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.1.1.14.3.2-1; Step 9, Annex A, Table A.4.1.4.3-1)

Derivation Path: Annex A, Table A.4.1.4.4-7

Table 5.1.1.14.3.3-5: Transmission Request from the SS-UE1 (MCVideo Client) (Steps 2, 10a1, 10a7 and 10b1, Table 5.1.1.14.3.2-1; Step 1, Annex A, Table A.3.7.3-1; Step 1, Annex A, Table A.3.11.3-1; Step 1, Annex A, Table A.3.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.1.1-1, with condition ACK				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Transmission Priority			ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.2	
Transmission Priority Value	"5"			

Table 5.1.1.14.3.3-6: Transmission Granted from the IUT (MCVideo Server) (Steps 2 and 10a9, Table 5.1.1.14.3.2-1; Steps 3b1 and 3a1, Annex A, Table A.3.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.1-1, with condition ACK IF the IUT (MCVideo Server) requests message acknowledgement.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Transmission Priority			ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.2	
Transmission Priority Value	"5"			

Table 5.1.1.14.3.3-7: Transmission Request from the SS-UE2 (MCVideo Client) (Steps 3a, 3b and 9, Table 5.1.1.14.3.2-1; Step 1, Annex A, Table A.3.7.3-1; Step 1, Annex A, Table A.3.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.1.1-1, with condition ACK				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Transmission Priority			ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.2	
Transmission Priority Value	"15"			

Table 5.1.1.14.3.3-8: Queue Position Info from the IUT (MCVideo Server) (Steps 3a and 3b, Table 5.1.1.14.3.2-1; Steps 3a1 and 3b1, Annex A, Table A.3.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.6-1, with condition ACK IF the IUT (MCVideo Server) requests message acknowledgement.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Queue Info		Defines the queue position and granted transmission control priority in the queue.	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.5	
Queue Priority Level	"15"		ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.2	

Table 5.1.1.14.3.3-9: Transmission Revoked from the IUT (MCVideo Server) (Step 4, Table 5.1.1.14.3.2-1; Steps 1aA and 1aB, Annex A, Table A.3.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.2-1, with condition ACK IF the IUT (MCVideo Server) requests message acknowledgement.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Reject Cause			ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.4	
Reject Cause Value	4		ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.10.2	
Reject Cause Phrase	"Media Burst pre-empted"	A text string encoded the text string in the SDES item CNAME.	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.10.2	

Table 5.1.1.14.3.3-10: Transmission End Request from the IUT (MCVideo Server) (Step 4, Table 5.1.1.14.3.2-1; Steps 1bA and 1bB, Annex A, Table A.3.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.1-1, with condition DOWNLINK. Condition ACK IF the IUT (MCVideo Server) requests message acknowledgement.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Reject Cause			ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.4	
Reject Cause Value	4		ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.10.2	
Reject Cause Phrase	"Media Burst pre-empted"	A text string encoded the text string in the SDES item CNAME.	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.10.2	

Table 5.1.1.14.3.3-11: Transmission End Notify from the IUT (MCVideo Server) (Step 5, Table 5.1.1.14.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.15-1, with condition ACK IF the IUT (MCVideo Server) requests message acknowledgement.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
User Id of the Transmitting User		Carries the identity of the user whose media transmission has been released	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.6	
User Id of the Transmitting User	px_MCVideo_ID_User_A			
Audio SSRC of the Transmitting User	Media/Audio SSRC of remote client (Client A)	SSRC of the user transmitting the media	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.16	

Table 5.1.1.14.3.3-12: Transmission Control Ack from the SS-UE1 (MCVideo Client) (Step 6, Table 5.1.1.14.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

Table 5.1.1.14.3.3-13: Transmission Granted from the IUT (MCVideo Server) (Steps 7 and 9, Table 5.1.1.14.3.2-1; Steps 3a1 and 3b1, Annex A, Table A.3.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.1-1, with condition ACK IF the IUT (MCVideo Server) requests message acknowledgement.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Transmission Priority			ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.2	
Transmission Priority Value	"15"			

Table 5.1.1.14.3.3-14: Media Transmission Notification from the IUT (MCVideo Server) (Steps 7 and 9, Table 5.1.1.14.3.2-1; Steps 3a3 and 3b2, Annex A, Table A.3.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.7-1, with condition ACK IF the IUT (MCVideo Server) requests message acknowledgement.

Table 5.1.1.14.3.3-15: Transmission End Request from the SS-UE2 (MCVideo Client) (Steps 8, 10a8 and 10b2, Table 5.1.1.14.3.2-1; Step 1, Annex A, Table A.3.8.3-1)

Derivation Path: Annex A, Table A.3.8.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
User Id of the Transmitting User		identity of the user whose media transmission is requested to be terminated.	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.6	
User Id of the Transmitting User	px_MCVideo_ID_User_B			

Table 5.1.1.14.3.3-16: Transmission End Notify from the IUT (MCVideo Server) (Steps 8, 10a8 and 10b2, Table 5.1.1.14.3.2-1; Steps 3a3 and 3b2, Annex A, Table A.3.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.15-1, with condition ACK IF the IUT (MCVideo Server) requests message acknowledgement.

Table 5.1.1.14.3.3-17: Queue Position Info from the IUT (MCVideo Server) (Steps 10a1, 10a2 and 10a7, Table 5.1.1.14.3.2-1; Steps 3a1 and 3b1, Annex A, Table A.3.12.3-1; Steps 3a1 and 3b1, Annex A, Table A.3.13.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.6-1, with condition ACK IF the IUT (MCVideo Server) requests message acknowledgement.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Queue Info		Defines the queue position and granted transmission control priority in the queue.	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.5	
Queue Priority Level	"5"		ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.2	

Table 5.1.1.14.3.3-18: Transmission End Request from the SS-UE1 (MCVideo Client) (Step 10a3, Table 5.1.1.14.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.1-1, with conditions ACK and UPLINK

Table 5.1.1.14.3.3-19: Transmission Control Ack from the IUT (MCVideo Server) (Steps 10a4, Table 5.1.1.14.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, with condition DOWNLINK.

Table 5.1.1.14.3.3-20: Transmission End Response from the IUT (MCVideo Server) (Step 10a5, Table 5.1.1.14.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.2-1, with condition ACK IF the IUT (MCVideo Server) requests message acknowledgement.

Table 5.1.1.14.3.3-21: Transmission Control Ack from the SS-UE1 (MCVideo Client) (Steps 10a6, Table 5.1.1.14.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, with condition UPLINK.

**Table 5.1.1.14.3.3-22: Transmission Revoked from the IUT (MCVideo Server)
(Step 12, Table 5.1.1.14.3.2-1;
Steps 1aA and 1aB, Annex A, Table A.3.14.3-1)**

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.2-1, with condition ACK IF the IUT (MCVideo Server) requests message acknowledgement.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Reject Cause			ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.4	
Reject Cause Value	2		ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.10.2	
Reject Cause Phrase	"Media burst too long"	A text string encoded the text string in the SDES item CNAME.	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.10.2	

**Table 5.1.1.14.3.3-23: Transmission End Request from the IUT (MCVideo Server)
(Step 12, Table 5.1.1.14.3.2-1;
Steps 1bA and 1bB, Annex A, Table A.3.14.3-1)**

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.1-1, with condition DOWNLINK. Condition ACK IF the IUT (MCVideo Server) requests message acknowledgement.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Reject Cause			ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.4	
Reject Cause Value	2		ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.10.2	
Reject Cause Phrase	"Media burst too long"	A text string encoded the text string in the SDES item CNAME.	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.10.2	

5.1.1.15 MCVideo Server - MCVideo Server / Pre-arranged Group Call / Reception Control State Transitions / Controlling server

5.1.1.15.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function initiates a pre-
arranged group call with implicit transmission control }
    then { IUT (MCVideo Server) accepts the call from initiator and establishes the call with all
affiliated users by sending call establishment requests to the SS (MCVideo Server) providing the
terminating participating function and sends transmission granted to the SS-UE1 (MCVideo Client) and
sends media transmission notification to the SS-UE2 (MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE2 (MCVideo Client) requests to receive SS-UE1 (MCVideo client) media transmission
}
    then { IUT (MCVideo Server) accepts SS-UE2 (MCVideo Client) media reception request and forwards
SS-UE1 (MCVideo Client) media transmission to SS-UE2 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE2 (MCVideo Client) requests to stop receiving SS-UE1 (MCVideo Client) media
transmission }
    then { IUT (MCVideo Server) accepts SS-UE2 media reception end request and stops forwarding
media }
            }
```

(4)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests to end the transmission control }
    then { IUT (MCVideo Server) sends transmission control idle to the SS-UE1 (MCVideo Client) and
stops forwarding media and sends transmission control idle to the SS-UE2 (MCVideo Client) }
            }
```

(5)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control }
    then { IUT (MCVideo Server) sends transmission granted to the SS-UE1 (MCVideo Client) and sends
media transmission notification to the SS-UE2 (MCVideo Client) }
            }
```

(6)

```
with { IUT (MCVideo Server) having established a pre-arranged group call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function ends the pre-
arranged group call }
    then { IUT (MCVideo Server) responds by sending a SIP 200 (OK) message to the SS (MCVideo
Server) ending the call and sends a SIP BYE message to the SS (MCVideo Server) providing the
terminating participating function }
            }
```

5.1.1.15.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.1.4.1.1, 9.2.1.4.1.2, 9.2.1.4.2, 9.2.1.4.3, 6.3.3.1.2 and 6.3.3.1.1, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.3.2.2, 6.3.3, 6.4.2, 6.4.3, 6.4.4, 9.2.1, 6.2.1, 6.2.2, 6.2.4, 6.3.3 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.1.1]

"This clause describes the procedures for inviting an MCVideo user to an MCVideo session. The procedure is initiated by the controlling MCVideo function as the result of an action in clause 9.2.1.4.2 or as the result of receiving a SIP 403 (Forbidden) response as described in this clause.

The controlling MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the public service identity of the terminating participating MCVideo function associated to the MCVideo user to be invited.;

NOTE 1: How the controlling MCVideo function finds the address of the terminating MCVideo participating function is out of the scope of the current release.

NOTE 2: If the terminating MCVideo user is part of a partner MCVideo system, then the public service identity can identify an entry point in the partner network that is able to identify the terminating participating MCVideo function.

3) shall set the P-Asserted-Identity header field to the public service identity of the controlling MCVideo function;

4) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcvideo-request-uri> element set to the MCVideo ID of the terminating user; and

b) the <mcvideo-calling-group-id> element set to the group identity;

NOTE 3: The <mcvideo-calling-user-id> is already included in the MIME body as a result of calling clause 6.3.3.1.2 in step 1).

5) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1;

6) if the in-progress emergency state of the group is set to a value of "true" the controlling MCVideo function:

a) shall include a Resource-Priority header field populated with the values for an MCVideo emergency group call as specified in clause 6.3.3.1.19;

b) if the received SIP INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "true":

i) shall include in the outgoing SIP INVITE request in the application/vnd.3gpp.mcvideo-info+xml MIME body an <emergency-ind> element set to a value of "true"; and

ii) if the <alert-ind> element is set to "true" in the received SIP INVITE request and the requesting MCVideo user and MCVideo group are authorised for the initiation of MCVideo emergency alerts as determined by the procedures of clause 6.3.3.1.13.1, shall populate the application/vnd.3gpp.mcvideo-info+xml MIME body and the application/vnd.3gpp.location-info+xml MIME body as specified in clause 6.3.3.1.12. Otherwise, shall set the <alert-ind> element to a value of "false"; and

c) if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <imminentperil-ind> element set to a value of "false";

7) if the in-progress emergency state of the group is set to a value of "false" and the in-progress imminent peril state of the group is set to a value of "true", the controlling MCVideo function:

a) shall include a Resource-Priority header field populated with the values for an MCVideo imminent peril group call as specified in clause 6.3.3.1.19; and

b) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true";

8) if:

a) an MCVideo GKTP document exists for the group identity contained in the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request; and

b) the MCVideo GKTP document contains a <MKFC-GKTPs> element;

then:

a) for each instance of <GKTP> element of the <MKFC-GKTPs> element of the MCVideo GKTP document:

i) shall perform the procedure in clause 6.3.3.6.2 to re-generate an I_MESSAGE; and

ii) if the procedure in clause 6.3.3.6.2 was successful, shall include the I_MESSAGE in a <GKTP> element of the <MKFC-GKTPs> element of an application/vnd.3gpp.mcvideo-info+xml MIME body included in the outgoing SIP INVITE request; and

9) shall send the SIP INVITE request towards the terminating network in accordance with 3GPP TS 24.229 [11].

Upon receiving a SIP 183 (Session Progress) response containing a Require header field with the option tag "100rel" and containing a P-Answer-State header field with the value "Unconfirmed" in response to the SIP INVITE request the controlling MCVideo function:

1) shall send a SIP PRACK request towards the MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCVideo function:

1) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

2) shall send a SIP NOTIFY request to all participants with a subscription to the conference event package as specified in clause 9.2.3.4; and

3) shall increment the local counter of the number of SIP 200 (OK) responses received from invited members, by 1.

NOTE 4: The notifications above could be sent prior to the SIP 200 (OK) response being sent to the inviting MCVideo client. These notifications received by MCVideo clients that are group members do not mean that the group session will be successfully established.

NOTE 5: The procedures executed by the controlling MCVideo function prior to sending a response to the inviting MCVideo client are specified in clause 9.2.1.4.2."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.1.2]

"The controlling MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the public service identity of the non-controlling MCVideo function serving the group identity of the MCVideo group owned by the partner MCVideo system;

3) shall set the P-Asserted-Identity to the public service identity of the controlling MCVideo function;

4) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcvideo-request-uri> element set to the group identity of the MCVideo group hosted by the non-controlling MCVideo function in the partner MCVideo system; and

b) the <mcvideo-calling-group-id> element set to the group identity of the group served by the controlling MCVideo function;

5) shall include the Recv-Info header field set to g.3gpp.mcvideo-transmission-request;

6) if:

a) an MCVideo GKTP document exists for the group identity contained in the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request; and

b) the MCVideo GKTP document contains a <MKFC-GKTPs> element;

then:

a) for each instance of <GKTP> element of the <MKFC-GKTPs> element of the MCVideo GKTP document:

i) shall perform the procedure in clause 6.3.3.6.2 to re-generate an I_MESSAGE; and

ii) if the procedure in clause 6.3.3.6.2 was successful, shall include the I_MESSAGE in a <GKTP> element of the <MKFC-GKTPs> element of an application/vnd.3gpp.mcvideo-info+xml MIME body included in the outgoing SIP INVITE request;

7) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1; and

8) shall send the SIP INVITE request towards the partner MCVideo system in accordance with 3GPP TS 24.229 [11].

Upon receiving SIP 403 (Forbidden) response for the SIP INVITE request, if according to local policy and if:

1) the response contains a Warning header field with the MCVideo warning code "128"; and

2) the response contains a P-Refused-URI-List header field and an application/resource-lists+xml MIME body as specified in IETF RFC 5318 [28];

NOTE 1: The application/resource-lists+xml MIME body contains MCVideo IDs identifying MCVideo users in a partner MCVideo system that needs to be invited to the prearranged group call in case of group regrouping using interrogating method.

Editor's Note:The above note currently isn't defined in the 23.280 and 23.281.

then the controlling MCVideo function:

1) shall check if the number of members of the MCVideo group exceeds the value contained in the <on-network-max-participant-count> element of the group document as specified in 3GPP TS 24.481 [24]. If exceeded, the controlling MCVideo function shall invite only <on-network-max-participant-count> members from the application/resource-lists+xml MIME body; and

NOTE 2: The <on-network-max-participant-count> element indicates the maximum number of participants allowed in the prearranged group session It is operator policy that determines which participants in the application/resource-lists+xml MIME body are invited to the group call.

2) shall invite MCVideo users as specified in this clause using the list of MCVideo IDs in URI-List.

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCVideo function:

1) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3; and

NOTE 3: The procedures executed by the controlling MCVideo function prior to sending a response to the inviting MCVideo client are specified in clause 9.2.1.4.2.

2) if at least one of the invited MCVideo clients has subscribed to the conference package, shall subscribe to the conference event package in the non-controlling MCVideo function as specified in clause 9.2.3.4.3."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.2]

"In the procedures in this clause:

1) MCVideo ID in an incoming SIP INVITE request refers to the MCVideo ID of the originating user from the <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;

2) group identity in an incoming SIP INVITE request refers to the group identity from the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;

3) MCVideo ID in an outgoing SIP INVITE request refers to the MCVideo ID of the called user in the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the outgoing SIP INVITE request;

4) indication of required group members in a SIP 183 (Session Progress) response refers to the <required> element of the application/vnd.3gpp.mcvideo-info+xml MIME body set to "true" in a SIP 183 (Session Progress) sent by the non-controlling MCVideo function of an MCVideo group;

5) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

6) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

Upon receipt of a "SIP INVITE request for controlling MCVideo function of an MCVideo group", the controlling MCVideo function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP INVITE request with a SIP 500 (Server Internal Error) response. The controlling MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15] and skip the rest of the steps;

NOTE 1: if the SIP INVITE request contains an emergency indication or an imminent peril indication set to a value of "true" and this is an authorised request for originating an MCVideo emergency group call as determined by clause 6.3.3.1.13.2, or for originating an MCVideo imminent peril group call as determined by clause 6.3.3.1.13.5, the controlling MCVideo function can according to local policy choose to accept the request.

2) shall determine if the media parameters are acceptable and the MCVideo codecs are offered in the SDP offer and if not reject the request with a SIP 488 (Not Acceptable Here) response and skip the rest of the steps;

3) shall reject the SIP request with a SIP 403 (Forbidden) response and not process the remaining steps if:

a) an Accept-Contact header field does not include the g.3gpp.mcvideo media feature tag; or

b) an Accept-Contact header field does not include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo";

4) if received SIP INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.3.1.17;

5) shall retrieve the necessary group document(s) from the group management server for the group identity contained in the SIP INVITE request and carry out initial processing as specified in clause 6.3.5.2;

6) if the result of the initial processing in clause 6.3.5.2 was:

a) that authorization of the MCVideo ID is required at a non-controlling MCVideo function of an MCVideo group is required, perform the actions in clause 6.3.3.1.13.7 and do not continue with the rest of the steps in this clause; and

b) that a SIP 3xx, 4xx, 5xx or 6xx response to the "SIP INVITE request for controlling MCVideo function of an MCVideo group" has been sent, do not continue with the rest of the steps in this clause;

7) shall perform the actions as described in clause 6.3.3.2.2;

8) shall maintain a local counter of the number of SIP 200 (OK) responses received from invited members and shall initialise this local counter to zero;

9) shall determine if an MCVideo group call for the group identity is already ongoing by determining if an MCVideo session identity has already been allocated for the group call and the MCVideo session is active;

10) if the SIP INVITE request contains an unauthorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2:

a) shall reject the SIP INVITE request with a SIP 403 (Forbidden) response as specified in clause 6.3.3.1.14; and

b) shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [11] and skip the rest of the steps;

11) if the SIP INVITE request contains an unauthorised request for an MCVideo imminent peril group call as determined by clause 6.3.3.1.13.5, shall reject the SIP INVITE request with a SIP 403 (Forbidden) response with the following clarifications:

a) shall include in the SIP 403 (Forbidden) response an application/vnd.3gpp.mcvideo-info+xml MIME body as specified in clause F.1 with the <mcvideoinfo> element containing the <mcvideo-Params> element with the <imminentperil-ind> element set to a value of "false"; and

b) shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [11] and skip the rest of the steps;

12) if a Resource-Priority header field is included in the SIP INVITE request:

a) if the Resource-Priority header field is set to the value indicated for emergency calls and the SIP INVITE request does not contain an emergency indication and the in-progress emergency state of the group is set to a value of "false", shall reject the SIP INVITE request with a SIP 403 (Forbidden) response and skip the rest of the steps; or

b) if the Resource-Priority header field is set to the value indicated for imminent peril calls and the SIP INVITE request does not contain an imminent peril indication and the in-progress imminent peril state of the group is set to a value of "false", shall reject the SIP INVITE request with a SIP 403 (Forbidden) response and skip the rest of the steps;

13) if the MCVideo group call is not ongoing then:

a) if:

i) the user identified by the MCVideo ID is not affiliated to the group identity contained in the SIP INVITE request as specified in clause 6.3.6;

ii) the group identity contained in the SIP INVITE request is not a constituent MCVideo group ID;

iii) the received SIP INVITE request does not contain an emergency indication or imminent peril indication; or

iv) the received SIP INVITE request is an authorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2 or MCVideo imminent peril group call as determined by steps clause 6.3.3.1.13.5 and is determined to not be eligible for implicit affiliation as specified in clause 8.2.2.3.6;

then shall return a SIP 403 (Forbidden) response with the warning text set to "120 user is not affiliated to this group" in a Warning header field as specified in clause 4.4, and skip the rest of the steps below;

b) if the user identified by the MCVideo ID is not authorised to initiate the prearranged group session as specified in clause 6.3.5.4, shall send a SIP 403 (Forbidden) response with the warning text set to: "119 user is not authorised to initiate the group call" in a Warning header field as specified in clause 4.4 and skip the rest of the steps below;

c) if the received SIP INVITE request contains an an authorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2 or MCVideo imminent peril group call as determined by clause 6.3.3.1.13.5 and the MCVideo user is eligible to be implicitly affiliated with the MCVideo group as determined as determined in step 13) a) iv) above, shall perform the implicit affiliation as specified in clause 8.2.2.3.7;

d) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [11];

e) shall create a prearranged group session and allocate an MCVideo session identity for the prearranged group call, and shall handle timer TNG3 (group call timer) as specified in clause 6.3.3.5;

f) if the group identity in the "SIP INVITE request for controlling MCVideo function of an MCVideo group" is a TGI:

i) shall for each of the constituent MCVideo groups homed on the primary MCVideo system:

A) if the controlling MCVideo function does not own the MCVideo group identified by the MCVideo group ID, then generate a SIP INVITE request towards the MCVideo server that owns the MCVideo group identity by following the procedures in clause 9.2.1.4.1.2; and

NOTE 2: The MCVideo server that the SIP INVITE request is sent to acts as a non-controlling MCVideo function;

B) if the controlling MCVideo function owns the MCVideo group identified by the MCVideo group ID then:

I) determine the members to invite to the prearranged MCVideo group call as specified in clause 6.3.5.5;

II) invite each group member determined in step A) above, to the group session, as specified in clause 9.2.1.4.1.1; and

III) interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3; and

ii) shall for each of the constituent MCVideo groups homed on the partner MCVideo system generate a SIP INVITE request for the MCVideo group identity homed on the partner MCVideo system as specified in clause 9.2.1.4.1.2; and

g) if the group identity in the SIP INVITE request for controlling MCVideo function of an MCVideo group is an MCVideo group ID:

i) shall determine the members to invite to the prearranged MCVideo group call as specified in clause 6.3.5.5;

ii) if necessary, shall start timer TNG1 (acknowledged call setup timer) according to the conditions stated in clause 6.3.3.3;

iii) if the received SIP INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element set to a value of "true":

A) shall cache the information that this MCVideo user has initiated an MCVideo emergency call;

B) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the information that this MCVideo user has initiated an MCVideo emergency alert; and

C) if the in-progress emergency state of the group is set to a value of "false":

I) shall set the value of the in-progress emergency state of the group to "true"; and

II) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16;

iv) if the in-progress emergency state of the group is set to a value of "false" and if the received SIP INVITE request contains an imminent peril indication set to a value of "true", the controlling MCVideo function shall:

A) shall cache the information that this MCVideo user has initiated an MCVideo imminent peril call; and

B) if the in-progress imminent peril state of the group is set to a value of "false", shall set the in-progress imminent peril state of the group to a value of "true";

v) shall invite each group member determined in step 13)g)i) above, to the group session, as specified in clause 9.2.1.4.1.1; and

vi) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3; and

14) if the MCVideo group call is ongoing then:

a) if:

i) the user identified by the MCVideo ID in the SIP INVITE request is not affiliated to the group identity contained in the SIP INVITE request as specified in clause 6.3.6;

ii) the group identity contained in the SIP INVITE request is not a constituent MCVideo group ID;

iii) the received SIP INVITE request does not contain an emergency indication or imminent peril indication; or

iv) the received SIP INVITE request is an authorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2 or MCVideo imminent peril group call as determined clause 6.3.3.1.13.5 and is determined to not be eligible for implicit affiliation as specified in clause 8.2.2.3.6;

then shall return a SIP 403 (Forbidden) response with the warning text set to "120 user is not affiliated to this group" in a Warning header field as specified in clause 4.4, and skip the rest of the steps below;

b) if the user identified by the MCVideo ID in the SIP INVITE request is not authorised to join the prearranged group session as specified in clause 6.3.5.3, shall send a SIP 403 (Forbidden) response with the warning text set to "121 user is not allowed to join the group call" in a Warning header field as specified in clause 4.4 and skip the rest of the steps below;

c) shall check if a Resource-Priority header field is included in the incoming SIP INVITE request and may apply any preferential treatment to the SIP request as specified in 3GPP TS 24.229 [11];

d) if <on-network-max-participant-count> as specified in 3GPP TS 24.481 [24] is already reached:

i) if, according to local policy, the user identified by the MCVideo ID in the SIP INVITE request is deemed to have a higher priority than an existing user in the group session, may remove a participant from the session by following clause 9.2.1.4.4.3, and skip the next step; and

NOTE 3: The local policy for deciding whether to admit a user to a call that has reached its maximum amount of participants can include the <user-priority> and the <participant-type> of the user as well as other information of the user from the group document as specified in 3GPP TS 24.481 [24]. The local policy decisions can also include taking into account whether the imminent-peril indicator or emergency indicator was received in the SIP INVITE request.

ii) shall return a SIP 486 (Busy Here) response with the warning text set to "122 too many participants" to the originating network as specified in clause 4.4 and skip the rest of the steps;

e) if the received SIP INVITE request contains an an authorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2 or MCVideo imminent peril group call as determined by clause 6.3.3.1.13.5 and the MCVideo user is eligible to be implicitly affiliated with the MCVideo group as determined in step 14) a) iv) above, shall perform the implicit affiliation as specified in clause 8.2.2.3.7;

f) if the received SIP INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element set to a value of "true":

i) shall cache the information that this MCVideo user has initiated an MCVideo emergency call;

ii) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall cache the information that this MCVideo user has initiated an MCVideo emergency alert;

iii) if the in-progress emergency state of the group is set to a value of "false":

A) shall set the value of the in-progress emergency state of the group to "true";

B) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16; and

C) shall generate SIP re-INVITE requests for the MCVideo emergency group call to the other call participants of the MCVideo group as specified in clause 6.3.3.1.6;

iv) if the in-progress imminent peril state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's imminent peril indication as specified in clause 6.3.3.1.11, setting the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11]; and

v) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5];

g) if the in-progress emergency state of the group is set to a value of "false" and if the SIP INVITE request contains an imminent peril indication set to a value of "true", the controlling MCVideo function:

i) shall cache the information that this MCVideo user has initiated an MCVideo imminent peril call; and

ii) if the in-progress imminent peril state of the group is set to a value of "false":

A) shall set the in-progress imminent peril state of the group to a value of "true";

B) shall generate SIP re-INVITE requests for the MCVideo imminent peril group call to the other call participants of the MCVideo group as specified in clause 6.3.3.1.15; and

C) upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

iii) if the in-progress imminent peril state of the group is set to a value of "true":

A) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's imminent peril indication as specified in clause 6.3.3.1.11, setting the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11];

h) shall generate a SIP 200 (OK) response as specified in the clause 6.3.3.2.4.2;

i) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

j) shall include in the SIP 200 (OK) response with the warning text set to "123 MCVideo session already exists" as specified in clause 4.4;

k) if the received SIP re-INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

l) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 4: In this case, the request was for an imminent peril call but a higher priority MCVideo emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

m) shall interact with media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 5: Resulting media plane processing is completed before the next step is performed.

n) shall send the SIP 200 (OK) response towards the inviting MCVideo client or inviting non-controlling MCVideo function according to 3GPP TS 24.229 [11];

o) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo User has joined in the MCVideo group session, as specified in clause 6.3.3.4;

NOTE 6: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

p) shall send a SIP NOTIFY request to each MCVideo client according to 3GPP TS 24.229 [11];

q) Upon receiving a SIP ACK to the above SIP 200 (OK) response and the SIP 200 (OK) response contained a Warning header field as specified in clause 4.4 with the warning text containing the mcvideo-warn-code set to "149", shall follow the procedures in clause 6.3.3.1.18; and

r) shall not continue with the rest of the clause.

Upon receiving a SIP 183 (Session Progress) response to the SIP INVITE request specified in clause 9.2.1.4.1 containing a P-Answer-State header field with the value "Unconfirmed" as specified in IETF RFC 4964 [30], the timer TNG1 (acknowledged call setup timer) is not running, the controlling MCVideo function supports media buffering and the SIP final response is not yet sent to the inviting MCVideo client:

1) *shall generate a SIP 200 (OK) response to SIP INVITE request as specified in the clause 6.3.3.2.3.2;*

2) *shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if the prearranged MCVideo group has more than <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [24];*

3) *shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;*

4) *shall include a P-Answer-State header field with the value "Unconfirmed";*

5) *if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;*

6) *if the received SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;*

7) *shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;*

NOTE 7: Resulting user plane processing is completed before the next step is performed.

8) *shall send the SIP 200 (OK) response towards the inviting MCVideo client according to 3GPP TS 24.229 [11];*

9) *shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo User has joined in the MCVideo group session, as specified in clause 6.3.3.4; and*

NOTE 8: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

10) *shall send a SIP NOTIFY request to each MCVideo client according to 3GPP TS 24.229 [11].*

Upon receiving a SIP 183 (Session Progress) response for a SIP INVITE request as specified in clause 9.2.1.4.1.2 containing an indication of required group members, the timer TNG1 (acknowledged call setup timer) is running and all SIP 200 (OK) responses have been received to all SIP INVITE requests sent to MCVideo clients specified in clause 9.2.1.4.1.1, then the controlling MCVideo function shall wait until the SIP 200 (OK) response has been received to the SIP INVITE request specified in clause 9.2.1.4.1.2 before generating a SIP 200 (OK) response to the "SIP INVITE request for controlling MCVideo function of an MCVideo group".

Upon receiving a SIP 200 (OK) response for a SIP INVITE request as specified in clause 9.2.1.4.1 that was sent to an affiliated and <on-network-required> group member as specified in 3GPP TS 24.481 [24]; and

1) *if the MCVideo ID in the SIP 200 (OK) response matches to the MCVideo ID in the corresponding SIP INVITE request;*

2) *there are no outstanding SIP 200 (OK) responses to SIP INVITE requests which were sent to affiliated and <on-network-required> group members as specified in 3GPP TS 24.481 [24]; and*

3) *there is no outstanding SIP 200 (OK) response to a SIP INVITE request sent in clause 9.2.1.4.1.2 where the SIP 183 (Session Progress) response contained an indication of required group members;*

the controlling MCVideo function:

1) *shall stop timer TNG1 (acknowledged call setup timer) as described in clause 6.3.3.3;*

2) shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;

3) shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if all members were not invited because the prearranged MCVideo group has been exceeded the <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [24];

4) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

5) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 9: Resulting media plane processing is completed before the next step is performed.

6) shall send a SIP 200 (OK) response to the inviting MCVideo client according to 3GPP TS 24.229 [11];

7) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo user has joined in the MCVideo group session, as specified in clause 6.3.3.4; and

NOTE 10: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

8) shall send the SIP NOTIFY request to the MCVideo clients according to 3GPP TS 24.229 [11].

Upon:

1) receiving a SIP 200 (OK) response for a SIP INVITE request as specified in clause 9.2.1.4.1;

2) the timer TNG1 (acknowledged call setup timer) is not running;

3) the local counter of the number of SIP 200 (OK) responses received from invited members is equal to the value of the <on-network-minimum-number-to-start> element of the group document;

4) the controlling MCVideo function supports media buffering; and

5) the SIP final response has not yet been sent to the inviting MCVideo client;

the controlling MCVideo function according to local policy:

1) shall generate SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.2;

2) shall include the warning text set to "122 too many participants" as specified in clause 4.4 in the SIP 200 (OK) response, if all members were not invited because the prearranged MCVideo group has exceeded the <max-participant-count> members as specified in 3GPP TS 24.481 [24];

3) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

4) if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

5) if the received SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

6) shall interact with the media plane as specified in 3GPP TS 24.581 [5] clause 6.3;

NOTE 11: Resulting media plane processing is completed before the next step is performed.

7) shall send a SIP 200 (OK) response to the inviting MCVideo client according to 3GPP TS 24.229 [11];

8) shall generate a notification to the MCVideo clients, which have subscribed to the conference state event package that the inviting MCVideo user has joined in the MCVideo group session, as specified in clause 6.3.3.4; and

NOTE 12: As a group document can potentially have a large content, the controlling MCVideo function can notify using content-indirection as defined in IETF RFC 4483 [29].

9) shall send the SIP NOTIFY request to the MCVideo clients according to 3GPP TS 24.229 [11].

Upon expiry of timer TNG1 (acknowledged call setup timer), if there are outstanding SIP 200 (OK) responses to SIP INVITE requests sent to affiliated and <on-network-required> group members as specified in 3GPP TS 24.481 [24], the controlling MCVideo function shall follow the procedures specified in clause 6.3.3.3.

If timer TNG1 (acknowledged call setup timer) is running and a final SIP 4xx, 5xx or 6xx response is received from an affiliated and <on-network-required> group member as specified in 3GPP TS 24.481 [24], the controlling MCVideo function shall follow the relevant procedures specified in clause 6.3.3.3.

If:

1) timer TNG1 (acknowledged call setup timer) is not running;

2) the local counter of the number of SIP 200 (OK) responses received from invited members is equal to the value of the <on-network-minimum-number-to-start> element of the group document; and

3) a final SIP 4xx, 5xx or 6xx response is received from an invited MCVideo client;

then the controlling MCVideo function shall perform one of the following based on policy:

1) send the SIP final response towards the inviting MCVideo client, according to 3GPP TS 24.229 [11], if a SIP final response was received from all the other invited MCVideo clients and the SIP 200 (OK) response is not yet sent; or

2) remove the invited MCVideo client from the MCVideo Session as specified in clause 6.3.3.1.5, if a SIP final response other than 2xx or 3xx was received from all the invited MCVideo clients and the SIP 200 (OK) response is already sent. The controlling MCVideo function may invite an additional member of the prearranged MCVideo group as specified in clause 9.2.1.4.1 that has not already been invited, if the prearranged MCVideo group has more than <on-network-max-participant-count> members as specified in 3GPP TS 24.481 [24], and all members have not yet been invited.

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCVideo client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.18."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.1.4.3]

"Upon receiving a SIP BYE request the controlling MCVideo function shall follow the procedures as specified in clause 6.3.3.2.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.2]

"This clause is referenced from other procedures.

The controlling MCVideo function shall generate an initial SIP INVITE request according to 3GPP TS 24.229 [11].

The controlling MCVideo function:

1) shall include in the Contact header field an MCVideo session identity for the MCVideo session with the g.3gpp.mcvideo media feature tag, the isfocus media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo" according to IETF RFC 3840 [22];

2) shall include an Accept-Contact header field containing the g.3gpp.mcvideo media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [20];

3) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcvideo" (coded as specified in 3GPP TS 24.229 [11]), in a P-Asserted-Service-Id header field according to IETF RFC 6050 [14] in the SIP INVITE request;

4) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo" along with parameters "require" and "explicit" according to IETF RFC 3841 [20];

5) shall include a Referred-By header field with the public user identity of the inviting MCVideo client;

6) should include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [7]. The refresher parameter shall be omitted;

7) shall include the Supported header field set to "timer";

8) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body to the outgoing INVITE request;

9) if the received SIP INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body containing an <ambient-viewing-type> element and:

a) if the Priv-Answer-Mode header field specified in IETF RFC 5373 [27] was included in the received SIP INVITE request with a value of "Auto" or if no Priv-Answer-Mode header field was received in the received SIP INVITE request; or

b) a Priv-Answer-Mode header field was received containing a value other than "Auto";

shall include a Priv-Answer-Mode header field set to a value of "Auto" in the outgoing SIP INVITE request; and

10) if the received SIP INVITE request did not contain an application/vnd.3gpp.mcvideo-info+xml MIME body containing an <ambient-viewing-type> element, shall include an unmodified Answer-Mode header field if present in the incoming SIP INVITE request."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.1]

"The SDP offer is generated based on the received SDP offer. The SDP offer generated by the controlling MCVideo function:

1) when initiating a new MCVideo session the SDP offer;

a) shall contain only one SDP media-level section for MCVideo video media stream as contained in the received SDP offer; and

b) shall contain an SDP media-level section for one media-transmission control entity, if present in the received SDP offer; and

2) when adding a new MCVideo user to an existing MCVideo Session, the SDP offer shall contain the media stream currently used in the MCVideo session.

When composing the SDP offer according to 3GPP TS 24.229 [11], the controlling MCVideo function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the controlling MCVideo function;

2) for the MCVideo video media stream, shall include all media-level attributes from the received SDP offer;

3) shall replace the IP address and port number for the offered media transmission control entity, if any, in the received SDP offer with the IP address and port number of the controlling MCVideo function; and

4) for the offered media transmission control entity, shall include the offered media transmission control entity 'fmtp' attributes as specified in 3GPP TS 24.581 [5] clause 14."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.2.2]

"When an MCVideo call is established a new instance of the transmission control server state machine for 'general transmission control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' is added.

When an MCVideo call is established a new instance of the transmission control server state machine for 'general reception control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic reception control operation towards the transmission participant' is added.

If the optional "mc_queueing" feature is supported and has been negotiated as specified in clause 14, the transmission control server could queue the implicit transmission control request for the media-transmission control entity.

The original initial SIP INVITE request or SIP REFER request to establish an MCVideo chat group call or to rejoin an ongoing MCVideo call is not handled as an implicit transmission control request message by the transmission control server unless explicitly stated in the SIP INVITE request or in the SIP REFER request.

The permission to send media to the inviting MCVideo client due to implicit transmission control request is applicable to both confirmed indication and unconfirmed indication.

When the first unconfirmed indication is received from the invited participating MCVideo function (see 3GPP TS 24.281 [2]) the transmission control server optionally can give an early indication to send RTP media packets, to the inviting MCVideo client.

If an early indication to send RTP media packets is given to the inviting MCVideo client, the transmission participant is granted the permission to send media and the MCVideo server buffers RTP media packets received from the MCVideo client at least until the first invited MCVideo client accepts the invitation or until the RTP media packet buffer exceeds it maximum limit to store RTP media packets.

If the MCVideo server does not support or does not allow media buffering then when an early indication to send RTP media packets is not given to the inviting MCVideo client, the transmission participant is granted the permission to send media when the first invited MCVideo client accepts the media.

Before the transmission control server sends the first transmission control message in the MCVideo call, the transmission control server has to assign itself a SSRC identifier to be included in media transmission control messages and quality feedback messages if the MCVideo server is supporting that option. A suitable algorithm to generate the SSRC identifier is described in IETF RFC 3550 [3].

The transmission participant and the transmission control server can negotiate the maximum priority level that the transmission participant is permitted to request. The transmission control server can pre-empt the current sender based on the negotiated maximum priority level that the transmission participant is permitted to request and the priority level included in the Transmission Media Request message.

NOTE: The maximum priority level that a transmission participant can use is negotiated as specified in clause 14.3.3 and is based on group configuration data retrieved by the controlling MCVideo function from the group management server as described in 3GPP TS 24.481 [12] and service configuration data retrieved by the controlling MCVideo function from the configuration management server as described in 3GPP TS 24.484 [13].

The transmission participant and the transmission control server can negotiate queueing of Transmission requests using the "mc_queueing" fmtp attribute as described in clause 14. If queueing is supported and negotiated, the transmission control server queues the transmission control request if a Transmission Media Request message is received when another transmission participant has the transmission and the priority of the current speaker is the same or higher."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.

Step 2 When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.

Step 2 When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.

The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.2]

"Upon receiving a transmission control message the participating MCVideo function:

1. *shall immediately forward the transmission control message to the transmission control server if the message is received from the transmission participant;*

2. *if an MBMS subchannel is not used for a transmission in the session the transmission control message is associated with, shall immediately forward the transmission control message to the transmission participant if the message is received from the transmission control server; and*

3. *if an MBMS subchannel is used for a transmission in the session the transmission control message is associated with:*

    a. *if*

        i. *the transmission control message is not a Transmission Idle message or a Media Transmission Notify message;*

        ii. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

        iii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report;*

    shall immediately forward the transmission control message to the transmission participant; and

    b. *if*

        i. *the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report; and*

        ii. *if the transmission control message is the Transmission Idle message or the Media Transmission Notify message,*

    shall perform actions as specified in clause 10.2.

NOTE: When the Transmit Idle or Media Transmission Notify messages are discarded the messages are sent to the MCVideo clients over the MBMS subchannel allocated for the transmission as specified in clause 10.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.3]

"Upon receiving RTP media packets the participating MCVideo function:

1. *shall immediately forward the RTP media packet to the controlling MCVideo function if the RTP packet is from an MCVideo client; and*

2. *if an MBMS subchannel is not used for a transmission in the session the RTP media packets are associated with, shall immediately forward the RTP media packets to the MCVideo client if the RTP packet is from the controlling MCVideo function or the non-controlling MCVideo function.*

3. *if an MBMS subchannel is used for a transmission in the session the RTP media packets are associated with and if RTP media packets are received from the controlling MCVideo function or the non-controlling MCVideo function:*

    a. *if*

        i. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

        ii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report,*

    shall immediately forward the RTP media packets to the MCVideo client; and

    b. *if the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report, shall perform actions as specified in clause 10.2."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.4]

"When the participating function receives an indication from the application and signalling plane that session release is initiated, the participating MCVideo function:

1. *shall stop sending transmission control messages towards the transmission participant and the transmission control server; and*

2. *shall stop sending RTP media packets towards the MCVideo client and towards the controlling MCVideo function.*

When the participating MCVideo function receives an indication from the application and signalling plane that the session is released, the participating MCVideo function:

1. *in case of a pre-established session, shall perform the actions in clause 9.3.2; and*

2. *in case of an on-demand session, shall release the media resources associated with the session."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.1]

"The transmission control messages are coded as described in clause 9.1.2 where the transmission control message is part of the application-dependent data.

For the transmission control protocol the ASCII name string is: MCVx (Mission Critical Video x) with x=0, 1 or 2 as specified in clause 9.1.2.

A list of transmission control messages can be found in clause 9.2.2.1.

The transmission control specific fields are specified in clause 9.2.3."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.2.1]

"Based on the negotiations during the call establishment specified in 3GPP TS 24.281 [2], a new instance of the 'Transmission participant state transition diagram for basic transmission control operation', as specified in clause 6.2.4 and a new instance of the 'Transmission participant state transition diagram for basic reception control operation' as specified in clause 6.2.5, shall be created for this call.

The SIP INVITE request sent by the application and signalling plane:

1. *shall be regarded an implicit Transmission request when an implicit Transmission request is negotiated; and*

2. *shall not be regarded as an implicit Transmission request in case of a rejoin to an already on-going group call.*

NOTE: The transmission participant can negotiate the use of prioritization of the Transmission Media Request message. In that case, the transmission participant can request permission to send media at a priority level that is either the same as or lower than the highest priority that was permitted to the participant in the MCVideo call initialization. If a transmission participant is authorized for pre-emptive priority in the MCVideo call it is good practise to always request permission to send RTP media packets at a priority level that is lower than pre-emptive priority unless the user explicitly requests to pre-empt the current RTP media packets sender."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.2.2]

"The MCVideo call release (whether it is initiated by the transmission participant or transmission control server) is a two-step procedure.

Step 1 *The transmission participant stops sending transmission control and reception control messages and the MCVideo client stops sending and receiving RTP media packets.*

Step 2 *When the application and signalling plane has determined that the MCVideo call is released, the corresponding instance of the 'Transmission participant state transition diagram for basic transmission control operation' as specified in clause 6.2.4 and the corresponding instance of the 'Transmission participant state transition diagram for basic reception control operation' as specified in clause 6.2.5 are terminated and the transmission participant releases all the used resources.*

The user plane can initiate the release step 1, but the application and signalling plane always initiates the release step 2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.2.5]

"The transmission participant shall behave according to the state diagrams and the state transitions specified in this clause.

Figure 6.2.5.1-1 shows the state diagram for 'Transmission participant state transition diagram for general reception control operation'.

…"

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.1.15.3 Test description

5.1.1.15.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the participating MCVideo functions.
- SS-UE1 (MCVideo Client) and SS-UE2 (MCVideo Client), simulated by the SS (MCVideo Server).

IUT:

- IUT (MCVideo Server), providing the controlling MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of all sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Transmission Control Server.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server).

- The SS-UE1 (MCVideo client) and the SS-UE2 (MCVideo client) are registered and authorized to use the MCVideo system.
- The SS-UE1 (MCVideo client) and the SS-UE2 (MCVideo client) are affiliated to the pre-arranged group and are authorized to initiate group calls.
- The IUT (MCVideo Server) is the controlling server of the pre-arranged group.

5.1.1.15.3.2 Test procedure sequence

Table 5.1.1.15.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E group call establishment' as described in Annex A, Table A.4.1.4.3-1 to establish a pre-arranged group call with implicit transmission control request according to option b.ii of note in Annex A, Table A.4.1.4.3-1, initiated by SS-UE1 (MCVideo Client)?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Media Reception Request and Response' as described in Annex A, Table A.3.9.3-1 to request SS-UE1 (MCVideo Client) media reception?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Media Reception End Request and Response' as described in Annex A, Table A.3.10.3-1 to request end of media reception?	-	-	3	P
4	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1 to end the transmission of SS-UE1 (MCVideo Client)?	-	-	4	P
5	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1 to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	5	P
6	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Media Reception Request and Response' as described in Annex A, Table A.3.9.3-1 to request SS-UE1 (MCVideo Client) media reception?	-	-	2	P
7	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1 to end the transmission of SS-UE1 (MCVideo Client), while SS-UE2 (MCVideo Client) is still permitted to receive media?	-	-	4	P
8	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.1.5.3-1, to release the call?	-	-	6	P

5.1.1.15.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step.

5.1.2 Chat Group Call

5.1.2.1 MCVideo Server - MCVideo Server / Chat Group Call / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group call / Cancel Imminent Peril State / Controlling server

5.1.2.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function initiates a chat
group call without implicit transmission control }
    then { IUT (MCVideo Server) accepts the call from initiator and sends transmission idle to SS-
UE1 (MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established a chat group call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function requests to join
the ongoing chat group call without implicit transmission control, on behalf of SS-UE2 (MCVideo
client) }
    then { IUT (MCVideo Server) accepts the call join request and sends transmission idle to SS-UE2
(MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established a chat group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests to end the transmission control }
    then { IUT (MCVideo Server) sends transmission control idle to SS-UE1 (MCVideo Client) and sends
transmission control idle to SS-UE2 (MCVideo Client) }
            }
```

(4)

```
with { IUT (MCVideo Server) having established a chat group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control }
    then { IUT (MCVideo Server) sends transmission granted to SS-UE1 (MCVideo Client) and sends
media transmission notification to SS-UE2 (MCVideo Client) }
            }
```

(5)

```
with { IUT (MCVideo Server) having established a chat group call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function upgrades the call
to emergency group call with implicit transmission control }
    then { IUT (MCVideo Server) accepts the upgrade from the initiator and updates the call for all
participants and sends transmission granted to SS-UE1 (MCVideo Client) and sends media transmission
notification to SS-UE2 (MCVideo Client) }
            }
```

(6)

```
with { IUT (MCVideo Server) having upgraded to an emergency chat group call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function requests to cancel
the ongoing emergency state }
    then { IUT (MCVideo Server) accepts the emergency state cancellation and updates the call for
all participants }
            }
```

(7)

```
with { IUT (MCVideo Server) having established a chat group call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function upgrades the call
to imminent peril group call with implicit transmission control }
    then { IUT (MCVideo Server) accepts the upgrade from the initiator and updates the call for all
participants and sends transmission granted to SS-UE1 (MCVideo Client) and sends media transmission
notification to SS-UE2 (MCVideo Client) }
            }
```

(8)

```
with { IUT (MCVideo Server) having upgraded to an imminent peril chat group call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function requests to cancel
the ongoing imminent peril state }
    then { IUT (MCVideo Server) accepts the imminent peril state cancellation and updates the call
for all participants }
            }
```

(9)

```
with { IUT (MCVideo Server) having established a chat group call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function ends the chat
group call }
    then { IUT (MCVideo Server) responds by sending a SIP 200 (OK) message to SS (MCVideo Server)
ending the call and sends a SIP BYE message to SS (MCVideo Server) providing the terminating
participating function }
            }
```

5.1.2.1.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.2.4.1, 9.2.2.4.2, 9.2.2.4.1.3, 9.2.2.4.2, 9.2.2.4.3.2, 9.2.2.4.3.3 and 9.2.2.2.1.1, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.3.1, 6.3.2.2, 9.2.1, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.4.1.1]

"In the procedures in this clause:

1) *MCVideo ID in an incoming SIP INVITE request refers to the MCVideo ID of the originating user from the <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;*

2) *group identity in an incoming SIP INVITE request refers to the group identity from the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;*

3) *MCVideo ID in an outgoing SIP INVITE request refers to the MCVideo ID of the called user in the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the outgoing SIP INVITE request;*

4) *emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and*

5) *alert indication in an incoming SIP INVITE request refers to the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.*

Upon receipt of a "SIP INVITE request for controlling MCVideo function of an MCVideo group" containing a group identity identifying a chat MCVideo group, the controlling MCVideo function:

…

3) *if received SIP INVITE request includes an application/vnd.3gpp.mcvideoinfo+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.3.1.17;*

4) shall retrieve the necessary group document(s) from the group management server for the group identity contained in the SIP INVITE request and carry out initial processing as specified in clause 6.3.5.2 and continue with the rest of the steps if the checks in clause 6.3.5.2 succeed;

…

6) if the SIP INVITE request contains unauthorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2:

a) shall reject the SIP INVITE request with a SIP 403 (Forbidden) response as specified in clause 6.3.3.1.14; and

b) shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [11] and skip the rest of the steps;

…

8) if a Resource-Priority header field is included in the SIP INVITE request:

a) if the Resource-Priority header field is set to the value indicated for emergency calls and the SIP INVITE request does not contain an emergency indication and the in-progress emergency state of the group is set to a value of "false", shall reject the SIP INVITE request with a SIP 403 (Forbidden) response and skip the remaining steps; and

b) if the Resource-Priority header field is set to the value indicated for imminent peril calls and the SIP INVITE request does not contain an imminent peril indication and the in-progress imminent peril state of the group is set to a value of "false", shall reject the SIP INVITE request with a SIP 403 (Forbidden) response; and skip the remaining steps;

9) shall determine if the media parameters are acceptable and the MCVideo codecs are offered in the SDP offer and if not reject the request with a SIP 488 (Not Acceptable Here) response and skip the rest of the steps;

10) shall create a chat group session and allocate an MCVideo session identity for the chat group session if the MCVideo chat group session identity does not already exist, and may handle timer TNG3 (group call timer) as specified in clause 6.3.3.5;

11) if the chat group session is ongoing and the <on-network-max-participant-count> as specified in 3GPP TS 24.481 [24] is already reached:

a) if, according to local policy, the user identified by the MCVideo ID in the SIP INVITE request is deemed to have a higher priority than an existing user in the chat group session, may remove a participant from the session by following clause 9.2.1.4.4.3, and skip the next step; and

NOTE 2: The local policy for deciding whether to admit a user to a call that has reached its maximum amount of participants can include the <user-priority> and the <participant-type> of the user as well as other information of the user from the group document as specified in 3GPP TS 24.481 [24]. The local policy decisions can also include taking into account whether the imminent-peril indicator or emergency indicator was received in the SIP INVITE request.

b) shall return a SIP 486 (Busy Here) response with the warning text set to "122 too many participants" to the originating network as specified in clause 4.4. Otherwise, continue with the rest of the steps;

12) if the received SIP INVITE request was determined to be eligible for implicit affiliation in step 5) and if clause 8.2.2.3.7 was not previously invoked in the present clause, shall perform the implicit affiliation as specified in clause 8.2.2.3.7;

13) if the SIP INVITE request contains an emergency indication set to a value of "true" or the in-progress emergency state of the group to "true" the controlling MCVideo function shall:

a) validate that the SIP INVITE request includes a Resource-Priority header field populated with the values for an MCVideo emergency group call as specified in clause 6.3.3.1.19, and if not:

i) perform the actions specified in clause 6.3.3.1.8;

ii) send the SIP UPDATE request generated in clause 6.3.3.1.8 towards the initiator of the SIP INVITE request according to 3GPP TS 24.229 [11]; and

iii) upon receiving a SIP 200 (OK) response to the SIP UPDATE request sent in clause 6.3.3.1.8, proceed with the rest of the steps.

NOTE 3: Verify that the Resource-Priority header is included and properly populated for both ongoing and newly-entered in-progress emergency states of the specified group.

b) if the in-progress emergency state of the group is set to a value of "true" and this MCVideo user is indicating a new emergency indication:

i) for each of the other affiliated members of the group generate a SIP MESSAGE request notification of the MCVideo user's emergency indication as specified in clause 6.3.3.1.11 with the following clarifications:

A) set the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true";

B) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, perform the procedures specified in clause 6.3.3.1.12; and

C) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11];

ii) cache the information that this MCVideo user has initiated an MCVideo emergency call; and

iii) if the SIP INVITE request contains an authorised request for an MCVideo emergency alert as determined in step i) B) above, cache the information that this MCVideo user has initiated an MCVideo emergency alert; and

c) if the in-progress emergency state of the group is set to a value of "false":

i) shall set the value of the in-progress emergency state of the group to "true";

ii) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16;

iii) shall generate SIP re-INVITE requests for the MCVideo emergency group call to the other affiliated and joined participants of the chat MCVideo group as specified in clause 6.3.3.1.6;

iv) shall generate SIP INVITE requests for the MCVideo emergency group call to the affiliated but not joined members of the chat MCVideo group as specified in clause 6.3.3.1.7;

A) for each affiliated but not joined member shall send the SIP INVITE request towards the MCVideo client as specified in 3GPP TS 24.229 [11]; and

B) upon receiving a SIP 200 (OK) response to the SIP INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5];

v) shall cache the information that this MCVideo user has initiated an MCVideo emergency call; and

vi) if the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body is set to *"true" and is an authorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall cache the information that this MCVideo user has initiated an MCVideo emergency alert; and*

vii) if the in-progress imminent peril state of the group is set to a value of "true", shall set it to a value of "false";

14) if the in-progress emergency state of the group is set to a value of "false" and if the SIP INVITE request contains an imminent peril indication set to a value of "true" or the in-progress imminent peril state of the group is set to "true", the controlling MCVideo function shall:

a) validate that the SIP INVITE request includes a Resource-Priority header field populated with the values for an MCVideo imminent peril group call as specified in clause 6.3.3.1.19, and if not:

i) perform the actions specified in clause 6.3.3.1.8;

ii) send the SIP UPDATE request generated in clause 6.3.3.1.8 towards the initiator of the SIP INVITE request according to 3GPP TS 24.229 [11]; and

iii) upon receiving a SIP 200 (OK) response to the SIP UPDATE request sent in clause 6.3.3.1.8 proceed with the rest of the steps.

NOTE 4: Verify that the Resource-Priority header is included and properly populated for both ongoing and newly-entered in-progress imminent peril states of the specified group.

b) if the in-progress imminent peril state of the group is set to a value of "true" and this MCVideo user is indicating a new imminent peril indication:

i) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's imminent peril indication as specified in clause 6.3.3.1.11 with the following clarifications;

A) set the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11]; and

ii) cache the information that this MCVideo user has initiated an MCVideo imminent peril call; and

c) if the in-progress imminent peril state of the group is set to a value of "false":

i) shall set the value of the in-progress imminent peril state of the group to "true";

ii) shall generate SIP re-INVITE requests for the MCVideo imminent peril group call to the other affiliated and joined participants of the chat MCVideo group as specified in clause 6.3.3.1.15;

iii) shall generate SIP INVITE requests for the MCVideo imminent peril call to the affiliated but not joined members of the chat MCVideo group as specified in clause 6.3.3.1.7;

A) for each affiliated but not joined member shall send the SIP INVITE request towards the MCVideo client as specified in 3GPP TS 24.229 [11]; and

B) Upon receiving a SIP 200 (OK) response to the SIP INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

iv) shall cache the information that this MCVideo user has initiated an MCVideo imminent peril call;

15) shall accept the SIP request and generate a SIP 200 (OK) response to the SIP INVITE request according to 3GPP TS 24.229 [11];

16) shall include in the SIP 200 (OK) response an SDP answer according to 3GPP TS 24.229 [11] with the clarifications specified in clause 6.3.3.2.1 unless the procedures of clause 6.3.3.1.8 were performed in step 13)a) or step 14)a) above;

17) should include the Session-Expires header field and start supervising the SIP session according to IETF RFC 4028 [23]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

18) shall include the "timer" option tag in a Require header field;

19) shall include the following in a Contact header field:

a) the g.3gpp.mcvideo media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo";

c) the MCVideo session identity; and

d) the media feature tag isfocus;

20) shall include the "tdialog" option tag in a Supported header field according to IETF RFC 4538 [32];

21) if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

22) if the received SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 5: In this case, the request was for an imminent peril call but a higher priority MCVideo emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

23) shall interact with media plane as specified in 3GPP TS 24.581 [5];

24) shall send the SIP 200 (OK) response to the MCVideo client according to 3GPP TS 24.229 [11]; and

25) if the chat group session was already ongoing and if at least one of the participants has subscribed to the conference event package, shall send a SIP NOTIFY request to all participants with a subscription to the conference event package as specified in clause 9.2.3.4.2.

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCVideo client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.18."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.4.1.2]

"In the procedures in this clause:

1) emergency indication in an incoming SIP re-INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

2) imminent peril indication in an incoming SIP re-INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

Upon receipt of a SIP re-INVITE request for an MCVideo session identity identifying a chat MCVideo group session, the controlling MCVideo function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP re-INVITE request with a SIP 500 (Server Internal Error) response. The controlling MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15] and skip the rest of the steps;

NOTE 1: if the SIP re-INVITE request contains an emergency indication or an imminent peril indication set to a value of "true" and this is an authorised request for originating an MCVideo emergency group call as determined by clause 6.3.3.1.13.2, or for originating an MCVideo imminent peril group call as determined by clause 6.3.3.1.13.5, the controlling MCVideo function can according to local policy choose to accept the request.

2) if the received SIP re-INVITE request includes an application/vnd.3gpp.mcvideo-info+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.3.1.17;

3) if the SIP re-INVITE request contains an unauthorised request for an MCVideo emergency call as determined by clause 6.3.3.1.13.2:

a) shall reject the SIP re-INVITE request with a SIP 403 (Forbidden) response as specified in clause 6.3.3.1.14; and

b) shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [11] and skip the rest of the steps;

4) if the SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "true" and is an authorised request to initiate an MCVideo emergency group call as determined by clause 6.3.3.1.13.2, the controlling MCVideo function shall:

a) validate that the SIP re-INVITE request includes a Resource-Priority header field is populated correctly for an MCVideo emergency group call as specified in clause 6.3.3.1.19, and if not:

i) shall perform the actions specified in clause 6.3.3.1.8; and

ii) upon receiving a SIP 200 (OK) response to the SIP UPDATE request sent in clause 6.3.3.18 shall proceed with the rest of the steps.

NOTE 2: Verify that the Resource-Priority header is included and properly populated for both ongoing and newly-entered in-progress emergency states of the specified group.

b) if the in-progress emergency state of the group is set to a value of "true" and this MCVideo user is indicating a new emergency indication:

i) shall cache the MCVideo ID of the MCVideo user that has initiated an MCVideo emergency call;

ii) if the SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <alert-ind> element set to a value of "true" and is an authorised request for an MCVideo emergency alert as determined by clause 6.3.3.1.13.1, shall cache the MCVideo ID of the MCVideo user that has initiated an MCVideo emergency alert; and

iii) for each of the other affiliated members of the group, generate a SIP MESSAGE request notification of the MCVideo user's emergency indication as specified in clause 6.3.3.1.11 with the following clarifications:

A) set the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true";

B) if the received SIP re-INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, perform the procedures specified in clause 6.3.3.1.12; and

C) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11]; and

c) if the in-progress emergency state of the group is set to a value of "false":

i) shall set the value of the in-progress emergency state of the group to "true";

ii) shall cache the MCVideo ID of the MCVideo user that has initiated an MCVideo emergency call;

iii) if the SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <alert-ind> element set to a value of "true" and this is an authorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall cache the MCVideo ID of the MCVideo user that has initiated an MCVideo emergency alert;

iv) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16;

v) shall generate SIP re-INVITE requests for the MCVideo emergency group call to the other affiliated and joined participants of the chat MCVideo group as specified in clause 6.3.3.1.6. The MCVideo controlling function:

A) for each affiliated and joined member shall send the SIP re-INVITE request towards the MCVideo client as specified in 3GPP TS 24.229 [11]; and

B) Upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

vi) *shall generate SIP INVITE requests for the MCVideo emergency group call to the affiliated but not joined members of the chat MCVideo group as specified in clause 6.3.3.1.7. The controlling MCVideo function:*

A) *for each affiliated but not joined member shall send the SIP INVITE request towards the MCVideo client as specified in 3GPP TS 24.229 [11]; and*

B) *Upon receiving a SIP 200 (OK) response to the SIP INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and*

vii) *if the in-progress imminent peril state of the group is set to a value of "true", shall set it to a value of "false";*

5) *if the SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "false" and is an unauthorised request for an MCVideo emergency group call cancellation as determined by clause 6.3.3.1.13.4:*

a) *shall reject the SIP re-INVITE request with a SIP 403 (Forbidden) response;*

b) *shall include in the SIP 403 (Forbidden) response an application/vnd.3gpp.mcvideo-info+xml MIME body as specified in annex F.1 with an <emergency-ind> element set to a value of "true";*

c) *if an <alert-ind> element of the mcvideoinfo MIME body is included set to "false" and there is an outstanding MCVideo emergency alert for this MCVideo user, shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body and <alert-ind> element set to a value of "true"; and*

d) *shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [11] and skip the rest of the steps;*

6) *if the SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "false" and is determined to be an authorised request for an MCVideo emergency call cancellation as specified in clause 6.3.3.1.13.4 and the in-progress emergency state of the group to is set to a value of "true" the controlling MCVideo function shall:*

a) *validate that the SIP re-INVITE request includes a Resource-Priority header field is populated correctly for a normal priority MCVideo group call as specified in clause 6.3.3.1.19, and if not:*

i) *shall perform the actions specified in clause 6.3.3.1.8; and*

ii) *upon receiving a SIP 200 (OK) response to the SIP UPDATE request sent in clause 6.3.3.1.8 shall proceed with the rest of the steps;*

NOTE 3: Verify that the Resource-Priority header is included and properly populated for an in-progress emergency state cancellation of the specified group.

b) *shall set the in-progress emergency group state of the group to a value of "false";*

c) *shall clear the cache of the MCVideo ID of the MCVideo user identified by the <originated-by> element as having an outstanding MCVideo emergency group call;*

d) *if an <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body is included and set to "false" and is determined to be an authorised request for an MCVideo emergency alert cancellation as specified in clause 6.3.3.1.13.3 and there is an outstanding MCVideo emergency alert for this MCVideo user shall:*

i) *if the received SIP re-INVITE request contains an <originated-by> element in the application/vnd.3gpp.mcvideo-info+xml MIME body, clear the cache of the MCVideo ID of the MCVideo user identified by the <originated-by> element as having an outstanding MCVideo emergency alert; and*

ii) *if the received SIP re-INVITE request does not contain an <originated-by> element in the application/vnd.3gpp.mcvideo-info+xml MIME body, clear the cache of the MCVideo ID of the sender of the SIP re-INVITE request as having an outstanding MCVideo emergency alert;*

e) shall generate SIP re-INVITE requests to the other affiliated and joined members of the MCVideo group as specified in clause 6.3.3.1.6. The MCVideo controlling function:

i) for each affiliated and joined member shall send the SIP re-INVITE request towards the MCVideo client as specified in 3GPP TS 24.229 [11]; and

ii) Upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

NOTE 4: Clause 6.3.3.1.5 will inform the affiliated and joined members of the cancellation of the MCVideo group's in-progress emergency state and the cancellation of the MCVideo emergency alert if applicable.

f) for each of the affiliated but not joined members of the group shall:

i) generate a SIP MESSAGE request notification of the cancellation of the MCVideo user's emergency call as specified in clause 6.3.3.1.11;

ii) set the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "false";

iii) if indicated above in step d), set the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "false"; and

iv) send the SIP MESSAGE request according to 3GPP TS 24.229 [11];

7) if a Resource-Priority header field is included in the SIP re-INVITE request:

a) if the Resource-Priority header field is set to the value indicated for emergency calls and the received SIP re-INVITE request does not contain an authorised request for an MCVideo emergency call as determined in step 4) above and the in-progress emergency state of the group is set to a value of "false", shall reject the SIP re-INVITE request with a SIP 403 (Forbidden) response and skip the remaining steps; or

b) if the Resource-Priority header field is set to the value indicated for imminent peril calls and the received SIP re-INVITE request does not contain an authorised request for an MCVideo imminent peril call as determined by the procedures of clause 6.3.3.1.13.5 and the in-progress imminent peril state of the group is set to a value of "false", shall reject the SIP re-INVITE request with a SIP 403 (Forbidden) response and skip the remaining steps;

8) if the received SIP re-INVITE request contains an imminent peril indication, shall perform the procedures specified in clause 9.2.2.4.1.3 and skip the rest of the steps;

9) shall include in the SIP 200 (OK) response an SDP answer according to 3GPP TS 24.229 [11] with the clarifications specified in clause 6.3.3.2.1 unless the procedures of clause 6.3.3.1.8 were performed in step 6) a) i) above;

10) shall include the "tdialog" option tag in a Supported header field according to IETF RFC 4538 [32];

11) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <alert-ind> element set to a value of "true" and if this is an unauthorised request for an MCVideo emergency alert as determined by clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

12) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <alert-ind> element set to a value of "false" and if this is an unauthorised request for an MCVideo emergency alert cancellation as determined by clause 6.3.3.1.13.3, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

13) if the received SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true", this is an authorised request for an MCVideo imminent peril group call and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 5: In this case, the request was for an imminent peril call but a higher priority MCVideo emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

14) shall interact with media plane as specified in 3GPP TS 24.581 [5]; and

15) shall send the SIP 200 (OK) response towards the MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCVideo client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.18."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.4.1.3]

"In the procedures in this clause:

1) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

When the controlling function receives a SIP re-INVITE request with and imminent peril indication, the controlling function:

1) if the SIP re-INVITE request contains an unauthorised request for an MCVideo imminent peril group call as determined by clause 6.3.3.1.13.5, shall reject the SIP re-INVITE request with a SIP 403 (Forbidden) response with the following clarifications:

a) shall include in the SIP 403 (Forbidden) response an application/vnd.3gpp.mcvideo-info+xml MIME body as specified in Annex F.1 with the <mcvideoinfo> element containing the <mcvideo-Params> element with the <imminentperil-ind> element set to a value of "false"; and

b) shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [11] and skip the rest of the steps;

2) if the in-progress emergency group state of the group is set to a value of "false" and if the SIP re-INVITE request contains an imminent peril indication set to a value of "true" or the in-progress imminent peril state of the group to "true", the controlling MCVideo function shall:

a) validate that the SIP re-INVITE request includes a Resource-Priority header field with the namespace set to the MCVideo-specific namespace and the priority set to the priority designated for imminent peril calls and if not:

i) perform the actions specified in clause 6.3.3.1.8;

ii) send the SIP UPDATE request generated in clause 6.3.3.1.8 towards the initiator of the SIP re-INVITE request according to 3GPP TS 24.229 [11]; and

iii) upon receiving a SIP 200 (OK) response to the SIP UPDATE request sent in clause 6.3.3.1.8 proceed with the rest of the steps.

NOTE 3: Verify that the Resource-Priority header is included and properly populated for both ongoing and newly-entered in-progress imminent peril states of the specified group.

b) if the in-progress imminent peril state of the group is set to a value of "true" and this MCVideo user is indicating a new imminent peril indication:

i) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's imminent peril indication as specified in clause 6.3.3.1.11 with the following clarifications;

A) set the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11]; and

ii) cache the information that this MCVideo user has initiated an MCVideo imminent peril call; and

c) if the in-progress imminent peril state of the group is set to a value of "false":

i) shall set the value of the in-progress imminent peril state of the group to "true";

ii) shall generate SIP re-INVITE requests for the MCVideo imminent peril group call to the other affiliated and joined participants of the chat MCVideo group as specified in clause 6.3.3.1.15;

iii) shall generate SIP INVITE requests for the MCVideo imminent peril group call to the affiliated but not joined members of the chat MCVideo group as specified in clause 6.3.3.1.7;

A) for each affiliated but not joined member shall send the SIP INVITE request towards the MCVideo client as specified in 3GPP TS 24.229 [11]; and

B) Upon receiving a SIP 200 (OK) response to the SIP INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

iv) shall cache the information that this MCVideo user has initiated an MCVideo imminent peril call;

3) if the SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "false" and is an unauthorised request for an MCVideo imminent peril group call cancellation as determined by clause 6.3.3.1.13.6 shall:

a) reject the SIP re-INVITE request with a SIP 403 (Forbidden) response to the SIP re-INVITE request; and

b) include in the SIP 403 (Forbidden) response:

i) include in the SIP 403 (Forbidden) response an application/vnd.3gpp.mcvideo-info+xml MIME body as specified in Annex F.1 with the <mcvideoinfo> element containing the <mcvideo-Params> element with the <imminentperil-ind> element set to a value of "false";

ii) send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [11]; and

iii) skip the rest of the steps;

4) if the SIP re-INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "false" and is determined to be an authorised request for an MCVideo imminent peril call cancellation as specified in clause 6.3.3.1.13.6 and the in-progress imminent peril state of the group to is set to a value of "true" the controlling MCVideo function shall:

a) validate that the SIP re-INVITE request includes a Resource-Priority header field with the namespace set to the MCVideo-specific namespace, and the priority set to the priority level designated for a normal priority MCVideo group call, and if not:

i) shall perform the actions specified in clause 6.3.3.1.8; and

ii) upon receiving a SIP 200 (OK) response to the SIP UPDATE request sent in clause 6.3.3.1.8 shall proceed with the rest of the steps;

NOTE 3: verify that the Resource-Priority header is included and properly populated for an in-progress emergency group state cancellation of the specified group.

b) shall set the in-progress imminent peril state of the group to a value of "false";

c) shall cache the information that this MCVideo user no longer has an outstanding MCVideo imminent peril group call;

d) shall generate SIP re-INVITES requests to the other affiliated and joined members of the MCVideo group as specified in clause 6.3.3.1.15. The MCVideo controlling function:

i) for each affiliated and joined member shall send the SIP re-INVITE request towards the MCVideo client as specified in 3GPP TS 24.229 [11]; and

ii) Upon receiving a SIP 200 (OK) response to the SIP re-INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

NOTE 4: clause 6.3.3.1.15 will inform the affiliated and joined members of the cancellation of the MCVideo group's in-progress emergency group state and the cancellation of the MCVideo emergency alert if applicable.

e) for each of the affiliated but not joined members of the group shall:

i) generate a SIP MESSAGE request notification of the cancellation of the MCVideo user's imminent peril call as specified in clause 6.3.3.1.11;

ii) set the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "false"; and

iii) send the SIP MESSAGE request according to 3GPP TS 24.229 [11];

5) shall include in the SIP 200 (OK) response an SDP answer according to 3GPP TS 24.229 [11] with the clarifications specified in clause 6.3.3.2.1 unless the procedures of clause 6.3.3.1.8 were performed in step 2) or 4) above;

6) shall include the "norefersub" option tag in a Supported header field according to IETF RFC 4488 [31];

7) shall include the "tdialog" option tag in a Supported header field according to IETF RFC 4538 [32];

8) shall interact with media plane as specified in 3GPP TS 24.581 [5]; and

9) shall send the SIP 200 (OK) response towards the MCVideo client according to 3GPP TS 24.229 [11]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.4.2]

"Upon receiving a SIP BYE request the controlling MCVideo function shall follow the procedures as specified in clause 6.3.3.2.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.4.3.2]

"When the MCVideo session for group call needs to be released as specified in clause 6.3.8.1, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.4.3.3]

"When an MCVideo client needs to be removed from the MCVideo session (e.g. due to de-affiliation or admitting a higher priority user), the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.4.

After successful removing the MCVideo client from the MCVideo session, the controlling MCVideo function may generate a notification to the MCVideo clients, which have subscribed to the conference state event package that an MCVideo user has been removed from the MCVideo session, as specified in clause 6.3.3.4 and send the SIP NOTIFY request to the MCVideo client according to 3GPP TS 24.229 [11]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.2.1.1]

"Upon receiving a request from an MCVideo user to establish an MCVideo group session using an MCVideo group identity, identifying a chat MCVideo group, the MCVideo client shall generate an initial SIP INVITE request by following the UE originating session procedures specified in 3GPP TS 24.229 [11], with the clarifications given below.

The MCVideo client:

1) if the MCVideo user has requested the origination of an MCVideo emergency group call or is originating an MCVideo chat group call and the MCVideo emergency state is already set, the MCVideo client shall comply with the procedures in clause 6.2.8.1.1;

2) if the MCVideo user has requested the origination of an MCVideo imminent peril group call, the MCVideo client shall comply with the procedures in clause 6.2.8.1.9;

3) shall include the g.3gpp.mcvideo media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo" in the Contact header field of the SIP INVITE request according to IETF RFC 3840 [22];

4) shall include an Accept-Contact header field containing the g.3gpp.mcvideo media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [20];

5) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcvideo" (coded as specified in 3GPP TS 24.229 [11]), in a P-Preferred-Service header field according to IETF RFC 6050 [14] in the SIP INVITE request;

6) shall include an Accept-Contact header field with the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [20];

7) should include the "timer" option tag in the Supported header field;

8) should include the Session-Expires header field according to IETF RFC 4028 [23]. It is recommended that the refresher parameter is omitted. If included, the refresher parameter shall be set to "uac";

9) shall set the Request-URI of the SIP INVITE request to the public service identity identifying the participating MCVideo function serving the MCVideo user;

NOTE 1: The MCVideo client is configured with public service identity identifying the participating MCVideo function serving the MCVideo user.

10) may include a P-Preferred-Identity header field in the SIP INVITE request containing a public user identity as specified in 3GPP TS 24.229 [11];

11) if the MCVideo emergency state is already set or the MCVideo client emergency group state for this group is set to "MVEG 2: in-progress", the MCVideo client shall comply with the procedures in clause 6.2.8.1.2;

12) if the MCVideo client imminent peril group state for this group is set to "MIG 2: in-progress" or "MVIG 3: confirm-pending" shall include the Resource-Priority header field and comply with the procedures in clause 6.2.8.1.12;

13) shall contain an application/vnd.3gpp.mcvideo-info+xml MIME body with the <mcvideoinfo> element containing the <mcvideo-Params> element with:

a) the <session-type> element set to a value of "chat";

b) the <mcvideo-request-uri> element set to the group identity; and

c) the <mcvideo-client-id> element set to the MCVideo client ID of the originating MCVideo client;

NOTE 2: The MCVideo ID of the originating MCVideo user is not included in the body, as this will be inserted into the body of the SIP INVITE request that is sent by the originating participating MCVideo function.

14) shall include in the SIP INVITE request an SDP offer according to 3GPP TS 24.229 [11] with the clarifications specified in clause 6.2.1;

15) if an implicit transmission request is required, shall indicate this as specified in clause 6.4; and

16) shall send the SIP INVITE request according to 3GPP TS 24.229 [11].

On receiving a SIP 2xx response to the SIP INVITE request, the MCVideo client:

1) shall interact with the user plane as specified in 3GPP TS 24.581 [5]; and

2) if the MCVideo emergency group call state is set to "MEGC 2: emergency-call-requested" or "MVEGC 3: emergency-call-granted" or the MCVideo imminent peril group call state is set to "MVIGC 2: imminent-peril-call-requested" or "MVIGC 3: imminent-peril-call-granted", the MCVideo client shall perform the actions specified in clause 6.2.8.1.4.

On receiving a SIP 4xx response, a SIP 5xx response or a SIP 6xx response to the SIP INVITE request:

1) if the MCVideo emergency group call state is set to "MVEGC 2: emergency-call-requested" or "MVEGC 3: emergency-call-granted"; or

2) if the MCVideo imminent peril group call state is set to "MVIGC 2: imminent-peril-call-requested" or "MVIGC 3: imminent-peril-call-granted";

the MCVideo client shall perform the actions specified in clause 6.2.8.1.5.

On receiving a SIP INFO request where the Request-URI contains an MCVideo session ID identifying an ongoing group session, the MCVideo client shall follow the actions specified in clause 6.2.8.1.13."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.2.1]

"The transmission control server arbitration logic in the transmission control server shall support the procedures in clauses 6.3.2 and 6.3.3 and shall behave according to the transmission control server state transition diagram for 'general transmission control operation' in clause 6.3.4.

The transmission control interface towards the MCVideo client in the transmission control server shall behave according to the transmission control server state transition diagram for 'basic transmission control operation towards the transmission participant' as specified in clause 6.3.5.

The reception control arbitration logic in the transmission control server shall support the procedures in clauses 6.3.2 and 6.3.3 and shall behave according to the reception control server state transition diagram for 'general reception control operation' in clause 6.3.6.

The transmission control interface towards the MCVideo client in the transmission control server shall behave according to the transmission control server state transition diagram for 'basic reception control operation towards the transmission participant' as specified in clause 6.3.7."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.2.2]

"When an MCVideo call is established a new instance of the transmission control server state machine for 'general transmission control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' is added.

When an MCVideo call is established a new instance of the transmission control server state machine for 'general reception control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic reception control operation towards the transmission participant' is added.

If the optional "mc_queueing" feature is supported and has been negotiated as specified in clause 14, the transmission control server could queue the implicit transmission control request for the media-transmission control entity.

The original initial SIP INVITE request or SIP REFER request to establish an MCVideo chat group call or to rejoin an ongoing MCVideo call is not handled as an implicit transmission control request message by the transmission control server unless explicitly stated in the SIP INVITE request or in the SIP REFER request.

The permission to send media to the inviting MCVideo client due to implicit transmission control request is applicable to both confirmed indication and unconfirmed indication.

When the first unconfirmed indication is received from the invited participating MCVideo function (see 3GPP TS 24.281 [2]) the transmission control server optionally can give an early indication to send RTP media packets, to the inviting MCVideo client.

If an early indication to send RTP media packets is given to the inviting MCVideo client, the transmission participant is granted the permission to send media and the MCVideo server buffers RTP media packets received from the MCVideo client at least until the first invited MCVideo client accepts the invitation or until the RTP media packet buffer exceeds it maximum limit to store RTP media packets.

If the MCVideo server does not support or does not allow media buffering then when an early indication to send RTP media packets is not given to the inviting MCVideo client, the transmission participant is granted the permission to send media when the first invited MCVideo client accepts the media.

Before the transmission control server sends the first transmission control message in the MCVideo call, the transmission control server has to assign itself a SSRC identifier to be included in media transmission control messages and quality feedback messages if the MCVideo server is supporting that option. A suitable algorithm to generate the SSRC identifier is described in IETF RFC 3550 [3].

The transmission participant and the transmission control server can negotiate the maximum priority level that the transmission participant is permitted to request. The transmission control server can pre-empt the current sender based on the negotiated maximum priority level that the transmission participant is permitted to request and the priority level included in the Transmission Media Request message.

NOTE: The maximum priority level that a transmission participant can use is negotiated as specified in clause 14.3.3 and is based on group configuration data retrieved by the controlling MCVideo function from the group management server as described in 3GPP TS 24.481 [12] and service configuration data retrieved by the controlling MCVideo function from the configuration management server as described in 3GPP TS 24.484 [13].

The transmission participant and the transmission control server can negotiate queueing of Transmission requests using the "mc_queueing" fmtp attribute as described in clause 14. If queueing is supported and negotiated, the transmission control server queues the transmission control request if a Transmission Media Request message is received when another transmission participant has the transmission and the priority of the current speaker is the same or higher."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.1]

"The transmission control messages are coded as described in clause 9.1.2 where the transmission control message is part of the application-dependent data.

For the transmission control protocol the ASCII name string is: MCVx (Mission Critical Video x) with x=0, 1 or 2 as specified in clause 9.1.2.

A list of transmission control messages can be found in clause 9.2.2.1.

The transmission control specific fields are specified in clause 9.2.3."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

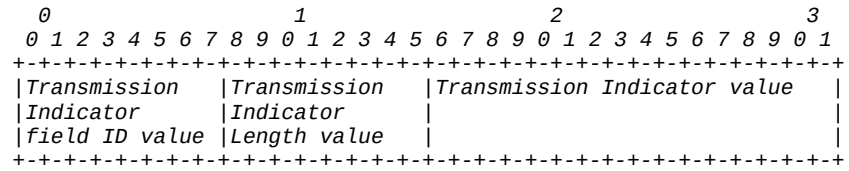
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

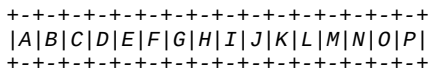


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. *if the active Transmission request queue is empty the transmission control server:*
    a. *shall send Transmission Idle message to all transmission participants. The Transmission Idle message:*
        i. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*
        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;*
    b. *shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;*
    c. *shall start timer T1 (Inactivity);*
    d. *shall set the general state to the 'G: Transmit Idle' state; and*
    e. *shall initialise counter Cx (Simultaneous transmission video) to 0.*

2. *if the active Transmission request queue is not empty the transmission control server:*

   a. *shall select a queued Transmission request from the top of the active Transmission request queue;*

   b. *shall remove that queued Transmission request from the active Transmission request queue; and*

   c. *shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:*

   a. *shall include the granted priority in the Transmission priority field;*

   b. *shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;*

   c. *if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

   d. *shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;*

2. *shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;*

3. *shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:*

   a. *shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;*

   b. *shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;*

   c. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;*

   d. *if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';*

   e. *if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';*

   f. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

   g. *shall initiate the 'general reception control operation' state machine;and*

4. *shall enter the 'G: Transmit Taken' state."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. *if counter Cx (Simultaneous transmission video) has not reached its upper limit:*

   a. *shall perform the actions specified in the clause 6.3.4.4.2;*

2. *if counter Cx (Simultaneous transmission video) has reached its upper limit:*

   a. *select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;*

   b. *shall stop timer T4 (Transmission Grant), if running;*

   c. *shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);*

   d. *shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;*

   e. *shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and*

   f. *shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:*

      i. *shall include the queue position and transmission priority in the Queue Info field; and*

      ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:*

   a. *shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and*

   b. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

2. *shall set the general state to 'G: pending Transmission Revoke'."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. *if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:*

   a. *shall initialize a general state machine as specified in clause 6.3.4.2.2; and*

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

   b. *shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;*

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.2.1.3 Test description

5.1.2.1.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the participating MCVideo functions.
- SS-UE1 (MCVideo Client) and SS-UE2 (MCVideo Client), simulated by the SS (MCVideo Server).

IUT:

- IUT (MCVideo Server), providing the controlling MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of all sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Transmission Control Server.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server).

- The SS-UE1 (MCVideo client) and the SS-UE2 (MCVideo client) are registered and authorized to use the MCVideo system.
- The SS-UE1 (MCVideo client) and the SS-UE2 (MCVideo client) are affiliated to the chat group and are authorized to initiate group calls.
- The IUT (MCVideo Server) is the controlling server of the chat group.

5.1.2.1.3.2 Test procedure sequence

Table 5.1.2.1.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.4.1.6.3-1 to establish a chat group call without implicit transmission control request according to option a of note in Annex A, Table A.4.1.6.3-1, initiated by SS-UE1 (MCVideo Client)?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.4.1.6.3-1 to join SS-UE2 (MCVideo Client) to a chat group call without implicit transmission control request according to option a of note in Annex A, Table A.4.1.6.3-1?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E session establishment/modification without provisional responses other than 100 Trying', as described in Annex A, Table A.4.1.1.3-1 to upgrade the call to emergency group call with implicit transmission request according to option b.ii of note in Annex A, Table A.4.1.1.3-1?	-	-	5	P
4	Check: Does the IUT (MCVideo Server) correctly perform procedure ' MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, to end SS-UE1 (MCVideo Client) transmission?	-	-	3	P
5	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	4	P
6	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, to end SS-UE1 (MCVideo Client) transmission?	-	-	3	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
7	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E session establishment/modification without provisional responses other than 100 Trying', as described in Annex A, Table A.4.1.1.3-1 to downgrade the call without implicit transmission request, with option a of note on Table A.4.1.1.3-1?	-	-	6	P
8	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	4	P
9	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, to end SS-UE1 (MCVideo Client) transmission?	-	-	3	P
10	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E session establishment/modification without provisional responses other than 100 Trying', as described in Annex A, Table A.4.1.1.3-1 to upgrade the call to imminent peril group call with implicit transmission request according to option b.ii of note in Annex A, Table A.4.1.1.3-1?	-	-	7	P
11	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, to end SS-UE1 (MCVideo Client) transmission?	-	-	3	P
12	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	4	P
13	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, to end SS-UE1 (MCVideo Client) transmission?	-	-	3	P
14	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E session establishment/modification without provisional responses other than 100 Trying', as described in Annex A, Table A.4.1.1.3-1 to downgrade the call without implicit transmission request, with option a of note on Table A.4.1.1.3-1?	-	-	8	P
15	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.1.5.3-1, to release the call?	-	-	9	P

5.1.2.1.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

**Table 5.1.2.1.3.3-1: SIP INVITE from SS (MCVideo Server)
(Step 1, Table 5.1.2.1.3.2-1; Step 1, Annex A, Table A.4.1.6.3-1)**

Derivation Path: Annex A, Table A.4.1.6.4-1, with condition CHAT-GROUP-CALL.

**Table 5.1.2.1.3.3-2: SIP 200 (OK) from IUT (MCVideo Server)
(Steps 1, 2, 3, 7, 10 and 14, Table 5.1.2.1.3.2-1; Step 3, Annex A, Table A.4.1.6.3-1;
Step 6, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.4.1.6.4-2, with condition CHAT-GROUP-CALL.

**Table 5.1.2.1.3.3-3: SIP INVITE from SS (MCVideo Server)
(Step 2, Table 5.1.2.1.3.2-1; Step 1, Annex A, Table A.4.1.6.3-1)**

Derivation Path: Annex A, Table A.4.1.6.4-1, with condition CHAT-GROUP-CALL.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.4.1.1.4-2			
MIME body part		**MCVideo-Info**		
MIME-part-body	MCVideo-Info as described in Table 5.1.2.1.3.3-3a			

Table 5.1.2.1.3.3-3a: MCVideo-Info in SIP INVITE (Table 5.1.2.1.3.3-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2, condition INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcvideoinfo				
mcvideo-Params				
mcvideo-calling-user-id	Encrypted (note) <mcvideo-calling-user-id> with mcvideoURI set to px_MCVideo_ID_User_B	The URI of the calling user		
mcvideo-client-id	Encrypted (note) <mcvideo-client-id> with mcvideoString set to px_MCX_Client_B_ID			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2A.				

**Table 5.1.2.1.3.3-4: SIP 200 (OK) from SS (MCVideo Server)
(Steps 3, 7, 10 and 14, Table 5.1.2.1.3.2-1; Step 5, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.4.1.1.4-7, with condition CHAT-GROUP-CALL.

**Table 5.1.2.1.3.3-5: SIP re-INVITE from SS (MCVideo Server)
(Step 3, Table 5.1.2.1.3.2-1; Step 1, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.4.1.1.4-1, with conditions CHAT-GROUP-CALL, re_INVITE and EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.1.3.3-6: SIP re-INVITE from IUT (MCVideo Server)
(Step 3, Table 5.1.2.1.3.2-1; Step 2, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.4.1.1.4-4, with condition CHAT-GROUP-CALL, re_INVITE and EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.1.3.3-7: SIP re-INVITE from SS (MCVideo Server)
(Steps 7 and 14, Table 5.1.2.1.3.2-1; Step 1, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.4.1.1.4-1, with conditions CHAT-GROUP-CALL and re_INVITE.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.1.3.3-8: SIP re-INVITE from IUT (MCVideo Server)
(Steps 7 and 14, Table 5.1.2.1.3.2-1; Step 2, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.4.1.1.4-4, with condition CHAT-GROUP-CALL and re_INVITE.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.1.3.3-9: SIP re-INVITE from SS (MCVideo Server)
(Step 10, Table 5.1.2.1.3.2-1; Step 1, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.4.1.1.4-1, with conditions CHAT-GROUP-CALL, re_INVITE and IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.1.3.3-10: SIP re-INVITE from IUT (MCVideo Server)
(Step 10, Table 5.1.2.1.3.2-1; Step 2, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.4.1.1.4-4, with conditions CHAT-GROUP-CALL, re_INVITE and IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.1.3.3-11: Transmission Idle from IUT (MCVideo Server)
(Steps 4 and 6, Table 5.1.2.1.3.2-1; Steps 4 and 5, Annex A, Table A.3.8.3-1)**

Derivation Path: Annex A, Table A.3.8.4-6, with condition EMERGENCY-CALL.

**Table 5.1.2.1.3.3-12: Transmission Request from SS (MCVideo Server)
(Step 5, Table 5.1.2.1.3.2-1; Step 1, Annex A, Table A.3.7.3-1)**

Derivation Path: Annex A, Table A.3.7.4-1, with condition EMERGENCY-CALL.

**Table 5.1.2.1.3.3-13: Transmission Granted from IUT (MCVideo Server)
(Steps 3 and 5, Table 5.1.2.1.3.2-1;
Step 3a1 or 3b1, Annex A, Table A.3.7.3-1; Step 9a1, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.3.7.4-2 or A.3.7.4-3 (note), with condition EMERGENCY-CALL.
NOTE: Depends on server implementation.

**Table 5.1.2.1.3.3-14: Media Transmission Notification from IUT (MCVideo Server)
(Steps 3 and 5, Table 5.1.2.1.3.2-1; Step 3a3 or 3b2, Annex A, Table A.3.7.3-1;
Step 10a1, Annex A, Table 4.1.1.3-1)**

Derivation Path: Annex A, Table A.3.7.4-4 or A.3.7.4-5 (note), with condition EMERGENCY-CALL.
NOTE: Depends on server implementation.

**Table 5.1.2.1.3.3-15: Transmission Idle from IUT (MCVideo Server)
(Steps 11 and 13, Table 5.1.2.1.3.2-1; Steps 4 and 5, Annex A, Table A.3.8.3-1)**

Derivation Path: Annex A, Table A.3.8.4-6, with condition IMMPERIL-CALL.

**Table 5.1.2.1.3.3-16: Transmission Request from SS (MCVideo Server)
(Step 12, Table 5.1.2.1.3.2-1; Step 1, Annex A, Table A.3.7.3-1)**

Derivation Path: Annex A, Table A.3.7.4-1, with condition IMMPERIL-CALL.

**Table 5.1.2.1.3.3-17: Transmission Granted from IUT (MCVideo Server)
(Steps 10 and 12, Table 5.1.2.1.3.2-1; Step 3a1 or 3b1, Annex A, Table A.3.7.3-1;
Step 9a1, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.3.7.4-2 or A.3.7.4-3 (note), with condition IMMPERIL-CALL.
NOTE: Depends on server implementation.

**Table 5.1.2.1.3.3-18: Media Transmission Notification from IUT (MCVideo Server)
(Steps 10 and 12, Table 5.1.2.1.3.2-1; Step 3a3 or 3b2, Annex A, Table A.3.7.3-1;
Step 10a1, Annex A, Table 4.1.1.3-1)**

Derivation Path: Annex A, Table A.3.7.4-4 or A.3.7.4-5 (note), with condition IMMPERIL-CALL.
NOTE: Depends on server implementation.

5.1.2.2 MCVideo Server - MCVideo Server / Chat Group Call / Originating Participating server

5.1.2.2.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS-UE1 (MCVideo Client) initiates a chat group call without implicit transmission
control }
    then { IUT (MCVideo Server) accepts the call from initiator and establishes the call with the SS
(MCVideo Server) providing the controlling function }
            }
```

(2)

```
with { IUT (MCVideo Server) having established a chat group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control }
    then { IUT (MCVideo Server) forwards the transmission request to SS (MCVideo Server) and
forwards the transmission granted received from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established a chat group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests to end the transmission control }
    then { IUT (MCVideo Server) forwards the transmission end request to SS (MCVideo Server) and
forwards the transmission end response to SS-UE1 (MCVideo Client) and forwards the transmission idle
received from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(4)

```
with { IUT (MCVideo Server) having established a chat group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) ends the chat group call }
    then { IUT (MCVideo Server) sends a SIP BYE message to SS (MCVideo Server) providing the
controlling function and responds by sending a SIP 200 (OK) message to SS-UE1 (MCVideo Client) }
          }
```

5.1.2.2.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.2.3.1.1, 6.3.2.1.3, 6.3.2.1.1.1, 6.3.2.1.5 and 6.3.2.1.2.1, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.4.1, 6.4.2, 6.4.3, 6.4.4, 6.3.3, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.3.1.1]

"In the procedures in this clause:

1) group identity in an incoming SIP INVITE request refers to the group identity from the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;

2) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body;

3) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

Upon receipt of a "SIP INVITE request for originating participating MCVideo function" for a group identity identifying a chat MCVideo group containing an application/vnd.3gpp.mcvideo-info+xml MIME body with the <session-type> element set to a value of "chat", the participating MCVideo function:

…

2) shall determine the MCVideo ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request, and authorise the calling user;

NOTE 2: The MCVideo ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

3) if through local policy in the originating participating MCVideo function, the user identified by the MCVideo ID is not authorised to make chat group calls, shall reject the "SIP INVITE request for originating participating MCVideo function" …;

4) shall determine if the media parameters are acceptable and the MCVideo codecs are offered in the SDP offer and if not, reject the request with a SIP 488 (Not Acceptable Here) response. Otherwise, continue with the rest of the steps;

5) shall check if the number of maximum simultaneous MCVideo group calls supported for the MCVideo user as specified in the <MaxSimultaneousCallsN6> element of the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) has been exceeded. If exceeded, the MCVideo function shall respond with a SIP 486 (Busy Here) response with the warning text set to "103 maximum simultaneous MCVideo group calls reached" in a Warning header field as specified in clause 4.4. Otherwise, continue with the rest of the steps;

NOTE 3: If the SIP INVITE request contains an emergency indication set to a value of "true" or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCVideo function can according to local policy choose to allow for an exception to the limit for the maximum simultaneous MCVideo sessions supported for the MCVideo user.

6) if the user identified by the MCVideo ID is not affiliated to the group identified in the "SIP INVITE request for originating participating MCVideo function" as determined by clause 8.2.2.2.11, shall perform the actions specified in clause 8.2.2.2.12 for implicit affiliation;

7) if the actions for implicit affiliation specified in step 6) above were performed but not successful in affiliating the MCVideo user due to the MCVideo user already having N2 simultaneous affiliations, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 486 (Busy Here) response with the warning text set to "102 too many simultaneous affiliations" in a Warning header field as specified in clause 4.4. and skip the rest of the steps.

NOTE 4: N2 is the total number of MCVideo groups that an MCVideo user can be affiliated to simultaneously as specified in 3GPP TS 23.281 [26].

NOTE 5: if the SIP INVITE request contains an emergency indication set to a value of "true" or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCVideo function can according to local policy choose to allow an exception to the N2 limit. Alternatively, a lower priority affiliation of the MCVideo user could be cancelled to allow for the new affiliation.

8) shall determine the public service identity of the controlling MCVideo function associated with the group identity in the SIP INVITE request;

NOTE 6: The public service identity can identify the controlling MCVideo function in the primary MCVideo system or a partner MCVideo system.

NOTE 7: How the participating MCVideo server discovers the public service identity of the controlling MCVideo function associated with the group identity is out of scope of the current document.

9) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3;

10) shall set the Request-URI to the public service identity of the controlling MCVideo function associated with the group identity present in the incoming SIP INVITE request;

11) shall include the MCVideo ID of the calling user in <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the SIP INVITE request;

12) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request as specified in clause 6.3.2.1.1.1;

13) if the received SIP INVITE request contains an application/vnd.3gpp.location-info+xml MIME body as specified in Annex F.3; and

a) if not already included, shall include a Content-Type header field set to "application/vnd.3gpp.location-info+xml"; and

b) if not already copied, shall copy the contents of the application/vnd.3gpp.location-info+xml MIME body received in the SIP INVITE request into an application/vnd.3gpp.location-info+xml MIME body included in the outgoing SIP request;

NOTE 8: Note that the application/vnd.3gpp.mcvideo-info+xml MIME body will already have been copied into the outgoing SIP INVITE request by clause 6.3.2.1.3.

14) if a Resource-Priority header field was included in the received SIP INVITE request, shall include a Resource-Priority header field according to rules and procedures of IETF RFC 4412 [33] set to the value indicated in the Resource-Priority header field of the SIP INVITE request from the MCVideo client; and

NOTE 9: The participating MCVideo function will leave verification of the Resource-Priority header field to the controlling MCVideo function.

15) shall forward the SIP INVITE request according to 3GPP TS 24.229 [11].

…

Upon receipt of a SIP 2xx response to the above SIP INVITE request in step 14) the participating MCVideo function:

1) if the SIP 2xx response contains an application/vnd.3gpp.mcvideo-info+xml MIME body with an <MKFC-GKTPs> element, shall perform the procedures in clause 6.3.2.3.2;

2) shall generate a SIP 200 (OK) response as specified in the clause 6.3.2.1.5.2;

3) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

4) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response;

5) shall include the public service identity received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response;

6) if the procedures of clause 8.2.2.2.12 for implicit affiliation were performed in the present clause, shall complete the implicit affiliation by performing the procedures of clause 8.2.2.2.13;

7) shall send the SIP 200 (OK) response to the MCVideo client according to 3GPP TS 24.229 [11]; and

8) shall interact with the media plane as specified in 3GPP TS 24.581 [5].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the above SIP INVITE request in step 14) the participating MCVideo function:

1) shall generate a SIP response according to 3GPP TS 24.229 [11];

2) shall include Warning header field(s) that were received in the incoming SIP response;

3) shall forward the SIP response to the MCVideo client according to 3GPP TS 24.229 [11]; and

4) if the implicit affiliation procedures of clause 8.2.2.2.12 were invoked in the current procedure, shall perform the procedures of clause 8.2.2.2.14."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.3]

"This clause is referenced from other procedures.

When generating an initial SIP INVITE request according to 3GPP TS 24.229 [11], on receipt of an incoming SIP INVITE request, the participating MCVideo function:

1) shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [20] if included in the incoming SIP INVITE request;

2) should include the Session-Expires header field according to IETF RFC 4028 [23]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

3) shall include the option tag "timer" in the Supported header field;

4) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP INVITE request to the P-Asserted-Identity header field of the outgoing SIP INVITE request;

5) shall include the g.3gpp.mcvideo media feature tag into the Contact header field of the outgoing SIP INVITE request;

6) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcvideo" (coded as specified in 3GPP TS 24.229 [11]), into the P-Asserted-Service header field of the outgoing SIP INVITE request;

7) if the incoming SIP INVITE request contained a MIME resource-lists body with the MCVideo ID of the invited MCVideo user, shall copy the MIME resource-lists body, according to rules and procedures of IETF RFC 5366 [37];

8) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the contents of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request to the outgoing SIP INVITE request; and

9) if the incoming SIP INVITE request contained an application/vnd.3gpp.location-info+xml MIME body, shall copy the contents of the application/vnd.3gpp.location-info+xml MIME body of the incoming SIP INVITE request to the outgoing SIP INVITE request."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.1.1]

"This clause is referenced from other clauses.

The SDP offer is generated based on the received SDP offer. The SDP offer generated by the participating MCVideo function:

1) shall contain two SDP media-level sections for MCVideo video media as contained in the received SDP offer; and

2) shall contain an SDP media-level section for one media-transmission control entity, if present in the received SDP offer.

When composing the SDP offer according to 3GPP TS 24.229 [11], the participating MCVideo function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the participating MCVideo function, if required;

NOTE 1: Requirements can exist for the participating MCVideo function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number for the offered media transmission control entity, if any, in the received SDP offer with the IP address and port number of the participating MCVideo function; and

NOTE 2: If the participating MCVideo function and the controlling MCVideo function are in the same MCVideo server, and the participating MCVideo function does not have a dedicated IP address or a dedicated port number for media transmission control or media stream, the replacement of the IP address or the port number is omitted.

3) shall contain an "a=key-mgmt" attribute field with a "mikey" attribute value, if present in the received SDP offer."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.5]

"Upon receiving a SIP BYE request from the MCVideo client, the participating MCVideo function:

1) shall interact with the media plane as specified in 3GPP TS 24.581 [5];

2) shall generate a SIP BYE request as specified in 3GPP TS 24.229 [11];

3) shall set the Request-URI to the MCVideo session identity as included in the received SIP BYE request;

4) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;

5) if the received SIP BYE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body into the outgoing SIP BYE request; and

6) shall send the SIP BYE request toward the controlling MCVideo function, according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response to the SIP BYE request the terminating MCVideo function shall forward a SIP 200 (OK) response to the MCVideo client and shall interact with the media plane as specified in 3GPP TS 24.581 [5] for releasing media plane resources associated with the SIP session with the controlling MCVideo function."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [11], the participating MCVideo function:

1) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCVideo function, for the accepted media stream in the received SDP offer, if required; and

NOTE 1: Requirements can exist for the participating MCVideo function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCVideo function, for the accepted media-transmission control entity, if present in the received SDP offer.

NOTE 2: If the participating MCVideo function and the controlling MCVideo function are in the same MCVideo server, and the participating MCVideo function does not have a dedicated IP address or a dedicated port number for media transmission control or media stream, the replacement of the IP address or the port number is omitted."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.1]

"Once an on-demand MCVideo session is established or a pre-established session is in use when the participating MCVideo function receives transmission control messages from the transmission participant in the MCVideo client or from the transmission control server in the controlling MCVideo function, the behaviour of the participating MCVideo function is described in the following clauses."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.2]

"Upon receiving a transmission control message the participating MCVideo function:

1. *shall immediately forward the transmission control message to the transmission control server if the message is received from the transmission participant;*

2. *if an MBMS subchannel is not used for a transmission in the session the transmission control message is associated with, shall immediately forward the transmission control message to the transmission participant if the message is received from the transmission control server; and*

3. *if an MBMS subchannel is used for a transmission in the session the transmission control message is associated with:*

   a. *if*

      i. *the transmission control message is not a Transmission Idle message or a Media Transmission Notify message;*

      ii. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

      iii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report;*

      shall immediately forward the transmission control message to the transmission participant; and

   b. *if*

      i. *the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report; and*

      ii *if the transmission control message is the Transmission Idle message or the Media Transmission Notify message,*

   shall perform actions as specified in clause 10.2.

NOTE: When the Transmit Idle or Media Transmission Notify messages are discarded the messages are sent to the MCVideo clients over the MBMS subchannel allocated for the transmission as specified in clause 10.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.3]

"Upon receiving RTP media packets the participating MCVideo function:

1. *shall immediately forward the RTP media packet to the controlling MCVideo function if the RTP packet is from an MCVideo client; and*

2. *if an MBMS subchannel is not used for a transmission in the session the RTP media packets are associated with, shall immediately forward the RTP media packets to the MCVideo client if the RTP packet is from the controlling MCVideo function or the non-controlling MCVideo function.*

3. *if an MBMS subchannel is used for a transmission in the session the RTP media packets are associated with and if RTP media packets are received from the controlling MCVideo function or the non-controlling MCVideo function:*

   a. *if*

      i. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

      ii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report,*

      shall immediately forward the RTP media packets to the MCVideo client; and

   b. *if the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report, shall perform actions as specified in clause 10.2."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.4]

"When the participating function receives an indication from the application and signalling plane that session release is initiated, the participating MCVideo function:

1. *shall stop sending transmission control messages towards the transmission participant and the transmission control server; and*

2. *shall stop sending RTP media packets towards the MCVideo client and towards the controlling MCVideo function.*

When the participating MCVideo function receives an indication from the application and signalling plane that the session is released, the participating MCVideo function:

1. *in case of a pre-established session, shall perform the actions in clause 9.3.2; and*

2. *in case of an on-demand session, shall release the media resources associated with the session."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

b. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.2.2.3 Test description

5.1.2.2.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the controlling MCVideo function.
- SS-UE1 (MCVideo Client)

IUT:

- IUT (MCVideo Server), providing the originating participating MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of some sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server and the HTTP Client.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server) and SS-UE1 (MCVideo Client).
- The SS-UE1 (MCVideo client) is registered and authorized to use the MCVideo system.
- The SS-UE1 (MCVideo client) is affiliated to the chat group and is authorized to initiate group calls.

5.1.2.2.3.2 Test procedure sequence

Table 5.1.2.2.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.4.2.1.3-1 to establish a chat group call without implicit transmission control request according to option a of note in Annex A, Table A.4.2.1.3-1, initiated by SS-UE1 (MCVideo Client?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Transmission Granted' as described in Annex A, Table A.4.2.4.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO Transmission End Request - Transmission End Response' as described in Annex A, Table A.4.2.5.3-1 to end SS-UE1 (MCVideo Client) transmission?	-	-	3	P
4	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO call release' as described in Annex A, Table A.4.2.3.3-1, to release the call?	-	-	4	P

5.1.2.2.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

**Table 5.1.2.2.3.3-1: SIP INVITE from SS-UE1 (MCVideo Client)
(Step 1, Table 5.1.2.2.3.2-1; Step 1, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-1, with condition CHAT-GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.2.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.2.2.3.2-1; Step 2, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-2, with condition CHAT-GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.2.3.3-3: SIP 200 (OK) from SS (MCVideo Server)
(Step 1, Table 5.1.2.2.3.2-1; Step 5, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-3, with condition CHAT-GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.2.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.1.2.2.3.2-1; Step 6, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-4, with condition CHAT-GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

5.1.2.3 MCVideo Server - MCVideo Server / Chat Group Call / Emergency Group Call / Controlling server

5.1.2.3.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function initiates an
emergency chat group call with implicit transmission control }
    then { IUT (MCVideo Server) accepts the call from initiator and establishes the call with all
affiliated users by sending call establishment requests to SS (MCVideo Server) providing the
terminating participating function and sends transmission granted to SS-UE1 (MCVideo Client) and
sends media transmission notification to SS-UE2 (MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established an emergency chat group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests to end the transmission control }
    then { IUT (MCVideo Server) sends transmission control idle to SS-UE1 (MCVideo Client) and sends
transmission control idle to SS-UE2 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established an emergency chat group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control }
    then { IUT (MCVideo Server) sends transmission granted to SS-UE1 (MCVideo Client) and sends
media transmission notification to SS-UE2 (MCVideo Client) }
            }
```

(4)

```
with { IUT (MCVideo Server) having established an emergency chat group call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function ends the chat
group call }
    then { IUT (MCVideo Server) responds by sending a SIP 200 (OK) message to SS (MCVideo Server)
ending the call and sends a SIP BYE message to SS (MCVideo Server) providing the terminating
participating function }
            }
```

5.1.2.3.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.2.4.1.1, 9.2.2.4.2, 9.2.2.4.3.2, 9.2.2.4.3.3, 6.3.3.1.6 and 6.3.3.2.4, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.3.1, 6.3.2.2, 9.2.1, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.4.1.1]

"In the procedures in this clause:

1) *MCVideo ID in an incoming SIP INVITE request refers to the MCVideo ID of the originating user from the <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;*

2) *group identity in an incoming SIP INVITE request refers to the group identity from the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;*

3) *MCVideo ID in an outgoing SIP INVITE request refers to the MCVideo ID of the called user in the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the outgoing SIP INVITE request;*

4) *emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and*

5) *alert indication in an incoming SIP INVITE request refers to the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.*

Upon receipt of a "SIP INVITE request for controlling MCVideo function of an MCVideo group" containing a group identity identifying a chat MCVideo group, the controlling MCVideo function:

1) *if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP INVITE request with a SIP 500 (Server Internal Error) response. The controlling MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15] and skip the rest of the steps;*

NOTE 1: If the SIP INVITE request contains an emergency indication set to a value of "true", the controlling MCVideo function can by means beyond the scope of this specification choose to accept the request.

2) *shall reject the SIP request with a SIP 403 (Forbidden) response and not process the remaining steps if:*

   a) *an Accept-Contact header field does not include the g.3gpp.mcvideo media feature tag;*

   b) *an Accept-Contact header field does not include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo"; or*

   c) *the isfocus media feature tag is present in the Contact header field;*

3) *if received SIP INVITE request includes an application/vnd.3gpp.mcvideoinfo+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.3.1.17;*

4) *shall retrieve the necessary group document(s) from the group management server for the group identity contained in the SIP INVITE request and carry out initial processing as specified in clause 6.3.5.2 and continue with the rest of the steps if the checks in clause 6.3.5.2 succeed;*

5) *if the MCVideo user identified by the MCVideo ID in the SIP INVITE request is not affiliated with the MCVideo group identified by the group identity in the SIP INVITE request as determined by the procedures of clause 6.3.6:*

   a) *shall check if the MCVideo user is eligible to be implicitly affiliated with the MCVideo chat group as determined by clause 8.2.2.3.6; and*

   b) *if the MCVideo user is not eligible for implicit affiliation, shall reject the SIP INVITE request with a SIP 403 (Forbidden) response with the warning text set to "120 user is not affiliated to this group" in a Warning header field as specified in clause 4.4 and skip the rest of the steps below;*

6) if the SIP INVITE request contains unauthorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2:

a) shall reject the SIP INVITE request with a SIP 403 (Forbidden) response as specified in clause 6.3.3.1.14; and

b) shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [11] and skip the rest of the steps;

7) if the SIP INVITE request contains an unauthorised request for an MCVideo imminent peril group call as determined by clause 6.3.3.1.13.6, shall reject the SIP INVITE request with a SIP 403 (Forbidden) response with the following clarifications:

a) shall include in the SIP 403 (Forbidden) response an application/vnd.3gpp.mcvideo-info+xml MIME body as specified in clause F.1 with the <mcvideoinfo> element containing the <mcvideo-Params> element with the <imminentperil-ind> element set to a value of "false"; and

b) shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [11] and skip the rest of the steps;

8) if a Resource-Priority header field is included in the SIP INVITE request:

a) if the Resource-Priority header field is set to the value indicated for emergency calls and the SIP INVITE request does not contain an emergency indication and the in-progress emergency state of the group is set to a value of "false", shall reject the SIP INVITE request with a SIP 403 (Forbidden) response and skip the remaining steps; and

b) if the Resource-Priority header field is set to the value indicated for imminent peril calls and the SIP INVITE request does not contain an imminent peril indication and the in-progress imminent peril state of the group is set to a value of "false", shall reject the SIP INVITE request with a SIP 403 (Forbidden) response; and skip the remaining steps;

9) shall determine if the media parameters are acceptable and the MCVideo codecs are offered in the SDP offer and if not reject the request with a SIP 488 (Not Acceptable Here) response and skip the rest of the steps;

10) shall create a chat group session and allocate an MCVideo session identity for the chat group session if the MCVideo chat group session identity does not already exist, and may handle timer TNG3 (group call timer) as specified in clause 6.3.3.5;

11) if the chat group session is ongoing and the <on-network-max-participant-count> as specified in 3GPP TS 24.481 [24] is already reached:

a) if, according to local policy, the user identified by the MCVideo ID in the SIP INVITE request is deemed to have a higher priority than an existing user in the chat group session, may remove a participant from the session by following clause 9.2.1.4.4.3, and skip the next step; and

NOTE 2: The local policy for deciding whether to admit a user to a call that has reached its maximum amount of participants can include the <user-priority> and the <participant-type> of the user as well as other information of the user from the group document as specified in 3GPP TS 24.481 [24]. The local policy decisions can also include taking into account whether the imminent-peril indicator or emergency indicator was received in the SIP INVITE request.

b) shall return a SIP 486 (Busy Here) response with the warning text set to "122 too many participants" to the originating network as specified in clause 4.4. Otherwise, continue with the rest of the steps;

12) if the received SIP INVITE request was determined to be eligible for implicit affiliation in step 5) and if clause 8.2.2.3.7 was not previously invoked in the present clause, shall perform the implicit affiliation as specified in clause 8.2.2.3.7;

13) if the SIP INVITE request contains an emergency indication set to a value of "true" or the in-progress emergency state of the group to "true" the controlling MCVideo function shall:

a) validate that the SIP INVITE request includes a Resource-Priority header field populated with the values for an MCVideo emergency group call as specified in clause 6.3.3.1.19, and if not:

i) perform the actions specified in clause 6.3.3.1.8;

ii) send the SIP UPDATE request generated in clause 6.3.3.1.8 towards the initiator of the SIP INVITE request according to 3GPP TS 24.229 [11]; and

iii) upon receiving a SIP 200 (OK) response to the SIP UPDATE request sent in clause 6.3.3.1.8, proceed with the rest of the steps.

NOTE 3: Verify that the Resource-Priority header is included and properly populated for both ongoing and newly-entered in-progress emergency states of the specified group.

b) if the in-progress emergency state of the group is set to a value of "true" and this MCVideo user is indicating a new emergency indication:

i) for each of the other affiliated members of the group generate a SIP MESSAGE request notification of the MCVideo user's emergency indication as specified in clause 6.3.3.1.11 with the following clarifications:

A) set the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true";

B) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, perform the procedures specified in clause 6.3.3.1.12; and

C) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11];

ii) cache the information that this MCVideo user has initiated an MCVideo emergency call; and

iii) if the SIP INVITE request contains an authorised request for an MCVideo emergency alert as determined in step i) B) above, cache the information that this MCVideo user has initiated an MCVideo emergency alert; and

c) if the in-progress emergency state of the group is set to a value of "false":

i) shall set the value of the in-progress emergency state of the group to "true";

ii) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16;

iii) shall generate SIP re-INVITE requests for the MCVideo emergency group call to the other affiliated and joined participants of the chat MCVideo group as specified in clause 6.3.3.1.6;

iv) shall generate SIP INVITE requests for the MCVideo emergency group call to the affiliated but not joined members of the chat MCVideo group as specified in clause 6.3.3.1.7;

A) for each affiliated but not joined member shall send the SIP INVITE request towards the MCVideo client as specified in 3GPP TS 24.229 [11]; and

B) upon receiving a SIP 200 (OK) response to the SIP INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5];

v) shall cache the information that this MCVideo user has initiated an MCVideo emergency call; and

vi) if the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body is set to "true" and is an authorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall cache the information that this MCVideo user has initiated an MCVideo emergency alert; and

vii) if the in-progress imminent peril state of the group is set to a value of "true", shall set it to a value of "false";

14) if the in-progress emergency state of the group is set to a value of "false" and if the SIP INVITE request contains an imminent peril indication set to a value of "true" or the in-progress imminent peril state of the group is set to "true", the controlling MCVideo function shall:

a) validate that the SIP INVITE request includes a Resource-Priority header field populated with the values for an MCVideo imminent peril group call as specified in clause 6.3.3.1.19, and if not:

i) perform the actions specified in clause 6.3.3.1.8;

ii) send the SIP UPDATE request generated in clause 6.3.3.1.8 towards the initiator of the SIP INVITE request according to 3GPP TS 24.229 [11]; and

iii) upon receiving a SIP 200 (OK) response to the SIP UPDATE request sent in clause 6.3.3.1.8 proceed with the rest of the steps.

NOTE 4: Verify that the Resource-Priority header is included and properly populated for both ongoing and newly-entered in-progress imminent peril states of the specified group.

b) if the in-progress imminent peril state of the group is set to a value of "true" and this MCVideo user is indicating a new imminent peril indication:

i) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's imminent peril indication as specified in clause 6.3.3.1.11 with the following clarifications;

A) set the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11]; and

ii) cache the information that this MCVideo user has initiated an MCVideo imminent peril call; and

c) if the in-progress imminent peril state of the group is set to a value of "false":

i) shall set the value of the in-progress imminent peril state of the group to "true";

ii) shall generate SIP re-INVITE requests for the MCVideo imminent peril group call to the other affiliated and joined participants of the chat MCVideo group as specified in clause 6.3.3.1.15;

iii) shall generate SIP INVITE requests for the MCVideo imminent peril call to the affiliated but not joined members of the chat MCVideo group as specified in clause 6.3.3.1.7;

A) for each affiliated but not joined member shall send the SIP INVITE request towards the MCVideo client as specified in 3GPP TS 24.229 [11]; and

B) Upon receiving a SIP 200 (OK) response to the SIP INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

iv) shall cache the information that this MCVideo user has initiated an MCVideo imminent peril call;

15) shall accept the SIP request and generate a SIP 200 (OK) response to the SIP INVITE request according to 3GPP TS 24.229 [11];

16) shall include in the SIP 200 (OK) response an SDP answer according to 3GPP TS 24.229 [11] with the clarifications specified in clause 6.3.3.2.1 unless the procedures of clause 6.3.3.1.8 were performed in step 13)a) or step 14)a) above;

17) should include the Session-Expires header field and start supervising the SIP session according to IETF RFC 4028 [23]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

18) shall include the "timer" option tag in a Require header field;

19) shall include the following in a Contact header field:

a) the g.3gpp.mcvideo media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo";

c) the MCVideo session identity; and

d) the media feature tag isfocus;

20) shall include the "tdialog" option tag in a Supported header field according to IETF RFC 4538 [32];

21) if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

22) if the received SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 5: In this case, the request was for an imminent peril call but a higher priority MCVideo emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

23) shall interact with media plane as specified in 3GPP TS 24.581 [5];

24) shall send the SIP 200 (OK) response to the MCVideo client according to 3GPP TS 24.229 [11]; and

25) if the chat group session was already ongoing and if at least one of the participants has subscribed to the conference event package, shall send a SIP NOTIFY request to all participants with a subscription to the conference event package as specified in clause 9.2.3.4.2.

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCVideo client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.18."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.4.2]

"Upon receiving a SIP BYE request the controlling MCVideo function shall follow the procedures as specified in clause 6.3.3.2.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.4.3.2]

"When the MCVideo session for group call needs to be released as specified in clause 6.3.8.1, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.4.3.3]

"When an MCVideo client needs to be removed from the MCVideo session (e.g. due to de-affiliation or admitting a higher priority user), the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.4.

After successful removing the MCVideo client from the MCVideo session, the controlling MCVideo function may generate a notification to the MCVideo clients, which have subscribed to the conference state event package that an MCVideo user has been removed from the MCVideo session, as specified in clause 6.3.3.4 and send the SIP NOTIFY request to the MCVideo client according to 3GPP TS 24.229 [11]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.6]

"This clause is referenced from other procedures.

This clause describes the procedures for inviting an MCVideo user to an MCVideo session associated with an MCVideo emergency group call or MCVideo imminent peril group call. The procedure is initiated by the controlling MCVideo function as the result of an action in clause 9.2.2.4.1.1.

The controlling MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

2) shall set the Request-URI to the address of the terminating participating MCVideo function associated with the MCVideo ID of the targeted MCVideo user;

3) shall include an application/vnd.3gpp.mcvideo-info+xml MIME body with the <mcvideoinfo> element containing the <mcvideo-Params> element populated as follows:

a) the <mcvideo-request-uri> element set to the value of the MCVideo ID of the targeted MCVideo user;

b) the <mcvideo-calling-user-id> element set to the value of the MCVideo ID of the calling MCVideo user; and

c) the <mcvideo-calling-group-id> element set to the value of the MCVideo group ID of the emergency group call.

4) shall include in the P-Asserted-Identity header field the public service identity of the controlling MCVideo function;

5) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1; and

6) if the in-progress emergency group state of the group is set to a value of "true" the controlling MCVideo function:

a) shall include a Resource-Priority header field populated with the values for an MCVideo emergency group call as specified in clause 6.3.3.1.18;

b) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <emergency-ind> element set to a value of "true";

c) if the <alert-ind> element is set to "true" in the received SIP INVITE request and the requesting MCVideo user and MCVideo group are authorised for the initiation of MCVideo emergency alerts as determined by the procedures of clause 6.3.3.1.12.1, shall populate the application/vnd.3gpp.mcvideo-info+xml MIME body and the application/vnd.3gpp.mcvideo-location-info+xml MIME body as specified in clause 6.3.3.1.11. Otherwise, shall set the <alert-ind> element to a value of "false" in the application/vnd.3gpp.mcvideo-info+xml MIME body; and

d) if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <imminentperil-ind> element set to a value of "false";

NOTE: If the imminent peril state of the group is true at this point, the controlling function will set it to false as part of the calling procedure.

7) if the in-progress emergency state of the group is set to a value of "false" and the in-progress imminent peril state of the group is set to a value of "true", the controlling MCVideo function:

a) shall include a Resource-Priority header field populated with the values for an MCVideo imminent peril group call as specified in clause 6.3.3.1.18; and

b) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true"; and

8) if:

a) an MCVideo GKTP document exists for the group identity contained in the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request; and

b) the MCVideo GKTP document contains an <MKFC-GKTPs> element;

then:

a) for each instance of <GKTP> element of the <MKFC-GKTPs> element of the MCVideo GKTP document:

i) shall perform the procedure in clause 6.3.3.6.2 to re-generate an I_MESSAGE; and

ii) if the procedure in clause 6.3.3.6.2 was successful, shall include the I_MESSAGE in a <GKTP> element of the <MKFC-GKTPs> element of an application/vnd.3gpp.mcvideo-info+xml MIME body included in the outgoing SIP INVITE request."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.2.4]

"Upon receiving a SIP BYE request the controlling MCVideo function:

1) shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.581 [5] for releasing the media plane resource associated with the SIP session towards the MCVideo client;

NOTE: The non-controlling MCVideo function is also regarded as a MCVideo client in a temporary MCVideo group session.

2) shall generate a SIP 200 (OK) response and send the SIP response towards the MCVideo client according to 3GPP TS 24.229 [11];

3) shall check the MCVideo session release policy as specified in clause 6.3.8.1 and clause 6.3.8.2 whether the MCVideo session needs to be released for each participant of the MCVideo session;

4) if release of the MCVideo session is required:

a) shall perform the procedures as specified in the clause 6.3.3.1.4 with the clarification that if the received SIP BYE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body, copy the application/vnd.3gpp.mcvideo-info+xml MIME body into the outgoing SIP BYE request; and

5) if a release of the MCVideo session is not required, shall send a SIP NOTIFY request to all remaining MCVideo clients in the MCVideo session with a subscription to the conference event package as specified in clause 9.2.3.4.2.

Upon receiving a SIP 200 (OK) response to the SIP BYE request the controlling MCVideo function shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.581 [5] for releasing media plane resources associated with the SIP session with the MCVideo participant."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.1]

"The transmission control server arbitration logic in the transmission control server shall support the procedures in clauses 6.3.2 and 6.3.3 and shall behave according to the transmission control server state transition diagram for 'general transmission control operation' in clause 6.3.4.

The transmission control interface towards the MCVideo client in the transmission control server shall behave according to the transmission control server state transition diagram for 'basic transmission control operation towards the transmission participant' as specified in clause 6.3.5.

The reception control arbitration logic in the transmission control server shall support the procedures in clauses 6.3.2 and 6.3.3 and shall behave according to the reception control server state transition diagram for 'general reception control operation' in clause 6.3.6.

The transmission control interface towards the MCVideo client in the transmission control server shall behave according to the transmission control server state transition diagram for 'basic reception control operation towards the transmission participant' as specified in clause 6.3.7."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.2.2]

"When an MCVideo call is established a new instance of the transmission control server state machine for 'general transmission control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' is added.

When an MCVideo call is established a new instance of the transmission control server state machine for 'general reception control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic reception control operation towards the transmission participant' is added.

If the optional "mc_queueing" feature is supported and has been negotiated as specified in clause 14, the transmission control server could queue the implicit transmission control request for the media-transmission control entity.

The original initial SIP INVITE request or SIP REFER request to establish an MCVideo chat group call or to rejoin an ongoing MCVideo call is not handled as an implicit transmission control request message by the transmission control server unless explicitly stated in the SIP INVITE request or in the SIP REFER request.

The permission to send media to the inviting MCVideo client due to implicit transmission control request is applicable to both confirmed indication and unconfirmed indication.

When the first unconfirmed indication is received from the invited participating MCVideo function (see 3GPP TS 24.281 [2]) the transmission control server optionally can give an early indication to send RTP media packets, to the inviting MCVideo client.

If an early indication to send RTP media packets is given to the inviting MCVideo client, the transmission participant is granted the permission to send media and the MCVideo server buffers RTP media packets received from the MCVideo client at least until the first invited MCVideo client accepts the invitation or until the RTP media packet buffer exceeds it maximum limit to store RTP media packets.

If the MCVideo server does not support or does not allow media buffering then when an early indication to send RTP media packets is not given to the inviting MCVideo client, the transmission participant is granted the permission to send media when the first invited MCVideo client accepts the media.

Before the transmission control server sends the first transmission control message in the MCVideo call, the transmission control server has to assign itself a SSRC identifier to be included in media transmission control messages and quality feedback messages if the MCVideo server is supporting that option. A suitable algorithm to generate the SSRC identifier is described in IETF RFC 3550 [3].

The transmission participant and the transmission control server can negotiate the maximum priority level that the transmission participant is permitted to request. The transmission control server can pre-empt the current sender based on the negotiated maximum priority level that the transmission participant is permitted to request and the priority level included in the Transmission Media Request message.

NOTE: The maximum priority level that a transmission participant can use is negotiated as specified in clause 14.3.3 and is based on group configuration data retrieved by the controlling MCVideo function from the group management server as described in 3GPP TS 24.481 [12] and service configuration data retrieved by the controlling MCVideo function from the configuration management server as described in 3GPP TS 24.484 [13].

The transmission participant and the transmission control server can negotiate queueing of Transmission requests using the "mc_queueing" fmtp attribute as described in clause 14. If queueing is supported and negotiated, the transmission control server queues the transmission control request if a Transmission Media Request message is received when another transmission participant has the transmission and the priority of the current speaker is the same or higher."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.1]

"The transmission control messages are coded as described in clause 9.1.2 where the transmission control message is part of the application-dependent data.

For the transmission control protocol the ASCII name string is: MCVx (Mission Critical Video x) with x=0, 1 or 2 as specified in clause 9.1.2.

A list of transmission control messages can be found in clause 9.2.2.1.

The transmission control specific fields are specified in clause 9.2.3."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

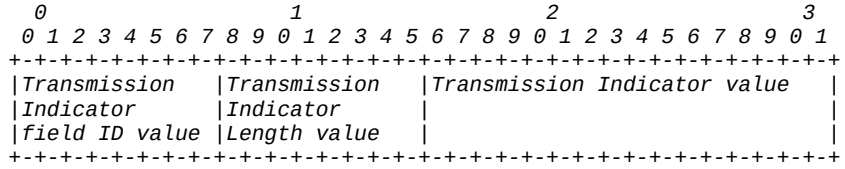
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

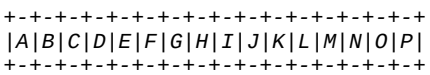


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. *if the active Transmission request queue is empty the transmission control server:*
    a. *shall send Transmission Idle message to all transmission participants. The Transmission Idle message:*
        i. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*
        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;*
    b. *shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;*
    c. *shall start timer T1 (Inactivity);*
    d. *shall set the general state to the 'G: Transmit Idle' state; and*
    e. *shall initialise counter Cx (Simultaneous transmission video) to 0.*
2. *if the active Transmission request queue is not empty the transmission control server:*
    a. *shall select a queued Transmission request from the top of the active Transmission request queue;*
    b. *shall remove that queued Transmission request from the active Transmission request queue; and*
    c. *shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:*
    a. *shall include the granted priority in the Transmission priority field;*
    b. *shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;*
    c. *if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*
    d. *shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;*
2. *shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;*
3. *shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:*
    a. *shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;*
    b. *shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;*

c. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;

d. if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';

e. if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

g. shall initiate the 'general reception control operation' state machine;and

4. shall enter the 'G: Transmit Taken' state."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. if counter Cx (Simultaneous transmission video) has not reached its upper limit:

a. shall perform the actions specified in the clause 6.3.4.4.2;

2. if counter Cx (Simultaneous transmission video) has reached its upper limit:

a. select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;

b. shall stop timer T4 (Transmission Grant), if running;

c. shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);

d. shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;

e. shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and

f. shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:

i. shall include the queue position and transmission priority in the Queue Info field; and

ii. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:

a. shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and

b. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

2. shall set the general state to 'G: pending Transmission Revoke'."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

b. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.2.3.3 Test description

5.1.2.3.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the participating MCVideo functions.
- SS-UE1 (MCVideo Client) and SS-UE2 (MCVideo Client), simulated by the SS (MCVideo Server).

IUT:

- IUT (MCVideo Server), providing the controlling MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of all sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Transmission Control Server.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server).
- The SS-UE1 (MCVideo client) and the SS-UE2 (MCVideo client) are registered and authorized to use the MCVideo system.
- The SS-UE1 (MCVideo client) and the SS-UE2 (MCVideo client) are affiliated to the chat group and are authorized to initiate group calls.
- The IUT (MCVideo Server) is the controlling server of the chat group.

5.1.2.3.3.2 Test procedure sequence

Table 5.1.2.3.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E group call establishment' as described in Annex A, Table A.4.1.4.3-1 to establish an emergency chat group call with implicit transmission control request according to option b.ii of note in Annex A, Table A.4.1.4.3-1, initiated by SS-UE1 (MCVideo Client)?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure ' MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	3	P
4	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.1.5.3-1, to release the call?	-	-	4	P

5.1.2.3.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

**Table 5.1.2.3.3.3-1: SIP INVITE from SS (MCVideo Server)
(Step 1, Table 5.1.2.3.3.2-1; Step 1, Annex A, Table A.4.1.4.3-1)**

Derivation Path: Annex A, Table A.4.1.4.4-1, with condition EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.3.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.2.3.3.2-1; Step 2, Annex A, Table A.4.1.4.3-1)**

Derivation Path: Annex A, Table A.4.1.4.4-2, with condition EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.3.3.3-3: SIP 200 (OK) from SS (MCVideo Server)
(Step 1, Table 5.1.2.3.3.2-1; Step 8, Annex A, Table A.4.1.4.3-1)**

Derivation Path: Annex A, Table A.4.1.4.4-6

**Table 5.1.2.3.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.1.2.3.3.2-1; Step 9, Annex A, Table A.4.1.4.3-1)**

Derivation Path: Annex A, Table A.4.1.4.4-7

**Table 5.1.2.3.3.3-5: Transmission Idle from IUT (MCVideo Server)
(Step 2, Table 5.1.2.3.3.2-1; Steps 4 and 5, Annex A, Table A.3.8.3-1)**

Derivation Path: Annex A, Table A.3.8.4-6, with condition EMERGENCY-CALL.

**Table 5.1.2.3.3.3-6: Transmission Request from SS (MCVideo Server)
(Step 3, Table 5.1.2.3.3.2-1; Step 1, Annex A, Table A.3.7.3-1)**

Derivation Path: Annex A, Table A.3.7.4-1, with condition EMERGENCY-CALL.

**Table 5.1.2.3.3.3-7: Transmission Granted from IUT (MCVideo Server)
(Steps 1 and 3, Table 5.1.2.3.3.2-1; Step 3a1 or 3b1, Annex A, Table A.3.7.3-1;
Step 9a1, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.3.7.4-2 or A.3.7.4-3 (note), with condition EMERGENCY-CALL.
NOTE: Depends on server implementation.

**Table 5.1.2.3.3.3-8: Media Transmission Notification from IUT (MCVideo Server)
(Steps 1 and 3, Table 5.1.2.3.3.2-1; Step 3a3 or 3b2, Annex A, Table A.3.7.3-1;
Step 10a1, Annex A, Table 4.1.1.3-1)**

Derivation Path: Annex A, Table A.3.7.4-4 or A.3.7.4-5 (note), with condition EMERGENCY-CALL.
NOTE: Depends on server implementation.

5.1.2.4 MCVideo Server - MCVideo Server / Chat Group Call / Emergency Group Call / Terminating Participating server

5.1.2.4.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the controlling function initiates an emergency chat
group call with automatic commencement }
    then { IUT (MCVideo Server) accepts the call and establishes the call with SS-UE2 (MCVideo
Client) and forwards the media transmission notification received from SS (MCVideo Server) to SS-UE2
(MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established an emergency chat group call }
ensure that {
  when { the SS (MCVideo Server) ends the media transmission }
    then { IUT (MCVideo Server) forwards the transmission end notify received from SS (MCVideo
Server) to SS-UE2 (MCVideo Client) and forwards the transmission idle received from SS (MCVideo
Server) to SS-UE2 (MCVideo Client) }
           }
```

(3)

```
with { IUT (MCVideo Server) having established an emergency chat group call }
ensure that {
  when { the SS (MCVideo Server) providing the controlling function ends the chat group call }
    then { IUT (MCVideo Server) sends a SIP BYE message to SS-UE2 (MCVideo Client) and responds by
sending a SIP 200 (OK) message to SS (MCVideo Server) }
           }
```

5.1.2.4.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.2.3.1.3, 6.3.2.1.5, 6.3.2.2.3, 6.3.2.2.8.1, 6.3.2.2.1, 6.3.2.2.4.2, 6.3.2.2.2.1 and 6.3.2.1.1.1, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.4.1, 6.4.2, 6.4.3, 6.4.4, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.3.1.3]

"This clause covers on-demand session.

Upon receipt of a "SIP INVITE request for terminating participating MCVideo function", for a terminating MCVideo client of a chat MCVideo group, the participating MCVideo function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the "SIP INVITE request for terminating participating MCVideo function" with a SIP 500 (Server Internal Error) response. The participating MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15]. Otherwise, continue with the rest of the steps;

NOTE: If the SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "true", the participating MCVideo function can by means beyond the scope of this specification choose to accept the request.

2) shall check the presence of the isfocus media feature tag in the URI of the Contact header field and if it is not present then the participating MCVideo function shall reject the request with a SIP 403 (Forbidden) response with the warning text set to "104 isfocus not assigned" in a Warning header field as specified in clause 4.4. Otherwise, continue with the rest of the steps;

3) shall generate a SIP INVITE request as specified in clause 6.3.2.2.3;

4) shall set the Request-URI to the public user identity associated with the MCVideo ID of the MCVideo user to be invited based on the contents of the Request-URI of the received "SIP INVITE request for terminating participating MCVideo function";

5) shall copy the contents of the P-Asserted-Identity header field of the incoming "SIP INVITE request for terminating participating MCVideo function" to the P-Asserted-Identity header field of the outgoing SIP INVITE request;

6) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received "SIP INVITE request for terminating participating MCVideo function" as specified in clause 6.3.2.2.1;

7) if the received SIP INVITE request contains a Resource-Priority header field, shall include a Resource-Priority header field with the contents set as in the received Resource-Priority header field;

8) shall perform the procedures specified in clause 6.3.2.2.9 to include any MIME bodies in the received SIP INVITE request; and

9) shall send the SIP INVITE request towards the MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response to the above SIP INVITE request sent to the MCVideo client, the participating MCVideo function:

1) *shall generate a SIP 200 (OK) response as described in the clause 6.3.2.2.4.2;*

2) *shall include in the SIP 200 (OK) response an SDP answer based on the SDP answer in the received SIP 200 (OK) response as specified in clause 6.3.2.2.2.1;*

3) *shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and*

4) *shall forward the SIP 200 (OK) response according to 3GPP TS 24.229 [11]."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.5]

"Upon receiving a SIP BYE request from the MCVideo client, the participating MCVideo function:

1) *shall interact with the media plane as specified in 3GPP TS 24.581 [5];*

2) *shall generate a SIP BYE request as specified in 3GPP TS 24.229 [11];*

3) *shall set the Request-URI to the MCVideo session identity as included in the received SIP BYE request;*

4) *shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;*

5) *if the received SIP BYE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body into the outgoing SIP BYE request; and*

6) *shall send the SIP BYE request toward the controlling MCVideo function, according to 3GPP TS 24.229 [11].*

Upon receiving a SIP 200 (OK) response to the SIP BYE request the terminating MCVideo function shall forward a SIP 200 (OK) response to the MCVideo client and shall interact with the media plane as specified in 3GPP TS 24.581 [5] for releasing media plane resources associated with the SIP session with the controlling MCVideo function."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.3]

"The participating MCVideo function shall generate an initial SIP INVITE request according to 3GPP TS 24.229 [11] and:

1) *shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [20] if included in the incoming SIP INVITE request;*

2) *should include the Session-Expires header field according to IETF RFC 4028 [23]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";*

3) *shall include the option tag "timer" in the Supported header field;*

4) *shall include the following in the Contact header field:*

   a) *the g.3gpp.mcvideo media feature tag;*

   b) *the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo";*

   c) *the isfocus media feature tag;*

   d) *an MCVideo session identity mapped to the MCVideo session identity provided in the Contact header field of the incoming SIP INVITE request; and*

   e) *any other uri-parameter provided in the Contact header field of the incoming SIP INVITE request;*

5) *shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [32];*

6) *shall include the option tag "norefersub" in a Supported header field according to rules and procedures of IETF RFC 4488 [31];*

7) may include a Resource-Share header field in accordance with clause 5.7.1.20.3 in 3GPP TS 24.229 [11]; and

8) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body to the outgoing SIP INVITE request."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.8.1]

"Upon receiving a SIP BYE request from the controlling MCVideo function, the participating MCVideo function:

1) shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.581 [5] for releasing media plane resource associated with the SIP session with the controlling MCVideo function;

2) shall generate a SIP BYE request according to 3GPP TS 24.229 [11];

3) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;

4) if the received SIP BYE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body into the outgoing SIP BYE request; and

5) shall send the SIP BYE request to the MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response to the SIP BYE request the participating MCVideo function:

1) shall send a SIP 200 (OK) response to the SIP BYE request received from the controlling MCVideo function according to 3GPP TS 24.229 [11]; and

2) shall interact with the media plane as specified in 3GPP TS 24.581 [5] for releasing media plane resources associated with the SIP session with the MCVideo client."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.1]

"The participating MCVideo function shall follow the procedures in clause 6.3.2.1.1."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.4.2]

"This clause is referenced from other procedures.

When sending SIP 200 (OK) responses, the participating MCVideo function shall generate a SIP 200 (OK) response according to 3GPP TS 24.229 [11] and:

1) shall include the option tag "timer" in a Require header field;

2) shall include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [23], "UAS Behavior". If no "refresher" parameter was included in the SIP INVITE request, the "refresher" parameter in the Session-Expires header field shall be set to "uas";

3) shall include the following in the Contact header field:

a) the g.3gpp.mcvideo media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo"; and

c) an MCVideo session identity mapped to the MCVideo session identity provided in the Contact header field of the received SIP INVITE request from the controlling MCVideo function;

4) shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [32]; and

5) if the incoming SIP response contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body to the outgoing SIP 200 (OK) response."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.2.1]

"The participating MCVideo function shall follow the procedures in clause 6.3.2.1.1."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.1.1]

"This clause is referenced from other clauses.

The SDP offer is generated based on the received SDP offer. The SDP offer generated by the participating MCVideo function:

1) shall contain two SDP media-level sections for MCVideo video media as contained in the received SDP offer; and

2) shall contain an SDP media-level section for one media-transmission control entity, if present in the received SDP offer.

When composing the SDP offer according to 3GPP TS 24.229 [11], the participating MCVideo function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the participating MCVideo function, if required;

NOTE 1: Requirements can exist for the participating MCVideo function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number for the offered media transmission control entity, if any, in the received SDP offer with the IP address and port number of the participating MCVideo function; and

NOTE 2: If the participating MCVideo function and the controlling MCVideo function are in the same MCVideo server, and the participating MCVideo function does not have a dedicated IP address or a dedicated port number for media transmission control or media stream, the replacement of the IP address or the port number is omitted.

3) shall contain an "a=key-mgmt" attribute field with a "mikey" attribute value, if present in the received SDP offer."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.1]

"Once an on-demand MCVideo session is established or a pre-established session is in use when the participating MCVideo function receives transmission control messages from the transmission participant in the MCVideo client or from the transmission control server in the controlling MCVideo function, the behaviour of the participating MCVideo function is described in the following clauses."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.2]

"Upon receiving a transmission control message the participating MCVideo function:

1. shall immediately forward the transmission control message to the transmission control server if the message is received from the transmission participant;

2. if an MBMS subchannel is not used for a transmission in the session the transmission control message is associated with, shall immediately forward the transmission control message to the transmission participant if the message is received from the transmission control server; and

3. if an MBMS subchannel is used for a transmission in the session the transmission control message is associated with:

a. if

i. the transmission control message is not a Transmission Idle message or a Media Transmission Notify message;

ii. the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or

iii. the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report;

shall immediately forward the transmission control message to the transmission participant; and

b. if

i. the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report; and

ii if the transmission control message is the Transmission Idle message or the Media Transmission Notify message,

shall perform actions as specified in clause 10.2.

NOTE: When the Transmit Idle or Media Transmission Notify messages are discarded the messages are sent to the MCVideo clients over the MBMS subchannel allocated for the transmission as specified in clause 10.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.3]

"Upon receiving RTP media packets the participating MCVideo function:

1. *shall immediately forward the RTP media packet to the controlling MCVideo function if the RTP packet is from an MCVideo client; and*
2. *if an MBMS subchannel is not used for a transmission in the session the RTP media packets are associated with, shall immediately forward the RTP media packets to the MCVideo client if the RTP packet is from the controlling MCVideo function or the non-controlling MCVideo function.*
3. *if an MBMS subchannel is used for a transmission in the session the RTP media packets are associated with and if RTP media packets are received from the controlling MCVideo function or the non-controlling MCVideo function:*

a. if

i. the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or

ii. the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report,

shall immediately forward the RTP media packets to the MCVideo client; and

b. if the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report, shall perform actions as specified in clause 10.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.4]

"When the participating function receives an indication from the application and signalling plane that session release is initiated, the participating MCVideo function:

1. *shall stop sending transmission control messages towards the transmission participant and the transmission control server; and*
2. *shall stop sending RTP media packets towards the MCVideo client and towards the controlling MCVideo function.*

When the participating MCVideo function receives an indication from the application and signalling plane that the session is released, the participating MCVideo function:

1. *in case of a pre-established session, shall perform the actions in clause 9.3.2; and*
2. *in case of an on-demand session, shall release the media resources associated with the session. "*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

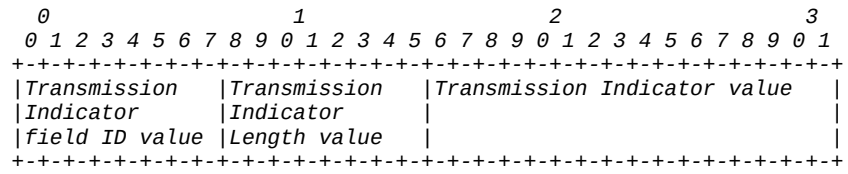
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2].".

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

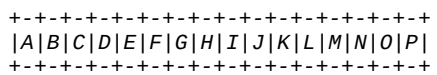


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. *if the active Transmission request queue is empty the transmission control server:*
    a. *shall send Transmission Idle message to all transmission participants. The Transmission Idle message:*
        i. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*
        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;*
    b. *shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;*
    c. *shall start timer T1 (Inactivity);*
    d. *shall set the general state to the 'G: Transmit Idle' state; and*
    e. *shall initialise counter Cx (Simultaneous transmission video) to 0.*
2. *if the active Transmission request queue is not empty the transmission control server:*
    a. *shall select a queued Transmission request from the top of the active Transmission request queue;*
    b. *shall remove that queued Transmission request from the active Transmission request queue; and*
    c. *shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:*
    a. *shall include the granted priority in the Transmission priority field;*
    b. *shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;*
    c. *if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*
    d. *shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;*
2. *shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;*
3. *shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:*
    a. *shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;*
    b. *shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;*

c. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;

d. if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';

e. if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

g. shall initiate the 'general reception control operation' state machine;and

4. shall enter the 'G: Transmit Taken' state."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. if counter Cx (Simultaneous transmission video) has not reached its upper limit:

a. shall perform the actions specified in the clause 6.3.4.4.2;

2. if counter Cx (Simultaneous transmission video) has reached its upper limit:

a. select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;

b. shall stop timer T4 (Transmission Grant), if running;

c. shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);

d. shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;

e. shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and

f. shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:

i. shall include the queue position and transmission priority in the Queue Info field; and

ii. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:

a. shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and

b. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

2. shall set the general state to 'G: pending Transmission Revoke'."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

b. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.2.4.3 Test description

5.1.2.4.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the controlling MCVideo function.
- SS-UE2 (MCVideo Client).

IUT:

- IUT (MCVideo Server), providing the terminating participating MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of some sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server and the HTTP Client.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server) and SS-UE2 (MCVideo Client).
- The SS-UE2 (MCVideo client) is registered and authorized to use the MCVideo system.
- The SS-UE2 (MCVideo client) is affiliated to the chat group and is authorized to initiate group calls.

5.1.2.4.3.2 Test procedure sequence

Table 5.1.2.4.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CT group/private call establishment, automatic commencement' as described in Annex A, Table A.4.3.3.3-1 to establish an emergency chat group call with automatic commencement mode and notify a media transmission from SS-UE1 (MCVideo Client)?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Notification - Idle' as described in Annex A, Table A.4.3.8.3-1 to notify the end of SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.3.5.3-1, to release the call?	-	-	3	P

5.1.2.4.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

**Table 5.1.2.4.3.3-1: SIP INVITE from SS (MCVideo Server)
(Step 1, Table 5.1.2.4.3.2-1; Step 1, Annex A, Table A.4.3.3.3-1)**

Derivation Path: Annex A, Table A.4.3.3.4-1, with conditions CHAT-GROUP-CALL and EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.4.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.2.4.3.2-1; Step 2, Annex A, Table A.4.3.3.3-1)**

Derivation Path: Annex A, Table A.4.3.3.4-2, with conditions CHAT-GROUP-CALL and EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.4.3.3-3: SIP 200 (OK) from SS-UE2 (MCVideo Client)
(Step 1, Table 5.1.2.4.3.2-1; Step 6, Annex A, Table A.4.3.3.3-1)**

Derivation Path: Annex A, Table A.4.3.3.4-7, with condition CHAT-GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.4.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.1.2.4.3.2-1; Step 7, Annex A, Table A.4.3.3.3-1)**

Derivation Path: Annex A, Table A.4.3.3.4-8, with condition CHAT-GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.4.3.3-5: Transmission Idle
(Step 2, Table 5.1.2.4.3.2-1; Steps 13 and 14, Annex A, Table A.4.3.8.3-1)**

Derivation Path: Annex A, Table A.4.3.8.4-6, with condition EMERGENCY-CALL.

**Table 5.1.2.4.3.3-6: Media Transmission Notification
(Step 1, Table 5.1.2.4.3.2-1; Steps 10a1 and 10a2, Annex A, Table 4.3.3.3-1)**

Derivation Path: Annex A, Table A.4.3.3.4-11, with condition EMERGENCY-CALL.

5.1.2.5 MCVideo Server - MCVideo Server / Chat Group Call / Emergency Group Call / Originating Participating server

5.1.2.5.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS-UE1 (MCVideo Client) initiates an emergency chat group call with implicit
transmission control }
    then { IUT (MCVideo Server) accepts the call from initiator and establishes the call with the SS
(MCVideo Server) providing the controlling function and forwards the transmission granted received
from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established an emergency chat group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests to end the transmission control }
    then { IUT (MCVideo Server) forwards the transmission end request to SS (MCVideo Server) and
forwards the transmission end response to SS-UE1 (MCVideo Client) and forwards the transmission idle
received from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established an emergency chat group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control }
    then { IUT (MCVideo Server) forwards the transmission request to SS (MCVideo Server) and
forwards the transmission granted received from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(4)

```
with { IUT (MCVideo Server) having established an emergency chat group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) ends the chat group call }
    then { IUT (MCVideo Server) sends a SIP BYE message to SS (MCVideo Server) providing the
controlling function and responds by sending a SIP 200 (OK) message to SS-UE1 (MCVideo Client) }
            }
```

5.1.2.5.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.2.3.1.1, 6.3.2.1.3, 6.3.2.1.1.1, 6.3.2.1.5 and 6.3.2.1.2.1, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.4.1, 6.4.2, 6.4.3, 6.4.4, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.3.1.1]

"In the procedures in this clause:

1) group identity in an incoming SIP INVITE request refers to the group identity from the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;

2) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body;

3) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

Upon receipt of a "SIP INVITE request for originating participating MCVideo function" for a group identity identifying a chat MCVideo group containing an application/vnd.3gpp.mcvideo-info+xml MIME body with the <session-type> element set to a value of "chat", the participating MCVideo function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP INVITE request with a SIP 500 (Server Internal Error) response. The participating MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15]. Otherwise, continue with the rest of the steps;

NOTE 1: if the SIP INVITE request contains an emergency indication set to a value of "true" or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCVideo function can according to local policy choose to accept the request.

2) shall determine the MCVideo ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request, and authorise the calling user;

NOTE 2: The MCVideo ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

3) if through local policy in the originating participating MCVideo function, the user identified by the MCVideo ID is not authorised to make chat group calls, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response to the SIP INVITE request, with warning text set to "108 user not authorised to make chat group calls" in a Warning header field as specified in clause 4.4;

4) shall determine if the media parameters are acceptable and the MCVideo codecs are offered in the SDP offer and if not, reject the request with a SIP 488 (Not Acceptable Here) response. Otherwise, continue with the rest of the steps;

5) shall check if the number of maximum simultaneous MCVideo group calls supported for the MCVideo user as specified in the <MaxSimultaneousCallsN6> element of the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) has been exceeded. If exceeded, the MCVideo function shall respond with a SIP 486 (Busy Here) response with the warning text set to "103 maximum simultaneous MCVideo group calls reached" in a Warning header field as specified in clause 4.4. Otherwise, continue with the rest of the steps;

NOTE 3: If the SIP INVITE request contains an emergency indication set to a value of "true" or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCVideo function can according to local policy choose to allow for an exception to the limit for the maximum simultaneous MCVideo sessions supported for the MCVideo user.

6) if the user identified by the MCVideo ID is not affiliated to the group identified in the "SIP INVITE request for originating participating MCVideo function" as determined by clause 8.2.2.2.11, shall perform the actions specified in clause 8.2.2.2.12 for implicit affiliation;

7) if the actions for implicit affiliation specified in step 6) above were performed but not successful in affiliating the MCVideo user due to the MCVideo user already having N2 simultaneous affiliations, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 486 (Busy Here) response with the warning text set to "102 too many simultaneous affiliations" in a Warning header field as specified in clause 4.4. and skip the rest of the steps.

NOTE 4: N2 is the total number of MCVideo groups that an MCVideo user can be affiliated to simultaneously as specified in 3GPP TS 23.281 [26].

NOTE 5: if the SIP INVITE request contains an emergency indication set to a value of "true" or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCVideo function can according to local policy choose to allow an exception to the N2 limit. Alternatively, a lower priority affiliation of the MCVideo user could be cancelled to allow for the new affiliation.

8) shall determine the public service identity of the controlling MCVideo function associated with the group identity in the SIP INVITE request;

NOTE 6: The public service identity can identify the controlling MCVideo function in the primary MCVideo system or a partner MCVideo system.

NOTE 7: How the participating MCVideo server discovers the public service identity of the controlling MCVideo function associated with the group identity is out of scope of the current document.

9) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3;

10) shall set the Request-URI to the public service identity of the controlling MCVideo function associated with the group identity present in the incoming SIP INVITE request;

11) shall include the MCVideo ID of the calling user in <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the SIP INVITE request;

12) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request as specified in clause 6.3.2.1.1.1;

13) if the received SIP INVITE request contains an application/vnd.3gpp.location-info+xml MIME body as specified in Annex F.3; and

a) if not already included, shall include a Content-Type header field set to "application/vnd.3gpp.location-info+xml"; and

b) if not already copied, shall copy the contents of the application/vnd.3gpp.location-info+xml MIME body received in the SIP INVITE request into an application/vnd.3gpp.location-info+xml MIME body included in the outgoing SIP request;

NOTE 8: Note that the application/vnd.3gpp.mcvideo-info+xml MIME body will already have been copied into the outgoing SIP INVITE request by clause 6.3.2.1.3.

14) if a Resource-Priority header field was included in the received SIP INVITE request, shall include a Resource-Priority header field according to rules and procedures of IETF RFC 4412 [33] set to the value indicated in the Resource-Priority header field of the SIP INVITE request from the MCVideo client; and

NOTE 9: The participating MCVideo function will leave verification of the Resource-Priority header field to the controlling MCVideo function.

15) shall forward the SIP INVITE request according to 3GPP TS 24.229 [11].

Upon receipt of a SIP 302 (Moved Temporarily) response to the above SIP INVITE request in step 14), the participating MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.2.1.10;

2) shall include an SDP offer based upon the SDP offer in the received SIP INVITE request from the MCVideo client as specified in clause 6.3.2.1.1.1; and

3) shall forward the SIP INVITE request according to 3GPP TS 24.229 [11];

Upon receipt of a SIP 2xx response to the above SIP INVITE request in step 14) the participating MCVideo function:

1) if the SIP 2xx response contains an application/vnd.3gpp.mcvideo-info+xml MIME body with an <MKFC-GKTPs> element, shall perform the procedures in clause 6.3.2.3.2;

2) shall generate a SIP 200 (OK) response as specified in the clause 6.3.2.1.5.2;

3) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

4) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response;

5) shall include the public service identity received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response;

6) if the procedures of clause 8.2.2.2.12 for implicit affiliation were performed in the present clause, shall complete the implicit affiliation by performing the procedures of clause 8.2.2.2.13;

7) shall send the SIP 200 (OK) response to the MCVideo client according to 3GPP TS 24.229 [11]; and

8) shall interact with the media plane as specified in 3GPP TS 24.581 [5].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the above SIP INVITE request in step 14) the participating MCVideo function:

1) shall generate a SIP response according to 3GPP TS 24.229 [11];

2) shall include Warning header field(s) that were received in the incoming SIP response;

3) shall forward the SIP response to the MCVideo client according to 3GPP TS 24.229 [11]; and

4) if the implicit affiliation procedures of clause 8.2.2.2.12 were invoked in the current procedure, shall perform the procedures of clause 8.2.2.2.14."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.3]

"This clause is referenced from other procedures.

When generating an initial SIP INVITE request according to 3GPP TS 24.229 [11], on receipt of an incoming SIP INVITE request, the participating MCVideo function:

1) shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [20] if included in the incoming SIP INVITE request;

2) should include the Session-Expires header field according to IETF RFC 4028 [23]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

3) shall include the option tag "timer" in the Supported header field;

4) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP INVITE request to the P-Asserted-Identity header field of the outgoing SIP INVITE request;

5) shall include the g.3gpp.mcvideo media feature tag into the Contact header field of the outgoing SIP INVITE request;

6) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcvideo" (coded as specified in 3GPP TS 24.229 [11]), into the P-Asserted-Service header field of the outgoing SIP INVITE request;

7) if the incoming SIP INVITE request contained a MIME resource-lists body with the MCVideo ID of the invited MCVideo user, shall copy the MIME resource-lists body, according to rules and procedures of IETF RFC 5366 [37];

8) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the contents of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request to the outgoing SIP INVITE request; and

9) if the incoming SIP INVITE request contained an application/vnd.3gpp.location-info+xml MIME body, shall copy the contents of the application/vnd.3gpp.location-info+xml MIME body of the incoming SIP INVITE request to the outgoing SIP INVITE request."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.1.1]

"This clause is referenced from other clauses.

The SDP offer is generated based on the received SDP offer. The SDP offer generated by the participating MCVideo function:

1) shall contain two SDP media-level sections for MCVideo video media as contained in the received SDP offer; and

2) shall contain an SDP media-level section for one media-transmission control entity, if present in the received SDP offer.

When composing the SDP offer according to 3GPP TS 24.229 [11], the participating MCVideo function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the participating MCVideo function, if required;

NOTE 1: Requirements can exist for the participating MCVideo function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number for the offered media transmission control entity, if any, in the received SDP offer with the IP address and port number of the participating MCVideo function; and

NOTE 2: If the participating MCVideo function and the controlling MCVideo function are in the same MCVideo server, and the participating MCVideo function does not have a dedicated IP address or a dedicated port number for media transmission control or media stream, the replacement of the IP address or the port number is omitted.

3) shall contain an "a=key-mgmt" attribute field with a "mikey" attribute value, if present in the received SDP offer."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.5]

"Upon receiving a SIP BYE request from the MCVideo client, the participating MCVideo function:

1) shall interact with the media plane as specified in 3GPP TS 24.581 [5];

2) shall generate a SIP BYE request as specified in 3GPP TS 24.229 [11];

3) shall set the Request-URI to the MCVideo session identity as included in the received SIP BYE request;

4) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;

5) if the received SIP BYE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body into the outgoing SIP BYE request; and

6) shall send the SIP BYE request toward the controlling MCVideo function, according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response to the SIP BYE request the terminating MCVideo function shall forward a SIP 200 (OK) response to the MCVideo client and shall interact with the media plane as specified in 3GPP TS 24.581 [5] for releasing media plane resources associated with the SIP session with the controlling MCVideo function."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [11], the participating MCVideo function:

1) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCVideo function, for the accepted media stream in the received SDP offer, if required; and

NOTE 1: Requirements can exist for the participating MCVideo function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCVideo function, for the accepted media-transmission control entity, if present in the received SDP offer.

NOTE 2: If the participating MCVideo function and the controlling MCVideo function are in the same MCVideo server, and the participating MCVideo function does not have a dedicated IP address or a dedicated port number for media transmission control or media stream, the replacement of the IP address or the port number is omitted."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.1]

"Once an on-demand MCVideo session is established or a pre-established session is in use when the participating MCVideo function receives transmission control messages from the transmission participant in the MCVideo client or from the transmission control server in the controlling MCVideo function, the behaviour of the participating MCVideo function is described in the following clauses."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.2]

"Upon receiving a transmission control message the participating MCVideo function:

1. shall immediately forward the transmission control message to the transmission control server if the message is received from the transmission participant;

2. if an MBMS subchannel is not used for a transmission in the session the transmission control message is associated with, shall immediately forward the transmission control message to the transmission participant if the message is received from the transmission control server; and

3. if an MBMS subchannel is used for a transmission in the session the transmission control message is associated with:

a. if

i. the transmission control message is not a Transmission Idle message or a Media Transmission Notify message;

ii. the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or

iii. the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report;

shall immediately forward the transmission control message to the transmission participant; and

b. if

i. the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report; and

ii if the transmission control message is the Transmission Idle message or the Media Transmission Notify message,

shall perform actions as specified in clause 10.2.

NOTE: When the Transmit Idle or Media Transmission Notify messages are discarded the messages are sent to the MCVideo clients over the MBMS subchannel allocated for the transmission as specified in clause 10.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.3]

"Upon receiving RTP media packets the participating MCVideo function:

1. *shall immediately forward the RTP media packet to the controlling MCVideo function if the RTP packet is from an MCVideo client; and*

2. *if an MBMS subchannel is not used for a transmission in the session the RTP media packets are associated with, shall immediately forward the RTP media packets to the MCVideo client if the RTP packet is from the controlling MCVideo function or the non-controlling MCVideo function.*

3. *if an MBMS subchannel is used for a transmission in the session the RTP media packets are associated with and if RTP media packets are received from the controlling MCVideo function or the non-controlling MCVideo function:*

   a. *if*

      i. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

      ii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report,*

      shall immediately forward the RTP media packets to the MCVideo client; and

   b. *if the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report, shall perform actions as specified in clause 10.2."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.4]

"When the participating function receives an indication from the application and signalling plane that session release is initiated, the participating MCVideo function:

1. *shall stop sending transmission control messages towards the transmission participant and the transmission control server; and*

2. *shall stop sending RTP media packets towards the MCVideo client and towards the controlling MCVideo function.*

When the participating MCVideo function receives an indication from the application and signalling plane that the session is released, the participating MCVideo function:

1. *in case of a pre-established session, shall perform the actions in clause 9.3.2; and*

2. *in case of an on-demand session, shall release the media resources associated with the session."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

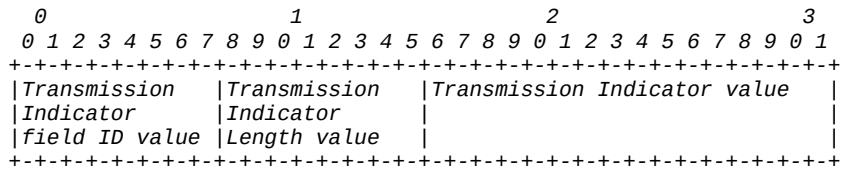
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

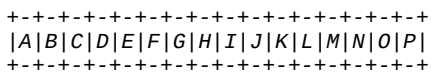


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. *if the active Transmission request queue is empty the transmission control server:*
    a. *shall send Transmission Idle message to all transmission participants. The Transmission Idle message:*
        i. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*
        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;*
    b. *shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;*
    c. *shall start timer T1 (Inactivity);*
    d. *shall set the general state to the 'G: Transmit Idle' state; and*
    e. *shall initialise counter Cx (Simultaneous transmission video) to 0.*
2. *if the active Transmission request queue is not empty the transmission control server:*
    a. *shall select a queued Transmission request from the top of the active Transmission request queue;*
    b. *shall remove that queued Transmission request from the active Transmission request queue; and*
    c. *shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:*
    a. *shall include the granted priority in the Transmission priority field;*
    b. *shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;*
    c. *if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*
    d. *shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;*
2. *shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;*
3. *shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:*
    a. *shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;*
    b. *shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;*
    c. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;*

d. if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';

e. if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

g. shall initiate the 'general reception control operation' state machine;and

4. shall enter the 'G: Transmit Taken' state."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to *imminent peril call, the transmission control arbitration logic in the transmission control server:*

1. if counter Cx (Simultaneous transmission video) has not reached its upper limit:

a. shall perform the actions specified in the clause 6.3.4.4.2;

2. if counter Cx (Simultaneous transmission video) has reached its upper limit:

a. select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;

b. shall stop timer T4 (Transmission Grant), if running;

c. shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);

d. shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;

e. shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and

f. shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:

i. shall include the queue position and transmission priority in the Queue Info field; and

ii. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:

a. shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and

b. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

2. shall set the general state to 'G: pending Transmission Revoke'."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

b. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.2.5.3 Test description

5.1.2.5.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the controlling MCVideo function.
- SS-UE1 (MCVideo Client).

IUT:

- IUT (MCVideo Server), providing the originating participating MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of some sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server and the HTTP Client.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server) and SS-UE1 (MCVideo Client).
- The SS-UE1 (MCVideo client) is registered and authorized to use the MCVideo system.
- The SS-UE1 (MCVideo client) is affiliated to the chat group and is authorized to initiate group calls.

5.1.2.5.3.2 Test procedure sequence

Table 5.1.2.5.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.4.2.1.3-1 to establish an emergency chat group call with implicit transmission control request according to option b.ii of note in Annex A, Table A.4.2.1.3-1, initiated by SS-UE1 (MCVideo Client?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO Transmission End Request - Transmission End Response' as described in Annex A, Table A.4.2.5.3-1 to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Transmission Granted' as described in Annex A, Table A.4.2.4.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	3	P
4	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO call release' as described in Annex A, Table A.4.2.3.3-1, to release the call?	-	-	4	P

5.1.2.5.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

Table 5.1.2.5.3.3-1: SIP INVITE from SS-UE1 (MCVideo Client)
(Step 1, Table 5.1.2.5.3.2-1; Step 1, Annex A, Table A.4.2.1.3-1)

Derivation Path: Annex A, Table A.4.2.1.4-1, with conditions CHAT-GROUP-CALL and EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.5.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.2.5.3.2-1; Step 2, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-2, with conditions CHAT-GROUP-CALL and EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.5.3.3-3: SIP 200 (OK) from SS (MCVideo Server)
(Step 1, Table 5.1.2.5.3.2-1; Step 5, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-3, with condition CHAT-GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.5.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.1.2.5.3.2-1; Step 6, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.1.4-4, with condition CHAT-GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.1.2.5.3.3-5: Transmission Idle
(Steps 2, Table 5.1.2.5.3.2-1; Steps 9 and 10, Annex A, Table A.4.2.5.3-1)**

Derivation Path: Annex A, Table A.4.2.5.4-3, with condition EMERGENCY-CALL.

**Table 5.1.2.5.3.3-6: Transmission Request
(Step 3, Table 5.1.2.5.3.2-1; Steps 1 and 2, Annex A, Table A.4.2.4.3-1)**

Derivation Path: Annex A, Table A.4.2.4.4-1, with condition EMERGENCY-CALL.

**Table 5.1.2.5.3.3-7: Transmission Granted
(Steps 1 and 3, Table 5.1.2.5.3.2-1; Steps 5 and 6, Annex A, Table A.4.2.4.3-1;
Steps 9a1 and 9a2, Annex A, Table A.4.2.1.3-1)**

Derivation Path: Annex A, Table A.4.2.4.4-2, with condition EMERGENCY-CALL.

5.1.2.6 MCVideo Server - MCVideo Server / Chat Group Call / Imminent Peril Group Call / Controlling server

5.1.2.6.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function initiates an
imminent peril chat group call with implicit transmission control }
    then { IUT (MCVideo Server) accepts the call from initiator and establishes the call with all
affiliated users by sending call establishment requests to SS (MCVideo Server) providing the
terminating participating function and sends transmission granted to SS-UE1 (MCVideo Client) and
sends media transmission notification to SS-UE2 (MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established an imminent peril chat group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests to end the transmission control }
    then { IUT (MCVideo Server) sends transmission control idle to SS-UE1 (MCVideo Client) and sends
transmission control idle to SS-UE2 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established an imminent peril chat group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control }
    then { IUT (MCVideo Server) sends transmission granted to SS-UE1 (MCVideo Client) and sends
media transmission notification to SS-UE2 (MCVideo Client) }
            }
```

(4)

```
with { IUT (MCVideo Server) having established an imminent peril chat group call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function ends the chat
group call }
    then { IUT (MCVideo Server) responds by sending a SIP 200 (OK) message to SS (MCVideo Server)
ending the call and sends a SIP BYE message to SS (MCVideo Server) providing the terminating
participating function }
            }
```

5.1.2.6.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.2.4.1.1, 9.2.2.4.2, 9.2.2.4.3.2, 9.2.2.4.3.3, 6.3.3.1.6 and 6.3.3.2.4, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.3.1, 6.3.2.2, 9.2.1, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.4.1.1]

"In the procedures in this clause:

1) MCVideo ID in an incoming SIP INVITE request refers to the MCVideo ID of the originating user from the <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;

2) group identity in an incoming SIP INVITE request refers to the group identity from the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;

3) MCVideo ID in an outgoing SIP INVITE request refers to the MCVideo ID of the called user in the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the outgoing SIP INVITE request;

4) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

5) alert indication in an incoming SIP INVITE request refers to the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

Upon receipt of a "SIP INVITE request for controlling MCVideo function of an MCVideo group" containing a group identity identifying a chat MCVideo group, the controlling MCVideo function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP INVITE request with a SIP 500 (Server Internal Error) response. The controlling MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15] and skip the rest of the steps;

NOTE 1: If the SIP INVITE request contains an emergency indication set to a value of "true", the controlling MCVideo function can by means beyond the scope of this specification choose to accept the request.

2) shall reject the SIP request with a SIP 403 (Forbidden) response and not process the remaining steps if:

a) an Accept-Contact header field does not include the g.3gpp.mcvideo media feature tag;

b) an Accept-Contact header field does not include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo"; or

c) the isfocus media feature tag is present in the Contact header field;

3) if received SIP INVITE request includes an application/vnd.3gpp.mcvideoinfo+xml MIME body with an <emergency-ind> element included or an <imminentperil-ind> element included, shall validate the request as described in clause 6.3.3.1.17;

4) shall retrieve the necessary group document(s) from the group management server for the group identity contained in the SIP INVITE request and carry out initial processing as specified in clause 6.3.5.2 and continue with the rest of the steps if the checks in clause 6.3.5.2 succeed;

5) if the MCVideo user identified by the MCVideo ID in the SIP INVITE request is not affiliated with the MCVideo group identified by the group identity in the SIP INVITE request as determined by the procedures of clause 6.3.6:

a) shall check if the MCVideo user is eligible to be implicitly affiliated with the MCVideo chat group as determined by clause 8.2.2.3.6; and

b) if the MCVideo user is not eligible for implicit affiliation, shall reject the SIP INVITE request with a SIP 403 (Forbidden) response with the warning text set to "120 user is not affiliated to this group" in a Warning header field as specified in clause 4.4 and skip the rest of the steps below;

6) if the SIP INVITE request contains unauthorised request for an MCVideo emergency group call as determined by clause 6.3.3.1.13.2:

a) shall reject the SIP INVITE request with a SIP 403 (Forbidden) response as specified in clause 6.3.3.1.14; and

b) shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [11] and skip the rest of the steps;

7) if the SIP INVITE request contains an unauthorised request for an MCVideo imminent peril group call as determined by clause 6.3.3.1.13.6, shall reject the SIP INVITE request with a SIP 403 (Forbidden) response with the following clarifications:

a) shall include in the SIP 403 (Forbidden) response an application/vnd.3gpp.mcvideo-info+xml MIME body as specified in clause F.1 with the <mcvideoinfo> element containing the <mcvideo-Params> element with the <imminentperil-ind> element set to a value of "false"; and

b) shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [11] and skip the rest of the steps;

8) if a Resource-Priority header field is included in the SIP INVITE request:

a) if the Resource-Priority header field is set to the value indicated for emergency calls and the SIP INVITE request does not contain an emergency indication and the in-progress emergency state of the group is set to a value of "false", shall reject the SIP INVITE request with a SIP 403 (Forbidden) response and skip the remaining steps; and

b) if the Resource-Priority header field is set to the value indicated for imminent peril calls and the SIP INVITE request does not contain an imminent peril indication and the in-progress imminent peril state of the group is set to a value of "false", shall reject the SIP INVITE request with a SIP 403 (Forbidden) response; and skip the remaining steps;

9) shall determine if the media parameters are acceptable and the MCVideo codecs are offered in the SDP offer and if not reject the request with a SIP 488 (Not Acceptable Here) response and skip the rest of the steps;

10) shall create a chat group session and allocate an MCVideo session identity for the chat group session if the MCVideo chat group session identity does not already exist, and may handle timer TNG3 (group call timer) as specified in clause 6.3.3.5;

11) if the chat group session is ongoing and the <on-network-max-participant-count> as specified in 3GPP TS 24.481 [24] is already reached:

a) if, according to local policy, the user identified by the MCVideo ID in the SIP INVITE request is deemed to have a higher priority than an existing user in the chat group session, may remove a participant from the session by following clause 9.2.1.4.4.3, and skip the next step; and

NOTE 2: The local policy for deciding whether to admit a user to a call that has reached its maximum amount of participants can include the <user-priority> and the <participant-type> of the user as well as other information of the user from the group document as specified in 3GPP TS 24.481 [24]. The local policy decisions can also include taking into account whether the imminent-peril indicator or emergency indicator was received in the SIP INVITE request.

b) shall return a SIP 486 (Busy Here) response with the warning text set to "122 too many participants" to the originating network as specified in clause 4.4. Otherwise, continue with the rest of the steps;

12) if the received SIP INVITE request was determined to be eligible for implicit affiliation in step 5) and if clause 8.2.2.3.7 was not previously invoked in the present clause, shall perform the implicit affiliation as specified in clause 8.2.2.3.7;

13) if the SIP INVITE request contains an emergency indication set to a value of "true" or the in-progress emergency state of the group to "true" the controlling MCVideo function shall:

a) validate that the SIP INVITE request includes a Resource-Priority header field populated with the values for an MCVideo emergency group call as specified in clause 6.3.3.1.19, and if not:

i) perform the actions specified in clause 6.3.3.1.8;

ii) send the SIP UPDATE request generated in clause 6.3.3.1.8 towards the initiator of the SIP INVITE request according to 3GPP TS 24.229 [11]; and

iii) upon receiving a SIP 200 (OK) response to the SIP UPDATE request sent in clause 6.3.3.1.8, proceed with the rest of the steps.

NOTE 3: Verify that the Resource-Priority header is included and properly populated for both ongoing and newly-entered in-progress emergency states of the specified group.

b) if the in-progress emergency state of the group is set to a value of "true" and this MCVideo user is indicating a new emergency indication:

i) for each of the other affiliated members of the group generate a SIP MESSAGE request notification of the MCVideo user's emergency indication as specified in clause 6.3.3.1.11 with the following clarifications:

A) set the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true";

B) if the received SIP INVITE contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, perform the procedures specified in clause 6.3.3.1.12; and

C) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11];

ii) cache the information that this MCVideo user has initiated an MCVideo emergency call; and

iii) if the SIP INVITE request contains an authorised request for an MCVideo emergency alert as determined in step i) B) above, cache the information that this MCVideo user has initiated an MCVideo emergency alert; and

c) if the in-progress emergency state of the group is set to a value of "false":

i) shall set the value of the in-progress emergency state of the group to "true";

ii) shall start timer TNG2 (in-progress emergency group call timer) and handle its expiry as specified in clause 6.3.3.1.16;

iii) shall generate SIP re-INVITE requests for the MCVideo emergency group call to the other affiliated and joined participants of the chat MCVideo group as specified in clause 6.3.3.1.6;

iv) shall generate SIP INVITE requests for the MCVideo emergency group call to the affiliated but not joined members of the chat MCVideo group as specified in clause 6.3.3.1.7;

A) for each affiliated but not joined member shall send the SIP INVITE request towards the MCVideo client as specified in 3GPP TS 24.229 [11]; and

B) upon receiving a SIP 200 (OK) response to the SIP INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5];

v) shall cache the information that this MCVideo user has initiated an MCVideo emergency call; and

vi) if the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body is set to "true" and is an authorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall cache the information that this MCVideo user has initiated an MCVideo emergency alert; and

vii) if the in-progress imminent peril state of the group is set to a value of "true", shall set it to a value of "false";

14) if the in-progress emergency state of the group is set to a value of "false" and if the SIP INVITE request contains an imminent peril indication set to a value of "true" or the in-progress imminent peril state of the group is set to "true", the controlling MCVideo function shall:

a) validate that the SIP INVITE request includes a Resource-Priority header field populated with the values for an MCVideo imminent peril group call as specified in clause 6.3.3.1.19, and if not:

i) perform the actions specified in clause 6.3.3.1.8;

ii) send the SIP UPDATE request generated in clause 6.3.3.1.8 towards the initiator of the SIP INVITE request according to 3GPP TS 24.229 [11]; and

iii) upon receiving a SIP 200 (OK) response to the SIP UPDATE request sent in clause 6.3.3.1.8 proceed with the rest of the steps.

NOTE 4: Verify that the Resource-Priority header is included and properly populated for both ongoing and newly-entered in-progress imminent peril states of the specified group.

b) if the in-progress imminent peril state of the group is set to a value of "true" and this MCVideo user is indicating a new imminent peril indication:

i) for each of the other affiliated member of the group generate a SIP MESSAGE request notification of the MCVideo user's imminent peril indication as specified in clause 6.3.3.1.11 with the following clarifications;

A) set the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true"; and

B) send the SIP MESSAGE request as specified in 3GPP TS 24.229 [11]; and

ii) cache the information that this MCVideo user has initiated an MCVideo imminent peril call; and

c) if the in-progress imminent peril state of the group is set to a value of "false":

i) shall set the value of the in-progress imminent peril state of the group to "true";

ii) shall generate SIP re-INVITE requests for the MCVideo imminent peril group call to the other affiliated and joined participants of the chat MCVideo group as specified in clause 6.3.3.1.15;

iii) shall generate SIP INVITE requests for the MCVideo imminent peril call to the affiliated but not joined members of the chat MCVideo group as specified in clause 6.3.3.1.7;

A) for each affiliated but not joined member shall send the SIP INVITE request towards the MCVideo client as specified in 3GPP TS 24.229 [11]; and

B) Upon receiving a SIP 200 (OK) response to the SIP INVITE request the controlling MCVideo function shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

iv) shall cache the information that this MCVideo user has initiated an MCVideo imminent peril call;

15) shall accept the SIP request and generate a SIP 200 (OK) response to the SIP INVITE request according to 3GPP TS 24.229 [11];

16) shall include in the SIP 200 (OK) response an SDP answer according to 3GPP TS 24.229 [11] with the clarifications specified in clause 6.3.3.2.1 unless the procedures of clause 6.3.3.1.8 were performed in step 13)a) or step 14)a) above;

17) should include the Session-Expires header field and start supervising the SIP session according to IETF RFC 4028 [23]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

18) shall include the "timer" option tag in a Require header field;

19) shall include the following in a Contact header field:

a) the g.3gpp.mcvideo media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo";

c) the MCVideo session identity; and

d) the media feature tag isfocus;

20) shall include the "tdialog" option tag in a Supported header field according to IETF RFC 4538 [32];

21) if the SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

22) if the received SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true" and if the in-progress emergency state of the group is set to a value of "true", shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 5: In this case, the request was for an imminent peril call but a higher priority MCVideo emergency call was already in progress on the group. Hence, the imminent peril call request aspect of the request is denied but the request is granted with emergency level priority.

23) shall interact with media plane as specified in 3GPP TS 24.581 [5];

24) shall send the SIP 200 (OK) response to the MCVideo client according to 3GPP TS 24.229 [11]; and

25) if the chat group session was already ongoing and if at least one of the participants has subscribed to the conference event package, shall send a SIP NOTIFY request to all participants with a subscription to the conference event package as specified in clause 9.2.3.4.2.

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCVideo client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.18."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.4.2]

"Upon receiving a SIP BYE request the controlling MCVideo function shall follow the procedures as specified in clause 6.3.3.2.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.4.3.2]

"When the MCVideo session for group call needs to be released as specified in clause 6.3.8.1, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.4.3.3]

"When an MCVideo client needs to be removed from the MCVideo session (e.g. due to de-affiliation or admitting a higher priority user), the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.4.

After successful removing the MCVideo client from the MCVideo session, the controlling MCVideo function may generate a notification to the MCVideo clients, which have subscribed to the conference state event package that an MCVideo user has been removed from the MCVideo session, as specified in clause 6.3.3.4 and send the SIP NOTIFY request to the MCVideo client according to 3GPP TS 24.229 [11]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.6]

"This clause is referenced from other procedures.

This clause describes the procedures for inviting an MCVideo user to an MCVideo session associated with an MCVideo emergency group call or MCVideo imminent peril group call. The procedure is initiated by the controlling MCVideo function as the result of an action in clause 9.2.2.4.1.1.

The controlling MCVideo function:

1) *shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;*

2) *shall set the Request-URI to the address of the terminating participating MCVideo function associated with the MCVideo ID of the targeted MCVideo user;*

3) *shall include an application/vnd.3gpp.mcvideo-info+xml MIME body with the <mcvideoinfo> element containing the <mcvideo-Params> element populated as follows:*

   a) *the <mcvideo-request-uri> element set to the value of the MCVideo ID of the targeted MCVideo user;*

   b) *the <mcvideo-calling-user-id> element set to the value of the MCVideo ID of the calling MCVideo user; and*

   c) *the <mcvideo-calling-group-id> element set to the value of the MCVideo group ID of the emergency group call.*

4) *shall include in the P-Asserted-Identity header field the public service identity of the controlling MCVideo function;*

5) *shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1; and*

6) *if the in-progress emergency group state of the group is set to a value of "true" the controlling MCVideo function:*

   a) *shall include a Resource-Priority header field populated with the values for an MCVideo emergency group call as specified in clause 6.3.3.1.18;*

   b) *shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <emergency-ind> element set to a value of "true";*

   c) *if the <alert-ind> element is set to "true" in the received SIP INVITE request and the requesting MCVideo user and MCVideo group are authorised for the initiation of MCVideo emergency alerts as determined by the procedures of clause 6.3.3.1.12.1, shall populate the application/vnd.3gpp.mcvideo-info+xml MIME body and the application/vnd.3gpp.mcvideo-location-info+xml MIME body as specified in clause 6.3.3.1.11. Otherwise, shall set the <alert-ind> element to a value of "false" in the application/vnd.3gpp.mcvideo-info+xml MIME body; and*

   d) *if the in-progress imminent peril state of the group is set to a value of "true" shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body an <imminentperil-ind> element set to a value of "false";*

NOTE: If the imminent peril state of the group is true at this point, the controlling function will set it to false as part of the calling procedure.

7) *if the in-progress emergency state of the group is set to a value of "false" and the in-progress imminent peril state of the group is set to a value of "true", the controlling MCVideo function:*

   a) *shall include a Resource-Priority header field populated with the values for an MCVideo imminent peril group call as specified in clause 6.3.3.1.18; and*

b) shall include in the application/vnd.3gpp.mcvideo-info+xml MIME body with the <imminentperil-ind> element set to a value of "true"; and

8) if:

a) an MCVideo GKTP document exists for the group identity contained in the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request; and

b) the MCVideo GKTP document contains an <MKFC-GKTPs> element;

then:

a) for each instance of <GKTP> element of the <MKFC-GKTPs> element of the MCVideo GKTP document:

i) shall perform the procedure in clause 6.3.3.6.2 to re-generate an I_MESSAGE; and

ii) if the procedure in clause 6.3.3.6.2 was successful, shall include the I_MESSAGE in a <GKTP> element of the <MKFC-GKTPs> element of an application/vnd.3gpp.mcvideo-info+xml MIME body included in the outgoing SIP INVITE request."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.2.4]

"Upon receiving a SIP BYE request the controlling MCVideo function:

1) shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.581 [5] for releasing the media plane resource associated with the SIP session towards the MCVideo client;

NOTE: The non-controlling MCVideo function is also regarded as a MCVideo client in a temporary MCVideo group session.

2) shall generate a SIP 200 (OK) response and send the SIP response towards the MCVideo client according to 3GPP TS 24.229 [11];

3) shall check the MCVideo session release policy as specified in clause 6.3.8.1 and clause 6.3.8.2 whether the MCVideo session needs to be released for each participant of the MCVideo session;

4) if release of the MCVideo session is required:

a) shall perform the procedures as specified in the clause 6.3.3.1.4 with the clarification that if the received SIP BYE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body, copy the application/vnd.3gpp.mcvideo-info+xml MIME body into the outgoing SIP BYE request; and

5) if a release of the MCVideo session is not required, shall send a SIP NOTIFY request to all remaining MCVideo clients in the MCVideo session with a subscription to the conference event package as specified in clause 9.2.3.4.2.

Upon receiving a SIP 200 (OK) response to the SIP BYE request the controlling MCVideo function shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.581 [5] for releasing media plane resources associated with the SIP session with the MCVideo participant."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.1]

"The transmission control server arbitration logic in the transmission control server shall support the procedures in clauses 6.3.2 and 6.3.3 and shall behave according to the transmission control server state transition diagram for 'general transmission control operation' in clause 6.3.4.

The transmission control interface towards the MCVideo client in the transmission control server shall behave according to the transmission control server state transition diagram for 'basic transmission control operation towards the transmission participant' as specified in clause 6.3.5.

The reception control arbitration logic in the transmission control server shall support the procedures in clauses 6.3.2 and 6.3.3 and shall behave according to the reception control server state transition diagram for 'general reception control operation' in clause 6.3.6.

The transmission control interface towards the MCVideo client in the transmission control server shall behave according to the transmission control server state transition diagram for 'basic reception control operation towards the transmission participant' as specified in clause 6.3.7."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.2.2]

"When an MCVideo call is established a new instance of the transmission control server state machine for 'general transmission control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' is added.

When an MCVideo call is established a new instance of the transmission control server state machine for 'general reception control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic reception control operation towards the transmission participant' is added.

If the optional "mc_queueing" feature is supported and has been negotiated as specified in clause 14, the transmission control server could queue the implicit transmission control request for the media-transmission control entity.

The original initial SIP INVITE request or SIP REFER request to establish an MCVideo chat group call or to rejoin an ongoing MCVideo call is not handled as an implicit transmission control request message by the transmission control server unless explicitly stated in the SIP INVITE request or in the SIP REFER request.

The permission to send media to the inviting MCVideo client due to implicit transmission control request is applicable to both confirmed indication and unconfirmed indication.

When the first unconfirmed indication is received from the invited participating MCVideo function (see 3GPP TS 24.281 [2]) the transmission control server optionally can give an early indication to send RTP media packets, to the inviting MCVideo client.

If an early indication to send RTP media packets is given to the inviting MCVideo client, the transmission participant is granted the permission to send media and the MCVideo server buffers RTP media packets received from the MCVideo client at least until the first invited MCVideo client accepts the invitation or until the RTP media packet buffer exceeds it maximum limit to store RTP media packets.

If the MCVideo server does not support or does not allow media buffering then when an early indication to send RTP media packets is not given to the inviting MCVideo client, the transmission participant is granted the permission to send media when the first invited MCVideo client accepts the media.

Before the transmission control server sends the first transmission control message in the MCVideo call, the transmission control server has to assign itself a SSRC identifier to be included in media transmission control messages and quality feedback messages if the MCVideo server is supporting that option. A suitable algorithm to generate the SSRC identifier is described in IETF RFC 3550 [3].

The transmission participant and the transmission control server can negotiate the maximum priority level that the transmission participant is permitted to request. The transmission control server can pre-empt the current sender based on the negotiated maximum priority level that the transmission participant is permitted to request and the priority level included in the Transmission Media Request message.

NOTE: The maximum priority level that a transmission participant can use is negotiated as specified in clause 14.3.3 and is based on group configuration data retrieved by the controlling MCVideo function from the group management server as described in 3GPP TS 24.481 [12] and service configuration data retrieved by the controlling MCVideo function from the configuration management server as described in 3GPP TS 24.484 [13].

The transmission participant and the transmission control server can negotiate queueing of Transmission requests using the "mc_queueing" fmtp attribute as described in clause 14. If queueing is supported and negotiated, the transmission control server queues the transmission control request if a Transmission Media Request message is received when another transmission participant has the transmission and the priority of the current speaker is the same or higher."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.1]

"The transmission control messages are coded as described in clause 9.1.2 where the transmission control message is part of the application-dependent data.

For the transmission control protocol the ASCII name string is: MCVx (Mission Critical Video x) with x=0, 1 or 2 as specified in clause 9.1.2.

A list of transmission control messages can be found in clause 9.2.2.1.

The transmission control specific fields are specified in clause 9.2.3."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

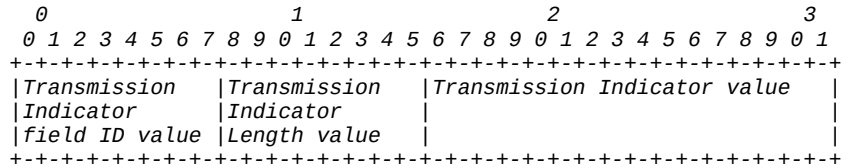
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding



The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P

When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. *if the active Transmission request queue is empty the transmission control server:*
    a. *shall send Transmission Idle message to all transmission participants. The Transmission Idle message:*
        i. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*
        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;*
    b. *shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;*
    c. *shall start timer T1 (Inactivity);*
    d. *shall set the general state to the 'G: Transmit Idle' state; and*
    e. *shall initialise counter Cx (Simultaneous transmission video) to 0.*
2. *if the active Transmission request queue is not empty the transmission control server:*
    a. *shall select a queued Transmission request from the top of the active Transmission request queue;*
    b. *shall remove that queued Transmission request from the active Transmission request queue; and*
    c. *shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:*
    a. *shall include the granted priority in the Transmission priority field;*
    b. *shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;*
    c. *if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*
    d. *shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;*
2. *shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;*
3. *shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:*
    a. *shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;*
    b. *shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;*
    c. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;*
    d. *if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';*
    e. *if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';*
    f. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*
    g. *shall initiate the 'general reception control operation' state machine;and*
4. *shall enter the 'G: Transmit Taken' state."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. *if counter Cx (Simultaneous transmission video) has not reached its upper limit:*
    a. *shall perform the actions specified in the clause 6.3.4.4.2;*
2. *if counter Cx (Simultaneous transmission video) has reached its upper limit:*
    a. *select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;*
    b. *shall stop timer T4 (Transmission Grant), if running;*
    c. *shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);*
    d. *shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;*

e. shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and

f. shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:

i. shall include the queue position and transmission priority in the Queue Info field; and

ii. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:

a. shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and

b. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

2. shall set the general state to 'G: pending Transmission Revoke'."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

b. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.2.6.3 Test description

5.1.2.6.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the participating MCVideo functions.
- SS-UE1 (MCVideo Client) and SS-UE2 (MCVideo Client), simulated by the SS (MCVideo Server).

IUT:

- IUT (MCVideo Server), providing the controlling MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of all sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Transmission Control Server.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server).
- The SS-UE1 (MCVideo client) and the SS-UE2 (MCVideo client) are registered and authorized to use the MCVideo system.
- The SS-UE1 (MCVideo client) and the SS-UE2 (MCVideo client) are affiliated to the chat group and are authorized to initiate group calls.
- The IUT (MCVideo Server) is the controlling server of the chat group.

5.1.2.6.3.2 Test procedure sequence

Table 5.1.2.6.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E group call establishment' as described in Annex A, Table A.4.1.4.3-1 to establish an imminent peril chat group call with implicit transmission control request according to option b.ii of note in Annex A, Table A.4.1.4.3-1, initiated by SS-UE1 (MCVideo Client)?	-	-	1	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
2	Check: Does the IUT (MCVideo Server) correctly perform procedure ' MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	3	P
4	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.1.5.3-1, to release the call?	-	-	4	P

5.1.2.6.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

Table 5.1.2.6.3.3-1: SIP INVITE from SS (MCVideo Server)
(Step 1, Table 5.1.2.6.3.2-1; Step 1, Annex A, Table A.4.1.4.3-1)

Derivation Path: Annex A, Table A.4.1.4.4-1, with condition IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.2.6.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.2.6.3.2-1; Step 2, Annex A, Table A.4.1.4.3-1)

Derivation Path: Annex A, Table A.4.1.4.4-2, with condition IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.2.6.3.3-3: SIP 200 (OK) from SS (MCVideo Server)
(Step 1, Table 5.1.2.6.3.2-1; Step 8, Annex A, Table A.4.1.4.3-1)

Derivation Path: Annex A, Table A.4.1.4.4-6

Table 5.1.2.6.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.1.2.6.3.2-1; Step 9, Annex A, Table A.4.1.4.3-1)

Derivation Path: Annex A, Table A.4.1.4.4-7

Table 5.1.2.6.3.3-5: Transmission Idle from IUT (MCVideo Server)
(Step 2, Table 5.1.2.6.3.2-1; Steps 4 and 5, Annex A, Table A.3.8.3-1)

Derivation Path: Annex A, Table A.3.8.4-6, with condition IMMPERIL-CALL.

Table 5.1.2.6.3.3-6: Transmission Request from SS (MCVideo Server)
(Step 3, Table 5.1.2.6.3.2-1; Step 1, Annex A, Table A.3.7.3-1)

Derivation Path: Annex A, Table A.3.7.4-1, with condition IMMPERIL-CALL.

Table 5.1.2.6.3.3-7: Transmission Granted from IUT (MCVideo Server) (Steps 1 and 3, Table 5.1.2.6.3.2-1; Step 3a1 or 3b1, Annex A, Table A.3.7.3-1; Step 9a1, Annex A, Table A.4.1.1.3-1)

Derivation Path: Annex A, Table A.3.7.4-2 or A.3.7.4-3 (note), with condition IMMPERIL-CALL.
NOTE: Depends on server implementation.

Table 5.1.2.6.3.3-8: Media Transmission Notification from IUT (MCVideo Server) (Steps 1 and 3, Table 5.1.2.6.3.2-1; Step 3a3 or 3b2, Annex A, Table A.3.7.3-1; Step 10a1, Annex A, Table 4.1.1.3-1)

Derivation Path: Annex A, Table A.3.7.4-4 or A.3.7.4-5 (note), with condition IMMPERIL-CALL.
NOTE: Depends on server implementation.

5.1.2.7 MCVideo Server - MCVideo Server / Chat Group Call / Imminent Peril Group Call / Terminating Participating server

5.1.2.7.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the controlling function initiates an imminent peril chat
group call with automatic commencement }
    then { IUT (MCVideo Server) accepts the call and establishes the call with SS-UE2 (MCVideo
Client) and forwards the media transmission notification received from SS (MCVideo Server) to SS-UE2
(MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established an imminent peril chat group call }
ensure that {
  when { the SS (MCVideo Server) ends the media transmission }
    then { IUT (MCVideo Server) forwards the transmission end notify received from SS (MCVideo
Server) to SS-UE2 (MCVideo Client) and forwards the transmission idle received from SS (MCVideo
Server) to SS-UE2 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established an imminent peril chat group call }
ensure that {
  when { the SS (MCVideo Server) providing the controlling function ends the chat group call }
    then { IUT (MCVideo Server) sends a SIP BYE message to SS-UE2 (MCVideo Client) and responds by
sending a SIP 200 (OK) message to SS (MCVideo Server) }
            }
```

5.1.2.7.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.2.3.1.3, 6.3.2.1.5, 6.3.2.2.3, 6.3.2.2.8.1, 6.3.2.2.1, 6.3.2.2.4.2, 6.3.2.2.2.1 and 6.3.2.1.1.1, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.4.1, 6.4.2, 6.4.3, 6.4.4, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.3.1.3]

"This clause covers on-demand session.

Upon receipt of a "SIP INVITE request for terminating participating MCVideo function", for a terminating MCVideo client of a chat MCVideo group, the participating MCVideo function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the "SIP INVITE request for terminating participating MCVideo function" with a SIP 500 (Server Internal Error) response. The participating MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15]. Otherwise, continue with the rest of the steps;

NOTE: If the SIP INVITE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body with the <emergency-ind> element set to a value of "true", the participating MCVideo function can by means beyond the scope of this specification choose to accept the request.

2) shall check the presence of the isfocus media feature tag in the URI of the Contact header field and if it is not present then the participating MCVideo function shall reject the request with a SIP 403 (Forbidden) response with the warning text set to "104 isfocus not assigned" in a Warning header field as specified in clause 4.4. Otherwise, continue with the rest of the steps;

3) shall generate a SIP INVITE request as specified in clause 6.3.2.2.3;

4) shall set the Request-URI to the public user identity associated with the MCVideo ID of the MCVideo user to be invited based on the contents of the Request-URI of the received "SIP INVITE request for terminating participating MCVideo function";

5) shall copy the contents of the P-Asserted-Identity header field of the incoming "SIP INVITE request for terminating participating MCVideo function" to the P-Asserted-Identity header field of the outgoing SIP INVITE request;

6) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received "SIP INVITE request for terminating participating MCVideo function" as specified in clause 6.3.2.2.1;

7) if the received SIP INVITE request contains a Resource-Priority header field, shall include a Resource-Priority header field with the contents set as in the received Resource-Priority header field;

8) shall perform the procedures specified in clause 6.3.2.2.9 to include any MIME bodies in the received SIP INVITE request; and

9) shall send the SIP INVITE request towards the MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response to the above SIP INVITE request sent to the MCVideo client, the participating MCVideo function:

1) shall generate a SIP 200 (OK) response as described in the clause 6.3.2.2.4.2;

2) shall include in the SIP 200 (OK) response an SDP answer based on the SDP answer in the received SIP 200 (OK) response as specified in clause 6.3.2.2.2.1;

3) shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

4) shall forward the SIP 200 (OK) response according to 3GPP TS 24.229 [11]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.5]

"Upon receiving a SIP BYE request from the MCVideo client, the participating MCVideo function:

1) shall interact with the media plane as specified in 3GPP TS 24.581 [5];

2) shall generate a SIP BYE request as specified in 3GPP TS 24.229 [11];

3) shall set the Request-URI to the MCVideo session identity as included in the received SIP BYE request;

4) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;

5) *if the received SIP BYE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body into the outgoing SIP BYE request; and*

6) *shall send the SIP BYE request toward the controlling MCVideo function, according to 3GPP TS 24.229 [11].*

Upon receiving a SIP 200 (OK) response to the SIP BYE request the terminating MCVideo function shall forward a SIP 200 (OK) response to the MCVideo client and shall interact with the media plane as specified in 3GPP TS 24.581 [5] for releasing media plane resources associated with the SIP session with the controlling MCVideo function."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.3]

"The participating MCVideo function shall generate an initial SIP INVITE request according to 3GPP TS 24.229 [11] and:

1) *shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [20] if included in the incoming SIP INVITE request;*

2) *should include the Session-Expires header field according to IETF RFC 4028 [23]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";*

3) *shall include the option tag "timer" in the Supported header field;*

4) *shall include the following in the Contact header field:*

   a) *the g.3gpp.mcvideo media feature tag;*

   b) *the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo";*

   c) *the isfocus media feature tag;*

   d) *an MCVideo session identity mapped to the MCVideo session identity provided in the Contact header field of the incoming SIP INVITE request; and*

   e) *any other uri-parameter provided in the Contact header field of the incoming SIP INVITE request;*

5) *shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [32];*

6) *shall include the option tag "norefersub" in a Supported header field according to rules and procedures of IETF RFC 4488 [31];*

7) *may include a Resource-Share header field in accordance with clause 5.7.1.20.3 in 3GPP TS 24.229 [11]; and*

8) *if the incoming SIP INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body to the outgoing SIP INVITE request."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.8.1]

"Upon receiving a SIP BYE request from the controlling MCVideo function, the participating MCVideo function:

1) *shall interact with the media plane as specified in clause 6.4 in 3GPP TS 24.581 [5] for releasing media plane resource associated with the SIP session with the controlling MCVideo function;*

2) *shall generate a SIP BYE request according to 3GPP TS 24.229 [11];*

3) *shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;*

4) *if the received SIP BYE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body into the outgoing SIP BYE request; and*

5) *shall send the SIP BYE request to the MCVideo client according to 3GPP TS 24.229 [11].*

Upon receiving a SIP 200 (OK) response to the SIP BYE request the participating MCVideo function:

1) *shall send a SIP 200 (OK) response to the SIP BYE request received from the controlling MCVideo function according to 3GPP TS 24.229 [11]; and*

2) *shall interact with the media plane as specified in 3GPP TS 24.581 [5] for releasing media plane resources associated with the SIP session with the MCVideo client."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.1]

"The participating MCVideo function shall follow the procedures in clause 6.3.2.1.1."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.4.2]

"This clause is referenced from other procedures.

When sending SIP 200 (OK) responses, the participating MCVideo function shall generate a SIP 200 (OK) response according to 3GPP TS 24.229 [11] and:

1) *shall include the option tag "timer" in a Require header field;*

2) *shall include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [23], "UAS Behavior". If no "refresher" parameter was included in the SIP INVITE request, the "refresher" parameter in the Session-Expires header field shall be set to "uas";*

3) *shall include the following in the Contact header field:*

   a) *the g.3gpp.mcvideo media feature tag;*

   b) *the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo"; and*

   c) *an MCVideo session identity mapped to the MCVideo session identity provided in the Contact header field of the received SIP INVITE request from the controlling MCVideo function;*

4) *shall include the option tag "tdialog" in a Supported header field according to rules and procedures of IETF RFC 4538 [32]; and*

5) *if the incoming SIP response contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body to the outgoing SIP 200 (OK) response."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.2.2.1]

"The participating MCVideo function shall follow the procedures in clause 6.3.2.1.1."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.1.1]

"This clause is referenced from other clauses.

The SDP offer is generated based on the received SDP offer. The SDP offer generated by the participating MCVideo function:

1) *shall contain two SDP media-level sections for MCVideo video media as contained in the received SDP offer; and*

2) *shall contain an SDP media-level section for one media-transmission control entity, if present in the received SDP offer.*

When composing the SDP offer according to 3GPP TS 24.229 [11], the participating MCVideo function:

1) *shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the participating MCVideo function, if required;*

NOTE 1: Requirements can exist for the participating MCVideo function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number for the offered media transmission control entity, if any, in the received SDP offer with the IP address and port number of the participating MCVideo function; and

NOTE 2: If the participating MCVideo function and the controlling MCVideo function are in the same MCVideo server, and the participating MCVideo function does not have a dedicated IP address or a dedicated port number for media transmission control or media stream, the replacement of the IP address or the port number is omitted.

3) shall contain an "a=key-mgmt" attribute field with a "mikey" attribute value, if present in the received SDP offer."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.2]

"Upon receiving a transmission control message the participating MCVideo function:

1. shall immediately forward the transmission control message to the transmission control server if the message is received from the transmission participant;

2. if an MBMS subchannel is not used for a transmission in the session the transmission control message is associated with, shall immediately forward the transmission control message to the transmission participant if the message is received from the transmission control server; and

3. if an MBMS subchannel is used for a transmission in the session the transmission control message is associated with:

a. if

i. the transmission control message is not a Transmission Idle message or a Media Transmission Notify message;

ii. the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or

iii. the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report;

shall immediately forward the transmission control message to the transmission participant; and

b. if

i. the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report; and

ii if the transmission control message is the Transmission Idle message or the Media Transmission Notify message,

shall perform actions as specified in clause 10.2.

NOTE: When the Transmit Idle or Media Transmission Notify messages are discarded the messages are sent to the MCVideo clients over the MBMS subchannel allocated for the transmission as specified in clause 10.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.3]

"Upon receiving RTP media packets the participating MCVideo function:

1. shall immediately forward the RTP media packet to the controlling MCVideo function if the RTP packet is from an MCVideo client; and

2. if an MBMS subchannel is not used for a transmission in the session the RTP media packets are associated with, shall immediately forward the RTP media packets to the MCVideo client if the RTP packet is from the controlling MCVideo function or the non-controlling MCVideo function.

3. *if an MBMS subchannel is used for a transmission in the session the RTP media packets are associated with and if RTP media packets are received from the controlling MCVideo function or the non-controlling MCVideo function:*

   a. *if*

      i. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

      ii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report,*

      shall immediately forward the RTP media packets to the MCVideo client; and

   b. *if the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report, shall perform actions as specified in clause 10.2."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.4]

"When the participating function receives an indication from the application and signalling plane that session release is initiated, the participating MCVideo function:

1. *shall stop sending transmission control messages towards the transmission participant and the transmission control server; and*

2. *shall stop sending RTP media packets towards the MCVideo client and towards the controlling MCVideo function.*

When the participating MCVideo function receives an indication from the application and signalling plane that the session is released, the participating MCVideo function:

1. *in case of a pre-established session, shall perform the actions in clause 9.3.2; and*

2. *in case of an on-demand session, shall release the media resources associated with the session."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

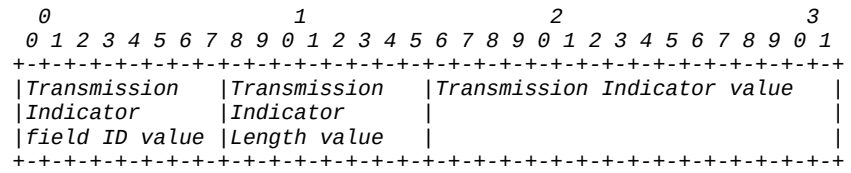
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

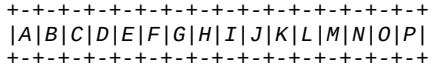


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. if the active Transmission request queue is empty the transmission control server:

a. shall send Transmission Idle message to all transmission participants. The Transmission Idle message:

i. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

ii. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

b. shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;

c. shall start timer T1 (Inactivity);

d. shall set the general state to the 'G: Transmit Idle' state; and

e. shall initialise counter Cx (Simultaneous transmission video) to 0.

2. if the active Transmission request queue is not empty the transmission control server:

a. shall select a queued Transmission request from the top of the active Transmission request queue;

b. shall remove that queued Transmission request from the active Transmission request queue; and

c. shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:

a. shall include the granted priority in the Transmission priority field;

b. shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;

c. if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

d. shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;

2. shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;

3. shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:

a. shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;

b. shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;

c. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;

d. if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';

e. if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

g. shall initiate the 'general reception control operation' state machine;and

4. shall enter the 'G: Transmit Taken' state."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. if counter Cx (Simultaneous transmission video) has not reached its upper limit:

a. shall perform the actions specified in the clause 6.3.4.4.2;

2. *if counter Cx (Simultaneous transmission video) has reached its upper limit:*

    a. *select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;*

    b. *shall stop timer T4 (Transmission Grant), if running;*

    c. *shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);*

    d. *shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;*

    e. *shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and*

    f. *shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:*

        i. *shall include the queue position and transmission priority in the Queue Info field; and*

        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:*

    a. *shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and*

    b. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

2. *shall set the general state to 'G: pending Transmission Revoke'."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. *if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:*

    a. *shall initialize a general state machine as specified in clause 6.3.4.2.2; and*

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

    b. *shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;*

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.2.7.3 Test description

5.1.2.7.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the controlling MCVideo function.
- SS-UE2 (MCVideo Client).

IUT:

- IUT (MCVideo Server), providing the terminating participating MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of some sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server and the HTTP Client.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server) and SS-UE2 (MCVideo Client).
- The SS-UE2 (MCVideo client) is registered and authorized to use the MCVideo system.

- The SS-UE2 (MCVideo client) is affiliated to the chat group and is authorized to initiate group calls.

5.1.2.7.3.2 Test procedure sequence

Table 5.1.2.7.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CT group/private call establishment, automatic commencement' as described in Annex A, Table A.4.3.3.3-1 to establish an imminent peril chat group call with automatic commencement mode and notify a media transmission from SS-UE1 (MCVideo Client)?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Notification - Idle' as described in Annex A, Table A.4.3.8.3-1 to notify the end of SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.3.5.3-1, to release the call?	-	-	3	P

5.1.2.7.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

Table 5.1.2.7.3.3-1: SIP INVITE from SS (MCVideo Server)
(Step 1, Table 5.1.2.7.3.2-1; Step 1, Annex A, Table A.4.3.3.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-1, with conditions CHAT-GROUP-CALL and IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.2.7.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.2.7.3.2-1; Step 2, Annex A, Table A.4.3.3.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-2, with conditions CHAT-GROUP-CALL and IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.2.7.3.3-3: SIP 200 (OK) from SS-UE2 (MCVideo Client)
(Step 1, Table 5.1.2.7.3.2-1; Step 6, Annex A, Table A.4.3.3.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-7, with condition CHAT-GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.2.7.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.1.2.7.3.2-1; Step 7, Annex A, Table A.4.3.3.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-8, with condition CHAT-GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.2.7.3.3-5: Transmission Idle
(Step 2, Table 5.1.2.7.3.2-1; Steps 13 and 14, Annex A, Table A.4.3.8.3-1)

Derivation Path: Annex A, Table A.4.3.8.4-6, with condition IMMPERIL-CALL.

Table 5.1.2.7.3.3-6: Media Transmission Notification
(Step 1, Table 5.1.2.7.3.2-1; Steps 10a1 and 10a2, Annex A, Table 4.3.3.3-1)

Derivation Path: Annex A, Table A.4.3.3.4-11, with condition IMMPERIL-CALL.

5.1.2.8 MCVideo Server - MCVideo Server / Chat Group Call / Imminent Peril Group Call / Originating Participating server

5.1.2.8.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS-UE1 (MCVideo Client) initiates an imminent peril chat group call with implicit
transmission control }
    then { IUT (MCVideo Server) accepts the call from initiator and establishes the call with the SS
(MCVideo Server) providing the controlling function and forwards the transmission granted received
from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established an imminent peril chat group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests to end the transmission control }
    then { IUT (MCVideo Server) forwards the transmission end request to SS (MCVideo Server) and
forwards the transmission end response to SS-UE1 (MCVideo Client) and forwards the transmission idle
received from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established an imminent peril chat group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control }
    then { IUT (MCVideo Server) forwards the transmission request to SS (MCVideo Server) and
forwards the transmission granted received from SS (MCVideo Server) to SS-UE1 (MCVideo Client) }
            }
```

(4)

```
with { IUT (MCVideo Server) having established an imminent peril chat group call }
ensure that {
  when { the SS-UE1 (MCVideo Client) ends the chat group call }
    then { IUT (MCVideo Server) sends a SIP BYE message to SS (MCVideo Server) providing the
controlling function and responds by sending a SIP 200 (OK) message to SS-UE1 (MCVideo Client) }
            }
```

5.1.2.8.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 9.2.2.3.1.1, 6.3.2.1.3, 6.3.2.1.1.1, 6.3.2.1.5 and 6.3.2.1.2.1, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.4.1, 6.4.2, 6.4.3, 6.4.4, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 9.2.2.3.1.1]

"In the procedures in this clause:

1) group identity in an incoming SIP INVITE request refers to the group identity from the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request;

2) emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body;

3) imminent peril indication in an incoming SIP INVITE request refers to the <imminentperil-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.

Upon receipt of a "SIP INVITE request for originating participating MCVideo function" for a group identity identifying a chat MCVideo group containing an application/vnd.3gpp.mcvideo-info+xml MIME body with the <session-type> element set to a value of "chat", the participating MCVideo function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the SIP INVITE request with a SIP 500 (Server Internal Error) response. The participating MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15]. Otherwise, continue with the rest of the steps;

NOTE 1: if the SIP INVITE request contains an emergency indication set to a value of "true" or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCVideo function can according to local policy choose to accept the request.

2) shall determine the MCVideo ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request, and authorise the calling user;

NOTE 2: The MCVideo ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

3) if through local policy in the originating participating MCVideo function, the user identified by the MCVideo ID is not authorised to make chat group calls, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response to the SIP INVITE request, with warning text set to "108 user not authorised to make chat group calls" in a Warning header field as specified in clause 4.4;

4) shall determine if the media parameters are acceptable and the MCVideo codecs are offered in the SDP offer and if not, reject the request with a SIP 488 (Not Acceptable Here) response. Otherwise, continue with the rest of the steps;

5) shall check if the number of maximum simultaneous MCVideo group calls supported for the MCVideo user as specified in the <MaxSimultaneousCallsN6> element of the <MCVideo-group-call> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) has been exceeded. If exceeded, the MCVideo function shall respond with a SIP 486 (Busy Here) response with the warning text set to "103 maximum simultaneous MCVideo group calls reached" in a Warning header field as specified in clause 4.4. Otherwise, continue with the rest of the steps;

NOTE 3: If the SIP INVITE request contains an emergency indication set to a value of "true" or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCVideo function can according to local policy choose to allow for an exception to the limit for the maximum simultaneous MCVideo sessions supported for the MCVideo user.

6) if the user identified by the MCVideo ID is not affiliated to the group identified in the "SIP INVITE request for originating participating MCVideo function" as determined by clause 8.2.2.2.11, shall perform the actions specified in clause 8.2.2.2.12 for implicit affiliation;

7) if the actions for implicit affiliation specified in step 6) above were performed but not successful in affiliating the MCVideo user due to the MCVideo user already having N2 simultaneous affiliations, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 486 (Busy Here) response with the warning text set to "102 too many simultaneous affiliations" in a Warning header field as specified in clause 4.4. and skip the rest of the steps.

NOTE 4: N2 is the total number of MCVideo groups that an MCVideo user can be affiliated to simultaneously as specified in 3GPP TS 23.281 [26].

NOTE 5: if the SIP INVITE request contains an emergency indication set to a value of "true" or an imminent peril indication set to a value of "true" and this is an authorised request for originating a priority call as determined by clause 6.3.2.1.8.1, the participating MCVideo function can according to local policy choose to allow an exception to the N2 limit. Alternatively, a lower priority affiliation of the MCVideo user could be cancelled to allow for the new affiliation.

8) shall determine the public service identity of the controlling MCVideo function associated with the group identity in the SIP INVITE request;

NOTE 6: The public service identity can identify the controlling MCVideo function in the primary MCVideo system or a partner MCVideo system.

NOTE 7: How the participating MCVideo server discovers the public service identity of the controlling MCVideo function associated with the group identity is out of scope of the current document.

9) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3;

10) shall set the Request-URI to the public service identity of the controlling MCVideo function associated with the group identity present in the incoming SIP INVITE request;

11) shall include the MCVideo ID of the calling user in <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the SIP INVITE request;

12) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request as specified in clause 6.3.2.1.1.1;

13) if the received SIP INVITE request contains an application/vnd.3gpp.location-info+xml MIME body as specified in Annex F.3; and

a) if not already included, shall include a Content-Type header field set to "application/vnd.3gpp.location-info+xml"; and

b) if not already copied, shall copy the contents of the application/vnd.3gpp.location-info+xml MIME body received in the SIP INVITE request into an application/vnd.3gpp.location-info+xml MIME body included in the outgoing SIP request;

NOTE 8: Note that the application/vnd.3gpp.mcvideo-info+xml MIME body will already have been copied into the outgoing SIP INVITE request by clause 6.3.2.1.3.

14) if a Resource-Priority header field was included in the received SIP INVITE request, shall include a Resource-Priority header field according to rules and procedures of IETF RFC 4412 [33] set to the value indicated in the Resource-Priority header field of the SIP INVITE request from the MCVideo client; and

NOTE 9: The participating MCVideo function will leave verification of the Resource-Priority header field to the controlling MCVideo function.

15) shall forward the SIP INVITE request according to 3GPP TS 24.229 [11].

Upon receipt of a SIP 302 (Moved Temporarily) response to the above SIP INVITE request in step 14), the participating MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.2.1.10;

2) shall include an SDP offer based upon the SDP offer in the received SIP INVITE request from the MCVideo client as specified in clause 6.3.2.1.1.1; and

3) shall forward the SIP INVITE request according to 3GPP TS 24.229 [11];

Upon receipt of a SIP 2xx response to the above SIP INVITE request in step 14) the participating MCVideo function:

1) if the SIP 2xx response contains an application/vnd.3gpp.mcvideo-info+xml MIME body with an <MKFC-GKTPs> element, shall perform the procedures in clause 6.3.2.3.2;

2) shall generate a SIP 200 (OK) response as specified in the clause 6.3.2.1.5.2;

3) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

4) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response;

5) shall include the public service identity received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response;

6) if the procedures of clause 8.2.2.2.12 for implicit affiliation were performed in the present clause, shall complete the implicit affiliation by performing the procedures of clause 8.2.2.2.13;

7) shall send the SIP 200 (OK) response to the MCVideo client according to 3GPP TS 24.229 [11]; and

8) shall interact with the media plane as specified in 3GPP TS 24.581 [5].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the above SIP INVITE request in step 14) the participating MCVideo function:

1) shall generate a SIP response according to 3GPP TS 24.229 [11];

2) shall include Warning header field(s) that were received in the incoming SIP response;

3) shall forward the SIP response to the MCVideo client according to 3GPP TS 24.229 [11]; and

4) if the implicit affiliation procedures of clause 8.2.2.2.12 were invoked in the current procedure, shall perform the procedures of clause 8.2.2.2.14."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.3]

"This clause is referenced from other procedures.

When generating an initial SIP INVITE request according to 3GPP TS 24.229 [11], on receipt of an incoming SIP INVITE request, the participating MCVideo function:

1) shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [20] if included in the incoming SIP INVITE request;

2) should include the Session-Expires header field according to IETF RFC 4028 [23]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

3) shall include the option tag "timer" in the Supported header field;

4) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP INVITE request to the P-Asserted-Identity header field of the outgoing SIP INVITE request;

5) shall include the g.3gpp.mcvideo media feature tag into the Contact header field of the outgoing SIP INVITE request;

6) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcvideo" (coded as specified in 3GPP TS 24.229 [11]), into the P-Asserted-Service header field of the outgoing SIP INVITE request;

7) if the incoming SIP INVITE request contained a MIME resource-lists body with the MCVideo ID of the invited MCVideo user, shall copy the MIME resource-lists body, according to rules and procedures of IETF RFC 5366 [37];

8) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the contents of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request to the outgoing SIP INVITE request; and

9) if the incoming SIP INVITE request contained an application/vnd.3gpp.location-info+xml MIME body, shall copy the contents of the application/vnd.3gpp.location-info+xml MIME body of the incoming SIP INVITE request to the outgoing SIP INVITE request."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.1.1]

"This clause is referenced from other clauses.

The SDP offer is generated based on the received SDP offer. The SDP offer generated by the participating MCVideo function:

1) shall contain two SDP media-level sections for MCVideo video media as contained in the received SDP offer; and

2) shall contain an SDP media-level section for one media-transmission control entity, if present in the received SDP offer.

When composing the SDP offer according to 3GPP TS 24.229 [11], the participating MCVideo function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the participating MCVideo function, if required;

NOTE 1: Requirements can exist for the participating MCVideo function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number for the offered media transmission control entity, if any, in the received SDP offer with the IP address and port number of the participating MCVideo function; and

NOTE 2: If the participating MCVideo function and the controlling MCVideo function are in the same MCVideo server, and the participating MCVideo function does not have a dedicated IP address or a dedicated port number for media transmission control or media stream, the replacement of the IP address or the port number is omitted.

3) shall contain an "a=key-mgmt" attribute field with a "mikey" attribute value, if present in the received SDP offer."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.5]

"Upon receiving a SIP BYE request from the MCVideo client, the participating MCVideo function:

1) shall interact with the media plane as specified in 3GPP TS 24.581 [5];

2) shall generate a SIP BYE request as specified in 3GPP TS 24.229 [11];

3) shall set the Request-URI to the MCVideo session identity as included in the received SIP BYE request;

4) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP BYE request to the P-Asserted-Identity header field of the outgoing SIP BYE request;

5) if the received SIP BYE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body into the outgoing SIP BYE request; and

6) shall send the SIP BYE request toward the controlling MCVideo function, according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response to the SIP BYE request the terminating MCVideo function shall forward a SIP 200 (OK) response to the MCVideo client and shall interact with the media plane as specified in 3GPP TS 24.581 [5] for releasing media plane resources associated with the SIP session with the controlling MCVideo function."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.2.1]

"When composing the SDP answer according to 3GPP TS 24.229 [11], the participating MCVideo function:

1) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCVideo function, for the accepted media stream in the received SDP offer, if required; and

NOTE 1: Requirements can exist for the participating MCVideo function to be always included in the path of the offered media stream, for example: for the support of features such as MBMS, lawful interception and recording. Other examples can exist.

2) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCVideo function, for the accepted media-transmission control entity, if present in the received SDP offer.

NOTE 2: If the participating MCVideo function and the controlling MCVideo function are in the same MCVideo server, and the participating MCVideo function does not have a dedicated IP address or a dedicated port number for media transmission control or media stream, the replacement of the IP address or the port number is omitted."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.1]

"Once an on-demand MCVideo session is established or a pre-established session is in use when the participating MCVideo function receives transmission control messages from the transmission participant in the MCVideo client or from the transmission control server in the controlling MCVideo function, the behaviour of the participating MCVideo function is described in the following clauses."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.2]

"Upon receiving a transmission control message the participating MCVideo function:

1. *shall immediately forward the transmission control message to the transmission control server if the message is received from the transmission participant;*

2. *if an MBMS subchannel is not used for a transmission in the session the transmission control message is associated with, shall immediately forward the transmission control message to the transmission participant if the message is received from the transmission control server; and*

3. *if an MBMS subchannel is used for a transmission in the session the transmission control message is associated with:*

   a. *if*

      i. *the transmission control message is not a Transmission Idle message or a Media Transmission Notify message;*

      ii. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

      iii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report;*

      shall immediately forward the transmission control message to the transmission participant; and

   b. *if*

      i. *the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report; and*

      ii *if the transmission control message is the Transmission Idle message or the Media Transmission Notify message,*

   shall perform actions as specified in clause 10.2.

NOTE: When the Transmit Idle or Media Transmission Notify messages are discarded the messages are sent to the MCVideo clients over the MBMS subchannel allocated for the transmission as specified in clause 10.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.3]

"Upon receiving RTP media packets the participating MCVideo function:

1. *shall immediately forward the RTP media packet to the controlling MCVideo function if the RTP packet is from an MCVideo client; and*

2. *if an MBMS subchannel is not used for a transmission in the session the RTP media packets are associated with, shall immediately forward the RTP media packets to the MCVideo client if the RTP packet is from the controlling MCVideo function or the non-controlling MCVideo function.*

3. *if an MBMS subchannel is used for a transmission in the session the RTP media packets are associated with and if RTP media packets are received from the controlling MCVideo function or the non-controlling MCVideo function:*

   a. *if*

      i. *the MCVideo client has not reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3; or*

      ii. *the MCVideo client has reported "not-listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report,*

      shall immediately forward the RTP media packets to the MCVideo client; and

   b. *if the MCVideo client has reported "listening" status as specified in 3GPP TS 24.281 [2] clause 14.2.3 in the latest received MBMS bearer listening status report, shall perform actions as specified in clause 10.2."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.4.4]

"When the participating function receives an indication from the application and signalling plane that session release is initiated, the participating MCVideo function:

1. *shall stop sending transmission control messages towards the transmission participant and the transmission control server; and*

2. *shall stop sending RTP media packets towards the MCVideo client and towards the controlling MCVideo function.*

When the participating MCVideo function receives an indication from the application and signalling plane that the session is released, the participating MCVideo function:

1. *in case of a pre-established session, shall perform the actions in clause 9.3.2; and*

2. *in case of an on-demand session, shall release the media resources associated with the session."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

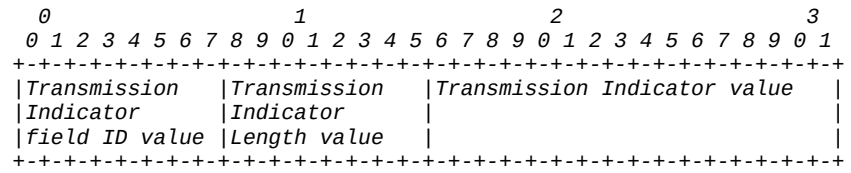
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

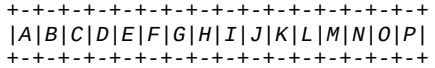


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. if the active Transmission request queue is empty the transmission control server:

a. shall send Transmission Idle message to all transmission participants. The Transmission Idle message:

i. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

ii. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

b. shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;

c. shall start timer T1 (Inactivity);

d. shall set the general state to the 'G: Transmit Idle' state; and

e. shall initialise counter Cx (Simultaneous transmission video) to 0.

2. if the active Transmission request queue is not empty the transmission control server:

a. shall select a queued Transmission request from the top of the active Transmission request queue;

b. shall remove that queued Transmission request from the active Transmission request queue; and

c. shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:

a. shall include the granted priority in the Transmission priority field;

b. shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;

c. if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

d. shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;

2. shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;

3. shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:

a. shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;

b. shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;

c. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;

d. if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';

e. if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';

f. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

g. shall initiate the 'general reception control operation' state machine;and

4. shall enter the 'G: Transmit Taken' state."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. if counter Cx (Simultaneous transmission video) has not reached its upper limit:

a. shall perform the actions specified in the clause 6.3.4.4.2;

2. *if counter Cx (Simultaneous transmission video) has reached its upper limit:*

   a. *select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;*

   b. *shall stop timer T4 (Transmission Grant), if running;*

   c. *shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);*

   d. *shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;*

   e. *shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and*

   f. *shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:*

      i. *shall include the queue position and transmission priority in the Queue Info field; and*

      ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:*

   a. *shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and*

   b. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

2. *shall set the general state to 'G: pending Transmission Revoke'."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. *if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:*

   a. *shall initialize a general state machine as specified in clause 6.3.4.2.2; and*

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

   b. *shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;*

2. if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and

a. if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;

b. if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and

ii. shall initialize a general state machine as specified in clause 6.3.4.2.2; and

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

c. if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.1.2.8.3 Test description

5.1.2.8.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the controlling MCVideo function.
- SS-UE1 (MCVideo Client)

IUT:

- IUT (MCVideo Server), providing the originating participating MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of some sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server and the HTTP Client.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server) and SS-UE1 (MCVideo Client).
- The SS-UE1 (MCVideo client) is registered and authorized to use the MCVideo system.

- The SS-UE1 (MCVideo client) is affiliated to the chat group and is authorized to initiate group calls.

5.1.2.8.3.2 Test procedure sequence

Table 5.1.2.8.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO session establishment/modification without provisional responses other than 100 Trying' as described in Annex A, Table A.4.2.1.3-1 to establish an imminent peril chat group call with implicit transmission control request according to option b.ii of note in Annex A, Table A.4.2.1.3-1, initiated by SS-UE1 (MCVideo Client?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO Transmission End Request - Transmission End Response' as described in Annex A, Table A.4.2.5.3-1 to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Transmission Granted' as described in Annex A, Table A.4.2.4.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	3	P
4	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo CO call release' as described in Annex A, Table A.4.2.3.3-1, to release the call?	-	-	4	P

5.1.2.8.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

Table 5.1.2.8.3.3-1: SIP INVITE from SS-UE1 (MCVideo Client)
(Step 1, Table 5.1.2.8.3.2-1; Step 1, Annex A, Table A.4.2.1.3-1)

Derivation Path: Annex A, Table A.4.2.1.4-1, with conditions CHAT-GROUP-CALL and IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.2.8.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.1.2.8.3.2-1; Step 2, Annex A, Table A.4.2.1.3-1)

Derivation Path: Annex A, Table A.4.2.1.4-2, with conditions CHAT-GROUP-CALL and IMMPERIL-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.2.8.3.3-3: SIP 200 (OK) from SS (MCVideo Server)
(Step 1, Table 5.1.2.8.3.2-1; Step 5, Annex A, Table A.4.2.1.3-1)

Derivation Path: Annex A, Table A.4.2.1.4-3, with condition CHAT-GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.2.8.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.1.2.8.3.2-1; Step 6, Annex A, Table A.4.2.1.3-1)

Derivation Path: Annex A, Table A.4.2.1.4-4, with condition CHAT-GROUP-CALL.
NOTE: Conditions are also applied to all body parts.

Table 5.1.2.8.3.3-5: Transmission Idle
(Steps 2, Table 5.1.2.8.3.2-1; Steps 9 and 10, Annex A, Table A.4.2.5.3-1)

Derivation Path: Annex A, Table A.4.2.5.4-3, with condition IMMPERIL-CALL.

Table 5.1.2.8.3.3-6: Transmission Request
(Step 3, Table 5.1.2.8.3.2-1; Steps 1 and 2, Annex A, Table A.4.2.4.3-1)

Derivation Path: Annex A, Table A.4.2.4.4-1, with condition IMMPERIL-CALL.

Table 5.1.2.8.3.3-7: Transmission Granted
(Steps 1 and 3, Table 5.1.2.8.3.2-1; Steps 5 and 6, Annex A, Table A.4.2.4.3-1;
Steps 9a1 and 9a2, Annex A, Table A.4.2.1.3-1)

Derivation Path: Annex A, Table A.4.2.4.4-2, with condition IMMPERIL-CALL.

5.2 Private Calls

5.2.1 MCVideo Server - MCVideo Server / Private Call / Automatic Commencement Mode / With Transmission Control / Upgrade to Emergency Call / Cancellation of Emergency on User Request / Controlling server

5.2.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function initiates a
private call with automatic commencement mode and implicit transmission control }
    then { IUT (MCVideo Server) accepts the call from initiator and establishes the call with the SS
(MCVideo Server) providing the terminating participating function and sends transmission granted to
SS-UE1 (MCVideo Client) and sends media transmission notification to SS-UE2 (MCVideo Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established a private call with automatic commencement and with
transmission control }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests to end the transmission control }
    then { IUT (MCVideo Server) sends transmission control idle to SS-UE1 (MCVideo Client) and sends
transmission control idle to SS-UE2 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established a private call with automatic commencement and with
transmission control }
ensure that {
  when { the SS-UE1 (MCVideo Client) requests the transmission control }
    then { IUT (MCVideo Server) sends transmission granted to SS-UE1 (MCVideo Client) and sends
media transmission notification to SS-UE2 (MCVideo Client) }
            }
```

(4)

```
with { IUT (MCVideo Server) having established a private call with automatic commencement and with
transmission control }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function upgrades the call
to emergency private call with implicit transmission control }
    then { IUT (MCVideo Server) accepts the upgrade from the initiator and updates the call for all
participants and sends transmission granted to SS-UE1 (MCVideo Client) and sends media transmission
notification to SS-UE2 (MCVideo Client) }
            }
```

(5)

```
with { IUT (MCVideo Server) having upgraded to an emergency private call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function requests to cancel
the ongoing emergency state }
    then { IUT (MCVideo Server) accepts the emergency state cancellation and updates the call for
all participants }
            }
```

(6)

```
with { IUT (MCVideo Server) having established a private call with automatic commencement and with
transmission control }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function ends the private
call }
    then { IUT (MCVideo Server) responds by sending a SIP 200 (OK) message to SS (MCVideo Server)
ending the call and sends a SIP BYE message to SS (MCVideo Server) providing the terminating
participating function }
            }
```

5.2.1.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 10.2.2.4.1, 10.2.2.4.2, 10.2.2.4.3, 10.2.2.4.4, 10.2.2.4.5, 10.2.2.4.6, 10.2.4.3.1, 6.3.3.2.4, 10.2.2.3.1.1, 10.2.2.3.2 and 6.2.3.1.1, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.3.1, 6.3.2.2, 9.2.1, 6.3.3, 9.2.3.11, 6.3.4.3.2, 6.3.4.4.2, 6.3.4.4.12, 6.3.4.5.2, 6.3.5.2.2 and 6.3.7.2.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.2.4.1]

"This clause describes the procedures for inviting an MCVideo user to an MCVideo session. The procedure is initiated by the controlling MCVideo function as the result of an action in clause 10.2.2.4.2

The controlling MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

NOTE 1: As a result of calling clause 6.3.3.1.2, the <mcvideo-calling-user-id> containing the calling user's MCVideo ID is copied into the outgoing SIP INVITE.

2) if the received SIP INVITE request contains an authorised request for an MCVideo emergency private call as determined by clause 6.3.3.1.13.2:

a) shall set the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true";

b) if the received SIP INVITE request contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, perform the procedures specified in clause 6.3.3.1.12; and

c) if the received SIP INVITE request did not contain an alert indication or contains an alert indication set to a value of "true" and is not an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall set the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "false";

3) shall copy the MCVideo ID of the MCVideo user listed in the MIME resources body of the incoming SIP INVITE request, into the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the outgoing SIP INVITE request;

4) shall set the Request-URI to the public service identity of the terminating participating MCVideo function associated to the MCVideo user to be invited;

NOTE 2: How the controlling MCVideo function finds the address of the terminating MCVideo participating function is out of the scope of the current release.

NOTE 3: If the terminating MCVideo user is part of a partner MCVideo system, then the public service identity can identify an entry point in the partner network that is able to identify the terminating participating MCVideo function.

5) shall copy the public user identity of the calling MCVideo user from the P-Asserted-Identity header field of the incoming SIP INVITE request into the P-Asserted-Identity header field of the SIP INVITE request;

6) shall include a Resource-Priority header field populated with the values for an MCVideo emergency private call as specified in clause 6.3.3.1.19, if either of the following conditions is met:

a) if the received SIP INVITE request contains an authorised request for an MCVideo emergency private call as determined in step 2 above; or

b) the originating MCVideo user is in an in-progress emergency private call state with the targeted MCVideo user;

7) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1;

8) shall send the SIP INVITE request towards the core network according to 3GPP TS 24.229 [11]; and

9) shall start a private call timer with a value set to the configured max private call duration for the user.

Upon receiving SIP 200 (OK) response for the SIP INVITE request the controlling MCVideo function:

1) shall cache the contact received in the Contact header field; and

2) shall interact with the media plane as specified in 3GPP TS 24.581 [5].

Upon expiry of the private call timer, the controlling MCVideo function shall follow the procedure for releasing private call session as specified in clause 10.2.5.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.2.4.2]

"In the procedures in this clause:

1) <emergency-ind> refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body;

2) <alert-ind> refers to the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and

3) <session-type> refers to the <session-type> element of an application/vnd.3gpp.mcvideo-info+xml MIME body.

Upon receipt of:

- a "SIP INVITE request for controlling MCVideo function of a private call"; or

the controlling MCVideo function:

1) if the <session-type> in the SIP INVITE request is set to "private":

a) shall check whether the public service identity contained in the Request-URI is allocated for private call and perform the actions specified in clause 6.3.7.1 if it is not allocated and skip the rest of the steps; and

b) shall perform actions to verify the MCVideo ID of the inviting MCVideo user in the <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the SIP INVITE request, and authorise the request according to local policy, and if it is not authorised the controlling MCVideo function shall return a SIP 403 (Forbidden) response with the warning text as specified in "Warning header field" and skip the rest of the steps;

2) if the incoming SIP INVITE request does not contain an application/resource-lists MIME body shall reject the SIP INVITE request with a SIP 403 (Forbidden) response including warning text set to "145 unable to determine called party" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

3) if the <session-type> is set to "private" and the application/resource-lists MIME body contains more than one <entry> element, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response including warning text set to "145 unable to determine called party" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

4) shall validate that the received SDP offer includes at least one media stream for which the media parameters and at least one codec or media format is acceptable by the controlling MCVideo function and if not, reject the request with a SIP 488 (Not Acceptable Here) response and skip the rest of the steps;

5) if received SIP INVITE request includes an <emergency-ind>, shall validate the request as described in clause 6.3.3.1.17;

6) if the received SIP INVITE request contains an unauthorised request for an MCVideo emergency private call as determined by clause 6.3.3.1.13.2:

a) shall reject the SIP INVITE request with a SIP 403 (Forbidden) response as specified in clause 6.3.3.1.14; and

b) shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [4] and skip the rest of the steps;

7) if a Resource-Priority header field is included in the received SIP INVITE request and if the Resource-Priority header field is set to the value indicated for emergency calls, shall reject the SIP INVITE request with a SIP 403 (Forbidden) response and skip the remaining steps if neither one of the following conditions are true:

a) the SIP INVITE request does not contain an authorised request for an MCVideo emergency call as determined in step 4 above; or

b) the originating MCVideo user is not in an in-progress emergency private call state with the targeted MCVideo user;

8) if:

a) the received SIP INVITE request contains an emergency indication set to a value of "true";

b) the originating MCVideo user is not in an in-progress emergency private call state with the targeted MCVideo user; and

c) if the <session-type> in the SIP INVITE request is set to "private";

then:

a) shall cache the information that the MCVideo user has initiated an MCVideo emergency private call to the targeted user; and

b) shall cache the information that the MCVideo user is in an in-progress emergency private call state with the targeted MCVideo user;

9) shall perform actions as described in clause 6.3.3.2.2;

10) shall allocate an MCVideo session identity for the MCVideo session; and

11) shall invite the MCVideo user(s) listed in the MIME resource-lists body of received SIP INVITE request as specified in clause 10.2.2.4.1.

Upon receiving a SIP 180 (Ringing) response and if the SIP 180 (Ringing) response or the SIP final response has not yet been sent to the inviting MCVideo client, the controlling MCVideo function:

1) if the SIP 180 (Ringing) response is associated with a SIP INVITE that contained a <session-type> set to "private", shall generate a SIP 180 (Ringing) response to the SIP INVITE request and send the SIP 180 (Ringing) response towards the inviting MCVideo client according to 3GPP TS 24.229 [11]; and

Upon receiving a SIP 200 (OK) response for the SIP INVITE request, the SIP dialog was established as a result of receiving a SIP INVITE request with a <session-type> element set to the value of "private" and the SIP final response has not yet been sent to the inviting MCVideo client, the controlling MCVideo function:

1) shall generate a SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;

2) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.2;

3) if the received SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 1: This is the case when the MCVideo user's request for an MCVideo emergency private call was granted but the request for the MCVideo emergency alert was denied.

4) shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

NOTE 2: Resulting media plane processing is completed before the next step is performed.

5) shall send a SIP 200 (OK) response towards the inviting MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request, the SIP dialog was established as a result of receiving a SIP INVITE request with a <session-type> element set to the value of "first-to-answer" and the SIP final response has not yet been sent to the inviting MCVideo client the controlling MCVideo function:

1) shall generate a SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;

2) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

3) the received SIP INVITE request contains an emergency indication set to a value of "true":

a) shall cache the information that the MCVideo user has initiated an MCVideo emergency private call to the targeted user; and

b) shall cache the information that the MCVideo user is in an in-progress emergency private call state with the targeted MCVideo user;

4) if the received SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 3: This is the case when the MCVideo user's request for an MCVideo emergency private call was granted but the request for the MCVideo emergency alert was denied.

5) shall interact with the media plane as specified in 3GPP TS 24.380 [5];

NOTE 4: Resulting media plane processing is completed before the next step is performed.

6) shall send a SIP 200 (OK) response towards the inviting MCVideo client according to 3GPP TS 24.229 [4]; and

7) if not successful in cancelling or terminating SIP dialogs in step 6) above, may repeat the SIP CANCEL and SIP BYE requests.

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCVideo client, where the SIP 200 (OK) response was sent with a Warning header field as specified in clause 4.4 with the warning text containing the mcvideo-warn-code set to "149", the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.18.

The controlling MCVideo function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [11]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.2.4.3]

"In the procedures in this clause:

1) *emergency indication in an incoming SIP re-INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and*

2) *alert indication in an incoming SIP re-INVITE request refers to the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.*

Upon receiving a SIP re-INVITE request with an emergency indication set to a value of "true", the controlling MCVideo function:

1) *shall validate that the received SDP offer includes at least one media stream for which the media parameters and at least one codec or media format is acceptable by the controlling MCVideo function and if not, reject the request with a SIP 488 (Not Acceptable Here) response and skip the rest of the steps;*

2) *shall validate the request as described in clause 6.3.3.1.17;*

3) *if the SIP re-INVITE request contains an unauthorised request for an MCVideo emergency private call as determined by clause 6.3.3.1.13.2:*

   a) *shall reject the SIP INVITE request with a SIP 403 (Forbidden) response as specified in clause 6.3.3.1.14; and*

   b) *shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [4] and skip the rest of the steps;*

4) *if a Resource-Priority header field is included in the received SIP re-INVITE request and if the Resource-Priority header field is set to the value indicated for emergency calls, shall reject the SIP re-INVITE request with a SIP 403 (Forbidden) response and skip the remaining steps if neither of the following conditions are true:*

   a) *the SIP re-INVITE request does contains an authorised request for an MCVideo emergency call as determined in step 2 above; or*

   b) *the originating MCVideo user is in an in-progress emergency private call state with the targeted MCVideo user;*

5) *if the SIP re-INVITE request contains an emergency indication set to a value of "true" and the originating MCVideo user is not in an in-progress emergency private call state with the targeted MCVideo user:*

   a) *shall cache the information that the MCVideo user is in an in-progress emergency private call state with the targeted MCVideo user; and*

   b) *if the SIP re-INVITE request contains an alert indication set to "true" and this is an authorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall cache the information that the MCVideo user has sent an MCVideo emergency alert to the targeted user; and*

6) *shall send a SIP re-INVITE invite towards the MCVideo user listed in the MIME resource-lists body of received SIP re-INVITE request as specified in clause 11.1.1.4.5.*

Upon receiving a SIP 200 (OK) response for the SIP re-INVITE request and if the SIP response has not yet been sent to the inviting MCVideo client, the controlling MCVideo function:

1) *shall generate a SIP 200 (OK) response to the SIP re-INVITE request as specified in the clause 6.3.3.2.3 before continuing with the rest of the steps;*

2) *shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP re-INVITE request containing the current media parameters used by the existing session;*

3) *if the received SIP re-INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4.*

NOTE: When a SIP 200 (OK) response sent to the originator as a response to a SIP INVITE request that contained authorised request(s) for an MCVideo emergency private call and optionally an MCVideo emergency alert, the originator will consider a SIP 200 (OK) response populated in this manner as confirmation that its request(s) for an upgrade to an MCVideo emergency private call and optionally an MCVideo emergency alert were accepted by the controlling function.

4) *shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and*

5) *shall send the SIP 200 (OK) response towards the inviting MCVideo client according to 3GPP TS 24.229 [4].*

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCVideo client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.18:

The controlling MCVideo function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [4]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.2.4.4]

"In the procedures in this clause:

1) *emergency indication in an incoming SIP INVITE request refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and*

2) *alert indication in an incoming SIP INVITE request refers to the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body.*

Upon receiving a SIP re-INVITE request with an emergency indication set to a value of "false", the controlling MCVideo function:

1) *shall validate that the received SDP offer includes at least one media stream for which the media parameters and at least one codec or media format is acceptable by the controlling MCVideo function and if not, reject the request with a SIP 488 (Not Acceptable Here) response and skip the rest of the steps;*

2) *shall validate the request as described in clause 6.3.3.1.17;*

3) *if the SIP re-INVITE request contains an unauthorised request for an MCVideo emergency private call cancellation as determined by clause 6.3.3.1.13.4:*

   a) *shall reject the SIP re-INVITE request with a SIP 403 (Forbidden) response;*

   b) *shall include in the SIP 403 (Forbidden) response an application/vnd.3gpp.mcvideo-info+xml MIME body as specified in annex F.1 with an <emergency-ind> element set to a value of "true";*

   c) *if the SIP re-INVITE request contains an alert indication set to "false" and this is an unauthorised request for an MCVideo emergency alert cancellation as specified in clause 6.3.3.1.13.3, shall include in the SIP 403 (Forbidden) response an application/vnd.3gpp.mcvideo-info+xml MIME body with an <alert-ind> element set to "true; and*

   d) *shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [4] and skip the rest of the steps;*

4) shall reject the SIP re-INVITE request with a SIP 403 (Forbidden) response if a Resource-Priority header field is included in the received SIP re-INVITE request set to the value configured for emergency calls, and skip the remaining steps; and

5) if the SIP re-INVITE request contains an authorised request for an MCVideo emergency private call cancellation as determined by clause 6.3.3.1.13.4:

a) shall clear the cache of the MCVideo ID of the originator of the MCVideo emergency private call that is no longer in an in-progress emergency private call state with the targeted MCVideo user; and

b) if the SIP re-INVITE request contains an alert indication set to "false" and this is an authorised request for an MCVideo emergency alert cancellation meeting the conditions specified in clause 6.3.3.1.13.3:

i) if the received SIP re-INVITE request contains an <originated-by> element in the application/vnd.3gpp.mcvideo-info+xml MIME body, shall clear the cache of the MCVideo ID of MCVideo user identified by the <originated-by> element as having an outstandingMCVideo emergency alert; and

ii) if the received SIP re-INVITE request does not contain an <originated-by> element in the application/vnd.3gpp.mcvideo-info+xml MIME body, clear the cache of the MCVideo ID of the sender of the SIP re-INVITE request as having an outstanding MCVideo emergency alert;

6) shall send a SIP re-INVITE request towards the MCVideo user listed in the MIME resource-lists body of received SIP re-INVITE request as specified in clause 11.1.1.4.6.

Upon receiving a SIP 200 (OK) response for the SIP re-INVITE request and if the SIP response has not yet been sent to the inviting MCVideo client, the controlling MCVideo function:

1) shall generate a SIP 200 (OK) response to the SIP re-INVITE request as specified in the clause 6.3.3.2.3 before continuing with the rest of the steps;

2) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP re-INVITE request as specified in the clause 6.3.3.2.2;

3) if the received SIP re-INVITE request contains an alert indication set to a value of "false" and this is an unauthorised request for an MCVideo emergency alert cancellation as specified in clause 6.3.3.1.13.3, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4.

NOTE: When a SIP 200 (OK) response sent to the originator as a response to a SIP INVITE request that contained authorised request(s) for an MCVideo emergency private call cancellation and optionally an MCVideo emergency alert cancellation, the originator will consider a SIP 200 (OK) response populated in this manner as confirmation that its request(s) for cancellation of an MCVideo emergency private call and optionally an MCVideo emergency alert were accepted by the controlling function.

4) shall interact with the media plane as specified in 3GPP TS 24.380 [5]; and

5) shall send the SIP 200 (OK) response towards the inviting MCVideo client according to 3GPP TS 24.229 [4].

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCVideo client, and the SIP 200 (OK) response was sent with the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.18.

The controlling MCVideo function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [4]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.2.4.5]

"This clause describes the procedures for sending a re-INVITE request to an MCVideo user in an MCVideo private call for the purpose of upgrading the session to an emergency private call session. The procedure is initiated by the controlling MCVideo function as the result of an action in clause 11.1.1.4.3.

The controlling MCVideo function:

1) shall generate a SIP re-INVITE request as specified in clause 6.3.3.1.9;

2) if the received SIP re-INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body to the outgoing re-INVITE request;

3) if the received SIP re-INVITE request contains an authorised request for an MCVideo emergency private call as determined by clause 6.3.3.1.13.2:

a) shall set the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true";

b) if the received SIP INVITE request contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, perform the procedures specified in clause 6.3.3.1.12; and

c) if the received SIP INVITE request did not contain an alert indication or contains an alert indication set to a value of "true" and is not an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall set the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "false";

4) shall include a Resource-Priority header field populated with the values for an MCVideo emergency private call as specified in clause 6.3.3.1.19, if the received SIP re-INVITE request contains an authorised request for an MCVideo emergency private call as determined in step 2 above; and

5) shall send the SIP re-INVITE request towards the core network according to 3GPP TS 24.229 [4].

Upon receiving SIP 200 (OK) response for the SIP re-INVITE request the controlling MCVideo function:

1) shall cache the contact received in the Contact header field."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.2.4.6]

"This clause describes the procedures for sending a re-INVITE request to an MCVideo user in an MCVideo emergency private call for the purpose of downgrading the session to a normal priority private call session. The procedure is initiated by the controlling MCVideo function as the result of an action in clause 11.1.1.4.4.

The controlling MCVideo function:

1) shall generate a SIP re-INVITE request as specified in clause 6.3.3.1.9;

2) if the received SIP re-INVITE request contained an application/vnd.3gpp.mcvideo-info+xml MIME body, shall copy the application/vnd.3gpp.mcvideo-info+xml MIME body to the outgoing re-INVITE request.

3) if the received SIP re-INVITE request contains an authorised request for an MCVideo emergency private call cancellation as determined by clause 6.3.3.1.13.4:

a) shall set the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "false";

b) if the received SIP INVITE request contains an alert indication set to a value of "false" and this is an authorised request for an MCVideo emergency alert cancellation meeting the conditions specified in clause 6.3.3.1.13.3:

i) shall set the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "false"; and

ii) if the received SIP request contains an <originated-by> element in the application/vnd.3gpp.mcvideo-info+xml MIME body, copy the contents of the received <originated-by> element to an <originated-by> element in the application/vnd.3gpp.mcvideo-info+xml MIME body in the outgoing SIP re-INVITE request;

c) if the received SIP INVITE request contains an alert indication set to a value of "false" and is not an authorised request for an MCVideo emergency alert cancellation meeting the conditions specified in clause 6.3.3.1.13.3, shall set the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true";

4) shall include a Resource-Priority header field populated with the values for a normal MCVideo private call as specified in clause 6.3.3.1.19, if the received SIP re-INVITE request contains an authorised request for an MCVideo emergency private call cancellation as determined in step 3 above; and

5) shall send the SIP re-INVITE request towards the core network according to 3GPP TS 24.229 [4].

Upon receiving SIP 200 (OK) response for the SIP re-INVITE request the controlling MCVideo function:

1) shall cache the contact received in the Contact header field."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.4.3.1]

"Upon receiving a SIP BYE request the controlling MCVideo function shall follow the procedures as specified in clause 6.3.3.2.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.2.4]

"Upon receiving a SIP BYE request the controlling MCVideo function:

1) shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.581 [5] for releasing the media plane resource associated with the SIP session towards the MCVideo client;

NOTE: The non-controlling MCVideo function is also regarded as a MCVideo client in a temporary MCVideo group session.

2) shall generate a SIP 200 (OK) response and send the SIP response towards the MCVideo client according to 3GPP TS 24.229 [11];

3) shall check the MCVideo session release policy as specified in clause 6.3.8.1 and clause 6.3.8.2 whether the MCVideo session needs to be released for each participant of the MCVideo session;

4) if release of the MCVideo session is required:

a) shall perform the procedures as specified in the clause 6.3.3.1.4 with the clarification that if the received SIP BYE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body, copy the application/vnd.3gpp.mcvideo-info+xml MIME body into the outgoing SIP BYE request; and

5) if a release of the MCVideo session is not required, shall send a SIP NOTIFY request to all remaining MCVideo clients in the MCVideo session with a subscription to the conference event package as specified in clause 9.2.3.4.2.

Upon receiving a SIP 200 (OK) response to the SIP BYE request the controlling MCVideo function shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.581 [5] for releasing media plane resources associated with the SIP session with the MCVideo participant."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.2.3.1.1]

"Upon receipt of a "SIP INVITE request for originating participating MCVideo function" containing an application/vnd.3gpp.mcvideo-info+xml MIME body with the <session-type> element set to a value of "private", the participating MCVideo function:

1) may reject the SIP INVITE request depending on the value of the Resource-Priority header field if the Resource-Priority header field is included in the received SIP INVITE request according to rules and procedures specified in IETF RFC 4412 [33] and shall not continue with the rest of the steps;

2) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 500 (Server Internal Error) response. The participating MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15] and shall not continue with the rest of the steps;

NOTE 1: If the received SIP INVITE request contains an emergency indication set to a value of "true", the participating MCVideo function can choose to accept the request.

NOTE 2: If the received SIP INVITE request contains an emergency indication set to a value of "true", the participating MCVideo function can choose to allow an exception to the limit on the number of private calls and accept the request.

3) shall determine the MCVideo ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request and shall authorise the user;

NOTE 3: The MCVideo ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

4) if the participating MCVideo function cannot find a binding between the public user identity and an MCVideo ID or if the validity period of an existing binding has expired, then the participating MCVideo function shall reject the SIP INVITE request with a SIP 404 (Not Found) response with the warning text set to "141 user unknown to the participating function" in a Warning header field as specified in clause 4.4, and shall not continue with any of the remaining steps;

5) shall:

a) if the <session-type> is set to "private", determine that the call is a private call;

6) if the call is a:

a) private call, determine the public service identity of the controlling MCVideo function for the private call service associated with the originating user's MCVideo ID identity;

7) if the participating MCVideo function is unable to identify the controlling MCVideo function for the private call service, it shall reject the SIP INVITE request with a SIP 404 (Not Found) response with the warning text "142 unable to determine the controlling function" in a Warning header field as specified in clause 4.4, and shall not continue with any of the remaining steps;

8) if the incoming SIP INVITE request does not contain an application/resource-lists MIME body, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response including warning text set to "145 unable to determine called party" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

9) if the call is a private call and the incoming SIP INVITE request contains an application/resource-lists MIME body with more than one <entry> element, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response including warning text set to "145 unable to determine called party" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

10) if the <allow-private-call> element of the <ruleset> element is not present in the MCVideo user profile document on the participating MCVideo function or is present with the value "false" (see the MCVideo user profile document in 3GPP TS 24.484 [25]), indicating that the user identified by the MCVideo ID is not authorised to initiate private calls, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response, with warning text set to "107 user not authorised to make private calls" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

11) if the call is a private call and:

a) if the received SIP INVITE request includes an Answer-Mode header field as specified in IETF RFC 5373 [27] with the value "Auto" and the <allow-automatic-commencement> element of the <ruleset> element is not present in the MCVideo user profile document on the participating MCVideo function or is present with the value "false" (see the MCVideo user profile document in 3GPP TS 24.484 [25]) indicating that the user identified by the MCVideo ID is not authorised to initiate private call with automatic commencement, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response including warning text set to "125 user not authorised to make private call with automatic commencement" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

b) if the received SIP INVITE request includes an Answer-Mode header field as specified in IETF RFC 5373 [27] with the value "Manual" and the <allow-manual-commencement> element of the <ruleset> element is not present in the MCVideo user profile document on the participating MCVideo function or is present with the value "false" (see the MCVideo user profile document in 3GPP TS 24.484 [25]), indicating that the user identified by the MCVideo ID is not authorised to initiate private call with manual commencement, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response including warning text set to "126 user not authorised to make private call with manual commencement" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

c) if the <PrivateCall> element exists in the MCVideo user profile document with one more <entry> elements (see the MCVideo user profile document in 3GPP TS 24.484 [25]) and:

i) if the "uri" attribute of the <entry> element of the application/resource-lists MIME body does not match with one of the <entry> elements of the <PrivateCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]); and

ii) if configuration is not set in the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) that allows the MCVideo user to make a private call to users not contained within the <entry> elements of the <PrivateCall> element;

then:

i) shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response including warning text set to "144 user not authorised to call this particular user" in a Warning header field as specified in clause 4.4 and shall not continue with the rest of the steps;

12) shall validate the media parameters and if the MCVideo video media codec is not offered in the "SIP INVITE request for originating participating MCVideo function" shall reject the request with a SIP 488 (Not Acceptable Here) response. Otherwise, continue with the rest of the steps;

13) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3 with the following clarifications:

a) if the conditions in step 12) above were executed and the participating MCVideo function determined that the "uri" attribute of only one of the <entry> elements of the application/resource-lists MIME body matched with an <entry> element of the <PrivateCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) then the <session-type> in the application/vnd.3gpp.mcvideo-info+xml MIME body of the SIP INVITE request generated in clause 6.3.2.1.3 is set to "private"; and

b) if the conditions in step 12) above were executed, then only the <entry> element(s) of the application/resource-lists MIME body that have a "uri" attribute that matched with an <entry> elements of the <PrivateCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) are included in the application/resource-lists MIME body in the SIP INVITE request generated in clause 6.3.2.1.3;

14) shall set the Request-URI to the public service identity of the controlling MCVideo function hosting the private call service as determined by step 6);

15) shall set the <mcvideo-calling-user-id> element in an application/vnd.3gpp.mcvideo-info+xml MIME body of the SIP INVITE request to the MCVideo ID of the calling user;

16) if the call is a private call and:

a) if a Priv-Answer-Mode header field specified in IETF RFC 5373 [27] was received in the incoming SIP INVITE request with a value of "Manual", shall not include a Priv-Answer-Mode header field in the outgoing SIP INVITE request;

b) if the <allow-force-auto-answer> element of the <ruleset> element is not present in the MCVideo user profile document on the participating MCVideo function or is present with the value "false" (see the MCVideo user profile document in 3GPP TS 24.484 [25]), and the Priv-Answer-Mode header field specified in IETF RFC 5373 [27] was received in the incoming SIP INVITE request with a value of "Auto", shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response including warning text set to "143 not authorised to force auto answer" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

c) if the <allow-force-auto-answer> element of the <ruleset> element is present in the MCVideo user profile document with the value "true" (see the MCVideo user profile document in 3GPP TS 24.484 [25]) on the participating MCVideo function, and the Priv-Answer-Mode header field specified in IETF RFC 5373 [27] was received in the incoming SIP INVITE request with a value of "Auto", shall include the Priv-Answer-Mode header field set to a value of "Auto" in the outgoing SIP INVITE request;

d) if a Priv-Answer-Mode header field containing the value of "Auto" has not been included in the outgoing SIP INVITE request as specified in step 17) above and the incoming "SIP INVITE request for originating participating MCVideo function" contained an Answer-Mode header field as specified in IETF RFC 5373 [27], then shall populate the Answer-Mode header field of the outgoing SIP INVITE request with the contents of the Answer-Mode header field from the incoming "SIP INVITE request for originating participating MCVideo function";

17) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received "SIP INVITE request for originating participating MCVideo function", as specified in clause 6.3.2.1.1.1;

18) shall include a Resource-Priority header field according to rules and procedures of 3GPP TS 24.229 [11] set to the value indicated in the Resource-Priority header field if included in the SIP INVITE request from the MCVideo client; and

19) shall forward the SIP INVITE request, according to 3GPP TS 24.229 [11].

Upon receiving a SIP 180 (Ringing) response, the participating MCVideo function:

1) shall generate a SIP 180 (Ringing) response to the SIP INVITE request as specified in the clause 6.3.2.1.5.1;

2) shall include the P-Asserted-Identity header field as received in the incoming SIP 180 (Ringing) response;

3) shall include Warning header field(s) received in the incoming SIP 180 (Ringing) response; and

4) shall forward the SIP 180 (Ringing) response to the MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response, the participating MCVideo function:

1) shall generate a SIP 200 (OK) response as specified in the clause 6.3.2.1.5.2;

2) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

3) shall include Warning header field(s) received in the incoming SIP 200 (OK) response;

4) shall include the P-Asserted-Identity header field received in the incoming SIP 200 (OK) response into the outgoing SIP 200 (OK) response;

5) shall include an MCVideo session identity mapped to the MCVideo session identity provided in the Contact header field of the received SIP 200 (OK) response;

6) shall send the SIP 200 (OK) response to the MCVideo client according to 3GPP TS 24.229 [11];

7) shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

8) shall start the SIP session timer according to rules and procedures of IETF RFC 4028 [23].

The participating MCVideo function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [11]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.2.3.2]

"This clause covers both on demand session and pre-established session.

Upon receipt of a "SIP INVITE request for terminating participating MCVideo function", the participating MCVideo function:

1) *if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the "SIP INVITE request for terminating participating MCVideo function" with a SIP 500 (Server Internal Error) response. The participating MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15], and shall not continue with the rest of the steps;*

2) *shall check the presence of the isfocus media feature tag in the URI of the Contact header field and if it is not present then the participating MCVideo function shall reject the request with a SIP 403 (Forbidden) response with the warning text set to "104 isfocus not assigned" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;*

3) *if the <session-type> element of the application/vnd.3gpp.mcvideo-info+xml MIME body is set to "private" and the Answer-Mode Indication in the application/poc-settings+xml MIME body has not yet been received from the invited MCVideo client as defined in clause 7.3.3 or clause 7.3.4, shall reject the request with a SIP 480 (Temporarily Unavailable) response with the warning text set to "146 T-PF unable to determine the service settings for the called user" in a Warning header field as specified in clause 4.4 and shall not continue with the rest of the steps;*

4) *shall use the MCVideo ID present in the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCVideo ID and public user identity;*

5) *if the binding between the MCVideo ID and public user identity does not exist, then the participating MCVideo function shall reject the SIP INVITE request with a SIP 404 (Not Found) response. Otherwise, continue with the rest of the steps;*

6) *when the called user identified by the MCVideo ID is unable to participate in private calls as identified in the called user's MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) on the terminating participating MCVideo function, shall reject the "SIP INVITE request for terminating participating MCVideo function" with a SIP 403 (Forbidden) response including warning text set to "127 user not authorised to be called in private call" in a Warning header field as specified in clause 4.4;*

7) *shall perform the automatic commencement procedures specified in clause 6.3.2.2.5.1 and according to IETF RFC 5373 [27] if one of the following conditions are met:*

   a) *"SIP INVITE request for terminating participating MCVideo function" contains an Answer-Mode header field with the value "Auto";*

   b) *"SIP INVITE request for terminating participating MCVideo function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCVideo client as per clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; or*

   c) *"SIP INVITE request for terminating participating MCVideo function" contains a Priv-Answer-Mode header field with the value "Auto"; and*

8) *shall perform the manual commencement procedures specified in clause 6.3.2.2.6.1 and according to IETF RFC 5373 [27] if either of the following conditions are met:*

   a) *"SIP INVITE request for terminating participating MCVideo function" contains an Answer-Mode header field with the value "Manual";*

   b) *"SIP INVITE request for terminating participating MCVideo function" does not contain an Answer-Mode header field and Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCVideo client as per clause 7.3.3 or clause 7.3.4 is set to "manual-answer"; or*

   c) *"SIP INVITE request for terminating participating MCVideo function" contains a Priv-Answer-Mode header field with the value "Manual"."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.2.3.1.1]

"When performing the automatic commencement mode procedures, the MCVideo client:

1) *shall accept the SIP INVITE request and generate a SIP 200 (OK) response according to rules and procedures of 3GPP TS 24.229 [11];*

2) *shall include the option tag "timer" in a Require header field of the SIP 200 (OK) response;*

3) *shall include the g.3gpp.mcvideo media feature tag in the Contact header field of the SIP 200 (OK) response;*

4) *shall include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo" in the Contact header field of the SIP 200 (OK) response;*

5) *shall include the Session-Expires header field in the SIP 200 (OK) response and start the SIP session timer according to IETF RFC 4028 [23]. The "refresher" parameter in the Session-Expires header field shall be set to "uas";*

6) *shall, if the incoming SIP INVITE request contains a Replaces header field, include in the SDP answer in the SIP 200 (OK) response to the SDP offer the parameters used for the pre-established session identified by the contents of the Replaces header field;*

7) *shall, if the incoming SIP INVITE request does not contain a Replaces header field, include an SDP answer in the SIP 200 (OK) response to the SDP offer in the incoming SIP INVITE request according to 3GPP TS 24.229 [11] with the clarifications given in clause 6.2.2;*

NOTE: In the case of a new emergency call where the terminating client is using a pre-established session, the SIP INVITE request containing a Replaces header is used to replace the pre-established session.

8) *shall send the SIP 200 (OK) response towards the MCVideo server according to rules and procedures of 3GPP TS 24.229 [11];*

9) *shall, if the incoming SIP INVITE request contains a Replaces header field, release the pre-established session identified by the contents of the Replaces header field; and*

10) *shall interact with the media plane as specified in 3GPP TS 24.581 [5].*

When NAT traversal is supported by the MCVideo client and when the MCVideo client is behind a NAT, generation of SIP responses is done as specified in this clause and as specified in IETF RFC 5626 [35]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.1]

"The transmission control server arbitration logic in the transmission control server shall support the procedures in clauses 6.3.2 and 6.3.3 and shall behave according to the transmission control server state transition diagram for 'general transmission control operation' in clause 6.3.4.

The transmission control interface towards the MCVideo client in the transmission control server shall behave according to the transmission control server state transition diagram for 'basic transmission control operation towards the transmission participant' as specified in clause 6.3.5.

The reception control arbitration logic in the transmission control server shall support the procedures in clauses 6.3.2 and 6.3.3 and shall behave according to the reception control server state transition diagram for 'general reception control operation' in clause 6.3.6.

The transmission control interface towards the MCVideo client in the transmission control server shall behave according to the transmission control server state transition diagram for 'basic reception control operation towards the transmission participant' as specified in clause 6.3.7."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.2.2]

"When an MCVideo call is established a new instance of the transmission control server state machine for 'general transmission control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' is added.

When an MCVideo call is established a new instance of the transmission control server state machine for 'general reception control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic reception control operation towards the transmission participant' is added.

If the optional "mc_queueing" feature is supported and has been negotiated as specified in clause 14, the transmission control server could queue the implicit transmission control request for the media-transmission control entity.

The original initial SIP INVITE request or SIP REFER request to establish an MCVideo chat group call or to rejoin an ongoing MCVideo call is not handled as an implicit transmission control request message by the transmission control server unless explicitly stated in the SIP INVITE request or in the SIP REFER request.

The permission to send media to the inviting MCVideo client due to implicit transmission control request is applicable to both confirmed indication and unconfirmed indication.

When the first unconfirmed indication is received from the invited participating MCVideo function (see 3GPP TS 24.281 [2]) the transmission control server optionally can give an early indication to send RTP media packets, to the inviting MCVideo client.

If an early indication to send RTP media packets is given to the inviting MCVideo client, the transmission participant is granted the permission to send media and the MCVideo server buffers RTP media packets received from the MCVideo client at least until the first invited MCVideo client accepts the invitation or until the RTP media packet buffer exceeds it maximum limit to store RTP media packets.

If the MCVideo server does not support or does not allow media buffering then when an early indication to send RTP media packets is not given to the inviting MCVideo client, the transmission participant is granted the permission to send media when the first invited MCVideo client accepts the media.

Before the transmission control server sends the first transmission control message in the MCVideo call, the transmission control server has to assign itself a SSRC identifier to be included in media transmission control messages and quality feedback messages if the MCVideo server is supporting that option. A suitable algorithm to generate the SSRC identifier is described in IETF RFC 3550 [3].

The transmission participant and the transmission control server can negotiate the maximum priority level that the transmission participant is permitted to request. The transmission control server can pre-empt the current sender based on the negotiated maximum priority level that the transmission participant is permitted to request and the priority level included in the Transmission Media Request message.

NOTE: The maximum priority level that a transmission participant can use is negotiated as specified in clause 14.3.3 and is based on group configuration data retrieved by the controlling MCVideo function from the group management server as described in 3GPP TS 24.481 [12] and service configuration data retrieved by the controlling MCVideo function from the configuration management server as described in 3GPP TS 24.484 [13].

The transmission participant and the transmission control server can negotiate queueing of Transmission requests using the "mc_queueing" fmtp attribute as described in clause 14. If queueing is supported and negotiated, the transmission control server queues the transmission control request if a Transmission Media Request message is received when another transmission participant has the transmission and the priority of the current speaker is the same or higher."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.1]

"The transmission control messages are coded as described in clause 9.1.2 where the transmission control message is part of the application-dependent data.

For the transmission control protocol the ASCII name string is: MCVx (Mission Critical Video x) with x=0, 1 or 2 as specified in clause 9.1.2.

A list of transmission control messages can be found in clause 9.2.2.1.

The transmission control specific fields are specified in clause 9.2.3."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.3]

"When an MCVideo client leaves an MCVideo call and the MCVideo call remains ongoing with the other MCVideo clients, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to the MCVideo client leaving the MCVideo call and .the MCVideo server discards transmission control messages and RTP media packets received from the MCVideo client leaving the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call with this transmission participant has been released, the corresponding instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' and 'basic reception control operation towards the transmission participant' is released.*

When an MCVideo call is released, the transmission control server follows a two-step procedure.

Step 1 *The MCVideo server stops sending transmission control messages and RTP media packets to all transmission participants in the MCVideo call.*

Step 2 *When the application and signalling plane has determined that the MCVideo call has been released, the corresponding instance of the transmission control server state machine for 'general transmission control operation' and 'general reception control operation' are also terminated, along with any 'basic transmission control operation towards the transmission participant' and any 'basic reception control operation towards the transmission participant' state machines for the transmission participants of this call.*

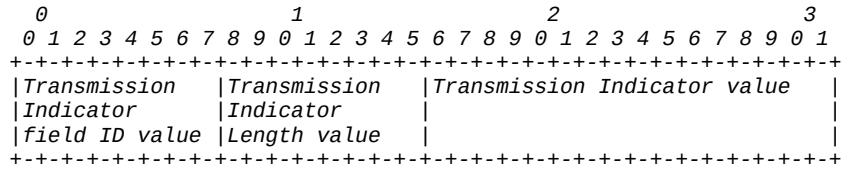
The transmission control server state machine for general transmission control operation initiates the MCVideo call release depending on the release policy specified in 3GPP TS 24.281 [2]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.3.11]

"The Transmission Indicator contains additional information about a received transmission control message.

Table 9.2.3.11-1 describes the coding of the Transmission Indicator field.

Table 9.2.3.11-1: Transmission Indicator field coding

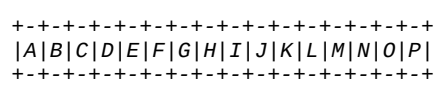


The <Transmission Indicator field ID> value is a binary value and is set according to table 9.2.3.1-1.

The <Transmission Indicator Length> value is a binary value and has the value '2'.

The <Transmission Indicator> value is a 16 bit bit-map named as shown in table 9.2.3.11-2:

Table 9.2.3.11-2: Transmission Indicator bit marking



When set to 1, the bit has the following meaning:

A = Normal call

B = Broadcast group call

C = System call

D = Emergency call

E = Imminent peril call

NOTE 1: The indicators C, D and E are only informative. There are no procedures specified for the C, D and E indicators in this release of the present document and the use of the indicators are implementation specific.

Bits F to P are reserved for future use and are set to 0.

There can be more than one bit set to 1 at the same time. The local policy in the transmission control server decides which combinations are possible and the priority of the indications."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.3.2]

"When entering this state from any state except the 'Start-stop' state and if no MCVideo client negotiated support of queueing Transmission requests as described in clause 14, the transmission control arbitration logic in the transmission control server:

1. *if the active Transmission request queue is empty the transmission control server:*
    a. *shall send Transmission Idle message to all transmission participants. The Transmission Idle message:*
        i. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*
        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;*
    b. *shall start timer T2 (Transmit Idle) and initialise counter C2 (Transmit Idle) to 1;*
    c. *shall start timer T1 (Inactivity);*
    d. *shall set the general state to the 'G: Transmit Idle' state; and*
    e. *shall initialise counter Cx (Simultaneous transmission video) to 0.*
2. *if the active Transmission request queue is not empty the transmission control server:*
    a. *shall select a queued Transmission request from the top of the active Transmission request queue;*
    b. *shall remove that queued Transmission request from the active Transmission request queue; and*
    c. *shall enter the 'G: Transmit Taken' state as specified in the clause 6.3.4.4.2 with respect to that transmission participant."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send a Transmission Grant message to the requesting transmission participant. The Transmission Grant message:*
    a. *shall include the granted priority in the Transmission priority field;*
    b. *shall increment counter Cx (Simultaneous transmission video) by 1 if counter Cx (Simultaneous transmission video) has not reached its upper limit;*
    c. *if a group call is a broadcast group call, system call, emergency call, an imminent peril call or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*
    d. *shall include the SSRC of the granted transmission participant to be used by the MCVideo user transmitting the media;*
2. *shall start timer T4 (Transmission Granted) if the Transmission request was queued and initialise the counter C4 (Transmission Grant) to 1;*

3. *shall send Media Transmission notify message to the reception control arbitration logic. The Media Transmission notification message:*

    a. *shall include the granted MCVideo user's MCVideo ID in the User ID field, if privacy is not requested;*

    b. *shall include the granted MCVideo user's SSRC in the SSRC of transmitter field;*

    c. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1;*

    d. *if the session is a broadcast group call, shall include the Permission to Request the Transmission field set to '0';*

    e. *if the session is not a broadcast group call, may include the Permission to Request the Transmission field set to '1';*

    f. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

    g. *shall initiate the 'general reception control operation' state machine;and*

4. *shall enter the 'G: Transmit Taken' state."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.4.12]

"Upon receiving an implicit Transmission request due to an upgrade to an emergency group call or due to an upgrade to imminent peril call, the transmission control arbitration logic in the transmission control server:

1. *if counter Cx (Simultaneous transmission video) has not reached its upper limit:*

    a. *shall perform the actions specified in the clause 6.3.4.4.2;*

2. *if counter Cx (Simultaneous transmission video) has reached its upper limit:*

    a. *select one of the transmission participants with permission to send media without the pre-emptive priority or low effective priority;*

    b. *shall stop timer T4 (Transmission Grant), if running;*

    c. *shall set the Reject Cause field in the Transmission Revoke message to #4 (Media Transmission pre-empted);*

    d. *shall enter the 'G: pending Transmission Revoke' state as specified in the clause 6.3.4.5.2;*

    e. *shall insert the transmission participant into the active Transmission request queue to the position in front of all queued requests, if not inserted yet or update the position of the transmission participant in the active Transmission request queue to the position in front of all other queued requests, if already inserted; and*

    f. *shall send a Transmission Queue Position Info message to the requesting transmission participant, if negotiated support of queueing Transmission requests as specified in clause 14. The Queue Position Request message:*

        i. *shall include the queue position and transmission priority in the Queue Info field; and*

        ii. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.4.5.2]

"When entering this state the transmission control arbitration logic in the transmission control server:

1. *shall send the Transmission Revoke or Transmission End Request message to the permitted transmission participant. The Transmission Revoke or Transmission End Request message:*

    a. *shall include the reason for sending the Transmission Revoke or Transmission End Request message in the <Reject Cause> value in the Reject Cause field; and*

    b. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

2. *shall set the general state to 'G: pending Transmission Revoke'."*

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.5.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. *if an MCVideo client initiates an MCVideo call with an implicit Transmission request, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:*

    a. *shall initialize a general state machine as specified in clause 6.3.4.2.2; and*

NOTE 2: In the clause 6.3.4.2.2 the 'general transmission control operation' state machine will continue with the initialization of the 'general transmission control operation' state machine.

    b. *shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;*

2. *if the associated MCVideo client rejoins an ongoing MCVideo call without an implicit Transmission request or initiates or joins a chat group call without an implicit Transmission request or attempts to initiate an already existing MCVideo call without an implicit Transmission request, and*

    a. *if an MCVideo call already exists but no MCVideo client has the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:*

        i. *should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:*

            A. *shall include a Message Sequence Number field with a Message Sequence Number value increased with 1; and*

            B. *if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and*

        ii. *shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2;*

    b. *if an MCVideo call is initiated, the transmission control interface towards the MCVideo client in the transmission control server:*

        i. *shall enter the state 'U: not permitted and Transmit Idle' as specified in the clause 6.3.5.5.2; and*

        ii. *shall initialize a general state machine as specified in clause 6.3.4.2.2; and*

NOTE 3: In the clause 6.3.4.2.2 the general state machine will continue with the initialization of the general state machine.

    c. *if other MCVideo clients have the permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:*

        i. *shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2;*

NOTE 3: For multiple MCVideo clients have the permission to send a media, the following Media Transmission notify messages will be handle as in 'U: not permitted and transmit taken' state as specified in the clause 6.3.5.4.2.

3. if the associated transmission participant attempts to initiate an already existing MCVideo call with an implicit Transmission request, and

a. if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.3.3; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;b. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has not reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall processes the implicit Transmission request as if a Transmission Media Request message was receive as specified in clause 6.3.4.4.12; and

ii. shall enter the state 'U: permitted' as specified in the clause 6.3.5.5.2;

c. if the MCVideo client negotiated support of queueing Transmission requests as specified in clause 14 and if other MCVideo clients have the permission to send media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall set the priority level to the negotiated maximum priority level that the MCVideo client is permitted to request, except for pre-emptive priority, when high priority is used;

NOTE 4: The maximum transmission priority the transmission participant is permitted to request is negotiated in the "mc_priority" fmtp attribute as specified in clause 14.

NOTE 5: The initial implicit Transmission request will not result in pre-emption when an MCVideo client is joining an ongoing MCVideo call. If the MCVideo client wants to pre-empt the current MCVideo client that are sending media, an explicit Transmission request with pre-emptive transmission priority is required.

ii. shall insert the MCVideo client into the active Transmission request queue to the position immediately following all queued Transmission requests with the same transmission priority;

iii. shall send a Transmission Queue Position Info message to the MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications;

iv. should send a Transmission Queue Position Info message with the updated status to the MCVideo clients in the active Transmission request queue which negotiated queueing of Transmission requests as specified in clause 14, which have requested the queue status, whose queue position has been changed since the previous Transmission Queue Position Info message and which is not the joining MCVideo client. The Transmission Queue Position Info message:

A. shall include the queue position and transmission priority in the Queue Info field; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

v. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

d. if the MCVideo client did not negotiate queueing of Transmission requests and if other MCVideo clients have the permission to send a media and if Cx (Simultaneous Transmission video) has reached it upper limit, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

4. if the MCVideo client is invited to the MCVideo call and

a. if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the 'U: not permitted and Transmit Taken' state as specified in the clause 6.3.5.4.2; and

b. if no other MCVideo client has the permission to send a media; the transmission control interface towards the MCVideo client in the transmission control server:

i. should send a Transmission Idle message to the MCVideo client. The Transmission Idle message:

A. shall include a Message Sequence Number field with a <Message Sequence Number> value increased with 1; and

B. if a group call is a broadcast group call, a system call, an emergency call, an imminent peril call, or a temporary group session, shall include the Transmission Indicator field with appropriate indications; and

ii. shall enter the 'U: not permitted and Transmit Idle' state as specified in the clause 6.3.5.3.2."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.7.2.2]

"When a SIP Session is established and if the session is a normal group call session:

NOTE 1: Temporary group call is not supported in this release. Normal group call contains pre-arranged group call, chat group call, broadcast group call.

1. if an MCVideo client initiate an MCVideo call, and the MCVideo call does not exist yet, the transmission control interface towards the MCVideo client in the transmission control server:

a. shall initialize a general state machine as specified in clause 6.3.6.2.2; and

NOTE 2: In the clause 6.3.6.2.2 the 'general reception control operation' state machine will continue with the initialization of the 'general reception control operation' state machine.

b. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

2. if the associated transmission participant attempts to initiate an already existing MCVideo call or MCVideo client rejoins an ongoing MCVideo call or if the MCVideo client is invited to the MCVideo call

a if no MCVideo client has the permission to send media, the transmission control interface towards the MCVideo client in the transmission control server

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

b if other MCVideo clients have permission to send a media, the transmission control interface towards the MCVideo client in the transmission control server:

i. shall enter the state 'U: not permitted to receive' as specified in the clause 6.3.7.3.2; and

ii. shall send media transmission notification as specified in the clause 6.3.7.3.3.

NOTE 3: The usecases, the MCVideo client initiating new call, joining ongoing call and re-joining ongoing call are applicable towards the originating MCVideo client and for other usecases it is applicable towards the terminating MCVideo client.

The maximum reception priority of the transmission participant is permitted to request is negotiated in the "mc_reception_priority" fmtp attribute as specified in clause 14."

5.2.1.3 Test description

5.2.1.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the participating MCVideo functions.
- SS-UE1 (MCVideo Client) and SS-UE2 (MCVideo Client), simulated by the SS (MCVideo Server).

IUT:

- IUT (MCVideo Server), providing the controlling MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of all sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Transmission Control Server.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server).

5.2.1.3.2 Test procedure sequence

Table 5.2.1.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E private call establishment, automatic commencement' as described in Annex A, Table A.4.1.3.3-1 to establish a private call with automatic commencement mode and implicit transmission control request according to option b.ii of note in Annex A, Table A.4.1.3.3-1, from SS-UE1 (MCVideo Client) to SS-UE2 (MCVideo Client)?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1 to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E session establishment/modification without provisional responses other than 100 Trying', as described in Annex A, Table A.4.1.1.3-1 to upgrade the call to emergency private call with implicit transmission request according to option b.ii of note in Annex A, Table A.4.1.1.3-1?	-	-	4	P
4	Check: Does the IUT (MCVideo Server) correctly perform procedure ' MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
5	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	3	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
6	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
7	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E session establishment/modification without provisional responses other than 100 Trying', as described in Annex A, Table A.4.1.1.3-1 to downgrade the call without implicit transmission request, with option a of note on Table A.4.1.1.3-1?	-	-	5	P
8	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1, to request permission to transmit for SS-UE1 (MCVideo Client)?	-	-	3	P
9	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission End Request - Idle and Transmission End Notify' as described in Annex A, Table A.3.8.3-1, to end SS-UE1 (MCVideo Client) transmission?	-	-	2	P
10	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.1.5.3-1, to release the call?	-	-	6	P

5.2.1.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

Table 5.2.1.3.3-1: SIP INVITE from SS (MCVideo Server)
(Step 1, Table 5.2.1.3.2-1; Step 1, Annex A, Table A.4.1.3.3-1)

Derivation Path: Annex A, Table A.4.1.3.4-1

Table 5.2.1.3.3-2: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.2.1.3.2-1; Step 2, Annex A, Table A.4.1.3.3-1)

Derivation Path: Annex A, Table A.4.1.3.4-2

Table 5.2.1.3.3-3: SIP 200 (OK) from SS (MCVideo Server)
(Steps 1, 3, and 7, Table 5.2.1.3.2-1; Step 8, Annex A, Table A.4.1.3.3-1;
Step 5, Annex A, Table A.4.1.1.3-1)

Derivation Path: Annex A, Table A.4.1.3.4-6

Table 5.2.1.3.3-4: SIP 200 (OK) from IUT (MCVideo Server)
(Steps 1, 3, and 7, Table 5.2.1.3.2-1; Step 9, Annex A, Table A.4.1.3.3-1;
Step 6, Annex A, Table A.4.1.1.3-1)

Derivation Path: Annex A, Table A.4.1.3.4-7

**Table 5.2.1.3.3-5: SIP re-INVITE from SS (MCVideo Server)
(Step 3, Table 5.2.1.3.2-1; Step 1, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.4.1.1.4-1, with conditions PRIVATE-CALL, re_INVITE and EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.2.1.3.3-6: SIP re-INVITE from IUT (MCVideo Server)
(Step 3, Table 5.2.1.3.2-1; Step 2, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.4.1.1.4-4, with condition PRIVATE-CALL, re_INVITE and EMERGENCY-CALL.
NOTE: Conditions are also applied to all body parts.

**Table 5.2.1.3.3-7: SIP re-INVITE from SS (MCVideo Server)
(Step 7, Table 5.2.1.3.2-1; Step 1, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.4.1.1.4-1, with conditions PRIVATE-CALL and re_INVITE.
NOTE: Conditions are also applied to all body parts.

**Table 5.2.1.3.3-8: SIP re-INVITE from IUT (MCVideo Server)
(Steps 7, Table 5.2.1.3.2-1; Step 2, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.4.1.1.4-4, with condition PRIVATE-CALL and re_INVITE.
NOTE: Conditions are also applied to all body parts.

**Table 5.2.1.3.3-9: Transmission Idle from IUT (MCVideo Server)
(Steps 4 and 6, Table 5.2.1.3.2-1; Steps 4 and 5, Annex A, Table A.3.8.3-1)**

Derivation Path: Annex A, Table A.3.8.4-6, with condition EMERGENCY-CALL.

**Table 5.2.1.3.3-10: Transmission Request from SS (MCVideo Server)
(Step 5, Table 5.2.1.3.2-1; Step 1, Annex A, Table A.3.7.3-1)**

Derivation Path: Annex A, Table A.3.7.4-1, with condition EMERGENCY-CALL.

**Table 5.2.1.3.3-11: Transmission Granted from IUT (MCVideo Server)
(Steps 3 and 5, Table 5.2.1.3.2-1; Step 3a1 or 3b1, Annex A, Table A.3.7.3-1;
Step 9a1, Annex A, Table A.4.1.1.3-1)**

Derivation Path: Annex A, Table A.3.7.4-2 or A.3.7.4-3 (NOTE 1), with condition EMERGENCY-CALL.
NOTE: Depends on server implementation.

**Table 5.2.1.3.3-12: Media Transmission Notification from IUT (MCVideo Server)
(Steps 3 and 5, Table 5.2.1.3.2-1;
Step 3a3 or 3b2, Annex A, Table A.3.7.3-1; Step 10a1, Annex A, Table 4.1.1.3-1)**

Derivation Path: Annex A, Table A.3.7.4-4 or A.3.7.4-5 (note), with condition EMERGENCY-CALL.
NOTE: Depends on server implementation.

5.2.2 MCVideo Server - MCVideo Server / Private Call / Manual Commencement Mode / Without Transmission Control / Controlling server

5.2.2.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function initiates a
private call with manual commencement mode and without transmission control }
    then { IUT (MCVideo Server) accepts the call from initiator and establishes the call with the SS
(MCVideo Server) providing the terminating participating function }
            }
```

(2)

```
with { IUT (MCVideo Server) having established a private call with manual commencement and without
transmission control }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function ends the private
call }
    then { IUT (MCVideo Server) responds by sending a SIP 200 (OK) message to SS (MCVideo Server)
ending the call and sends a SIP BYE message to SS (MCVideo Server) providing the terminating
participating function }
            }
```

5.2.2.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 10.2.3.1, 10.2.3.3.1, 10.2.3.3.2, 10.2.2.4.1, 10.2.2.4.2, 10.2.4.3.1, 6.3.3.2.4, 10.2.2.3.1.1, 10.2.2.3.2 and 6.2.3.2.1, and ETSI TS 124 581 [6] (3GPP TS 24.581), clauses 6.3.1, 6.3.2.2 and 9.2.1. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.3.1]

"When the MCVideo user wants to make an on-demand private call without transmission control or first-to-answer call without transmission control, the MCVideo client shall follow the procedures in clause 10.2.2.1.1 with the following exceptions:

1) *in step 8) of clause 10.2.2.1.1, the MCVideo client shall not offer a media-level section for a media-transmission control entity; and*

2) *step 9) of clause 10.2.2.1.1 shall be ignored.*

Upon receipt of an initial SIP INVITE request for the private call an SDP offer not including a media-level section for a media-transmission control entity, the MCVideo client shall consider it as the request for private call without transmission control and shall follow the procedures as specified in clause 10.2.2.1.2 for on-demand session."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.3.3.1]

"The controlling MCVideo function shall follow the procedures as specified in clause 10.2.2.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.3.3.2]

"Upon receiving of a "SIP INVITE request for controlling MCVideo function of a private call" or a "SIP INVITE request for controlling MCVideo function of a first-to-answer call", with SDP offer not including media-level section for media-transmission control entity, the controlling MCVideo function shall consider it as the request for the private call without transmission control or first-to-answer call without transmission control, and shall follow the procedures as specified in clause 10.2.2.4.2."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.2.4.1]

"This clause describes the procedures for inviting an MCVideo user to an MCVideo session. The procedure is initiated by the controlling MCVideo function as the result of an action in clause 10.2.2.4.2

The controlling MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

NOTE 1: As a result of calling clause 6.3.3.1.2, the <mcvideo-calling-user-id> containing the calling user's MCVideo ID is copied into the outgoing SIP INVITE.

2) if the received SIP INVITE request contains an authorised request for an MCVideo emergency private call as determined by clause 6.3.3.1.13.2:

a) shall set the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true";

b) if the received SIP INVITE request contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, perform the procedures specified in clause 6.3.3.1.12; and

c) if the received SIP INVITE request did not contain an alert indication or contains an alert indication set to a value of "true" and is not an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall set the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "false";

3) shall copy the MCVideo ID of the MCVideo user listed in the MIME resources body of the incoming SIP INVITE request, into the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the outgoing SIP INVITE request;

4) shall set the Request-URI to the public service identity of the terminating participating MCVideo function associated to the MCVideo user to be invited;

NOTE 2: How the controlling MCVideo function finds the address of the terminating MCVideo participating function is out of the scope of the current release.

NOTE 3: If the terminating MCVideo user is part of a partner MCVideo system, then the public service identity can identify an entry point in the partner network that is able to identify the terminating participating MCVideo function.

5) shall copy the public user identity of the calling MCVideo user from the P-Asserted-Identity header field of the incoming SIP INVITE request into the P-Asserted-Identity header field of the SIP INVITE request;

6) shall include a Resource-Priority header field populated with the values for an MCVideo emergency private call as specified in clause 6.3.3.1.19, if either of the following conditions is met:

a) if the received SIP INVITE request contains an authorised request for an MCVideo emergency private call as determined in step 2 above; or

b) the originating MCVideo user is in an in-progress emergency private call state with the targeted MCVideo user;

7) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1;

8) shall send the SIP INVITE request towards the core network according to 3GPP TS 24.229 [11]; and

9) shall start a private call timer with a value set to the configured max private call duration for the user.

Upon receiving SIP 200 (OK) response for the SIP INVITE request the controlling MCVideo function:

1) shall cache the contact received in the Contact header field; and

2) shall interact with the media plane as specified in 3GPP TS 24.581 [5].

Upon expiry of the private call timer, the controlling MCVideo function shall follow the procedure for releasing private call session as specified in clause 10.2.5.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.2.4.2]

"In the procedures in this clause:

1) *<emergency-ind> refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body;*

2) *<alert-ind> refers to the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and*

3) *<session-type> refers to the <session-type> element of an application/vnd.3gpp.mcvideo-info+xml MIME body.*

Upon receipt of:

- *a "SIP INVITE request for controlling MCVideo function of a private call"; or*

the controlling MCVideo function:

1) *if the <session-type> in the SIP INVITE request is set to "private":*

   a) *shall check whether the public service identity contained in the Request-URI is allocated for private call and perform the actions specified in clause 6.3.7.1 if it is not allocated and skip the rest of the steps; and*

   b) *shall perform actions to verify the MCVideo ID of the inviting MCVideo user in the <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the SIP INVITE request, and authorise the request according to local policy, and if it is not authorised the controlling MCVideo function shall return a SIP 403 (Forbidden) response with the warning text as specified in "Warning header field" and skip the rest of the steps;*

2) *if the incoming SIP INVITE request does not contain an application/resource-lists MIME body shall reject the SIP INVITE request with a SIP 403 (Forbidden) response including warning text set to "145 unable to determine called party" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;*

3) *if the <session-type> is set to "private" and the application/resource-lists MIME body contains more than one <entry> element, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response including warning text set to "145 unable to determine called party" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;*

4) *shall validate that the received SDP offer includes at least one media stream for which the media parameters and at least one codec or media format is acceptable by the controlling MCVideo function and if not, reject the request with a SIP 488 (Not Acceptable Here) response and skip the rest of the steps;*

5) *if received SIP INVITE request includes an <emergency-ind>, shall validate the request as described in clause 6.3.3.1.17;*

6) *if the received SIP INVITE request contains an unauthorised request for an MCVideo emergency private call as determined by clause 6.3.3.1.13.2:*

   a) *shall reject the SIP INVITE request with a SIP 403 (Forbidden) response as specified in clause 6.3.3.1.14; and*

   b) *shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [4] and skip the rest of the steps;*

7) *if a Resource-Priority header field is included in the received SIP INVITE request and if the Resource-Priority header field is set to the value indicated for emergency calls, shall reject the SIP INVITE request with a SIP 403 (Forbidden) response and skip the remaining steps if neither one of the following conditions are true:*

   a) *the SIP INVITE request does not contain an authorised request for an MCVideo emergency call as determined in step 4 above; or*

   b) *the originating MCVideo user is not in an in-progress emergency private call state with the targeted MCVideo user;*

8) if:

a) the received SIP INVITE request contains an emergency indication set to a value of "true";

b) the originating MCVideo user is not in an in-progress emergency private call state with the targeted MCVideo user; and

c) if the <session-type> in the SIP INVITE request is set to "private";

then:

a) shall cache the information that the MCVideo user has initiated an MCVideo emergency private call to the targeted user; and

b) shall cache the information that the MCVideo user is in an in-progress emergency private call state with the targeted MCVideo user;

9) shall perform actions as described in clause 6.3.3.2.2;

10) shall allocate an MCVideo session identity for the MCVideo session; and

11) shall invite the MCVideo user(s) listed in the MIME resource-lists body of received SIP INVITE request as specified in clause 10.2.2.4.1.

Upon receiving a SIP 180 (Ringing) response and if the SIP 180 (Ringing) response or the SIP final response has not yet been sent to the inviting MCVideo client, the controlling MCVideo function:

1) if the SIP 180 (Ringing) response is associated with a SIP INVITE that contained a <session-type> set to "private", shall generate a SIP 180 (Ringing) response to the SIP INVITE request and send the SIP 180 (Ringing) response towards the inviting MCVideo client according to 3GPP TS 24.229 [11]; and

Upon receiving a SIP 200 (OK) response for the SIP INVITE request, the SIP dialog was established as a result of receiving a SIP INVITE request with a <session-type> element set to the value of "private" and the SIP final response has not yet been sent to the inviting MCVideo client, the controlling MCVideo function:

1) shall generate a SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;

2) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.2;

3) if the received SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 1: This is the case when the MCVideo user's request for an MCVideo emergency private call was granted but the request for the MCVideo emergency alert was denied.

4) shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

NOTE 2: Resulting media plane processing is completed before the next step is performed.

5) shall send a SIP 200 (OK) response towards the inviting MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request, the SIP dialog was established as a result of receiving a SIP INVITE request with a <session-type> element set to the value of "first-to-answer" and the SIP final response has not yet been sent to the inviting MCVideo client the controlling MCVideo function:

1) shall generate a SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;

2) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

3) the received SIP INVITE request contains an emergency indication set to a value of "true":

a) shall cache the information that the MCVideo user has initiated an MCVideo emergency private call to the targeted user; and

b) shall cache the information that the MCVideo user is in an in-progress emergency private call state with the targeted MCVideo user;

4) if the received SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 3: This is the case when the MCVideo user's request for an MCVideo emergency private call was granted but the request for the MCVideo emergency alert was denied.

5) shall interact with the media plane as specified in 3GPP TS 24.380 [5];

NOTE 4: Resulting media plane processing is completed before the next step is performed.

6) shall send a SIP 200 (OK) response towards the inviting MCVideo client according to 3GPP TS 24.229 [4]; and

7) if not successful in cancelling or terminating SIP dialogs in step 6) above, may repeat the SIP CANCEL and SIP BYE requests.

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCVideo client, where the SIP 200 (OK) response was sent with a Warning header field as specified in clause 4.4 with the warning text containing the mcvideo-warn-code set to "149", the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.18.

The controlling MCVideo function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [11]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.4.3.1]

"Upon receiving a SIP BYE request the controlling MCVideo function shall follow the procedures as specified in clause 6.3.3.2.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.2.4]

"Upon receiving a SIP BYE request the controlling MCVideo function:

1) shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.581 [5] for releasing the media plane resource associated with the SIP session towards the MCVideo client;

NOTE: The non-controlling MCVideo function is also regarded as a MCVideo client in a temporary MCVideo group session.

2) shall generate a SIP 200 (OK) response and send the SIP response towards the MCVideo client according to 3GPP TS 24.229 [11];

3) shall check the MCVideo session release policy as specified in clause 6.3.8.1 and clause 6.3.8.2 whether the MCVideo session needs to be released for each participant of the MCVideo session;

4) if release of the MCVideo session is required:

a) shall perform the procedures as specified in the clause 6.3.3.1.4 with the clarification that if the received SIP BYE request contains an application/vnd.3gpp.mcvideo-info+xml MIME body, copy the application/vnd.3gpp.mcvideo-info+xml MIME body into the outgoing SIP BYE request; and

5) if a release of the MCVideo session is not required, shall send a SIP NOTIFY request to all remaining MCVideo clients in the MCVideo session with a subscription to the conference event package as specified in clause 9.2.3.4.2.

Upon receiving a SIP 200 (OK) response to the SIP BYE request the controlling MCVideo function shall interact with the media plane as specified in clause 6.3 in 3GPP TS 24.581 [5] for releasing media plane resources associated with the SIP session with the MCVideo participant."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.2.3.1.1]

"Upon receipt of a "SIP INVITE request for originating participating MCVideo function" containing an application/vnd.3gpp.mcvideo-info+xml MIME body with the <session-type> element set to a value of "private", the participating MCVideo function:

1) may reject the SIP INVITE request depending on the value of the Resource-Priority header field if the Resource-Priority header field is included in the received SIP INVITE request according to rules and procedures specified in IETF RFC 4412 [33] and shall not continue with the rest of the steps;

2) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 500 (Server Internal Error) response. The participating MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15] and shall not continue with the rest of the steps;

NOTE 1: If the received SIP INVITE request contains an emergency indication set to a value of "true", the participating MCVideo function can choose to accept the request.

NOTE 2: If the received SIP INVITE request contains an emergency indication set to a value of "true", the participating MCVideo function can choose to allow an exception to the limit on the number of private calls and accept the request.

3) shall determine the MCVideo ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP INVITE request and shall authorise the user;

NOTE 3: The MCVideo ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in clause 7.3.

4) if the participating MCVideo function cannot find a binding between the public user identity and an MCVideo ID or if the validity period of an existing binding has expired, then the participating MCVideo function shall reject the SIP INVITE request with a SIP 404 (Not Found) response with the warning text set to "141 user unknown to the participating function" in a Warning header field as specified in clause 4.4, and shall not continue with any of the remaining steps;

5) shall:

a) if the <session-type> is set to "private", determine that the call is a private call;

6) if the call is a:

a) private call, determine the public service identity of the controlling MCVideo function for the private call service associated with the originating user's MCVideo ID identity;

7) if the participating MCVideo function is unable to identify the controlling MCVideo function for the private call service, it shall reject the SIP INVITE request with a SIP 404 (Not Found) response with the warning text "142 unable to determine the controlling function" in a Warning header field as specified in clause 4.4, and shall not continue with any of the remaining steps;

8) if the incoming SIP INVITE request does not contain an application/resource-lists MIME body, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response including warning text set to "145 unable to determine called party" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

9) if the call is a private call and the incoming SIP INVITE request contains an application/resource-lists MIME body with more than one <entry> element, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response including warning text set to "145 unable to determine called party" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

10) if the <allow-private-call> element of the <ruleset> element is not present in the MCVideo user profile document on the participating MCVideo function or is present with the value "false" (see the MCVideo user profile document in 3GPP TS 24.484 [25]), indicating that the user identified by the MCVideo ID is not authorised to initiate private calls, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response, with warning text set to "107 user not authorised to make private calls" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

11) if the call is a private call and:

a) if the received SIP INVITE request includes an Answer-Mode header field as specified in IETF RFC 5373 [27] with the value "Auto" and the <allow-automatic-commencement> element of the <ruleset> element is not present in the MCVideo user profile document on the participating MCVideo function or is present with the value "false" (see the MCVideo user profile document in 3GPP TS 24.484 [25]) indicating that the user identified by the MCVideo ID is not authorised to initiate private call with automatic commencement, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response including warning text set to "125 user not authorised to make private call with automatic commencement" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

b) if the received SIP INVITE request includes an Answer-Mode header field as specified in IETF RFC 5373 [27] with the value "Manual" and the <allow-manual-commencement> element of the <ruleset> element is not present in the MCVideo user profile document on the participating MCVideo function or is present with the value "false" (see the MCVideo user profile document in 3GPP TS 24.484 [25]), indicating that the user identified by the MCVideo ID is not authorised to initiate private call with manual commencement, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response including warning text set to "126 user not authorised to make private call with manual commencement" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

c) if the <PrivateCall> element exists in the MCVideo user profile document with one more <entry> elements (see the MCVideo user profile document in 3GPP TS 24.484 [25]) and:

i) if the "uri" attribute of the <entry> element of the application/resource-lists MIME body does not match with one of the <entry> elements of the <PrivateCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]); and

ii) if configuration is not set in the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) that allows the MCVideo user to make a private call to users not contained within the <entry> elements of the <PrivateCall> element;

then:

i) shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response including warning text set to "144 user not authorised to call this particular user" in a Warning header field as specified in clause 4.4 and shall not continue with the rest of the steps;

12) shall validate the media parameters and if the MCVideo video media codec is not offered in the "SIP INVITE request for originating participating MCVideo function" shall reject the request with a SIP 488 (Not Acceptable Here) response. Otherwise, continue with the rest of the steps;

13) shall generate a SIP INVITE request as specified in clause 6.3.2.1.3 with the following clarifications:

a) if the conditions in step 12) above were executed and the participating MCVideo function determined that the "uri" attribute of only one of the <entry> elements of the application/resource-lists MIME body matched with an <entry> element of the <PrivateCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) then the <session-type> in the application/vnd.3gpp.mcvideo-info+xml MIME body of the SIP INVITE request generated in clause 6.3.2.1.3 is set to "private"; and

b) if the conditions in step 12) above were executed, then only the <entry> element(s) of the application/resource-lists MIME body that have a "uri" attribute that matched with an <entry> elements of the <PrivateCall> element of the MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) are included in the application/resource-lists MIME body in the SIP INVITE request generated in clause 6.3.2.1.3;

14) shall set the Request-URI to the public service identity of the controlling MCVideo function hosting the private call service as determined by step 6);

15) shall set the <mcvideo-calling-user-id> element in an application/vnd.3gpp.mcvideo-info+xml MIME body of the SIP INVITE request to the MCVideo ID of the calling user;

16) if the call is a private call and:

a) if a Priv-Answer-Mode header field specified in IETF RFC 5373 [27] was received in the incoming SIP INVITE request with a value of "Manual", shall not include a Priv-Answer-Mode header field in the outgoing SIP INVITE request;

b) if the <allow-force-auto-answer> element of the <ruleset> element is not present in the MCVideo user profile document on the participating MCVideo function or is present with the value "false" (see the MCVideo user profile document in 3GPP TS 24.484 [25]), and the Priv-Answer-Mode header field specified in IETF RFC 5373 [27] was received in the incoming SIP INVITE request with a value of "Auto", shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response including warning text set to "143 not authorised to force auto answer" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

c) if the <allow-force-auto-answer> element of the <ruleset> element is present in the MCVideo user profile document with the value "true" (see the MCVideo user profile document in 3GPP TS 24.484 [25]) on the participating MCVideo function, and the Priv-Answer-Mode header field specified in IETF RFC 5373 [27] was received in the incoming SIP INVITE request with a value of "Auto", shall include the Priv-Answer-Mode header field set to a value of "Auto" in the outgoing SIP INVITE request;

d) if a Priv-Answer-Mode header field containing the value of "Auto" has not been included in the outgoing SIP INVITE request as specified in step 17) above and the incoming "SIP INVITE request for originating participating MCVideo function" contained an Answer-Mode header field as specified in IETF RFC 5373 [27], then shall populate the Answer-Mode header field of the outgoing SIP INVITE request with the contents of the Answer-Mode header field from the incoming "SIP INVITE request for originating participating MCVideo function";

17) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received "SIP INVITE request for originating participating MCVideo function", as specified in clause 6.3.2.1.1.1;

18) shall include a Resource-Priority header field according to rules and procedures of 3GPP TS 24.229 [11] set to the value indicated in the Resource-Priority header field if included in the SIP INVITE request from the MCVideo client; and

19) shall forward the SIP INVITE request, according to 3GPP TS 24.229 [11].

Upon receiving a SIP 180 (Ringing) response, the participating MCVideo function:

1) shall generate a SIP 180 (Ringing) response to the SIP INVITE request as specified in the clause 6.3.2.1.5.1;

2) shall include the P-Asserted-Identity header field as received in the incoming SIP 180 (Ringing) response;

3) shall include Warning header field(s) received in the incoming SIP 180 (Ringing) response; and

4) shall forward the SIP 180 (Ringing) response to the MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response, the participating MCVideo function:

1) shall generate a SIP 200 (OK) response as specified in the clause 6.3.2.1.5.2;

2) shall include in the SIP 200 (OK) response an SDP answer as specified in the clause 6.3.2.1.2.1;

3) shall include Warning header field(s) received in the incoming SIP 200 (OK) response;

4) shall include the P-Asserted-Identity header field received in the incoming SIP 200 (OK) response into the outgoing SIP 200 (OK) response;

5) shall include an MCVideo session identity mapped to the MCVideo session identity provided in the Contact header field of the received SIP 200 (OK) response;

6) shall send the SIP 200 (OK) response to the MCVideo client according to 3GPP TS 24.229 [11];

7) shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

8) shall start the SIP session timer according to rules and procedures of IETF RFC 4028 [23].

The participating MCVideo function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [11]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.2.3.2]

"This clause covers both on demand session and pre-established session.

Upon receipt of a "SIP INVITE request for terminating participating MCVideo function", the participating MCVideo function:

1) if unable to process the request due to a lack of resources or a risk of congestion exists, may reject the "SIP INVITE request for terminating participating MCVideo function" with a SIP 500 (Server Internal Error) response. The participating MCVideo function may include a Retry-After header field to the SIP 500 (Server Internal Error) response as specified in IETF RFC 3261 [15], and shall not continue with the rest of the steps;

2) shall check the presence of the isfocus media feature tag in the URI of the Contact header field and if it is not present then the participating MCVideo function shall reject the request with a SIP 403 (Forbidden) response with the warning text set to "104 isfocus not assigned" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

3) if the <session-type> element of the application/vnd.3gpp.mcvideo-info+xml MIME body is set to "private" and the Answer-Mode Indication in the application/poc-settings+xml MIME body has not yet been received from the invited MCVideo client as defined in clause 7.3.3 or clause 7.3.4, shall reject the request with a SIP 480 (Temporarily Unavailable) response with the warning text set to "146 T-PF unable to determine the service settings for the called user" in a Warning header field as specified in clause 4.4 and shall not continue with the rest of the steps;

4) shall use the MCVideo ID present in the <mcvideo-request-uri> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCVideo ID and public user identity;

5) if the binding between the MCVideo ID and public user identity does not exist, then the participating MCVideo function shall reject the SIP INVITE request with a SIP 404 (Not Found) response. Otherwise, continue with the rest of the steps;

6) when the called user identified by the MCVideo ID is unable to participate in private calls as identified in the called user's MCVideo user profile document (see the MCVideo user profile document in 3GPP TS 24.484 [25]) on the terminating participating MCVideo function, shall reject the "SIP INVITE request for terminating participating MCVideo function" with a SIP 403 (Forbidden) response including warning text set to "127 user not authorised to be called in private call" in a Warning header field as specified in clause 4.4;

7) shall perform the automatic commencement procedures specified in clause 6.3.2.2.5.1 and according to IETF RFC 5373 [27] if one of the following conditions are met:

a) "SIP INVITE request for terminating participating MCVideo function" contains an Answer-Mode header field with the value "Auto";

b) "SIP INVITE request for terminating participating MCVideo function" does not contain an Answer-Mode header field and the Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCVideo client as per clause 7.3.3 or clause 7.3.4 is set to "auto-answer"; or

c) "SIP INVITE request for terminating participating MCVideo function" contains a Priv-Answer-Mode header field with the value "Auto"; and

8) shall perform the manual commencement procedures specified in clause 6.3.2.2.6.1 and according to IETF RFC 5373 [27] if either of the following conditions are met:

a) "SIP INVITE request for terminating participating MCVideo function" contains an Answer-Mode header field with the value "Manual";

b) "SIP INVITE request for terminating participating MCVideo function" does not contain an Answer-Mode header field and Answer-Mode Indication received in the application/poc-settings+xml MIME body received from the invited MCVideo client as per clause 7.3.3 or clause 7.3.4 is set to "manual-answer"; or

c) "SIP INVITE request for terminating participating MCVideo function" contains a Priv-Answer-Mode header field with the value "Manual"."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.2.3.2.1]

"When performing the manual commencement mode procedures:

1) if the MCVideo user declines the MCVideo session invitation the MCVideo client shall send a SIP 480 (Temporarily Unavailable) response towards the MCVideo server with the warning text set to: "110 user declined the call invitation" in a Warning header field as specified in clause 4.4, and not continue with the rest of the steps in this clause.

The MCVideo client:

1) shall accept the SIP INVITE request and generate a SIP 180 (Ringing) response according to rules and procedures of 3GPP TS 24.229 [11];

2) shall include the option tag "timer" in a Require header field of the SIP 180 (Ringing) response;

3) shall include the g.3gpp.mcvideo media feature tag in the Contact header field of the SIP 180 (Ringing) response;

4) shall include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcvideo" in the Contact header field of the SIP 180 (Ringing) response; and

5) shall send the SIP 180 (Ringing) response to the MCVideo server.

When sending the SIP 200 (OK) response to the incoming SIP INVITE request, the MCVideo client shall follow the procedures in clause 6.2.3.1.1.

When NAT traversal is supported by the MCVideo client and when the MCVideo client is behind a NAT, generation of SIP responses is done as specified in this clause and as specified in IETF RFC 5626 [35]."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.1]

"The transmission control server arbitration logic in the transmission control server shall support the procedures in clauses 6.3.2 and 6.3.3 and shall behave according to the transmission control server state transition diagram for 'general transmission control operation' in clause 6.3.4.

The transmission control interface towards the MCVideo client in the transmission control server shall behave according to the transmission control server state transition diagram for 'basic transmission control operation towards the transmission participant' as specified in clause 6.3.5.

The reception control arbitration logic in the transmission control server shall support the procedures in clauses 6.3.2 and 6.3.3 and shall behave according to the reception control server state transition diagram for 'general reception control operation' in clause 6.3.6.

The transmission control interface towards the MCVideo client in the transmission control server shall behave according to the transmission control server state transition diagram for 'basic reception control operation towards the transmission participant' as specified in clause 6.3.7."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 6.3.2.2]

"When an MCVideo call is established a new instance of the transmission control server state machine for 'general transmission control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic transmission control operation towards the transmission participant' is added.

When an MCVideo call is established a new instance of the transmission control server state machine for 'general reception control operation' is created.

For each MCVideo client added to the MCVideo call, a new instance of the transmission control server state machine for 'basic reception control operation towards the transmission participant' is added.

If the optional "mc_queueing" feature is supported and has been negotiated as specified in clause 14, the transmission control server could queue the implicit transmission control request for the media-transmission control entity.

The original initial SIP INVITE request or SIP REFER request to establish an MCVideo chat group call or to rejoin an ongoing MCVideo call is not handled as an implicit transmission control request message by the transmission control server unless explicitly stated in the SIP INVITE request or in the SIP REFER request.

The permission to send media to the inviting MCVideo client due to implicit transmission control request is applicable to both confirmed indication and unconfirmed indication.

When the first unconfirmed indication is received from the invited participating MCVideo function (see 3GPP TS 24.281 [2]) the transmission control server optionally can give an early indication to send RTP media packets, to the inviting MCVideo client.

If an early indication to send RTP media packets is given to the inviting MCVideo client, the transmission participant is granted the permission to send media and the MCVideo server buffers RTP media packets received from the MCVideo client at least until the first invited MCVideo client accepts the invitation or until the RTP media packet buffer exceeds it maximum limit to store RTP media packets.

If the MCVideo server does not support or does not allow media buffering then when an early indication to send RTP media packets is not given to the inviting MCVideo client, the transmission participant is granted the permission to send media when the first invited MCVideo client accepts the media.

Before the transmission control server sends the first transmission control message in the MCVideo call, the transmission control server has to assign itself a SSRC identifier to be included in media transmission control messages and quality feedback messages if the MCVideo server is supporting that option. A suitable algorithm to generate the SSRC identifier is described in IETF RFC 3550 [3].

The transmission participant and the transmission control server can negotiate the maximum priority level that the transmission participant is permitted to request. The transmission control server can pre-empt the current sender based on the negotiated maximum priority level that the transmission participant is permitted to request and the priority level included in the Transmission Media Request message.

NOTE: The maximum priority level that a transmission participant can use is negotiated as specified in clause 14.3.3 and is based on group configuration data retrieved by the controlling MCVideo function from the group management server as described in 3GPP TS 24.481 [12] and service configuration data retrieved by the controlling MCVideo function from the configuration management server as described in 3GPP TS 24.484 [13].

The transmission participant and the transmission control server can negotiate queueing of Transmission requests using the "mc_queueing" fmtp attribute as described in clause 14. If queueing is supported and negotiated, the transmission control server queues the transmission control request if a Transmission Media Request message is received when another transmission participant has the transmission and the priority of the current speaker is the same or higher."

[ETSI TS 124 581 [6] (3GPP TS 24.581), clause 9.2.1]

"The transmission control messages are coded as described in clause 9.1.2 where the transmission control message is part of the application-dependent data.

For the transmission control protocol the ASCII name string is: MCVx (Mission Critical Video x) with x=0, 1 or 2 as specified in clause 9.1.2.

A list of transmission control messages can be found in clause 9.2.2.1.

The transmission control specific fields are specified in clause 9.2.3."

5.2.2.3 Test description

5.2.2.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the participating MCVideo functions.
- SS-UE1 (MCVideo Client) and SS-UE2 (MCVideo Client), simulated by the SS (MCVideo Server).

IUT:

- IUT (MCVideo Server), providing the controlling MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of all sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Transmission Control Server.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server).

5.2.2.3.2 Test procedure sequence

Table 5.2.2.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E private call establishment, manual commencement' as described in Annex A, Table A.4.1.2.3-1 to establish a private call with manual commencement mode and without transmission control according to option c of note in Annex A, Table A.4.1.2.3-1, from SS-UE1 (MCVideo Client) to SS-UE2 (MCVideo Client)?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.1.5.3-1, to release the call?	-	-	2	P

5.2.2.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step.

5.3 Video Push

5.3.1 MCVideo Server - MCVideo Server / Video push Call / One-to-one video push call / Controlling server

5.3.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function initiates a one-
to-one video push call with transmission control }
    then { IUT (MCVideo Server) accepts the video push call from initiator and establishes the call
with the SS (MCVideo Server) providing the terminating participating function and sends transmission
granted to SS-UE1 (MCVideo Client) and sends media transmission notification to SS-UE2 (MCVideo
Client) }
            }
```

(2)

```
with { IUT (MCVideo Server) having established a one-to-one video push call }
ensure that {
  when { the SS-UE2 (MCVideo client) sends the receive media request }
    then { IUT (MCVideo Server) accepts the request by sending a receive media response message and
SS-UE1 (MCVideo Client) starts pushing a video to SS-UE2 (MCVideo Client)}
            }
```

(3)

```
with { IUT (MCVideo Server) having established a one-to-one video push call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function ends the video
push call }
    then { IUT (MCVideo Server) responds by sending a SIP 200 (OK) message to SS (MCVideo Server)
ending the call and sends a SIP BYE message to SS (MCVideo Server) providing the terminating
participating function }
            }
```

5.3.1.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 13.2.3.2.1, 13.2.3.2.2, 10.2.2.4.1 and 10.2.5.4. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 13.2.3.2.1]

"Upon receipt of a "SIP INVITE request for controlling MCVideo function" containing an application/vnd.3gpp.mcvideo-info+xml MIME body with the <session-type> element set to a value of "one-to-one video push", the controlling MCVideo function:

1) *shall check whether the public service identity contained in the Request-URI is allocated for one-to-one video push call and perform the actions specified in clause 6.3.7.1 if it is not allocated and skip the rest of the steps;*

2) *shall check whether the MCVideo ID of the MCVideo calling user in the <mcvideo-calling-user-id> is authorised to initiate the one-to-one video push call, and if it is not authorised the controlling MCVideo function shall return a SIP 403 (Forbidden) response with the warning text as specified in "Warning header field" and skip the rest of the steps;*

3) *shall check whether the MCVideo ID of the MCVideo called party user in the <mcvideo-called-party-id> is authorised to receive the one-to-one video push call, and if it is not authorised the controlling MCVideo function shall return a SIP 403 (Forbidden) response with the warning text as specified in "Warning header field" and skip the rest of the steps;*

4) *shall allocate an MCVideo session identity for the MCVideo session; and*

5) *shall invite the MCVideo user listed in the MIME resource-lists body of received SIP INVITE request as specified in clause 10.2.2.4.1."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 13.2.3.2.2]

"When the MCVideo session for one-to-one video push call needs to be released, the controlling MCVideo function shall perform the procedures of clause 10.2.5.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.2.4.1]

"This clause describes the procedures for inviting an MCVideo user to an MCVideo session. The procedure is initiated by the controlling MCVideo function as the result of an action in clause 10.2.2.4.2

The controlling MCVideo function:

1) shall generate a SIP INVITE request as specified in clause 6.3.3.1.2;

NOTE 1: As a result of calling clause 6.3.3.1.2, the <mcvideo-calling-user-id> containing the calling user's MCVideo ID is copied into the outgoing SIP INVITE.

2) if the received SIP INVITE request contains an authorised request for an MCVideo emergency private call as determined by clause 6.3.3.1.13.2:

a) shall set the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "true";

b) if the received SIP INVITE request contains an alert indication set to a value of "true" and this is an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, perform the procedures specified in clause 6.3.3.1.12; and

c) if the received SIP INVITE request did not contain an alert indication or contains an alert indication set to a value of "true" and is not an authorised request for an MCVideo emergency alert meeting the conditions specified in clause 6.3.3.1.13.1, shall set the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body to a value of "false";

3) shall copy the MCVideo ID of the MCVideo user listed in the MIME resources body of the incoming SIP INVITE request, into the <mcvideo-request-uri> element in the application/vnd.3gpp.mcvideo-info+xml MIME body of the outgoing SIP INVITE request;

4) shall set the Request-URI to the public service identity of the terminating participating MCVideo function associated to the MCVideo user to be invited;

NOTE 2: How the controlling MCVideo function finds the address of the terminating MCVideo participating function is out of the scope of the current release.

NOTE 3: If the terminating MCVideo user is part of a partner MCVideo system, then the public service identity can identify an entry point in the partner network that is able to identify the terminating participating MCVideo function.

5) shall copy the public user identity of the calling MCVideo user from the P-Asserted-Identity header field of the incoming SIP INVITE request into the P-Asserted-Identity header field of the SIP INVITE request;

6) shall include a Resource-Priority header field populated with the values for an MCVideo emergency private call as specified in clause 6.3.3.1.19, if either of the following conditions is met:

a) if the received SIP INVITE request contains an authorised request for an MCVideo emergency private call as determined in step 2 above; or

b) the originating MCVideo user is in an in-progress emergency private call state with the targeted MCVideo user;

7) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating network according to the procedures specified in clause 6.3.3.1.1;

8) shall send the SIP INVITE request towards the core network according to 3GPP TS 24.229 [11]; and

9) shall start a private call timer with a value set to the configured max private call duration for the user.

Upon receiving SIP 200 (OK) response for the SIP INVITE request the controlling MCVideo function:

1) shall cache the contact received in the Contact header field; and

2) shall interact with the media plane as specified in 3GPP TS 24.581 [5].

Upon expiry of the private call timer, the controlling MCVideo function shall follow the procedure for releasing private call session as specified in clause 10.2.5.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.5.4]

"When the MCVideo session for private call needs to be released as specified in clause 6.3.8.2, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.5."

5.3.1.3 Test description

5.3.1.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the participating MCVideo functions.
- SS-UE1 (MCVideo Client) and SS-UE2 (MCVideo Client), simulated by the SS (MCVideo Server).
- SS-UE1 (MCVideo Client) has a pre-recorded video that is available to be pushed to another client.

IUT:

- IUT (MCVideo Server), providing the controlling MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of all sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Transmission Control Server.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server).

5.3.1.3.2 Test procedure sequence

Table 5.3.1.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E private call establishment, automatic commencement' as described in Annex A, Table A.4.1.3.3-1 to push a video from SS-UE1 (MCVideo Client) to SS-UE2 (MCVideo Client), with implicit transmission control request according to option b.i of note in Annex A, Table A.4.1.3.3-1?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Media Reception Request and Response' as described in Annex A, Table A.3.9.3-1 to start receiving media from SS-UE1 (MCVideo Client), in SS-UE2 (MCVideo Client)?	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.1.5.3-1, to release the call?	-	-	3	P

5.3.1.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

Table 5.3.1.3.3-1: SIP INVITE from SS (MCVideo Server)
(Step 1, Table 5.3.1.3.2-1; Step 1, Annex A, Table A.4.1.3.3-1)

Derivation Path: Annex A, Table A.4.1.3.4-1.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.4.1.1.4-2			
MIME body part		**MCVideo-Info**		
MIME-part-body	MCVideo-Info as described in Table 5.3.1.3.3-2			

Table 5.3.1.3.3-2: MCVideo-Info in SIP INVITE (Table 5.3.1.3.3-1)

Derivation Path: Annex A, Table A.4.1.1.4-3, with condition PRIVATE-CALL.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcvideoinfo				
mcvideo-Params				
session-type	"one-to-one video push"			

**Table 5.3.1.3.3-3: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.3.1.3.2-1; Step 2, Annex A, Table A.4.1.3.3-1)**

Derivation Path: Annex A, Table A.4.1.3.4-2.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.4.1.1.4-5			
MIME body part		**MCVideo-Info**		
MIME-part-body	MCVideo-Info as described in Table 5.3.1.3.3-4			

Table 5.3.1.3.3-4: MCVideo-Info in SIP INVITE (Table 5.3.1.3.3-3)

Derivation Path: Annex A, Table A.4.1.1.4-6, with condition PRIVATE-CALL.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcvideoinfo				
mcvideo-Params				
session-type	"one-to-one video push"			

**Table 5.3.1.3.3-5: SIP 200 (OK) from SS (MCVideo Server)
(Step 1, Table 5.3.1.3.2-1; Step 8, Annex A, Table A.4.1.3.3-1)**

Derivation Path: Annex A, Table A.4.1.3.4-6

**Table 5.3.1.3.3-6: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.3.1.3.2-1; Step 9, Annex A, Table A.4.1.3.3-1)**

Derivation Path: Annex A, Table A.4.1.3.4-7

5.4 Video Pull

5.4.1 MCVideo Server - MCVideo Server / Video pull Call / One-to-one video pull call / Controlling server

5.4.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCVideo Server) connected to SS (MCVideo Server) }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function initiates a one-
to-one video pull call without implicit transmission control }
    then { IUT (MCVideo Server) accepts the video pull call from initiator and establishes the call
with the SS (MCVideo Server) providing the terminating participating function }
            }
```

(2)

```
with { IUT (MCVideo Server) having established a one-to-one video pull call }
ensure that {
  when { the SS-UE2 (MCVideo Client) sends the transmission request to the IUT (MCVideo Server) }
    then { IUT (MCVideo Server) accepts the request by sending a transmission granted message and
sends media transmission notification to SS-UE1 (MCVideo Client) }
            }
```

(3)

```
with { IUT (MCVideo Server) having established a one-to-one video pull call }
ensure that {
  when { the SS-UE1 (MCVideo client) sends the receive media request }
    then { IUT (MCVideo Server) accepts the request by sending a receive media response message and
SS-UE1 (MCVideo Client) starts pulling a video from SS-UE2 (MCVideo Client)}
            }
```

(4)

```
with { IUT (MCVideo Server) having established a one-to-one video pull call }
ensure that {
  when { the SS (MCVideo Server) providing the originating participating function ends the video
pull call }
    then { IUT (MCVideo Server) responds by sending a SIP 200 (OK) message to SS (MCVideo Server)
ending the call and sends a SIP BYE message to SS (MCVideo Server) providing the terminating
participating function }
            }
```

5.4.1.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 281 [5] (3GPP TS 24.281), clauses 12.2.3.2.2, 12.2.3.2.3 and 10.2.5.4. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 12.2.3.2.2]

"Upon receipt of a "SIP INVITE request for controlling MCVideo function" containing an application/vnd.3gpp.mcvideo-info+xml MIME body with the <session-type> element set to a value of "one-to-one video pull", the controlling MCVideo function determine that the call is a private call and shall perform the procedures of the clause 10.2.2.4.2, with following clarifications:

1) *shall check whether the MCVideo ID of the MCVideo calling user in the <mcvideo-calling-user-id> is authorised to initiate the one-to-one video pull call, and if it is not authorised the controlling MCVideo function shall return a SIP 403 (Forbidden) response with the warning text as specified in "Warning header field" and skip the rest of the steps; and*

2) *shall check whether the MCVideo ID of the MCVideo called party user in the <mcvideo-called-party-id> is authorised to receive the one-to-one video pull call, and if it is not authorised the controlling MCVideo function shall return a SIP 403 (Forbidden) response with the warning text as specified in "Warning header field" and skip the rest of the steps."*

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.2.4.2]

"In the procedures in this clause:

1) *<emergency-ind> refers to the <emergency-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body;*

2) *<alert-ind> refers to the <alert-ind> element of the application/vnd.3gpp.mcvideo-info+xml MIME body; and*

3) *<session-type> refers to the <session-type> element of an application/vnd.3gpp.mcvideo-info+xml MIME body.*

Upon receipt of:

- *a "SIP INVITE request for controlling MCVideo function of a private call"; or*

the controlling MCVideo function:

1) *if the <session-type> in the SIP INVITE request is set to "private":*

   a) *shall check whether the public service identity contained in the Request-URI is allocated for private call and perform the actions specified in clause 6.3.7.1 if it is not allocated and skip the rest of the steps; and*

b) shall perform actions to verify the MCVideo ID of the inviting MCVideo user in the <mcvideo-calling-user-id> element of the application/vnd.3gpp.mcvideo-info+xml MIME body of the SIP INVITE request, and authorise the request according to local policy, and if it is not authorised the controlling MCVideo function shall return a SIP 403 (Forbidden) response with the warning text as specified in "Warning header field" and skip the rest of the steps;

2) if the incoming SIP INVITE request does not contain an application/resource-lists MIME body shall reject the SIP INVITE request with a SIP 403 (Forbidden) response including warning text set to "145 unable to determine called party" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

3) if the <session-type> is set to "private" and the application/resource-lists MIME body contains more than one <entry> element, shall reject the "SIP INVITE request for originating participating MCVideo function" with a SIP 403 (Forbidden) response including warning text set to "145 unable to determine called party" in a Warning header field as specified in clause 4.4, and shall not continue with the rest of the steps;

4) shall validate that the received SDP offer includes at least one media stream for which the media parameters and at least one codec or media format is acceptable by the controlling MCVideo function and if not, reject the request with a SIP 488 (Not Acceptable Here) response and skip the rest of the steps;

5) if received SIP INVITE request includes an <emergency-ind>, shall validate the request as described in clause 6.3.3.1.17;

6) if the received SIP INVITE request contains an unauthorised request for an MCVideo emergency private call as determined by clause 6.3.3.1.13.2:

a) shall reject the SIP INVITE request with a SIP 403 (Forbidden) response as specified in clause 6.3.3.1.14; and

b) shall send the SIP 403 (Forbidden) response as specified in 3GPP TS 24.229 [4] and skip the rest of the steps;

7) if a Resource-Priority header field is included in the received SIP INVITE request and if the Resource-Priority header field is set to the value indicated for emergency calls, shall reject the SIP INVITE request with a SIP 403 (Forbidden) response and skip the remaining steps if neither one of the following conditions are true:

a) the SIP INVITE request does not contain an authorised request for an MCVideo emergency call as determined in step 4 above; or

b) the originating MCVideo user is not in an in-progress emergency private call state with the targeted MCVideo user;

8) if:

a) the received SIP INVITE request contains an emergency indication set to a value of "true";

b) the originating MCVideo user is not in an in-progress emergency private call state with the targeted MCVideo user; and

c) if the <session-type> in the SIP INVITE request is set to "private";

then:

a) shall cache the information that the MCVideo user has initiated an MCVideo emergency private call to the targeted user; and

b) shall cache the information that the MCVideo user is in an in-progress emergency private call state with the targeted MCVideo user;

9) shall perform actions as described in clause 6.3.3.2.2;

10) shall allocate an MCVideo session identity for the MCVideo session; and

11) shall invite the MCVideo user(s) listed in the MIME resource-lists body of received SIP INVITE request as specified in clause 10.2.2.4.1.

Upon receiving a SIP 180 (Ringing) response and if the SIP 180 (Ringing) response or the SIP final response has not yet been sent to the inviting MCVideo client, the controlling MCVideo function:

1) if the SIP 180 (Ringing) response is associated with a SIP INVITE that contained a <session-type> set to "private", shall generate a SIP 180 (Ringing) response to the SIP INVITE request and send the SIP 180 (Ringing) response towards the inviting MCVideo client according to 3GPP TS 24.229 [11]; and

Upon receiving a SIP 200 (OK) response for the SIP INVITE request, the SIP dialog was established as a result of receiving a SIP INVITE request with a <session-type> element set to the value of "private" and the SIP final response has not yet been sent to the inviting MCVideo client, the controlling MCVideo function:

1) shall generate a SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;

2) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.2;

3) if the received SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 1: This is the case when the MCVideo user's request for an MCVideo emergency private call was granted but the request for the MCVideo emergency alert was denied.

4) shall interact with the media plane as specified in 3GPP TS 24.581 [5]; and

NOTE 2: Resulting media plane processing is completed before the next step is performed.

5) shall send a SIP 200 (OK) response towards the inviting MCVideo client according to 3GPP TS 24.229 [11].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request, the SIP dialog was established as a result of receiving a SIP INVITE request with a <session-type> element set to the value of "first-to-answer" and the SIP final response has not yet been sent to the inviting MCVideo client the controlling MCVideo function:

1) shall generate a SIP 200 (OK) response to the SIP INVITE request as specified in the clause 6.3.3.2.3.2 before continuing with the rest of the steps;

2) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the clause 6.3.3.2.1;

3) the received SIP INVITE request contains an emergency indication set to a value of "true":

a) shall cache the information that the MCVideo user has initiated an MCVideo emergency private call to the targeted user; and

b) shall cache the information that the MCVideo user is in an in-progress emergency private call state with the targeted MCVideo user;

4) if the received SIP INVITE request contains an alert indication set to a value of "true" and this is an unauthorised request for an MCVideo emergency alert as specified in clause 6.3.3.1.13.1, shall include in the SIP 200 (OK) response the warning text set to "149 SIP INFO request pending" in a Warning header field as specified in clause 4.4;

NOTE 3: This is the case when the MCVideo user's request for an MCVideo emergency private call was granted but the request for the MCVideo emergency alert was denied.

5) shall interact with the media plane as specified in 3GPP TS 24.380 [5];

NOTE 4: Resulting media plane processing is completed before the next step is performed.

6) shall send a SIP 200 (OK) response towards the inviting MCVideo client according to 3GPP TS 24.229 [4]; and

7) if not successful in cancelling or terminating SIP dialogs in step 6) above, may repeat the SIP CANCEL and SIP BYE requests.

Upon receiving a SIP ACK to the SIP 200 (OK) response sent towards the inviting MCVideo client, where the SIP 200 (OK) response was sent with a Warning header field as specified in clause 4.4 with the warning text containing the mcvideo-warn-code set to "149", the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.18.

The controlling MCVideo function shall forward any other SIP response that does not contain SDP, including any MIME bodies contained therein, along the signalling path to the originating network according to 3GPP TS 24.229 [11]."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 12.2.3.2.3]

"When the MCVideo session for one-to-one video pull call needs to be released, the controlling MCVideo function shall perform the procedures of the clause 10.2.5.4."

[ETSI TS 124 281 [5] (3GPP TS 24.281), clause 10.2.5.4]

"When the MCVideo session for private call needs to be released as specified in clause 6.3.8.2, the controlling MCVideo function shall follow the procedures in clause 6.3.3.1.5."

5.4.1.3 Test description

5.4.1.3.1 Pre-test conditions

System Simulator:

- SS (MCVideo Server), providing the participating MCVideo functions.
- SS-UE1 (MCVideo Client) and SS-UE2 (MCVideo Client), simulated by the SS (MCVideo Server).
- SS-UE2 (MCVideo Client) has a pre-recorded video that is available to be pulled from another client.

IUT:

- IUT (MCVideo Server), providing the controlling MCVideo function.
- The IUT (MCVideo Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCVideo Server) also consists of all sub-systems of the MCVideo Server, including the Media Distribution Function, the MCVideo User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Transmission Control Server.

Preamble:

- The IUT (MCVideo Server) is connected to SS (MCVideo Server).

5.4.1.3.2 Test procedure sequence

Table 5.4.1.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E private call establishment, automatic commencement' as described in Annex A, Table A.4.1.3.3-1 to pull a video from SS-UE2 (MCVideo Client), without implicit transmission control request according to option a of note in Annex A, Table A.4.1.3.3-1?	-	-	1	P
2	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Transmission Request - Granted and Media Transmission Notification' as described in Annex A, Table A.3.7.3-1, to request permission to transmit for SS-UE2 (MCVideo Client)? (Note)	-	-	2	P
3	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo Media Reception Request and Response' as described in Annex A, Table A.3.9.3-1 to start receiving media from SS-UE2 (MCVideo Client), in SS-UE1 (MCVideo Client)?	-	-	3	P
4	Check: Does the IUT (MCVideo Server) correctly perform procedure 'MCVideo E2E call release' as described in Annex A, Table A.4.1.5.3-1, to release the call?	-	-	4	P

NOTE 1: It is assumed that the client considers the establishment of the video pull call as implicit transmission request, i.e. that it automatically enters the 'U: pending request to transmit' state (ETSI TS 124 581 [6] (3GPP TS 24.581) clause 6.2.4.4).

NOTE 2: It is not clear in ETSI TS 124 581 [6] (3GPP TS 24.581) what exactly is meant with "implicit transmit media request" in context of CT call establishment (as stated in ETSI TS 124 281 [5] (3GPP TS 24.281)) and how it shall be handled.

5.4.1.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) and Annex A, in the subclauses referenced in each step, with the following clarifications.

Table 5.4.1.3.3-1: SIP INVITE from SS (MCVideo Server)
(Step 1, Table 5.4.1.3.2-1; Step 1, Annex A, Table A.4.1.3.3-1)

Derivation Path: Annex A, Table A.4.1.3.4-1.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.4.1.1.4-2			
MIME body part		**MCVideo-Info**		
MIME-part-body	MCVideo-Info as described in Table 5.4.1.3.3-2			

Table 5.4.1.3.3-2: MCVideo-Info in SIP INVITE (Table 5.4.1.3.3-1)

Derivation Path: Annex A, Table A.4.1.1.4-3, with condition PRIVATE-CALL.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcvideoinfo				
mcvideo-Params				
session-type	"one-to-one video pull"			

Table 5.4.1.3.3-3: SIP INVITE from IUT (MCVideo Server)
(Step 1, Table 5.4.1.3.2-1; Step 2, Annex A, Table A.4.1.3.3-1)

Derivation Path: Annex A, Table A.4.1.3.4-2.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.4.1.1.4-5			
MIME body part		**MCVideo-Info**		
MIME-part-body	MCVideo-Info as described in Table 5.4.1.3.3-4			

Table 5.4.1.3.3-4: MCVideo-Info in SIP INVITE (Table 5.4.1.3.3-3)

Derivation Path: Annex A, Table A.4.1.1.4-6, with condition PRIVATE-CALL.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcvideoinfo				
mcvideo-Params				
session-type	"one-to-one video pull"			

Table 5.4.1.3.3-5: SIP 200 (OK) from SS (MCVideo Server)
(Step 1, Table 5.4.1.3.2-1; Step 8, Annex A, Table A.4.1.3.3-1)

Derivation Path: Annex A, Table A.4.1.3.4-6

Table 5.4.1.3.3-6: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table 5.4.1.3.2-1; Step 9, Annex A, Table A.4.1.3.3-1)

Derivation Path: Annex A, Table A.4.1.3.4-7

Table 5.4.1.3.3-7: Media Transmission Notification
(Step 2, Table 5.4.1.3.2-1; Step 3b2, Annex A, Table A.3.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.7-1

Table 5.4.1.3.3-8: Media Transmission Notification
(Step 2, Table 5.4.1.3.2-1; Step 3a3, Annex A, Table A.3.7.3-1)

Derivation Path: Table 5.4.1.3.3-7, with condition ACK

Annex A (normative): Common Test Environment

A.1 Introduction

This annex specifies additional generic test procedures to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), to be used by the test cases wherever applicable.

A.2 Generic test procedures for Server-Client MCS operation

A.2.1 General

The purpose of the procedures specified in the following clauses is to facilitate test description by providing procedure sequences which can be referred from the relevant TCs specified in the present document.

The procedures specified are required to ensure that any MC service can take place between a MC client and a MC server or specific MC relevant pre-conditions are met before a test case can be executed.

A.2.2 Void

A.2.3 MCX Server Originated notifications regarding temporary group creation or tear down

A.2.3.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.2.3.2 Definition of system information messages

N/A.

A.2.3.3 Procedure

Table A.2.3.3-1: MCX Server originated notifications regarding temporary group creation or tear down

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCX Server send a SIP NOTIFY to the SS-UE informing about change of group A's configuration document?	<--	SIP NOTIFY	-	P
2	The SS-UE (MCX Client) sends a SIP 200 (OK) message to the MCX Server.	-->	SIP 200 (OK)	-	-
3	The SS-UE (MCX Client) sends an HTTP GET Request message to the MCX Server that contains the access token and the XCAP-URI of the Group Configuration document.	-->	HTTP GET	-	-
4	Check: Does the MCX Server send a HTTP 200 (OK) message including the updated Group Document?	<--	HTTP 200 (OK)	-	P
5	Check: Does the MCX Server send a SIP NOTIFY message to the SS-UE (MCX Client) containing the group key transport payloads (GKTP) document including the group keys?	<-	SIP NOTIFY	-	P
6	The SS-UE (MCX Client) sends a SIP 200 (OK) message.	-->	SIP 200 (OK)	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.3.4 Specific message contents

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).
- Clarifications specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.3.22.4 shall also be followed.

Table A.2.3.4-1: SIP NOTIFY (Step 1, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-1

Table A.2.3.4-2: SIP 200 (OK) (Steps 2, 6, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-2

Table A.2.3.4-3: HTTP GET (Step 3, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-3

Table A.2.3.4-4: HTTP 200 (OK) (Step 4, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-4

Table A.2.3.4-5: SIP NOTIFY (Step 5, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-6

A.2.4 MCX Server Terminated Group Creation

A.2.4.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.4.2 Definition of system information messages

N/A.

A.2.4.3 Procedure

Table A.2.4.3-1: MCX Server Terminated Group Creation procedure

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a HTTP PUT to the Group Management Server for requesting the creation of the new group. (Note)	-->	HTTP PUT	-	-
2	Check: Does the MCX Server respond with a HTTP 201 (Created) response message?	<--	HTTP 201 (Created)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.4.4 Specific message contents

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.4.4-1: HTTP PUT (Step 1, Table A.2.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.4-1, condition GROUPCREATE

A.2.5 MCX Server Terminated Temporary Group Creation

A.2.5.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.5.2 Definition of system information messages

N/A.

A.2.5.3 Procedure

A.2.5.3-1: MCX Participating Terminated Temporary Group Creation procedure

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a HTTP POST to the MCX Server to request for creation of a temporary group. (Note)	-->	HTTP POST	-	-
2	Check: Does the MCX Server send a HTTP 200 (OK) containing the GMOP group-regroup-creation-response?	<--	HTTP 200 (OK)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.5.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.5.4-1: HTTP POST (Step 1, Table A.2.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.3-1, condition TEMPGROUP

Table A.2.5.4-2: HTTP 200 (OK) (Step 2, Table A.2.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, condition TEMPGROUP

A.2.6 MCX Server Terminated Temporary Group Tear Down

A.2.6.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.6.2 Definition of system information messages

N/A.

A.2.6.3 Procedure

Table A.2.6.3-1: MCX Server Terminated Temporary Group Tear Down procedure

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a HTTP DELETE to the MCX Server to request for tear down of a temporary group. (Note)	-->	HTTP DELETE	-	-
2	Check: Does the MCX Server send a HTTP 200 (OK) response message?	<--	HTTP 200 (OK)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.6.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.6.4-1: HTTP DELETE (Step 1, Table A.2.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.5-1, condition TEMPGROUP

A.2.7 MCX Subscription and Notification

A.2.7.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.7.2 Definition of system information messages

N/A.

A.2.7.3 Procedure

Table A.2.7.3-1: MCX Subscription and Notification

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP SUBSCRIBE message request.	-->	SIP SUBSCRIBE	-	-
2	Check: Does the MCX Server respond to the SIP SUBSCRIBE message with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	P
3	Check: Does the MCX Server send a SIP NOTIFY message?	<--	SIP NOTIFY	-	P
4	The SS-UE (MCX Client) responds with a SIP 200 (OK) message.	-->	SIP 200 (OK)		
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.7.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.7.4-1: SIP SUBSCRIBE (Step 1, Table A.2.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires				
Value	"4294967295"	to receive the current status and later notification	ETSI TS 124 379 [3] (3GPP TS 24.379) clauses 9.2.1.3 and 9A.2.1.3; ETSI TS 124 281 [5] (3GPP TS 24.281) clause 8.2.1.3; ETSI TS 124 282 [19] (3GPP TS 24.282) clauses 8.2.3 and 22.2.1.3	
Message-body				
MIME body part		**MCPTT Info**	ETSI TS 124 379 [3] (3GPP TS 24.379) clause 9.2.1.3 and 9A.2.1.3	MCPTT
MIME-part-body	MCPTT-Info as described in Table A.2.7.4-2			
MIME body part		**MCVIDEO Info**	ETSI TS 124 281 [5] (3GPP TS 24.281) clause 8.2.1.3	MCVIDEO
MIME-part-body	MCVIDEO-Info as described in Table A.2.7.4-3			
MIME body part		**MCDATA Info**	ETSI TS 124 282 [19] (3GPP TS 24.282) clauses 8.2.3 and 22.2.1.3	MCDATA
MIME-part-body	MCDATA-Info as described in Table A.2.7.4-4			

Table A.2.7.4-2: MCPTT-Info in SIP SUBSCRIBE (Table A.2.7.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Mcpttinfo				
mcptt-Params				
mcptt-request-uri	encrypted (note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [3] (3GPP TS 24.379) clauses 9.2.1.3 and 9A.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table A.2.7.4-3: MCVideo-Info in SIP SUBSCRIBE (Table A.2.7.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Mcvideoinfo				
mcvideo-Params				
mcvideo-request-uri	encrypted (note) <mcvideo-request-uri> with mcvideoURI set to px_MCVideo_ID_User_A		ETSI TS 124 281 [5] (3GPP TS 24.281) clause 8.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2A.				

Table A.2.7.4-4: MCData-Info in SIP SUBSCRIBE (Table A.2.7.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Mcdatainfo				
mcdata-Params				
mcdata-request-uri	encrypted (note) <mcdata-request-uri> with mcdataURI set to px_MCDATA_ID_User_A		ETSI TS 124 282 [19] (3GPP TS 24.282) clauses 8.2.3 and 22.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3A.				

Table A.2.7.4-5: SIP 200 (OK) (Step 2, Table A.2.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition SUBSCRIBE-RSP

A.2.8 MCX Participating Terminated SIP MESSAGE

A.2.8.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData

A.2.8.2 Definition of system information messages

N/A.

A.2.8.3 Procedure

Table A.2.8.3-1: MCX Participating Terminated SIP MESSAGE

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) send a SIP MESSAGE message (note)	-->	SIP MESSAGE	-	-
2	Check: Does the MCX Server responds with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.8.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.9 MCX Participating Originated SIP MESSAGE

A.2.9.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData

A.2.9.2 Definition of system information messages

N/A.

A.2.9.3 Procedure

Table A.2.9.3-1: MCX Participating Originated SIP MESSAGE

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCX Server send a SIP MESSAGE message?	<--	SIP MESSAGE	-	P
2	The SS-UE (MCX Client) responds with a SIP 200 (OK) message (note)	→	SIP 200 (OK)	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.9.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.10 MCX E2E SIP MESSAGE

A.2.10.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo

A.2.10.2 Definition of system information messages

N/A.

A.2.10.3 Procedure

Table A.2.10.3-1: MCX E2E SIP MESSAGE

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP MESSAGE message	-->	SIP MESSAGE	-	-
2	Check: Does the MCX Server send a SIP MESSAGE to SS-UE2?	<--	SIP MESSAGE	-	P
3	The SS-UE2 (MCX Client) responds with a SIP 200 (OK) message	-->	SIP 200 (OK)	-	-
4	Check: Does the MCX Server respond with a SIP 200 (OK) message to SS-UE1?	<--	SIP 200 (OK)	-	P

A.2.10.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.11 MCX E2E SIP MESSAGE Request - Accept

A.2.11.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo

A.2.11.2 Definition of system information messages

N/A.

A.2.11.3 Procedure

Table A.2.11.3-1: MCX E2E SIP MESSAGE Request - Accept Participating Originated

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP MESSAGE message.	-->	SIP MESSAGE	-	-
2	Check: Does the MCX Server respond with a SIP 200 (OK) message	<--	SIP 200 (OK)	-	P
3	Check: Does the MCX server send a SIP MESSAGE accepting the request to SS-UE1?	<--	SIP MESSAGE	-	P
4	The SS-UE1 (MCX Client) responds with a SIP 200 (OK) message.	-->	SIP 200 (OK)	-	-
-	EXCEPTION: Depending on the MCX Server implementation, the following steps may be performed in parallel to steps 3-4.	-	-	-	-
5	Check: Does the MCX Server send a SIP MESSAGE message to SS-UE2?	<--	SIP MESSAGE	-	P
6	The SS-UE2 (MCX Client) responds with a SIP 200 (OK) message.	-->	SIP 200 (OK)	-	-

A.2.11.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.12 MCX Presence Event Status Change

A.2.12.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData

A.2.12.2 Definition of system information messages

N/A.

A.2.12.3 Procedure

Table A.2.12.3-1: MCX Presence Event Status Change

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP PUBLISH message.	-->	SIP PUBLISH	-	P
2	Check: Does the MCX Server respond to the SIP PUBLISH message with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	-
3	Check: Does the MCX Server send a SIP NOTIFY message informing about the status change progress?	<--	SIP NOTIFY	-	P
4	The SS-UE (MCX Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
5	Check: Does the MCX Server send a SIP NOTIFY informing about the final status?	<--	SIP NOTIFY	-	P
6	The SS-UE responds with a SIP 200 (OK).	-->	SIP 200 (OK)	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.12.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.12.4-1: SIP PUBLISH (Step 1, Table A.2.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.11-1, condition PRESENCE-EVENT

Table A.2.12.4-2: MCPTT-Info in SIP PUBLISH (Table A.2.12.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Mcpttinfo				
mcptt-Params				
mcptt-request-uri	encrypted (note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [3] (3GPP TS 24.379) clauses 9.2.1.3 and 9A.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table A.2.12.4-3: MCVideo-Info in SIP PUBLISH (Table A.2.12.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Mcvideoinfo				
mcvideo-Params				
mcvideo-request-uri	encrypted (note) <mcvideo-request-uri> with mcvideoURI set to px_MCVideo_ID_User_A		ETSI TS 124 281 [5] (3GPP TS 24.281) clause 8.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2A.				

Table A.2.12.4-4: MCData-Info in SIP PUBLISH (Table A.2.12.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Mcdatainfo				
mcdata-request-uri	encrypted (note) <mcdata-request-uri> with mcdataURI set to px_MCDATA_ID_User_A		ETSI TS 124 282 [19] (3GPP TS 24.282) clauses 8.2.3 and 22.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3A.				

Table A.2.12.4-5: SIP 200 (OK) (Step 2, Table A.2.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, condition PUBLISH-RSP

Table A.2.12.4-6: SIP NOTIFY (Steps 3 and 5, Table A.2.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, condition PRESENCE-EVENT

Table A.2.12.4-7: SIP 200 (OK) (Steps 4 and 6, Table A.2.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1

A.2.13 MCX Participating Terminated pre-established session establishment

A.2.13.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo

A.2.13.2 Definition of system information messages

N/A.

A.2.13.3 Procedure

Table A.2.13.3-1: MCX Participating Terminated pre-established session establishment

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP INVITE message in order to create a pre-established session	-->	SIP INVITE	-	-
-	EXCEPTION: Step 2a1 describes behaviour that depends on the MCX Server implementation; the "lower case letter" identifies a step sequence that take place if the MCX Server responds to a SIP INVITE with a SIP 100 (Trying)	-	-	-	-
2a1	Check: Does the MCX Server respond with a SIP 100 (Trying) message?	<--	SIP 100 Trying	-	P
3	Check: Does the MCX Server respond with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	P
4	The SS-UE (MCX Client) responds with a SIP ACK message.	-->	SIP ACK	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.13.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.13.4-1: SIP INVITE (Step 1, Table A.2.13.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.3.4-1

Table A.2.13.4-2: SIP 200 (OK) (Step 3, Table A.2.13.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.3.4-2

A.2.14 MCX Participating Terminated session establishment/modification without provisional responses other than 100 Trying

A.2.14.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo.

A.2.14.2 Definition of system information messages

N/A.

A.2.14.3 Procedure

Table A.2.14.3-1: MCX Participating Terminated session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP INVITE requesting the establishment/modification of a call (note)	-->	SIP INVITE	-	-
-	EXCEPTION: Step 2a1 describes behaviour that depends on the MCX Server implementation; the "lower case letter" identifies a step sequence that take place if the MCX Server responds to a SIP INVITE with a SIP 100 (Trying)	-	-	-	-
2a1	Check: Does the MCX Server respond with a SIP 100 Trying?	<--	SIP 100 (Trying)	-	P
3	Check: Does the MCX Server respond with a SIP 200 (OK)?	<--	SIP 200 (OK)	-	P
4	The SS-UE (MCX Client) sends a SIP ACK to acknowledge the session establishment/modification (NOTE 1)	-->	SIP ACK	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.14.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).

Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.14.4-1: SIP INVITE (Step 1, Table A.2.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1 (note)
NOTE: With condition re_INVITE for session modifications

Table A.2.14.4-1: SIP 200 (OK) (Step 3, Table A.2.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition INVITE-RSP

A.2.15 MCX Participating Terminated call release

A.2.15.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.15.2 Definition of system information messages

N/A.

A.2.15.3 Procedure

Table A.2.15.3-1: MCX Participating Terminated call release

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP BYE request to terminate the MCX session (note)	-->	SIP BYE	-	-
2	Check: Does the MCX Server respond with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.15.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.16 MCX Participating Originated session establishment/modification without provisional responses other than 100 Trying

A.2.16.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo.

A.2.16.2 Definition of system information messages

N/A.

A.2.16.3 Procedure

Table A.2.16.3-1: MCX Participating Originated session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCX Server send a SIP INVITE requesting the establishment/modification of a call?	<--	SIP INVITE	-	P
2	The SS-UE (MCX Client) sends a SIP 100 Trying (note)	-->	SIP 100 (Trying)	-	-
3	The SS-UE (MCX Client) responds with a SIP 200 (OK) (note)	-->	SIP 200 (OK)	-	-
4	Check: Does the MCX server send a SIP ACK to acknowledge the session establishment/modification?	<--	SIP ACK	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.16.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.16.4-1: SIP INVITE (Step 1, Table A.2.16.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (NOTE 1).
NOTE 1: With condition re_INVITE for session modifications

Table A.2.16.4-2: SIP 200 (OK) (Step 3, Table A.2.16.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition INVITE-RSP

A.2.17 MCX Participating Originated call release

A.2.17.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.17.2 Definition of system information messages

N/A.

A.2.17.3 Procedure

Table A.2.17.3-1: MCX Participating Originated call release

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCX Server send a SIP BYE request to terminate the MCX session?	<--	SIP BYE	-	P
2	The SS-UE (MCX Client) responds with a SIP 200 (OK) message (note).	-->	SIP 200 (OK)	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.17.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5. and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.18 MCX E2E session establishment/modification without provisional responses other than 100 Trying

A.2.18.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo.

A.2.18.2 Definition of system information messages

N/A.

A.2.18.3 Procedure

Table A.2.18.3-1: MCX E2E session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP INVITE requesting the establishment/modification of a call	-->	SIP INVITE	-	-
2	Check: Does the MCX Server forward the SIP INVITE requesting the establishment/modification of a call to SS-UE2 (MCX Client)?	<--	SIP INVITE	-	P
3	The SS-UE2 (MCX Client) sends a SIP 100 Trying	-->	SIP 100 (Trying)	-	-
4	Check: Does the MCX Server respond with a SIP 100 Trying to SS-UE1 (MCX Client)?	<--	SIP 100 (Trying)	-	P
5	The SS-UE2 (MCX Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
6	Check: Does the MCX Server respond with a SIP 200 (OK) to SS-UE1 (MCX Client)?	<--	SIP 200 (OK)	-	P
7	The SS-UE1 (MCX Client) sends a SIP ACK to acknowledge the session establishment/modification	-->	SIP ACK	-	-
8	Check: Does the MCX server send a SIP ACK to acknowledge the session establishment/modification to SS-UE2?	<--	SIP ACK	-	P

A.2.18.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.18.4-1: SIP INVITE (Step 1, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1 (note)
NOTE: With condition re_INVITE for session modifications

Table A.2.18.4-2: SIP INVITE (Step 2, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (note)
NOTE: With condition re_INVITE for session modifications

Table A.2.18.4-1: SIP 200 (OK) (Step 5, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition INVITE-RSP

Table A.2.18.4-1: SIP 200 (OK) (Step 6, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition INVITE-RSP

A.2.19 MCX E2E call release

A.2.19.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.19.2 Definition of system information messages

N/A.

A.2.19.3 Procedure

Table A.2.19.3-1: MCX E2E call release

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP BYE request to terminate the MCX session	-->	SIP BYE	-	-
2	Check: Does the MCX Server respond a SIP BYE request to terminate the MCX session to SS-UE2 (MCX Client)?	<--	SIP BYE	-	P
3	The SS-UE2 (MCX Client) responds with a SIP 200 (OK) message.	-->	SIP 200 (OK)	-	-
4	Check: Does the MCX Server respond a SIP 200 (OK) message to SS-UE1?	<--	SIP 200 (OK)	-	P

A.2.19.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.19.4-1: SIP BYE from the SS-UE1 (Step 1, Table A.2.19.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.1-1

Table A.2.19.4-2: SIP BYE from the MCPTT Server (Step 2, Table A.2.19.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1

A.2.20 MCX E2E group call establishment, manual commencement

A.2.20.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.20.2 Definition of system information messages

N/A.

A.2.20.3 Procedure

Table A.2.20.3-1: MCX E2E group call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP INVITE requesting the establishment/modification of a call	-->	SIP INVITE	-	-
2	Check: Does the MCX Server forward the SIP INVITE requesting the establishment/modification of a call to SS-UE2 (MCX Client)?	<--	SIP INVITE	-	P
3	The SS-UE2 (MCX Client) sends a SIP 100 Trying	-->	SIP 100 (Trying)	-	-
4	Check: Does the MCX Server respond with a SIP 100 Trying to SS-UE1 (MCX Client)?	<--	SIP 100 (Trying)	-	P
5	The SS-UE2 (MCX Client) sends a SIP 183 (Session Progress) reliably.	-->	SIP 183 (Session Progress)	-	-
6	Check: Does the MCX Server acknowledge the receipt of SIP 183 (Session Progress) from SS-UE2?	<--	PRACK	-	P
7	The SS-UE2 (MCX Client) responds a SIP 200 (OK) to PRACK message	-->	SIP 200 (OK)	-	-
8	The SS-UE2 (MCX Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
9	Check: Does the MCX Server respond with a SIP 200 (OK) to SS-UE1 (MCX Client)?	<--	SIP 200 (OK)	-	P
10	The SS-UE1 (MCX Client) sends a SIP ACK to acknowledge the session establishment/modification	-->	SIP ACK	-	-
11	Check: Does the MCX server send a SIP ACK to acknowledge the session establishment/modification to SS-UE2?	<--	SIP ACK	-	P

A.2.20.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with conditions GROUP-CALL (note) and MANUAL; and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).

- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

NOTE: For chat group calls, condition CHAT-GROUP-CALL shall be used instead of GROUP-CALL.

Table A.2.20.4-1: SIP 200 (OK) (Step 8, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition INVITE-RSP

Table A.2.20.4-2: SIP 200 (OK) (Step 9, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition INVITE-RSP

A.2.21 MCX E2E private call establishment, manual commencement

A.2.21.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.21.2 Definition of system information messages

N/A.

A.2.21.3 Procedure

Table A.2.21.3-1: MCX E2E private call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP INVITE requesting the establishment/modification of a call	-->	SIP INVITE	-	-
2	Check: Does the MCX Server forward the SIP INVITE requesting the establishment/modification of a call to SS-UE2 (MCX Client)?	<--	SIP INVITE	-	P
3	The SS-UE2 (MCX Client) sends a SIP 100 Trying	-->	SIP 100 (Trying)	-	-
4	Check: Does the MCX Server respond with a SIP 100 Trying to SS-UE1 (MCX Client)?	<--	SIP 100 (Trying)	-	P
5	The SS-UE2 (MCX Client) sends a SIP 180 (Ringing) unreliably.	-->	SIP 180 (Ringing)	-	-
6	Check: Does the MCX Server respond with a unreliable SIP 180 (Ringing) to SS-UE1 (MCX Client)?	<--	SIP 180 (Ringing)	-	P
7	The SS-UE2 (MCX Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
8	Check: Does the MCX Server respond with a SIP 200 (OK) to SS-UE1 (MCX Client)?	<--	SIP 200 (OK)	-	P
9	The SS-UE1 (MCX Client) sends a SIP ACK to acknowledge the session establishment/modification	-->	SIP ACK	-	-
10	Check: Does the MCX server send a SIP ACK to acknowledge the session establishment/modification to SS-UE2?	<--	SIP ACK	-	P

A.2.21.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with conditions PRIVATE-CALL and MANUAL; and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.21.4-1: SIP 200 (OK) (Step 7, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition INVITE-RSP

Table A.2.21.4-1: SIP 200 (OK) (Step 8, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition INVITE-RSP

A.3 Generic test procedures for MCVideo Server-Client operation

A.3.0 General

The purpose of the procedures specified in the following clauses is to facilitate test description by providing MCVideo procedure sequences which can be referred from the relevant TCs specified in the present document.

The procedures specified are required to ensure that MCVideo service can take place between a MCVideo Client and a MCVideo server or specific MCVideo relevant pre-conditions are met before a test case can be executed.

A.3.1 MCVideo E2E session establishment/modification without provisional responses other than 100 Trying

A.3.1.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.1.2 Definition of system information messages

N/A.

A.3.1.3 Procedure

Table A.3.1.3-1: MCVideo E2E session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform test procedure MCX E2E session establishment/modification without provisional responses other than 100 Trying' for as described in Table A.2.18.3-1 with MCVideo as MCX?	-	-	-	P
-	EXCEPTION: Step 2a1-2a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS-UE1 requests implicit transmission control in the SIP INVITE request but it is not implicitly granted. (Note)	-	-	-	-
2a1	Check: Does the IUT (MCVideo Server) send a Transmission Granted message to SS-UE1?	<--	Transmission Granted	-	P
2a2	IF the Transmission Granted message requires acknowledgement, the SS-UE1 sends a Transmission Control ACK message.	-->	Transmission Control Ack	-	-
-	EXCEPTION: Step 3a1-3a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is granted to SS-UE1, either implicit or explicitly. (Note)	-	-	-	-
3a1	Check: Does the IUT (MCVideo Server) send a Media Transmission Notification message to SS-UE2?	<--	Media Transmission Notification	-	P
3a2	IF the Media Transmission Notification message requires acknowledgement, the SS-UE2 sends a Transmission Control ACK message.	-->	Transmission Control Ack	-	-
-	EXCEPTION: Steps 4a1-4a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is not granted to SS-UE1. (Note)	-	-	-	-
4a1	Check: Does the IUT (MCVideo Server) send a Transmission Idle message to SS-UE1?	<--	Transmission Idle	-	P
4a2	Check: Does the IUT (MCVideo Server) send a Transmission Idle message to SS-UE2?	<--	Transmission Idle	-	P

NOTE: Possibilities in SDP-offer/answer depend on the test case requirements:

a. SS-UE1 sends SDP offer with media description for transmission control but without implicit transmission request.

b. SS-UE1 sends SDP offer with media description for transmission control and with implicit transmission request:

i. SDP answer from MCVideo Server contains "mc_implicit_request" and "mc_granted" (Transmission is implicitly granted).

ii. SDP answer from MCVideo Server contains "mc_implicit request" and but no "mc_granted" (Transmission needs to be explicitly granted at step 2a1).

iii. SDP answer from MCVideo Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the transmission).

c. SS-UE1 sends SDP offer without media description for transmission control.

A.3.1.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCVIDEO and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.1.4-1: SIP INVITE from SS-UE1
(Step 1, Table A.3.1.3-1; Step 1, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1 (note), with condition MCVIDEO				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires			IETF RFC 4028 [8]	
generic-param	"1800"	The recommended initial value is 1 800 in IETF RFC 4028 [8].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-2			
MIME body part		MCVideo-Info		
MIME-part-body	MCVideo-Info as described in Table A.3.1.4-3			
NOTE: With condition re_INVITE for session modifications.				

Table A.3.1.4-2: SDP in SIP INVITE (Table A.3.1.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	352			
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [10]	
port	port number assigned by the SS-UE1 (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [11] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE1			
Bandwidth		b= line		
"AS"	37		ETSI TS 126 114 [9] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [12]	
"RR"	2000		IETF RFC 3556 [12]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		**Media description for video**		
media description		m= line media = video SDP media-level section for a media-transmission control entity		
port	port number of the audio stream incremented by 2 (resulting in even integer)	The transport port from which the media stream is sent		
fmt	"100"			
Connection Data		c= line		
connection-address	IP address of the SS-UE1			
Bandwidth		b= line		
"AS"	315			
"RS"	0		IETF RFC 3556 [12]	
"RR"	2500		IETF RFC 3556 [12]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"100"			
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"100"			
format specific parameters		Parameters the H264 codec	IETF RFC 6184 [18]	
packetization-mode	"0"			
profile-level-id	"42e00c"			
Media description[3]		Media description for media control		NOT WITHOUT_TRANSMISSIONCONTROL
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE1 being different than the port number of the audio and video channels (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE1			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCVideo"			
format specific parameters				
mc_queueing	not present			
	present	Parameter has no value	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 14.2.2	pc_MCVideo_TransmissionRequestQueueing
mc_priority	The priority value specified in the MCVideo Group configuration	The value of the <user-priority> element for the user in the MCVideo Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.2-1) See note 4	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 14.2.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	The max priority value specified in the MCVideo Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCVideo Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1)	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 14.2.3	PRIVATE-CALL
mc_reception_priority	The priority value specified in the MCVideo Group configuration	The value of the <user-reception-priority> element for the user in the MCVideo Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.2-1) See note 5	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 14.2.6	GROUP-CALL OR CHAT-GROUP-CALL
mc_reception_priority	The max priority value specified in the MCVideo Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCVideo Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1)	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 14.2.6	PRIVATE-CALL

NOTE 1: If the SS-UE requests a call with media transmission control and implicit transmission (Option b of note on Table A.3.1.3-1), apply condition IMPLICIT_GRANT_REQUESTED.
NOTE 2: If the SS-UE requests a call without media transmission (Option c of note on Table A.3.1.3-1), apply condition WITHOUT_TRANSMISSIONCONTROL
NOTE 3: With condition INITIAL_SDP_OFFER if condition re_INVITE is not in use for Table A.3.1.4-1. Otherwise, apply condition SDP_OFFER.
NOTE 4: The <num-levels-priority-hierarchy> element for on-network in the MCVideo Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1).
NOTE 5: The <num-levels-priority-hierarchy> element for on-network in the MCVideo Service Configuration has higher value than any <user-reception-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1).

Table A.3.1.4-3: MCVideo-Info in SIP INVITE (Table A.3.1.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2, condition INVITE_REFER

Table A.3.1.4-4: SIP INVITE from the IUT (MCVideo-Server)
(Step 1, Table A.3.1.3-1; Step 2 Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (note), with condition MCVIDEO				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Resource-Priority			IETF RFC 4412 [13] IETF RFC 7134 [15] IETF RFC 8101 [14]	EMERGENCY-CALL or IMMPERIL-CALL
r-value				EMERGENCY-CALL
namespace	value of the <resource-priority-namespace> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
r-priority	value of the <resource-priority-priority> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
r-value				IMMPERIL-CALL
namespace	value of the <resource-priority-namespace> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
r-priority	value of the <resource-priority-priority> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
Session-Expires			IETF RFC 4028 [8]	
delta-seconds	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-5			
MIME body part		**MCVideo-Info**		
MIME-part-body	MCVideo-Info as described in Table A.3.1.4-6			
NOTE: With condition re_INVITE for session modifications.				

Table A.3.1.4-5: SDP in SIP INVITE (Table A.3.1.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[3]				NOT WITHOUT_ TRANSMIS SIONCONT ROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCVideo"			
format specific parameters				
mc_priority	Any value lower or equal to the priority specified in the MCVideo Group configuration	The value of the <user-priority> element for the user in the MCVideo Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.2-1) See note 3	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 14.2.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	Any value lower or equal to the priority specified in the MCVideo Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCVideo Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1)	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 14.2.3	PRIVATE-CALL
NOTE 1: If the SS-UE1 requests a call without media transmission control (Option c of NOTE 1 on Table A.3.1.3-1), apply condition WITHOUT_TRANSMISSIONCONTROL. NOTE 2: With condition INITIAL_SDP_OFFER if condition re_INVITE is not in use for Table A.3.1.4-3. Otherwise, apply condition SDP_OFFER. NOTE 3: The <num-levels-priority-hierarchy> element for on-network in the MCVideo Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1).				

Table A.3.1.4-6: MCVideo-Info in SIP INVITE (Table A.3.1.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcvideoinfo				
mcvideo-Params				
mcvideo-request-uri	Encrypted (note) <mcvideo-request-uri> with mcvideoURI set to px_MCVideo_ID_User_B	The URI of the called user		
mcvideo-calling-user-id	Encrypted (note) <mcvideo-calling-user-id> with mcvideoURI set to px_MCVideo_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-2A				

Table A.3.1.4-7: SIP 200 (OK) from SS-UE2
(Step 1, Table A.3.1.3-1; Step 5, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1, with condition MCVIDEO and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type			**IETF** RFC 5621 [**16**]	
value	"application/sdp"			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-8			

Table A.3.1.4-8: SDP in SIP 200 (OK) (Table A.3.1.4-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-2, condition SDP_ANSWER.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	352			
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [10]	
port	port number assigned by the SS-UE2 (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [11] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE2			
Bandwidth		b= line		
"AS"	37		ETSI TS 126 114 [9] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [12]	
"RR"	2000		IETF RFC 3556 [12]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-2, condition SDP_ANSWER.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		**Media description for video**		
media description		m= line media = video SDP media-level section for a media-transmission control entity		
port	port number of the audio stream incremented by 2 (resulting in even integer)	The transport port from which the media stream is sent		
fmt	"100"			
Connection Data		c= line		
connection-address	IP address of the SS-UE2			
Bandwidth		b= line		
"AS"	315			
"RS"	0		IETF RFC 3556 [12]	
"RR"	2500		IETF RFC 3556 [12]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"100"			
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"100"			
format specific parameters	same parameters and values as sent in the corresponding SDP offer	Parameters the H264 codec		
Media description[3]		Media description for media control		NOT WITHOUT_ TRANSMIS SIONCONT ROL
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE2 being different than the port number of the audio and video channels (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE2			
NOTE: If the SS-UE1 requests a call without media transmission control (Option c of note on Table A.3.1.3-1), apply condition WITHOUT_TRANSMISSIONCONTROL.				

**Table A.3.1.4-9: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table A.3.1.3-1; Step 6, Table A.2.18.3-1)**

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition MCVIDEO and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires				
generic-param	Same value as session expires header in SIP INVITE or lower			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-10			

Table A.3.1.4-10: SDP in SIP 200 (OK) (Table A.3.1.4-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-2, condition SDP_ANSWER.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[3]				NOT WITHOUT_TRANSMISSIONCONTROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCVideo"			
format specific parameters				
mc_priority	if a value is provided in the offer: the value defined in the MCVideo Group Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <user-priority> element for the user in the MCVideo Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.2-1)	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 14.2.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	if a value is provided in the offer: the value defined in the MCVideo Service Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <num-levels-priority-hierarchy> element for on-network in the MCVideo Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1)	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 14.2.3	PRIVATE-CALL

NOTE 1: If the SS-UE requests an implicit transmission (Option b of note on Table A.3.1.3-1), more conditions may apply:
- i. If the server implicitly grants the transmission, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied.
- ii. If the transmission needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies.

NOTE 2: If the SS-UE requests a call without media transmission (Option c of note on Table A.3.1.3-1) apply condition WITHOUT_TRANSMISSIONCONTROL.
NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the MCVideo Server.

Table A.3.1.4-11: Transmission Granted (Step 2a1, Table A.3.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.1-1, with condition ACK IF the IUT (MCVideo Server) requests message acknowledgement.

Table A.3.1.4-12: Media Transmission Notification (Step 3a1, Table A.3.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.7-1, with condition ACK IF the IUT (MCVideo Server) requests message acknowledgement.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
User Id of the Transmitting User		User ID of the user transmitting the media	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.6	
User Id of the Transmitting User	px_MCVideo_ID_User_A			
Audio SSRC of the Transmitting User	Media/Audio SSRC of remote client (Client A)	SSRC of the user transmitting the media	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.16	
Functional Alias	Not present			
	px_MCVideo_ID_FA_A	functional alias URI of the transmitting user	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.21	FA

Table A.3.1.4-13: Transmission Idle (Steps 4a1 and 4a2, Table A.3.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.16-1.

Table A.3.1.4-14: Transmission Control Ack (Steps 2a2 and 3a2, Table A.3.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

A.3.2 MCVideo E2E group call establishment, manual commencement

A.3.2.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.2.2 Definition of system information messages

N/A.

A.3.2.3 Procedure

Table A.3.2.3-1: MCVideo E2E group call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform test procedure MCX E2E group call establishment, manual commencement' as described in Table A.2.20.3-1 with MCVideo as MCX?	-	-	-	P
-	EXCEPTION: Step 2a1-2a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS-UE1 requests implicit transmission control in the SIP INVITE request but it is not implicitly granted (NOTE1)	-	-	-	-
2a1	Check: Does the IUT (MCVideo Server) send a Transmission Granted message to SS-UE1?	<--	Transmission Granted	-	P
2a2	IF the Transmission Granted message requires acknowledgement, the SS-UE1 sends a Transmission Control ACK message.	-->	Transmission Control Ack	-	-
-	EXCEPTION: Step 3a1-3a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is granted to SS-UE1, either implicit or explicitly. (NOTE1)	-	-	-	-
3a1	Check: Does the IUT (MCVideo Server) send a Media Transmission Notification message to SS-UE2?	<--	Media Transmission Notification	-	P
3a2	IF the Media Transmission Notification message requires acknowledgement, the SS-UE2 sends a Transmission Control ACK message.	-->	Transmission Control Ack	-	-
-	EXCEPTION: Steps 4a1-4a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is not granted to SS-UE1. (NOTE1)	-	-	-	-
4a1	Check: Does the IUT (MCVideo Server) send a Transmission Idle message to SS-UE1?	<--	Transmission Idle	-	P
4a2	Check: Does the IUT (MCVideo Server) send a Transmission Idle message to SS-UE2?	<--	Transmission Idle	-	P

NOTE: Possibilities in SDP-offer/answer depend on the test case requirements:

a. SS-UE1 sends SDP offer with media description for transmission control but without implicit transmission request.

b. SS-UE1 sends SDP offer with media description for transmission control and with implicit transmission request:

i. SDP answer from MCVideo Server contains "mc_implicit_request" and "mc_granted" (Transmission media request is implicitly granted).

ii. SDP answer from MCVideo Server contains "mc_implicit request" and but no "mc_granted" (Transmission media request needs to be explicitly granted at step 2a1).

iii. SDP answer from MCVideo Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly send the Transmission media request).

A.3.2.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCVIDEO and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.2.4-1: SIP INVITE from the SS-UE1
(Step 1, Table A.3.2.3-1; Step 1, Table A.2.20.3-1)

Derivation Path: Table A.3.1.4-1, with condition MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-2			
MIME body part		**MCVideo-Info**		
MIME-part-body	MCVideo-Info as described in Table A.3.2.4-2			

Table A.3.2.4-2: MCVideo-Info in SIP INVITE (Table A.3.2.4-1)

Derivation Path: Table A.3.1.4-5, with condition GROUP-CALL (note).
NOTE: For chat group calls CHAT-GROUP-CALL condition shall be used instead of GROUP-CALL.

Table A.3.2.4-3: SIP INVITE from the MCVideo-Server
(Step 1, Table A.3.2.3-1; Step 2, Table A.2.20.3-1)

Derivation Path: Table A.3.1.4-4, with condition MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-5			
MIME body part		**MCVideo-Info**		
MIME-part-body	MCVideo-Info as described in Table A.3.2.4-4			

Table A.3.2.4-4: MCVideo-Info in SIP INVITE (Table A.3.2.4-3)

Derivation Path: Table A.3.1.4-6, with condition GROUP-CALL (note).
NOTE: For chat group calls CHAT-GROUP-CALL condition shall be used instead of GROUP-CALL.

Table A.3.2.4-5: SIP 200 (OK) from the SS-UE2
(Step 1, Table A.3.2.3-1; Step 8, Table A.2.20.3-1)

Derivation Path: Table A.3.1.4-7, with condition GROUP-CALL (note).
NOTE: For chat group calls CHAT-GROUP-CALL condition shall be used instead of GROUP-CALL.

Table A.3.2.4-6: SIP 200 (OK) from the MCVideo Server
(Step 1, Table A.3.2.3-1; Step 9, Table A.2.20.3-1)

Derivation Path: Table A.3.1.4-9

Table A.3.2.4-7: SIP 183 (Session Progress) from the SS-UE2
(Step 1, Table A.3.2.3-1; Step 5, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.3.1-1, with condition MCVIDEO and 100rel

Table A.3.2.4-8: Transmission Granted (Step 2a1, Table A.3.2.3-1)

Derivation Path: Table A.3.1.4-11

Table A.3.2.4-9: Media Transmission Notification (Step 3a1, Table A.3.2.3-1)

Derivation Path: Table A.3.1.4-12

Table A.3.2.4-10: Transmission Idle (Steps 4a1 and 4a2, Table A.3.2.3-1)

Derivation Path: Table A.3.1.4-13

Table A.3.2.4-11: Transmission Control Ack (Steps 2a2 and 3a2, Table A.3.2.3-1)

Derivation Path: Table A.3.1.4-14

A.3.3 MCVideo E2E private call establishment, manual commencement

A.3.3.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.3.2 Definition of system information messages

N/A.

A.3.3.3 Procedure

Table A.3.3.3-1: MCVideo E2E private call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the IUT (MCVideo Server) correctly perform test procedure MCX E2E private call establishment, manual commencement' as described in Table A.2.21.3-1 with MCVideo as MCX?	-	-	-	P
-	EXCEPTION: Step 2a1-2a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS-UE1 requests implicit transmission control in the SIP INVITE request but it is not implicitly granted (note)	-	-	-	-
2a1	Check: Does the IUT (MCVideo Server) send a Transmission Granted message to SS-UE1?	<--	Transmission Granted	-	P
2a2	IF the Transmission Granted message requires acknowledgement, the SS-UE1 sends a Transmission Control ACK message.	-->	Transmission Control Ack	-	-
-	EXCEPTION: Step 3a1-3a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is granted to SS-UE1, either implicit or explicitly. (note)	-	-	-	-
3a1	Check: Does the IUT (MCVideo Server) send a Media Transmission Notification message to SS-UE2?	<--	Media Transmission Notification	-	P
3a2	IF the Media Transmission Notification message requires acknowledgement, the SS-UE2 sends a Transmission Control ACK message.	-->	Transmission Control Ack	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
-	EXCEPTION: Steps 4a1-4a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is not granted to SS-UE1. (note)	-	-	-	-
4a1	Check: Does the IUT (MCVideo Server) send a Transmission Idle message to SS-UE1?	<--	Transmission Idle	-	P
4a2	Check: Does the IUT (MCVideo Server) send a Transmission Idle message to SS-UE2?	<--	Transmission Idle	-	P

NOTE: Possibilities in SDP-offer/answer depend on the test case requirements:
a. SS-UE1 sends SDP offer with media description for transmission control but without implicit transmission request.
b. SS-UE1 sends SDP offer with media description for transmission control and with implicit transmission request:
i. SDP answer from MCVideo Server contains "mc_implicit_request" and "mc_granted" (Transmission is implicitly granted).
ii. SDP answer from MCVideo Server contains "mc_implicit request" and but no "mc_granted" (Transmission needs to be explicitly granted at step 2a1).
iii. SDP answer from MCVideo Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the transmission).
c. SS-UE1 sends SDP offer without media description for transmission control

A.3.3.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCVIDEO and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.3.4-1: SIP INVITE from the SS-UE1
(Step 1, Table A.3.3.3-1; Step 1, Table A.2.21.3-1)

Derivation Path: Table A.3.1.4-1, with condition PRIVATE-CALL and MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.3.4-2			
MIME body part		**MCVideo-Info**		
MIME-part-body	MCVideo-Info as described in Table A.3.3.4-3			

Table A.3.3.4-2: SDP in SIP INVITE (Table A.3.3.4-1)

Derivation Path: Table A.3.1.4.2-2, with condition PRIVATE-CALL

Table A.3.3.4-3: MCVideo-Info in SIP INVITE (Table A.3.3.4-1)

Derivation Path: Table A.3.1.4-3, with condition PRIVATE-CALL

Table A.3.3.4-4: SIP INVITE from the MCVideo-Server to SS-UE2 (Step 1, Table A.3.3.3-1; Step 2, Table A.2.21.3-1)

Derivation Path: Table A.3.1.4-4, with condition MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-5			
MIME body part		**MCVideo-Info**		
MIME-part-body	MCVideo-Info as described in Table A.3.3.4-5			

Table A.3.3.4-5: MCVideo-Info in SIP INVITE (Table A.3.3.4-4)

Derivation Path: Table A.3.1.4-6, with condition PRIVATE-CALL

Table A.3.3.4-6: SIP 200 (OK) from the SS-UE2 to MCVideo Server (Step 1, Table A.3.3.3-1; Step 7, Table A.2.21.3-1)

Derivation Path: Table A.3.1.4-7

Table A.3.3.4-7: SIP 200 (OK) from the MCVideo Server to SS-UE1 (Step 1, Table A.3.3.3-1; Step 8, Table A.2.21.3-1)

Derivation Path: Table A.3.1.4-9

Table A.3.3.4-8: SIP 180 (Ringing) response (Step 1, Table A.3.3.3-1; Steps 5, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.2.1-1, with condition MCVIDEO

Table A.3.3.4-9: SIP 180 (Ringing) response (Step 1, Table A.3.3.3-1; Steps 6, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.2.2-1, with condition MCVIDEO

Table A.3.3.4-10: Transmission Granted (Step 2a1, Table A.3.3.3-1)

Derivation Path: Table A.3.1.4-11

Table A.3.3.4-11: Media Transmission Notification (Step 3a1, Table A.3.3.3-1)

Derivation Path: Table A.3.1.4-12

Table A.3.3.4-12: Transmission Idle (Steps 4a1 and 4a2, Table A.3.3.3-1)

Derivation Path: Table A.3.1.4-13

Table A.3.3.4-13: Transmission Control Ack (Steps 2a2 and 3a2, Table A.3.3.3-1)

Derivation Path: Table A.3.1.4-14

A.3.4 MCVideo Participating Terminated session establishment/modification without provisional responses other than 100 Trying

A.3.4.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.4.2 Definition of system information messages

N/A.

A.3.4.3 Procedure

Table A.3.4.3-1: MCVideo Participating Terminated session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCVideo server correctly perform test procedure 'MCX Participating Terminated session establishment/modification without provisional responses other than 100 Trying' for as described in Table A.2.14.3-1 with MCVideo as MCX? (NOTE 2)	-	-	-	P
-	EXCEPTION: Step 2a1-2a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS-UE requests implicit transmission control in the SIP INVITE request but it is not implicitly granted (NOTE 1)	-	-	-	-
2a1	Check: Does the MCVideo server send a Transmission Granted message to SS-UE?	<--	Transmission Granted	-	P
2a2	IF the Transmission Granted message requires acknowledgement, the SS-UE sends a Transmission Control ACK message.	-->	Transmission Control Ack	-	-
-	EXCEPTION: Step 3a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is not granted to SS-UE. (NOTE 1)	-	-	-	-
3a1	Check: Does the MCVideo server send a Transmission Idle message to SS-UE?	<--	Transmission Idle	-	P

NOTE 1: Possibilities in SDP-offer/answer depend on the test case requirements:
- a. SS-UE sends SDP offer with media description for transmission control but without implicit transmission request.
- b. SS-UE sends SDP offer with media description for transmission control and with implicit transmission request:
  - i. SDP answer from MCVideo Server contains "mc_implicit_request" and "mc_granted" (Transmission is implicitly granted).
  - ii. SDP answer from MCVideo Server contains "mc_implicit request" and but no "mc_granted" (Transmission needs to be explicitly granted at step 2a1).
  - iii. SDP answer from MCVideo Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the transmission).
- c. SS-UE sends SDP offer without media description for transmission control.

NOTE 2: The SS-UE being used is specified in the test case calling the procedure.

A.3.4.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCVIDEO and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.4.4-1: SIP INVITE from SS-UE
(Step 1, Table A.3.4.3-1; Step 1, Table A.2.14.3-1)

Derivation Path: Table A.3.1.4-1

Table A.3.4.4-2: SIP 200 (OK) from IUT (MCVideo Server)
(Step 1, Table A.3.4.3-1; Step 3, Table A.2.14.3-1)

Derivation Path: Table A.3.1.4-9

Table A.3.4.4-3: Transmission Granted (Step 2a1, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.1-1, with condition ACK IF the IUT (MCVideo Server) requests message acknowledgement.

Table A.3.4.4-4: Transmission Idle (Step 3a1, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.16-1.

Table A.3.4.4-5: Transmission Control Ack (Step 2a2, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

A.3.5 MCVideo Group Affiliation Status Determination and Subscription

A.3.5.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.5.2 Definition of system information messages

N/A.

A.3.5.3 Procedure

Table A.3.5.3-1: MCVideo Group affiliation status determination and subscription

Same procedure as defined in: Table A.2.7.3-1

A.3.5.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCVideo Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCVideo Client) or SS-UE2 (MCVideo Client).

Table A.3.5.4-1: SIP SUBSCRIBE (Step 1, Table A.3.5.3-1)

Derivation Path: Table A.2.7.4-1, conditions MCVIDEO and PRESENCE-EVENT

Table A.3.5.4-2: SIP NOTIFY (Step 3, Table A.3.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, conditions MCVIDEO and PRESENCE-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 281 [5] (3GPP TS 24.281) clause 8.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.5.4-3			

Table A.3.5.4-3: PIDF in SIP NOTIFY (Table A.3.5.4-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-2 (note)
NOTE: PIDF document contains tuple with empty <status> element.

A.3.6 MCVideo Group Affiliation Status Change

A.3.6.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.6.2 Definition of system information messages

N/A.

A.3.6.3 Procedure

Table A.3.6.3-1: MCVideo Group Affiliation Status Change

Same procedure as defined in Table A.2.12.3-1

A.3.6.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCVideo Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCVideo Client) or SS-UE2 (MCVideo Client).

Table A.3.6.4-1: SIP PUBLISH (Step 1, Table A.3.6.3-1)

Derivation Path: Table A.2.12.4-1, condition MCVIDEO

Table A.3.6.4-2: PIDF in SIP PUBLISH (Table A.3.6.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.1-2, condition AFFILIATION

Table A.3.6.4-3: SIP 200 (OK) (Step 2, Table A.3.6.3-1)

Derivation Path: Table A.2.12.4-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires			IETF RFC 3261 [7] IETF RFC 3903 [17]	
value	same value as in the request or lower			

Table A.3.6.4-4: SIP NOTIFY (Step 3, Table A.3.6.3-1)

Derivation Path: Table A.2.12.4-6, condition MCVIDEO				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 281 [5] (3GPP TS 24.281) clause 8.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.6.4-5			

Table A.3.6.4-5: PIDF in SIP NOTIFY (Table A.3.6.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-2, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	"affiliating"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 281 [5] (3GPP TS 24.281) clause 8.2.2.3.5	
..p-id	same value as received in the SIP PUBLISH message.			

Table A.3.6.4-6: SIP NOTIFY (Step 5, Table A.3.6.3-1)

Derivation Path: Table A.2.12.4-6, condition MCVIDEO				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 281 [5] (3GPP TS 24.281) clause 8.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.6.4-7			

Table A.3.6.4-7: PIDF in SIP NOTIFY (Table A.3.6.4-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-2, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	"affiliated"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 281 [5] (3GPP TS 24.281) clause 8.2.2.3.5	
..p-id	same value as received in the SIP PUBLISH message.			

A.3.7 MCVideo Transmission Request - Granted and Media Transmission Notification

A.3.7.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.7.2 Definition of system information messages

N/A.

A.3.7.3 Procedure

Table A.3.7.3-1: MCVideo Transmission Request - Granted and Media Transmission Notification

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCVideo Client) sends a Transmission Request message, with acknowledgement. (Note)	-->	Transmission Request	-	-
2	Check: Does the IUT (MCVideo Server) send a Transmission Control Ack message?	<--	Transmission Control Ack	-	P
-	EXCEPTION: Steps 3a1-3b2 describe behaviour that depends on the MCVideo Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCVideo Server requests or not acknowledgement of sent transmission control messages.	-	-	-	-
3a1	Check: Does the IUT (MCVideo Server) send a Transmission Granted message to SS-UE1 with acknowledgement? (Note)	<--	Transmission Granted	-	P
3a2	The SS-UE1 (MCVideo Client) sends a Transmission Control Ack message in response to the Transmission Granted message. (Note)	-->	Transmission Control Ack	-	-
3a3	Check: Does the IUT (MCVideo Server) send a Media Transmission Notification message to SS-UE2 with acknowledgement? (Note)	<--	Media Transmission Notification	-	P
3a4	The SS-UE2 (MCVideo Client) sends a Transmission Control Ack message in response to the Media Transmission Notification message. (Note)	-->	Transmission Control Ack	-	-
3b1	Check: Does the IUT (MCVideo Server) send a Transmission Granted message to SS-UE1 with no acknowledgement?	<--	Transmission Granted	-	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
3b2	Check: Does the IUT (MCVideo Server) send a Media Transmission Notification message to SS-UE2 with no acknowledgement? (Note)	<--	Media Transmission Notification	-	P
NOTE: SS-UEs may switch, depending on the test case calling this procedure.					

A.3.7.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.7.4-1: Transmission Request (Step 1, Table A.3.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.1.1-1, with condition ACK

Table A.3.7.4-2: Transmission Granted (Step 3b1, Table A.3.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.1-1

Table A.3.7.4-3: Transmission Granted (Step 3a1, Table A.3.7.3-1)

Derivation Path: Table A.3.7.4-2, with condition ACK

Table A.3.7.4-4: Media Transmission Notification (Step 3b2, Table A.3.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.7-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
User Id of the Transmitting User		User ID of the user transmitting the media	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.6	
User Id of the Transmitting User	px_MCVideo_ID_User_A			
Audio SSRC of the Transmitting User	Media/Audio SSRC of remote client (Client A)	SSRC of the user transmitting the media	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.16	
Functional Alias	Not present			
	px_MCVideo_ID_FA_A	functional alias URI of the transmitting user	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.21	FA

Table A.3.7.4-5: Media Transmission Notification (Step 3a3, Table A.3.7.3-1)

Derivation Path: Table A.3.7.4-4, with condition ACK

Table A.3.7.4-6: Transmission Control Ack (Steps 3a2 and 3a4, Table A.3.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

Table A.3.7.4-7: Transmission Control Ack (Steps 2, Table A.3.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition DOWNLINK

A.3.8 MCVideo Transmission End Request - Idle and Transmission End Notify

A.3.8.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.8.2 Definition of system information messages

N/A.

A.3.8.3 Procedure

Table A.3.8.3-1: MCVideo Transmission End Request - Idle and Transmission End Notify

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCVideo Client) sends a Transmission End Request message, with acknowledgement. (Note 1)	-->	Transmission End Request	-	-
2	Check: Does the IUT (MCVideo Server) send a Transmission Control Ack message?	<--	Transmission Control Ack	-	P
-	EXCEPTION: Steps 3a1-3b2 describe behaviour that depends on the MCVideo Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCVideo Server requests or not acknowledgement of sent transmission control messages.	-	-	-	-
3a1	Check: Does the IUT (MCVideo Server) send a Transmission End Response message to SS-UE1 (MCVideo Client) with acknowledgement? (Note 1)	<--	Transmission End Response	-	P
3a2	The SS-UE1 (MCVideo Client) sends a Transmission Control Ack message in response to the Transmission End Response message. (Note 1)	-->	Transmission Control Ack	-	-
3a3	Check: Does the IUT (MCVideo Server) send a Transmission End Notify message to SS-UE2 (MCVideo Client) with acknowledgement? (Notes 1 and 2)	<--	Transmission End Notify	-	P
3a4	The SS-UE2 (MCVideo Client) sends a Transmission Control Ack message in response to the Transmission End Notify message. (Note 1)	-->	Transmission Control Ack	-	-
-	EXCEPTION: Steps 3aA1-3aA4 describe behaviour that depends on the test case calling this procedure; the "upper case letter" identifies a step sequence that takes place if SS-UE2 has previously requested media reception by performing procedure 'MCVideo Media Reception Request and Response', as described in Table A.3.9.3-1.	-	-	-	-
3aA1	Check: Does the IUT (MCVideo Server) send a Media Reception End Request message to SS-UE2 (MCVideo Client) with acknowledgement? (Note 1)	<--	Media Reception End Request	-	P
3aA2	The SS-UE2 (MCVideo Client) sends a Transmission Control Ack message in response to the Media Reception End Request message. (Note 1)	-->	Transmission Control Ack	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
3aA3	The SS-UE2 (MCVideo Client) sends a Media Reception End Response message, with acknowledgement. (Note 1)	-->	Media Reception End Response	-	-
3aA4	Check: Does the IUT (MCVideo Server) send a Transmission Control Ack message?	<--	Transmission Control Ack	-	P
3b1	Check: Does the IUT (MCVideo Server) send a Transmission End Response message to SS-UE1 (MCVideo Client) with no acknowledgement? (Note 1)	<--	Transmission End Response	-	P
3b2	Check: Does the IUT (MCVideo Server) send a Transmission End Notify message to SS-UE2 (MCVideo Client) with no acknowledgement? (Notes 1 and 2)	<--	Transmission End Notify	-	P
-	EXCEPTION: Steps 3bA1-3bA3 describe behaviour that depends on the test case calling this procedure; the "upper case letter" identifies a step sequence that takes place if SS-UE2 (MCVideo Client) has previously requested media reception by performing procedure 'MCVideo Media Reception Request and Response', as described in Table A.3.9.3-1.	-	-	-	-
3bA1	Check: Does the IUT (MCVideo Server) send a Media Reception End Request message to SS-UE2 (MCVideo Client) with no acknowledgement? (Note 1)	<--	Media Reception End Request	-	P
3bA2	The SS-UE2 (MCVideo Client) sends a Media Reception End Response message, with acknowledgement. (Note 1)	-->	Media Reception End Response	-	-
3bA3	Check: Does the IUT (MCVideo Server) send a Transmission Control Ack message?	<--	Transmission Control Ack	-	P
4	Check: Does the IUT (MCVideo Server) send a Transmission Idle message to SS-UE1 (MCVideo Client)?	<--	Transmission Idle	-	P
5	Check: Does the IUT (MCVideo Server) send a Transmission Idle message to SS-UE2 (MCVideo Client)?	<--	Transmission Idle	-	P
NOTE 1: SS-UEs may switch, depending on the test case calling this procedure. NOTE 2: According to ETSI TS 124 581 [6] (3GPP TS 24.581), sections 6.3.7.3.7 and 6.3.7.4.7, the MCVideo Server sends a Transmission End Notify message when it has stopped receiving RTP media packets from the transmitter participant.					

A.3.8.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.8.4-1: Transmission End Request (Step 1, Table A.3.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.1-1, with conditions ACK and UPLINK

Table A.3.8.4-2: Transmission End Response (Step 3b1, Table A.3.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.2-1, with condition DOWNLINK

Table A.3.8.4-3: Transmission End Response (Step 3a1, Table A.3.8.3-1)

Derivation Path: Table A.3.8.4-2, with condition ACK

Table A.3.8.4-4: Transmission End Notify (Step 3b2, Table A.3.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.15-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
User Id of the Transmitting User		Carries the identity of the user whose media transmission has been released	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.6	
User Id of the Transmitting User	px_MCVideo_ID_User_A			
Audio SSRC of the Transmitting User	Media/Audio SSRC of remote client (Client A)	SSRC of the user transmitting the media	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.16	

Table A.3.8.4-5: Transmission End Notify (Step 3a3, Table A.3.8.3-1)

Derivation Path: Table A.3.8.4-4, with condition ACK

Table A.3.8.4-6: Transmission Idle (Steps 4 and 5, Table A.3.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.16-1

Table A.3.8.4-7: Transmission Control Ack (Steps 3a2, 3a4 and 3aA2, Table A.3.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

Table A.3.8.4-8: Transmission Control Ack (Steps 2, 3aA4 and 3bA3, Table A.3.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition DOWNLINK

Table A.3.8.4-9: Media Reception End Response (Steps 3aA3 and 3bA2, Table A.3.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.4-1, with conditions ACK and UPLINK

Table A.3.8.4-10: Media Reception End Request (Step 3bA1, Table A.3.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.3-1, with condition DOWNLINK

Table A.3.8.4-11: Media Reception End Request (Step 3aA1, Table A.3.8.3-1)

Derivation Path: Table A.3.8.4-10, with condition ACK

A.3.9 MCVideo Media Reception Request and Response

A.3.9.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.9.2 Definition of system information messages

N/A.

A.3.9.3 Procedure

Table A.3.9.3-1: MCVideo Media Reception Request and Response

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCVideo Client) sends a Receive Media Request, with acknowledgement	-->	Receive Media Request	-	-
2	Check: Does the IUT (MCVideo Server) send a Transmission Control Ack message?	<--	Transmission Control Ack	-	P
-	EXCEPTION: Steps 3a1-3b1 describe behaviour that depends on the MCVideo Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCVideo Server requests or not acknowledgement of sent transmission control messages.	-	-	-	-
3a1	Check: Does the IUT (MCVideo Server) send a Receive Media Response message to the SS-UE, with acknowledgement?	<--	Receive Media Response	-	P
3a2	The SS-UE (MCVideo Client) sends a Transmission Control Ack message in response to the Receive Media Response message.	-->	Transmission Control Ack	-	-
3b1	Check: Does the IUT (MCVideo Server) send a Receive Media Response message to the SS-UE, with no acknowledgement?	<--	Receive Media Response	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.3.9.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.9.4-1: Receive Media Request (Step 1, Table A.3.9.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.1.4-1, with condition ACK

Table A.3.9.4-2: Receive Media Response (Step 3b1, Table A.3.9.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.8-1

Table A.3.9.4-3: Receive Media Response (Step 3a1, Table A.3.9.3-1)

Derivation Path: Table A.3.9.4-2, with condition ACK

Table A.3.9.4-4: Transmission Control Ack (Step 3a2, Table A.3.9.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

Table A.3.9.4-5: Transmission Control Ack (Steps 2, Table A.3.9.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition DOWNLINK

A.3.10 MCVideo Media Reception End Request and Response

A.3.10.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.10.2 Definition of system information messages

N/A.

A.3.10.3 Procedure

Table A.3.10.3-1: MCVideo Media Reception Request and Response

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCVideo Client) sends a Media Reception End Request, with acknowledgement	-->	Media Reception End Request	-	-
2	Check: Does the IUT (MCVideo Server) send a Transmission Control Ack message?	<--	Transmission Control Ack	-	P
-	EXCEPTION: Steps 3a1-3b1 describe behaviour that depends on the MCVideo Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCVideo Server requests or not acknowledgement of sent transmission control messages.	-	-	-	-
3a1	Check: Does the IUT (MCVideo Server) send a Media Reception End Response message to the SS-UE, with acknowledgement?	<--	Media Reception End Response	-	P
3a2	The SS-UE (MCVideo Client) sends a Transmission Control Ack message in response to the Media Reception End Response message.	-->	Transmission Control Ack	-	-
3b1	Check: Does the IUT (MCVideo Server) send a Media Reception End Response message to the SS-UE, with no acknowledgement?	<--	Media Reception End Response	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.3.10.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.10.4-1: Media Reception End Request (Step 1, Table A.3.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.3-1, with conditions ACK and UPLINK

Table A.3.10.4-2: Media Reception End Response (Step 3b1, Table A.3.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.4-1, with condition DOWNLINK

Table A.3.10.4-3: Media Reception End Response (Step 3a1, Table A.3.10.3-1)

Derivation Path: Table A.3.10.4-2, with condition ACK

Table A.3.10.4-4: Transmission Control Ack (Step 3a2, Table A.3.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

Table A.3.10.4-5: Transmission Control Ack (Steps 2, Table A.3.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition DOWNLINK

A.3.11 MCVideo Transmission Request - Transmission Rejected

A.3.11.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.11.2 Definition of system information messages

N/A.

A.3.11.3 Procedure

Table A.3.11.3-1: MCVideo Transmission Request - Transmission Rejected

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCVideo Client) sends a Transmission Request message, with acknowledgement. (Note)	-->	Transmission Request	-	-
2	Check: Does the IUT (MCVideo Server) send a Transmission Control Ack message?	<--	Transmission Control Ack	-	P
-	EXCEPTION: Steps 3a1-3b1 describe behaviour that depends on the MCVideo Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCVideo Server requests or not acknowledgement of sent transmission control messages.	-	-	-	-
3a1	Check: Does the IUT (MCVideo Server) send a Transmission Rejected message to the SS-UE, with acknowledgement?	<--	Transmission Rejected	-	P
3a2	The SS-UE (MCVideo Client) sends a Transmission Control Ack message in response to the Transmission Rejected message.	-->	Transmission Control Ack	-	-
3b1	Check: Does the IUT (MCVideo Server) send a Transmission Rejected message to the SS-UE, with no acknowledgement?	<--	Transmission Rejected	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCVideo Client) will be assumed.					

A.3.11.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.11.4-1: Transmission Request (Step 1, Table A.3.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.1.1-1, with condition ACK

Table A.3.11.4-2: Transmission Rejected (Step 3b1, Table A.3.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.2-1

Table A.3.11.4-3: Transmission Rejected (Step 3a1, Table A.3.11.3-1)

Derivation Path: Table A.3.7.4-2, with condition ACK

Table A.3.11.4-4: Transmission Control Ack (Step 3a2, Table A.3.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

Table A.3.11.4-5: Transmission Control Ack (Steps 2, Table A.3.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition DOWNLINK

A.3.12 MCVideo Transmission Request - Queue Position Info

A.3.12.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.12.2 Definition of system information messages

N/A.

A.3.12.3 Procedure

Table A.3.12.3-1: MCVideo Transmission Request - Queue Position Info

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCVideo Client) sends a Transmission Request message, with acknowledgement. (Note)	-->	Transmission Request	-	-
2	Check: Does the IUT (MCVideo Server) send a Transmission Control Ack message?	<--	Transmission Control Ack	-	P
-	EXCEPTION: Steps 3a1-3b1 describe behaviour that depends on the MCVideo Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCVideo Server requests or not acknowledgement of sent transmission control messages.	-	-	-	-
3a1	Check: Does the IUT (MCVideo Server) send a Queue Position Info message to the SS-UE, with acknowledgement?	<--	Queue Position Info	-	P
3a2	The SS-UE (MCVideo Client) sends a Transmission Control Ack message in response to the Queue Position Info message.	-->	Transmission Control Ack	-	-
3b1	Check: Does the IUT (MCVideo Server) send a Queue Position Info message to the SS-UE, with no acknowledgement?	<--	Queue Position Info	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCVideo Client) will be assumed.					

A.3.12.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.12.4-1: Transmission Request (Step 1, Table A.3.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.1.1-1, with condition ACK

Table A.3.12.4-2: Queue Position Info (Step 3b1, Table A.3.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.6-1

Table A.3.12.4-3: Queue Position Info (Step 3a1, Table A.3.12.3-1)

Derivation Path: Table A.3.7.4-2, with condition ACK

Table A.3.12.4-4: Transmission Control Ack (Step 3a2, Table A.3.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

Table A.3.12.4-5: Transmission Control Ack (Steps 2, Table A.3.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition DOWNLINK

A.3.13 MCVideo Queue Position Request - Queue Position Info

A.3.13.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.13.2 Definition of system information messages

N/A.

A.3.13.3 Procedure

Table A.3.13.3-1: MCVideo Queue Position Request - Queue Position Info

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCVideo Client) sends a Queue Position Request message, with acknowledgement. (Note)	-->	Queue Position Request	-	-
2	Check: Does the IUT (MCVideo Server) send a Transmission Control Ack message?	<--	Transmission Control Ack	-	P
-	EXCEPTION: Steps 3a1-3b1 describe behaviour that depends on the MCVideo Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCVideo Server requests or not acknowledgement of sent transmission control messages.	-	-	-	-
3a1	Check: Does the IUT (MCVideo Server) send a Queue Position Info message to the SS-UE, with acknowledgement?	<--	Queue Position Info	-	P
3a2	The SS-UE (MCVideo Client) sends a Transmission Control Ack message in response to the Queue Position Info message.	-->	Transmission Control Ack	-	-
3b1	Check: Does the IUT (MCVideo Server) send a Queue Position Info message to the SS-UE, with no acknowledgement?	<--	Queue Position Info	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCVideo Client) will be assumed.					

A.3.13.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.13.4-1: Queue Position Request (Step 1, Table A.3.13.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.1.3-1, with condition ACK

Table A.3.13.4-2: Queue Position Info (Step 3b1, Table A.3.13.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.6-1

Table A.3.13.4-3: Queue Position Info (Step 3a1, Table A.3.13.3-1)

Derivation Path: Table A.3.7.4-2, with condition ACK

Table A.3.13.4-4: Transmission Control Ack (Step 3a2, Table A.3.13.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

Table A.3.13.4-5: Transmission Control Ack (Steps 2, Table A.3.13.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition DOWNLINK

A.3.14 MCVideo Transmission Revoked - Transmission End Request

A.3.14.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.14.2 Definition of system information messages

N/A.

A.3.14.3 Procedure

Table A.3.14.3-1: MCVideo Transmission Revoked - Transmission End Request

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
-	EXCEPTION: Steps 1aA-3bB describe behaviour that depends on the MCVideo Server implementation; the "uppercase case letter" identifies a step sequence that takes place if the MCVideo Server requests or not acknowledgement of sent transmission control messages; the "lower case letter" identifies a step sequence that takes place if the MCVideo Server uses Transmission Revoked or Transmission End Request message to revoke the transmission control grant.	-	-	-	-
1aA	Check: Does the IUT (MCVideo Server) send a Transmission Revoked message to the SS-UE, with acknowledgement?	<--	Transmission Revoked	-	P
2aA	The SS-UE (MCVideo Client) sends a Transmission Control Ack message in response to the Transmission Revoked message.	-->	Transmission Control Ack	-	-
3aA	The SS-UE (MCVideo Client) sends a Transmission End Request message, with acknowledgement. (Note)	-->	Transmission End Request	-	-
4aA	Check: Does the IUT (MCVideo Server) send a Transmission Control Ack message?	<--	Transmission Control Ack	-	P
5aA	Check: Does the IUT (MCVideo Server) send a Transmission End Response message to SS-UE with acknowledgement? (Note)	<--	Transmission End Response	-	P
6aA	The SS-UE (MCVideo Client) sends a Transmission Control Ack message in response to the Transmission End Response message. (Note)	-->	Transmission Control Ack	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1bA	Check: Does the IUT (MCVideo Server) send a Transmission End Request message to SS-UE with acknowledgement? (Note)	<--	Transmission End Request	-	P
2bA	The SS-UE (MCVideo Client) sends a Transmission Control Ack message in response to the Transmission End Request message. (Note)	-->	Transmission Control Ack	-	-
3bA	The SS-UE (MCVideo Client) sends a Transmission End Response message, with acknowledgement. (Note)	-->	Transmission End Response	-	-
4bA	Check: Does the IUT (MCVideo Server) send a Transmission Control Ack message?	<--	Transmission Control Ack	-	P
1aB	Check: Does the IUT (MCVideo Server) send a Transmission Revoked message to the SS-UE, with no acknowledgement?	<--	Transmission Revoked	-	P
2aB	The SS-UE (MCVideo Client) sends a Transmission End Request message, with acknowledgement. (Note)	-->	Transmission End Request	-	-
3aB	Check: Does the IUT (MCVideo Server) send a Transmission Control Ack message?	<--	Transmission Control Ack	-	P
4aB	Check: Does the IUT (MCVideo Server) send a Transmission End Response message to SS-UE with no acknowledgement? (Note)	<--	Transmission End Response	-	P
1bB	Check: Does the IUT (MCVideo Server) send a Transmission End Request message to SS-UE with no acknowledgement? (Note)	<--	Transmission End Request	-	P
2bB	The SS-UE (MCVideo Client) sends a Transmission End Response message, with acknowledgement. (Note)	-->	Transmission End Response	-	-
3bB	Check: Does the IUT (MCVideo Server) send a Transmission Control Ack message?	<--	Transmission Control Ack	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCVideo Client) will be assumed.					

A.3.14.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.14.4-1: Transmission Revoked (Step 1aB, Table A.3.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.2-1

Table A.3.14.4-2: Transmission Revoked (Step 1aA, Table A.3.14.3-1)

Derivation Path: Table A.3.14.4-1, with condition ACK

Table A.3.14.4-3: Transmission End Request (Steps 3aA and 2aB, Table A.3.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.1-1, with conditions ACK and UPLINK

Table A.3.14.4-4: Transmission End Response (Step 4aB, Table A.3.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.2-1, with condition DOWNLINK

Table A.3.14.4-5: Transmission End Response (Step 5aA, Table A.3.14.3-1)

Derivation Path: Table A.3.14.4-4, with condition ACK

Table A.3.14.4-6: Transmission End Request (Step 1bB, Table A.3.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.1-1, with condition DOWNLINK

Table A.3.14.4-7: Transmission End Request (Step 1bA, Table A.3.14.3-1)

Derivation Path: Table A.3.14.4-6, with condition ACK

Table A.3.14.4-8: Transmission End Response (Steps 3bA and 2bB, Table A.3.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.2-1, with conditions ACK and UPLINK

Table A.3.14.4-9: Transmission Control Ack (Steps 2aA, 6aA and 2bA, Table A.3.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

Table A.3.14.4-10: Transmission Control Ack (Steps 4aA, 4bA, 3aB and 3bB, Table A.3.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition DOWNLINK

A.4 Generic test procedures for MCVideo Server-Server operation

A.4.0 General

The purpose of the procedures specified in the following clauses is to facilitate test description by providing MCVideo procedure sequences which can be referred from the relevant TCs specified in the present document.

The procedures specified are required to ensure that MCVideo service can take place between MCVideo servers or specific MCVideo relevant pre-conditions are met before a test case can be executed.

A.4.1 Generic test procedures for MCVideo Controlling Server function

A.4.1.1 MCVideo E2E session establishment/modification without provisional responses other than 100 Trying

A.4.1.1.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.1.1.2 Definition of system information messages

N/A.

A.4.1.1.3 Procedure

Table A.4.1.1.3-1: MCVideo E2E session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCVideo Server) providing the originating participating function sends a SIP INVITE requesting the establishment/modification of a call from SS-UE1 (MCVideo Client).	<--	SIP INVITE	-	-
2	Check: Does the IUT (MCVideo Server) forward the SIP INVITE request to the SS (MCVideo Server) providing the terminating participating function, to establish/modify a call with SS-UE2 (MCVideo Server)?	-->	SIP INVITE	-	P
3	The SS (MCVideo Server) providing the terminating participating function sends a SIP 100 Trying to the IUT (MCVideo Server).	<--	SIP 100 (Trying)	-	-
4	Check: Does the IUT (MCVideo Server) respond with a SIP 100 Trying to the SS (MCVideo Server) providing the originating participating function?	-->	SIP 100 (Trying)	-	P
5	The SS (MCVideo Server) providing the terminating participating function responds with a SIP 200 (OK).	<--	SIP 200 (OK)	-	-
6	Check: Does the IUT (MCVideo Server) respond with a SIP 200 (OK) to the SS (MCVideo Server) providing the originating participating function?	-->	SIP 200 (OK)	-	P
7	The SS (MCVideo Server) providing the originating participating function sends a SIP ACK to acknowledge the session establishment/modification to the IUT (MCVideo Server).	<--	SIP ACK	-	-
8	Check: Does the IUT (MCVideo Server) send a SIP ACK to acknowledge the session establishment/modification to the SS (MCVideo Server) providing the terminating participating function?	-->	SIP ACK	-	P
-	EXCEPTION: Step 9a1-9a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS (MCVideo Server) providing originating participating function requests implicit transmission control for SS-UE1 (MCVideo Client), in the SIP INVITE request, but it is not implicitly granted (note).	-	-	-	-
9a1	Check: Does the IUT (MCVideo Server) send a Transmission Granted message to the SS (MCVideo Server) providing originating participating function, to implicitly grant the transmission to SS-UE1 (MCVideo Client)?	-->	Transmission Granted	-	P
9a2	IF the Transmission Granted message requires acknowledgement, the SS (MCVideo Server) providing originating participating function sends a Transmission Control ACK message.	<--	Transmission Control Ack	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
-	EXCEPTION: Step 10a1-10a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is granted to SS-UE1 (MCVideo Client), either implicit or explicitly. (Note)	-	-	-	-
10a1	Check: Does the IUT (MCVideo Server) send a Media Transmission Notification message to the SS (MCVideo Server) providing terminating participating function, to inform SS-UE2 (MCVideo Client) that the transmission request has been granted to SS-UE1 (MCVideo Client)?	-->	Media Transmission Notification	-	P
10a2	IF the Media Transmission Notification message requires acknowledgement, the SS (MCVideo Server) providing terminating participating function sends a Transmission Control ACK message.	<--	Transmission Control Ack	-	-
-	EXCEPTION: Steps 11a1-11a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is not granted to SS-UE1 (MCVideo Client). (Note)	-	-	-	-
11a1	Check: Does the IUT (MCVideo Server) send a Transmission Idle message to the SS (MCVideo Server) providing originating participating function?	-->	Transmission Idle	-	P
11a2	Check: Does the IUT (MCVideo Server) send a Transmission Idle message to the SS (MCVideo Server) providing terminating participating function?	-->	Transmission Idle	-	P

NOTE: Possibilities in SDP-offer/answer depend on the test case requirements:
a. SS (MCVideo Server) providing originating participating function sends SDP offer with media description for transmission control but without implicit transmission request.
b. SS (MCVideo Server) providing originating participating function sends SDP offer with media description for transmission control and with implicit transmission request:
   i. SDP answer from MCVideo Server contains "mc_implicit_request" and "mc_granted" (Transmission is implicitly granted).
   ii. SDP answer from MCVideo Server contains "mc_implicit request" and but no "mc_granted" (Transmission needs to be explicitly granted at step 9a1).
   iii. SDP answer from MCVideo Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the transmission)
c. SS (MCVideo Server) providing originating participating function sends SDP offer without media description for transmission control.

A.4.1.1.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCVIDEO and in the test case calling the procedure with the following clarifications.

Table A.4.1.1.4-1: SIP INVITE from SS (MCVideo Server) (Step 1, Table A.4.1.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (note), with condition MCVIDEO				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCVideo_PublicServiceId_B	The URI of the MCVideo Server providing the controlling function.		
Request-URI	same URI as the IUT has sent earlier in the Contact header of a response within the same dialog	Contact URI of the IUT providing the controlling function.		re_INVITE

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (note), with condition MCVIDEO				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
via-branch	Value assigned by the SS starting with 'z9hG4bK'			
Record-Route	not present			
Require	Not present			
Proxy-Require	Not present			
From				
addr-spec	tsc_MCVideo_PublicServiceId_A	The URI of the MCVideo Server which is simulated by the SS, providing originating participating function.		
To				
addr-spec	tsc_MCVideo_PublicServiceId_B	The URI of the MCVideo Server providing the controlling function.		
Contact				
user-info and host	tsc_MCVideo_PublicServiceId_A	The URI of the MCVideo Server which is simulated by the SS, providing originating participating function.		
feature-param	"isfocus" not present			
P-Asserted-Service				
Service-ID	"urn:urn-7:3gpp-service.ims.icsi.mcvideo"		ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.2.1.3	
Call-ID			IETF RFC 3261 [7]	
callid	Value assigned by the SS providing the originating participating function			
Call-ID			IETF RFC 3261 [7]	re_INVITE
callid	same value as in INVITE creating the dialog			
Session-Expires			IETF RFC 4028 [8]	
generic-param	"1800"	The recommended initial value is 1800 in IETF RFC 4028 [8].		
Answer-Mode	not present			GROUP-CALL OR CHAT-GROUP-CALL
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.4.1.1.4-2			
MIME body part		**MCVideo-Info**		
MIME-part-body	MCVideo-Info as described in Table A.4.1.1.4-3			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (note), with condition MCVIDEO				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
MIME body part		**Resource list**		PRIVATE-CALL AND NOT re_INVITE
MIME-part-headers				
Content-Type	"application/resource-lists+xml"			
Content-ID	Unique id in format of a Message-ID assigned by the SS	Unique URL identifying the Resource-lists XML MIME body; used as reference in the signature MIME body		
MIME-part-body	Resource list as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-2			
NOTE: With condition re_INVITE for session modifications.				

Table A.4.1.1.4-2: SDP in SIP INVITE (Table A.4.1.1.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-2.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Connection Data		c= line		
Addrtype	"IP4" or "IP6" depending on IP address	This depends on the connection address		
connection-address	IP address of the SS providing the originating participating function			
Bandwidth		b= line		
"AS"	352			
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [10]	
port	port number assigned by the SS providing the originating participating function (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [11] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS providing the originating participating function			
Bandwidth		b= line		
"AS"	37			
"RS"	0		IETF RFC 3556 [12]	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-2.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
"RR"	2000		IETF RFC 3556 [12]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		**Media description for video**		
media description		m= line media = video SDP media-level section for a media-transmission control entity		
port	port number of the audio stream incremented by 2 (resulting in even integer)	The transport port from which the media stream is sent		
fmt	"100"			
Connection Data		c= line		
connection-address	IP address of the SS providing the originating participating function			
Bandwidth		b= line		
"AS"	315			
"RS"	0		IETF RFC 3556 [12]	
"RR"	2500		IETF RFC 3556 [12]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"100"			
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"100"			
format specific parameters		Parameters the H264 codec	IETF RFC 6184 [18]	
packetization-mode	"0"			
profile-level-id	"42e00c"			
sprop-parameter-sets	"J0LgDJWgUH6Af1A=,KM46gA=="			
Media description[3]		Media description for media control		NOT WITHOUT_TRANSMISSIONCONTROL

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-2.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS providing the originating participating function. Different than the port number of the audio and video channels (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS providing the originating participating function			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCVideo"			
format specific parameters				
mc_queueing	not present			
	present	Parameter has no value	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 14.2.2	pc_MCVideo_TransmissionRequestQueueing
mc_priority	The priority value specified in the MCVideo Group configuration	The value of the <user-priority> element for the user in the MCVideo Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.2-1) See note 4	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 14.2.3	GROUP-CALL OR CHAT-GROUP-CALL
	The max priority value specified in the MCVideo Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCVideo Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1)	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 14.2.3	PRIVATE-CALL
mc_reception_priority	The priority value specified in the MCVideo Group configuration	The value of the <user-reception-priority> element for the user in the MCVideo Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.2-1) See note 5	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 14.2.6	GROUP-CALL OR CHAT-GROUP-CALL
	The max priority value specified in the MCVideo Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCVideo Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1)	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 14.2.6	PRIVATE-CALL

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-2.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
NOTE 1: If the SS providing the originating participating function requests a call with media transmission control and implicit transmission (Option b of note on Table A.4.1.1.3-1), apply condition IMPLICIT_GRANT_REQUESTED.				
NOTE 2: If the SS providing the originating participating function requests a call without media transmission (Option c of note on Table A.4.1.1.3-1), apply condition WITHOUT_TRANSMISSIONCONTROL.				
NOTE 3: With condition INITIAL_SDP_OFFER if condition re_INVITE is not in use for Table A.4.1.1.4-1. Otherwise, condition SDP_OFFER is applied.				
NOTE 4: The <num-levels-priority-hierarchy> element for on-network in the MCVideo Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1).				
NOTE 5: The <num-levels-priority-hierarchy> element for on-network in the MCVideo Service Configuration has higher value than any <user-reception-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1).				

Table A.4.1.1.4-3: MCVideo-Info in SIP INVITE (Table A.4.1.1.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2, condition INVITE_REFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcvideoinfo				
mcvideo-Params				
mcvideo-calling-user-id	Encrypted (note) <mcvideo-calling-user-id> with mcvideoURI set to px_MCVideo_ID_User_A	The URI of the calling user		
mcvideo-client-id	Encrypted (note) <mcvideo-client-id> with mcvideoString set to px_MCX_Client_A_ID			GROUP-CALL OR CHAT-GROUP-CALL
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2A.				

Table A.4.1.1.4-4: SIP INVITE from the IUT (MCVideo-Server) (Step 2, Table A.4.1.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (note), with condition MCVIDEO				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCVideo_PublicServiceId_A	The URI of the MCVideo Server providing the terminating participating function.		
Request-URI	same URI as the SS providing the terminating participating function has sent earlier in the Contact header of a response within the same dialog	Contact URI of the SS providing the terminating participating function		re_INVITE
Via				
sent-protocol	"SIP/2.0/UDP"			
sent-by		Address of the other endpoint (the caller)		
host	Host name of the SIP URI being used in the From header			
port	Same port number as in Contact-header	Caller's port number		
Record-Route	not present			
Require	Not present			
Proxy-Require	Not present			
Referred-By			ETSI TS 124 281 [5] (3GPP TS 24.281) clauses 6.3.3.1.2	
value	Default public user id (px_MCX_SIP_PublicUserId_A_1)			
From				

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (note), with condition MCVIDEO				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
addr-spec	tsc_MCVideo_PublicServiceId_B	The URI of the MCVideo Server which is simulated by the SS, providing controlling function.		
To				
addr-spec	tsc_MCVideo_PublicServiceId_A	The URI of the MCVideo Server providing the terminating participating function.		
Resource-Priority			IETF RFC 4412 [13] IETF RFC 7134 [15] IETF RFC 8101 [14] ETSI TS 124 281 [5] (3GPP TS 24.281) clauses 9.2.1.4.1.1 and 10.2.2.4.1	EMERGENCY-CALL or IMMPERIL-CALL
r-value				EMERGENCY-CALL
namespace	value of the <resource-priority-namespace> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-priority	value of the <resource-priority-priority> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-value				IMMPERIL-CALL
namespace	value of the <resource-priority-namespace> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
r-priority	value of the <resource-priority-priority> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVideo		
Call-ID			IETF RFC 3261 [7]	
callid	any allowed value			
Call-ID			IETF RFC 3261 [7]	re_INVITE
callid	same value as in INVITE creating the dialog			
CSeq			IETF RFC 3261 [7]	
value	any allowed value			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (note), with condition MCVIDEO				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
value	value of CSeq sent by the endpoint within its previous request in the same dialog but increased by one			re_INVITE
Max-Forwards			IETF RFC 3261 [7]	
value	any allowed value			
Session-Expires			IETF RFC 4028 [8]	
delta-seconds	any allowed value			
Answer-Mode	not present			GROUP-CALL OR CHAT-GROUP-CALL
P-Asserted-Service				
Service-ID	"urn:urn-7:3gpp-service.ims.icsi.mcvideo"		ETSI TS 124 281 [5] (3GPP TS 24.281), clause 6.3.3.1.2	
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.4.1.1.4-5			
MIME body part		**MCVideo-Info**		
MIME-part-body	MCVideo-Info as described in Table A.4.1.1.4-6			
NOTE: With condition re_INVITE for session modifications.				

Table A.4.1.1.4-5: SDP in SIP INVITE (Table A.4.1.1.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[3]				NOT WITHOUT_ TRANSMIS SIONCONT ROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCVideo"			
format specific parameters				
mc_priority	Any value lower or equal to the priority specified in the MCVideo Group configuration	The value of the <user-priority> element for the user in the MCVideo Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.2-1) See note 4	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 14.2.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	Any value lower or equal to the priority specified in the MCVideo Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCVideo Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1)	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 14.2.3	PRIVATE-CALL

NOTE 1: If the SS providing the originating participating function requests a call without media transmission control (Option c of note on Table A.4.1.1.3-1), apply condition WITHOUT_TRANSMISSIONCONTROL.
NOTE 2: With condition INITIAL_SDP_OFFER if condition re_INVITE is not in use for Table A.4.1.1.4-3. Otherwise, condition SDP_OFFER is applied.
NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the IUT (MCVideo Server).
NOTE 4: The <num-levels-priority-hierarchy> element for on-network in the MCVideo Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1).

Table A.4.1.1.4-6: MCVideo-Info in SIP INVITE (Table A.4.1.1.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcvideoinfo				
mcvideo-Params				
mcvideo-request-uri	Encrypted (note) <mcvideo-request-uri> with mcvideoURI set to px_MCVideo_ID_User_B	The URI of the called user		
mcvideo-calling-user-id	Encrypted (note) <mcvideo-calling-user-id> with mcvideoURI set to px_MCVideo_ID_User_A	The URI of the calling user		
mcvideo-client-id	Encrypted (note) <mcvideo-client-id> with mcvideoString set to px_MCX_Client_A_ID	According to ETSI TS 124 281 [5] (3GPP TS 24.281) clause 6.3.3.1.2, the controlling function shall copy the MCVideo-Info received in the incoming INVITE.	ETSI TS 124 281 [5] (3GPP TS 24.281) clause 6.3.3.1.2	GROUP-CALL OR CHAT-GROUP-CALL

NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-2A

Table A.4.1.1.4-7: SIP 200 (OK) from SS (MCVideo Server) (Step 5, Table A.4.1.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, with condition MCVIDEO and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Supported			ETSI TS 124 281 [5] (3GPP TS 24.281) clause 6.3.2.2.4.2	
option-tag	"tdialog"			
Record-Route	not present			
Session-Expires				
generic-param	3600			
refresher	"uas"		ETSI TS 124 281 [5] (3GPP TS 24.281) clause 6.3.2.2.4.2	
Content-Type			IETF RFC 5621 [16]	
value	"application/sdp"			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.4.1.1.4-8			

Table A.4.1.1.4-8: SDP in SIP 200 (OK) (Table A.4.1.1.4-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-2, condition SDP_ANSWER.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"22222222"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"22222222"			
Session Name	" "	s= line single empty space indicating no session name		
Connection Data		c= line		
Addrtype	"IP4" or "IP6" depending on IP address	This depends on the connection address		
connection-address	IP address of the SS providing the terminating participating function			
Bandwidth		b= line		
"AS"	352			
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [10]	
port	port number assigned by the SS providing the terminating participating function (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [11] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS providing the terminating participating function			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-2, condition SDP_ANSWER.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Bandwidth		b= line		
"AS"	37		ETSI TS 126 114 [9] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [12]	
"RR"	2000		IETF RFC 3556 [12]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		**Media description for video**		
media description		m= line media = video SDP media-level section for a media-transmission control entity		
port	port number of the audio stream incremented by 2 (resulting in even integer)	The transport port from which the media stream is sent		
fmt	"100"			
Connection Data		c= line		
connection-address	IP address of the SS providing the terminating participating function			
Bandwidth		b= line		
"AS"	315			
"RS"	0		IETF RFC 3556 [12]	
"RR"	2500		IETF RFC 3556 [12]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"100"			
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"100"			
format specific parameters	same parameters and values as sent in the corresponding SDP offer	Parameters the H264 codec		
Media description[3]		Media description for media control		NOT WITHOUT_ TRANSMIS SIONCONT ROL

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-2, condition SDP_ANSWER.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS providing the terminating participating function. Different than the port number of the audio and video channels (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS providing the terminating participating function			
NOTE: If the SS providing the originating participating function requests a call without media transmission (Option c of note on Table A.4.1.1.3-1) apply condition WITHOUT_TRANSMISSIONCONTROL.				

Table A.4.1.1.4-9: SIP 200 (OK) from IUT (MCVideo Server) (Step 6, Table A.4.1.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition MCVIDEO and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity			ETSI TS 124 281 [5] (3GPP TS 24.281) clause 6.3.3.2.3.2	
addr-spec				
user-info and host	tsc_MCVideo_PublicServiceId_B	The URI of the IUT (MCVideo Server), providing controlling function.		
port	not present			
Supported			ETSI TS 124 281 [5] (3GPP TS 24.281) clause 6.3.3.2.3.2	
option-tag	"tdialog"			
option-tag	"norefersub"			
option-tag	"nosub"			
Record-Route	not present			
Session-Expires				
generic-param	Same value as session expires header in SIP INVITE or lower			
refresher	"uac"		ETSI TS 124 281 [5] (3GPP TS 24.281) clause 6.3.3.2.3.2	
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.4.1.1.4-10			

Table A.4.1.1.4-10: SDP in SIP 200 (OK) (Table A.4.1.1.4-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-2, condition SDP_ANSWER.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[3]				NOT WITHOUT_TRANSMISSIONCONTROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCVideo"			
format specific parameters				
mc_priority	if a value is provided in the offer: the value defined in the MCVideo Group Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <user-priority> element for the user in the MCVideo Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.7.2-1)	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 14.2.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	if a value is provided in the offer: the value defined in the MCVideo Service Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <num-levels-priority-hierarchy> element for on-network in the MCVideo Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1)	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 14.2.3	PRIVATE-CALL
NOTE 1: If the SS providing the originating participating function requests an implicit transmission (Option b of note on Table A.4.1.1.3-1), more conditions may apply: i. If the IUT (MCVideo Server) implicitly grants the transmission, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied. ii. If the transmission needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies. NOTE 2: If the SS providing the originating participating function requests a call without media transmission (Option c of note on Table A.4.1.1.3-1) apply condition WITHOUT_TRANSMISSIONCONTROL. NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the IUT (MCVideo Server).				

Table A.4.1.1.4-11: Transmission Granted (Step 9a1, Table A.4.1.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.1-1, with condition ACK IF the IUT (MCVideo Server) requests message acknowledgement.

Table A.4.1.1.4-12: Media Transmission Notification (Step 10a1, Table A.4.1.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.7-1, with condition ACK IF the IUT (MCVideo Server) requests message acknowledgement.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
User Id of the Transmitting User		User ID of the user transmitting the media	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.6	
User Id of the Transmitting User	px_MCVideo_ID_User_A			
Audio SSRC of the Transmitting User	Media/Audio SSRC of remote client (Client A)	SSRC of the user transmitting the media	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.16	
Functional Alias	Not present			
	px_MCVideo_ID_FA_A	functional alias URI of the transmitting user	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.21	FA

Table A.4.1.1.4-13: Transmission Idle (Steps 11a1 and 11a2, Table A.4.1.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.16-1.

Table A.4.1.1.4-14: Transmission Control Ack (Steps 9a2 and 10a2, Table A.4.1.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

Table A.4.1.1.4-15: SIP ACK from SS (MCVideo Server) (Step 7, Table A.4.1.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.1.2-1.
NOTE: Elements whose values mention the UE shall be understood as if they were referring to the IUT (MCVideo Server), and SS as SS (MCVideo Server) providing originating participating function.

Table A.4.1.1.4-16: SIP ACK from IUT (MCVideo Server) (Step 8, Table A.4.1.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.1.2-1.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Max-Forwards			IETF RFC 3261 [7]	
value	any allowed value	Non-zero value		
NOTE: Elements whose values mention the UE shall be understood as if they were referring to the SS (MCVideo Server) providing terminating participating function, and SS as the IUT (MCVideo Server).				

A.4.1.2 MCVideo E2E private call establishment, manual commencement

A.4.1.2.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.1.2.2 Definition of system information messages

N/A.

A.4.1.2.3 Procedure

Table A.4.1.2.3-1: MCVideo E2E private call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCVideo Server) providing the originating participating function sends a SIP INVITE requesting the establishment of a private call from SS-UE1 (MCVideo Client) to SS-UE2 (MCVideo Client), with manual commencement.	<--	SIP INVITE	-	-
2	Check: Does the IUT (MCVideo Server) forward the SIP INVITE request to the SS (MCVideo Server) providing the terminating participating function, to establish a private call with SS-UE2 (MCVideo Server)?	-->	SIP INVITE	-	P
3	The SS (MCVideo Server) providing the terminating participating function sends a SIP 100 Trying to the IUT (MCVideo Server).	<--	SIP 100 (Trying)	-	-
4	Check: Does the IUT (MCVideo Server) respond with a SIP 100 Trying to the SS (MCVideo Server) providing the originating participating function?	-->	SIP 100 (Trying)	-	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
5	The SS (MCVideo Server) providing the terminating participating function sends a SIP 180 Ringing to the IUT (MCVideo Server).	<--	SIP 180 (Ringing)	-	-
6	Check: Does the IUT (MCVideo Server) respond with a SIP 180 Ringing to the SS (MCVideo Server) providing the originating participating function?	-->	SIP 180 (Ringing)	-	P
7	The SS (MCVideo Server) providing the terminating participating function responds with a SIP 200 (OK).	<--	SIP 200 (OK)	-	-
8	Check: Does the IUT (MCVideo Server) respond with a SIP 200 (OK) to the SS (MCVideo Server) providing the originating participating function?	-->	SIP 200 (OK)	-	P
9	The SS (MCVideo Server) providing the originating participating function sends a SIP ACK to acknowledge the session establishment to the IUT (MCVideo Server).	<--	SIP ACK	-	-
10	Check: Does the IUT (MCVideo Server) send a SIP ACK to acknowledge the session establishment to the SS (MCVideo Server) providing the terminating participating function?	-->	SIP ACK	-	P
-	EXCEPTION: Step 11a1-11a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS (MCVideo Server) providing originating participating function requests implicit transmission control for SS-UE1 (MCVideo Client), in the SIP INVITE request, but it is not implicitly granted (note).	-	-	-	-
11a1	Check: Does the IUT (MCVideo Server) send a Transmission Granted message to the SS (MCVideo Server) providing originating participating function, to implicitly grant the transmission to SS-UE1 (MCVideo Client)?	-->	Transmission Granted	-	P
11a2	IF the Transmission Granted message requires acknowledgement, the SS (MCVideo Server) providing originating participating function sends a Transmission Control ACK message.	<--	Transmission Control Ack	-	-
-	EXCEPTION: Step 12a1-12a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is granted to SS-UE1 (MCVideo Client), either implicit or explicitly. (Note)	-	-	-	-
12a1	Check: Does the IUT (MCVideo Server) send a Media Transmission Notification message to the SS (MCVideo Server) providing terminating participating function, to inform SS-UE2 (MCVideo Client) that the transmission request has been granted to SS-UE1 (MCVideo Client)?	-->	Media Transmission Notification	-	P
12a2	IF the Media Transmission Notification message requires acknowledgement, the SS (MCVideo Server) providing terminating participating function sends a Transmission Control ACK message.	<--	Transmission Control Ack	-	-
-	EXCEPTION: Steps 13a1-13a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is not granted to SS-UE1 (MCVideo Client). (Note)	-	-	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
13a1	Check: Does the IUT (MCVideo Server) send a Transmission Idle message to the SS (MCVideo Server) providing originating participating function?	-->	Transmission Idle	-	P
13a2	Check: Does the IUT (MCVideo Server) send a Transmission Idle message to the SS (MCVideo Server) providing terminating participating function?	-->	Transmission Idle	-	P

NOTE: Possibilities in SDP-offer/answer depend on the test case requirements:
a. SS (MCVideo Server) providing originating participating function sends SDP offer with media description for transmission control but without implicit transmission request.
b. SS (MCVideo Server) providing originating participating function sends SDP offer with media description for transmission control and with implicit transmission request:
   i. SDP answer from MCVideo Server contains "mc_implicit_request" and "mc_granted" (Transmission is implicitly granted).
   ii. SDP answer from MCVideo Server contains "mc_implicit request" and but no "mc_granted" (Transmission needs to be explicitly granted at step 9a1).
   iii. SDP answer from MCVideo Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the transmission)
c. SS (MCVideo Server) providing originating participating function sends SDP offer without media description for transmission control.

A.4.1.2.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCVIDEO and in the test case calling the procedure with the following clarifications.

Table A.4.1.2.4-1: SIP INVITE from SS (MCVideo Server) (Step 1, Table A.4.1.2.3-1)

Derivation Path: Table A.4.1.1.4-1, with conditions PRIVATE-CALL and MANUAL.
NOTE: Conditions are also applied to all body parts.

Table A.4.1.2.4-2: SIP INVITE from IUT (MCVideo Server) (Step 2, Table A.4.1.2.3-1)

Derivation Path: Table A.4.1.1.4-4, with conditions PRIVATE-CALL and MANUAL.
NOTE: Conditions are also applied to all body parts.

Table A.4.1.2.4-3: SIP 180 (Ringing) from SS (MCVideo Server) (Step 5, Table A.4.1.2.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.2.2-1, with conditions MCVIDEO

Table A.4.1.2.4-4: SIP 180 (Ringing) from IUT (MCVideo Server) (Step 6, Table A.4.1.2.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.2.2-1, with conditions MCVIDEO

Table A.4.1.2.4-5: SIP 200 (OK) from SS (MCVideo Server) (Step 7, Table A.4.1.2.3-1)

Derivation Path: Table A.4.1.1.4-7, with condition PRIVATE-CALL.
NOTE: Conditions are also applied to all body parts.

Table A.4.1.2.4-6: SIP 200 (OK) from IUT (MCVideo Server) (Step 8, Table A.4.1.2.3-1)

Derivation Path: Table A.4.1.1.4-9, with condition PRIVATE-CALL.
NOTE: Conditions are also applied to all body parts.

Table A.4.1.2.4-7: SIP ACK from SS (MCVideo Server) (Step 9, Table A.4.1.2.3-1)

Derivation Path: Table A.4.1.1.4-15

Table A.4.1.2.4-8: SIP ACK from IUT (MCVideo Server) (Step 10, Table A.4.1.2.3-1)

Derivation Path: Table A.4.1.1.4-16

Table A.4.1.2.4-9: Transmission Granted (Step 11a1, Table A.4.1.2.3-1)

Derivation Path: Table A.4.1.1.4-11

Table A.4.1.2.4-10: Media Transmission Notification (Step 12a1, Table A.4.1.2.3-1)

Derivation Path: Table A.4.1.1.4-12

Table A.4.1.2.4-11: Transmission Idle (Steps 13a1 and 13a2, Table A.4.1.2.3-1)

Derivation Path: Table A.4.1.1.4-13

Table A.4.1.2.4-12: Transmission Control Ack (Steps 11a2 and 12a2, Table A.4.1.2.3-1)

Derivation Path: Table A.4.1.1.4-14

A.4.1.3 MCVideo E2E private call establishment, automatic commencement

A.4.1.3.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.1.3.2 Definition of system information messages

N/A.

A.4.1.3.3 Procedure

Table A.4.1.3.3-1: MCVideo E2E private call establishment, automatic commencement mode

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCVideo Server) providing the originating participating function sends a SIP INVITE requesting the establishment of a private call from SS-UE1 (MCVideo Client) to SS-UE2 (MCVideo client), with automatic commencement.	<--	SIP INVITE	-	-
2	Check: Does the IUT (MCVideo Server) forward the SIP INVITE request to the SS (MCVideo Server) providing the terminating participating function, to establish a private call with SS-UE2 (MCVideo Server)?	-->	SIP INVITE	-	P
3	The SS (MCVideo Server) providing the terminating participating function sends a SIP 100 Trying to the IUT (MCVideo Server).	<--	SIP 100 (Trying)	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
4	Check: Does the IUT (MCVideo Server) respond with a SIP 100 Trying to the SS (MCVideo Server) providing the originating participating function?	-->	SIP 100 (Trying)	-	P
5	The SS (MCVideo Server) providing the terminating participating function sends a SIP 183 (Session Progress) reliably.	<--	SIP 183 (Session Progress)	-	-
6	Check: Does the IUT (MCVideo Server) acknowledge the receipt of SIP 183 (Session Progress) from SS (MCVideo Server)?	-->	PRACK	-	P
7	The SS (MCVideo Server) providing the terminating participating function responds a SIP 200 (OK) to PRACK message	<--	SIP 200 (OK)	-	-
8	The SS (MCVideo Server) providing the terminating participating function responds with a SIP 200 (OK).	<--	SIP 200 (OK)	-	-
9	Check: Does the IUT (MCVideo Server) respond with a SIP 200 (OK) to the SS (MCVideo Server) providing the originating participating function?	-->	SIP 200 (OK)	-	P
10	The SS (MCVideo Server) providing the originating participating function sends a SIP ACK to acknowledge the session establishment to the IUT (MCVideo Server).	<--	SIP ACK	-	-
11	Check: Does the IUT (MCVideo Server) send a SIP ACK to acknowledge the session establishment to the SS (MCVideo Server) providing the terminating participating function?	-->	SIP ACK	-	P
-	EXCEPTION: Step 12a1-12a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS (MCVideo Server) providing originating participating function requests implicit transmission control for SS-UE1 (MCVideo Client), in the SIP INVITE request, but it is not implicitly granted (note).	-	-	-	-
12a1	Check: Does the IUT (MCVideo Server) send a Transmission Granted message to the SS (MCVideo Server) providing originating participating function, to implicitly grant the transmission to SS-UE1 (MCVideo Client)?	-->	Transmission Granted	-	P
12a2	IF the Transmission Granted message requires acknowledgement, the SS (MCVideo Server) providing originating participating function sends a Transmission Control ACK message.	<--	Transmission Control Ack	-	-
-	EXCEPTION: Step 13a1-13a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is granted to SS-UE1 (MCVideo Client), either implicit or explicitly. (Note)	-	-	-	-
13a1	Check: Does the IUT (MCVideo Server) send a Media Transmission Notification message to the SS (MCVideo Server) providing terminating participating function, to inform SS-UE2 (MCVideo Client) that the transmission request has been granted to SS-UE1 (MCVideo Client)?	-->	Media Transmission Notification	-	P
13a2	IF the Media Transmission Notification message requires acknowledgement, the SS (MCVideo Server) providing terminating participating function sends a Transmission Control ACK message.	<--	Transmission Control Ack	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
-	EXCEPTION: Steps 14a1-14a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is not granted to SS-UE1 (MCVideo Client). (Note)	-	-	-	-
14a1	Check: Does the IUT (MCVideo Server) send a Transmission Idle message to the SS (MCVideo Server) providing originating participating function?	-->	Transmission Idle	-	P
14a2	Check: Does the IUT (MCVideo Server) send a Transmission Idle message to the SS (MCVideo Server) providing terminating participating function?	-->	Transmission Idle	-	P
NOTE: Possibilities in SDP-offer/answer depend on the test case requirements: a. SS (MCVideo Server) providing originating participating function sends SDP offer with media description for transmission control but without implicit transmission request. b. SS (MCVideo Server) providing originating participating function sends SDP offer with media description for transmission control and with implicit transmission request: i. SDP answer from MCVideo Server contains "mc_implicit_request" and "mc_granted" (Transmission is implicitly granted). ii. SDP answer from MCVideo Server contains "mc_implicit request" and but no "mc_granted" (Transmission needs to be explicitly granted at step 9a1). iii. SDP answer from MCVideo Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the transmission). c. SS (MCVideo Server) providing originating participating function sends SDP offer without media description for transmission control.				

A.4.1.3.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCVIDEO and in the test case calling the procedure with the following clarifications.

Table A.4.1.3.4-1: SIP INVITE from SS (MCVideo Server) (Step 1, Table A.4.1.3.3-1)

Derivation Path: Table A.4.1.1.4-1, with condition PRIVATE-CALL.
NOTE: Conditions are also applied to all body parts.

Table A.4.1.3.4-2: SIP INVITE from IUT (MCVideo Server) (Step 2, Table A.4.1.3.3-1)

Derivation Path: Table A.4.1.1.4-4, with condition PRIVATE-CALL.
NOTE: Conditions are also applied to all body parts.

Table A.4.1.3.4-3: SIP 183 (Session Progress) from SS (MCVideo Server) (Step 5, Table A.4.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.3.2-1, with conditions MCVIDEO and 100rel

Table A.4.1.3.4-4: SIP PRACK from IUT (MCVideo Server) (Step 6, Table A.4.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.10.2-1.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Max-Forwards			IETF RFC 3261 [7]	
value	any allowed value	Non-zero value		
NOTE: Elements whose values mention the UE shall be understood as if they were referring to the SS (MCVideo Server) providing terminating participating function.				

Table A.4.1.3.4-5: SIP 200 (OK) to PRACK from SS (MCVideo Server) (Step 7, Table A.4.1.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1

Table A.4.1.3.4-6: SIP 200 (OK) from SS (MCVideo Server) (Step 8, Table A.4.1.3.3-1)

Derivation Path: Table A.4.1.1.4-7, with condition PRIVATE-CALL.
NOTE: Conditions are also applied to all body parts.

Table A.4.1.3.4-7: SIP 200 (OK) from IUT (MCVideo Server) (Step 9, Table A.4.1.3.3-1)

Derivation Path: Table A.4.1.1.4-9, with condition PRIVATE-CALL.
NOTE: Conditions are also applied to all body parts.

Table A.4.1.3.4-8: SIP ACK from SS (MCVideo Server) (Step 10, Table A.4.1.3.3-1)

Derivation Path: Table A.4.1.1.4-15

Table A.4.1.3.4-9: SIP ACK from IUT (MCVideo Server) (Step 11, Table A.4.1.3.3-1)

Derivation Path: Table A.4.1.1.4-16

Table A.4.1.3.4-10: Transmission Granted (Step 12a1, Table A.4.1.3.3-1)

Derivation Path: Table A.4.1.1.4-11

Table A.4.1.3.4-11: Media Transmission Notification (Step 13a1, Table A.4.1.3.3-1)

Derivation Path: Table A.4.1.1.4-12

Table A.4.1.3.4-12: Transmission Idle (Steps 14a1 and 14a2, Table A.4.1.3.3-1)

Derivation Path: Table A.4.1.1.4-13

Table A.4.1.3.4-13: Transmission Control Ack (Steps 12a2 and 13a2, Table A.4.1.3.3-1)

Derivation Path: Table A.4.1.1.4-14

A.4.1.4 MCVideo E2E group call establishment

A.4.1.4.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.1.4.2 Definition of system information messages

N/A.

A.4.1.4.3 Procedure

Table A.4.1.4.3-1: MCVideo E2E group call establishment

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCVideo Server) providing the originating participating function sends a SIP INVITE requesting the establishment of a group call from SS-UE1 (MCVideo Client).	<--	SIP INVITE	-	-
2	Check: Does the IUT (MCVideo Server) forward the SIP INVITE request to the SS (MCVideo Server) providing the terminating participating function, to establish a group call with SS-UE2 (MCVideo Server)?	-->	SIP INVITE	-	P
3	The SS (MCVideo Server) providing the terminating participating function sends a SIP 100 Trying to the IUT (MCVideo Server).	<--	SIP 100 (Trying)	-	-
4	Check: Does the IUT (MCVideo Server) respond with a SIP 100 Trying to the SS (MCVideo Server) providing the originating participating function?	-->	SIP 100 (Trying)	-	P
5	The SS (MCVideo Server) providing the terminating participating function sends a SIP 183 (Session Progress) reliably.	<--	SIP 183 (Session Progress)	-	-
6	Check: Does the IUT (MCVideo Server) acknowledge the receipt of SIP 183 (Session Progress) from SS (MCVideo Server)?	-->	PRACK	-	P
7	The SS (MCVideo Server) providing the terminating participating function responds a SIP 200 (OK) to PRACK message	<--	SIP 200 (OK)	-	-
8	The SS (MCVideo Server) providing the terminating participating function responds with a SIP 200 (OK).	<--	SIP 200 (OK)	-	-
9	Check: Does the IUT (MCVideo Server) respond with a SIP 200 (OK) to the SS (MCVideo Server) providing the originating participating function?	-->	SIP 200 (OK)	-	P
10	The SS (MCVideo Server) providing the originating participating function sends a SIP ACK to acknowledge the session establishment to the IUT (MCVideo Server).	<--	SIP ACK	-	-
11	Check: Does the IUT (MCVideo Server) send a SIP ACK to acknowledge the session establishment to the SS (MCVideo Server) providing the terminating participating function?	-->	SIP ACK	-	P
-	EXCEPTION: Step 12a1-12a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS (MCVideo Server) providing originating participating function requests implicit transmission control for SS-UE1 (MCVideo Client), in the SIP INVITE request, but it is not implicitly granted (note).	-	-	-	-
12a1	Check: Does the IUT (MCVideo Server) send a Transmission Granted message to the SS (MCVideo Server) providing originating participating function, to implicitly grant the transmission to SS-UE1 (MCVideo Client)?	-->	Transmission Granted	-	P
12a2	IF the Transmission Granted message requires acknowledgement, the SS (MCVideo Server) providing originating participating function sends a Transmission Control ACK message.	<--	Transmission Control Ack	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
-	EXCEPTION: Step 13a1-13a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is granted to SS-UE1 (MCVideo Client), either implicit or explicitly. (Note)	-	-	-	-
13a1	Check: Does the IUT (MCVideo Server) send a Media Transmission Notification message to the SS (MCVideo Server) providing terminating participating function, to inform SS-UE2 (MCVideo Client) that the transmission request has been granted to SS-UE1 (MCVideo Client)?	-->	Media Transmission Notification	-	P
13a2	IF the Media Transmission Notification message requires acknowledgement, the SS (MCVideo Server) providing terminating participating function sends a Transmission Control ACK message.	<--	Transmission Control Ack	-	-
-	EXCEPTION: Steps 14a1-14a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is not granted to SS-UE1 (MCVideo Client). (Note)	-	-	-	-
14a1	Check: Does the IUT (MCVideo Server) send a Transmission Idle message to the SS (MCVideo Server) providing originating participating function?	-->	Transmission Idle	-	P
14a2	Check: Does the IUT (MCVideo Server) send a Transmission Idle message to the SS (MCVideo Server) providing terminating participating function?	-->	Transmission Idle	-	P

NOTE: Possibilities in SDP-offer/answer depend on the test case requirements
a. SS (MCVideo Server) providing originating participating function sends SDP offer with media description for transmission control but without implicit transmission request.
b. SS (MCVideo Server) providing originating participating function sends SDP offer with media description for transmission control and with implicit transmission request:
i. SDP answer from MCVideo Server contains "mc_implicit_request" and "mc_granted" (Transmission is implicitly granted).
ii. SDP answer from MCVideo Server contains "mc_implicit request" and but no "mc_granted" (Transmission needs to be explicitly granted at step 9a1).
iii. SDP answer from MCVideo Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the transmission).
c. SS (MCVideo Server) providing originating participating function sends SDP offer without media description for transmission control.

A.4.1.4.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCVIDEO and in the test case calling the procedure with the following clarifications.

Table A.4.1.4.4-1: SIP INVITE from SS (MCVideo Server) (Step 1, Table A.4.1.4.3-1)

Derivation Path: Table A.4.1.1.4-1, with condition GROUP -CALL.
NOTE 1: Conditions are also applied to all body parts.
NOTE 2: For chat group calls CHAT-GROUP-CALL condition shall be used instead of GROUP-CALL.

Table A.4.1.4.4-2: SIP INVITE from IUT (MCVideo Server) (Step 2, Table A.4.1.4.3-1)

Derivation Path: Table A.4.1.1.4-4, with condition GROUP -CALL.
NOTE 1: Conditions are also applied to all body parts.
NOTE 2: For chat group calls CHAT-GROUP-CALL condition shall be used instead of GROUP-CALL.

Table A.4.1.4.4-3: SIP 183 (Session Progress) from SS (MCVideo Server) (Step 5, Table A.4.1.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.3.2-1, with conditions MCVIDEO and 100rel.

Table A.4.1.4.4-4: SIP PRACK from IUT (MCVideo Server) (Step 6, Table A.4.1.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.10.2-1.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Max-Forwards			IETF RFC 3261 [7]	
value	any allowed value	Non-zero value		
NOTE: Elements whose values mention the UE shall be understood as if they were referring to the SS (MCVideo Server) providing terminating participating function.				

Table A.4.1.4.4-5: SIP 200 (OK) to PRACK from SS (MCVideo Server) (Step 7, Table A.4.1.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1.

Table A.4.1.4.4-6: SIP 200 (OK) from SS (MCVideo Server) (Step 8, Table A.4.1.4.3-1)

Derivation Path: Table A.4.1.1.4-7, with condition GROUP -CALL.
NOTE 1: Conditions are also applied to all body parts. NOTE 2: For chat group calls CHAT-GROUP-CALL condition shall be used instead of GROUP-CALL.

Table A.4.1.4.4-7: SIP 200 (OK) from IUT (MCVideo Server) (Step 9, Table A.4.1.4.3-1)

Derivation Path: Table A.4.1.1.4-9, with condition GROUP-CALL.
NOTE 1: Conditions are also applied to all body parts. NOTE 2: For chat group calls CHAT-GROUP-CALL condition shall be used instead of GROUP-CALL.

Table A.4.1.4.4-8: SIP ACK from SS (MCVideo Server) (Step 10, Table A.4.1.4.3-1)

Derivation Path: Table A.4.1.1.4-15

Table A.4.1.4.4-9: SIP ACK from IUT (MCVideo Server) (Step 11, Table A.4.1.4.3-1)

Derivation Path: Table A.4.1.1.4-16

Table A.4.1.4.4-10: Transmission Granted (Step 12a1, Table A.4.1.4.3-1)

Derivation Path: Table A.4.1.1.4-11

Table A.4.1.4.4-11: Media Transmission Notification (Step 13a1, Table A.4.1.4.3-1)

Derivation Path: Table A.4.1.1.4-12

Table A.4.1.4.4-12: Transmission Idle (Steps 14a1 and 14a2, Table A.4.1.4.3-1)

Derivation Path: Table A.4.1.1.4-13

Table A.4.1.4.4-13: Transmission Control Ack (Steps 12a2 and 13a2, Table A.4.1.4.3-1)

Derivation Path: Table A.4.1.1.4-14

A.4.1.5 MCVideo E2E call release

A.4.1.5.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.1.5.2 Definition of system information messages

N/A.

A.4.1.5.3 Procedure

Table A.4.1.5.3-1: MCVideo E2E call release

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCVideo Server) providing the originating participating function sends a SIP BYE to terminate the call.	<--	SIP BYE	-	-
2	Check: Does the IUT (MCVideo Server) forward the SIP BYE request to the SS (MCVideo Server) providing the terminating participating function?	-->	SIP BYE	-	P
3	The SS (MCVideo Server) providing the terminating participating function responds with a 200 (OK) to the IUT (MCVideo Server).	<--	SIP 200 (OK)	-	-
4	Check: Does the IUT (MCVideo Server) forward the SIP 200 (OK) response to the SS (MCVideo Server) providing the originating participating function?	-->	SIP 200 (OK)	-	P

A.4.1.5.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCVIDEO and in the test case calling the procedure with the following clarifications.

Table A.4.1.5.4-1: SIP BYE from the SS (MCVideo Server) (Step 1, Table A.4.1.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1.
NOTE: References to UE in subclauses relate to the IUT (MCVideo Server) and SS references to the SS (MCVideo Server) providing the originating participating function.

Table A.4.1.5.4-2: SIP BYE from the IUT (MCVideo Server) (Step 2, Table A.4.1.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1.
NOTE: References to UE in subclauses relate to the SS (MCVideo Server) providing the terminating participating function and SS references to the IUT (MCVideo Server).

Table A.4.1.5.4-3: SIP 200 OK from the SS (MCVideo Server) (Step 3, Table A.4.1.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1.

Table A.4.1.5.4-4: SIP 200 OK from the IUT (MCVideo Server) (Step 4, Table A.4.1.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1.

A.4.1.6 MCVideo CO session establishment/modification without provisional responses other than 100 Trying

A.4.1.6.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.1.6.2 Definition of system information messages

N/A.

A.4.1.6.3 Procedure

Table A.4.1.6.3-1: MCVideo CO session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCVideo Server) providing the originating participating function sends a SIP INVITE requesting the establishment/modification of a call from SS-UE1 (MCVideo Client). (Note 2)	<--	SIP INVITE	-	-
2	Check: Does the IUT (MCVideo Server) respond with a SIP 100 Trying to the SS (MCVideo Server) providing the originating participating function?	-->	SIP 100 (Trying)	-	P
3	Check: Does the IUT (MCVideo Server) respond with a SIP 200 (OK) to the SS (MCVideo Server) providing the originating participating function?	-->	SIP 200 (OK)	-	P
4	The SS (MCVideo Server) providing the originating participating function sends a SIP ACK to acknowledge the session establishment/modification to the IUT (MCVideo Server).	<--	SIP ACK	-	-
-	EXCEPTION: Step 5a1-5a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS (MCVideo Server) providing originating participating function requests implicit transmission control for SS-UE (MCVideo Client), in the SIP INVITE request, but it is not implicitly granted (Notes 1 and 2).	-	-	-	-
5a1	Check: Does the IUT (MCVideo Server) send a Transmission Granted message to the SS (MCVideo Server) providing originating participating function, to implicitly grant the transmission to SS-UE (MCVideo Client)? (Note 2)	-->	Transmission Granted	-	P
5a2	IF the Transmission Granted message requires acknowledgement, the SS (MCVideo Server) providing originating participating function sends a Transmission Control ACK message.	<--	Transmission Control Ack	-	-
-	EXCEPTION: Step 6a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is not granted to SS-UE (MCVideo Client). (Notes 1 and 2)	-	-	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
6a1	Check: Does the IUT (MCVideo Server) send a Transmission Idle message to the SS (MCVideo Server) providing originating participating function?	-->	Transmission Idle	-	P

NOTE 1: Possibilities in SDP-offer/answer depend on the test case requirements:
- a. SS (MCVideo Server) providing originating participating function sends SDP offer with media description for transmission control but without implicit transmission request.
- b. SS (MCVideo Server) providing originating participating function sends SDP offer with media description for transmission control and with implicit transmission request:
  - i. SDP answer from MCVideo Server contains "mc_implicit_request" and "mc_granted" (Transmission is implicitly granted).
  - ii. SDP answer from MCVideo Server contains "mc_implicit request" and but no "mc_granted" (Transmission needs to be explicitly granted at step 9a1).
  - iii. SDP answer from MCVideo Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the transmission).
- c. SS (MCVideo Server) providing originating participating function sends SDP offer without media description for transmission control.

NOTE 2: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCVideo Client) will be assumed.

A.4.1.6.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCVIDEO and in the test case calling the procedure with the following clarifications.

Table A.4.1.6.4-1: SIP INVITE from SS (MCVideo Server) (Step 1, Table A.4.1.6.3-1)

Derivation Path: Table A.4.1.1.4-1.

Table A.4.1.6.4-2: SIP 200 (OK) from IUT (MCVideo Server) (Step 3, Table A.4.1.6.3-1)

Derivation Path: Table A.4.1.1.4-9.

Table A.4.1.6.4-3: SIP ACK from SS (MCVideo Server) (Step 4, Table A.4.1.6.3-1)

Derivation Path: Table A.4.1.1.4-15

Table A.4.1.6.4-4: Transmission Granted (Step 5a1, Table A.4.1.6.3-1)

Derivation Path: Table A.4.1.1.4-11

Table A.4.1.6.4-5: Transmission Idle (Step 6a1, Table A.4.1.6.3-1)

Derivation Path: Table A.4.1.1.4-13

Table A.4.1.6.4-6: Transmission Control Ack (Step 5a2, Table A.4.1.6.3-1)

Derivation Path: Table A.4.1.1.4-14

A.4.2 Generic test procedures for MCVideo Originating Participating Server function

A.4.2.1 MCVideo CO session establishment/modification without provisional responses other than 100 Trying

A.4.2.1.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.2.1.2 Definition of system information messages

N/A.

A.4.2.1.3 Procedure

Table A.4.2.1.3-1: MCVideo CO session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCVideo Client) sends a SIP INVITE requesting the establishment/modification of a call.	<--	SIP INVITE	-	-
2	Check: Does the IUT (MCVideo Server) forward the SIP INVITE request to the SS (MCVideo Server) providing the controlling function, to establish/modify a call?	-->	SIP INVITE	-	P
3	The SS (MCVideo Server) providing the controlling function sends a SIP 100 Trying to the IUT (MCVideo Server).	<--	SIP 100 (Trying)	-	-
4	Check: Does the IUT (MCVideo Server) respond with a SIP 100 Trying to the SS-UE1 (MCVideo Client)?	-->	SIP 100 (Trying)	-	P
5	The SS (MCVideo Server) providing the controlling function responds with a SIP 200 (OK).	<--	SIP 200 (OK)	-	-
6	Check: Does the IUT (MCVideo Server) respond with a SIP 200 (OK) to the SS-UE1 (MCVideo Client)?	-->	SIP 200 (OK)	-	P
7	The SS-UE1 (MCVideo Client) sends a SIP ACK to acknowledge the session establishment/modification to the IUT (MCVideo Server).	<--	SIP ACK	-	-
8	Check: Does the IUT (MCVideo Server) send a SIP ACK to acknowledge the session establishment/modification to the SS (MCVideo Server) providing the controlling function?	-->	SIP ACK	-	P
-	EXCEPTION: Step 9a1-9a4 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS-UE1 (MCVideo Client) requests implicit transmission control in the SIP INVITE request, but it is not implicitly granted (Note).	-	-	-	-
9a1	The SS (MCVideo Server) providing the controlling function sends a Transmission Granted message with request for acknowledgement to the IUT (MCVideo Server), to grant the transmission to SS-UE1 (MCVideo Client).	<--	Transmission Granted	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
9a2	Check: Does the IUT (MCVideo Server) forward the Transmission Granted message to the SS-UE1 (MCVideo Client)?	-->	Transmission Granted	-	P
9a3	The SS-UE1 (MCVideo Client) sends a Transmission Control ACK message.	<--	Transmission Control Ack	-	-
9a4	Check: Does the IUT (MCVideo Server) forward the acknowledgement of SS-UE1 (MCVideo Client) to SS (MCVideo Server)?	-->	Transmission Control Ack	-	P
-	EXCEPTION: Steps 10a1-10a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is not granted to SS-UE1 (MCVideo Client). (Note)	-	-	-	-
10a1	The SS (MCVideo Server) providing the controlling function sends a Transmission Idle message to the IUT (MCVideo Server).	<--	Transmission Idle	-	-
10a2	Check: Does the IUT (MCVideo Server) forward the Transmission Idle message to the SS-UE1 (MCVideo Client)?	-->	Transmission Idle	-	P

NOTE: Possibilities in SDP-offer/answer depend on the test case requirements:
a. SS (MCVideo Server) providing originating participating function sends SDP offer with media description for transmission control but without implicit transmission request.
b. SS (MCVideo Server) providing originating participating function sends SDP offer with media description for transmission control and with implicit transmission request:
   i. SDP answer from MCVideo Server contains "mc_implicit_request" and "mc_granted" (Transmission is implicitly granted).
   ii. SDP answer from MCVideo Server contains "mc_implicit request" and but no "mc_granted" (Transmission needs to be explicitly granted at step 9a1).
   iii. SDP answer from MCVideo Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the transmission).
c. SS (MCVideo Server) providing originating participating function sends SDP offer without media description for transmission control.

A.4.2.1.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCVIDEO and in the test case calling the procedure with the following clarifications.

Table A.4.2.1.4-1: SIP INVITE from SS-UE1 (MCVideo Client) (Step 1, Table A.4.2.1.3-1)

Derivation Path: Table A.3.1.4-1

Table A.4.2.1.4-2: SIP INVITE from IUT (MCVideo Server) (Step 2, Table A.4.2.1.3-1)

Derivation Path: Table A.4.1.1.4-1

Table A.4.2.1.4-3: SIP 200 (OK) from SS (MCVideo Server) (Step 5, Table A.4.2.1.3-1)

Derivation Path: Table A.4.1.1.4-9

Table A.4.2.1.4-4: SIP 200 (OK) from IUT (MCVideo Server) (Step 6, Table A.4.2.1.3-1)

Derivation Path: Table A.3.1.4-9

Table A.4.2.1.4-5: SIP ACK from SS-UE1 (MCVideo Client) (Step 7, Table A.4.2.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.1.1-1.
NOTE: Elements whose values mention the UE shall be understood as if they were referring to the SS-UE1 (MCVideo Client), and SS as IUT (MCVideo Server) providing the originating participating function.

Table A.4.2.1.4-6: SIP ACK from IUT (MCVideo Server) (Step 8, Table A.4.2.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.1.2-1.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Max-Forwards			IETF RFC 3261 [7]	
value	any allowed value	Non-zero value		
NOTE: Elements whose values mention the UE shall be understood as if they were referring to the SS (MCVideo Server) providing the controlling function, and SS as the IUT (MCVideo Server).				

Table A.4.2.1.4-7: Transmission Granted (Steps 9a1 and 9a2, Table A.4.2.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.1-1, with condition ACK

Table A.4.2.1.4-8: Transmission Idle (Steps 10a1 and 10a2, Table A.4.2.1.3-1)

Derivation Path: Table A.4.1.1.4-13

Table A.4.2.1.4-9: Transmission Control Ack (Steps 9a3 and 9a4, Table A.4.2.1.3-1)

Derivation Path: Table A.4.1.1.4-14

A.4.2.2 MCVideo CO private call establishment, manual commencement

A.4.2.2.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.2.2.2 Definition of system information messages

N/A.

A.4.2.2.3 Procedure

Table A.4.2.2.3-1: MCVideo CO private call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCVideo Client) sends a SIP INVITE requesting the establishment of a private call with SS-UE2 (MCVideo Client).	<--	SIP INVITE	-	-
2	Check: Does the IUT (MCVideo Server) forward the SIP INVITE request to the SS (MCVideo Server) providing the controlling function, to establish a call with SS-UE2 (MCVideo Client)?	-->	SIP INVITE	-	P
3	The SS (MCVideo Server) providing the controlling function sends a SIP 100 Trying to the IUT (MCVideo Server).	<--	SIP 100 (Trying)	-	-
4	Check: Does the IUT (MCVideo Server) respond with a SIP 100 Trying to the SS-UE1 (MCVideo Client)?	-->	SIP 100 (Trying)	-	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
5	The SS (MCVideo Server) providing the controlling function sends a SIP 180 Ringing to the IUT (MCVideo Server).	<--	SIP 180 (Ringing)	-	-
6	Check: Does the IUT (MCVideo Server) respond with a SIP 180 Ringing to the SS-UE1 (MCVideo Client)?	-->	SIP 180 (Ringing)	-	P
7	The SS (MCVideo Server) providing the controlling function responds with a SIP 200 (OK).	<--	SIP 200 (OK)	-	-
8	Check: Does the IUT (MCVideo Server) respond with a SIP 200 (OK) to the SS-UE1 (MCVideo Client)?	-->	SIP 200 (OK)	-	P
9	The SS-UE1 (MCVideo Client) sends a SIP ACK to acknowledge the session establishment to the IUT (MCVideo Server).	<--	SIP ACK	-	-
10	Check: Does the IUT (MCVideo Server) send a SIP ACK to acknowledge the session establishment to the SS (MCVideo Server) providing the controlling function?	-->	SIP ACK	-	P
-	EXCEPTION: Step 11a1-11a4 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS-UE1 (MCVideo Client) requests implicit transmission control in the SIP INVITE request, but it is not implicitly granted (Note).	-	-	-	-
11a1	The SS (MCVideo Server) providing the controlling function sends a Transmission Granted message with request for acknowledgement to the IUT (MCVideo Server), to grant the transmission to SS-UE1 (MCVideo Client).	<--	Transmission Granted	-	P
11a2	Check: Does the IUT (MCVideo Server) forward the Transmission Granted message to the SS-UE1 (MCVideo Client)?	-->	Transmission Granted	-	P
11a3	The SS-UE1 (MCVideo Client) sends a Transmission Control ACK message.	<--	Transmission Control Ack	-	-
11a4	Check: Does the IUT (MCVideo Server) forward the acknowledgement of SS-UE1 (MCVideo Client) to SS (MCVideo Server)?	-->	Transmission Control Ack	-	P
-	EXCEPTION: Steps 12a1-12a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is not granted to SS-UE1 (MCVideo Client). (Note)	-	-	-	-
12a1	The SS (MCVideo Server) providing the controlling function sends a Transmission Idle message to the IUT (MCVideo Server).	<--	Transmission Idle	-	-
12a2	Check: Does the IUT (MCVideo Server) forward the Transmission Idle message to the SS-UE1 (MCVideo Client)?	-->	Transmission Idle	-	P

NOTE 1: Possibilities in SDP-offer/answer depend on the test case requirements:

a. SS (MCVideo Server) providing originating participating function sends SDP offer with media description for transmission control but without implicit transmission request.

b. SS (MCVideo Server) providing originating participating function sends SDP offer with media description for transmission control and with implicit transmission request:

   i. SDP answer from MCVideo Server contains "mc_implicit_request" and "mc_granted" (Transmission is implicitly granted).

   ii. SDP answer from MCVideo Server contains "mc_implicit request" and but no "mc_granted" (Transmission needs to be explicitly granted at step 9a1).

   iii. SDP answer from MCVideo Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the transmission).

c. SS (MCVideo Server) providing originating participating function sends SDP offer without media description for transmission control.

A.4.2.2.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCVIDEO and in the test case calling the procedure with the following clarifications.

Table A.4.2.2.4-1: SIP INVITE from SS-UE1 (MCVideo Client) (Step 1, Table A.4.2.2.3-1)

Derivation Path: Table A.4.2.1.4-1, with conditions PRIVATE-CALL and MANUAL.
NOTE: Conditions are also applied to all body parts.

Table A.4.2.2.4-2: SIP INVITE from IUT (MCVideo Server) (Step 2, Table A.4.2.2.3-1)

Derivation Path: Table A.4.2.1.4-2, with conditions PRIVATE-CALL and MANUAL.
NOTE: Conditions are also applied to all body parts.

Table A.4.2.2.4-3: SIP 180 (Ringing) from SS (MCVideo Server) (Step 5, Table A.4.2.2.3-1)

Derivation Path: Table A.4.1.2.4-4

Table A.4.2.2.4-4: SIP 180 (Ringing) from IUT (MCVideo Server) (Step 6, Table A.4.2.2.3-1)

Derivation Path: Table A.3.3.4-9

Table A.4.2.2.4-5: SIP 200 (OK) from SS (MCVideo Server) (Step 7, Table A.4.2.2.3-1)

Derivation Path: Table A.4.2.1.4-3, with condition PRIVATE-CALL.
NOTE: Conditions are also applied to all body parts.

Table A.4.2.2.4-6: SIP 200 (OK) from IUT (MCVideo Server) (Step 8, Table A.4.2.2.3-1)

Derivation Path: Table A.4.2.1.4-4, with condition PRIVATE-CALL.
NOTE: Conditions are also applied to all body parts.

Table A.4.2.2.4-7: SIP ACK from SS-UE1 (MCVideo Client) (Step 9, Table A.4.2.2.3-1)

Derivation Path: Table A.4.2.1.4-5

Table A.4.2.2.4-8: SIP ACK from IUT (MCVideo Server) (Step 10, Table A.4.2.2.3-1)

Derivation Path: Table A.4.2.1.4-6

Table A.4.2.2.4-9: Transmission Granted (Steps 11a1 and 11a2, Table A.4.2.2.3-1)

Derivation Path: Table A.4.2.1.4-7

Table A.4.2.2.4-10: Transmission Idle (Steps 12a1 and 12a2, Table A.4.2.2.3-1)

Derivation Path: Table A.4.2.1.4-8

Table A.4.2.2.4-11: Transmission Control Ack (Steps 11a3 and 11a4, Table A.4.2.2.3-1)

Derivation Path: Table A.4.2.1.4-9

A.4.2.3 MCVideo CO call release

A.4.2.3.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.2.3.2 Definition of system information messages

N/A.

A.4.2.3.3 Procedure

Table A.4.2.3.3-1: MCVideo CO call release

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCVideo Client) sends a SIP BYE to terminate the call.	<--	SIP BYE	-	-
2	Check: Does the IUT (MCVideo Server) forward the SIP BYE request to the SS (MCVideo Server) providing the controlling function?	-->	SIP BYE	-	P
3	The SS (MCVideo Server) providing the controlling function responds with a 200 (OK) to the IUT (MCVideo Server).	<--	SIP 200 (OK)	-	-
4	Check: Does the IUT (MCVideo Server) forward the SIP 200 (OK) response to the SS-UE1 (MCVideo Client)?	-->	SIP 200 (OK)	-	P

A.4.2.3.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCVIDEO and in the test case calling the procedure with the following clarifications.

Table A.4.2.3.4-1: SIP BYE from the SS-UE1 (Step 1, Table A.4.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.1-1.
NOTE: References to UE in subclauses relate to the SS-UE1 (MCVideo Client) and SS references to the IUT (MCVideo Server).

Table A.4.2.3.4-2: SIP BYE from the IUT (MCVideo Server) (Step 2, Table A.4.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1.
NOTE: References to UE in subclauses relate to the SS (MCVideo Server) providing the controlling function and SS references to the IUT (MCVideo Server).

Table A.4.2.3.4-3: SIP 200 OK from the SS (MCVideo Server) (Step 3, Table A.4.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1.

Table A.4.2.3.4-4: SIP 200 OK from the IUT (MCVideo Server) (Step 4, Table A.4.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1.

A.4.2.4 MCVideo Transmission Request - Transmission Granted

A.4.2.4.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.2.4.2 Definition of system information messages

N/A.

A.4.2.4.3 Procedure

Table A.4.2.4.3-1: MCVideo Transmission Request - Transmission Granted

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCVideo Client) sends a Transmission Request message, with request for acknowledgement. (Note)	<--	Transmission Request	-	-
2	Check: Does the IUT (MCVideo Server) forward the Transmission Request message to the SS (MCVideo Server) providing the controlling function?	-->	Transmission Request	-	P
3	The SS (MCVideo Server) acknowledges the Transmission Request message.	<--	Transmission Control Ack	-	-
4	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
5	The SS (MCVideo Server) providing the controlling function sends a Transmission Granted message with request for acknowledgement to IUT (MCVideo Server).	<--	Transmission Granted	-	-
6	Check: Does the IUT (MCVideo Server) forward the Transmission Granted message to SS-UE (MCVideo Client)? (Note)	-->	Transmission Granted	-	P
7	The SS-UE (MCVideo Client) acknowledges the Transmission Granted message. (Note)	<--	Transmission Control Ack	-	-
8	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCVideo Client) will be assumed.					

A.4.2.4.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications.

Table A.4.2.4.4-1: Transmission Request (Steps 1 and 2, Table A.4.2.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.1.1-1, with condition ACK

Table A.4.2.4.4-2: Transmission Granted (Steps 5 and 6, Table A.4.2.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.1-1, with condition ACK

Table A.4.2.4.4-3: Transmission Control Ack (Steps 3 and 4, Table A.4.2.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition DOWNLINK

Table A.4.2.4.4-4: Transmission Control Ack (Steps 7 and 8, Table A.4.2.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

A.4.2.5 MCVideo CO Transmission End Request - Transmission End Response

A.4.2.5.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.2.5.2 Definition of system information messages

N/A.

A.4.2.5.3 Procedure

Table A.4.2.5.3-1: MCVideo CO Transmission End Request - Transmission End Response

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCVideo Client) sends a Transmission End Request message, with request for acknowledgement. (Note)	<--	Transmission End Request	-	-
2	Check: Does the IUT (MCVideo Server) forward the Transmission End Request message to the SS (MCVideo Server) providing the controlling function?	-->	Transmission End Request	-	P
3	The SS (MCVideo Server) acknowledges the Transmission End Request message.	<--	Transmission Control Ack	-	-
4	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
5	The SS (MCVideo Server) providing the controlling function sends a Transmission End Response message with request for acknowledgement to IUT (MCVideo Server).	<--	Transmission End Response	-	-
6	Check: Does the IUT (MCVideo Server) forward the Transmission End Response message to SS-UE (MCVideo Client)? (Note)	-->	Transmission End Response	-	P
7	The SS-UE (MCVideo Client) acknowledges the Transmission End Response message. (Note)	<--	Transmission Control Ack	-	-
8	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
9	The SS (MCVideo Server) providing the controlling function sends a Transmission Idle message to IUT (MCVideo Server).	<--	Transmission Idle	-	-
10	Check: Does the IUT (MCVideo Server) forward the Transmission Idle message to SS-UE (MCVideo Client)? (Note)	-->	Transmission Idle	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCVideo Client) will be assumed.					

A.4.2.5.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications.

Table A.4.2.5.4-1: Transmission End Request (Steps 1 and 2, Table A.4.2.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.1-1, with conditions ACK and UPLINK

Table A.4.2.5.4-2: Transmission End Response (Steps 5 and 6, Table A.4.2.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.2-1, with conditions ACK and DOWNLINK

Table A.4.2.5.4-3: Transmission Idle (Steps 9 and 10, Table A.4.2.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.16-1

Table A.4.2.5.4-4: Transmission Control Ack (Steps 3 and 4, Table A.4.2.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition DOWNLINK

Table A.4.2.5.4-5: Transmission Control Ack (Steps 7 and 8, Table A.4.2.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

A.4.2.6 MCVideo CT Transmission End Request - Transmission End Response

A.4.2.6.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.2.6.2 Definition of system information messages

N/A.

A.4.2.6.3 Procedure

Table A.4.2.6.3-1: MCVideo CT Transmission End Request - Transmission End Response

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCVideo Server) providing the controlling function sends a Transmission End Request message, with request for acknowledgement.	<--	Transmission End Request	-	-
2	Check: Does the IUT (MCVideo Server) forward the Transmission End Request message to the SS-UE (MCVideo Client)? (Note)	-->	Transmission End Request	-	P
3	The SS-UE (MCVideo Client) acknowledges the Transmission End Request message. (Note)	<--	Transmission Control Ack	-	-
4	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
5	The SS-UE (MCVideo Client) sends a Transmission End Response message with request for acknowledgement to IUT (MCVideo Server). (Note)	<--	Transmission End Response	-	-
6	Check: Does the IUT (MCVideo Server) forward the Transmission End Response message to SS (MCVideo Server)?	-->	Transmission End Response	-	P
7	The SS (MCVideo Server) acknowledges the Transmission End Response message.	<--	Transmission Control Ack	-	-
8	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
9	The SS (MCVideo Server) providing the controlling function sends a Transmission Idle message to IUT (MCVideo Server).	<--	Transmission Idle	-	-
10	Check: Does the IUT (MCVideo Server) forward the Transmission Idle message to SS-UE (MCVideo Client)? (Note)	-->	Transmission Idle	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCVideo Client) will be assumed.					

A.4.2.6.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications.

Table A.4.2.6.4-1: Transmission End Request (Steps 1 and 2, Table A.4.2.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.1-1, with conditions ACK and DOWNLINK

Table A.4.2.6.4-2: Transmission End Response (Steps 5 and 6, Table A.4.2.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.2-1, with conditions ACK and UPLINK

Table A.4.2.6.4-3: Transmission Idle (Steps 9 and 10, Table A.4.2.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.16-1

Table A.4.2.6.4-4: Transmission Control Ack (Steps 3 and 4, Table A.4.2.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

Table A.4.2.6.4-5: Transmission Control Ack (Steps 7 and 8, Table A.4.2.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition DOWNLINK

A.4.2.7 MCVideo Transmission Request - Transmission Rejected

A.4.2.7.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.2.7.2 Definition of system information messages

N/A.

A.4.2.7.3 Procedure

Table A.4.2.7.3-1: MCVideo Transmission Request - Transmission Rejected

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCVideo Client) sends a Transmission Request message, with request for acknowledgement. (Note)	<--	Transmission Request	-	-
2	Check: Does the IUT (MCVideo Server) forward the Transmission Request message to the SS (MCVideo Server) providing the controlling function?	-->	Transmission Request	-	P
3	The SS (MCVideo Server) acknowledges the Transmission Request message.	<--	Transmission Control Ack	-	-
4	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
5	The SS (MCVideo Server) providing the controlling function sends a Transmission Rejected message with request for acknowledgement to IUT (MCVideo Server).	<--	Transmission Rejected	-	-
6	Check: Does the IUT (MCVideo Server) forward the Transmission Rejected message to SS-UE (MCVideo Client)? (Note)	-->	Transmission Rejected	-	P
7	The SS-UE (MCVideo Client) acknowledges the Transmission Rejected message. (Note)	<--	Transmission Control Ack	-	-
8	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCVideo Client) will be assumed.					

A.4.2.7.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications.

Table A.4.2.7.4-1: Transmission Request (Steps 1 and 2, Table A.4.2.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.1.1-1, with condition ACK

Table A.4.2.7.4-2: Transmission Rejected (Steps 5 and 6, Table A.4.2.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.2-1, with condition ACK

Table A.4.2.7.4-3: Transmission Control Ack (Steps 3 and 4, Table A.4.2.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition DOWNLINK

Table A.4.2.7.4-4: Transmission Control Ack (Steps 7 and 8, Table A.4.2.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

A.4.2.8 MCVideo Transmission Request - Queue Position Info

A.4.2.8.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.2.8.2 Definition of system information messages

N/A.

A.4.2.8.3 Procedure

Table A.4.2.8.3-1: MCVideo Transmission Request - Queue Position Info

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCVideo Client) sends a Transmission Request message, with request for acknowledgement. (Note)	<--	Transmission Request	-	-
2	Check: Does the IUT (MCVideo Server) forward the Transmission Request message to the SS (MCVideo Server) providing the controlling function?	-->	Transmission Request	-	P
3	The SS (MCVideo Server) acknowledges the Transmission Request message.	<--	Transmission Control Ack	-	-
4	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
5	The SS (MCVideo Server) providing the controlling function sends a Queue Position Info message with request for acknowledgement to IUT (MCVideo Server).	<--	Queue Position Info	-	-
6	Check: Does the IUT (MCVideo Server) forward the Queue Position Info message to SS-UE (MCVideo Client)? (Note)	-->	Queue Position Info	-	P
7	The SS-UE (MCVideo Client) acknowledges the Queue Position Info message. (Note)	<--	Transmission Control Ack	-	-
8	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCVideo Client) will be assumed.					

A.4.2.8.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications.

Table A.4.2.8.4-1: Transmission Request (Steps 1 and 2, Table A.4.2.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.1.1-1, with condition ACK

Table A.4.2.8.4-2: Queue Position Info (Steps 5 and 6, Table A.4.2.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.6-1, with condition ACK

Table A.4.2.8.4-3: Transmission Control Ack (Steps 3 and 4, Table A.4.2.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition DOWNLINK

Table A.4.2.8.4-4: Transmission Control Ack (Steps 7 and 8, Table A.4.2.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

A.4.2.9 MCVideo Queue Position Request - Queue Position Info

A.4.2.9.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.2.9.2 Definition of system information messages

N/A.

A.4.2.9.3 Procedure

Table A.4.2.9.3-1: MCVideo Transmission Request - Queue Position Info

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCVideo Client) sends a Queue Position Request message, with request for acknowledgement. (Note)	<--	Queue Position Request	-	-
2	Check: Does the IUT (MCVideo Server) forward the Queue Position Request message to the SS (MCVideo Server) providing the controlling function?	-->	Queue Position Request	-	P
3	The SS (MCVideo Server) acknowledges the Queue Position Request message.	<--	Transmission Control Ack	-	-
4	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
5	The SS (MCVideo Server) providing the controlling function sends a Queue Position Info message with request for acknowledgement to IUT (MCVideo Server).	<--	Queue Position Info	-	-
6	Check: Does the IUT (MCVideo Server) forward the Queue Position Info message to SS-UE (MCVideo Client)? (Note)	-->	Queue Position Info	-	P
7	The SS-UE (MCVideo Client) acknowledges the Queue Position Info message. (Note)	<--	Transmission Control Ack	-	-
8	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCVideo Client) will be assumed.					

A.4.2.9.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications.

Table A.4.2.9.4-1: Queue Position Request (Steps 1 and 2, Table A.4.2.9.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.1.3-1, with condition ACK

Table A.4.2.9.4-2: Queue Position Info (Steps 5 and 6, Table A.4.2.9.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.6-1, with condition ACK

Table A.4.2.9.4-3: Transmission Control Ack (Steps 3 and 4, Table A.4.2.9.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition DOWNLINK

Table A.4.2.9.4-4: Transmission Control Ack (Steps 7 and 8, Table A.4.2.9.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

A.4.3 Generic test procedures for MCVideo Terminating Participating Server function

A.4.3.1 MCVideo CT session establishment/modification without provisional responses other than 100 Trying

A.4.3.1.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.3.1.2 Definition of system information messages

N/A.

A.4.3.1.3 Procedure

Table A.4.3.1.3-1: MCVideo CT session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCVideo Server) providing the controlling function sends a SIP INVITE requesting the establishment/modification of a call initiated by SS-UE1 (MCVideo Client), to the IUT (MCVideo Server) providing the terminating participating function.	<--	SIP INVITE	-	-
2	Check: Does the IUT (MCVideo Server) forward the SIP INVITE request to establish/modify a call with SS-UE2 (MCVideo Client)?	-->	SIP INVITE	-	P
3	The SS-UE2 (MCVideo Client) sends a SIP 100 Trying to the IUT (MCVideo Server).	<--	SIP 100 (Trying)	-	-
4	Check: Does the IUT (MCVideo Server) forward the SIP 100 Trying to the SS (MCVideo Server) providing the controlling function?	-->	SIP 100 (Trying)	-	P
5	The SS-UE2 (MCVideo Client) responds with a SIP 200 (OK).	<--	SIP 200 (OK)	-	-
6	Check: Does the IUT (MCVideo Server) forward the SIP 200 (OK) to the SS (MCVideo Server) providing the controlling function?	-->	SIP 200 (OK)	-	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
7	The SS (MCVideo Server) providing the controlling function sends a SIP ACK to acknowledge the session establishment/modification to the IUT (MCVideo Server).	<--	SIP ACK	-	-
8	Check: Does the IUT (MCVideo Server) forward the SIP ACK to acknowledge the session establishment/modification to the SS-UE2 (MCVideo Client)?	-->	SIP ACK	-	P
-	EXCEPTION: Step 9a1-9a4 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is granted to SS-UE1 (MCVideo Client).	-	-	-	-
9a1	The SS (MCVideo Server) providing the controlling function sends a Media Transmission Notification message with request for acknowledgement to the IUT (MCVideo Server), to inform SS-UE2 (MCVideo Client) that the transmission request has been granted to SS-UE1 (MCVideo Client)?	<--	Media Transmission Notification	-	-
9a2	Check: Does the IUT (MCVideo Server) forward the Media Transmission Notification message to SS-UE2 (MCVideo Client)?	-->	Media Transmission Notification	-	P
9a3	The SS-UE2 (MCVideo Client) sends a Transmission Control ACK message.	<--	Transmission Control Ack	-	-
9a4	Check: Does the IUT (MCVideo Server) forward the acknowledgement of SS-UE2 (MCVideo Client) to SS (MCVideo Server)?	-->	Transmission Control Ack	-	P
-	EXCEPTION: Steps 10a1-10a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is not granted to SS-UE1 (MCVideo Client). (NOTE 1)	-	-	-	-
10a1	The SS (MCVideo Server) providing the controlling function sends a Transmission Idle message to the IUT (MCVideo Server).	<--	Transmission Idle	-	-
10a2	Check: Does the IUT (MCVideo Server) forward the Transmission Idle message to the SS-UE2 (MCVideo Client)?	-->	Transmission Idle	-	P

A.4.3.1.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCVIDEO and in the test case calling the procedure with the following clarifications.

Table A.4.3.1.4-1: SIP INVITE from SS (MCVideo Server) (Step 1, Table A.4.3.1.3-1)

Derivation Path: Table A.4.1.1.4-4

Table A.4.3.1.4-2: SIP INVITE from IUT (MCVideo Server) (Step 2, Table A.4.3.1.3-1)

Derivation Path: Table A.3.1.4-4

Table A.4.3.1.4-3: SIP 200 (OK) from SS-UE2 (MCVideo Client) (Step 5, Table A.4.3.1.3-1)

Derivation Path: Table A.3.1.4-7

Table A.4.3.1.4-4: SIP 200 (OK) from IUT (MCVideo Server) (Step 6, Table A.4.3.1.3-1)

Derivation Path: Table A.4.1.1.4-7

Table A.4.3.1.4-5: SIP ACK from SS (MCVideo Server) (Step 7, Table A.4.3.1.3-1)

Derivation Path: Table A.4.1.1.4-16

Table A.4.3.1.4-6: SIP ACK from IUT (MCVideo Server) (Step 8, Table A.4.3.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.1.2-1.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Max-Forwards			IETF RFC 3261 [7]	
value	any allowed value	Non-zero value		
NOTE: Elements whose values mention the UE shall be understood as if they were referring to the SS-UE2 (MCVideo Client), and SS as the IUT (MCVideo Server).				

Table A.4.3.1.4-7: Media Transmission Notification (Steps 9a1 and 9a2, Table A.4.3.1.3-1)

Derivation Path: Table A.4.1.1.4-12, with condition ACK

Table A.4.3.1.4-8: Transmission Idle (Steps 10a1 and 10a2, Table A.4.3.1.3-1)

Derivation Path: Table A.4.1.1.4-13

Table A.4.3.1.4-9: Transmission Control Ack (Steps 9a3 and 9a4, Table A.4.3.1.3-1)

Derivation Path: Table A.4.1.1.4-14

A.4.3.2 MCVideo CT private call establishment, manual commencement

A.4.3.2.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.3.2.2 Definition of system information messages

N/A.

A.4.3.2.3 Procedure

Table A.4.3.2.3-1: MCVideo CT private call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCVideo Server) providing the controlling function sends a SIP INVITE requesting the establishment of a private call from SS-UE1 (MCVideo Client) to SS-UE2 (MCVideo Client).	<--	SIP INVITE	-	-
2	Check: Does the IUT (MCVideo Server) forward the SIP INVITE request to establish a private call with SS-UE2 (MCVideo Client)?	-->	SIP INVITE	-	P
3	The SS-UE2 (MCVideo Client) sends a SIP 100 Trying to the IUT (MCVideo Server).	<--	SIP 100 (Trying)	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
4	Check: Does the IUT (MCVideo Server) forward the SIP 100 Trying to the SS (MCVideo Server) providing the controlling function?	-->	SIP 100 (Trying)	-	P
5	The SS-UE2 (MCVideo Client) sends a SIP 180 Ringing to the IUT (MCVideo Server).	<--	SIP 180 (Ringing)	-	-
6	Check: Does the IUT (MCVideo Server) forward the SIP 180 Ringing to the SS (MCVideo Server) providing the controlling function?	-->	SIP 180 (Ringing)	-	P
7	The SS-UE2 (MCVideo Client) responds with a SIP 200 (OK).	<--	SIP 200 (OK)	-	-
8	Check: Does the IUT (MCVideo Server) forward the SIP 200 (OK) to the SS (MCVideo Server) providing the controlling function?	-->	SIP 200 (OK)	-	P
9	The SS (MCVideo Server) providing the controlling function sends a SIP ACK to acknowledge the session establishment to the IUT (MCVideo Server).	<--	SIP ACK	-	-
10	Check: Does the IUT (MCVideo Server) forward the SIP ACK to acknowledge the session establishment to the SS-UE2 (MCVideo Client)?	-->	SIP ACK	-	P
-	EXCEPTION: Step 11a1-11a4 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is granted to SS-UE1 (MCVideo Client).	-	-	-	-
11a1	The SS (MCVideo Server) providing the controlling function sends a Media Transmission Notification message with request for acknowledgement to the IUT (MCVideo Server), to inform SS-UE2 (MCVideo Client) that the transmission request has been granted to SS-UE1 (MCVideo Client)?	<--	Media Transmission Notification	-	-
11a2	Check: Does the IUT (MCVideo Server) forward the Media Transmission Notification message to SS-UE2 (MCVideo Client)?	-->	Media Transmission Notification	-	P
11a3	The SS-UE2 (MCVideo Client) sends a Transmission Control ACK message.	<--	Transmission Control Ack	-	-
11a4	Check: Does the IUT (MCVideo Server) forward the acknowledgement of SS-UE2 (MCVideo Client) to SS (MCVideo Server)?	-->	Transmission Control Ack	-	P
-	EXCEPTION: Steps 12a1-12a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is not granted to SS-UE1 (MCVideo Client). (NOTE 1)	-	-	-	-
12a1	The SS (MCVideo Server) providing the controlling function sends a Transmission Idle message to the IUT (MCVideo Server).	<--	Transmission Idle	-	-
12a2	Check: Does the IUT (MCVideo Server) forward the Transmission Idle message to the SS-UE2 (MCVideo Client)?	-->	Transmission Idle	-	P

A.4.3.2.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCVIDEO and in the test case calling the procedure with the following clarifications.

Table A.4.3.2.4-1: SIP INVITE from SS (MCVideo Server) (Step 1, Table A.4.3.2.3-1)

Derivation Path: Table A.4.3.1.4-1, with conditions PRIVATE-CALL.
NOTE: Conditions are also applied to all body parts.

Table A.4.3.2.4-2: SIP INVITE from IUT (MCVideo Server) (Step 2, Table A.4.3.2.3-1)

Derivation Path: Table A.4.3.1.4-2, with conditions PRIVATE-CALL and MANUAL.
NOTE: Conditions are also applied to all body parts.

Table A.4.3.2.4-3: SIP 180 (Ringing) from SS-UE2 (MCVideo Client) (Step 5, Table A.4.3.2.3-1)

Derivation Path: Table A.3.3.4-8

Table A.4.3.2.4-4: SIP 180 (Ringing) from IUT (MCVideo Server) (Step 6, Table A.4.3.2.3-1)

Derivation Path: Table A.4.1.2.4-3

Table A.4.3.2.4-5: SIP 200 (OK) from SS-UE2 (MCVideo Client) (Step 7, Table A.4.3.2.3-1)

Derivation Path: Table A.4.3.1.4-3, with condition PRIVATE-CALL.
NOTE: Conditions are also applied to all body parts.

Table A.4.3.2.4-6: SIP 200 (OK) from IUT (MCVideo Server) (Step 8, Table A.4.3.2.3-1)

Derivation Path: Table A.4.3.1.4-4, with condition PRIVATE-CALL.
NOTE: Conditions are also applied to all body parts.

Table A.4.3.2.4-7: SIP ACK from SS (MCVideo Server) (Step 9, Table A.4.3.2.3-1)

Derivation Path: Table A.4.3.1.4-5

Table A.4.3.2.4-8: SIP ACK from IUT (MCVideo Server) (Step 10, Table A.4.3.2.3-1)

Derivation Path: Table A.4.3.1.4-6

Table A.4.3.2.4-9: Media Transmission Notification (Steps 11a1 and 11a2, Table A.4.3.2.3-1)

Derivation Path: Table A.4.3.1.4-7

Table A.4.3.2.4-10: Transmission Idle (Steps 12a1 and 12a2, Table A.4.3.2.3-1)

Derivation Path: Table A.4.3.1.4-8

Table A.4.3.2.4-11: Transmission Control Ack (Steps 11a3 and 11a4, Table A.4.3.2.3-1)

Derivation Path: Table A.4.3.1.4-9

A.4.3.3 MCVideo CT group/private call establishment, automatic commencement

A.4.3.3.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.3.3.2 Definition of system information messages

N/A.

A.4.3.3.3 Procedure

Table A.4.3.3.3-1: MCVideo CT group/private call establishment, automatic commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCVideo Server) providing the controlling function sends a SIP INVITE requesting the establishment of a call from SS-UE1 (MCVideo Client).	<--	SIP INVITE	-	-
2	Check: Does the IUT (MCVideo Server) forward the SIP INVITE request to establish a call with SS-UE2 (MCVideo Client)?	-->	SIP INVITE	-	P
3	The SS-UE2 (MCVideo Client) sends a SIP 100 Trying to the IUT (MCVideo Server).	<--	SIP 100 (Trying)	-	-
4	Check: Does the IUT (MCVideo Server) forward the SIP 100 Trying to the SS (MCVideo Server) providing the controlling function?	-->	SIP 100 (Trying)	-	P
-	EXCEPTION: Steps 5a1 to 5b3 describe behaviour that depends on the IUT (MCVideo Server) implementation; the "lower case letter" identifies a step sequence that may take place if the IUT (MCVideo Server) responds reliably or unreliably to a SIP INVITE with a SIP 183 (Session Progress)	-	-	-	-
5a1	Check: Does the IUT (MCVideo Server) send a SIP 183 (Session Progress) unreliably to the SS (MCVideo Server) providing the controlling function?	-->	SIP 183 (Session Progress)	-	P
5b1	Check: Does the IUT (MCVideo Server) send a SIP 183 (Session Progress) reliably to the SS (MCVideo Server) providing the controlling function?	-->	SIP 183 (Session Progress)	-	P
5b2	The SS (MCVideo Server) acknowledges the receipt of SIP 183 (Session Progress).	<--	PRACK	-	-
5b3	Check: Does the IUT (MCVideo Server) respond with SIP 200 (OK) to PRACK message?	-->	SIP 200 (OK)	-	P
6	The SS-UE2 (MCVideo Client) responds with a SIP 200 (OK) to SIP INVITE message.	<--	SIP 200 (OK)	-	-
7	Check: Does the IUT (MCVideo Server) forward the SIP 200 (OK) to the SS (MCVideo Server) providing the controlling function?	-->	SIP 200 (OK)	-	P
8	The SS (MCVideo Server) providing the controlling function sends a SIP ACK to acknowledge the session establishment to the IUT (MCVideo Server).	<--	SIP ACK	-	-
9	Check: Does the IUT (MCVideo Server) forward the SIP ACK to acknowledge the session establishment to the SS-UE2 (MCVideo Client)?	-->	SIP ACK	-	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
-	EXCEPTION: Step 10a1-10a4 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is granted to SS-UE1 (MCVideo Client).	-	-	-	-
10a1	The SS (MCVideo Server) providing the controlling function sends a Media Transmission Notification message with request for acknowledgement to the IUT (MCVideo Server), to inform SS-UE2 (MCVideo Client) that the transmission request has been granted to SS-UE1 (MCVideo Client)?	<--	Media Transmission Notification	-	-
10a2	Check: Does the IUT (MCVideo Server) forward the Media Transmission Notification message to SS-UE2 (MCVideo Client)?	-->	Media Transmission Notification	-	P
10a3	The SS-UE2 (MCVideo Client) sends a Transmission Control ACK message.	<--	Transmission Control Ack	-	-
10a4	Check: Does the IUT (MCVideo Server) forward the acknowledgement of SS-UE2 (MCVideo Client) to SS (MCVideo Server)?	-->	Transmission Control Ack	-	P
-	EXCEPTION: Steps 11a1-11a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is not granted to SS-UE1 (MCVideo Client).	-	-	-	-
11a1	The SS (MCVideo Server) providing the controlling function sends a Transmission Idle message to the IUT (MCVideo Server).	<--	Transmission Idle	-	-
11a2	Check: Does the IUT (MCVideo Server) forward the Transmission Idle message to the SS-UE2 (MCVideo Client)?	-->	Transmission Idle	-	P

A.4.3.3.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCVIDEO and in the test case calling the procedure with the following clarifications.

Table A.4.3.3.4-1: SIP INVITE from SS (MCVideo Server) (Step 1, Table A.4.3.3.3-1)

Derivation Path: Table A.4.3.1.4-1.

Table A.4.3.3.4-2: SIP INVITE from IUT (MCVideo Server) (Step 2, Table A.4.3.3.3-1)

Derivation Path: Table A.4.3.1.4-2.

Table A.4.3.3.4-3: SIP 183 (Session Progress) from IUT (MCVideo Server) (Step 5a1, Table A.4.3.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.3.2-1, with conditions MCVIDEO.

Table A.4.3.3.4-4: SIP 183 (Session Progress) from IUT (MCVideo Server) (Step 5b1, Table A.4.3.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.3.2-1, with conditions MCVIDEO and 100rel

Table A.4.3.3.4-5: SIP PRACK from SS (MCVideo Server) (Step 5b2, Table A.4.3.3.3-1)

Derivation Path: Table A.4.1.4.4-4

Table A.4.3.3.4-6: SIP 200 (OK) to PRACK from IUT (MCVideo Server) (Step 5b3, Table A.4.3.3.3-1)

Derivation Path: Table A.4.1.4.4-5.

Table A.4.3.3.4-7: SIP 200 (OK) from SS-UE2 (MCVideo Client) (Step 6, Table A.4.3.3.3-1)

Derivation Path: Table A.4.3.1.4-3

Table A.4.3.3.4-8: SIP 200 (OK) from IUT (MCVideo Server) (Step 7, Table A.4.3.3.3-1)

Derivation Path: Table A.4.3.1.4-4

Table A.4.3.3.4-9: SIP ACK from SS (MCVideo Server) (Step 8, Table A.4.3.3.3-1)

Derivation Path: Table A.4.3.1.4-5

Table A.4.3.3.4-10: SIP ACK from IUT (MCVideo Server) (Step 9, Table A.4.3.3.3-1)

Derivation Path: Table A.4.3.1.4-6

Table A.4.3.3.4-11: Media Transmission Notification (Steps 10a1 and 10a2, Table A.4.3.3.3-1)

Derivation Path: Table A.4.3.1.4-7

Table A.4.3.3.4-12: Transmission Idle (Steps 11a1 and 11a2, Table A.4.3.3.3-1)

Derivation Path: Table A.4.3.1.4-8

Table A.4.3.3.4-13: Transmission Control Ack (Steps 10a3 and 10a4, Table A.4.3.3.3-1)

Derivation Path: Table A.4.3.1.4-9

A.4.3.4 MCVideo CT group call establishment, manual commencement

A.4.3.4.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.3.4.2 Definition of system information messages

N/A.

A.4.3.4.3 Procedure

Table A.4.3.4.3-1: MCVideo CT group call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCVideo Server) providing the controlling function sends a SIP INVITE requesting the establishment of a group call.	<--	SIP INVITE	-	-
2	Check: Does the IUT (MCVideo Server) forward the SIP INVITE request to establish a group call with SS-UE2 (MCVideo Client)?	-->	SIP INVITE	-	P
3	The SS-UE2 (MCVideo Client) sends a SIP 100 Trying to the IUT (MCVideo Server).	<--	SIP 100 (Trying)	-	-
4	Check: Does the IUT (MCVideo Server) forward the SIP 100 Trying to the SS (MCVideo Server) providing the controlling function?	-->	SIP 100 (Trying)	-	P
5	The SS-UE2 (MCVideo Client) sends a SIP 183 (Session Progress) reliably to the IUT (MCVideo Server).	<--	SIP 183 (Session Progress)	-	-
6	Check: Does the IUT (MCVideo Server) forward the SIP 183 (Session Progress) message to the SS (MCVideo Server) providing the controlling function?	-->	SIP 183 (Session Progress)	-	P
7	The SS (MCVideo Server) acknowledges the receipt of SIP 183 (Session Progress).	<--	PRACK	-	-
8	Check: Does the IUT (MCVideo Server) forward the SIP PRACK message to SS-UE2 (MCVideo Client)?	-->	PRACK	-	P
9	The SS-UE2 (MCVideo Client) responds with SIP 200 (OK) to PRACK message.	<--	SIP 200 (OK)	-	-
10	Check: Does the IUT (MCVideo Server) forward the SIP 200 (OK) to the SS (MCVideo Server) providing the controlling function?	-->	SIP 200 (OK)	-	P
11	The SS-UE2 (MCVideo Client) responds with a SIP 200 (OK) to SIP INVITE message.	<--	SIP 200 (OK)	-	-
12	Check: Does the IUT (MCVideo Server) forward the SIP 200 (OK) to the SS (MCVideo Server) providing the controlling function?	-->	SIP 200 (OK)	-	P
13	The SS (MCVideo Server) providing the controlling function sends a SIP ACK to acknowledge the session establishment to the IUT (MCVideo Server).	<--	SIP ACK	-	-
14	Check: Does the IUT (MCVideo Server) forward the SIP ACK to acknowledge the session establishment to the SS-UE2 (MCVideo Client)?	-->	SIP ACK	-	P
-	EXCEPTION: Step 15a1-15a4 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is granted to SS-UE1 (MCVideo Client).	-	-	-	-
15a1	The SS (MCVideo Server) providing the controlling function sends a Media Transmission Notification message with request for acknowledgement to the IUT (MCVideo Server), to inform SS-UE2 (MCVideo Client) that the transmission request has been granted to SS-UE1 (MCVideo Client)?	<--	Media Transmission Notification	-	-
15a2	Check: Does the IUT (MCVideo Server) forward the Media Transmission Notification message to SS-UE2 (MCVideo Client)?	-->	Media Transmission Notification	-	P
15a3	The SS-UE2 (MCVideo Client) sends a Transmission Control ACK message.	<--	Transmission Control Ack	-	-
15a4	Check: Does the IUT (MCVideo Server) forward the acknowledgement of SS-UE2 (MCVideo Client) to SS (MCVideo Server)?	-->	Transmission Control Ack	-	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
-	EXCEPTION: Steps 16a1-16a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the transmission is not granted to SS-UE1 (MCVideo Client).	-	-	-	-
16a1	The SS (MCVideo Server) providing the controlling function sends a Transmission Idle message to the IUT (MCVideo Server).	<--	Transmission Idle	-	-
16a2	Check: Does the IUT (MCVideo Server) forward the Transmission Idle message to the SS-UE2 (MCVideo Client)?	-->	Transmission Idle	-	P

A.4.3.4.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCVIDEO and in the test case calling the procedure with the following clarifications.

Table A.4.3.4.4-1: SIP INVITE from SS (MCVideo Server) (Step 1, Table A.4.3.4.3-1)

Derivation Path: Table A.4.3.1.4-1, with conditions GROUP-CALL.
NOTE 1: Conditions are also applied to all body parts. NOTE 2: For chat group calls CHAT-GROUP-CALL condition shall be used instead of GROUP-CALL.

Table A.4.3.4.4-2: SIP INVITE from IUT (MCVideo Server) (Step 2, Table A.4.3.4.3-1)

Derivation Path: Table A.4.3.1.4-2, with conditions GROUP -CALL and MANUAL.
NOTE 1: Conditions are also applied to all body parts. NOTE 2: For chat group calls CHAT-GROUP-CALL condition shall be used instead of GROUP-CALL.

Table A.4.3.4.4-3: SIP 183 (Session Progress) from SS-UE2 (MCVideo Client) (Step 5, Table A.4.3.4.3-1)

Derivation Path: Table A.3.2.4-7.

Table A.4.3.4.4-4: SIP 183 (Session Progress) from IUT (MCVideo Server) (Step 6, Table A.4.3.4.3-1)

Derivation Path: Table A.4.1.4.4-3

Table A.4.3.4.4-5: SIP PRACK from SS (MCVideo Server) (Step 7, Table A.4.3.4.3-1)

Derivation Path: Table A.4.1.4.4-4

Table A.4.3.4.4-6: SIP PRACK from IUT (MCVideo Server) (Step 8, Table A.4.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.10.2-1.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Max-Forwards			IETF RFC 3261 [7]	
Value	any allowed value	Non-zero value		
NOTE: Elements whose values mention the UE shall be understood as if they were referring to the SS-UE2 (MCVideo Client), and SS as IUT (MCVideo Server).				

Table A.4.3.4.4-7: SIP 200 (OK) to PRACK from SS-UE2 (MCVideo Client) (Step 9, Table A.4.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1

Table A.4.3.4.4-8: SIP 200 (OK) to PRACK from IUT (MCVideo Server) (Step 10, Table A.4.3.4.3-1)

Derivation Path: Table A.4.1.4.4-5.

Table A.4.3.4.4-9: SIP 200 (OK) from SS-UE2 (MCVideo Client) (Step 11, Table A.4.3.4.3-1)

Derivation Path: Table A.4.3.1.4-3, with condition GROUP -CALL.
NOTE: Conditions are also applied to all body parts.

Table A.4.3.4.4-10: SIP 200 (OK) from IUT (MCVideo Server) (Step 12, Table A.4.3.4.3-1)

Derivation Path: Table A.4.3.1.4-4, with condition GROUP -CALL.
NOTE: Conditions are also applied to all body parts.

Table A.4.3.4.4-11: SIP ACK from SS (MCVideo Server) (Step 13, Table A.4.3.4.3-1)

Derivation Path: Table A.4.3.1.4-5

Table A.4.3.4.4-12: SIP ACK from IUT (MCVideo Server) (Step 14, Table A.4.3.4.3-1)

Derivation Path: Table A.4.3.1.4-6

Table A.4.3.4.4-13: Media Transmission Notification (Steps 15a1 and 15a2, Table A.4.3.4.3-1)

Derivation Path: Table A.4.3.1.4-7

Table A.4.3.4.4-14: Transmission Idle (Steps 16a1 and 16a2, Table A.4.3.4.3-1)

Derivation Path: Table A.4.3.1.4-8

Table A.4.3.4.4-15: Transmission Control Ack (Steps 15a3 and 15a4, Table A.4.3.4.3-1)

Derivation Path: Table A.4.3.1.4-9

A.4.3.5 MCVideo CT call release

A.4.3.5.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.3.5.2 Definition of system information messages

N/A.

A.4.3.5.3 Procedure

Table A.4.3.5.3-1: MCVideo CO call release

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCVideo Server) providing the controlling function sends a SIP BYE to terminate the call.	<--	SIP BYE	-	-
2	Check: Does the IUT (MCVideo Server) forward the SIP BYE request to the SS-UE2 (MCVideo Client)?	-->	SIP BYE	-	P
3	The SS-UE2 (MCVideo Client) responds with a 200 (OK) to the IUT (MCVideo Server).	<--	SIP 200 (OK)	-	-
4	Check: Does the IUT (MCVideo Server) forward the SIP 200 (OK) response to the SS (MCVideo Server) providing the controlling function?	-->	SIP 200 (OK)	-	P

A.4.3.5.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCVIDEO and in the test case calling the procedure with the following clarifications.

Table A.4.3.5.4-1: SIP BYE from the SS (MCVideo Server) (Step 1, Table A.4.3.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1.
NOTE: References to UE in subclauses relate to the SS (MCVideo Server) providing the controlling function and SS references to the IUT (MCVideo Server).

Table A.4.3.5.4-2: SIP BYE from the IUT (MCVideo Server) (Step 2, Table A.4.3.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1
NOTE: References to UE in subclauses relate to the SS-UE2 (MCVideo Client) and SS references to the IUT (MCVideo Server).

Table A.4.3.5.4-3: SIP 200 OK from the SS-UE2 (MCVideo Client) (Step 3, Table A.4.3.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1.

Table A.4.3.5.4-4: SIP 200 OK from the IUT (MCVideo Server) (Step 4, Table A.4.3.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1.

A.4.3.6 MCVideo Media Transmission Notification - Receive Media Request

A.4.3.6.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.3.6.2 Definition of system information messages

N/A.

A.4.3.6.3 Procedure

Table A.4.3.6.3-1: MCVideo Media Transmission Notification - Receive Media Request

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCVideo Server) providing the controlling function sends a Media Transmission Notification message with request for acknowledgement to IUT (MCVideo Server).	<--	Media Transmission Notification	-	-
2	Check: Does the IUT (MCVideo Server) forward the Media Transmission Notification message to SS-UE (MCVideo Client)? (Note)	-->	Media Transmission Notification	-	P
3	The SS-UE (MCVideo Client) acknowledges the Media Transmission Notification message. (Note)	<--	Transmission Control Ack	-	-
4	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
5	The SS-UE (MCVideo Client) sends a Receive Media Request message, with request for acknowledgement. (Note)	<--	Receive Media Request	-	-
6	Check: Does the IUT (MCVideo Server) forward the Receive Media Request message to the SS (MCVideo Server) providing the controlling function?	-->	Receive Media Request	-	P
7	The SS (MCVideo Server) acknowledges the Receive Media Request message.	<--	Transmission Control Ack	-	-
8	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
9	The SS (MCVideo Server) providing the controlling function sends a Receive Media Response message with request for acknowledgement to IUT (MCVideo Server).	<--	Receive Media Response	-	-
10	Check: Does the IUT (MCVideo Server) forward the Receive Media Response message to SS-UE (MCVideo Client)? (Note)	-->	Receive Media Response	-	P
11	The SS-UE (MCVideo Client) acknowledges the Receive Media Response message. (Note)	<--	Transmission Control Ack	-	-
12	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE2 (MCVideo Client) will be assumed.					

A.4.3.6.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications.

Table A.4.3.6.4-1: Media Transmission Notification (Steps 1 and 2, Table A.4.3.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.7-1, with condition ACK				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
User Id of the Transmitting User		User ID of the user transmitting the media	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.6	
User Id of the Transmitting User	px_MCVideo_ID_User_A			
Audio SSRC of the Transmitting User	Media/Audio SSRC of remote client (Client A)	SSRC of the user transmitting the media	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.16	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.7-1, with condition ACK				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Functional Alias	Not present			
	px_MCVideo_ID_FA_A	functional alias URI of the transmitting user	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.21	FA

Table A.4.3.6.4-2: Receive Media Request (Steps 5 and 6, Table A.4.3.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.1.4-1, with condition ACK

Table A.4.3.6.4-3: Receive Media Response (Steps 9 and 10, Table A.4.3.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.8-1, with condition ACK

Table A.4.3.6.4-4: Transmission Control Ack (Steps 7 and 8, Table A.4.3.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition DOWNLINK

Table A.4.3.6.4-5: Transmission Control Ack (Steps 3, 4, 11 and 12, Table A.4.3.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

A.4.3.7 MCVideo Media Reception End Request - Response

A.4.3.7.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.3.7.2 Definition of system information messages

N/A.

A.4.3.7.3 Procedure

Table A.4.3.7.3-1: MCVideo Media Reception End Request - Response

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCVideo Client) sends a Media Reception End Request message, with request for acknowledgement. (Note)	<--	Media Reception End Request	-	-
2	Check: Does the IUT (MCVideo Server) forward the Media Reception End Request message to the SS (MCVideo Server) providing the controlling function?	-->	Media Reception End Request	-	P
3	The SS (MCVideo Server) acknowledges the Media Reception End Request message.	<--	Transmission Control Ack	-	-
4	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
5	The SS (MCVideo Server) providing the controlling function sends a Media Reception End Response message with request for acknowledgement to IUT (MCVideo Server).	<--	Media Reception End Response	-	-
6	Check: Does the IUT (MCVideo Server) forward the Media Reception End Response message to SS-UE (MCVideo Client)? (Note)	-->	Media Reception End Response	-	P
7	The SS-UE (MCVideo Client) acknowledges the Media Reception End Response message. (Note)	<--	Transmission Control Ack	-	-
8	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE2 (MCVideo Client) will be assumed.					

A.4.3.7.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications.

Table A.4.3.7.4-1: Media Reception End Request (Steps 1 and 2, Table A.4.3.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.3-1, with conditions ACK and UPLINK

Table A.4.3.7.4-2: Media Reception End Response (Steps 5 and 6, Table A.4.3.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.4-1, with conditions ACK and DOWNLINK

Table A.4.3.7.4-3: Transmission Control Ack (Steps 3 and 4, Table A.4.3.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition DOWNLINK

Table A.4.3.7.4-4: Transmission Control Ack (Steps 7 and 8, Table A.4.3.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

A.4.3.8 MCVideo Transmission End Notification - Idle

A.4.3.8.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.3.8.2 Definition of system information messages

N/A.

A.4.3.8.3 Procedure

Table A.4.3.8.3-1: MCVideo Transmission End Notification - Idle

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS (MCVideo Server) providing the controlling function sends a Transmission End Notify message with request for acknowledgement to IUT (MCVideo Server).	<--	Transmission End Notify	-	-
2	Check: Does the IUT (MCVideo Server) forward the Transmission End Notify message to SS-UE (MCVideo Client)? (Note)	-->	Transmission End Notify	-	P
3	The SS-UE (MCVideo Client) acknowledges the Transmission End Notify message. (Note)	<--	Transmission Control Ack	-	-
4	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
-	EXCEPTION: Steps 5-12 describe behaviour that depends on the test case calling this procedure; it identifies a step sequence that takes place if SS-UE2 has previously requested media reception by performing procedure 'MCVideo Media Transmission Notification - Receive Media Request', as described in Table A.4.3.6.3-1.	-	-	-	-
5	The SS (MCVideo Server) providing the controlling function sends a Media Reception End Request message with request for acknowledgement to IUT (MCVideo Server).	<--	Media Reception End Request	-	-
6	Check: Does the IUT (MCVideo Server) forward the Media Reception End Request message to SS-UE (MCVideo Client)? (Note)	-->	Media Reception End Request	-	P
7	The SS-UE (MCVideo Client) acknowledges the Media Reception End Request message. (Note)	<--	Transmission Control Ack	-	-
8	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
9	The SS-UE (MCVideo Client) sends a Media Reception End Response message, with request for acknowledgement. (Note)	<--	Media Reception End Response	-	-
10	Check: Does the IUT (MCVideo Server) forward the Media Reception End Response message to the SS (MCVideo Server) providing the controlling function?	-->	Media Reception End Response	-	P
11	The SS (MCVideo Server) acknowledges the Media Reception End Response message.	<--	Transmission Control Ack	-	-
12	Check: Does the IUT (MCVideo Server) forward the Transmission Control Ack message?	-->	Transmission Control Ack	-	P
13	The SS (MCVideo Server) providing the controlling function sends a Transmission Idle message to IUT (MCVideo Server).	<--	Transmission Idle	-	-
14	Check: Does the IUT (MCVideo Server) forward the Transmission Idle message to SS-UE (MCVideo Client)? (Note)	-->	Transmission Idle	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE2 (MCVideo Client) will be assumed.					

A.4.3.8.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications.

Table A.4.3.8.4-1: Transmission End Notify (Steps 1 and 2, Table A.4.3.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.15-1, with condition ACK				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
User Id of the Transmitting User		Carries the identity of the user whose media transmission has been released	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.6	
User Id of the Transmitting User	px_MCVideo_ID_User_A			
Audio SSRC of the Transmitting User	Media/Audio SSRC of remote client (Client A)	SSRC of the user transmitting the media	ETSI TS 124 581 [6] (3GPP TS 24.581) clause 9.2.3.16	

Table A.4.3.8.4-2: Media Reception End Request (Steps 5 and 6, Table A.4.3.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.3-1, with conditions ACK and DOWNLINK

Table A.4.3.8.4-3: Media Reception End Response (Steps 9 and 10, Table A.4.3.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.4-1, with conditions ACK and UPLINK

Table A.4.3.8.4-4: Transmission Control Ack (Steps 11 and 12, Table A.4.3.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition DOWNLINK

Table A.4.3.8.4-5: Transmission Control Ack (Steps 3, 4, 7 and 8, Table A.4.3.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.3.5-1, condition UPLINK

Table A.4.3.8.4-6: Transmission Idle (Steps 13 and 14, Table A.4.3.8.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.11.2.16-1

History

Version	Date	Status
V1.1.1	February 2026	Publication